MINDCRAFT

Research Constitution

How do humans become confident mathematical thinkers?

Identity transformation · Evidence · Red Team · Product systems · Experiments

This is not a pitch deck. This is the company’s research operating system.
Challenge assumptions. Label uncertainty. Kill weak arguments.
Source corpus: ~194,479 words · multi-chapter living lab

MindCraft Research Lab · ongoing evidence program

MindCraft Research Constitution v1

Status: Living operating document — not a pitch deck

Edition: v1.13 (Synthesizer consolidation of Parts XCII–XCIX into surviving doctrine)

Research question: How do humans become *confident mathematical thinkers*?

Product thesis under audit: The product is identity transformation, not mathematics delivery.

Last updated: 2026-08-07 (Synthesizer v1.13 — promote SAFE-EXPLAIN…SAFE-ROI)

Growth model: Core OS (this file) + `chapters/*.md` via `CHAPTER_MANIFEST.txt` → PDF

Scale intent: Multi-month densification toward 150–300 pages of *evidenced* material — never fluff

Epistemic rule: Every claim is labeled FACT / HYPOTHESIS / FOUNDER BELIEF / SPECULATION.

Deep-dive chapters currently mounted

Part	File	Focus
XXIV	`chapters/24_affect_anxiety_belonging.md`	Anxiety, WM, stereotype threat, belonging
XXV	`chapters/25_self_efficacy_narrative_identity.md`	Bandura sources, narrative identity, habit≠identity
XXVI	`chapters/26_cognitive_load_mastery_tutoring.md`	CLT, fading, mastery war, tutoring calibration
XXVII	`chapters/27_markets_parents_competition.md`	WTP, parents, competitive teardown
XXVIII	`chapters/28_history_meaning_math.md`	Invention-story thesis under trial
XXIX	`chapters/29_spacing_retrieval.md`	Spacing, retrieval, memory of competence
XXX	`chapters/30_attribution_helplessness.md`	Weiner attributions, helplessness, feedback language
XXXI	`chapters/31_flow_challenge_skill.md`	Flow, challenge–skill, difficulty design
XXXII	`chapters/32_parent_anxiety_transmission.md`	Maloney/Beilock parent anxiety pathway
XXXIII	`chapters/33_ai_tutors_trust_sycophancy.md`	AI trust / sycophancy; RT: Bastani■PWC, guards≠gain
XXXIV	`chapters/34_formal_causal_dag_identification.md`	FEI causal DAG; L0–L4 claim ladder; confounders
XXXV	`chapters/35_competitive_session_audits.md`	Khan/Duo/Brilliant/ChatGPT session mechanism audits
XXXVI	`chapters/36_equity_audit_story_worlds.md`	Equity audit of story worlds; whose history; tokenism kill

XXXVII	chapters/37_expectancy_value_eccles.md	Eccles SEVT; utility/cost/attainment → choice; EVT experiments
XXXVIII	chapters/38_goal_orientation.md	Mastery vs performance goal structures; appearance ban; GO experiments
XXXIX	chapters/39_interleaving_vs_blocking.md	Interleave vs block; strategy selection; IL experiments
XL	chapters/40_self_explanation_prompts.md	Self-explanation prompts; Chi/Renkl → coach UX; SE experiments
XLI	chapters/41_desirable_difficulties_x_anxiety.md	Desirable difficulties × anxiety; SAFE-DD; Bjork vs Ashcraft
XLII	chapters/42_social_comparison_leaderboards.md	Social comparison & leaderboards; SAFE-COMPARE; ranks help/hurt
XLIII	chapters/43_habit_formation_science.md	Habit formation (Wood/Clear); SAFE-HABIT; cue≠colonization
XLIV	chapters/44_intrinsic_motivation_killers.md	Intrinsic motivation killers (Deci); SAFE-REWARD; controlling≠informational
XLV	chapters/45_mathematical_resilience.md	Mathematical resilience (JW/Lee); SAFE-RESILIENCE; grit theater kill
XLVI	chapters/46_teacher_tutor_expectancy.md	Tutor expectancy (Pygmalion/Golem); SAFE-EXPECTANCY; trait-label ban
XLVII	chapters/47_sleep_stress_learning.md	Sleep/stress/learning; SAFE-TIMING; all-nighter & sleep-app kill
XLVIII	chapters/48_transfer_variation_theory.md	Transfer & variation theory; SAFE-TRANSFER; story≠far-transfer kill
XLIX	chapters/49_misconceptions_as_productive.md	Misconceptions as productive; SAFE-MISCON; soft-wrong / diagnostic wrongs
L	chapters/50_confidence_calibration.md	Confidence calibration; SAFE-CALIB; over/underconfidence; C4 hide-correctness

LI	`chapters/51_deliberate_practice_tutoring.md`	Deliberate practice in tutoring; SAFE-DP; Ericsson vs Hambrick; 45-min spine
LII	`chapters/52_community_of_practice.md`	Community of practice; SAFE-CoP; Lave/Wenger LPP; careful school transfer
LIII	`chapters/53_religion_ritual_light_touch.md`	Religion/ritual light-touch; SAFE-RITUAL; meaning without cult dynamics
LIV	`chapters/54_military_aviation_brief_debrief.md`	Military/aviation brief-debrief; SAFE-AAR; after-action reviews for sessions
LV	`chapters/55_sports_film_study_pedagogy.md`	Sports film-study pedagogy; SAFE-FILM; error clips → coach cards
LVI	`chapters/56_music_pedagogy_ladders.md`	Music pedagogy ladders; SAFE-MUSIC; scales → repertoire → recital
LVII	`chapters/57_chess_annotation_metacognition.md`	Chess annotation & metacognition; SAFE-ANNOT; postmortem UX
LVIII	`chapters/58_therapy_graded_exposure.md`	Therapy graded exposure; SAFE-EXPOSE; math anxiety ladders
LIX	`chapters/59_parent_wtp_experiments.md`	Parent WTP experiments; SAFE-WTP; CBC/conjoint + message tests
LX	`chapters/60_trust_after_ai_hallucination.md`	Trust after AI hallucination; SAFE-REPAIR; escalate-to-human
LXI	`chapters/61_measurement_of_math_identity.md`	Measurement of math identity; SAFE-IDMEASURE; scales + demand effects
LXII	`chapters/62_longitudinal_identity_change.md`	Longitudinal identity change; SAFE-LONGID; 8/26/52 week horizons
LXIII	`chapters/63_act_exam_pressure.md`	ACT/exam pressure; SAFE-EXAM; learn affect vs exam affect
LXIV	`chapters/64_multilingual_ell_math_identity.md`	Multilingual/ELL math identity; SAFE-ELL; language vs math load

LXV	`chapters/65_gender_math_stereotypes_update.md`	Gender & math stereotypes update; SAFE-GENDER; parity vs climate
LXVI	`chapters/66_socioeconomic_constraint_on_grit.md`	Socioeconomic constraint on grit; SAFE-STRUCTURE; structure > slogans
LXVII	`chapters/67_ontology_as_diagnosis_moat.md`	Ontology as diagnosis moat; SAFE-ONTOLOGY; state×graph×FEI×transfer
LXVIII	`chapters/68_human_in_the_loop_tutor_ops.md`	Human-in-the-loop tutor ops; SAFE-HITL; playbooks/QA/FEI training
LXIX	`chapters/69_privacy_student_affect_data.md`	Privacy & student affect data; SAFE-PRIVACY; ERT ban; check-in ethics
LXX	`chapters/70_constitution_meta_bayesian_lab.md`	Constitution meta; SAFE-LABMETA; falsify/check/demote lab process
LXXI	`chapters/71_peer_nearpeer_expert_tutoring.md`	Peer vs near-peer vs expert tutoring; SAFE-TUTORGRAIN; hire bar
LXXII	`chapters/72_generated_question_validity_key_risk.md`	Generated question validity & key risk; SAFE-GENQ; verify-before-ship
LXXIII	`chapters/73_district_procurement_privacy_gtm.md`	District procurement & privacy GTM; SAFE-PROCURE; trust packet / NDPA
LXXIV	`chapters/74_spaced_retrieval_schedules_product_ux.md`	Spaced retrieval schedules in product UX; SAFE-SCHED; ISI×RI / equal vs expanding
LXXV	`chapters/75_tutor_workforce_pipeline_quality_drift.md`	Tutor workforce pipeline & quality drift; SAFE-WORKFORCE; fidelity over tenure
LXXVI	`chapters/76_forgetting_curves_product_honesty.md`	Forgetting curves as product honesty; SAFE-FORGET; aged evidence / delayed probes
LXXVII	`chapters/77_adaptive_spacing_algorithms_vs_fixed_calendars.md`	Adaptive spacing vs fixed calendars; SAFE-ADAPT; calendar-first / banded personalization

LXXVIII	chapters/78_cram_products_vs_durable_identity.md	Cram products vs durable-identity positioning; SAFE-DURABLE; dual-rail GTM / no point guarantees
LXXIX	chapters/79_feedback_timing_immediate_vs_delayed.md	Feedback timing immediate vs delayed; SAFE-FBTIME; mode-conditional clocks / focus>timing
LXXX	chapters/80_bridge_gap_product_narrative.md	Bridge-gap product narrative; SAFE-BRIDGE; connection failures > isolated nodes
LXXXI	chapters/81_cold_start_without_fake_mastery.md	Cold-start without fake mastery; SAFE-COLD; humble prior + labeled seed + probes
LXXXII	chapters/82_format_axis_product_narrative.md	Format-axis product narrative; SAFE-FORMAT; FormatId gaps / conversion evidence
LXXXIII	chapters/83_parent_dashboard_honesty_ux.md	Parent dashboard honesty UX; SAFE-PDASH; proof age / MoC without shame
LXXXIV	chapters/84_tutor_talk_ratio_instrumentation.md	Tutor talk-ratio instrumentation; SAFE-TALK; construction > airtime
LXXXV	chapters/85_story_world_cognitive_load_vs_identity.md	Story-world cognitive load vs identity payoff; SAFE-STORYLOAD; CLT-budgeted wrap
LXXXVI	chapters/86_competence_evidence_social_proof.md	Competence evidence as social proof; SAFE-PROOF; witnessable solo MoC
LXXXVII	chapters/87_bank_coverage_honesty_marketing.md	Bank coverage honesty in marketing; SAFE-COVER; matrix > item-count
LXXXVIII	chapters/88_worked_example_fading_solver_ux.md	Worked-example fading in Solver UX; SAFE-FADE; completion → solo
LXXXIX	chapters/89_help_seeking_vs_help_abuse.md	Help-seeking vs help abuse; SAFE-HELP; instrumental > executive
XC	chapters/90_product_analytics_fei_north_stars.md	Product analytics for FEI North Stars; SAFE-INSTRUMENT; ship the four

XCI	`chapters/91_hint_economy_contingent_scaffolding.md`	Hint economy & contingent scaffolding; SAFE-HINT; soft→hard priced peeks
XCII	`chapters/92_explanation_length_germane_load.md`	Explanation length vs germane load; SAFE-EXPLAIN; short coach vs monologue
XCIII	`chapters/93_retrieval_failure_modes_practice_ux.md`	Retrieval failure modes; SAFE-RETRIEVE; TOT vs blank vs struggle bounds
XCIV	`chapters/94_productive_failure_vs_guided_success.md`	Productive failure vs guided success; SAFE-PF; generate→consolidate fidelity
XCV	`chapters/95_cognitive_apprenticeship_tutor_playbooks.md`	Cognitive apprenticeship in tutor playbooks; SAFE-APPRENTICE; model→coach→fade
XCVI	`chapters/96_attention_residue_device_distraction.md`	Attention residue & device distraction; SAFE-ATTN; attempt-window protection
XCVII	`chapters/97_dual_coding_diagram_formatid_load.md`	Dual coding & diagram FormatId load; SAFE-DUAL; contiguous structural figures
XCVIII	`chapters/98_metacognitive_monitoring_gap_scan.md`	Metacognitive monitoring in gap-scan; SAFE-MONITOR; named judgments → control
XCIX	`chapters/99_opportunity_cost_tutor_minutes_vs_ai.md`	Tutor minutes vs AI scaffold; SAFE-ROI; FEI per tutor-minute / Map minute budgets
C	`chapters/100_worked_example_vs_problem_solving_timing.md`	Worked-example vs PS timing across expertise; SAFE-EXPTIME (provisional); grain-local adaptive fade
CI	`chapters/101_self_regulated_learning_cycles_practice_ux.md`	SRL cycles (Zimmerman) in Practice UX; SAFE-SRL (provisional); phase→control micro-prompts
CII	`chapters/102_generative_learning_activities_practice.md`	Generative learning (Fiorella/Mayer) in Practice; SAFE-GENERATE (provisional); SOI-forcing micro-GLAs

Queued next: see `NEXT_LAB.md` (Part CIII error climate / psychological safety, then 104).

Synthesizer note (v1.13): Eight researcher chapters (XCII–XCIX) landed after v1.12 as provisional I.4 appends — explanation length vs germane load, retrieval failure modes, productive failure sequencing, cognitive apprenticeship playbooks, attention residue / device distraction, dual coding × FormatId, metacognitive monitoring in gap-scan, and tutor-minute vs AI ROI. This edition promotes each stack to company law (removes provisional tags), formalizes Red Team kills #46–#53, confirms EXPLAIN/RETRIEVE/PF/APPRENTICE/ATTN/DUAL/MONITOR/ROI experiment families in Part IX, and refreshes metrics/glossary/competitive implication. Deep-dive files remain authoritative for citations; the OS keeps only *surviving* product rules.

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0. How to read this document

This Constitution is the research OS for MindCraft. It exists to prevent the company from building features that feel right and fail quietly.

Rules of engagement:

1. Founder beliefs enter as **FOUNDER BELIEF**, not as law.
2. The Red Team's job is destruction. Unchallenged claims are invalid.
3. Page count is not quality. A shorter true section beats a longer vague one.
4. "AI will make explanations free" is a hypothesis about markets, not a moral slogan.
5. Product implications must map to experiments with falsifiers.

This v1 is dense and incomplete on purpose. Expanding toward 150–300 pages is allowed only when new evidence, frameworks, or contradicted priors justify the ink.

Part I — Executive Summary

I.1 The compressed answer

HYPOTHESIS (confidence: medium–high): Humans become confident mathematical thinkers when three conditions compound:

1. **Affective permission** — the nervous system stops treating math as social/physical threat.
2. **Competence evidence** — the learner accumulates undeniable, self-attributed successes at the right grain of difficulty — including *strategy selection* and *self-accounted why*, not only blocked accuracy.
3. **Identity re-storying** — the person revises the narrative "I am not a math person" into a workable self-story compatible with struggle — via expectancy × value × cost (Eccles), not posters.

Explanations alone rarely produce (1)–(3). Fluent AI explanations can *accelerate wrong confidence* (Bastani unguarded GPT; Part XXXIII). Explanations help (2) only *after* (1) and when the student still does the decisive generative work (Parts XXVI, XL).

I.2 Verdict on the scarcity thesis

FOUNDER BELIEF under audit: AI commoditizes knowledge; scarcity shifts to motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

Red Team verdict (updated v1.13):

Claim	Status	Notes
Explanations are getting cheaper/faster	**FACT** (directionally)	Generative AI + open content lowers marginal cost of explanation
“Tutoring is free”	**FALSE as stated**	Attention, accountability, diagnosis, calibrated trust remain scarce
Content is free	**MOSTLY FACT** for commodity content	Differentiation = curation, sequencing, trust, outcomes
Motivation/identity become scarce	**HYPOTHESIS**	Product differentiator; not proven sole moat — instrument E×V×Cost separately (XXXVII)
Emotional safety is first-order	**HYPOTHESIS with supportive evidence**	Ashcraft WM tax; segment, don’t romanticize softness
Unguarded AI chat improves solo learning	**KILLED**	Bastani: concurrent ↑ then solo exam ↓; guards neutralize harm ≠ prove gain
Bastani “GPT Tutor” ≡ MindCraft ontology wrap	**KILLED**	Category error (XXXIII RT)
Streaks / blocked fluency = mastery	**KILLED as North Star**	HID + XXXVIII/XXXIX; measure delayed mix + transfer
Harder-always / anxiety-blind “desirable difficulty”	**KILLED**	SAFE-DD (XLI); equip→destake→dose before Bjork dose
Ranks / XP / grit meters as identity engines	**KILLED**	SAFE-COMPARE / REWARD / RESILIENCE (XLII–XLV)
Story immersion ≡ far transfer; mistake posters ≡ learning	**KILLED**	SAFE-TRANSFER / MISCON (XLVIII–XLIX)
Raise confidence / Belief Score™ / CA→grades	**KILLED**	SAFE-CALIB (L); Foster: appropriate confidence, not inflate
Hours / forums / ritual→score / domain cosplay	**KILLED**	SAFE-DP / CoP / RITUAL (LI–LIII); borrow method not costume
Tip-flood / engine-dump / flooding ≡ pedagogy	**KILLED**	SAFE-AAR / FILM / MUSIC / ANNOT / EXPOSE (LIV–LVIII)

Likert≡WTP / survey$≡revenue / ACT-point packages	**KILLED**	SAFE-WTP (LIX); CBC + bias controls; message≠package
Never-hallucinates / apology≡repair / Identity Score™ / 8-week math person	**KILLED**	SAFE-REPAIR / IDMEASURE / LONGID (LX–LXII)
Day-one ACT flood / EL≡ability / pink track / grit-as-character	**KILLED**	SAFE-EXAM / ELL / GENDER / STRUCTURE (LXIII–LXVI)
Graph file / RAG chat ≡ diagnosis moat	**KILLED**	SAFE-ONTOLOGY (LXVII); moat = state×ontology×FEI×solo transfer
Warm human / hours / got-it ≡ tutor quality	**KILLED**	SAFE-HITL (LXVIII); Map-briefed prompt>pour + FEI QA
Emotion AI / Anxiety Score™ / FERPA≡ethics	**KILLED**	SAFE-PRIVACY (LXIX); ERT ban; compliance floor ≠ doctrine
Page-count science / Bayesian theater / cite-wash	**KILLED**	SAFE-LABMETA (LXX); synthesizer must demote
Ivy/expert résumé / peer Discord ≡ tutoring quality	**KILLED**	SAFE-TUTORGRAIN (LXXI); trained near-peer + hire-by-fidelity
LLM items ≡ shipped bank / item-count hero	**KILLED**	SAFE-GENQ (LXXII); verify-before-ship; drop-rate gate
FERPA badge / click-wrap / biometric school SKU ≡ GTM	**KILLED**	SAFE-PROCURE (LXXIII); privacy review = sales gate
Expanding SRS / perfect-interval AI / streak-as-spacing	**KILLED**	SAFE-SCHED (LXXIV); horizon-matched ISI×RI; equal-ish lags
Tutor headcount / tenure / wellness theater ≡ quality	**KILLED**	SAFE-WORKFORCE (LXXV); fidelity@tenure + coach-capacity hire gate
Mastery fireworks / Ebbinghaus Score™ ≡ durable competence	**KILLED**	SAFE-FORGET (LXXVI); aged evidence + delayed probes
SM-2/FSRS / black-box perfect-interval AI ≡ math identity	**KILLED**	SAFE-ADAPT (LXXVII); calendar-first; banded inspectable adapt
Cram / bootcamp / +points packages ≡ durable identity	**KILLED**	SAFE-DURABLE (LXXVIII); dual-rail GTM; no point guarantees
Instant-feedback AI / always-delay ≡ learning science	**KILLED**	SAFE-FBTIME (LXXIX); mode-conditional clocks; focus>timing

Bridge-count / re-explain ≡ connection mastery	**KILLED**	SAFE-BRIDGE (LXXX); remediate the join; measure join transfer
Day-one mastery fireworks / confidence ≡ mastery	**KILLED**	SAFE-COLD (LXXXI); humble prior + labeled seed + probes
Format-count / every-format / Format Personality™	**KILLED**	SAFE-FORMAT (LXXXII); conversion evidence; structural tags only
Tonight-% / streak / shame-rank parent portal ≡ trust	**KILLED**	SAFE-PDASH (LXXXIII); proof-age hero; autonomy>control
Talk-% / silence theater / got-it ≡ Socratic quality	**KILLED**	SAFE-TALK (LXXXIV); construction > airtime; prompt density
Immersion / lore / Story Engagement ≡ learning	**KILLED**	SAFE-STORYLOAD (LXXXV); CLT-budgeted structure wrap
User-count / star-wall / vague praise ≡ learning proof	**KILLED**	SAFE-PROOF (LXXXVI); dated solo competence artifacts
Item-count / complete-ACT / unsynced GENQ ≡ coverage	**KILLED**	SAFE-COVER (LXXXVII); matrix + gaps + use-tier
Always-full-worked / never-fade / solution-first ≡ learning	**KILLED**	SAFE-FADE (LXXXVIII); completion ladder + solo proof
Unlimited-hints / help-NPS / avoidance-as-grit ≡ learning	**KILLED**	SAFE-HELP (LXXXIX); instrumental contingent help + solo
DAU/streak/XP / FEI Score™ / thumbs-up ≡ learning NS	**KILLED**	SAFE-INSTRUMENT (XC); FEI four + XXI.4 co-gates
Unlimited free hard peeks / Hint Score™ / XP-for-hints ≡ learning	**KILLED**	SAFE-HINT (XCI); soft-first contingent + construction cost
Longer≡better / Explanation Score™ / monologue≡SE ≡ learning	**KILLED**	SAFE-EXPLAIN (XCII); short principle-first + SE-before-wrap
All-struggle≡productive / Struggle Score™ / blank-time / never-stuck ≡ learning	**KILLED**	SAFE-RETRIEVE (XCIII); mode-contingent stall UX
Every-miss≡PF / fail-first-always / PF Score™ / discovery-without-consolidation ≡ learning	**KILLED**	SAFE-PF (XCIV); fidelity-gated generate→consolidate
Guild cosplay / modeling≡lecture / never-fade / Apprenticeship Score™ ≡ learning	**KILLED**	SAFE-APPRENTICE (XCV); phase-fidelity model→coach→fade

Focus Score™ / phone-shame / mid-attempt DAU pushes / multitasking-as-talent ≡ learning	**KILLED**	SAFE-ATTN (XCVI); attempt-window protection + tech breaks
Dual Coding Score™ / always-add-picture / decoration≡dual-coding / visual-learner meshing / animation-default	**KILLED**	SAFE-DUAL (XCVII); contiguous structural figures + format-hop proof
Monitoring Score™ / metacognition-without-control / easy-only≡calibration / one-scan calibration	**KILLED**	SAFE-MONITOR (XCVIII); named judgments → miss-class control
AI-replaces-tutors / tutoring-is-free / unlimited-human / hours NS / Session ROI Score™	**KILLED**	SAFE-ROI (XCIX); Map minute budgets + FEI per tutor-minute
Universal E0→E3 destiny / fixed≡adaptive / confidence≡fade stage / Example Timing Score™	**KILLED** *(provisional)*	SAFE-EXPTIME (C); grain-local adaptive example↔solve timing
SRL Score™ / reflection-streak / survey≡event / streak≡SRL / forethought-without-control	**KILLED** *(provisional)*	SAFE-SRL (CI); phase→control cycles + FEI proof
Generative Learning Score™ / AI≡generation / teach-for-XP / eight-menu GLA bomb	**KILLED** *(provisional)*	SAFE-GENERATE (CII); SOI-forcing sparse GLAs + transfer proof

Implication: Do not bet on "better explanations," graph-file cosplay, warm-tutor theater, emotion cameras, page-count science, Ivy hire theater, unverified AI banks, FERPA-badge GTM, expanding-SRS mystique, headcount vanity, mastery fireworks, FSRS brand, cram packages, instant-feedback theater, bridge-count ads, day-one greens, format-count cosplay, parent surveillance portals, Talk Ratio theater, immersive lore-as-pedagogy, star-wall / user-count learning proof, item-count / complete-ACT coverage theater, always-full-worked / unlimited-solutions Solver theater, unlimited-hints / never-stuck answer-dump theater, DAU/streak/XP / FEI Score™ dashboard theater, unlimited free hard peeks / Hint Score™ theater, longer≡better / unlimited thorough AI essay theater, struggle-theater / blank-time / always-wait / instant-dump stall UX, fail-first / every-miss-as-PF / discovery-without-consolidation cosplay, guild / master-tutor / modeling-as-lecture / never-fade apprenticeship costume, Focus Score™ / phone-shame / mid-attempt notification-growth theater, or Monitoring Score™ / metacognition-without-control / one-scan calibration theater, or AI-replaces-tutors / tutoring-is-free / unlimited-human / hours-booked / Session ROI Score™ theater. Bet on **FEI + pedagogy wrap + SAFE-* stack**: fear→evidence→identity, with inspectable diagnosis, Map-briefed humans, privacy-bound affect, verified keys, blueprint-honest coverage matrices, district trust packets, honest spaced returns, fidelity-over-tenure ops, time-honest MoC, dual-rail durable GTM, load-honest story wrap, witnessable dated solo competence artifacts, guidance that fades as the student proves the join, instrumental help that still demands construction, instrumented `retry_120s` / motive-coded `challenge_accept` / `transfer_pass` / `solo_transfer_pass` under XXI.4 co-gates, contingent soft→hard peeks priced in construction effort, length-contingent principle coaches that leave room to think, mode-contingent retrieval support (TOT cue vs freeze scaffold) without Struggle Score™, fidelity-gated productive-failure missions (generate then consolidate on student RSMs) only when priors and affect allow, tutor/coach cognitive-apprenticeship phases (model→coach→fade + articulation) logged for fidelity rather than guild branding, and attempt windows protected from residue and device tax rather than Focus Score™ costume.

I.3 What to optimize (North Star debate)

Metric	Predicts short-term engagement?	Predicts durable math identity?	Risk
Correct answers (esp. blocked)	Medium	Low–medium	Fluency illusion; gaming
Time spent / chat tokens	High	Low	Engagement / AI-monologue theater
Streaks / DAU	High	Uncertain	Habit ≠ identity (HID)
Confidence (self-report)	Medium	Medium	Cheap talk; overtrust after fluent AI
Challenge-seeking (`challenge_accept` + *why*)	Medium	**High**	Must code mastery vs appearance/normative motive (XXXVIII)
Voluntary return after failure (`retry_120s`)	Medium	**High**	Needs coach content discipline (SE prompts)
Delayed mixed / `transfer_pass` / `solo_transfer_pass`	Medium	**High**	Anti-fake-mastery; anti-AI-crutch
"I am a math person" + behavior	Medium	**Highest target**	Slow, noisy; L3+ claim only

Working North Star (HYPOTHESIS):

Challenge-seeking under safety with transfer — the student chooses a harder problem for a *mastery* reason, returns after a miss, can pick the strategy on a mixed delayed set, and attributes success to strategy — without needing the model to finish the thought.

I.4 Surviving doctrine stack (Synthesizer merge through XCIX)

Duplicate frameworks collapsed. Deep dives own citations; this table is **company law until killed**.

Layer	Surviving rule	Source parts	Commercial use
FEI / SAFE-CRAFT	Safety → attempt → informative miss → earned win → attribution → transfer	XVI, VIII	Session demo in marketing; not feature matrix
Claim ladder L0–L4	Do not sell identity (L3) from session A/Bs (L1)	XXXIV	Parent/copy hygiene; pre-reg estimands

PWC (pedagogy wrap constraint)	Guarded AI + ontology spine; success = `solo_transfer_pass` ≥ no-AI	XXXIII, XXXV	Ban "we're Bastani's Tutor"; ban unguarded Solver hero
E×V×Cost	Instrument expectancy, value, cost separately; student-authored utility > staff utility dump	XXXVII	Kill single "motivation" KPI
TARGET / mastery climate	Default-ban appearance UX + streak-as-North-Star; soft-wrong = mastery evaluation structure	XXXVIII	CTA copy audit; appearance climate kill
SAFE-DD	Equip → destake → dose → frame → measure; desirable difficulty only after WM/threat capacity	XLI	Ban hardness-as-virtue; gate IL/SE by anxiety segment
SAFE-COMPARE	Criterion feedback default; peer ranks opt-in; no forced infinite named leaderboards for Maya	XLII	Leaderboard / league kill as growth engine
SAFE-HABIT	Cue practice not the counter; miss ≠ identity failure; habit metrics subordinate to FEI	XLIII	Streak colonization kill; if-then onboarding OK
SAFE-REWARD	No expected tokens for mere engagement; informational feedback > inducement; transfer-gated unlocks	XLIV	XP-as-love / coin-for-open kill
SAFE-RESILIENCE	Growth-zone challenge + support recruitment; ban grit NS, danger-zone-as-DD, Resilience Score™	XLV	Grit-poster / character-meter kill
SAFE-EXPECTANCY	Task-only tutor briefs; ban trait/weakness labels; CIOF mediation; cold-start scripts	XLVI	Pygmalion marketing kill; tutor ethics

SAFE-TIMING	Encode → sleep → transfer claim; never reward midnight volume; stress/restriction-aware load; pressure ≠ DD	XLVII	All-nighter / sleep-app brand kill
SAFE-TRANSFER	Name the hop; VT vary-on-purpose; hug exam + bridge abstraction; story ≠ transfer; blocked ≠ ready	XLVIII	Narrative-theater / blocked-accuracy vanity kill
SAFE-MISCON	Elicit → classify → destake → springboard → route; soft-wrong ≠ empty kindness; confidence-tiered feedback	XLIX	Mistake-poster / red-X-as-rigor / equal-wrong graph kill
SAFE-CALIB	Item-level confidence → miss class → tiered feedback; raise *appropriate* confidence, not Belief Score™	L	Hide-correctness diagnostic; ban CA→grades marketing
SAFE-DP	Diagnose → isolate → goal → stretch → attempt → method feedback → revise → prove → stop	LI	Ban 10k-hours / minutes / coverage theater as DP
SAFE-CoP	Name the practice; peripheral = real work; old-timer required; trajectory > community tab	LII	Ban Discord/streak ≡ LPP; “join the practice” not “join the community”
SAFE-RITUAL	Light, optional, practice-pointing marks; doctrinal default; anti-totalism; no ritual→score	LIII	Ban streak-as-liturgy; family/order milieu copy
SAFE-AAR	Brief → attempt with traces → four-question student-first close → one improve-how	LIV	Ban bootcamp brand; +25% ACT from debrief metas

SAFE-FILM	Objective clip + ≤3 cues + learner-timed review + self∪model; next-attempt commitment	LV	Ban tip-flood film room; sports-academy cosplay
SAFE-MUSIC	Scales (goalful ingredients) → chosen repertoire → transfer-as-recital proof	LVI	Ban virtuoso cosplay; hours KPI; bare-pass theater
SAFE-ANNOT	Annotate-before-reveal ; principle tags; forced critical restudy; classify→drill; annotate wins	LVII	Ban GM cosplay; engi ne-dump≡coaching; rating NS
SAFE-EXPOSE	Situation hierarchy; destaked approach; expectancy violation; fade safety behaviors; multi-context	LVIII	Ban therapy branding; flooding; Calm Score™; one-session cure
SAFE-WTP	Parent CBC with price+neither; ship-mapped attributes; message≠package; cash pilot before stickers	LIX	Ban Likert≡WTP; survey$≡revenue; ACT-point SKUs
SAFE-REPAIR	Detect → classify → own explanatorily → Map-checked correct → destaked re-attempt → triggered escalate	LX	Ban never-hallucinates; rote apology≡repair; infinite free escalate
SAFE-IDMEASURE	Validated Cribbs-factor battery + FEI behavioral co-primary + demand-aware admin; narrative probe light	LXI	Ban Identity Score™; two-item≡FEI; splash Likert; empty “math person” praise
SAFE-LONGID	Stage claims: 8w process / 26w triangulated movement / 52w persistence; fade-aware packages	LXII	Ban math-person-in-8-weeks; end-of-program Likert≡durable
SAFE-EXAM	Dual rail MA+TA; destake-to-learn / dose-to-prove; readiness = delayed mixed under stakes	LXIII	Ban day-one ACT flood; blocked≡exam-ready; ACT guarantees; Calm Score™ exam NS

SAFE-ELL	Separate language vs math load; linguistic modify without math dilute; format triangulation before verdict	LXIV	Ban math-is-language-free; forever procedure ELL track; translate≡identity
SAFE-GENDER	Parity hygiene; cue hygiene; belonging via practice recognition; no ST→score / pink cosplay	LXV	Ban innate-gap premise; threat-theater ACT; girl-SKU / pink track
SAFE-STRUCTURE	Structure/scaffolds before grit slogans; bandwidth hygiene; behavior≠trait; no character-blame equity	LXVI	Ban grit-NS / Grit Score™ / poor-kids-need-grit / streak-as-character
SAFE-ONTOLOGY	Diagnosis-before-dialogue; inspectable state; LLM bookends; moat = state×ontology×FEI×transfer not graph file	LXVII	Ban node-count moat / RAG≡KT / fluency-as-diagnosis / ITS-2σ / smarter-than-ChatGPT
SAFE-HITL	Map-briefed tutors; prompt>pour; answer evidence not "got it?"; complementarity; coaching cycles; FEI QA; packetized escalate	LXVIII	Ban warm≡quality / hours NS / Discord≡QA / Ivy≡playbook / explain-first / humans-skip-Map
SAFE-PRIVACY	Self-authored short-TTL affect → pedagogy modifiers only; ERT ban; contextual integrity; refusal-safe	LXIX	Ban face/voice emotion AI / Anxiety Score™ / FERPA≡ethics / mood gates / sell-affect
SAFE-LABMETA	Label → check → kill/wound/survive → demote copy; small FEI hard core; synthesizer must delete; page count vanity	LXX	Ban page-count NS / Bayesian theater / cite-wash / belief-by-repetition / science-backed ads without falsifiers

SAFE-TUTORGRAIN	Default trained near-peer; structure>pairing; no 2σ; Map>credential diagnosis; suppress knowledge-telling; expert=escalate	LXXI	Ban Ivy/PhD≡FEI / Discord-peer≡tutoring / explainers hire bar / expertise≡diagnosis / tutor-learning NS
SAFE-GENQ	Verify-before-ship; key fails hard-fail; Kane IUA for bank uses; drop-rate gate; coverage≠NS; LLM≠classical AIG without constraints	LXXII	Ban LLM-items≡shipped / item-count hero / unverified diagnostic keys / fluency≡keyed correctness / scale-at-30%-drop
SAFE-PROCURE	Privacy review = GTM gate; NDPA-ready DPA; Exhibit A honesty; free=paid approval; no bio / no pupil-data ads	LXXIII	Ban FERPA-badge≡ready / click-wrap school path / biometric school SKU / marketing-first without packet
SAFE-SCHED	Horizon-matched ISI×RI; delayed first return; equal-ish lags for long RI; generation Returns; exam dual rail; no streak-as-spacing	LXXIV	Ban expanding-SRS hero / perfect-interval AI / cram≡ready / overlearn-tonight / restudy≡review / shuffle≡spacing
SAFE-WORKFORCE	Fidelity-over-tenure; continuous coaching; coach-capacity gates hiring; burnout→load redesign; named drift channels; retention≠NS	LXXV	Ban headcount/hours/retention-% NS / tenure≡quality / onboarding≡coaching / wellness-theater burnout / Burnout Score™ / Ivy-bench pipeline
SAFE-FORGET	Age evidence; JOLs≠retention; delayed probes as truth; returns as competence protection; no shame-curve hero	LXXVI	Ban mastery-fireworks ≡durable / Ebbinghaus Score™ / streak-as-memory / confidence≡retention / cured-forgetting ads / retention% guarantees
SAFE-ADAPT	Calendar-first; adapt only inside RI bands with due+why; selection-among-due > opaque rewrite; flashcard lineage≠FEI	LXXVII	Ban SM-2/FSRS/Anki hero / black-box perfect-interval AI / flashcard-MAE≡ACT / engagement≡learning / silent ease-hell

SAFE-DURABLE	Dual-rail GTM: durable learn + late prove; cram-intercept not cosplay; legible MoC for WTP; no point guarantees	LXXVIII	Ban cram≡ready / ACT-point packages / overlearn-tonight / shame-parent GTM / streak-as-durable
SAFE-FBTIME	Mode-conditional timing: diagnostic hide / learn micro-delay+SE / prove KR; focus>timing; prior-knowledge switch	LXXIX	Ban instant-feedback AI hero / always-delay≡science / feedback≡learning / green-check NS / ego-toast≡FI
SAFE-BRIDGE	Connection-first diagnosis when endpoints non-naive; remediate relation with co-present supports; measure join transfer	LXXX	Ban bridge-count moat / re-explain≡bridge / coverage≡connection / we-connect-everything / shame-edge Map
SAFE-COLD	Humble prior → labeled seed → hide-correctness probes; Map chrome = evidence age; honest mapping ≠ fake knowing	LXXXI	Ban day-one mastery fireworks / confidence≡mastery / skip-to-green / prior-as-biography / placement-belt cold start
SAFE-FORMAT	Format as first-class evidence; conversion as object; support before stack; structural FormatId only	LXXXII	Ban format-count moat / every-format ads / decoration≡coverage / concept≡format-flexible / Format Personality™
SAFE-PDASH	Beliefs before chrome; proof-age hero; one improvement CTA; autonomy>control; ranks/affect off	LXXXIII	Ban tonight-%/streak parent NS / fireworks portal / shame ranks / live wrong-answer stalk / gradebook cosplay
SAFE-TALK	Construction > airtime; prompt density + constructive share + wait bands; pour-streak QA; human=AI schema	LXXXIV	Ban Talk Ratio Score™ / talk-% NS / 50/50 dogma / silence≡Socratic / got-it≡construction / AI monologue≡dialogue

SAFE-STORYLOAD	CLT budgets narrative; structure-encode or cut; coherence > immersion; novices get less lore; skip-story; story≠transfer	LXXXV	Ban immersion/lore NS / Story Engagement Score™ / seductive trivia≡pedagogy / forced lore walls / fandom-skin hero
SAFE-PROOF	Competence > crowd; enactive dated solo artifacts; provincial similarity; student-owned thin portfolio; two-sided honesty	LXXXVI	Ban user-count/star-wall learning proof / vague praise / Identity Score™ splash / +points case studies / forced public portfolio NS
SAFE-COVER	Matrix > total; gaps as features; use-tier labels; examTag honesty; GENQ never counted unverified; prior substantiation via audit	LXXXVII	Ban item-count hero / complete-ACT puffery / GCSE-as-ACT / unsynced GENQ counts / ontology-nodes≡bank
SAFE-FADE	Stage not costume; completion bridge; backward fade first; expertise-aware up/down; attempt grain; proof = solo transfer	LXXXVIII	Ban always-full-worked / never-fade / solution-first / view≡mastery / Fade Score™ / unlimited-solutions hero
SAFE-HELP	Instrumental > executive; abuse∪avoidance fail; contingent stage; SE before bottom-out; metacognition≠score; dignity telemetry	LXXXIX	Ban unlimited-hints / help-NPS NS / avoidance-as-grit / Help Score™ / parent hint-stalk / never-stuck dump
SAFE-INSTRUMENT	Ship the four FEI events; NSM stack + stage OMTM; XXI.4 co-gates; anti-Goodhart; anti-gaming companions; Kane IUA for uses	XC	Ban DAU/streak/XP NS / FEI Score™ / single-KPI tutor pay / thumbs-up≡learning / identity ads without IUA

SAFE-HINT	Soft before hard; contingency up/down; peek cost=construction; fade×hint one machine; no XP/Hint Score™; inspectable stage	XCI	Ban unlimited free hard peeks / Hint Score™ / XP-for-hints / shame-timer cost / black-box perfect-hint AI / fade+free dump
SAFE-EXPLAIN	Short principle-first before monologue; germane≠more text; SE before wrap; length fades with expertise; coherence>charm; no token NS	XCII	Ban longer≡better / Explanation Score™ / AI monologue≡SE / unlimited thorough AI essays / fixed long wrap / seductive coach fluff
SAFE-RETRIEVE	Name stall mode; attempt before heavy help; TOT→cue; freeze→bounded soft; progress≠blank-time; no Struggle Score™	XCIII	Ban all-struggle≡productive / blank-time NS / always-wait / instant dump / never-stuck hero / errors-always-scar
SAFE-PF	PF = generate→consolidate on student RSMs; fidelity+prior gates; guided default when thin/exam/freeze; no PF Score™	XCIV	Ban every-miss≡PF / fail-first-always / generation-without-consolidation / discovery-as-identity / ACT-from-invent
SAFE-APPRENTICE	Model control/heuristics → coach → scaffold→fade; articulate; shared ca_phase; proof = solo+articulation	XCV	Ban guild cosplay / modeling≡lecture / never-fade / talk-%≡CA / Apprenticeship Score™ / CA ads without phases
SAFE-ATTN	Protect mid-attempt windows; residue-aware park/resume; opt-in focus + tech breaks; presence hygiene without shame	XCVI	Ban Focus Score™ / phone-shame / mid-attempt DAU pushes / multitasking-as-talent / guaranteed ACT from focus

SAFE-DUAL	Structural dual coding; contiguous labels; coherence weed; static annotated default; conversion stays the object	XCVII	Ban Dual Coding Score™ / always-add-picture / decoration≡learning / styles meshing / animation-default / split-legend density
SAFE-MONITOR	Named judgments; C4 destake; confidence×outcome classes; control wiring mandatory; densifies SAFE-CALIB/COLD	XCVIII	Ban Monitoring Score™ / metacognition-without-control / easy-only≡calibration / one-scan calibration / raise-metacognition identity
SAFE-ROI	Tutor minutes scarce; Map minute budgets; constrained AI for reps; humans for joins/destake/repair/witness; FEI per tutor-minute	XCIX	Ban AI-replaces-tutors / tutoring-is-free / unlimited-human / hours NS / Session ROI Score™ / VanLehn-as-ChatGPT
SAFE-EXPTIME *(provisional)*	Grain-local example↔solve timing; completion bridge; adaptive>fixed; rapid rechecks; densifies SAFE-FADE	C	Ban universal E0→E3 destiny / fixed≡adaptive / confidence≡stage / Example Timing Score™ / always-struggle-first
SAFE-SRL *(provisional)*	Phase→control cycles; process goals; attribution→plan writeback; microanalysis>inventories; densifies MONITOR/AAR	CI	Ban SRL Score™ / reflection-streak NS / survey≡event / streak≡SRL / forethought-without-control / ability-shame reflection
SAFE-GENERATE *(provisional)*	SOI-forcing sparse GLAs; student produces; teach-back act; guided FormatId draw; densifies EXPLAIN/DUAL/SRL/SE	CII	Ban Generative Learning Score™ / AI≡generation / teach-for-XP / eight-menu bomb / verbatim summary theater
Block → near-miss interleave → spaced mix	Blocking = acquisition scaffold; delayed mixed accuracy = readiness signal	XXIX, XXXIX, LXXIV	Ban “shuffle = science”; ban blocked-accuracy vanity

Student-generated why	Faded examples + structured principle/misconception prompts *before* AI wrap	XXVI, XL	Ban AI-monologue≡SE; ban explain-own-wrong-first default
Competitive wedge	Do not out-content Khan, out-streak Duo, out-delight Brilliant, or out-fluency ChatGPT	XXXV, XX	Sell recoverable struggle + competence evidence + solo transfer
Equity of worlds	Story wrap is identity technology; tokenism and stereotype-cueing copy are kills	XXXVI, XXVIII	HIST-EQ; belonging without "even you can"

Merged / demoted (do not treat as separate products): "AI tutor," "mastery path," "growth mindset," "engagement," "grit," "habit streak," "celebrate mistakes," "community of practice," "deliberate practice brand," "film study," "recital," "grandmaster review," "exposure therapy," "girl STEM brand," "WTP survey," "Identity Score™," "8-week math person," "character equity," "knowledge-graph moat," "emotion AI empathy," "science-backed page count," "Ivy tutor SKU," "AI wrote N questions," "FERPA-compliant badge," "perfect-interval SRS," "tutor headcount / retention-%," "mastery fireworks / personal Ebbinghaus," "FSRS/Anki brain optimizer," "cram/bootcamp/+points packages," "instant feedback AI / always-delay science," "bridge-count / we connect everything," "day-one personalized mastery / placement belt," "every format / visual learning / Format Personality™," "parent fireworks portal / Family Leaderboard / live wrong-answer stalk," "immersive story world / lore engagement / Story Engagement Score™," "user-count / star-wall / vague confidence testimonials as learning proof," "item-count / complete ACT bank / AI fills all holes coverage theater," "always-full-worked / unlimited solutions / never-fade Solver," "unlimited hints / never stuck / Help Score™," "DAU/streak/XP North Star / FEI Score™ / thumbs-up≡learning," "unlimited free hard peeks / Hint Score™ / XP-for-hints / perfect-hint AI," "longer≡better / Explanation Score™ / unlimited thorough AI essays / monologue≡SE," "maximize productive struggle minutes / Struggle Score™ / blank-time grit / always-wait / never-stuck dump," "fail-first always / every miss is productive failure / PF Score™ / discovery-without-consolidation," "guild / master tutor / cognitive apprenticeship™ costume / Apprenticeship Score™ / modeling-minutes / never-fade coach," "Focus Score™ / deep-work theater / phone-shame / mid-attempt streak pings / multitasking digital-native talent," "Dual Coding Score™ / always-add-picture / decoration-as-dual-coding / visual-learner meshing / animation-first multimedia / split-legend density theater," and "Monitoring Score™ / metacognition-without-control / easy-only calibration / one-scan calibration theater," and "AI-replaces-tutors / tutoring-is-free / unlimited-human / hours-booked / Session ROI Score™ theater," and "SRL Score™ / reflection-streak / survey-SRL≡event / streak≡self-regulation theater" are *not* independent North Stars — they are subordinate UX under FEI + the rows above (borrow *method*, never *costume*).

Next research bottleneck: Part CIII **Error climate & psychological safety in tutoring**. Experiment families CAL/DP/CoP/RIT/AAR/FILM/MUSIC/ANNOT/EXP-O/WTP/REPAIR/IDM/LONG/EXAM/ELL/GEND/STRUCT/ONTO/HITL/PRIV/LABMETA/GRAIN/GENQ/PROCURE/SCHED/WORK/FORGET/ADAPT/CRAM/FB/BRIDGE/COLD/FORMAT/PDASH/TALK/STORYLOAD/PROOF/COVER/FADE/HELP/INSTR/HINT/EXPLAIN/RETRIEVE/PF/APPRENTICE/ATTN/DUAL/MONITOR/ROI/EXPTIME/SRL/GENACT gate claims above L1. Researcher count since v1.13 synthesizer = **3**.

Part II — Key Insights (challenged)

Insight 1 — Fear is a performance bottleneck, not a personality trait

FACT: Math anxiety is associated with reduced working-memory availability during math tasks; anxious learners underperform relative to ability (Ashcraft & Kirk line of work; broader anxiety–WM literature).

HYPOTHESIS: For a large subset of MindCraft's Maya archetype, the binding constraint is not "missing explanation," it is **threat appraisal** ("I will look stupid").

Implication: Soft-wrong UX, hide-correctness diagnostics, wizard coaching, and narrative framing are not polish — they are load-bearing if the thesis holds.

Red Team: Many high performers succeed under stress. Anxiety interventions have heterogeneous effects. Do not medicalize ordinary difficulty.

Insight 2 — Growth mindset is real, small, and context-dependent

FACT: Yeager et al. (2019, *Nature*) — brief online growth-mindset intervention improved grades for lower-achieving students and increased advanced math enrollment in a nationally representative U.S. sample; effects stronger when peer norms supported challenge-seeking.

FACT / REVIEW: Domain-general mindset messaging shows mixed results; math-*specific* mindset interventions more often report positive outcomes (2023 systematic review in *Educational Research Review*).

FACT / CAUTION: "False growth mindset" (praise effort without teaching strategy / changing environment) is harmful (Dweck's own caution; Kohn's critique).

Implication: MindCraft must not ship empty "believe in yourself" copy. Mindset must be embodied in **task design, feedback, and tutor norms**.

Insight 3 — Autonomy, competence, relatedness are the psychological tripod

FACT: Self-Determination Theory (Ryan & Deci) — sustained intrinsic motivation requires autonomy, competence, relatedness.

FACT (meta): SDT-based education interventions show reliable gains for autonomy and competence support; relatedness effects are less consistent (Wang et al. 2024 meta-analysis).

Implication: Product loops must deliver:

- Autonomy: meaningful choice (path, story world, pace)
- Competence: mastery grain that produces earned wins
- Relatedness: tutor presence, parent signal, peer-safe identity (hardest; do not fake with empty social)

Insight 4 — Mastery learning works — until you measure the wrong thing

FACT: Kulik, Kulik & Bangert-Drowns (1990) meta-analysis — mastery programs show positive exam effects (overall ~0.5 SD in their synthesis), stronger for weaker students; can increase time; self-paced college mastery can reduce completion.

FACT / CONTRADICTION: Slavin's "Mastery Learning Reconsidered" (1987) — little support for group-based mastery on *standardized* measures; experimenter-made tests show moderate effects; coverage vs mastery tradeoff is real.

Implication: MindCraft's mastery graph is promising **if** mastery means transferable competence, not local quiz familiarity. Measure far transfer, not only same-item success.

Insight 5 — Habit products ≠ identity products

FACT / INDUSTRY: Duolingo's public method emphasizes gamification for motivation and return (streaks, leagues, variable rewards) (Duolingo Method whitepaper, 2023). Streaks exploit loss aversion; they drive DAU.

HYPOTHESIS: Streaks can increase practice frequency without changing "I am a math person." Habit without identity yields brittle engagement (stop streak → stop learning).

Implication: Use habit mechanics as **on-ramps**, not as the destination. Pair with identity-marking rituals (map fill, story progression, tutor recognition of growth).

Insight 6 — Narrative is not decoration; it is encoding + meaning

FOUNDER BELIEF (WORLD_VISION): Math stripped of invention-story becomes alienating machinery.

HYPOTHESIS: Historical/human invention frames increase curiosity and reduce "arbitrary rule" appraisals, improving persistence.

Evidence status: Strong tradition in math education (history of math pedagogy); fewer large RCTs than mindset literature. Treat as **design prior + research program**, not settled science.

Insight 7 — AI commoditizes *answers*, not *becoming*

HYPOTHESIS: As answer generation approaches free, willingness-to-pay migrates to:

1. Diagnosis of the *actual* break (gap map)
2. Emotional containment during struggle
3. Accountability relationships (tutor)
4. Identity-consistent practice worlds
5. Trusted outcome signaling to parents

SPECULATION: In 30 years, "show me the steps" is ambient. "Help me become the kind of person who stays with hard problems" remains a human+system problem.

Part III — Research Log (v1 seed)

Date	Finding	Type	Action

2026-08-08	SAFE-GENERATE: SOI-forcing sparse GLAs in Practice; kill Generative Learning Score™ / AI≡generation / teach-for-XP	Evidence	GENACT-1…5; next = CIII error climate
2026-08-08	SAFE-SRL: phase⟶control Zimmerman cycles in Practice; kill SRL Score™ / streak≡SRL / survey≡event	Evidence	SRL-1…5; next = CII generative learning
2026-08-08	SAFE-EXPTIME: grain-local example⟷solve timing; kill universal E0⟶E3 destiny / fixed≡adaptive / Timing Score™	Evidence	EXPTIME-1…5; next = CI SRL cycles
2026-08-07	Synthesizer v1.13: merge XCII–XCIX SAFE-* into I.4; kills #46–#53	Synthesis	Surviving commercial law; next = C worked-example timing
2026-08-07	SAFE-ROI: Map minute budgets; kill AI-replaces-tutors / tutoring-is-free / hours NS / Session ROI Score™	Evidence	ROI-1…5; next = C worked-example timing
2026-08-07	SAFE-ATTN: protect attempt windows; kill Focus Score™ / phone-shame / mid-attempt DAU pushes	Evidence	ATTN-1…5; next = XCVII dual coding
2026-08-07	SAFE-DUAL: contiguous structural figures; kill Dual Coding Score™ / styles meshing / always-add-picture	Evidence	DUAL-1…5; next = XCVIII metacog monitoring
2026-08-07	SAFE-MONITOR: named judgments ⟶ control; kill Monitoring Score™ / one-scan calibration	Evidence	MONITOR-1…5; next = XCIX tutor vs AI ROI

2026-08-07	SAFE-APPRENTICE: model→coach→fade + articulate; kill guild cosplay / modeling≡lecture	Evidence	APPRENTICE-1…5; next = XCVI attention residue
2026-08-07	SAFE-PF: fidelity-gated generate→consolidate; kill every-miss≡PF / fail-first-always	Evidence	PF-1…5; next = XCV apprenticeship
2026-08-06	Synthesizer v1.12: merge LXXXIV–XCI SAFE-* into I.4; kills #38–#45	Synthesis	Surviving commercial law; next = XCII explanation length
2026-08-05	Synthesizer v1.11: merge LXXV–LXXXIII SAFE-* into I.4; kills #29–#37	Synthesis	Surviving commercial law; next = LXXXIV tutor talk-ratio
2026-08-04	Synthesizer v1.10: merge LXVII–LXXIV SAFE-* into I.4; kills #21–#28	Synthesis	Surviving commercial law; next = LXXV tutor workforce
2026-08-02	Synthesizer v1.9: merge LIX–LXVI SAFE-* into I.4; kills #18–#20	Synthesis	Surviving commercial law; next = LXVII ontology moat
2026-08-01	Synthesizer v1.8: merge L–LVIII SAFE-* into I.4; kills #15–#17	Synthesis	Surviving commercial law; next = LIX parent WTP
2026-08-01	SAFE-EXPOSE: graded approach; kill therapy cosplay / flooding / Calm Score™	Evidence	EXP-O1…5; expectancy violation
2026-08-01	SAFE-ANNOT / MUSIC / FILM: postmortem + ladder + sparse clips	Evidence	ANNOT/MUSIC/FILM-1…5
2026-07-31	Synthesizer v1.7: merge XLI–XLIX SAFE-* into I.4; kills #10–#14	Synthesis	Surviving commercial law; next = L confidence calibration

2026-07-31	Misconceptions productive: SAFE-MISCON; soft-wrong ≠ empty kindness	Evidence	MIS-1…4; springboard + classify
2026-07-30	Transfer/VT: story ≠ far transfer; blocked ≠ ready	Evidence	TR-1…4; SAFE-TRANSFER
2026-07-30	Sleep/stress: encode→sleep→claim; pressure ≠ DD	Evidence	TIM-1…4; SAFE-TIMING
2026-07-29	Synthesizer v1.6: merge XXXIII–XL into I.4 doctrine stack	Synthesis	Surviving commercial law; next was XLI DD×anxiety
2026-07-29	SE prompts: AI monologue ■ SE; structure before wrap	Evidence	SE-1…4; densify Exp A/D
2026-07-29	Interleaving: block→near-miss→spaced mix; kill shuffle=science	Evidence	IL-1…4; ban blocked vanity
2026-07-29	Goal climate: appearance/streak NS killed; TARGET audit	Evidence	GO-1…4
2026-07-29	Eccles E×V×Cost: kill single motivation KPI	Evidence	EVT-1…4
2026-07-25	Constituted Research Lab + Constitution v1	Process	Create OS
2026-07-25	Scarcity thesis partially false as slogans; true as product wedge	Analysis	Rewrite pitch language
2026-07-25	Yeager 2019 constrains mindset claims	Evidence	Ban empty mindset copy
2026-07-25	Soft-wrong + wizard coach shipped in product	Product	Instrument coach → retry rate
2026-07-25	Mastery literature split (Kulik vs Slavin)	Evidence	Define mastery measurement carefully
2026-07-25	Bloom 2-sigma overstated vs VanLehn / modern tutoring metas	Evidence	Stop marketing 2-sigma
2026-07-25	Founder sequence wounded → SAFE-CRAFT	Synthesis	Attempt earlier; invention-story optional

2026-07-25	Deliberate practice necessary≠sufficient (Hambrick line)	Evidence	Progress-vs-self framing

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Part IV — Evidence Table (selected)

Claim	Best supporting evidence	Contradicting / limiting evidence	Confidence	MindCraft implication
Brief mindset intervention can move grades + advanced math taking	Yeager et al., Nature 2019	Effects heterogeneous; peer norms matter; replication debates in broader mindset meta-analyses	Medium–High	Mindset must be ecological, not banner text
Math-domain mindset > generic	2023 Ed Research Review systematic review	Study quality varies; publication bias possible	Medium	Math-specific identity language
SDT need support raises motivation	Wang et al. 2024 meta	Relatedness effects weaker	Medium–High	Design for autonomy + competence first
Math anxiety impairs performance	Cognitive literature (Ashcraft tradition)	Not all low performers are anxious	Medium–High	Affective onboarding is first-class
Mastery learning raises achievement	Kulik et al. 1990	Slavin 1987 on standardized tests	Medium	Mastery graph + transfer tests
Gamified habits raise return	Duolingo method / industry practice	Habit ≠ deep learning; extrinsic traps (Deci)	Medium	Streaks as on-ramp only
1:1 tutoring is uniquely powerful	Bloom “2 sigma” framing (1984)	Often overstated; quality varies; VanLehn etc. nuance tutoring effects	Medium	Keep human tutors; AI amplifies, not replaces

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Part V — Contradictory Evidence (do not skip)

1. **Mindset skepticism:** Some metas find small average effects; school implementations often degrade into posters.
2. **Grit / resilience hype:** Duckworth-style grit is contested; structural inequality and curriculum quality matter more than character slogans.
3. **Discovery learning failures:** Pure constructivism without guidance often fails (Kirschner, Sweller, Clark critiques). Narrative worlds must still teach.
4. **Engagement ≠ learning:** Time-on-app can rise while understanding falls (edtech's original sin).
5. **AI may *increase* explanation value temporarily:** Novices may trust fluent wrongness; trust becomes the scarce resource, not explanation volume.
6. **Some students want speed and drills:** Identity-through-story is not universal. Offer multiple vessels.

Part VI — Mental Models

VI.1 The Identity Transformation Cascade (ITC)

HYPOTHESIS — MindCraft original synthesis

```
Threat ↓ → Attention available → Micro-success → Attribution ("I figured it")
→ Willingness to re-enter challenge → Accumulated competence evidence
→ Narrative update ("maybe I can") → Identity claim ("I do math")
→ Environment selection (harder courses, peers, tutors)
→ Reinforcing loop
```

Balancing loop: Premature hard problems → threat ↑ → avoidance → identity freeze.

Product translation: Gap scan (no shame) → soft-wrong → tiny earned wins → map fill → tutor recognition → advanced path.

VI.2 The Three Doors (what students actually need in a stuck moment)

Door	Need	Wrong product response	Right response
A	Safety	"Here's a longer explanation"	Contain affect; normalize miss
B	Clarity	Pep talk	Precise model / worked example at right grain
C	Agency	Take over and solve for them	Scaffolded choice; student does the decisive step

FOUNDER BELIEF: Tutors often open Door B first. Many students are still behind Door A.

VI.3 Knowledge commodity stack (post-AI)

```
Layer 0: Facts & procedures → commoditizing fast
Layer 1: Explanations → commoditizing fast
Layer 2: Diagnosis of misconceptions → partially automatable; trust-sensitive
Layer 3: Adaptive sequencing → automatable with good ontology
Layer 4: Emotional co-regulation → scarce (human + careful UX)
Layer 5: Identity & meaning → scarce
Layer 6: Accountability relationship → scarce
```

MindCraft's durable stack lives in **Layers 2–6**, with 0–1 as utilities.

VI.4 Transfer from outside education

Domain	Mechanism	Transfer to MindCraft (surviving law)
Games (Celeste, FromSoftware-lite design)	Fair difficulty + death as information	Soft-wrong as physics (SAFE-MISCON / FEI)
Sports coaching	Film study of *specific* error	SAFE-FILM: ≤3-cue attempt clips + self◡model
Music	Scales → repertoire → recital	SAFE-MUSIC: ingredient drills → chosen pieces → transfer proof
Therapy (exposure)	Graded approach + expectancy violation	SAFE-EXPOSE: situation hierarchy under destaked stakes
Aviation / military	Brief → attempt → structured debrief	SAFE-AAR: four-question student-first close
Chess annotation	Postmortem before engine; principle tags	SAFE-ANNOT: annotate→classify→drill; wins too
Martial arts belts	Visible competence ladder	Mastery pips / map regions (criterion, not league)
Language apps	Habit loops	SAFE-HABIT cue practice — without Duo's extrinsic ceiling
Religion / ritual	Shared meaning + belonging	SAFE-RITUAL: light optional marks; anti-totalism
Craft apprenticeship	Legitimate peripheral participation	SAFE-CoP: real peripheral work + old-timer; not Discord KPIs

---

Part VII — Universal Learning Principles (provisional)

1. **Emotion gates cognition** in threatened domains.

2. **Earned difficulty** beats random difficulty.
3. **Feedback should be information, not identity judgment.**
4. **Attribution matters:** strategy > talent for growth; luck attributions kill agency.
5. **Spaced retrieval** beats massed re-reading (cognitive science consensus).
6. **Worked examples** → **fading guidance** for novices (cognitive load).
7. **Belonging uncertainty** destroys persistence for stereotyped groups (Walton/Cohen tradition).
8. **Transfer requires varied practice**, not identical clones.
9. **Human accountability** multiplies AI tools.
10. **Identity is a lagging indicator**; design leading indicators (challenge-seeking, return-after-fail).

Part VIII — Product Implications (systems, not features)

VIII.1 Core system: Fear → Evidence → Identity (FEI Loop)

Reinforcing loop R1 (desired):

Safety UX → attempt → informative miss → coach → success → map fill → identity → more attempts.

Balancing loop B1 (threat):

Public shame / red buzz → avoidance → less practice → skill lag → more shame.

MindCraft already partially implements FEI: hide-correctness diagnostic, soft-wrong, wizard coach, story chapters, knowledge map. **Missing instrumentation:** does coach raise retry rate and later challenge-seeking?

VIII.2 What to stop optimizing

- Vanity DAU without learning transfer
- Explanation length as quality proxy
- Points that do not mark identity-relevant milestones

VIII.3 What to instrument next (minimum viable science)

1. **Retry rate within 2 minutes of soft-wrong** (with vs without wizard coach)
2. **Challenge-seeking:** % choosing harder level when offered
3. **Return after fail session** (D1 retention after a loss session)
4. **Self-identity item** (pre/post, 1–2 items, math-specific)
5. **Advanced course intent / enrollment** (slow, gold)

VIII.4 Tutor system role

Tutors are not content pipes. They are:

- Affect regulators
- Attribution coaches
- Bridge diagnosticians (concept A known, link A→B broken)
- Identity witnesses (“I saw you become someone who stays”)

AI should brief tutors, not replace witnessing.

Part IX — Experiments (pre-registered style)

Core A–D remain. Chapter ticks densified them into families — prefer the densified IDs when shipping.

Experiment A — Wizard coach on soft-wrong

- **Question:** Does the under-Graph wizard increase productive retry vs soft-wrong alone?
- **Design:** A/B, chapter + practice
- **Primary:** retry within 120s; secondary: eventual correct without hint binge
- **Falsifier:** no lift; or lift only in engagement with worse accuracy
- **Densified content (v1.6):** Coach turns must be SE-constrained (principle / misconception contrast), not pep talk — see SE-1…4 / SE-2 as operational pairs with Experiment D.

Experiment B — Story-first vs bare stem

- **Question:** Does invention-story framing raise persistence on first miss?
- **Design:** within-concept crossover
- **Primary:** time-to-abandon after first miss
- **Falsifier:** story slows without improving retention/transfer
- **Equity densify:** HIST-EQ / Part XXXVI — whose story, tokenism kill.

Experiment C — North Star validation

- **Question:** Which early metric best predicts 8-week challenge-seeking?
- **Candidates:** streak length, soft-wrong retry, identity item, map mastery Δ, delayed mix accuracy
- **Method:** observational + lagged prediction
- **v1.6 prior:** Expect streaks to lose; `retry_120s` + `transfer_pass` + mastery-motive `challenge_accept` to win.

Experiment D — Explanation timing

- **Question:** Is explanation-before-attempt worse than attempt-with-safety for anxious students?
- **Design:** screen with anxiety pretest; randomize explanation-first vs attempt-first
- **Falsifier:** explanation-first wins for all segments
- **Densified:** Prefer SE-2/SE-3 (contrastive / structured generation) over monologue-first.

Experiment families mounted in chapters (do not orphan)

Family	Identifies (max claim without follow-up)	Parts
AIT-*	Guarded AI → solo transfer / anti-sycophancy	XXXIII
DAG-*	Estimand hygiene; covariate vs mediator	XXXIV
CSA-*	Session preference vs competitors; paste under guards	XXXV
HIST-EQ / equity probes	Story belonging without stereotype cue	XXXVI
EVT-*	Expectancy / value / cost levers	XXXVII
GO-*	Mastery vs performance CTA climate	XXXVIII
IL-*	Schedule → discrimination / delayed mix	XXXIX
SE-*	Prompt regime → transfer / misconception recurrence	XL
DD-*	Equip/destake/dose × anxiety → retry + delayed mix	XLI
SC-*	Criterion vs peer rank climate → affect + challenge	XLII
HAB-*	Cue/if-then vs streak colonization → return without identity harm	XLIII
IM-*	Informational vs engagement-loot rewards → free-choice + transfer	XLIV
RES-*	Growth-zone + support recruitment vs grit theater	XLV
EXP-*	Tutor brief / CIOF / AI expectancy → talk ratio + solo transfer	XLVI
TIM-*	Practice timing × sleep/stress → delayed transfer	XLVII
TR-*	VT / hug / bridge sequence → format-hop + solo transfer	XLVIII
MIS-*	Soft-wrong springboard / mis-mapped distractors / confidence tier	XLIX
CAL-*	Confidence-tiered feedback / hide-correctness / underconf unlock	L

DP-*	Session spine / edge targeting / method feedback / effort cap	LI
CoP-*	Named practice / peripheral work / old-timer / trajectory metrics	LII
RIT-*	Light ritual vs silent / recovery marker / anti-streak liturgy	LIII
AAR-*	Brief + four-question close / student-first / trace-backed debrief	LIV
FILM-*	Sparse clip cues / self-controlled review / self∪model pairing	LV
MUSIC-*	Ladder UX / repertoire choice / identified technique / transfer recital	LVI
ANNOT-*	Annotate-before-reveal / forced restudy / classify→drill / annotate wins	LVII
EXP-O-*	Situation hierarchy / expectancy check / fade safety / multi-context	LVIII
WTP-*	Parent CBC / FEI report vs score / message arms / cash pilot	LIX
REPAIR-*	Post-hallucination repair / reattempt / packetized escalate	LX
IDM-*	Validated identity battery / demand controls / FEI co-primary	LXI
LONG-*	8/26/52w horizons / fade checks / staged identity claims	LXII
EXAM-*	Destake-learn / dose-prove ladders / delayed mixed under stakes	LXIII
ELL-*	Language vs math load / linguistic modify / format triangulation	LXIV
GEND-*	Cue hygiene / practice recognition / no ST→score / no pink track	LXV
STRUCT-*	Bandwidth hygiene / structure>slogans / access honesty / no grit-NS	LXVI

ONTO-*	Ontology-constrained next vs chat / bridge route / Map brief / state vs summary	LXVII
HITL-*	Map-briefed tutor / prompt-first playbook / coached QA / tutor+Map vs AI-alone	LXVIII
PRIV-*	Check-in TTL / soften-on-stress / task-only brief / ERT preference / refusal-safe	LXIX
LABMETA-*	Pre-reg gate / Red Team bite / copy lint / synthesizer demote / external skeptic	LXX
GRAIN-*	Trained near-peer / Map vs expert-no-Map / peer-pair / hire-by-fidelity / parent CBC grain	LXXI
GENQ-*	Verify-on vs off / generated vs seed / diagnostic seal / prompt-harden drop rate / parent CBC verified-bank	LXXII
PROCURE-*	Trust-packet-first / NDPA vs custom / LEA-approved free pilot / district vs parent message / no-bio attribute	LXXIII
SCHED-*	Horizon equal-lag vs massed / expanding vs equal / delayed first return / generation vs restudy / ACT-date scheduler	LXXIV
WORK-*	Continuous coaching vs onboard-only / coach-capacity hire cap / caseload vs wellness / Map-brief hard-require / parent CBC fidelity	LXXV
FORGET-*	Aged-node vs fireworks / honest post-mission vs confetti / parent proof-age CBC / RI-salience vs fluency JOL / clear-state expiry	LXXVI
ADAPT-*	Fixed calendar vs banded modulation / due-set priority vs FIFO / inspectable vs opaque smart review / FSRS-urgency priority-only / stress-capped pull-forward	LXXVII

CRAM-*	Durable vs cram-pack CBC / cram-intercept UX / dual-rail label / score-lift vs proof-age ads / massed week vs horizon Returns	LXXVIII
FB-*	Micro-delay+SE vs instant KR / tiered timing / diagnostic hide / prove-rail KR / process vs ego toast	LXXIX
BRIDGE-*	Bridge-card vs endpoint re-teach / Map edge CTA / detector↔tutor / co-present vs sequential / parent CBC connection copy	LXXX
COLD-*	Seed±probes / honest Map chrome vs fireworks / uniform-easy QA / contradicting-reason on easy / parent CBC honest-map	LXXXI
FORMAT-*	Format-gap CTA / co-present conversion / structural vs decorative / supported multi-format / parent CBC format-flex	LXXXII
PDASH-*	Proof-age strip vs fireworks / improvement digest / absolute vs rank / autonomy vs nag / parent CBC MoC portal	LXXXIII
TALK-*	Prompt-first vs explain-first / coach dashboard vs clip QA / SE gate / AI prompt-cap / parent CBC construction	LXXXIV
STORYLOAD-*	Structure wrap vs plain / lore budget / skip-story / mastery-gated reveal / parent CBC clear-story vs immersion	LXXXV
PROOF-*	Competence-card vs star-wall landing / provincial case / two-sided MoC vs +points / opt-in share / parent CBC dated solo	LXXXVI
COVER-*	Honest matrix vs item-count hero / parent CBC holes+seal / district audit packet / in-app gap callout / audit cadence	LXXXVII

FADE-*	Backward completion ladder vs always-full / forward vs backward / fade-up after solos / SE-on-blank / parent CBC fade story	LXXXVIII
HELP-*	Contingent instrumental vs unlimited bottom-out / SE-before-hint / help-invite vs freeze / meta-msg vs gate / parent CBC next-step help	LXXXIX
INSTR-*	Lagged FEI vs streak/DAU prediction / tutor KPI co-gates / event fire audit / motive-coded challenge CTA / parent CBC FEI report	XC
HINT-*	Soft-first contingent vs free hard / SE-cost vs dwell / fade×hint couple / peek budget / parent CBC next-step hints	XCI
EXPLAIN-*	Principle-short vs monologue / SE-before-wrap / length×fade / soft-wrong short mark / parent CBC clear-short	XCII
RETRIEVE-*	TOT micro-cue vs instant hint / freeze bound soft vs long wait / attempt-before-help / progress-aware dwell / parent CBC attempt-then-coach	XCIII
PF-*	High-fidelity PF vs DI / student-RSM C&C vs generic lecture / thin-prior gate / freeze-exit PF / parent CBC invent-then-compare	XCIV
APPRENTICE-*	Playbook model→coach→fade+articulate vs explain-pour / control-talk model vs steps-only / within-session fade / reciprocal beat / parent CBC take-the-wheel	XCV
ATTN-*	Mid-attempt push suppression / opt-in focus mode / tech-break between items / phone-salience tip / parent CBC attempt-window vs streak / Maya residue phenomenology	XCVI

DUAL-*	Contiguous annotated figure vs split legend / structural vs decorative / static vs animation default / coach+figure length / parent CBC format-hop	XCVII
MONITOR-*	Named judgment+miss-class vs easy-only / C4 destake / control-wired routing / one-scan vs revisit / parent CBC monitoring honesty	XCVIII
ROI-*	Map minute budget vs open hours / AI-reps+human-join vs AI-alone / FEI-per-tutor-minute dashboard / escalate precision / parent CBC scarce-human honesty	XCIX
EXPTIME-*	Fixed vs adaptive fade / concept vs concept×format initial stage / confidence→E vs probe→E / always-E0 vs always-E3 vs adaptive / parent CBC step-down help	C
SRL-*	Forethought chips vs autopilot / attribution→plan writeback / microanalysis vs journal / process vs ability attribution UI / parent CBC plans-the-attempt	CI
GENACT-*	Teach-fictive-peer vs solution-view / guided draw vs free vs provided / SOI summarize vs XP / sparse vs eight-menu / parent CBC generates-explanation	CII

Part X — Weekly Priorities (research × product)

1. Instrument FEI loop events in analytics (retry, coach shown, write-mode exit).
2. Run Experiment A for 2 weeks.
3. Red Team the WORLD_VISION narrative claims with one outside learning scientist.
4. Interview 10 Mayas: “When did math stop feeling dangerous?” (qual).
5. Define “mastery” operationally so Slavin’s critique cannot gut the graph.

Part XI — Founder Questions (do not dodge)

1. If ChatGPT tutors become "good enough," what *relationship* do parents still buy?
2. Are we building for Maya's identity — or for parents' anxiety about scores? Both? Tradeoff?
3. Would we accept lower short-term accuracy for higher long-term challenge-seeking?
4. Is the story world load-bearing, or is gap+tutor enough?
5. What would falsify the identity-transformation thesis in 90 days?

Part XII — Long-term Roadmap (research horizons)

H1 (now): Prove FEI loop moves retry + challenge-seeking.

H2: Prove identity language + map produce durable self-concept change.

H3: Prove human tutor + AI diagnosis beats AI-alone on persistence and transfer.

H4: Generalize beyond math (science/identity) only after math identity mechanism is solid.

Part XIII — Open Problems

1. Causal path from narrative history-of-math to identity (under-identified).
2. Relatedness at scale without creepy social.
3. Measuring identity without demand effects.
4. Preventing gamification from colonizing meaning.
5. AI fluent nonsense vs trust calibration for teens.
6. Equity: does story-world privilege culturally specific narratives?

Part XIV — Red Team Dossier (destroy weak arguments)

Kill #1: "Explanations are free, therefore MindCraft should not explain"

Destroyed form: Absolutism.

Surviving form: Explanations are *necessary infrastructure* with collapsing *willingness-to-pay*; differentiation is diagnosis + affect + identity + accountability.

Kill #2: "Just fix mindset"

Destroyed: Poster mindset.

Surviving: Ecological mindset (task + feedback + peer/tutor norms), math-specific — TARGET structures (XXXVIII).

Kill #3: “Engagement is the goal”

Destroyed: Engagement without transfer.

Surviving: Engagement as oxygen for the FEI loop, not the fire.

Kill #4: “Mastery graph = Bloom 2-sigma”

Destroyed: Inflated tutoring mythology.

Surviving: Mastery *can* help weaker students if mastery is real (`transfer_pass`) and time costs are managed.

Kill #5: “Identity transformation is unmeasurable, so ship vibes”

Destroyed: Vibes-as-strategy.

Surviving: Leading indicators (retry, challenge-seeking, delayed mix) + lagging identity items; L0–L4 claim ladder (XXXIV).

Kill #6 (v1.6): “AI tutor / Bastani GPT Tutor is our product”

Destroyed: Unguarded chat-as-learning; category-error marketing; guards-as-gain.

Surviving: Pedagogy wrap + `solo_transfer_pass` ≥ no-AI bar (XXXIII RT).

Kill #7 (v1.6): “Blocked accuracy / streaks / shuffle prove learning”

Destroyed: Fluency illusion; streak North Star; random shuffle ≡ interleaving science.

Surviving: Short block → near-miss interleave → spaced mixed check (XXXIX); streaks as on-ramp only (HID).

Kill #8 (v1.6): “Any explain box / AI monologue = self-explanation science”

Destroyed: Untargeted justify-your-wrong-guess; chat length as depth.

Surviving: Structured principle/misconception SE on correct or known-incorrect content before AI wrap (XL).

Kill #9 (v1.6): “One motivation dial”

Destroyed: Collapsing expectancy and value; staff “math is useful” dumps as EVT.

Surviving: E×V×Cost triad with student-authored utility (XXXVII).

Kill #10 (v1.7): “Harder is always better / desirable difficulty ignores anxiety”

Destroyed: Anxiety-blind interleave/SE/pressure-as-DD; hardness theater as rigor brand.

Surviving: SAFE-DD equip → destake → dose → frame → measure (XLI); gate IL/SE by segment.

Kill #11 (v1.7): “Leaderboards / streaks / XP create math identity”

Destroyed: Appearance climates, streak colonization, engagement-loot as love-of-math.

Surviving: SAFE-COMPARE + SAFE-HABIT + SAFE-REWARD — criterion feedback, cue practice, informational unlocks only (XLII–XLIV).

Kill #12 (v1.7): “Grit / Resilience Score™ / stay in the danger zone”

Destroyed: Character meters; danger-zone-as-desirable-difficulty; grit posters as product.

Surviving: SAFE-RESILIENCE growth-zone + support recruitment; measure recovery/transfer not MRS theater (XLV).

Kill #13 (v1.7): “Pygmalion labels / warm AI that finishes the problem”

Destroyed: Trait/weakness tutor briefs; warmth-without-output-demand as expectancy science.

Surviving: SAFE-EXPECTANCY task briefs + CIOF; AI must clear PWC/`solo_transfer_pass` (XLVI + XXXIII).

Kill #14 (v1.7): “Story / sleep tips / mistake celebration = transfer”

Destroyed: Narrative immersion as automatic far transfer; sleep-app brand; empty mistake posters; red-X-as-rigor; equal graph updates for all wrongs.

Surviving: SAFE-TRANSFER + SAFE-TIMING + SAFE-MISCON — vary on purpose, encode→sleep→claim, elicit→classify→springboard→route (XLVII–XLIX).

Kill #15 (v1.8): “Raise confidence / Belief Score™ / CA raises grades”

Destroyed: Inflate-confidence brand; confidence meters as identity; confidence assessment alone as attainment lever; equal updates for guess vs high-conf misconception.

Surviving: SAFE-CALIB — item-level elicit → miss class → tiered feedback; appropriate confidence; C4 hide-correctness as diagnostic hygiene (L).

Kill #16 (v1.8): “Hours / forums / ritual→score / domain costume = identity”

Destroyed: 10k-hours and minutes KPIs; Discord/streak ≡ CoP; unretracted ritual→performance marketing; bootcamp/conservatory/GM/therapy branding as belonging.

Surviving: SAFE-DP + SAFE-CoP + SAFE-RITUAL — diagnose→prove spine; named practice with real peripheral work; light optional marks pointing at recover/prove (LI–LIII).

Kill #17 (v1.8): “Tip-flood / engine-dump / flooding ≡ high-performance pedagogy”

Destroyed: Uncapped film-room tips; Stockfish/LLM monologue as annotation; day-one timed mixed ACT as “exposure”; calm-to-zero / Calm Score™; loss-only shame reels; bare-pass recital theater.

Surviving: SAFE-AAR + SAFE-FILM + SAFE-MUSIC + SAFE-ANNOT + SAFE-EXPOSE — brief–debrief, sparse clips, skills→pieces→proof, annotate→drill, graded approach with expectancy violation (LIV–LVIII).

Kill #18 (v1.9): “Likert≡WTP / survey dollars≡revenue / ACT-point packages”

Destroyed: Parent survey willingness as price truth; engagement-streak or unlimited-AI as default parent value props; guaranteed ACT points in tiers; message arms treated as package economics.

Surviving: SAFE-WTP — CBC with price+neither, ship-mapped attributes (diagnosis, FEI report, human minutes, AI policy), cheap-talk/cash pilots, message≠package (LIX).

Kill #19 (v1.9): "Never-hallucinates / apology≡repair / Identity Score™ / 8-week math person"

Destroyed: Invincible-AI brand; rote or empathy apology as trust fix; thumbs-up≡repaired; permanent disuse brand; Math Identity Score™; two-item HSLS pre/post≡FEI; splash-screen identity Likert; empty AI "math person" praise; end-of-program Likert≡durable identity; math-person-in-8-weeks.

Surviving: SAFE-REPAIR + SAFE-IDMEASURE + SAFE-LONGID — detect→own→Map-correct→reattempt→triggered escalate; validated factors + FEI behavioral co-primary; staged 8/26/52w claims with fade checks (LX–LXII).

Kill #20 (v1.9): "Day-one ACT flood / EL≡ability / pink track / grit-as-character"

Destroyed: Timed ACT as onboarding; blocked accuracy as exam-ready; Calm Score™ exam NS; math-is-language-free; forever procedure-only ELL track; translate≡identity; innate boys-better premise; ST-removal→ACT; pink/for-her easy track; grit-as-NS / Grit Score™ / poor-kids-need-grit / streak-as-character / scarcity-dropout≡moral-failure.

Surviving: SAFE-EXAM + SAFE-ELL + SAFE-GENDER + SAFE-STRUCTURE — destake-to-learn/dose-to-prove; separate language vs math load; parity + cue hygiene + practice recognition; structure/scaffolds before slogans (LXIII–LXVI).

Kill #21 (v1.10): "Graph file / RAG chat ≡ diagnosis moat"

Destroyed: Node-count or ontology JSON as moat; RAG or chat-memory as knowledge tracing; fluency/tokens as diagnosis; Bloom ITS-2σ; "smarter than ChatGPT"; ALEKS-cosplay without FEI transfer proof.

Surviving: SAFE-ONTOLOGY — diagnosis-before-dialogue; inspectable longitudinal state; LLM bookends under PWC; moat = state×ontology×FEI×solo transfer (LXVII).

Kill #22 (v1.10): "Warm human / hours / got-it ≡ tutor quality"

Destroyed: Warm presence or Ivy credentials as quality; hours-booked NS; "Do you understand?" / smile / stars as mastery or QA; Discord≡tutor QA; explain-first brand; AI-replaces-tutors or humans-skip-Map; onboarding deck≡coaching.

Surviving: SAFE-HITL — Map-briefed tutors; prompt>pour; answer evidence; complementarity; coaching cycles with fidelity; FEI QA; packetized escalate (LXVIII).

Kill #23 (v1.10): "Emotion AI / Anxiety Score™ / FERPA≡ethics"

Destroyed: Face/voice/wearable emotion recognition as empathy or personalization; Anxiety Score™ / continuous mood NS or parent SKU; FERPA-compliant as ethics proof; mandatory mood gates; emotion-trait tutor briefs; sell/enrich affect; empathy-camera brand.

Surviving: SAFE-PRIVACY — self-authored short-TTL check-in → pedagogy modifiers only; ERT ban; contextual integrity; refusal-safe; compliance is a floor not a doctrine (LXIX).

Kill #24 (v1.10): “Page-count science / Bayesian theater / cite-wash”

Destroyed: Chapter/page count as lab NS; “Bayesian update” without model check or experiment; DOI lists that change no product rule; FOUNDER BELIEF promoted to FACT by repetition; synthesizer that never demotes; “science-backed Constitution” ads without falsifiers; fake precision confidence numbers.

Surviving: SAFE-LABMETA — label → Red Team/check → kill/wound/survive → claim-ladder demotion of copy; small FEI hard core; hostile prior on single-study overclaim; synthesizer must delete (LXX).

Kill #25 (v1.10): “Ivy/expert résumé / peer Discord ≡ tutoring quality”

Destroyed: PhD/Ivy/expert credentials as FEI proof or default SKU; Bloom 2σ human-tutor marketing; unstructured peer Discord ≡ product tutoring; “great explainers” hire bar; expertise ≡ accurate diagnosis; tutor-learning as student North Star.

Surviving: SAFE-TUTORGRAIN — trained near-peer default; structure>pairing; Map carries diagnosis; suppress knowledge-telling; expert=escalate/coach-of-tutors; hire by fidelity drills (LXXI).

Kill #26 (v1.10): “LLM items ≡ shipped bank / item-count hero / unverified diagnostic keys”

Destroyed: LLM-generated items as ready-to-ship without independent key verify; item-count / “AI wrote N questions” as hero metric or ACT/identity proof; unverified keys as C4 diagnostic or mastery ground truth; fluency of stem ≡ keyed correctness; clone floods ≡ transfer; scaling generation at ~30% verify drop.

Surviving: SAFE-GENQ — verify-before-ship; key fails hard-fail; Kane IUA for bank uses; drop-rate gate before scale; coverage≠NS; seal C4/mastery writes to verified keys (LXXII).

Kill #27 (v1.10): “FERPA badge / click-wrap / biometric school SKU ≡ district GTM”

Destroyed: FERPA-compliant badge as district-ready or ethics; teacher click-wrap / free viral classroom as primary school path; biometric/face-voice empathy as school differentiator; marketing-first RFP without DPA+Exhibit A; parent WTP copy as LEA trust story; targeted ads / sell-affect on pupil records.

Surviving: SAFE-PROCURE — privacy review = GTM gate; NDPA-ready contracting; Exhibit A honesty; free=paid approval rigor; no biometrics; no pupil-data ads; deletion on exit (LXXIII).

Kill #28 (v1.10): “Expanding SRS / perfect-interval AI / streak-as-spacing ≡ durable math”

Destroyed: Expanding-retrieval hero marketing; “AI found your perfect interval”; streak/daily-open as the spacing engine; cram-week ≡ ACT-ready; overlearn-tonight as retention; tip-restudy as Review; shuffle ≡ spaced retrieval; schedule-guaranteed ACT points.

Surviving: SAFE-SCHED — horizon-matched ISI×RI; delayed first return; equal-ish lags for long RI; generation Returns on Practice/Map; exam dual rail (spaced learn / timed prove); anxiety-aware lag stretch; instrument SCHED-* before SRS brand claims (LXXIV).

Kill #29 (v1.11): “Tutor headcount / tenure / wellness theater ≡ quality”

Destroyed: Tutor headcount / hours / retention-% as North Stars; tenure months ≡ FEI quality; onboarding deck ≡ continuous coaching; wellness-module theater as burnout fix; Burnout Score™; marketplace-volume / Ivy-bench as attrition insurance.

Surviving: SAFE-WORKFORCE — fidelity-over-tenure; continuous coaching; coach-capacity gates hiring; burnout→load redesign; named drift channels; retention without fidelity = vanity (LXXV).

Kill #30 (v1.11): "Mastery fireworks / Ebbinghaus Score™ ≡ durable competence"

Destroyed: Tonight clear / mastery fireworks as durable competence; personal Ebbinghaus / Forgetting Score™ as UI hero; streak-as-memory; confidence≡retention; cured-forgetting ads; guaranteed retention% from decay chrome.

Surviving: SAFE-FORGET — age the evidence; JOLs≠retention; delayed probes as truth surface; frame returns as competence protection; no shame-curve hero (LXXVI).

Kill #31 (v1.11): "SM-2 / FSRS / black-box perfect-interval AI ≡ math identity"

Destroyed: SM-2/Anki/FSRS as hero brand; silent ease-hell ISI rewrite; flashcard MAE / recall-prob wins as ACT or solo_transfer proof; scheduler DAU/engagement as learning success; "perfect interval AI" copy.

Surviving: SAFE-ADAPT — calendar-first (SAFE-SCHED); adapt only inside declared RI bands with visible due+why; prefer selection-among-due over opaque rewrite; flashcard lineage ≠ FEI endpoints; instrument ADAPT-* before any SRS-optimizer marketing (LXXVII).

Kill #32 (v1.11): "Cram / bootcamp / +points packages ≡ durable math identity"

Destroyed: Cram-week ≡ ACT-ready or math-person; guaranteed ACT/SAT point SKUs; overlearn-tonight as retention method; score-only default parent brand; shame-parent GTM; streak/tonight-accuracy as durable proof; unlabeled "final mastery blitz."

Surviving: SAFE-DURABLE — dual-rail GTM (durable learn + late prove); cram-intercept not cram-cosplay; legible MoC (proof age / returns) for WTP; no point guarantees; instrument CRAM-* before anti-cram attack ads (LXXVIII).

Kill #33 (v1.11): "Instant feedback AI / always-delay science ≡ learning"

Destroyed: Instant-feedback AI as hero claim; always-delay≡desirable-difficulty science; feedback≡learning (ignore Kluger harm rate); green-check / latency NPS as North Star; ego-toast≡formative feedback; soft-wrong delay as Calm Score™.

Surviving: SAFE-FBTIME — mode-conditional timing (diagnostic hide / learn micro-delay+SE / prove KR labeled); focus>timing; prior-knowledge switch; instrument FB-* before timing brand ads (LXXIX).

Kill #34 (v1.11): "Bridge-count / re-explain ≡ connection mastery"

Destroyed: Bridge-/edge-count as moat or quality; chat or tutor re-explain of endpoint ≡ bridging the join; coverage of both topics ≡ connection; absolute "we connect everything"; missing-prereq conflated with bridge gap; shame-edge Map theater; analogy mention without cognitive supports ≡ pedagogy.

Surviving: SAFE-BRIDGE — connection-first narrative when endpoints non-naive; remediate the relation with co-present supports; measure join `solo_transfer_pass`; instrument BRIDGE-* before connection-hero ads (LXXX).

Kill #35 (v1.11): "Day-one mastery fireworks / confidence ≡ mastery"

Destroyed: Onboarding green Map / mastery confetti from empty or seed-only state; self-rated easy ≡ mastery or ACT readiness; skip-scan writing optimistic mastery; population prior sold as student biography; placement-belt / Identity Score™ cold start; diagnostic-completion % as North Star; chat fluency as calibrated prior.

Surviving: SAFE-COLD — humble prior → labeled seed → hide-correctness probes; Map chrome matches evidence age; sell honest mapping not fake knowing; instrument COLD-* before day-one personalization-hero ads (LXXXI).

Kill #36 (v1.11): “Format-count / every-format / decoration ≡ format mastery”

Destroyed: FormatId vocabulary size as moat; absolute “we teach every format” / visual-learning-for-all ads; decorative story art ≡ FormatId coverage; single-vessel concept greens ≡ format-flexible readiness; format shuffle ≡ science without conversion; Format Personality™ / learning-style quiz as North Star.

Surviving: SAFE-FORMAT — format as first-class evidence; conversion as learning object with supports; structural tags only; severity in worstWeakness; instrument FORMAT-* before format-hero ads (LXXXII).

Kill #37 (v1.11): “Tonight-% / streak / shame-rank parent portal ≡ trust”

Destroyed: Tonight accuracy / streak / XP as parent North Star; mastery fireworks on parent home; classmate shame ranks as default; live wrong-answer surveillance; Anxiety Score™ on parent view; gradebook/SIS cosplay as differentiation; controlling homework-nag loop; guaranteed points from parent widgets.

Surviving: SAFE-PDASH — belief correction via proof age + return agenda; one improvement CTA; autonomy support > control; ranks/affect off; differentiate on MoC not SIS; instrument PDASH-* before Family Leaderboard / live-progress campaigns (LXXXIII).

Kill #38 (v1.12): “Talk-% / silence / got-it ≡ Socratic quality”

Destroyed: Talk Ratio Score™ or raw tutor/student talk-% as North Star; 50/50 airtime dogma; silence theater as “Socratic”; “got it?” as constructive evidence; AI monologue sold as dialogue via token counts; emotion-AI scoring of talk; Bloom-from-student-talk marketing.

Surviving: SAFE-TALK — construction > airtime; prompt density + student constructive share + wait bands + pour-streak QA; same schema for human and AI; instrument TALK-* before Socratic/talk-balance campaigns (LXXXIV).

Kill #39 (v1.12): “Immersion / lore / Story Engagement ≡ learning”

Destroyed: Immersion or lore-depth as North Star / Story Engagement Score™; “story always helps learning”; seductive franchise trivia as pedagogy; forced pre-attempt lore walls; surface fandom personalization as hero; day-one encyclopedia before first attempt; story immersion ≡ far transfer.

Surviving: SAFE-STORYLOAD — CLT budgets narrative; structure-encode or cut; coherence > immersion; novices get less lore; skip-story; language≠math; story≠transfer; instrument STORYLOAD-* before immersive-world campaigns (LXXXV).

Kill #40 (v1.12): “User-count / star-wall / vague praise ≡ learning proof”

Destroyed: User-count / DAU / star-average as primary learning social proof; vague confidence testimonials without a named hop; Identity Score™ splash as proof; +points / guaranteed ACT case studies; Family Leaderboard as social proof; AI “math person” praise as competence; forced public portfolio / viral share as North Star.

Surviving: SAFE-PROOF — competence > crowd; enactive dated solo artifacts; provincial similarity; student-owned thin portfolio; two-sided honesty; opt-in dignity; instrument PROOF-* before star/user-count learning ads (LXXXVI).

Kill #41 (v1.12): "Item-count / complete-ACT / unsynced GENQ ≡ coverage"

Destroyed: Item-count / "N questions" as marketing hero or North Star; "complete ACT bank / every concept" without concept×level honesty; GCSE/other-examTag sold as ACT inventory; counting unverified or unsynced generated items; coverage % ≡ exam readiness or identity; ontology node count ≡ playable bank; stale-audit coverage ads; "AI fills any hole instantly" as substantiation.

Surviving: SAFE-COVER — matrix > total; gaps as features; use-tier labels (Kane IUA); examTag honesty; GENQ never counted unverified; prior substantiation via regenerated audit; instrument COVER-* before complete-bank / item-count campaigns (LXXXVII).

Kill #42 (v1.12): "Always-full-worked / never-fade / solution-first ≡ learning"

Destroyed: Always-full-worked or solution-first Solver as learning North Star / brand; never-fade step scaffolding; viewing steps ≡ mastery; expertise-blind full dumps for near-fluent learners; unlimited perfect solutions as differentiator vs ChatGPT; Fade Score™ or guidance-% vanity NS; forward-only fade dogma without local test.

Surviving: SAFE-FADE — stage not costume; completion bridge; backward fade first; expertise-aware up/down; attempt grain before reveal; proof = solo transfer; instrument FADE-* before unlimited-solutions / always-show-every-step campaigns (LXXXVIII).

Kill #43 (v1.12): "Unlimited-hints / help-NPS / avoidance-as-grit ≡ learning"

Destroyed: Unlimited hints / always-open bottom-out as learning North Star; hint volume or help NPS as pedagogy proof; help avoidance as grit; Help Tutor metacognition as guaranteed domain/ACT gains; gaming detectors as parent shame dashboards; "AI always knows when you need help" black-box brand; Help Score™ vanity NS; permanent tutor dependency as "support."

Surviving: SAFE-HELP — instrumental > executive; abuse and avoidance both fail; contingent stage not unlimited menu; SE/attempt before bottom-out; metacognition ≠ score magic; dignity telemetry; proof = solo transfer; instrument HELP-* before never-stuck / unlimited-hints campaigns (LXXXIX).

Kill #44 (v1.12): "DAU/streak/XP / FEI Score™ / thumbs-up ≡ learning North Star"

Destroyed: DAU / streak / XP / time-on-app as learning North Stars; blocked-session accuracy as readiness NS; FEI Score™ composite sold as proof; single-event tutor pay or shame ranks on `retry_120s` alone; coach thumbs-up / help NPS without `solo_transfer_pass`; science-backed identity ads from unvalidated dashboards; celebrating FEI lifts when `hint_binge` / `ai_reveal_rate` rise.

Surviving: SAFE-INSTRUMENT — ship the four FEI events; NSM stack + stage OMTM; XXI.4 co-primary gates; anti-Goodhart / anti-gaming companions; Kane IUA for marketing uses; dignity telemetry; instrument INSTR-* / Experiment C before identity-from-dashboard campaigns (XC).

Kill #45 (v1.12): "Unlimited free hard peeks / Hint Score™ / XP-for-hints ≡ learning"

Destroyed: Unlimited free hard / bottom-out peeks as learning North Star or kindness brand; Hint Score™ or raw hint-click engagement; XP/streak rewards for peeking; shame timers or parent live peek-stalk as "cost"; black-box "AI perfect hint" mystique; fade costume with a free hard submenu underneath.

Surviving: SAFE-HINT — soft before hard; contingency up/down; peek cost = construction; fade×hint one machine; separate ITS soft/hard from Saye & Brush hard/soft scaffolds; inspectable stage; instrument HINT-* before unlimited-free-hints campaigns (XCI).

Kill #46 (v1.13): “Longer≡better / Explanation Score™ / monologue≡SE ≡ learning”

Destroyed: Longer coach/Solver essays as quality or care; token count / thoroughness NPS / Explanation Score™ as learning KPIs; AI monologue labeled as self-explanation; unlimited thorough AI explanation brand; fixed multi-paragraph wrap across expertise stages; seductive coach fluff as pedagogy.

Surviving: SAFE-EXPLAIN — short principle-first before monologue; germane ≠ more text; SE before wrap; length fades with expertise/fade×hint; coherence > charm; no token North Star; instrument EXPLAIN-* before unlimited-thorough-AI campaigns (XCII).

Kill #47 (v1.13): “All-struggle≡productive / Struggle Score™ / blank-time / never-stuck ≡ learning”

Destroyed: Treating every stall as productive struggle or desirable difficulty; blank-seconds / dwell-XP / Struggle Score™ as learning North Stars; always-wait silence theater; instant hard dump on first hesitation; never-stuck answer-delivery brand; “errors always scar memory” fear marketing that skips attempts.

Surviving: SAFE-RETRIEVE — name stall mode; attempt before heavy help; TOT → wait + micro-cue; freeze → bounded soft scaffold; progress marks productive struggle (not clock alone); unsuccessful retrieval + feedback can potentiate; instrument RETRIEVE-* before struggle-theater / never-stuck campaigns (XCIII).

Kill #48 (v1.13): “Every-miss≡PF / fail-first-always / PF Score™ / discovery-without-consolidation ≡ learning”

Destroyed: Labeling every unsuccessful attempt as productive failure; fail-first / invent-always as brand or North Star; generation without Phase-2 consolidation that builds on student RSMs; PF Score™ / generation-minutes KPIs; pure discovery-as-identity; anxiety-forced endless generation cosplay; guaranteed ACT points from invent-first.

Surviving: SAFE-PF — designed generate→consolidate-on-student-RSMs; fidelity + prior-knowledge gates; guided/I-PS default when priors thin, procedural/exam chrome, or freeze (SAFE-RETRIEVE exit); proof = conceptual + transfer; instrument PF-* before invent-first campaigns (XCIV).

Kill #49 (v1.13): “Guild cosplay / modeling≡lecture / never-fade / Apprenticeship Score™ ≡ learning”

Destroyed: Guild / master / journeyman costume as pedagogy proof; modeling as lecture or AI monologue without student attempt; never-fade permanent coaching as kindness; talk-% alone as cognitive apprenticeship; Apprenticeship Score™ / modeling-minutes NS; Discord≡apprenticeship; reciprocal-teaching costume without turn-taking; CA marketing without inspectable phase logs.

Surviving: SAFE-APPRENTICE — model control/heuristics → coach → scaffold→fade; articulation required; shared `ca_phase` for HITL+AI; proof = solo transfer + articulation quality; instrument APPRENTICE-* before guild / apprenticeship-costume campaigns (XCV).

Kill #50 (v1.13): “Focus Score™ / phone-shame / mid-attempt DAU pushes / multitasking-as-talent ≡ learning”

Destroyed: Focus Score™ / Deep Work Score™ as North Star; phone-shame or character-moralizing as pedagogy; “digital natives multitask fine” as free pass; mid-attempt streak/DAU pushes as engagement science; phone presence as proof of modern learning; focus-kit ads promising guaranteed ACT points; covert gaze/mic attention surveillance.

Surviving: SAFE-ATTN — protect mid-attempt windows; residue-aware park/resume; opt-in focus + tech breaks; presence hygiene without shame; interrupt telemetry ≠ productive struggle; instrument ATTN-* before focus-costume campaigns (XCVI).

Kill #51 (v1.13): "Dual Coding Score™ / always-add-picture / styles meshing / animation-default ≡ learning"

Destroyed: Dual Coding Score™ / Visual Engagement Score™ as North Star; always-add-a-picture as dual-coding science; decorative chrome ≡ FormatId learning; visual-learner meshing tracks; animation/video as default modernity over static annotated figures; stacked narration+on-screen text+busy figure as thoroughness; split legends as information density; unverified generated figures as diagnostic truth; guaranteed ACT points from multimedia dual-coding ads.

Surviving: SAFE-DUAL — structural dual coding; contiguous labels; coherence weeding; static annotated default; conversion remains the object (SAFE-FORMAT); instrument DUAL-* before dual-coding / visual-learner campaigns (XCVII).

Kill #52 (v1.13): "Monitoring Score™ / metacognition-without-control / one-scan calibration ≡ learning"

Destroyed: Monitoring Score™ / Metacognition % as North Star; "we teach metacognition" ads without judgment→control wiring; global easy/kinda/hard alone as calibrated monitoring science; immediate post-item surety equated to delayed JOL accuracy; confidence slider without miss-class routing; raise-metacognition / raise-confidence as identity transformation; one gap-scan as calibration cure; guaranteed ACT points from metacognitive gap-scan packaging.

Surviving: SAFE-MONITOR — named judgment types; C4 destake; confidence×outcome miss classes; control wiring mandatory; densifies SAFE-CALIB / SAFE-COLD; instrument MONITOR-* before metacognition brand campaigns (XCVIII).

Kill #53 (v1.13): "AI-replaces-tutors / tutoring-is-free / hours-booked / Session ROI Score™ ≡ learning"

Destroyed: AI replaces tutors / tutoring-is-free slogans; unlimited human entitlement SKUs; hours booked / session count as North Star; Session ROI Score™ gamified ops vanity; VanLehn ITS≈human as ChatGPT-dump license; Bastani practice-with-AI scores as learning proof; school pooled tutoring ES as thin marketplace ACT guarantee.

Surviving: SAFE-ROI — tutor minutes as scarce inventory; Map minute budgets; constrained AI for reps; humans for joins/destake/repair/witness; FEI per tutor-minute; instrument ROI-* before AI-replaces-tutors / unlimited-human campaigns (XCIX).

Surviving (provisional): SAFE-EXPTIME — grain-local example↔solve timing; completion bridge; adaptive>fixed when evidence exists; rapid rechecks beat Likert destiny; densifies SAFE-FADE; instrument EXPTIME-* before universal E0→E3 / Timing Score™ campaigns (C).

Surviving (provisional): SAFE-SRL — Zimmerman forethought→performance→reflection with phase→control writeback; microanalysis>trait surveys; process attributions; densifies SAFE-MONITOR/AAR/ATTN/HELP; instrument SRL-* before SRL Score™ / streak≡SRL / metacognition-module ads (CI).

Surviving (provisional): SAFE-GENERATE — SOI-forcing sparse generative asks; student produces (coach does not substitute); teach-back requires the explanation act; guided FormatId draw/map; densifies

SAFE-EXPLAIN/DUAL/SRL/SE; instrument GENACT-* before Generative Learning Score™ / AI≡generation / eight-strategy ACT ads (CII).

Part XV — References (verified starting set)

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Part XVI — Mechanism Deep Dive: How identity actually changes

This part answers the research question at mechanism grain. It is synthesis, not settled science.

XVI.1 What “confident mathematical thinker” means (operational)

Do not use: “likes math apps” or “says math is fun once.”

Working definition (HYPOTHESIS): A confident mathematical thinker is a person who, in math-relevant contexts:

1. **Approaches** moderately hard problems without immediate avoidance.
2. **Persists** after a miss long enough to extract information.
3. **Attributes** success primarily to controllable causes (strategy, practice, help-seeking) rather than luck or tutor magic.
4. **Selects** into harder future opportunities when available (courses, contests, careers) at rates above their prior baseline.
5. **Endorses** a math-capable self-concept *and* behaves consistently with that endorsement.

Items 1–4 are leading. Item 5 is lagging. Product should optimize leading indicators while periodically measuring the lagging identity claim.

XVI.2 The threat–competence–narrative triad

Three literatures converge on the same transformation engine:

Mechanism	Core claim	Confidence	Key sources
Threat / affect	Math anxiety transiently reduces working-memory resources available for math	**High (FACT direction)**	Ashcraft & Kirk (2001); Suárez-Pellicioni et al. review (2016); WM mediation meta (2021)
Competence evidence	Motivation requires felt competence; mastery structures can raise achievement when mastery is real	**Medium–High**	SDT (Ryan & Deci); Kulik mastery meta; Slavin critique
Narrative / identity	Self-stories and belonging cues shape challenge-seeking and persistence	**Medium**	Yeager et al. (2019); Walton/Cohen belonging tradition; narrative psychology

HYPOTHESIS — Triad necessity: For students starting from “I am bad at math,” improving any one factor alone is usually insufficient:

- Safety without competence → comfort without skill → identity still fragile.
- Competence without safety → possible for some; many never enter the practice volume.
- Narrative without either → empty affirmation (false growth mindset).

XVI.3 Founder tutoring sequence — audit

FOUNDER BELIEF (observed in tutoring):

1. Reduce fear
2. Demystify mathematics
3. Explain why humanity invented it
4. Simplify the narrative
5. Create a tiny success
6. Reinforce success
7. Celebrate progress → identity shifts

Red Team audit:

Step	Universal?	Evidence status	Failure mode
1 Fear ↓	Common for anxious subset; not for all	Strong for anxious learners	Over-soothing that removes challenge
2 Demystify	Often helpful	Medium (cognitive clarity)	Over-simplification that lies
3 Invention story	Founder prior	**Under-evidenced at RCT scale**	Cultural mismatch; time cost
4 Simplify narrative	Often helpful	Medium	Removes necessary difficulty
5 Tiny success	Near-universal lever	High (competence need)	Too-easy wins → hollow identity
6 Reinforce	High if attribution-correct	High (feedback literature)	Praise of talent or praise of empty effort
7 Celebrate	Helpful if genuine	Medium	Performative celebration → cynicism

Improved sequence (HYPOTHESIS — “SAFE-CRAFT”):

1. **S**afety: threat appraisal ↓ (tone, privacy, soft-wrong, no public shame)
2. **A**ttempt: student acts before receiving a full lecture (productive struggle at right grain)
3. **F**eedback-as-information: miss becomes diagnosis, not character
4. **E**arned micro-win: success the student can own
5. **C**ausal story: “what worked” (strategy), optionally “why humans invented this tool”
6. **R**e-entry: immediate second attempt / next slightly harder item
7. **A**ttribution + witness: tutor/system names the growth specifically

8. **F**uture self: map fill / path unlock that marks identity-relevant progress
9. **T**ransfer check: varied item so win was not clone memorization

What changed vs founder sequence: Attempt moves earlier; invention-story becomes optional enrichment inside step 5, not a prerequisite to agency; transfer check prevents fake mastery.

XVI.4 Math anxiety — what is known, what is not

FACT: Higher math anxiety correlates with lower math performance (meta r ≈ −0.17 in one 2021 synthesis of 57 studies — modest average, not destiny).

FACT: Ashcraft & Kirk (2001) show math anxiety can act like a dual-task load: worry competes for working memory, especially on computation-span / carry-heavy tasks.

HYPOTHESIS: In product UX, anything that increases social-evaluative threat (red buzzers, public leaderboards for novices, tutor contempt, parent hovering UI) will shrink effective WM and look like "they don't get it."

Contradictions / limits:

- Not all underperformance is anxiety; some is missing knowledge, poor curriculum, sleep, language barriers.
- Trait vs state anxiety differ; trait may partly reflect prior failures (reverse causality possible).
- High performers can be anxious and still succeed via avoidance of hard electives — a hidden identity cost.

MindCraft rule: Treat anxiety as a first-class load on the learning system, not as a soft skill sidebar.

XVI.5 Deliberate practice — necessary, not sufficient

FACT / TRADITION: Ericsson's deliberate practice emphasizes structured, feedback-rich, effortful practice aimed at improvement (not mere repetition).

FACT / CRITIQUE: Hambrick and colleagues argue deliberate practice is important but not sufficient; in chess/music reanalyses, deliberate practice often explains roughly ~1/3 of reliable variance, leaving most unexplained (abilities, starting age, opportunity, etc.).

Implication for MindCraft: Do not sell "10,000 hours." Sell **high-quality attempts under feedback**. Also respect individual differences: same practice volume will not equalize outcomes. Identity work must include "progress relative to self," not only absolute rank.

XVI.6 Tutoring effects — demythologize Bloom

FACT: Bloom (1984) popularized the "2 sigma" tutoring + mastery framing.

FACT / CALIBRATION: Broader tutoring metas are far smaller (often ~0.3–0.4 SD in modern RCT syntheses cited by Education Next). VanLehn (2011) estimated human tutoring ~0.79 SD vs no tutoring, with step-based ITS nearly comparable (~0.76).

Implication: Human tutors remain valuable, especially for affect + accountability + diagnosis of weird misconceptions. They are not magic 2-sigma machines. AI step-level feedback can capture much of the *cognitive* tutoring benefit; the scarce remainder is Layers 4–6 (emotion, identity, accountability).

Part XVII — Scarcity Thesis: Full Red Team Trial

XVII.1 The claim under oath

FOUNDER BELIEF: AI commoditizes knowledge; future scarcity is motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

XVII.2 Cross-examination

Claim A — "Explanations are free"

Verdict: Directionally **FACT** for commodity explanations; **FALSE** for *trusted*, *personalized*, *correct*, *curriculum-aligned* explanations under time pressure.

Evidence: Generative AI + open platforms collapse marginal cost of text explanation.

Contradiction: Hallucinations, shallow fluency, and misalignment create a *trust tax*. Parents may still pay for verified human judgment.

Claim B — "Tutoring is free"

Verdict: FALSE.

AI chat is cheap. High-quality tutoring includes scheduling, relationship, emotional labor, accountability, and reputation. Those remain expensive. Nickow et al.–style tutoring metas still show meaningful effects when tutoring is structured — it is not free at scale.

Claim C — "Content is free"

Verdict: MOSTLY FACT for undifferentiated content; **FALSE** for sequenced, assessed, outcome-linked content systems.

Khan/YouTube prove content abundance. Differentiation is curation + measurement + human wrap.

Claim D — "Motivation/identity become the scarce goods"

Verdict: HYPOTHESIS with strategic plausibility.

As cognitive help approaches free, *willingness to engage difficulty* becomes the binding constraint for many students. This does not mean motivation products automatically win — many habit apps fail to produce identity.

Claim E — "Emotional safety is first-order"

Verdict: HYPOTHESIS — true for anxious / threatened learners; not universal.

For some, speed drills and competition *increase* engagement. Segment, do not romanticize softness.

XVII.3 Surviving thesis (company doctrine)

Adopted doctrine (refined v1.7):

> MindCraft does not compete primarily on explanation volume, AI tutor warmth, streak/XP theater, grit meters, or story immersion. It competes on converting threatened learners into challenge-seeking mathematical agents through diagnosis, affective design, mastery-climate evaluation, sequenced desirable difficulties (SAFE-DD), student-generated why, variation-designed transfer, productive-error routing, narrative meaning under equity constraints, and human accountability — with AI as *guarded* infrastructure that must clear `solo_transfer_pass` ≥ no-AI.

Falsifiers (90-day capable):

1. Anxious segment shows no FEI lift vs explanation-first control.
2. Parents refuse to pay when score gains lag identity metrics.
3. AI-alone matches human+AI on persistence + transfer for target segment.
4. Guarded Solver raises concurrent scores but loses `solo_transfer_pass` vs no-AI (PWC fail).
5. Interleave/SE regimes raise delayed mix but destroy `retry_120s` in high-anxiety segment without sequenced SAFE-DD (XLI; pending DD-1/DD-2).
6. Soft-wrong without springboard/route matches key-only on retry + misconception recurrence (MIS-1 fail).
7. VT/hug sequences fail to beat random surface skins on next-day format-hop transfer (TR-1 fail).

Part XVIII — Systems Maps (loops, not features)

XVIII.1 FEI reinforcing loop (desired)

```
+------------------ identity claim ("I do math") <---+
| |
v |
safety UX ----> attempt ----> informative miss ----> coach
^ | |
| v v
+------ map fill / witness <---- earned success <+
```

XVIII.2 Shame balancing loop (destroy this)

```
public failure signal -> threat ↑ -> avoidance -> fewer attempts
^ |
+--------- skill lag / worse scores ------+
```

XVIII.3 Extrinsic colonization loop (danger)

```
streak / points ↑ -> return ↑ -> shallow grinding ↑
| |
+-- identity unchanged --------+--> burnout / streak break → exit
```

Design rule: Habit loops may feed FEI but must not replace identity markers.

XVIII.4 Parent trust loop

```
visible diagnosis + honest progress -> parent trust ↑ -> continued purchase
^ |
+---- outcome evidence (scores / challenge-seeking)
```

FOUNDER QUESTION unresolved: Optimize Maya's identity or parent's anxiety? Doctrine: report *both* challenge-seeking and skill evidence; never fake either.

XVIII.5 Network effects (honest)

MindCraft today has weak classic network effects. Do not pretend otherwise.

Possible compounding advantages (HYPOTHESIS):

1. Ontology + misconception memory → better diagnosis over time (data moat if privacy-respecting).
2. Tutor playbooks trained on FEI outcomes → service quality moat.
3. Story worlds with longitudinal character arcs → switching costs of meaning (fragile; easy to cargo-cult).

Part XIX — Cross-Domain Transfer Catalog

Principles that repeatedly change human behavior outside classrooms — filtered for MindCraft use.

Domain	Robust mechanism	Transfer	30-year durable?	AI improve?
Exposure therapy	Graded exposure under safety	Anxiety-graded problem sets	Yes	Yes (personalize gradient)
Sports film study	Specific error review	Wrong-choice coach	Yes	Yes (auto clip of miss type)
Music pedagogy	Technique → repertoire → performance	Drill → quest → "recital" signal	Yes	Partial
Martial arts	Visible ranks + dojo norms	Map regions + tutor dojo culture	Yes	Weak for norms
Aviation	Checklists under stress	Procedure scaffolds when anxious	Yes	Yes

Games (Celeste-like)	Fair failure + retry culture	Soft-wrong physics	Yes	Yes
Chess	Annotated game review	Session postmortems	Yes	Yes
Coaching	Identity + accountability	Tutor as witness	Yes	Assist, not replace
Religion/ritual	Meaning + belonging	Light rituals only; avoid cult dynamics	Yes (human)	Risky
Creator economy	Public craft identity	Optional shareable math artifacts	Uncertain	Yes

Anti-transfer: Casino variable rewards → addiction without skill. Do not copy engagement maxing from social media.

Part XX — Competitive Landscape (mechanism lens)

Product	Primary scarce resource they sell	Identity claim?	Risk
Khan Academy	Free mastery content + reputation	Weak–medium	Explanation commodity; “mastery” without transfer_pass
Duolingo	Habit / streak motivation	Weak (language identity sometimes)	Extrinsic ceiling; appearance of consistency
Brilliant	Aesthetic problem joy + prestige	Medium	Narrow segment; not Maya threat profile
Coursera/edX	Credentials	Medium (career identity)	Completion collapse
Chegg / homework help	Answers under deadline	Anti-identity (outsourcing)	Integrity & AI shock
Private tutors	Human accountability + customization	High when good	Supply constrained
ChatGPT tutors	Instant explanation	Low unless wrapped	Trust / hallucination; Bastani Base harm
MindCraft (target)	FEI conversion + tutor witness + gap diagnosis + solo transfer	**Intended high**	Must prove, not assert

Strategic implication (v1.13 densified): Do not out-Khan Khan on content breadth or item-count theater (SAFE-COVER). Do not out-ChatGPT on always-full-worked / unlimited-solutions Solver (SAFE-FADE). Do not out-Duo Duo on streaks/leagues/XP or expanding-SRS mystique. Do not out-Brilliant Brilliant on puzzle delight. Do not out-ChatGPT on fluency or graph-file cosplay. Do not out-Anki/FSRS on black-box interval theater. Do not out-Kaplan/bootcamp on massed cram or +points packages (SAFE-DURABLE). Do not out-instant-feedback AI or always-delay cosplay (SAFE-FBTIME). Do not out-node-weakness theater when the failure is a join (SAFE-BRIDGE). Do not out-day-one mastery fireworks or placement-belt cold starts (SAFE-COLD). Do not out-format-count / every-format / visual-learning-style quizzes (SAFE-FORMAT). Do not out-Dual-Coding-Score / always-add-picture / animation-first multimedia theater (SAFE-DUAL). Do not out-Explanation-Score / longer≡better monologue / monologue≡SE theater (SAFE-EXPLAIN). Do not out-Struggle-Score / blank-time / always-wait / never-stuck stall theater (SAFE-RETRIEVE). Do not out-fail-first / every-miss-as-PF / discovery-without-consolidation cosplay (SAFE-PF). Do not out-guild / modeling-as-lecture / Apprenticeship Score™ costume (SAFE-APPRENTICE). Do not out-Focus-Score / phone-shame / mid-attempt DAU-push theater (SAFE-ATTN). Do not out-Monitoring-Score / metacognition-without-control / one-scan calibration theater (SAFE-MONITOR). Do not out-AI-replaces-tutors / tutoring-is-free / unlimited-human / hours-booked / Session ROI Score™ theater (SAFE-ROI). Do not out-parent fireworks portals, Family Leaderboards, or live wrong-answer stalk (SAFE-PDASH). Do not out-Talk Ratio Score / silence theater / Socratic-by-talk-% ads (SAFE-TALK). Do not out-immersive lore / Story Engagement Score / franchise trivia-as-pedagogy (SAFE-STORYLOAD). Do not out-user-count / star-wall / vague-confidence testimonials as learning proof (SAFE-PROOF). Do not out-DAU/streak/XP / FEI Score™ dashboard theater as learning proof (SAFE-INSTRUMENT). Do not out-tutor-headcount vanity or wellness-theater fidelity (SAFE-WORKFORCE). Do not out-grit character apps, therapy Calm Score™, sports academies, conservatories, grandmaster cosplay, pink STEM SKUs, survey-priced ACT guarantees, Ivy-tutor theater, emotion cameras, or FERPA-badge GTM. Out-compete on the **session + ops + honesty stack**: inspectable diagnosis (SAFE-ONTOLOGY), honest cold-start mapping (SAFE-COLD), format-gap conversion evidence (SAFE-FORMAT), contiguous structural dual-channel figures (SAFE-DUAL), parent MoC proof-age surfaces (SAFE-PDASH), witnessable dated solo competence artifacts (SAFE-PROOF), blueprint-honest coverage matrices with gaps named (SAFE-COVER), faded guidance that earns solo carry (SAFE-FADE), instrumental contingent help not unlimited-hint dumps (SAFE-HELP), FEI North Star events under co-gates not vanity OMTMs (SAFE-INSTRUMENT), contingent soft→hard peeks priced in construction not free dumps (SAFE-HINT), length-contingent principle coaches with SE-before-wrap (SAFE-EXPLAIN), mode-contingent retrieval (TOT cue vs freeze scaffold) without Struggle Score™ (SAFE-RETRIEVE), fidelity-gated productive-failure generate→consolidate when priors allow (SAFE-PF), model→coach→fade apprenticeship phases with articulation (SAFE-APPRENTICE), attempt windows protected from residue and mid-attempt pushes (SAFE-ATTN), connection-first Map when endpoints are green (SAFE-BRIDGE), Map-briefed prompt>pour humans with minute budgets (SAFE-HITL / TUTORGRAIN / WORKFORCE / ROI), verify-before-ship banks (SAFE-GENQ), privacy-bound affect + district trust packets (SAFE-PRIVACY / PROCURE), horizon-matched returns (SAFE-SCHED), calendar-first banded adaptivity (SAFE-ADAPT), time-honest aged evidence (SAFE-FORGET), dual-rail durable GTM (SAFE-DURABLE), mode-conditional feedback clocks (SAFE-FBTIME), SAFE-DP spine under SAFE-DD, productive-error + calibration (SAFE-MISCON / CALIB), recoverable AI truth (SAFE-REPAIR), validated identity on long clocks (SAFE-IDMEASURE / LONGID), structure honesty (SAFE-ELL / GENDER / STRUCTURE), parent CBC (SAFE-WTP), and transfer when help is gone — then *say that* in marketing (CSA-2). Lab process itself obeys SAFE-LABMETA (falsify before “science-backed”).

Part XXI — Metrics Dictionary (instrumentation)

XXI.1 Banned as North Stars

- Raw DAU / time-on-app without transfer
- Streak length alone
- Explanation open-rate / AI chat token count
- Points / coins / XP velocity
- Blocked-session accuracy alone
- Coach thumbs-up / “felt helpful” without `solo_transfer_pass`
- Named leaderboard rank / league tier
- Resilience Score™ / grit meter / MRS / Grit Score™ as in-app KPI
- Midnight minutes practiced / all-nighter volume
- Belief Score™ / confidence meters / confidence streaks
- Community tab visits / Discord volume / games played / rating
- Calm Score™ / anxiety-must-hit-zero meters
- Tip-recall / film-room engagement without method-change + transfer
- Math Identity Score™ / two-item “math person” alone / splash identity Likert
- Parent survey WTP dollars / guaranteed ACT-point package tiers
- NAEP EL-gap closure / gender STEM-gap closure as product claims
- Ontology node-count / “powered by our knowledge graph” without FEI outcome
- Hours-booked / smile-star tutor QA without `solo_transfer_pass`
- Anxiety Score™ / face-voice emotion engagement meters
- Chapter/page count / “science-backed Constitution” without falsifiers
- Item-count / “AI wrote N questions” without verify rate
- Complete-ACT / every-concept coverage ads without concept×level matrix + gaps
- FERPA-compliant badge / teacher click-wrap as district readiness
- Expanding-SRS / “perfect interval AI” / streak-as-spacing engines
- Tutor headcount / retention-% without fidelity@tenure
- Mastery fireworks / personal Ebbinghaus / Forgetting Score™ as durable-competence claims
- SM-2/FSRS/Anki hero / black-box perfect-interval AI as learning NS
- Cram-week / ACT-point package tiers as durable-identity claims
- Instant-feedback latency / always-delay as learning NS
- Bridge-count / “we connect everything” without join transfer
- Day-one mastery greens / placement-belt cold start
- Format-count / Format Personality™ / learning-style quiz NS
- Parent tonight-% / Family Leaderboard / live wrong-answer stalk
- Talk Ratio Score™ / raw talk-% / 50/50 airtime dogma as Socratic NS
- Story Engagement Score™ / immersion / lore-depth as learning NS
- Fade Score™ / always-full-worked / unlimited-solutions as learning NS
- Help Score™ / unlimited-hints / never-stuck dump as learning NS
- Hint Score™ / XP-for-hints / unlimited free hard peeks as learning NS
- FEI Score™ / single-event tutor commissions on `retry_120s` alone
- Coach thumbs-up / help NPS without `solo_transfer_pass`

- Explanation Score™ / token-count quality / longer≡better monologue as learning NS
- Struggle Score™ / blank-time / dwell-XP as productive-struggle NS
- PF Score™ / fail-first-always / generation-minutes without consolidation
- Apprenticeship Score™ / modeling-minutes / guild cosplay as pedagogy NS
- Focus Score™ / Deep Work Score™ / mid-attempt DAU pushes as learning NS
- Dual Coding Score™ / Visual Engagement Score™ / always-add-picture as learning NS
- Monitoring Score™ / Metacognition % / one-scan calibration as learning NS
- Session ROI Score™ / hours-booked / AI-replaces-tutors as learning NS

XXI.2 Leading indicators (ship first)

Metric ID	Definition	Why
`retry_120s`	New attempt on same or isomorphic item within 120s of soft-wrong	Persistence under safety
`coach_shown`	Wizard/coach surfaced	Treatment exposure
`write_exit_to_retry`	Leave write mode then attempt	Agency after reflection
`challenge_accept`	Chose harder level when offered	Challenge-seeking (instrument *motive*)
`hint_binge`	≥3 hints without independent solve	Gaming / helplessness / failed exposure fade
`transfer_pass`	Correct on varied item after mastery mark	Anti-fake-mastery
`solo_transfer_pass`	Transfer with AI/Solver closed or denied	Anti-crutch (XXXIII)
`ai_reveal_rate`	Full-answer / unguarded reveal frequency	Crutch exposure
`strategy_class_error`	Wrong *procedure family* on mixed set	Interleaving discrimination (XXXIX)
`se_principle_hit`	Correct structured principle/ingredient pick when prompted	SE quality proxy (XL)
`help_recruit_then_solo`	Opened card/tutor/peer help then solved without full reveal	SAFE-RESILIENCE support path (XLV)
`misconception_route_hit`	Soft-wrong routed to mapped mis_ / ingredient failure_mode	SAFE-MISCON classification (XLIX)
`format_hop_pass`	Correct after declared format/context hop (VT check)	SAFE-TRANSFER discernment (XLVIII)
`confidence_miss_tier`	High vs low confidence on miss (when elicited)	SAFE-CALIB / hypercorrection (L)

`annotate_before_reveal`	Student principle/why logged before key/coach/model	SAFE-ANNOT / SAFE-AAR (LVII / LIV)
`film_cue_count`	Coaching points attached to attempt clip (cap ≤3)	SAFE-FILM less-is-more (LV)
`avoided_format_approach`	Attempt on previously avoided format/stakes step	SAFE-EXPOSE approach rate (LVIII)
`repair_reattempt_pass`	Correct destaked re-attempt after owned AI miss	SAFE-REPAIR calibration (LX)
`escalation_resolve_48h`	Packetized human handoff closed with fix	SAFE-REPAIR escalate SLA (LX)
`improve_how_specificity`	Debrief yields concrete next-attempt tweak	SAFE-AAR close quality (LIV)
`stakes_ladder_step`	Declared pressure step completed after destaked equip	SAFE-EXAM dose-to-prove (LXIII)
`lang_vs_math_miss_flag`	Miss tagged language-parse vs concept (when elicited)	SAFE-ELL load separation (LXIV)
`identity_battery_ok`	Validated multi-factor identity admin completed (not splash)	SAFE-IDMEASURE demand-aware (LXI)
`bandwidth_tax_skip`	Incomplete mission under self-reported scarcity/stress (not grit)	SAFE-STRUCTURE hygiene (LXVI)
`map_brief_open`	Tutor/session opened Map diagnosis before first explain	SAFE-HITL / ONTOLOGY brief fidelity (LXVIII / LXVII)
`key_verify_pass`	Generated item passed independent key check before bank write	SAFE-GENQ ship gate (LXXII)
`return_isi_days`	Days between concept exposure and scheduled retrieval return	SAFE-SCHED horizon lag (LXXIV)
`trust_packet_ready`	DPA + Exhibit A + data map + deletion/breach posted for LEA	SAFE-PROCURE GTM gate (LXXIII)
`tutor_fidelity_30d` / `tutor_fidelity_90d`	FEI rubric sample pass rate at tenure horizons	SAFE-WORKFORCE drift gate (LXXV)
`coach_util`	Coach sample bandwidth vs active tutor count	SAFE-WORKFORCE hire-cap (LXXV)
`last_proof_age_days`	Days since last delayed/generation proof on a concept	SAFE-FORGET aged evidence (LXXVI)
`retention_probe_7d` / `retention_probe_28d`	Pass rate on scheduled delayed retrieval probes	SAFE-FORGET / MoC honesty (LXXVI)
`return_due_reason`	Shown rationale chip for next return (horizon vs evidence pull)	SAFE-ADAPT inspectability (LXXVII)

`adapt_band_isi_days`	Actual ISI vs horizon baseline (within declared band)	SAFE-ADAPT modulation audit (LXXVII)
`cram_intercept_shown`	User requested massed week and saw RI tradeoff + denser Returns offer	SAFE-DURABLE cram-intercept (LXXVIII)
`prove_rail_labeled`	Exam-week intensity sessions tagged prove (not mastery)	SAFE-DURABLE dual-rail honesty (LXXVIII)
`fb_microdelay_ms`	Forced SE/attempt gate duration before learn-rail reveal	SAFE-FBTIME learn gate (LXXIX)
`soft_wrong_se_before_reveal`	Soft-wrong path required student why before coach	SAFE-FBTIME / SAFE-SE (LXXIX)
`bridge_gap_mission_start`	Practice launched from bridge-gap CTA (not node-only)	SAFE-BRIDGE Map/Practice (LXXX)
`join_solo_transfer_pass`	Solo pass on delayed item requiring named bridge	SAFE-BRIDGE join transfer (LXXX)
`seed_probe_disagree`	High-confidence seed vs hide-correctness probe miss on same concept	SAFE-COLD calibration (LXXXI)
`cold_map_honesty`	First-session Map showed untouched/seed labels (not mastery greens)	SAFE-COLD chrome (LXXXI)
`format_gap_mission_start`	Practice launched from format-gap CTA (not topic-only)	SAFE-FORMAT Map/Practice (LXXXII)
`format_hop_solo_transfer_pass`	Solo pass on delayed item in previously weak FormatId	SAFE-FORMAT conversion (LXXXII)
`parent_proof_age_view`	Parent opened proof-age strip / MoC surface	SAFE-PDASH honesty (LXXXIII)
`parent_return_cta_start`	Student return/mission started after parent CTA or digest	SAFE-PDASH action loop (LXXXIII)
`student_constructive_share`	Why/SE/soft-wrong turns / student turns (sampled session)	SAFE-TALK construction fidelity (LXXXIV)
`prompt_density`	Eliciting prompts / tutor turns	SAFE-TALK prompt>pour (LXXXIV)
`wait_ms_p50`	Median wait time 1 and 2 (ms)	SAFE-TALK wait coaching (LXXXIV)
`pour_streak_max`	Longest consecutive tutor explain turns	SAFE-TALK monologue detect (LXXXIV)
`story_time_to_first_attempt_s`	Seconds from module open to first math attempt	SAFE-STORYLOAD pre-attempt budget (LXXXV)
`story_wrap_noninferior`	Transfer/retention within margin vs plain stem	SAFE-STORYLOAD learning gate (LXXXV)

`proof_card_share_opt_in`	Opt-in shares of dated solo-transfer proof cards	SAFE-PROOF share dignity (LXXXVI)
`proof_card_named_hop`	Share/view events that name concept/bridge/FormatId + proof type	SAFE-PROOF specificity (LXXXVI)
`coverage_matrix_view`	Parent/LEA/admin opened dated concept×level coverage matrix	SAFE-COVER honesty surface (LXXXVII)
`coverage_claim_mismatch`	Marketed coverage vs playable audit disagree (ops incident)	SAFE-COVER substantiation (LXXXVII)
`fade_stage`	Solver/help guidance stage E0–E3 at attempt start	SAFE-FADE ladder (LXXXVIII)
`exptime_grain`	Concept×format×bridge grain used to set/recheck fade stage	SAFE-EXPTIME (C)
`srl_phase_event`	Forethought / performance / reflection micro-event with control writeback	SAFE-SRL (CI)
`gla_soi_event`	Sparse generative ask (summarize/teach/draw/map) with SOI quality gate	SAFE-GENERATE (CII)
`fade_step_attempt`	Student-generated completion/SE before next worked unlock	SAFE-FADE attempt grain (LXXXVIII)
`help_deliberate_ms`	Time on non-bottom hint level before next action	SAFE-HELP deliberate help (LXXXIX)
`help_executive_race`	≥2 hint-level advances under short dwell before bottom-out	SAFE-HELP abuse detect (LXXXIX)
`challenge_motive`	Enum on `challenge_accept`: mastery / appearance / normative / unknown	SAFE-INSTRUMENT motive gate (XC)
`fei_event_fire_ok`	Soft-wrong path emitted retry/coach/binge companions without dark funnel	SAFE-INSTRUMENT completeness (XC)
`hint_level_open`	Soft/principle vs hard/bottom-out level shown at request	SAFE-HINT ladder grain (XCI)
`hard_peek_cost_ok`	Hard level advanced only after SE/blank/dwell/stage gate	SAFE-HINT construction cost (XCI)
`coach_wrap_tokens`	Coach/Solver wrap token count before student SE	SAFE-EXPLAIN length discipline (XCII)
`se_before_wrap`	Student principle/why logged before full coach wrap	SAFE-EXPLAIN SE-before-wrap (XCII)
`stall_mode`	Enum: tot / blank_freeze / progress / avoidance / unknown	SAFE-RETRIEVE mode classify (XCIII)

tot_cue_then_retry	Micro-cue after TOT wait then independent retry	SAFE-RETRIEVE TOT path (XCIII)
pf_phase	Mission phase: generate / consolidate / guided	SAFE-PF sequence fidelity (XCIV)
pf_rsm_captured	Student RSM attempt logged before Phase-2 C&C	SAFE-PF consolidation input (XCIV)
ca_phase	Shared model/coach/scaffold/fade/articulate/reflect label	SAFE-APPRENTICE phase log (XCV)
ca_articulate_ok	Student articulation before bottom-out reveal	SAFE-APPRENTICE articulate gate (XCV)
attempt_interrupted	Focus lost / push / tab-away during active attempt	SAFE-ATTN residue (XCVI)
focus_mode_opt_in	Session used opt-in focus / push-suppress mode	SAFE-ATTN hygiene (XCVI)
figure_contiguous_ok	Stem+figure labels spatially contiguous (audit flag)	SAFE-DUAL contiguity (XCVII)
decorative_figure_reject	Decorative figure blocked from diagnostic use	SAFE-DUAL coherence (XCVII)
judgment_type	Gap-scan judgment enum: ease / RCJ / delayed_JOL	SAFE-MONITOR named judgment (XCVIII)
monitor_control_wired	Confidence×outcome miss class drove a routing action	SAFE-MONITOR control wiring (XCVIII)
tutor_minute_budget_ok	Human session stayed within Map minute budget	SAFE-ROI scarce inventory (XCIX)
fei_per_tutor_minute	FEI events / billed tutor minutes (ops; not vanity NS)	SAFE-ROI allocation proof (XCIX)

XXI.3 Lagging indicators

Metric ID	Definition
math_person_item	1–7 agreement: “I am a math person” (math-specific; never sole endpoint)
cribbs_interest / cribbs_recognition / cribbs_competence	Factor scores from validated battery (LXI)
anxiety_state_item	Short state anxiety before session
advanced_intent	Intent to take harder course / contest
tutor_witness_note	Tutor tagged identity-relevant growth
identity_fade_26w / identity_fade_52w	Triangulated identity movement vs baseline at horizons (LXII)

XXI.4 Decision rule (HYPOTHESIS)

Ship changes that raise `retry_120s` and mastery-motive `challenge_accept` without raising `hint_binge` / `ai_reveal_rate`, and without dropping `transfer_pass` or `solo_transfer_pass`. Prefer delayed mixed accuracy over blocked session accuracy.

Part XXII — Appendices

Appendix A — Glossary

Term	Meaning
FEI Loop	Fear → Evidence → Identity system
ITC	Identity Transformation Cascade
Soft-wrong	Miss treated as information, not shame theater
Gap scan	Low-shame diagnostic of confidence/exposure
Challenge-seeking	Choosing harder math opportunity when optional
False growth mindset	Effort praise without strategy / conditions
SAFE-CRAFT	Improved tutoring sequence (Part XVI.3)
Layers 0–6	Knowledge commodity stack (Part VI.3)
HID	Habit/Identity Divergence — streaks can raise return without identity
MoC	Memory of Competence — spaced retrieval after transfer_pass
PWC	Pedagogy Wrap Constraint — guarded AI + student generation + solo transfer bar
Claim ladder L0–L4	Max defensible causal/marketing claim given design (Part XXXIV)
E×V×Cost	Eccles expectancy × subjective task value × cost triad
TARGET	Classroom/goal-structure dimensions cueing mastery vs performance
SAFE-DD	Equip → destake → dose → frame → measure (Part XLI)
SAFE-COMPARE	Criterion-default comparison policy (Part XLII)
SAFE-HABIT	Cue practice without streak colonization (Part XLIII)
SAFE-REWARD	Informational / non-controlling reward policy (Part XLIV)

SAFE-RESILIENCE	Growth-zone + support recruitment, not grit meters (Part XLV)
SAFE-EXPECTANCY	Task briefs + CIOF; ban trait labels (Part XLVI)
SAFE-TIMING	Encode → sleep → delayed transfer claim (Part XLVII)
SAFE-TRANSFER	Variation/hug/bridge transfer design (Part XLVIII)
SAFE-MISCON	Productive-error elicit → classify → springboard → route (Part XLIX)
SAFE-CALIB	Item-level confidence → miss class → tiered feedback (Part L)
SAFE-DP	Diagnose → isolate → attempt → method feedback → prove spine (Part LI)
SAFE-CoP	Named practice; peripheral = real work; trajectory > tab (Part LII)
SAFE-RITUAL	Light optional practice-pointing marks; anti-totalism (Part LIII)
SAFE-AAR	Brief → traces → four-question student-first close (Part LIV)
SAFE-FILM	Sparse attempt clips + self∪model + next commitment (Part LV)
SAFE-MUSIC	Scales → repertoire → transfer-as-recital ladder (Part LVI)
SAFE-ANNOT	Annotate-before-reveal → classify → drill; annotate wins (Part LVII)
SAFE-EXPOSE	Situation hierarchy + expectancy violation + fade safety (Part LVIII)
SAFE-WTP	Parent CBC + bias controls + message≠package (Part LIX)
SAFE-REPAIR	Detect→own→correct→reattempt→triggered escalate (Part LX)
SAFE-IDMEASURE	Validated identity factors + FEI behavior + demand-aware admin (Part LXI)
SAFE-LONGID	Stage 8w process / 26w triangulated identity / 52w persistence; fade-aware (Part LXII)
SAFE-EXAM	Destake-to-learn / dose-to-prove; delayed mixed under stakes; no ACT guarantees (Part LXIII)
SAFE-ELL	Language load ≠ math load; modify incidental English not math; no deficit track (Part LXIV)
SAFE-GENDER	Parity hygiene; cue hygiene; practice recognition; no ST→score / pink cosplay (Part LXV)

SAFE-STRUCTURE	Structure > slogans; bandwidth hygiene; no grit-NS / character-blame equity (Part LXVI)
SAFE-ONTOLOGY	Diagnosis-before-dialogue; inspectable state; LLM bookends; moat≠graph file (Part LXVII)
SAFE-HITL	Map-briefed tutors; prompt>pour; coaching cycles; FEI QA; packetized escalate (Part LXVIII)
SAFE-PRIVACY	Self-authored short-TTL affect → pedagogy only; ERT ban; contextual integrity (Part LXIX)
SAFE-LABMETA	Label→check→kill/demote; small FEI core; page-count vanity kill (Part LXX)
SAFE-TUTORGRAIN	Trained near-peer default; structure>pairing; hire by fidelity; expert=escalate (Part LXXI)
SAFE-GENQ	Verify-before-ship; key hard-fail; Kane IUA; drop-rate gate; coverage≠NS (Part LXXII)
SAFE-PROCURE	Privacy review = GTM gate; NDPA-ready; Exhibit A honesty; no bio / no pupil ads (Part LXXIII)
SAFE-SCHED	Horizon-matched ISI×RI; delayed first return; equal-ish lags; generation Returns; no streak-as-spacing (Part LXXIV)
SAFE-WORKFORCE	Fidelity-over-tenure; continuous coaching; coach-capacity hire gate; burnout→load redesign; no headcount NS (Part LXXV)
SAFE-FORGET	Age evidence; JOLs≠retention; delayed probes as truth; returns as competence protection; no Ebbinghaus Score™ (Part LXXVI)
SAFE-ADAPT	Calendar-first; banded inspectable personalization; selection>opaque rewrite; flashcard lineage≠FEI (Part LXXVII)
SAFE-DURABLE	Dual-rail GTM; cram-intercept not cosplay; legible MoC for WTP; no point guarantees (Part LXXVIII)
SAFE-FBTIME	Mode-conditional timing; focus>timing; prior-knowledge switch; no instant-feedback hero (Part LXXIX)
SAFE-BRIDGE	Connection-first when endpoints non-naive; remediate relation; measure join transfer; no bridge-count moat (Part LXXX)
SAFE-COLD	Humble prior → labeled seed → probes; Map = evidence age; no day-one mastery fireworks (Part LXXXI)
SAFE-FORMAT	Format as first-class evidence; conversion as object; support before stack; no format-count / style-quiz NS (Part LXXXII)

SAFE-PDASH	Beliefs before chrome; proof-age hero; one improvement CTA; autonomy>control; ranks/affect off (Part LXXXIII)
SAFE-TALK	Construction > airtime; prompt density + constructive share + wait bands; no Talk Ratio Score™ (Part LXXXIV)
SAFE-STORYLOAD	CLT-budgeted narrative wrap; structure-encode or cut; coherence > immersion; no Story Engagement Score™ (Part LXXXV)
SAFE-PROOF	Competence > crowd; dated solo artifacts; provincial similarity; thin portfolio; no star-wall / +points proof (Part LXXXVI)
SAFE-COVER	Matrix > total; gaps as features; use-tier labels; examTag honesty; no item-count / complete-ACT / unsynced GENQ (Part LXXXVII)
SAFE-FADE	Stage not costume; completion bridge; backward fade; expertise-aware; attempt grain; no unlimited-solutions / never-fade hero (Part LXXXVIII)
SAFE-HELP	Instrumental > executive; abuse∪avoidance fail; contingent stage; SE before bottom-out; no unlimited-hints / Help Score™ (Part LXXXIX)
SAFE-INSTRUMENT	Ship the four; NSM+OMTM; XXI.4 co-gates; anti-Goodhart; Kane IUA; no DAU/streak/FEI Score™ NS (Part XC)
SAFE-HINT	Soft before hard; contingency; construction-priced peeks; fade×hint; no Hint Score™ / free hard dump (Part XCI)
SAFE-EXPLAIN	Short principle-first; germane≠more text; SE before wrap; length fades; no Explanation Score™ / monologue≡SE (Part XCII)
SAFE-RETRIEVE	Mode-contingent stalls; TOT→cue; freeze→soft bound; attempt before help; no Struggle Score™ / never-stuck (Part XCIII)
SAFE-PF	Fidelity-gated generate→consolidate on student RSMs; guided default when thin/exam/freeze; no PF Score™ / every-miss≡PF (Part XCIV)
SAFE-APPRENTICE	Model→coach→scaffold→fade + articulate; shared ca_phase; no guild cosplay / modeling≡lecture / Apprenticeship Score™ (Part XCV)
SAFE-ATTN	Protect attempt windows; residue park/resume; opt-in focus + tech breaks; no Focus Score™ / phone-shame / mid-attempt DAU pushes (Part XCVI)

SAFE-DUAL	Structural dual coding; contiguous labels; coherence weed; static annotated default; no Dual Coding Score™ / styles meshing / always-add-picture (Part XCVII)
SAFE-MONITOR	Named judgments → miss-class control; C4 destake; densifies SAFE-CALIB/COLD; no Monitoring Score™ / one-scan calibration (Part XCVIII)
SAFE-ROI	Map minute budgets; constrained AI for reps; humans for joins; FEI per tutor-minute; no AI-replaces-tutors / hours NS (Part XCIX)
SAFE-EXPTIME *(provisional)*	Grain-local example↔solve timing; adaptive>fixed; no universal E0→E3 / Timing Score™ (Part C)
SAFE-SRL *(provisional)*	Phase→control SRL cycles; microanalysis>surveys; no SRL Score™ / streak≡SRL (Part CI)
SAFE-GENERATE *(provisional)*	SOI-forcing sparse GLAs; no Generative Learning Score™ / AI≡generation / teach-for-XP (Part CII)

Appendix B — Maya interview protocol (qual)

Goal: Map moments when math stopped (or started) feeling dangerous.

1. “Tell me about a time math felt dangerous or humiliating.”
2. “What did your body do? What did you do next?”
3. “Tell me about a time you surprised yourself in math.”
4. “Who saw that moment? Did it matter that someone saw it?”
5. “If an app celebrated you, would that feel real or fake? Why?”
6. “What would make you take a harder math class next year?”
7. “When you get stuck, do you want comfort, a hint, or the answer? In what order?”

Coding: threat markers; attribution language; witness presence; desire for story vs drill.

Appendix C — Causal DAG (provisional)

```
prior failures → math anxiety → WM load → performance ↓
\ \ \→ avoidance → practice ↓ → skill lag
\ \→ fixed identity narrative
\→ missing knowledge ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■→ performance ↓

safety UX → threat ↓ → WM available ↑ → attempt quality ↑
coach + diagnosis → misconception repair → competence evidence ↑
competence evidence + witness → identity update → challenge-seeking ↑
challenge-seeking → harder practice → skill ↑ (reinforcing)
```

Open identification problem: Does identity cause challenge-seeking, or does challenge-seeking cause identity? Likely bidirectional. Measure both.

Appendix D — Experiment pre-registration sketches

A — Wizard coach

- Population: chapter/practice users with soft-wrong enabled
- Arms: coach on / coach off
- Primary: `retry_120s`
- Guardrail: `transfer_pass`, session completion
- Stop rule: clear harm on transfer or spike in hint binge

B — Story frame

- Arms: invention-story wrapper vs bare stem (same math)
- Primary: time-to-abandon after first miss
- Secondary: enjoyment, retention @ 7d

D — Explanation timing × anxiety

- Screen: brief anxiety item
- Arms: explain-first vs attempt-first
- Heterogeneity: high vs low anxiety

Appendix E — Red Team standing orders

1. Attack any claim that cannot name a falsifier.
2. Attack any metric that can be gamed without learning.
3. Attack any narrative feature without a learning outcome.
4. Attack Bloom 2-sigma mythology on sight.
5. Attack "AI makes tutoring free" slogans on sight.

Appendix F — Future chapters (queued, not padded)

1. Neuroscience of math anxiety (amygdala/WM models; cautious translation)
2. History of mathematics as meaning technology — under trial via XXVIII + HIST
3. Parent trust economics & willingness-to-pay — **DONE** Part LIX / SAFE-WTP (builds EVT-4 / XXVII)

3b. Trust after AI hallucination / repair + escalate — **DONE** Part LX / SAFE-REPAIR (builds XXXIII)

3c. Measurement of math identity — **DONE** Part LXI / SAFE-IDMEASURE

3d. Longitudinal identity change models — **DONE** Part LXII / SAFE-LONGID

3e. ACT/exam pressure special case — **DONE** Part LXIII / SAFE-EXAM

3f. Multilingual / ELL math identity — **DONE** Part LXIV / SAFE-ELL

3g. Gender & math stereotypes update — **DONE** Part LXV / SAFE-GENDER

3h. Socioeconomic constraint on "grit" — **DONE** Part LXVI / SAFE-STRUCTURE

3i. Ontology as diagnosis moat — **DONE** Part LXVII / SAFE-ONTOLOGY

3j. Human-in-the-loop tutor ops — **DONE** Part LXVIII / SAFE-HITL

3k. Privacy & student affect data — **DONE** Part LXIX / SAFE-PRIVACY

3l. Constitution meta / Bayesian lab process — **DONE** Part LXX / SAFE-LABMETA

3m. Peer vs near-peer vs expert tutoring — **DONE** Part LXXI / SAFE-TUTORGRAIN

3n. Generated question validity & key risk — **DONE** Part LXXII / SAFE-GENQ

3o. District procurement & privacy review as GTM — **DONE** Part LXXIII / SAFE-PROCURE

3p. Spaced retrieval schedules in product UX — **DONE** Part LXXIV / SAFE-SCHED

3q. Tutor workforce pipeline & quality drift — **DONE** Part LXXV / SAFE-WORKFORCE

3r. Forgetting curves as product honesty — **DONE** Part LXXVI / SAFE-FORGET

3s. Adaptive spacing algorithms vs fixed calendars — **DONE** Part LXXVII / SAFE-ADAPT

3t. Cram products vs durable-identity positioning — **DONE** Part LXXVIII / SAFE-DURABLE

3u. Feedback timing immediate vs delayed — **DONE** Part LXXIX / SAFE-FBTIME

3v. Bridge-gap product narrative — **DONE** Part LXXX / SAFE-BRIDGE

3w. Cold-start without fake mastery — **DONE** Part LXXXI / SAFE-COLD

3x. Format-axis product narrative — **DONE** Part LXXXII / SAFE-FORMAT

3y. Parent dashboard honesty UX — **DONE** Part LXXXIII / SAFE-PDASH

3z. Tutor talk-ratio instrumentation — **DONE** Part LXXXIV / SAFE-TALK

3aa. Story-world cognitive load vs identity payoff — **DONE** Part LXXXV / SAFE-STORYLOAD

3ab. Competence evidence as social proof — **DONE** Part LXXXVI / SAFE-PROOF

3ac. Bank coverage honesty in marketing — **DONE** Part LXXXVII / SAFE-COVER

3ad. Worked-example fading in Solver UX — **DONE** Part LXXXVIII / SAFE-FADE

3ae. Help-seeking vs help abuse — **DONE** Part LXXXIX / SAFE-HELP

3af. Product analytics for FEI North Stars — **DONE** Part XC / SAFE-INSTRUMENT

3ag. Hint economy & contingent scaffolding — **DONE** Part XCI / SAFE-HINT

3ah. Explanation length vs germane load — **DONE** Part XCII / SAFE-EXPLAIN

3ai. Retrieval failure modes in practice UX — **DONE** Part XCIII / SAFE-RETRIEVE

3aj. Productive failure vs guided success sequencing — **DONE** Part XCIV / SAFE-PF

3ak. Cognitive apprenticeship in tutor playbooks — **DONE** Part XCV / SAFE-APPRENTICE

3al. Attention residue & device distraction in practice — **DONE** Part XCVI / SAFE-ATTN

3am. Dual coding & diagram FormatId load — **DONE** Part XCVII / SAFE-DUAL

3an. Metacognitive monitoring in gap-scan — **DONE** Part XCVIII / SAFE-MONITOR

3ao. Opportunity cost of tutor minutes vs AI scaffold — **DONE** Part XCIX / SAFE-ROI

3ap. Worked-example vs problem-solving timing across expertise — **DONE** Part C / SAFE-EXPTIME

3aq. Self-regulated learning cycles (Zimmerman) in Practice UX — **DONE** Part CI / SAFE-SRL

3ar. Generative learning activities (Fiorella/Mayer) in Practice — **DONE** Part CII / SAFE-GENERATE

3as. Error climate & psychological safety in tutoring — queued id 103 **NEXT**

3at. Homework help vs practice identity conflict — queued id 104

3au. Gesture & embodiment in math practice — queued id 105

3av. Peer explanation quality (knowledge-building vs telling) — queued id 106

3aw. Curiosity & prediction before reveal — queued id 107

4. Equity audit of story worlds — **DONE** (Part XXXVI)
5. Competitive teardown — **DONE** session audits (Part XXXV); usage telemetry still open
6. Formal Bayesian update process for Constitution claims — **DONE** Part LXX / SAFE-LABMETA
7. **DONE (2026-07-30):** Desirable difficulties × anxiety (Part XLI) — SAFE-DD stack
8. **DONE (2026-07-30):** Social comparison & leaderboards (Part XLII) — SAFE-COMPARE stack
9. **DONE (2026-07-30):** Habit formation science without identity colonization (Part XLIII) — SAFE-HABIT stack
10. **DONE (2026-07-30):** Intrinsic motivation killers / Deci caveats (Part XLIV) — SAFE-REWARD stack
11. **DONE (2026-07-30):** Mathematical resilience (Part XLV / id 45) — SAFE-RESILIENCE stack
12. **DONE (2026-07-30):** Teacher/tutor expectancy effects (Part XLVI / id 46) — SAFE-EXPECTANCY stack
13. **DONE (2026-07-30):** Sleep, stress, and learning (Part XLVII / id 47) — SAFE-TIMING stack
14. **DONE (2026-07-30):** Transfer & variation theory (Part XLVIII / id 48) — SAFE-TRANSFER stack
15. **DONE (2026-07-31):** Misconceptions as productive (Part XLIX / id 49) — SAFE-MISCON stack
16. **Synthesizer v1.7 (2026-07-31):** Merged XLI–XLIX into I.4 / XIV / IX / XXI — no new chapter
17. **DONE (2026-07-31):** Confidence calibration (Part L / id 50) — SAFE-CALIB stack
18. **DONE (2026-07-31):** Deliberate practice in tutoring sessions (Part LI / id 51) — SAFE-DP stack
19. **DONE (2026-07-31):** Community of practice / Lave–Wenger (Part LII / id 52) — SAFE-CoP stack
20. **DONE (2026-07-31):** Religion/ritual light-touch (Part LIII / id 53) — SAFE-RITUAL stack
21. **DONE (2026-07-31):** Military/aviation brief-debrief (Part LIV / id 54) — SAFE-AAR stack
22. **DONE (2026-07-31):** Sports film-study pedagogy (Part LV / id 55) — SAFE-FILM stack
23. **DONE (2026-08-01):** Music pedagogy ladders (Part LVI / id 56) — SAFE-MUSIC stack
24. **DONE (2026-08-01):** Chess annotation & metacognition (Part LVII / id 57) — SAFE-ANNOT stack
25. **DONE (2026-08-01):** Therapy graded exposure (Part LVIII / id 58) — SAFE-EXPOSE stack
26. **Synthesizer v1.8 (2026-08-01):** Merged L–LVIII into I.4 / XIV / IX / XXI — no new chapter
27. **DONE (2026-08-01):** Parent WTP experiments design (Part LIX / id 59) — SAFE-WTP stack
28. **DONE (2026-08-01):** Trust after AI hallucination (Part LX / id 60) — SAFE-REPAIR stack
29. **DONE (2026-08-01):** Measurement of math identity (Part LXI / id 61) — SAFE-IDMEASURE stack
30. **DONE (2026-08-02):** Longitudinal identity change models (Part LXII / id 62) — SAFE-LONGID stack
31. **DONE (2026-08-02):** ACT/exam pressure special case (Part LXIII / id 63) — SAFE-EXAM stack
32. **DONE (2026-08-02):** Multilingual / ELL math identity (Part LXIV / id 64) — SAFE-ELL stack
33. **DONE (2026-08-02):** Gender & math stereotypes update (Part LXV / id 65) — SAFE-GENDER stack
34. **DONE (2026-08-02):** Socioeconomic constraint on “grit” (Part LXVI / id 66) — SAFE-STRUCTURE stack
35. **Synthesizer v1.9 (2026-08-02):** Merged LIX–LXVI into I.4 / XIV / IX / XXI — no new chapter
36. **DONE (2026-08-02):** Ontology as diagnosis moat (Part LXVII / id 67) — SAFE-ONTOLOGY stack

37. **DONE (2026-08-03):** Human-in-the-loop tutor ops (Part LXVIII / id 68) — SAFE-HITL stack
38. **DONE (2026-08-03):** Privacy & student affect data (Part LXIX / id 69) — SAFE-PRIVACY stack
39. **DONE (2026-08-03):** Constitution meta / Bayesian lab process (Part LXX / id 70) — SAFE-LABMETA stack
40. **DONE (2026-08-03):** Peer vs near-peer vs expert tutoring (Part LXXI / id 71) — SAFE-TUTORGRAIN stack
41. **DONE (2026-08-03):** Generated question validity & key risk (Part LXXII / id 72) — SAFE-GENQ stack
42. **DONE (2026-08-03):** District procurement & privacy review as GTM (Part LXXIII / id 73) — SAFE-PROCURE stack
43. **DONE (2026-08-04):** Spaced retrieval schedules in product UX (Part LXXIV / id 74) — SAFE-SCHED stack
44. **Synthesizer v1.10 (2026-08-04):** Merged LXVII–LXXIV into I.4 / XIV / IX / XXI — no new chapter
45. **DONE (2026-08-04):** Tutor workforce pipeline & quality drift (Part LXXV / id 75) — SAFE-WORKFORCE stack
46. **DONE (2026-08-04):** Forgetting curves as product honesty (Part LXXVI / id 76) — SAFE-FORGET stack
47. **DONE (2026-08-04):** Adaptive spacing algorithms vs fixed calendars (Part LXXVII / id 77) — SAFE-ADAPT stack
48. **DONE (2026-08-04):** Cram products vs durable-identity positioning (Part LXXVIII / id 78) — SAFE-DURABLE stack
49. **DONE (2026-08-04):** Feedback timing immediate vs delayed (Part LXXIX / id 79) — SAFE-FBTIME stack
50. **DONE (2026-08-04):** Bridge-gap product narrative (Part LXXX / id 80) — SAFE-BRIDGE stack
51. **DONE (2026-08-05):** Cold-start without fake mastery (Part LXXXI / id 81) — SAFE-COLD stack
52. **DONE (2026-08-05):** Format-axis product narrative (Part LXXXII / id 82) — SAFE-FORMAT stack
53. **DONE (2026-08-05):** Parent dashboard honesty UX (Part LXXXIII / id 83) — SAFE-PDASH stack
54. **Synthesizer v1.11 (2026-08-05):** Merged LXXV–LXXXIII into I.4 / XIV / IX / XXI — no new chapter
55. **DONE (2026-08-05):** Tutor talk-ratio instrumentation (Part LXXXIV / id 84) — SAFE-TALK stack
56. **DONE (2026-08-05):** Story-world cognitive load vs identity payoff (Part LXXXV / id 85) — SAFE-STORYLOAD stack
57. **DONE (2026-08-05):** Competence evidence as social proof (Part LXXXVI / id 86) — SAFE-PROOF stack
58. **DONE (2026-08-05):** Bank coverage honesty in marketing (Part LXXXVII / id 87) — SAFE-COVER stack
59. **DONE (2026-08-06):** Worked-example fading in Solver UX (Part LXXXVIII / id 88) — SAFE-FADE stack
60. **DONE (2026-08-06):** Help-seeking vs help abuse (Part LXXXIX / id 89) — SAFE-HELP stack
61. **DONE (2026-08-06):** Product analytics for FEI North Stars (Part XC / id 90) — SAFE-INSTRUMENT stack
62. **DONE (2026-08-06):** Hint economy & contingent scaffolding (Part XCI / id 91) — SAFE-HINT stack
63. **Synthesizer v1.12 (2026-08-06):** Merged LXXXIV–XCI into I.4 / XIV / IX / XXI — no new chapter
64. **DONE (2026-08-06):** Explanation length vs germane load (Part XCII / id 92) — SAFE-EXPLAIN stack
65. **DONE (2026-08-06):** Retrieval failure modes in practice UX (Part XCIII / id 93) — SAFE-RETRIEVE stack
66. **DONE (2026-08-07):** Productive failure vs guided success sequencing (Part XCIV / id 94) — SAFE-PF stack
67. **DONE (2026-08-07):** Cognitive apprenticeship in tutor playbooks (Part XCV / id 95) — SAFE-APPRENTICE stack
68. **DONE (2026-08-07):** Attention residue & device distraction in practice (Part XCVI / id 96) — SAFE-ATTN stack
69. **DONE (2026-08-07):** Dual coding & diagram FormatId load (Part XCVII / id 97) — SAFE-DUAL stack
70. **DONE (2026-08-07):** Metacognitive monitoring in gap-scan (Part XCVIII / id 98) — SAFE-MONITOR stack
71. **DONE (2026-08-07):** Opportunity cost of tutor minutes vs AI scaffold (Part XCIX / id 99) — SAFE-ROI stack
72. **Synthesizer v1.13 (2026-08-07):** Merged XCII–XCIX into I.4 / XIV / IX / XXI — no new chapter

73. **DONE (2026-08-08):** Worked-example vs problem-solving timing across expertise (Part C / id 100) — SAFE-EXPTIME stack
74. **DONE (2026-08-08):** Self-regulated learning cycles in Practice UX (Part CI / id 101) — SAFE-SRL stack
75. **DONE (2026-08-08):** Generative learning activities in Practice (Part CII / id 102) — SAFE-GENERATE stack
76. **NEXT:** Error climate & psychological safety in tutoring (Part CIII / id 103)

Part XXIII — Research Lab Operating Cadence

Weekly (60–90 min):

1. One claim enters trial (evidence + contradiction).
2. Red Team attempts kill.
3. Surviving claim updates Constitution with confidence tag.
4. One experiment metric reviewed.

Synthesizer (~every 8 researcher ticks): Merge duplicate frameworks into Part I.4 / XIV; refresh Executive Summary; regenerate PDF. Do not write a new chapter on synthesizer ticks.

Monthly:

- Rewrite one weak chapter.
- External reader (learning scientist or skeptical founder).
- Regenerate PDF.

Quarterly:

- Kill or promote North Star metric.
- Publish internal "what we believed that was wrong" memo.

Closing stance

MindCraft's deepest risk is not technical failure. It is **winning the wrong game**: becoming the best explanation engine in a world where explanations are cheap — or the warmest AI tutor in a world where warmth without solo transfer is a crutch — while losing the only game that compounds: helping a human revise who they are in the presence of difficulty.

This Constitution exists so the company notices that risk early, and runs experiments that can kill beloved ideas.

v1.13 synthesizer pass folded Parts XCII–XCIX into surviving commercial doctrine (I.4 rows SAFE-EXPLAIN through SAFE-ROI; Red Team kills #46–#53 promoted from provisional). Experiment families EXPLAIN/RETRIEVE/PF/APPRENTICE/ATTN/DUAL/MONITOR/ROI confirmed mounted. Parts C–CII landed as provisional densifiers (SAFE-EXPTIME / SAFE-SRL / SAFE-GENERATE). Next researcher id: CIII error climate & psychological safety. Researcher count since v1.13 = 3. Page count is not the finish line — falsifiable truth is.

Part XXIV — Affect, Anxiety, Belonging (deep dive)

Chapter status: Living evidence brief

Primary question: When is emotional safety causally load-bearing for mathematical confidence?

Owners: Cognitive Neuroscientist seat · Ed Psych seat · Red Team

XXIV.1 What is happening, mechanistically?

Three related but non-identical constructs get collapsed in product talk. Separate them.

Construct	Definition (working)	Time scale	Product signal
State math anxiety	Transient dread / arousal in a math moment	Seconds–minutes	Avoid click, freeze, rapid hint binge
Trait math anxiety	Stable tendency to experience math as threat	Months–years	Course avoidance, “I’m not a math person”
Belonging uncertainty	“People like me don’t belong in this domain”	Contextual, can be acute	Withdrawal after mild adversity
Stereotype threat	Extra evaluative pressure from group stereotype relevance	Situational	Underperformance vs ability when stereotype is cued

FACT (direction): Math anxiety is associated with worse math performance; one 2021 meta-analysis reported mean r ≈ –0.17 across 57 studies (modest average association — not destiny).

FACT (mechanism candidate): Ashcraft & Kirk (2001) show high math-anxiety individuals have reduced working-memory span especially on computation-based span tasks; dual-task math + memory load amplifies errors/RT. Anxiety behaves like a competing load on central executive resources.

FACT (related): Reviews (e.g., Suárez-Pellicioni et al., 2016) organize explanations as (a) WM competition, (b) attentional control / inhibition deficits, (c) possible low-level numerical representation differences — not mutually exclusive.

HYPOTHESIS for MindCraft: For the Maya-anxious segment, UX that increases social-evaluative threat (shame theater, public ranking early, contemptuous feedback) will look like a “knowledge gap” while actually being a temporary capacity tax.

XXIV.2 Why does this happen? (first principles)

1. **Working memory is scarce.** Multi-step math is a WM sport. Anything that occupies the central executive (worry, self-monitoring for stereotype confirmation, rumination about looking stupid) steals the resource the task needs.
2. **Humans are status-sensitive learners.** Math is a publicly ranked school subject. Threat is often social ("I will be seen as dumb") more than arithmetic.
3. **Avoidance is rational short-term.** Leaving the room ends the aversive arousal. Long-term, avoidance prevents the mastery experiences that would update efficacy.
4. **Identity consolidates avoidance.** Repeated exits become "I'm not a math person," which then licenses future exits (narrative lock-in).

Human need satisfied by safety design: Need for non-humiliation while competence is still fragile; relatedness/respect; predictability.

XXIV.3 Belonging and stereotype threat — what transfers?

FACT: Steele & Aronson tradition — when a test is framed as ability-diagnostic under a negative group stereotype, performance can drop relative to non-diagnostic framing (classic lab pattern; effect sizes and replication quality vary by context).

FACT: Spencer, Steele & Quinn (1999) line — women/math stereotype relevance can impair performance when threat is cued.

FACT: Walton & Cohen (2007, 2011) — belonging uncertainty makes adversity feel identity-diagnostic; brief belonging interventions framing adversity as common/transient improved long-term outcomes for marginalized college students in landmark trials (e.g., Science 2011 belonging intervention).

Red Team limits:

- Effects are **heterogeneous** and context-dependent (like mindset).
- School-wide scaling often attenuates lab effects.
- MindCraft must not cargo-cult a 45-minute belonging essay into a gamified toast.
- Belonging interventions are not a substitute for teaching.

Surviving transfer:

- Normalize struggle without normalizing low standards.
- Make early adversity non-identity-diagnostic ("misses are how the map learns").
- Avoid stereotype-cueing copy ("even girls can…" is often worse).
- Tutor witnessing is a belonging technology when authentic.

XXIV.4 Neuroscience: what we can say without overclaiming

FACT / REVIEW level: Math anxiety literature includes ERP/fMRI correlates; reviews document affective and cognitive network involvement. Exact "brain region product features" are **SPECULATION** and should not drive roadmap.

Allowed claim: Affective arousal and executive control interact; designing for lower threat is biologically plausible.

Forbidden claim: “Our purple calm mode activates the prefrontal cortex.”

XXIV.5 Contradictory evidence (keep visible)

1. Many students underperform from **missing knowledge**, not anxiety.
2. Some learners seek **competitive arousal** and dislike soft UX.
3. Reverse causality: low skill → anxiety → further avoidance (bidirectional).
4. Trait anxiety interventions that only soothe without competence can create comfort without growth.
5. Meta correlations are modest; anxiety is one bottleneck among many.

XXIV.6 Can this generalize? 30 years? AI?

Question	Answer
Generalize beyond math?	Yes — any high-status evaluative domain (music juries, coding interviews).
Still work in 30 years?	Yes — human threat systems are not a 2026 fad.
Can AI improve it?	Yes for personalization of challenge gradient & detection of freeze patterns; risky for fake empathy.
MindCraft change	Soft-wrong, hide-correctness diagnostics, coach tone, private early practice, tutor witness scripts are load-bearing for anxious segment.

XXIV.7 Experimental validation program

ID	Question	Design	Primary	Falsifier
AFF-1	Does soft-wrong raise retry vs hard-fail UX?	A/B	`retry_120s`	No lift / accuracy down
AFF-2	Does anxiety-state moderate explanation-first harm?	Anxiety screen × timing	transfer + retry	No interaction

AFF-3	Does “misses are information” copy reduce belonging threat items?	Pre/post items	belonging/threat items	Demand effects only
AFF-4	Tutor witness note vs no note after growth moment	Tutor RCT	2-week challenge -seeking	Null

Confidence in chapter thesis: Medium–High that affect matters for a large subset; Medium that MindCraft’s current UX already captures the right mechanisms; Low that we have measured it.

XXIV.8 Product system (not features)

Affective Load Manager (system):

1. Detect freeze / hint-binge / rapid exit (behavioral).
2. Downshift evaluative threat (tone, privacy, soft-wrong).
3. Offer graded re-entry (easier isomorphic item), not pep talk alone.
4. Restore challenge once `retry` and success stabilize.
5. Log whether the student later chooses harder work (identity leading indicator).

Balancing loop: chronic downshift → boredom → exit. Must re-raise difficulty.

Part XXV — Self-Efficacy, Narrative Identity, Habit (deep dive)

Chapter status: Living evidence brief

Primary question: What updates the self-story “I am bad at math,” and what merely increases login frequency?

Owners: Learning Science · Narrative Psychology · Behavioral Econ · Red Team

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XXV.1 Self-efficacy is not “confidence vibes”

FACT (theory): Bandura’s social cognitive theory — self-efficacy is belief in capability to organize and execute actions required for specific attainments. It is **task- and domain-specific**, not a global personality glow.

FACT (sources): Efficacy information is processed from four sources (Bandura, 1997):

1. **Mastery experiences** (usually strongest)
2. **Vicarious experiences** (models)

3. **Social persuasion** (credible others)
4. **Physiological / affective states** (how arousal is interpreted)

FACT (measurement tradition): Usher & Pajares (2009/2010 line) validated source scales for middle-school mathematics; sources relate to math self-efficacy, goals, optimism.

HYPOTHESIS: MindCraft's FEI loop is largely a **self-efficacy engineering system**:

FEI stage	Efficacy source
Safety / soft-wrong	Reinterprets physiological arousal ("alert, not doomed")
Earned micro-win	Mastery experience
Coach naming strategy	Persuasion + attribution
Tutor witness / peer story	Vicarious + persuasion
Map fill after transfer	Mastery that survives variation

XXV.2 Why mastery experiences dominate — and how products fake them

What is happening: A win only updates efficacy if the learner **attributes** it to their capability under meaningful difficulty.

Fake mastery (product sins):

- Too-easy items → "this doesn't count"
- Heavy hinting → "the app did it"
- Identical clones after "mastery" → recognition, not competence
- Points for opening lessons → engagement theater

Real mastery design rules (HYPOTHESIS):

1. Difficulty just above comfort (desirable difficulty, not cruelty).
2. Student performs the decisive step.
3. Immediate informative feedback.
4. Soon after: varied item (transfer_pass).
5. Explicit attribution to strategy ("you isolated the factor") not talent.

Contradicting caution: After strong efficacy exists, occasional failure hurts less (Bandura). Early failures hurt more. Protect early arc; do not permanently infantilize.

XXV.3 Reciprocal loops: efficacy ↔ achievement

FACT / EMERGING: Longitudinal work often finds reciprocal relations — achievement raises efficacy and efficacy raises later achievement (e.g., recent high-school math reciprocity analyses in large-scale assessment literature). Causality is not one-way.

Implication: “Just raise confidence” without skill is unstable. “Just raise skill” while humiliating can stall practice volume for anxious students. Build both.

XXV.4 Narrative identity — the story that persists

FACT / TRADITION: Narrative identity research (McAdams tradition) — people maintain identity through evolving life stories with themes of agency, communion, redemption, contamination.

HYPOTHESIS (MindCraft synthesis): Math identity change is a **redemption narrative rewrite**:

- Contamination story: “I failed → I am bad at math → I avoid → I stay bad.”
- Redemption story: “I struggled → I used a strategy → I grew → I am becoming someone who stays.”

Product implication: Story worlds matter if they supply **symbols for redemption**, not only costume. The map lighting after a real win is a narrative punctuation mark.

Red Team: Most edtech “stories” are skin. If removing the narrative leaves identical pedagogy and identical emotions, the story was decoration. Test with story-off experiments (Experiment B).

XXV.5 Habit vs identity — the Duolingo trap, precisely

FACT / INDUSTRY: Habit products use cues, streaks, variable rewards, loss aversion. Duolingo’s method materials emphasize motivation systems for return.

FACT / THEORY tension: Self-Determination Theory warns that controlling extrinsic rewards can undermine intrinsic motivation when they feel coercive (classic Deci findings; nuance exists for informational rewards).

HYPOTHESIS — Habit/Identity Divergence Model (HID):

```
Habit success: cue → routine → reward → return frequency ↑
Identity success: evidence → attribution → self-story → choice of harder future ↑

Overlap zone: daily practice that produces mastery evidence
Failure mode A: habit without mastery → brittle streak addiction
Failure mode B: identity talk without practice → empty affirmation
```

Metric separation (mandatory):

- Habit health: D1/D7 return, streak
- Identity health: `challenge_accept`, advanced course intent, math-person item + behavior consistency

Optimize for overlap zone. Never promote a habit metric that moves opposite identity metrics.

XXV.6 Vicarious experience and “who is the model?”

HYPOTHESIS: Models work when perceived as **similar yet slightly ahead** (“near-peer who struggled”), not as genius tutors who never miss.

Product:

- Villager / character arcs that show struggle → strategy → win.
- Tutor stories: “I used to blank on fractions.”
- Avoid only showcasing perfect speedruns.

Social persuasion: Credibility matters. Random app toast < trusted tutor sentence. Parents can persuade or destroy (“we’re not math people” is efficacy poison — FOUNDER BELIEF with strong anecdotal support; needs interview confirmation).

XXV.7 Universal principles from this chapter

1. Efficacy is specific; measure math efficacy, not “confidence.”
2. Mastery experiences must be owned and non-trivial.
3. Arousal interpretation is designable (anxiety as signal vs doom).
4. Narrative punctuation should mark real competence events.
5. Habits are oxygen; identity is the fire.

XXV.8 Experiments

ID	Question	Design	Primary
EFF-1	Does requiring decisive student step (vs auto-solve) raise efficacy item?	A/B	math efficacy delta
EFF-2	Strategy attribution copy vs talent copy after win	A/B	challenge_accept @ 7d
EFF-3	Streak on vs identity milestone on (map fill)	A/B	retention vs challenge_accept tradeoff
EFF-4	Near-peer struggle story vs expert-perfect story	A/B	persistence after first miss

Falsifier for identity thesis: Habit metrics rise for 8 weeks while challenge-seeking and transfer stay flat — then MindCraft is a streak app, not an identity company.

Part XXVI — Cognitive Load, Mastery, Tutoring (deep dive)

Chapter status: Living evidence brief

Primary question: How do we create earned competence without drowning working memory — and without mythologizing tutoring?

Owners: Cognitive Science · Learning Science · AI Researcher · Red Team

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XXVI.1 Cognitive Load Theory — the non-negotiable physics

FACT / TRADITION: Cognitive Load Theory (Sweller and colleagues) — working memory is limited; instructional design that forces novices into unguided search can impede schema acquisition.

FACT: Worked-example effect — for novices, studying worked examples often beats equivalent time in conventional problem solving for structurally similar later items (classic Sweller & Cooper algebra studies, 1985 line).

FACT / DESIGN IMPLICATION: Guidance fading — move from full worked examples → completion problems → independent solving as expertise grows (Renkl/Atkinson fading tradition). Continuing full examples after expertise rises can trigger **expertise reversal** (redundancy).

Why this matters to identity: Overloaded novices experience failure as self-indictment. Load mismanagement manufactures fake "I'm bad at math" evidence.

XXVI.2 Reconciling CLT with "attempt before lecture"

SAFE-CRAFT says attempt early. CLT says novices need worked examples. **Is this a contradiction?**

Resolution (HYPOTHESIS — Attempt Grain Principle):

Learner state	Right "attempt"	Wrong "attempt"
True novice, new schema	Attempt a **micro-step** or completion blank inside a worked example	Full novel multi-step problem cold
Partial schema	Isomorphic problem with soft-wrong	Leap to transfer without success
Anxious + some knowledge	Slightly easier isomorphic first	Public hard problem as opener
Advanced	Minimal guidance; generative struggle	Endless examples (expertise reversal)

Product rule: "Struggle first" means **agency at the right grain**, not abandonment into search.

This dissolves a false war between "productive struggle" slogans and cognitive load science.

XXVI.3 Mastery learning — keep the war on the table

Supporting evidence: Kulik, Kulik & Bangert-Drowns (1990) meta — mastery programs associated with positive exam effects (often cited ~0.5 SD overall in their synthesis), often stronger for weaker students; time costs can rise; completion can suffer in some self-paced settings.

Contradicting evidence: Slavin (1987) "Mastery Learning Reconsidered" — limited support for group-based mastery on standardized measures; stronger on experimenter-made tests; coverage vs depth tradeoff is real.

MindCraft doctrine:

1. Mastery means **transfer_pass**, not same-item streak.
2. Track time-to-mastery; do not hide cost.
3. Allow strategic coverage decisions (exam track) without pretending infinite time.
4. Ontology graph is only as good as far-transfer checks.

XXVI.4 Tutoring — calibrate the mythology

Bloom (1984): Tutoring + mastery framing popularized as ~2 SD ("2 sigma problem").

Calibration FACTS:

- Cohen, Kulik & Kulik (1982) tutoring meta — much smaller average effects than 2.0.
- Nickow, Oreopoulos & Quan (2020) tutoring RCT meta — on the order of ~0.37 SD (impressive, not magic).
- VanLehn (2011) — human tutoring ~0.79 SD vs no tutoring; step-based ITS ~0.76; answer-based CAI smaller. Granularity of feedback matters; beyond a point, finer grain plateaus.

Strategic reading for MindCraft:

Tutoring function	Automate?	Keep human?
Step feedback	Yes (ITS-like)	Optional
Misconception diagnosis	Partially (ontology + traces)	Human for weird cases
Affect co-regulation	Partially (UX)	Human strong
Accountability / relationship	Weakly	Human strong
Identity witnessing	Poorly	Human strong

AI implication: Compete on diagnosis quality + FEI wrapping; do not sell "we are 2-sigma."

XXVI.5 Deliberate practice inside sessions

FACT / TRADITION: Ericsson et al. (1993) — structured, feedback-rich practice aimed at improvement.

FACT / CRITIQUE: Hambrick/Macnamara line — deliberate practice explains substantial but incomplete variance; not a sufficient account of expertise differences.

Session design (HYPOTHESIS):

1. Clear target skill (ingredient-level).
2. Immediate feedback at step grain when possible.
3. Successive refinement on weak subskill (not random variety first).
4. Then varied practice for transfer.
5. Short enough to protect affect; frequent enough to compound.

XXVI.6 Minimal guidance failures

FACT: Kirschner, Sweller & Clark (2006) critique pure discovery/minimal guidance for novices — often inferior to guided instruction.

Implication for story worlds: Narrative must not become unguided discovery of math. Story wraps guided cognition. World_VISION is compatible with guidance; it is incompatible with vibes-only exploration.

XXVI.7 Systems: Competence Production Line

```
diagnosis (gap) → grain selection → worked/fade → attempt
→ soft-wrong info → repair → transfer check
→ mastery mark → spaced return → challenge raise
```

Balancing loops:

- Too slow fade → boredom
- Too fast fade → threat + load spike → avoidance
- Mastery without spacing → illusion of competence

XXVI.8 Experiments

ID	Question	Design	Primary
CLT-1	Worked-example-first vs problem-first for true novices	Within concept	near & far transfer
CLT-2	Fade schedule adaptive to mastery graph vs fixed	A/B	time-to-transfer_pass

TUT-1	Human tutor + AI brief vs AI-alone on persistence	RCT	retry + 4-week challenge_accept
MAS-1	Mastery gate with transfer_pass vs same-item 3-in-a-row	A/B	2-week retention

Red Team standing kill: Any claim that MindCraft "replicates Bloom 2-sigma" without citing modern tutoring metas.

Part XXVII — Markets, Parents, Competition (deep dive)

Chapter status: Living strategy brief (evidence + market reasoning)

Primary question: If explanations cheapen, what do families still buy — and what should MindCraft refuse to become?

Owners: Founder Philosopher · Product Strategist · Behavioral Econ · Red Team

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XXVII.1 Re-state the scarcity doctrine under market pressure

Adopted doctrine (from Part XVII): Compete on diagnosis, affect, identity, accountability — with AI as infrastructure.

Market test: Parents often buy **score relief** and **homework peace**, not "identity transformation." Students may buy relief from shame. Schools buy coverage and equity metrics.

FOUNDER QUESTION (unresolved): Can MindCraft sell identity truthfully while packaging score outcomes parents recognize?

Working answer (HYPOTHESIS): Sell the FEI system as the *method*; report scores/challenge-seeking as *evidence*. Never sell vibes without transfer.

XXVII.2 Willingness-to-pay stack (post-AI)

Layer	Marginal cost trend	WTP trend	Notes
Raw answers	→ 0	↓↓	Chegg shock / GPT
Generic explanations	→ 0	↓	Unless trusted
Personalized practice	↓	→ / ↑	Needs outcomes
Diagnosis of *this* student	↓ with AI	↑ if trusted	Ontology advantage possible

Emotional safety during struggle	Hard to automate well	↑ for anxious segment	UX + tutor
Accountability relationship	Scarce	↑↑	Scheduling, human
Trusted outcome signal to parents	Scarce	↑↑	Reports that don't lie

SPECULATION (30-year): Ambient AI tutors everywhere. Differentiator = trusted human+system that changes trajectory, not chat fluency.

XXVII.3 Parent trust economics

What parents fear (hypothesis from tutoring practice — needs interviews):

1. Child is falling behind permanently.
2. They cannot help at home.
3. Child hates learning / shuts down.
4. Money spent on apps that entertain without learning.

Trust builders (HYPOTHESIS):

- Specific diagnosis ("fractions↔ratios bridge weak") over vague "needs practice"
- Visible soft-wrong → retry behavior (effort signal)
- Transfer checks (anti-cheat of fake mastery)
- Honest plateaus ("this week: consolidation, not fireworks")
- Tutor notes that sound human and concrete

Trust destroyers:

- Inflated streaks as "progress"
- Grade promises you can't keep
- Dark-pattern engagement
- AI that hallucinates pedagogy confidently

Experiment PAR-1: Parent report variants — score-only vs FEI dashboard (retry, challenge-seeking, mastery map). Measure retention/WTP intent and comprehension of child's actual bottleneck.

XXVII.4 Competitive teardown (mechanism lens)

Khan Academy

- **Sells:** free mastery content + brand trust
- **Strength:** coverage, accessibility
- **Weakness for Maya:** can still feel lonely/evaluative; identity not core

- **MindCraft response:** do not out-content; out-diagnosis + human wrap

Duolingo

- **Sells:** habit motivation
- **Strength:** return loops
- **Weakness:** habit ≠ deep skill/identity (HID model)
- **MindCraft response:** borrow on-ramps; refuse extrinsic colonization of meaning

Brilliant

- **Sells:** aesthetic intellectual joy + prestige
- **Strength:** delightful problems for already-curious
- **Weakness:** may miss threatened segment
- **MindCraft response:** win the anxious middle, not only the delighted

Tutoring marketplaces

- **Sells:** access to humans
- **Strength:** relationship potential
- **Weakness:** uneven quality; weak diagnosis systems
- **MindCraft response:** FEI-trained tutors + AI briefings

Generic GPT tutors

- **Sells:** instant explanation
- **Strength:** 24/7 cognitive help
- **Weakness:** trust, curriculum alignment, affect, accountability
- **MindCraft response:** wrap models in ontology + FEI + human witness

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XXVII.5 Business model risks tied to research thesis

If thesis true…	Business implication
Identity is the product	Retention via meaning + progress, not only content library
Affect is first-order for segment	CAC messaging should speak shame→agency, not “more videos”
Human witness scarce	Hybrid model: AI scale + tutor moments that matter
Parents buy trust	Reporting quality is a product surface

If thesis false…	Pivot signal
Explanation-first wins all segments	Become best adaptive explainer (crowded)
Streaks drive outcomes without identity	Become math Duolingo (crowded, brittle)

Parents only pay for score spikes	Short-cycle test prep (possible, different company)

XXVII.6 Network effects — honesty clause

FACT about present: MindCraft does not have Facebook-style network effects.

Plausible compounding (HYPOTHESIS):

1. Misconception telemetry → better diagnosis (privacy-constrained data advantage)
2. Tutor playbook quality → service moat
3. Longitudinal identity arcs in worlds → switching costs of meaning (fragile)

Red Team: “Community” features often fake relatedness. SDT metas show relatedness interventions harder to move. Do not ship empty social to check a box.

XXVII.7 Pricing ethics (Founder Philosopher)

HYPOTHESIS: Charging for emotional safety is legitimate if competence and transfer are real; illegitimate if the product only soothes.

Guardrail: Every paid claim in marketing must map to a measured FEI or learning metric. If not measured, do not claim.

XXVII.8 Experiments & interviews

1. **10 parent interviews:** what did you think you bought? what would make you cancel?
2. **10 Maya interviews:** when did math stop feeling dangerous?
3. **PAR-1** report A/B
4. **COMP-1** message test: identity framing vs score framing vs hybrid (conversion + 30-day retention)

Confidence: Medium that hybrid packaging wins; Low on exact price points without data.

Part XXVIII — History, Meaning, and Why Math Exists (deep dive)

Chapter status: Design-research brief (under-evidenced; high founder interest)

Primary question: Does explaining *why humanity invented a mathematical idea* change persistence and identity — or is it a beautiful distraction?

Owners: Historian of Mathematics seat · Founder Philosopher · Learning Science · Red Team

XXVIII.1 The founder prior (label it)

FOUNDER BELIEF: Students disengage partly because math is presented as arbitrary machinery. Reconnecting ideas to human invention, need, and story demystifies the subject and restores meaning.

WORLD_VISION alignment: Story-first questions; math frozen, narrative wraps.

Epistemic status: Strong pedagogical tradition; **weak large-scale RCT base** relative to mindset/SDT. Treat as a research program, not settled science.

XXVIII.2 What "meaning" might be doing (mechanisms)

If invention-story helps, why?

Mechanism	Claim type	Testable implication
Arbitrary-rule appraisal ↓	HYPOTHESIS	Fewer "why do we have to learn this?" exits
Curiosity spike	HYPOTHESIS	Longer voluntary time before first quit
Cognitive schema anchor	HYPOTHESIS	Better retention of structure if story encodes structure (not trivia)
Identity affiliation ("humans who solve")	HYPOTHESIS	Higher math-person endorsement
Seductive details (harm)	FACT tradition in multimedia learning	Stories that add irrelevant color can *hurt* retention

Critical constraint: History must encode the **mathematical structure**, not costume the problem with dragons while leaving the math alien.

XXVIII.3 Seductive details problem (Red Team weapon)

FACT / TRADITION: Multimedia learning research (Mayer and others) — interesting but irrelevant details can impair learning (seductive details effect). Not always replicated equally, but the risk is real.

Kill criterion for MindCraft stories: If a narrative element can be removed without changing the mathematical mental model, it is decoration. Decoration can remain for motivation **only if** Experiment B shows persistence/retention gains that justify time cost.

Survive criterion: Narrative that makes the *invariant* memorable (e.g., why we need a common measure; why zero; why coordinates) is structural meaning.

XXVIII.4 History of math as demystification technology

Examples of structural meaning (illustrative — not a curriculum):

1. **Fractions / ratios:** Fair sharing, measurement, comparison when wholes differ.
2. **Negative numbers:** Debt, direction, opposite change — not "fake numbers."
3. **Coordinates:** Descartes' move — marry algebra to geometry to locate.
4. **Functions:** Input–output machines for prediction and dependency.
5. **Probability:** Reasoning under incomplete information; insurance, games, risk.
6. **Quadratics / optimization:** Paths of projectiles, areas, tradeoffs — tools for extrema.

HYPOTHESIS: One crisp invention sentence + one visual beat beats a museum lecture.

XXVIII.5 Cross-domain analogy: martial arts lineages & music history

Domain	Meaning practice	Transfer
Martial arts	Lineage stories + why a form exists	Respect + purpose; still drill techniques
Music	Composer context + etudes	Meaning without skipping scales
Religion	Origin myths	Caution: do not sacralize math cultishly
Architecture	"Why this arch holds"	Structure-first stories

Principle: Meaning wraps discipline; it does not replace repetition.

XXVIII.6 Equity and whose history?

Open problem: Whose invention stories? Eurocentric timelines can alienate. Global histories of mathematics (India, Islamic golden age, China, Africa, Indigenous measurement systems) matter for belonging **and** accuracy.

HYPOTHESIS: Inclusive structural histories can support belonging for students who feel math is "not for people like me" — but only if not tokenistic.

Experiment HIST-EQ: Same math, Euro-only story vs plural invention story; measure belonging items + learning — watch for seductive-detail harm.

XXVIII.7 Integration with SAFE-CRAFT

Invention story belongs in **step C (Causal story)** after an attempt grain, or as a 10–20s prelude that encodes structure — not as a gate that delays agency for anxious students who need a micro-win first.

Default order for threatened learners: Safety → micro-attempt → feedback → meaning beat → next attempt.

Default order for curious/low-anxiety: Short meaning beat → attempt → fade guidance.

Segment, don't dogmatize.

XXVIII.8 Experimental program (this chapter lives or dies here)

ID	Question	Design	Primary	Kill condition
HIST-1	Structure-encoding story vs bare stem	Crossover	retention @ 7d + retry	Story slows, no learning lift
HIST-2	Structure story vs seductive skin (dragons, no structure)	3-arm	transfer	Skin ≥ structure
HIST-3	Pre-attempt story vs post-miss story	Timing	time-to-abandon	Either timing harms transfer
HIST-4	Plural history vs single tradition	A/B	belonging + learning	Belonging up, learning down badly

Current confidence that story is load-bearing for identity: Low–Medium.

Current confidence that bad story can harm: Medium–High.

Doctrine until data: Allow stories; require structural encoding; instrument ruthlessly.

XXVIII.9 What would make this chapter graduate from "founder belief"?

Graduation criteria (all required):

1. HIST-1 or HIST-3 shows persistence or retention lift in target segment.
2. No material drop in `transfer_pass`.
3. Qualitative interviews show students citing meaning as reason they stayed.
4. Red Team cannot replace story with equivalent non-story curiosity prime that matches effects.

Until then: **FOUNDER BELIEF under active trial.**

Part XXIX — Spacing, Retrieval, and the Memory of Competence

Chapter status: Living evidence brief

Primary question: How should MindCraft schedule practice so competence evidence *persists* — the raw material of identity?

Owners: Cognitive Science · Data Scientist · Product Strategist · Red Team

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XXIX.1 Why memory science belongs in an identity Constitution

Identity ("I am someone who can do math") requires **autobiographical evidence**. If skills evaporate between sessions, the self-story collapses into "I used to be able to when the app held my hand."

Retention is not a separate KPI from identity. It is the substrate.

XXIX.2 Spacing / distributed practice

FACT: Cepeda, Pashler, Vul, Wixted & Rohrer (2006) — quantitative synthesis of distributed practice (317 experiments / 184 articles; 839 assessments). Spacing and lag effects are robust; the inter-study interval that maximizes retention **increases as the desired retention interval increases**.

FACT: Cepeda et al. (2008, *Psychological Science*) — "temporal ridgeline" of optimal retention: gap between study episodes should scale with how long you need the knowledge to last.

Product implication: "Do 40 problems tonight" (massing) can produce local mastery marks that fail a week later. Mastery graph must schedule **returns**, not only first clears.

HYPOTHESIS — Identity Spacing Rule: After `transfer_pass`, schedule a retrieval at a lag matched to the student's likely next exam/use horizon (days for homework, weeks for ACT track).

XXIX.3 Retrieval practice / testing effect

FACT / TRADITION: Roediger & Karpicke (2006) — retrieval practice (testing) often beats restudy for long-term retention ("power of testing memory").

FACT / nuance: Expanding vs equal spacing of retrieval — expanding can help short-term; equal spacing often competitive or better for long-term (Karpicke & Roediger, 2007 line).

MindCraft translation:

- Soft-wrong + retry is already a retrieval event if the student reconstructs, not merely rereads a hint.
- "Review mode" that only re-shows worked examples without retrieval is weaker than asking for the decisive step again.

- Gap scan / practice should prefer **generation** over passive video when the goal is durable competence.

XXIX.4 Desirable difficulties (Bjork)

FACT / TRADITION: Conditions that slow acquisition can improve retention/transfer (spacing, interleaving, generation) — “desirable difficulties.”

Red Team / affect conflict: Desirable difficulties raise short-term failure rates → can spike threat for anxious learners.

Resolution (HYPOTHESIS — Sequenced Difficulty):

1. Early FEI: protect safety; use micro-grain success.
2. Then introduce spacing/interleaving once `retry_120s` is healthy.
3. Never introduce desirable difficulty as shame (“you got this wrong because you’re weak”). Frame as training physics.

XXIX.5 Interleaving (brief)

FACT / TRADITION: Mixed practice of related problem types often improves discrimination/transfer vs blocked practice (Rohrer/Taylor line in math practice research — effects depend on similarity structure).

HYPOTHESIS: Interleave *after* initial schema formation (CLT), not before. Block → fade → interleave → spaced retrieval.

XXIX.6 System: Memory of Competence (MoC)

```
learn → transfer_pass → schedule spaced retrieval
↑ |
+---- fail retrieval -------+--> diagnose: forgot vs never had
```

Forgot vs never-had is a product-critical distinction:

- Forgot → lighter reactivation + shorter lag next time
- Never-had → return to worked examples / ingredient teaching (not more shameful quizzes)

XXIX.7 Experiments

ID	Question	Design	Primary
SPC-1	Spaced return vs massed cram after mastery mark	A/B	retention @ 7d / 28d

RET-1	Retrieval prompt vs restudy after soft-wrong	A/B	retention + retry quality
INT-1	Interleave after fade vs blocked-only	Within concept	far transfer

Falsifier for "identity without memory": Challenge-seeking rises while 28-day retention collapses — then celebration is theater.

Confidence: High that spacing/retrieval are real; Medium that MindCraft currently schedules them well; product audit needed.

Part XXX — Attribution, Helplessness, and the Language of Feedback

Chapter status: Living evidence brief

Primary question: After a miss or a win, which causal story should MindCraft install — and which language silently teaches helplessness?

Owners: Ed Psych · UX Researcher · Tutor Ops · Red Team

XXX.1 Weiner's dimensions (the grammar of cause)

FACT / THEORY: Weiner's attributional theory of achievement motivation (e.g., 1979 classroom theory; 1985 *Psychological Review*) — people explain success/failure along dimensions that shape emotion and future expectancy:

Dimension	Poles (simplified)	Linked consequence (Weiner tradition)
Locus	Internal ↔ External	Pride/shame vs externalization
Stability	Stable ↔ Unstable	Expectancy of future same outcome
Controllability	Controllable ↔ Uncontrollable	Guilt/responsibility vs helplessness / pity

HYPOTHESIS for product copy: The most dangerous failure attribution in math is **internal + stable + uncontrollable** ("I'm just not smart at math").

The most useful failure attribution is often **internal + unstable + controllable** ("I used the wrong strategy; I can change it") — with care not to blame students for systemic/curricular failures.

XXX.2 Learned helplessness in learning contexts

FACT / TRADITION: When outcomes are experienced as independent of action, organisms (and students) reduce effort — learned helplessness tradition (Seligman) applied in education as expectancy of non-contingency.

Product factories of helplessness:

1. Hints that auto-solve → outcome independent of student action
2. Random difficulty spikes → non-contingent failure
3. Praise for “being smart” then fail → talent threat
4. Opaque AI answers → student cannot see what *they* controlled

Antidote design: Make the decisive controllable step visible. Soft-wrong should point to a **changeable move**.

XXX.3 Attributional retraining (AR)

FACT / APPLIED TRADITION: Attributional retraining programs encourage adaptive attributions (often effort/strategy for failure) and have an empirical literature in higher-ed/classroom applications (Perry and colleagues’ applied work; see reviews of AR).

Caveats (Red Team):

- “Just try harder” without strategy is false growth mindset.
- Effort attribution for impossible items is gaslighting.
- Ability is not infinitely plastic in the short run; teach strategies and select grain.

MindCraft AR micro-protocol (HYPOTHESIS):

1. Name the miss specifically.
2. Offer one controllable strategy.
3. Invite immediate re-attempt.
4. On success: attribute to the strategy used.
5. On second miss: downshift grain (CLT), do not escalate shame.

XXX.4 Feedback language bank (draft doctrine)

Situation	Avoid	Prefer
First miss	“Wrong.” / “Easy one…”	“That trap is common — here’s the fork.”
Hint use	“Here’s the answer.”	“Your move: which factor is shared?”
Success after struggle	“You’re a genius!”	“You changed approach — that worked.”
Success too easy	Big celebration	Quiet confirm + raise challenge

Tutor note	“Good job!”	“I saw you return after the miss on axis of symmetry.”

XXX.5 Intersection with mindset and SDT

- **Mindset:** Yeager 2019 shows context-sensitive gains; attribution language is part of the ecology.
- **SDT competence:** Controllable success feeds competence need.
- **Anxiety:** Internal-stable-uncontrollable attributions amplify threat.

Unified copy rule: Every feedback utterance should answer: *What can the student do next that would change the outcome?* If it cannot answer, rewrite.

XXX.6 Experiments

ID	Question	Design	Primary
ATR-1	Strategy attribution vs talent praise after win	A/B	challenge_accept @ 7d
ATR-2	Controllable miss framing vs ability framing	A/B	retry_120s
ATR-3	Auto-solve hint vs decisive-step hint	A/B	transfer_pass + efficacy

Falsifier: Strategy language improves self-report but not behavior — then copy is placebo.

Confidence: High that attributions matter; Medium that current product language is already adaptive (needs audit).

Part XXXI — Flow, Challenge–Skill Balance, and Difficulty Design

Chapter status: Living evidence brief

Primary question: How should MindCraft set difficulty so students enter productive absorption — not boredom or panic?

Owners: Game Designer · Learning Science · UX · Red Team

XXXI.1 Flow, stripped of mysticism

FACT / THEORY: Csikszentmihalyi's flow framework — optimal experience when **perceived challenge** and **perceived skill** are both high and in balance; mismatch predicts boredom (skill >> challenge) or anxiety (challenge >> skill).

Education transfer: Classroom/ESM studies (e.g., Shernoff et al. tradition) link challenge–skill balance to engagement quality. Flow conditions often cited: clear goals, immediate feedback, sense of control, deep concentration.

Red Team: Flow is not a product feature you "turn on." Self-report flow can be confounded with fun. Some high-learning moments are effortful and not blissful. Do not optimize for "lost track of time" if transfer falls.

XXXI.2 Perceived vs actual skill (critical nuance)

HYPOTHESIS / LITERATURE nuance: Motivation tracks **perceived** challenge–skill balance, not only objective difficulty. Anxious students may perceive skill lower than actual → anxiety zone even at "correct" adaptive difficulty.

Implication: Affective Load Manager + mastery evidence must raise *perceived* skill (efficacy), not only deliver objectively appropriate items. Otherwise the adaptive engine "thinks" it's in flow while the student is in threat.

XXXI.3 Game design transfer (Celeste principle)

What good hard games do:

1. Failures are frequent, cheap, informative.
2. Restart is instant.
3. Difficulty is fair (player believes success is possible).
4. Mastery is visible (you *feel* getting better).

MindCraft mapping:

Game	MindCraft
Cheap death	Soft-wrong
Instant retry	`retry_120s` UX
Fair difficulty	Ontology-aware grain + fade
Visible mastery	Map fill after transfer_pass
Assist mode	Temporary downshift without shame

Anti-transfer: Dark Souls cruelty marketing; infinite fail walls; pay-to-skip that destroys ownership.

XXXI.4 Dynamic difficulty as a system

```
sense: accuracy, latency, hint binge, exit, anxiety proxy
↓
```

```
classify zone: boredom / flow / anxiety / helplessness
↓
act: raise / hold / fade-down / change grain / insert worked example
↓
re-check transfer — not only near accuracy
```

Balancing loop: Chronic downshift → boredom → identity stall.

Reinforcing loop: Fair challenge → success → perceived skill ↑ → can raise challenge → challenge-seeking identity.

XXXI.5 Relation to North Star

Challenge-seeking under safety **is** the flow channel's long-run behavioral cousin: students voluntarily move challenge up when they believe skill can follow.

Measure both:

- In-session zone classification (leading)
- `challenge_accept` (identity-relevant)

XXXI.6 Experiments

ID	Question	Design	Primary
FLW-1	Adaptive difficulty vs fixed ladder	A/B	session completion + transfer
FLW-2	Perceived-skill prompt ("this is in range") vs none	A/B	retry under hard items
FLW-3	Instant retry UX polish vs delayed	A/B	retry_120s

Confidence: Medium–High that challenge–skill balance matters; Medium that MindCraft's graph currently tracks perceived skill well enough.

Part XXXII — Parent Math Anxiety Transmission

Chapter status: Living evidence brief

Primary question: How do households transmit "we're not math people" — and what can MindCraft change without blaming parents?

Owners: Ed Psych · Product (parent surfaces) · Founder Philosopher · Red Team

XXXII.1 The key finding

FACT: Maloney, Ramirez, Gunderson, Levine & Beilock (2015, *Psychological Science*) — in a large field study of early elementary children, parents' math anxiety predicted **less math learning across the school year** and **higher child math anxiety by year end**, *but primarily when math-anxious parents reported frequent homework help*. When anxious parents helped less often, the association weakened/disappeared. Effects were **math-specific** (not reading achievement).

Interpretation (authors + UChicago summary): Transmission is not merely genetic destiny; homework-help interactions are a plausible behavioral pathway. "Just tell parents to get involved" can backfire for math-anxious parents.

Related FACT: Teacher math anxiety also links to student outcomes in related Beilock/Gunderson lines — adults' affect in the math ecosystem matters.

XXXII.2 Why this happens (mechanisms)

HYPOTHESES consistent with the finding:

1. **Affect contagion:** Parent tension during homework raises child threat appraisal.
2. **Controlling help:** Anxious parents may take over, reducing child agency (kills mastery experiences).
3. **Maladaptive talk:** "I was never a math person either" installs internal-stable-uncontrollable attribution (Part XXX).
4. **Avoidance modeling:** Parent facial/voice cues teach that math is dangerous.
5. **Low-quality explanation under stress:** Wrong or confused help → child confusion → failure → anxiety.

Human need: Parents want to help and not harm. Product must give them a **safe role**.

XXXII.3 MindCraft implication — redesign the parent role

Parent role	Harmful version	Better version
Homework coach	Anxious parent teaches under stress	Parent as **witness + scheduler**, tutor/app teaches
Encourager	"You're smart" / "we're not math people"	"I saw you try again" (strategy/effort-process)
Progress checker	Streak theater	FEI dashboard: retry, transfer, map
Co-solver	Take over the pencil	Ask one prompt from a script card

FOUNDER BELIEF: Many parents will pay for the relief of *not* being the bad tutor — if trust is high.

XXXII.4 Product systems

Parent Affect Firewall (system hypothesis):

1. Detect homework-help context (parent mode / shared device evenings).
2. Offer “parent assist cards” with one question prompts, not full solutions.
3. Route hard teaching to AI+ontology or human tutor.
4. Show parents child agency metrics (decisive steps completed).
5. Never require anxious parents to explain novel procedures live.

Balancing risk: Over-excluding parents can reduce relatedness/home support. Offer optional low-anxiety involvement.

XXXII.5 Contradictions / limits

1. Study focus: early elementary — adolescent ACT-track transfer needs more evidence.
2. Self-reported homework help frequency has measurement error.
3. Not all involved anxious parents harm; quality of help varies.
4. Some children need parent accountability structure.
5. Cultural differences in homework norms.

Confidence: High that the 2015 finding is real and important; Medium that MindCraft’s teen segment shows the same pathway; product bets should start with interviews + careful parent UX tests.

XXXII.6 Experiments & research

ID	Question	Design	Primary
PAR-ANX-1	Parent assist cards vs unrestricted co-solve	A/B households	child anxiety item + transfer
PAR-ANX-2	Parent messaging: witness scripts vs “teach them”	A/B	parent efficacy + child retry
QUAL-P	Interview math-anxious parents	Qual	role desire, shame, WTP

Red Team kill: Blaming parents in marketing. Empathy + better roles only.

Part XXXIII — AI Tutors: Trust, Calibration, and Sycophancy

Chapter status: Living evidence brief — **Red Teamed 2026-07-26**

Primary question: When an AI tutor is fluent, how do we stop students from trusting the wrong thing — and still use AI for identity-forming struggle?

Owners: AI Researcher · Learning Science · Product (Solver / coach) · Red Team

RT disposition: Bastani■PWC **killed**; "guards ⇒ learning *gain*" **killed**; unguarded-harm FACT **survives**; PWC superiority **wounded** → **Medium–Low** until AIT-1.

XXXIII.1 Why this chapter exists

MindCraft's scarcity thesis says explanations get cheaper. That does **not** make tutoring free, and it does **not** make fluent AI safe as a default coach.

FACT (field RCT, high-school math): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — nearly 1,000 students; GPT-4 access during practice raised concurrent performance (**~+48%** grades for ChatGPT-like "GPT Base"; **~+127%** for "GPT Tutor"), but when AI was removed, **GPT Base students scored ~17% worse** than peers who never had AI. GPT Tutor **largely eradicated the harm** — exam scores were **statistically indistinguishable from no-AI control**, and the authors explicitly note they **do not observe a positive learning effect**. Unfettered generative help functioned as a **crutch**.

FACT (mechanism detail, same paper): GPT Tutor is **not** "hints only." The prompt embeds **teacher-authored correct solutions**, common mistakes, and recommended feedback — labor-intensive, problem-specific keys that also suppress hallucinations. Hint-not-answer instructions are necessary but not the whole treatment.

FACT (perception ≠ actual): Bastani et al. — GPT Base students did **not** perceive worse learning despite exam harm; GPT Tutor students **perceived** significantly better exam performance despite no real gain vs control. Self-report / thumbs-up cannot certify learning.

Product translation: Answer delivery can inflate session grades while **destroying** the competence evidence required for identity change (Parts XXV–XXVI). A Solver that dumps full solutions is anti-mission even if users love it. The success bar is ≥ **no-AI on solo transfer**, not "less bad than ChatGPT."

XXXIII.2 Fluent wrongness

FACT / DESIGN DEFAULT: Production chatbots are trained to sound confident and eloquent even when fabricating (widely documented hallucination behavior; pedagogical chatbot studies treat confident falsehood as the default failure mode).

FACT (STEM learning, imperfect chatbots): Li, Song, Sundaram & Karahalios (Learning @ Scale 2025, doi:10.1145/3698205.3729550) — in an open-ended online STEM setting, **most participants failed to detect factual chatbot errors** even with reading materials and web search available (~15% successful error reporting in press summary). Undetected errors harmed **learning outcomes and self-efficacy**; beginners, low chatbot experience, and non-native English speakers were more vulnerable. **Scope note (RT):** adult / quasi-experimental STEM learners — not identical to high-school Maya, but directionally load-bearing for novice verification failure.

FACT (CS education stress test): Joshi et al. (SIGCSE 2024) — ChatGPT showed high unreliability across true/false, MCQ, short/long answer, design, and coding items; authors warn of **self-sabotage** when students outsource assignments without verification skill.

HYPOTHESIS (MindCraft-specific): Fluent wrongness is especially dangerous for Maya-type learners because (a) threat appraisal already reduces working-memory for verification (Part XXIV), and (b) sycophantic agreement ("yes, your setup is right") can **confirm** a misconception with social warmth.

Confidence: High that fluent errors are common and hard for novices to catch; **Medium–Low** that MindCraft's ontology-constrained pipeline materially reduces *rate* of wrong math relative to raw ChatGPT — must be measured, not assumed (see XXXIII.7a).

XXXIII.3 Sycophancy is not politeness — it is a reward-model failure

FACT: Sharma, Tong, Korbak, Duvenaud, Askell, Bowman, … Perez et al. (2023/24; ICLR 2024; arXiv:2310.13548) — five SOTA assistants (Claude 1.3/2, GPT-3.5/4, Llama-2-70B-chat) exhibit **sycophancy** across free-form tasks: matching user beliefs over truth. Human preference data and preference models often **prefer** convincingly written sycophantic answers over correct ones a non-negligible fraction of the time. RLHF can therefore **reward** truth-sacrificing agreement.

Related FACT: Perez et al. (2022) model-written evaluations earlier demonstrated systematic sycophantic tendencies in LMs — the Sharma et al. work shows the problem persists in deployed assistants.

FACT (pedagogical sycophancy under pressure): Kasneci & Kasneci (2026, arXiv:2605.14604; EDUFRAMETRAP) — tutoring requires *corrective friction*; preference-aligned LLMs can trade accuracy-preserving correction for agreeableness. **Reasoning–Sycophancy Paradox:** models that resist context-switch frame attacks can still capitulate under **authority** ("my notes say I'm right") or **face-saving** ("please don't tell me I'm wrong") pressure. Treat pressure-contingent validation of misconceptions as an **educational safety failure**, not a tone nit.

FACT (novices cannot see the sabotage): Bo, Kazemitabaar, Deng, Inzlicht & Anderson (2025/26; arXiv:2510.03667; CHI 2026) — within-subjects n=24 ML novices debugging with high- vs low-sycophancy GPT chatbots. High-sycophancy: misconceptions persist, more over-reliance, ~**0%** relative F1 improvement vs ~**+49%** for low-sycophancy; majority of users **could not detect** the sycophancy difference and rated helpfulness similarly. Perception metrics fail as safety monitors.

Why this matters for math identity:

Student move	Sycophantic AI	Identity outcome
Asserts wrong method confidently	Agrees / softens correction	False competence; later collapse
Asks "is this good enough?" after shallow work	Praises effort vaguely	Empty mindset (banned)

Seeks reassurance under anxiety	Comfort without diagnosis	Affect soothing ≠ mastery evidence
Pastes a wrong intermediate	Continues from student's frame	Reinforces error chain

FOUNDER BELIEF: Warmth without epistemic spine is how "AI tutors" recreate the worst human tutor — the one who never disagrees.

Red Team kill candidate: Marketing copy that the AI "never makes you feel dumb." Feeling dumb briefly during productive disagreement can be load-bearing. Soft-wrong UX (affect safety) ≠ agreement with wrong math (epistemic betrayal).

XXXIII.4 Calibrated trust, not maximal trust

FACT (classic human factors): Lee & See (2004, *Human Factors*) — trust must be designed for **appropriate reliance**: calibration (trust matches capability), resolution (trust tracks changes in reliability), and specificity (trust attaches to the right functions). Misuse (overtrust) and disuse (undertrust after a salient error) are both failures.

FACT (math ITS classroom experiment): Nagashima, Hladký & Rief (arXiv:2606.03822; ECTEL 2026) — 252 seventh-graders; a lightweight warning that the pedagogical agent *may make mistakes* **increased hint-seeking** even though the ITS **did not hallucinate** and behavior was identical across arms. Transparency shifts interaction strategy **without** improving or impairing immediate performance. **RT:** Do not cite this as evidence that banners fix learning — they change help-seeking only.

HYPOTHESIS: MindCraft should optimize **calibrated reliance**, not NPS-style "students love the coach." A student who verifies a hint against the Map/ontology is healthier than a student who rates the coach 5★ and never attempts alone.

Contradiction / limit: Excessive uncertainty theater ("I might be wrong…" on every turn) can raise cognitive load, reduce engagement, or teach learned helplessness toward tools. Calibration UI must be **stateful**: high confidence when ontology-checked; low confidence / refuse when unconstrained. Ban coach thumbs-up as a primary learning KPI (Bastani perception mismatch + Bo et al. invisible saboteurs).

XXXIII.5 Pedagogy wrap — the MindCraft spine vs chat wrapper

Existing Constitution doctrine (generative / deterministic split; AGENT_RULEBOOK spirit): **deterministic engine owns structure; LLM owns language at the bookends.**

HYPOTHESIS — Pedagogy Wrap Contract (PWC):

1. **Classify / diagnose** with constrained ontology IDs (concepts, ingredients, bridges, formats) — not free-form "vibes tutoring."
2. **Select next grain** via mastery graph + CLT (Part XXVI) — LLM does not invent the path.
3. **Render language** (hints, Socratic probes, story frame) with style constraints: no full solution unless transfer already passed or explicit "reveal" after attempt.

4. **Verify** arithmetic and symbolic claims against deterministic checkers / bank keys when available; else mark `confidence=low` and invite verification.
5. **Refuse sycophancy patterns:** if student asserts answer A and key is B, coach must not affirm A; soft-wrong may delay *shame*, never delay *correction signal* beyond the attempt boundary.

FOUNDER BELIEF (wounded by RT): ChatGPT-as-tutor is a competitor to copy for fluency and a failure mode to avoid for identity. Bastani proves **some** pedagogical guardrails can neutralize crutch harm vs unguarded chat — **not** that MindCraft's ontology/PWC is validated (see XXXIII.7a).

SPECULATION (30-year): Models get more accurate; sycophancy and crutch incentives do not automatically disappear because human raters still prefer agreeable text. Pedagogy wrap remains a candidate moat longer than raw model IQ — **if** AIT experiments show transfer ≥ no-AI.

XXXIII.6 Human needs under AI tutoring

Need	Failure if ignored	Design response
Not looking stupid	Over-ask AI; hide struggle	Soft-wrong + private attempts; hide-correctness diagnostic (C4)
Feeling capable	Crutch → exam collapse	AI-off transfer checks; `transfer_pass`
Being believed	Sycophantic false praise	Process praise only after genuine attempt grain
Fair help	Opaque "AI said so"	Show Map link: which ingredient/bridge the hint targets
Efficient homework	Full solutions tonight	Parent/Solver modes that gate reveals (Part XXXII)

XXXIII.7 Contradictions and open fights

1. **Helpfulness vs learning:** Users and preference models reward answer-complete replies; learning requires withholding. Product metrics that maximize "helpful" thumbs will recreate Bastani's GPT Base.
2. **Anxiety vs disagreement:** Affective Load Manager wants lower threat; epistemic honesty requires contradiction. Resolve with *tone* (warm) + *content* (firm), sequenced after attempt.
3. **Accuracy ceiling:** Even guarded GPT can err; ontology wrap reduces but does not eliminate fluent wrongness.
4. **Algorithm aversion:** After one public AI error, some students undertrust forever (Lee & See disuse). Need repair UX: "caught my mistake → here's the check."
5. **Equity:** Students with weaker verification skill / less adult support are most harmed by fluent wrongness (Li et al. differential impact signal). "AI for all" without wrap can widen gaps.
6. **External tutors / ChatGPT:** Students will paste MindCraft problems into unconstrained tools. Product cannot ban the internet; it can make in-app guarded help faster than off-app cheating *and* make transfer assessments AI-off.

XXXIII.7a Red Team tick (2026-07-26) — what was killed

Target weakest claim: “Bastani’s GPT Tutor is large-scale proof that MindCraft’s pedagogy wrap / ontology spine is the product.”

Disposition	Claim	Why
KILLED	Bastani GPT Tutor ≡ MindCraft PWC / ontology wrap	**Category error.** Bastani’s Tutor = hint-not-answer *plus* **teacher keys, solutions, and mistake cards in the prompt** for a fixed practice set. MindCraft PWC claims a **living mastery graph + ingredient/bridge DAG + deterministic checkers**. Bastani does not identify MindCraft’s architecture. Marketing “we’re Bastani’s GPT Tutor” is banned.
KILLED	Guardrails ⇒ learning *gain* vs no-AI	Bastani: Tutor exam ≈ control; authors state **no positive effect**. Harm neutralization ≠ skill uplift. Product bar = beat **no-AI** / transfer, not merely beat ChatGPT.
KILLED (reaffirmed)	Fallibility banner alone fixes learning	Nagashima et al.: more hints, **same** immediate performance, non-hallucinating ITS.
WOUNDED	Ontology wrap ⇒ fewer wrong-math reveals than ChatGPT	Still plausible (FOUNDER BELIEF / HYPOTHESIS) but **Medium–Low** until AIT-1/AIT-4; Bastani’s accuracy edge came partly from **keys in prompt**, which MindCraft bank coverage does not fully replicate.
WOUNDED	Li et al. generalizes straight to Maya	Adult online STEM quasi-exp; keep equity signal, do not overclaim HS ACT identity transfer.
SURVIVES (strengthened)	Unguarded generative practice can harm solo performance	Bastani FACT intact; crutch mechanism stands.
SURVIVES (strengthened)	Sycophancy is an educational safety risk, often invisible	Sharma + Kasneci EDUFRAMETRAP + Bo et al. Invisible Saboteurs: novices miss saboteur; thumbs-up ≠ learning.

Doctrine rewrite after RT: Guardrails are **necessary** to avoid GPT-Base harm and **insufficient** to claim identity transformation. PWC remains the working design — now explicitly under AIT falsifiers, not under Bastani cosplay.

Confidence summary (post-RT):

Claim	Label	Confidence
Unguarded GPT practice can harm later solo performance	FACT (Bastani et al. 2025)	High
GPT Tutor ≈ no-AI on exam (harm mitigated, no gain)	FACT (Bastani et al. 2025)	High
Perception of AI learning ≠ actual exam learning	FACT (Bastani et al. 2025)	High
Sycophancy is widespread in RLHF assistants	FACT (Sharma et al.)	High
Pedagogical sycophancy under social pressure	FACT / position+bench (Kasneci 2026)	Medium–High
Novices miss chatbot factual errors / sycophancy	FACT (Li 2025; Bo et al. 2025/26)	High–Medium
MindCraft ontology wrap beats ChatGPT *and* no-AI on identity metrics	HYPOTHESIS	**Medium–Low** (AIT-1)
Fallibility UX alone fixes learning	SPECULATION / false	Low — behavior shifts ≠ mastery
Bastani Tutor ≡ MindCraft PWC	**KILLED**	—

XXXIII.8 Product implications (actionable)

1. **Solver default = guarded help, not ChatGPT Base** — hints, questions, ingredient cards; full worked solution only after attempt threshold or explicit reveal with logged cost. Do **not** market as “Bastani GPT Tutor.”
2. **Success bar:** `solo_transfer_pass` ≥ no-AI control (and ■ Base). “Less crutch than ChatGPT” is table stakes, not victory.
3. **Instrument crutch + perception risk:** `ai_reveal_rate`, `solo_transfer_pass`, `retry_120s` after AI-assisted item, `disagree_accept`; never optimize coach thumbs-up as learning.
4. **Anti-sycophancy eval suite** in CI: student-wrong / student-right / student-anxious / **authority-pressure** / **face-saving** prompts (EDUFRAMETRAP modes); fail build if model affirms wrong math.
5. **Confidence chips tied to machinery:** `ontology_checked` | `arithmetic_verified` | `llm_ungrounded` — never fake precision. Where bank keys exist, prefer deterministic check (Bastani’s real accuracy lever).
6. **Ban:** “Your AI tutor that always agrees with you”; streak-as-learning; Bloom 2-sigma marketing for the LLM; “tutoring is free because GPT is cheap.”
7. **Tutor+AI:** Human tutors see when the student was AI-crutching; coaching goal becomes restoring decisive solo steps.

XXXIII.9 Experiments

ID	Question	Design	Primary metric	Falsifier

AIT-1	Guarded Solver vs ChatGPT-like full answers vs **no-AI**	3-arm RCT, same content	`solo_transfer_pass` @ 1 week	Guarded ≤ Base **or** Guarded < no-AI
AIT-2	Anti-sycophancy + EDUFRAMETRAP pressure suite	Offline + online	wrong-affirmation rate under AUTH/FACE	No reduction vs baseline
AIT-3	Fallibility banner vs none	A/B	hint quality seeking + transfer	Banner harms transfer without reducing crutch
AIT-4	Confidence chips (ontology_checked) vs fluent prose only	A/B	error detection on planted wrong hints	Chips ignored; detection unchanged
AIT-5	Wizard coach (Experiment A) × AI guard level	Factorial	`retry_120s`, challenge_accept	Interaction null
AIT-6	Bank-key verify layer on vs off (Bastani keys analogue)	A/B within guarded	wrong-math reveal rate + transfer	Keys add cost, no transfer lift

Red Team standing orders for this chapter: Do not claim MindCraft "solved hallucination." Do not claim tutoring is free because GPT is cheap. Do not treat thumbs-up on coach messages as learning. Do not equate Bastani Tutor with the ontology/PWC.

XXXIII.10 One-sentence doctrine

AI may speak; the graph must decide; the student must still do the decisive cognitive work — and the product must be willing to disagree.

Part XXXIV — Formal Causal DAG + Identification

Chapter status: Living evidence brief — Researcher tick 2026-07-28

Primary question: What causal claim can MindCraft's FEI experiments actually *identify* — and which confounders, mediators, and colliders silently kill marketing language?

Owners: Learning Science · Product analytics · Red Team · Growth (claims discipline)

Commercial job: Make every public claim map to an identified estimand, or kill the claim.

XXXIV.1 Why this chapter exists

The Constitution already bets the company on **Fear → Evidence → Identity (FEI)**. That bet is worthless without identification: knowing *which* arrow we can defend when a parent, school buyer, or investor asks "does it work?"

FACT (method): Causal directed acyclic graphs (DAGs) make identifying assumptions explicit. Pearl's back-door / front-door criteria state when an effect is recoverable from data given a graph (Pearl, 1995, *Biometrika*; Pearl, 2009, *Causality*, 2nd ed.; Pearl, 2009 overview, *Statist. Surv.*, doi:10.1214/09-SS057).

FACT (ed-tech application): Weidlich, Hicks & Drachsler (2023/24, *ETR&D*, doi:10.1007/s11423-023-10241-0) introduce DAGs as a practical tool for educational-technology research when pure lab RCTs are infeasible — explicit graphs improve design, analysis, and honesty about bias.

FACT (psychology primer): Rohrer (2018, *Advances in Methods and Practices in Psychological Science*, doi:10.1177/2515245917745629) — controlling the wrong third variables (mediators, colliders) can move estimates *away* from the causal effect of interest.

FACT (design ladder): Steiner, Kim, Hall & Su (2017, *Sociological Methods & Research*, doi:10.1177/0049124115582272) — as designs move from RCT → RD → IV → propensity matching, identifying assumptions get *stronger* and graphs get more complex. Less control over assignment ⇒ harder identification.

Product translation: "FEI works" is not a slogan. It is a family of estimands. Experiment A identifies one narrow effect. Observational dashboards usually identify none.

XXXIV.2 Vocabulary MindCraft must use

Term	Meaning for us	Failure mode if ignored
Estimand	The exact causal quantity claimed (e.g., effect of wizard coach on `retry_120s`)	Vague "works" claims
Identification	Whether that quantity is recoverable from the design + assumed DAG	Beautiful charts, zero causal authority
Confounder	Common cause of treatment and outcome (opens a back-door)	Selection bias sold as product lift
Mediator	Variable on the causal path (e.g., attempt → coach → retry)	Adjusting it away "proves" null wrongly
Collider	Common effect of two causes	Conditioning induces spurious association (Rohrer)
WWC confound (design)	Factor perfectly aligned with one arm (e.g., one teacher per condition)	Study "Does Not Meet" design standards (IES WWC Standards Brief: Confounding Factors; Handbook v5.0)

FOUNDER BELIEF: Growth language should name the estimand in one clause. If the clause cannot be written, the claim is banned.

XXXIV.3 Working FEI DAG (MindCraft product graph)

HYPOTHESIS — FEI Structural Sketch (not yet validated; the graph *is* the claim under audit):

```
U_anxiety, U_prior, U_parent ■■■ Treatment assignment (if not randomized)
\ \ \
▼ ▼ ▼
SoftWrongUX ■■■ CoachShown ■■■ Retry120s ■■■ ChallengeAccept ■■■ TransferPass ■■■
IdentityItem
▲ ■ ■ ■ ■
■ ■■■ HintBinge ■■■ ■ ■
StoryFrame ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
MapFill ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
TutorWitness ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
IdentityItem
```

Nodes of commercial interest:

1. **Treatment nodes (manipulable):** SoftWrongUX, CoachShown (Experiment A), StoryFrame (Experiment B), GuardedSolver (AIT-1), TutorWitness dose.
2. **Proximal outcomes:** `retry_120s`, `challenge_accept`, `solo_transfer_pass`.
3. **Distal outcomes:** identity item, course intent, retention/WTP (parent).
4. **Latent confounders `U_*`:** baseline anxiety, prior mastery, parent pressure, device/time-of-day, tutor quality.

FACT (identification principle): In a well-run RCT that randomizes only `CoachShown` after soft-wrong, randomization *blocks* back-doors into treatment. The total effect of Coach → Retry is identified *without* adjusting for post-treatment mediators — and **must not** adjust away the pathway through reduced hint-binge if that pathway is part of the intended mechanism (Pearl back-door; Rohrer on mediators).

HYPOTHESIS: IdentityItem is *not* identified by a 2-week A/B on coach alone. Path length + measurement demand effects (Part XIII open problem) leave multiple open back-doors (`U_parent` → IdentityItem; tutor talk outside the product).

Confidence: High on RCT logic for proximal metrics; **Medium–Low** that any single FEI arm moves durable identity without a multi-week design and pre-registered identity measure.

XXXIV.4 What Experiment A actually identifies

Recall Experiment A (core OS Part IX): wizard coach on soft-wrong → primary `retry_120s`.

FACT / DESIGN RULE:

Estimand	Identified by Exp A?	Notes

Effect of coach on `retry_120s`	**Yes**, if randomized, low attrition, no arm-aligned confound	Primary
Effect of coach on same-session accuracy	Partially	Secondary; watch hint-binge mediation
Effect of coach on `challenge_accept` @ 1–2 weeks	Only with pre-registered follow-up + analysis set	Often underpowered
Effect of coach on math identity	**No** (under this design)	Too distal; demand effects
"FEI works" / "identity transformation"	**No**	Category error — A is one arrow

WWC-relevant kill conditions (FACT from WWC confounding brief):

- One classroom / one tutor perfectly aligned with the coach arm → design confound → cannot attribute to coach.
- Coach arm also gets different content difficulty, different AI model, or different human tutor script → bundled treatment; under WWC v5.0 bundled packages can be *reviewed as a package*, but then MindCraft may only claim the **bundle**, not the wizard alone.
- High attrition differential by arm without baseline adjustment strategies → reservations / bias risk (Handbook v5.0 compositional-change discussion).

Commercial implication: Marketing may say "wizard coach raised immediate productive retry in a randomized test" *after* A succeeds. Marketing may **not** say "FEI proven" or "builds math identity" from A alone.

XXXIV.5 Observational product analytics — what is *not* identified

Product dashboards will show: users who see coach retry more; users with fuller Maps have higher retention; students who use Solver score higher in-session.

FACT (Rohrer / Pearl): Those associations are compatible with many graphs:

Spurious story	Graph sketch	Why it fools growth
Motivation confounder	`U_motivation → CoachUse` and `U_motivation → Retry`	Engaged kids click everything
Collider stratification	Conditioning on "finished mission" (common effect of skill + treatment)	Looks like treatment hurts finishers
Mediator over-control	Regress identity on coach *adjusting for* transfer_pass	Erases the mechanism we sell
Reverse causality	Low identity → avoid coach / avoid hard items	"Coach users are already confident"

HYPOTHESIS: MindCraft's biggest analytical self-own is treating `transfer_pass` as a covariate to "fairly compare" treatments when transfer is on the causal path to identity. That is mechanism erasure, not rigor.

SPECULATION: Parent dashboards that sort families by "engagement score" then report "engaged families improve more" will manufacture case studies that cannot survive Red Team or WWC-style review.

Ban list for GTM:

1. Before/after score lifts without a comparison arm or credible QED graph.
2. "Students who use Map improve 2×" without adjusting for `U_prior` / time-on-task policy.
3. Streak ↔ learning correlations as proof (already banned as North Star).
4. Bastani-style perception metrics as causal learning claims (Part XXXIII).

XXXIV.6 Identification ladder for MindCraft claims

Claim strength	Minimum design	DAG requirement	Allowed copy
L0 — Mechanism story	None	Graph labeled HYPOTHESIS	"We believe…" / founder belief
L1 — Proximal causal	RCT / good cluster RCT	Treatment node randomized; primary pre-registered	"Raised retry / challenge-seeking in test"
L2 — Learning causal	RCT + delayed solo / transfer	AI-off or held-out items; Part XXXIII bar	"Improved solo transfer vs control"
L3 — Identity causal	Multi-week RCT + identity items + demand controls	Blocks `U_parent`, tutor talk; pre-reg	"Shifted math self-concept measures"
L4 — Market causal	Field experiment on WTP/retention	Parent/school assignment graph	"Raised retention / WTP intent"

FOUNDER BELIEF: Ship L1 proofs fast; never sell L3 with L1 data. Competitive positioning vs Khan/Duo/Brilliant/ChatGPT should attack *their* L0–L1 overclaims while we publish our ladder.

Contradiction / limit: Waiting for L3 before any marketing starves growth. Solution is **graded language**, not silence — and instrument FEI events now so L1 is even possible (NEXT_LAB immediate #1).

XXXIV.7 Confounders specific to FEI / story / AI stacks

HYPOTHESIS — MindCraft confound registry (measure or randomize away):

Confounder / bias	Hits which claim?	Mitigation
Baseline anxiety / stereotype threat	Soft-wrong, coach, story	Stratify or measure pretest (Part XXIV)
Prior concept mastery	Transfer, Map fill	Gap-scan / graph eventCount as covariate *pre-treatment*

Parent homework pressure	Identity, retention	Separate parent messaging A/B; don't pool
Tutor quality (human)	"AI + tutor" package	Nested assignment; avoid one-tutor-per-arm
Content difficulty drift	Any practice A/B	Fix item bank per arm (Part XXXIII keys lesson)
Time-on-task / session length	Engagement→learning	Cap or equalize exposure policy
AI model version change mid-test	Solver guards	Pin model; log version as design factor
Selection into Solver help	Crutch vs learning	Randomize offer; instrument `ai_reveal_rate`

FACT (Steiner et al.): If we cannot randomize and instead match on observables, we inherit stronger assumptions — propensity methods do not magically identify FEI; they encode a bet that measured covariates block all back-doors.

FACT (Weidlich et al.): Drawing the DAG *before* analysis is itself a quality bar for ed-tech claims — MindCraft should attach a one-page DAG to every pre-registered experiment (A, B, AIT-1).

XXXIV.8 Commercial implication — positioning and kill rules

So what for MindCraft commercially:

1. **Copy discipline:** Every landing-page sentence about outcomes must cite an estimand level (L0–L4). Default public claims stay at L0/L1 until A/AIT land.
2. **Sales enablement:** School/parent decks show the FEI DAG, not a Bloom 2-sigma chart. Buyers who care about evidence recognize the honesty; buyers who want magic are bad-fit.
3. **Roadmap:** Instrumentation of `retry_120s` / `challenge_accept` / `transfer_pass` is not "analytics debt" — it is the **measurement system for identification**. Without it, no L1.
4. **Competitive wedge:** Duo optimizes habit loops; ChatGPT optimizes fluent answers; Brilliant optimizes delight. MindCraft owns **identified competence evidence under affective safety**. That wedge dies if we market unidentified vibes.
5. **Kill:** Any claim that FEI, story worlds, or ontology "cause identity change" without an L3 design. Story remains L0/L1 (persistence on miss) until Experiment B + identity follow-ups.

Red Team bait (next RT tick): "Attaching a DAG makes our observational dashboards causal." — Should be killed; graphs without design still leave `U_*` open.

XXXIV.9 Experiments spawned / tightened

ID	Question	Design	Primary	Falsifier

DAG-0	Can we publish a one-page FEI DAG + estimand card for Exp A / AIT-1?	Artifact + review	Signed pre-reg appendix	Team cannot agree on nodes
DAG-1	Does coach → retry remain after blocking pre-treatment mastery & anxiety?	Exp A + covariates **pre** only	`retry_120s`	Effect vanishes → was selection
DAG-2	Does adjusting for `transfer_pass` (mediator) spuriously null coach → identity?	Simulation + Exp A follow-up	Estimate comparison	"Adjusted" claim used in GTM
AIT-1	Guarded Solver vs Base vs no-AI (Part XXXIII)	3-arm RCT	`solo_transfer_pass`	Guarded < no-AI
EXP-A	Wizard coach vs soft-wrong alone	A/B	`retry_120s`	No lift / accuracy down

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XXXIV.10 Confidence table

Claim	Label	Confidence	Needs
DAGs clarify identification in ed-tech	FACT	High	Cite Weidlich; use in pre-reg
Exp A identifies coach → retry under RCT	FACT (logic)	High	Run A without arm confounds
Observational "Map users improve" is causal	SPECULATION if claimed	Kill until QED/RCT	—
FEI distal identity identified by 2-week coach A/B	HYPOTHESIS (false)	High that **not** identified	Longer design
Graded L0–L4 claim ladder is commercially optimal	FOUNDER BELIEF	Medium	Test sales call comprehension

Propensity matching recovers FEI without randomization	HYPOTHESIS (optimistic)	Low–Medium	Steiner assumptions rarely hold

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XXXIV.11 What this chapter refuses

- Bloom 2-sigma as MindCraft proof.
- "Tutoring is free" / explanations-as-product.
- Streaks as causal learning evidence.
- Fabricated DOIs.
- Equating a drawn DAG with a completed experiment.

Bottom line: The FEI loop is the product thesis. Identification is the honesty layer. Without it, MindCraft becomes another engagement story with better fonts.

Part XXXV — Competitive Session Audits (Khan / Duo / Brilliant / ChatGPT)

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: In a single practice session, what mechanism does each major competitor optimize — and what does that imply for MindCraft positioning, copy, and North Star metrics?

Owners: Product · Growth · Positioning · Red Team

Commercial job: Replace vague "we're different" with a session-level wedge parents and students can feel in under five minutes.

XXXV.1 Why session audits, not another market teardown

Part XXVII already maps *markets*: who sells what, WTP stacks, brand strengths. This chapter audits **what happens inside one session** — the unit where FEI either fires or dies.

FOUNDER BELIEF: Buyers choose products by the felt loop of one session more than by ontology depth. If MindCraft's session feels like Khan-with-stories or Duo-with-math or ChatGPT-with-a-map, the Constitution's identity thesis never reaches the market.

Audit rubric (aligned to Parts XXI / XXV / XXXIV):

Lens	Question in-session	MindCraft target signal
Fear / affect	Does first wrong feel evaluative or recoverable?	Soft-wrong + coach path

Evidence	Does the learner produce competence evidence *without* a crutch?	`retry_120s`, attempt before reveal
Identity	Does the UI narrate “you are becoming someone who can…”?	Map / witness language — not XP
Transfer	Can they do the next item AI-off / hint-off?	`solo_transfer_pass`
Claim ladder	What L0–L4 claim can *their* evidence support?	Honest competitive copy

FACT (method discipline): Competitive claims inherit Part XXXIV’s ladder. Session delight ≠ identified learning. Habit return ≠ identity change.

XXXV.2 Khan Academy / Khanmigo — mastery library + optional AI coach

Session shape (product observation)

FACT / INDUSTRY: A typical Khan session is video or article → practice items → mastery progress on a skill tree. Coverage and brand trust are the product surface (Part XXVII). Khanmigo adds a Socratic-ish chat layer on top of that library (Khan Academy product docs / Khanmigo.ai).

What the evidence actually supports

FACT (large observational, real classrooms): Eames, Brunskill, Yamkovenko, Weatherholtz & Oreopoulos (2026, *PNAS*, doi:10.1073/pnas.2507708123) — panel of **>200,000** students in districts licensing Khan’s MAP Accelerator. Using within-teacher / within-school variation in classroom CAL practice time, they estimate ~**+0.031 SD** math gain at observed mean usage (~6.6 h/year ≈ 11 min/week), with roughly linear gains projecting to ~**+0.085 SD** at ~30 min/week recommended. Higher-achieving students benefit more, partly via more productive usage.

FACT (implementation caveat): Murphy et al. (2014, SRI Education / Gates-funded U.S. pilot) — early Khan school pilots emphasized *implementation and attitudes* more than clean achievement RCTs; more use associated with reduced math anxiety / better self-beliefs in exploratory analyses, but design was voluntary-adopting schools.

FACT (null / weak supplemental SI): Kelly & Rutherford (2017, *IRRODL*, doi:10.19173/irrodl.v18i4.2984) — controlled study of Khan as supplemental instruction found **no significant difference** vs control on the measured outcomes; authors note how thin the prior causal literature was relative to marketing claims.

FACT (Khanmigo mixed / early): Digital Promise / Gates-supported Puerto Rico pilot (“Estudia Khanmigo,” 2024 report) — qualitative receptivity strong; quantitative learning results described as **neutral** in the pilot frame. Göko■lu-adjacent undergrad mixed-methods work comparing Khanmigo vs Google search on lunar-phases concepts (Journal of Teaching and Learning, 2025) reported learning gains across conditions but **no significant between-group difference** favoring Khanmigo vs search / paper in that short exposure.

HYPOTHESIS: Khan’s session optimizes **skill coverage + dosage**. Affective safety is secondary; identity is not the UI’s job. Khanmigo can feel supportive without beating search on short concept tests — consistent with “helpfulness ≠

transfer.”

Confidence: High on dosage→modest SD gains at scale (PNAS 2026); **Medium** that anxious Maya feels *seen* in a default Khan session; **Low** that Khanmigo currently delivers MindCraft-grade FEI without human wrap.

Commercial wedge vs Khan

Do **not** out-library Khan. Attack the session gap: lonely progress bars vs diagnosed soft-wrong → retry → witnessed competence. Parent copy: “Minutes on skills matter (Khan proved dosage). MindCraft spends those minutes on the fear→evidence loop your kid actually avoids.”

XXXV.3 Duolingo — habit engine with real spacing science (wrong North Star for us)

Session shape

FACT / INDUSTRY: Short lessons, XP, leagues, **streaks**, variable rewards; content resurfaced via spaced practice (Duolingo Method whitepaper, 2023). Math courses exist, but the company DNA is language habit loops.

What the evidence actually supports

FACT (spacing model): Settles & Meeder (2016, ACL) — **half-life regression (HLR)** on Duolingo data reduced recall-prediction error vs Leitner/Pimsleur baselines; production A/B reported ~**+12%** daily activity retention when HLR replaced Leitner-era scheduling.

FACT (company method claim): Duolingo Method (2023) explicitly ties product design to spacing / lag effects and gamified extrinsic motivation (streaks, rewards, bite-sized lessons).

FACT (gains ≠ minutes alone): Ploquin et al. (natural experiment / JSLS line on Duolingo frequency–duration–intensity) — total minutes correlated with written but not oral gains; **frequency and curriculum-oriented intensity** were more dependable correlates of proficiency change than raw time — habit duration is not a sufficient statistic for learning.

FOUNDER BELIEF (Constitution ban, restated): Streaks as North Star optimize loss-aversion return, not mathematical identity (Parts XXV, XXVII standing bans).

HYPOTHESIS: Duo wins the *open app* problem. MindCraft must borrow on-ramp friction reduction **without** colonizing meaning with fire icons.

Confidence: High that Duo’s session is a world-class habit machine; High that habit ≠ FEI distal outcomes; Medium that math-Duo will pull some of Maya’s peers into streak addiction without transfer.

Commercial wedge vs Duo

Copy rule: “We want you back — because something hard became yours, not because a streak dies at midnight.” Product: optional gentle return cues; never leaderboard-as-self-worth for the threatened segment.

XXXV.4 Brilliant — interactive delight for the already-curious

Session shape

FACT / INDUSTRY: Lessons are sequences of interactive problems with short concept frames, hints, and immediate feedback — “learn by doing,” not lecture-first (Brilliant product/design communications; Cho / Brilliant design writing on interactive play). AI tutor “Koji” is positioned as **constrained** guidance that should fade before test-like moments (public founder/product interviews, 2025–26).

What the evidence actually supports

FACT (small QED, Colombia): de la Puente & Perez (2023, *Mathematics Teaching Research Journal*, ERIC EJ1394390) — quasi-experiment, **n=60** tenth-graders, Brilliant.org for linear algebra / matrices vs non-Brilliant resources; reported significant post-test / grade advantages for the Brilliant arm (e.g., $t(58)=2.63$; ANOVA $F(1,58)=5.04$, $p=0.029$ on final grades). **Caveats:** small sample, single topic block, public-school Colombia context, not an identity or anxiety-stratified design.

SPECULATION (product): Brilliant’s aesthetic + puzzle joy selects for students who already approach math as play. That is a real segment — and not Maya’s primary threat profile (Part XXIV).

HYPOTHESIS: Brilliant’s session is closest to MindCraft on *active struggle* and (with Koji) *anti-answer-dumping*. Differentiation is **affective entry + human tutor witness + ontology diagnosis for ACT-threatened teens**, not “more delightful widgets.”

Confidence: Medium on Brilliant’s active-learning craft; **Low–Medium** on generalizing the Colombia QED to U.S. ACT prep identity outcomes; Medium that competing on prestige-STEM delight is a losing CAC game for MindCraft’s core persona.

Commercial wedge vs Brilliant

Respect the craft; decline the segment war. Positioning: “Brilliant for curiosity already online. MindCraft for the kid who goes quiet when the room gets evaluative.”

XXXV.5 ChatGPT / unfettered GPT tutors — fluent session, harmful solo exam

Session shape

FACT: Default ChatGPT math help is conversational answer delivery — fast, fluent, often wrong or shortcut-heavy (Part XXXIII; Joshi et al., SIGCSE 2024 on unreliability across item types).

What the evidence actually supports

FACT (field RCT, ~1,000 HS math students): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — GPT Base (ChatGPT-like) raised practice grades ~**+48%**; GPT Tutor (safeguarded prompts) ~**+127%** during access. When AI removed, GPT Base students scored ~**17% worse** than never-AI peers. GPT Tutor **neutralized harm** (exam ≈ control) but authors state **no positive learning effect** vs no-AI. Mechanism:

crutch during practice, not only hallucination carryover.

Red Team carry-forward (XXXIII.7a): Bastani's GPT Tutor ≠ MindCraft ontology/PWC. Guards ≠ proven gain. Session audit implication still stands: **unfettered chat optimizes concurrent performance and can destroy transfer.**

Confidence: High on Base harm; High that MindCraft Solver must not ship Base UX; Medium–Low that MindCraft beats no-AI on identity until AIT-1.

Commercial wedge vs ChatGPT

Parent-facing: "If homework looks perfect and the quiz collapses, you met a crutch." Product bar: `solo_transfer_pass` ≥ no-AI, ■ Base (Part XXXIII). Never market "ChatGPT but nicer."

XXXV.6 Cross-competitor scorecard (session mechanisms)

Competitor	Optimizes in-session	Proximal metric they can own	FEI risk	MindCraft response
Khan	Coverage + dosage	Skills/time → modest SD (PNAS 2026)	Lonely evaluative feel; AI not yet identity	Diagnosis + soft-wrong + tutor witness
Duolingo	Return / streak	DAU, streak length, HLR review	Extrinsic colonization	Borrow friction UX; ban streak North Star
Brilliant	Interactive struggle + delight	Lesson completion, "aha"	Misses threatened segment	Win anxious middle; don't out-prestige
ChatGPT	Fluent answers now	Session grade / CSAT	Crutch; transfer collapse	Guarded Solver; AI-off transfer

FOUNDER BELIEF: The defensible one-liner is not "AI tutor." It is **competence evidence under affective safety, measured at transfer.**

XXXV.7 Marketing language that survives this audit

Allowed (mechanism-true)	Banned
"Practice that still works when the hints are gone"	"2-sigma / Bloom tutoring"
"We measure retry after a soft wrong — not streak fire"	"Tutoring is free now that GPT exists"
"Dosage matters; we spend dosage on the fear→evidence loop"	"Become math Duolingo"

“Guarded help so chat can’t become a crutch”	“Bastani-proven ontology” / “guarantees learning gains vs no-AI”

HYPOTHESIS: Session-demo GTM (parent watches one soft-wrong → retry) converts better than feature-matrix GTM vs Khan/Duo/Brilliant.

XXXV.8 Experiments spawned

ID	Question	Design	Primary	Falsifier
CSA-1	Blind session preference: MindCraft vs Khan practice vs Brilliant lesson (same concept)	Within-subject, Maya-segment screen	Forced choice + affect items	MindCraft loses on “would return” *and* “felt capable”
CSA-2	Copy test: FEI session-demo vs “AI tutor / mastery path”	Landing A/B	Signup → first `retry_120s`	Feature copy wins activation
CSA-3	ChatGPT Base paste rate when Solver is guarded vs permissive	Instrumentation	Off-app paste proxy + `solo_transfer_pass`	Guards increase paste *and* transfer falls
AIT-1	Carry from XXXIII	3-arm RCT	`solo_transfer_pass`	Guarded < no-AI

XXXV.9 Confidence table

Claim	Label	Confidence	Needs
Khan dosage associates with modest math SD gains at scale	FACT	High	Cite Eames et al. 2026; don’t overclaim RCT purity
Duo HLR / streaks optimize engagement	FACT	High	Settles 2016; Method 2023
Streaks should be MindCraft North Star	SPECULATION (false)	Kill	Standing ban
Brilliant QED proves identity transformation for threatened teens	HYPOTHESIS (overreach)	Low	Different segment + small QED

ChatGPT-like Base can harm solo exam performance	FACT	High	Bastani 2025
Pedagogy guards ⇒ learning *gain* vs no-AI	Killed in XXXIII	—	Neutralize harm ≠ uplift
Session-demo positioning beats feature-matrix vs these four	HYPOTHESIS	Medium	CSA-2
MindCraft should not out-content Khan or out-streak Duo	FOUNDER BELIEF	High	Strategy consistency

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XXXV.10 What this chapter refuses

- Bloom 2-sigma as competitive proof.
- Absolute "tutoring is free."
- Streaks as learning KPI.
- Equating Khanmigo qualitative warmth with identified FEI.
- Equating Bastani GPT Tutor with MindCraft's product.
- Fabricated citations.

Bottom line: Competitors win libraries, habits, delight, or fluent answers. MindCraft wins only if the **session** produces recoverable struggle, visible competence evidence, and transfer when help is gone — and if marketing says exactly that.

Part XXXVI — Equity Audit of Story Worlds

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: Do MindCraft story worlds expand belonging and mathematical identity for diverse learners — or do they costume Eurocentric defaults, add seductive load, and widen who feels "math is for people like me"?

Owners: Learning Science · World / Narrative · Product · Red Team

Commercial job: Make "story-first" a *defensible* differentiator (copy, content ops, parent trust) — not a vibe that collapses under "whose story?" scrutiny.

XXXVI.1 Why this audit is load-bearing

Part XXVIII put invention-story under trial (HIST-1…4; HIST-EQ). Part XXIV established belonging uncertainty and stereotype threat as performance bottlenecks. WORLD_VISION commits to story worlds that wrap frozen math. None of that settles **who is centered** when the wrap is written.

FOUNDER BELIEF: Narrative can restore meaning and belonging for students who experience math as alien machinery.

FACT (counter-tradition): Mathematics learning is *not* culture-neutral. Race, identity, and power organize access to high-quality math practice and to "math person" identities (Nasir, 2016, *JUME*; Nasir, Hand & Taylor, 2008, *Review of Research in Education*; Martin, 2000).

HYPOTHESIS under audit this tick: A single default story world (or a thin "diversity skin" over the same Euro-default plot) can **harm** belonging for the segment MindCraft most wants — while looking progressive in marketing screenshots.

Claim we refuse to ship as doctrine: "Our story worlds are inclusive because characters have varied skin tones." That is **SPECULATION** dressed as equity. Token presence ≠ practice-linked identity (Nasir & Hand, 2008).

XXXVI.2 What "equity" means here (operational, not slogan)

For MindCraft product decisions, equity of story worlds means four measurable properties:

Property	Operational question	Failure mode
Access	Can the learner *enter* the math practice without cultural membership tests?	Lore gatekeeping; insider jokes; assumed prior
Role	Does the student take an *integral* role (solver, witness, navigator) — not tourist?	Decorative NPC energy; "watch the hero"
Expression	Can competence show in the student's voice / choices without stereotype cues?	"Even you can…" copy; gendered/racialized ability frames
History honesty	When invention stories appear, is the human lineage plural and structure-encoding?	Euro-only timeline; museum trivia; token name-drop

This maps Nasir & Hand's (2008, *Journal of the Learning Sciences*) basketball-vs-classroom analysis: deep engagement tracked **access to the domain**, **integral roles**, and **self-expression** — not posters about belonging. Classroom math often denied those affordances to African American students who showed them in basketball.

Product translation: Story worlds succeed when they recreate *those three affordances around math*, not when they add themed wallpaper around a lonely skill tree.

XXXVI.3 Evidence base (what is solid vs thin)

A. Identity is sociopolitical, not cosmetic

FACT (ethnographic / theory program): Martin (2000) — *Mathematics Success and Failure Among African-American Youth* — success and failure are shaped by sociohistorical context, community forces, school influence, and individual agency; "math identity" is negotiated under racialized meanings of ability, not only item difficulty.

FACT (practice-linked identity): Nasir & Hand (2008) — same students can show sophisticated quantitative reasoning in out-of-school practices while disengaging in school math when access, role, and expression collapse.

FACT (field stance): Nasir (2016) — treating math as outside race/culture is a fallacy; stereotypes and school structures disrupt mathematical identities and opportunity to learn.

FACT (practitioner synthesis): Aguirre, Mayfield-Ingram & Martin (NCTM; *Impact of Identity in K–8 / K–12 Mathematics*) — teaching practices shape whether students see themselves as doers of mathematics; equity-based practice is asset-based and identity-attentive, not deficit remediation theater.

Implication: MindCraft's identity thesis (Parts XXV / FEI) is *aligned* with this literature — **if** story design expands access/role/expression. Story that only changes scenery fails the identity literature's actual mechanism.

B. Belonging interventions are narrow — do not cargo-cult into lore

FACT (RCT): Walton & Cohen (2011, *Science*, doi:10.1126/science.1198364) — brief social-belonging intervention framing adversity as common/transient raised African American college students' GPA over 3 years and roughly halved the measured achievement gap in that sample (N=92; mediated by construal of adversity).

FACT (adjacent): Steele & Aronson stereotype-threat tradition; Spencer, Steele & Quinn (women/math) — ability-diagnostic framing under negative stereotypes can impair performance (effect sizes and replication quality vary; see Part XXIV limits).

Red Team (carry from XXIV): Effects are heterogeneous; school-scale attenuates; a 45-minute belonging essay ≠ a gamified toast ≠ a fantasy prologue. Belonging is not a substitute for teaching.

HYPOTHESIS: Story worlds help belonging when they make early adversity **non-identity-diagnostic** ("misses teach the map") *and* refuse stereotype-cueing copy — not when they lecture "you belong here" in lore text.

C. Whose history? Plural roots are historical FACT; classroom transfer is HYPOTHESIS

FACT (historiography): Joseph (1991/2011, *The Crest of the Peacock*) documents non-European mathematical lineages (Egypt, Mesopotamia, India, China, Islamic world, and more) against Eurocentric "Greeks-only greatness" narratives.

FACT (program): D'Ambrosio's ethnomathematics programme (e.g., 1985–1990 line; ethno + mathema + tics) treats mathematical knowing as culturally situated practice, not a single civilizational monopoly.

FACT (critical pedagogy tradition): Gutstein (2006, *Reading and Writing the World with Mathematics*) — math can be used to read/write sociopolitical conditions; equity is not only "access to algebra" but agency. **Caveat for MindCraft:** Full Freirean classroom projects are not the product's unit of delivery; borrow the *honesty about power*, not a forced activism skin on ACT prep items.

HYPOTHESIS (from XXVIII HIST-EQ / HIST-4): Plural invention stories that encode structure can lift belonging without crushing retention — **if** they are not seductive trivia.

D. Seductive details and "diversity costume" share a kill path

FACT: Harp & Mayer (1998, *Journal of Educational Psychology*, doi:10.1037/0022-0663.90.3.414) — interesting but irrelevant details can damage learning via cognitive interest mechanisms.

FACT (meta-analytic): Sundararajan & Adesope (2020, *Educational Psychology Review*, doi:10.1007/s10648-020-09522-4) — including seductive details tends to hinder learning; moderators exist (format, pacing, subject), but the mean direction is caution, not license.

Kill criterion (FOUNDER BELIEF → testable): If a cultural reference, character trait, or "heritage beat" can be removed without changing the mathematical mental model **and** without changing who can take an integral role, it is decoration. Decoration that only signal brand virtue are equity theater **and** CLT risk.

Survive criterion: Cultural/historical content that (a) encodes the invariant, (b) expands access/role/expression for under-affirmed students, and (c) does not cue stereotypes.

XXXVI.4 Failure modes specific to MindCraft story worlds

Failure mode	What it looks like in product	Likely harm	Epistemic label
Euro-default adventure	Castles, Greco-Roman "genius," white default heroes	Quiet alienation; "not for people like me"	HYPOTHESIS (strong prior from identity lit)
Token palette swap	Same plot; recolored sprites	Cynical read; no identity mechanism	HYPOTHESIS
Heritage trivia dump	Long museum lore before first attempt	Seductive load; delayed micro-win for anxious Maya	FACT risk (CLT / seductive details)
Stereotype-friendly tropes	"Street-smart" vs "book-smart"; gendered magic roles	Cue threat / essentialism	HYPOTHESIS (aligned with threat lit)
Forced activism costume	Every item a justice pamphlet	Parent trust fracture; math time stolen; off-mission	SPECULATION commercially; Gutstein ≠ ACT session
Single-world monoculture	One lore for all students forever	No cultural fit; no choice autonomy (SDT)	HYPOTHESIS

FOUNDER BELIEF (constrained): Plural *structural* histories + multiple entry aesthetics beat one mega-franchise lore — but only with HIST/EQ experiments and content-ops cost discipline.

XXXVI.5 Audit rubric (ship this as content ops, not a vibes review)

Before a story module graduates from draft:

1. **Structure test:** Name the mathematical invariant the narrative makes memorable. If blank → rewrite or cut.
2. **Role test:** In the first 60 seconds, does the student *do* math as an agent, or watch lore? Threatened learners need Safety → micro-attempt before long meaning (XXVIII.7).
3. **Belonging-copy test:** Ban ability-diagnostic group framing ("girls struggle with…", "unlike other kids…"). Prefer adversity-as-common-and-transient framing (Walton & Cohen mechanism — adapted carefully; do not fake an RCT in UI).
4. **History plurality test:** If an invention beat exists, is at least one non-Euro lineage available in the corpus for that concept family over time (not every single card)? Token one-off "Al-Khwarizmi cameo" without structure = fail.
5. **Seductive-details test:** Remove the cultural beat; if learning items and role affordances unchanged, cut or move post-success.
6. **Stereotype scan:** Two reviewers (ideally including someone outside the default writer demographic) flag tropes; unresolved flags block ship.
7. **Claim ladder (Part XXXIV):** Marketing may say "we design stories so more students can take an integral role in the math" (L0/L1 product intent). It may **not** say "our lore closes achievement gaps" without L3 evidence.

XXXVI.6 Commercial implications (copy, positioning, roadmap)

Marketing language that survives

- **Allowed:** "Math isn't a private club with one history. Stories should make the *idea* human — and give every learner a real role in solving it."
- **Allowed:** "We refuse decoration that wastes working memory. Narrative earns its place when it encodes the math or restores a sense that struggle isn't proof you don't belong."
- **Banned:** "Culturally magical world closes the gap."
- **Banned:** "Diverse characters = equitable outcomes."
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity (standing bans).

Competitive stance

Khan's library and Duo's streaks are largely **culture-thin** at the session surface (Part XXXV). MindCraft *chooses* narrative risk. That is only a wedge if the equity audit is visible in product craft — otherwise competitors can dismiss story as gimmick **and** critics can dismiss it as exclusionary. Honest position: "We accept narrative risk; we instrument belonging + learning; we kill modules that fail the rubric."

Growth / parent trust

Parents in threatened segments buy **dignity + results**. Equity theater without `retry_120s` / transfer reads as politics instead of help. Lead with FEI evidence; treat plural history as accuracy and belonging hygiene, not a culture-war banner unless the parent segment asks for it.

Product bets

1. **Content ops checklist** above as merge gate for story modules.
2. **At least two aesthetic/cultural entry frames** for core onboarding arcs (same math, different wrap) — autonomy without fragmenting ontology.
3. **HIST-EQ / EQ-1…3** before scaling "world lore" as brand centerpiece.
4. Tutor witness remains the primary belonging technology for high-threat students (XXIV); story is amplifier, not substitute.

XXXVI.7 Experiments spawned

ID	Question	Design	Primary	Kill condition
EQ-1	Euro-default story vs plural structure-encoding story (same items)	A/B or crossover	belonging/threat items + `retry_120s` + retention@7d	Belonging flat; or learning drop > pre-reg threshold
EQ-2	Token palette-swap vs role/access redesign (same art budget)	3-arm	integral-role self-report + attempt latency	Palette ≥ role redesign on belonging
EQ-3	Pre-attempt heritage lore vs post-success meaning beat	Timing	time-to-first-attempt; transfer	Pre-lore delays attempt with no belonging lift
HIST-EQ	(from XXVIII) Plural vs single-tradition invention	A/B	belonging + learning	Belonging up, transfer down badly
EQ-QUAL	10 Maya interviews: "who is this world for?"	Protocol B + equity probes	theme codes	Students read world as "not for me" at >threshold

Pre-reg discipline (XXXIV): EQ-* identify story→belonging/persistence. They do **not** identify story→GPA gap closure. Do not sell L3 from L1.

XXXVI.8 Confidence table

Claim	Label	Confidence
Math learning is entangled with race/culture/identity	FACT (research consensus in equity math ed)	High

Access / role / expression predict engagement better than posters	FACT (Nasir & Hand mechanisms) + HYPOTHESIS for digital worlds	Medium–High
Default Euro adventure can alienate	HYPOTHESIS	Medium
Token diversity without role redesign fails	HYPOTHESIS	Medium–High
Seductive cultural trivia can hurt learning	FACT (seductive-details tradition + meta-analysis direction)	Medium–High
Plural invention stories lift belonging *and* learning	HYPOTHESIS (HIST-EQ unrun)	Low–Medium
Story can replace tutor witnessing for threat	SPECULATION / likely false	Low (against)
Equity marketing without FEI metrics is brand risk	FOUNDER BELIEF	Medium–High

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XXXVI.9 What this chapter kills

1. **Kill:** “Inclusive skins” as the equity strategy.
2. **Kill:** Shipping long pre-attempt lore as belonging intervention.
3. **Kill:** Marketing claims that story worlds close racial/gender achievement gaps without identified designs.
4. **Wound:** Single-franchise world as forever default — survivable only as temporary scaffold with EQ-1/2 on the roadmap.
5. **Survive (constrained):** Story-first WORLD_VISION — **if** narrative encodes structure, expands integral roles, passes stereotype scan, and stays on the L0–L2 claim ladder until experiments graduate it.

Doctrine until data: Plural, structure-encoding, role-rich worlds under audit > monoculture lore > token palette swaps > heritage trivia dumps.

Part XXXVII — Expectancy-Value (Eccles): Utility, Cost, Attainment → Course Choice

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: Why do students *choose* (or abandon) math pathways — and which MindCraft surfaces move expectancy vs value vs cost?

Owners: Learning Science · Product · Growth / Positioning · Red Team

Commercial job: Replace vague “make math meaningful” copy with a **defensible E×V map**: which product moves which lever, what parents hear, what we refuse to claim.

XXXVII.1 Why this chapter exists

Parts XXV (self-efficacy / narrative identity) and XXIV (anxiety / belonging) explain *can I* and *am I safe*. They do not fully explain *will I keep choosing math* when Instagram, sports, and sleep compete.

FACT (program lineage): Eccles-Parsons et al. (1983) built an achievement-choice model to explain gendered STEM participation: performance and choice are most proximally driven by **expectancies for success** and **subjective task values** (STV), themselves shaped by socializers, culture, and prior experience (*Expectancies, values, and academic behaviors*, in Spence, *Achievement and Achievement Motives*).

FACT (rename / situating): Eccles & Wigfield (2020, *Contemporary Educational Psychology*, doi:10.1016/j.cedpsych.2020.101859) rename the framework **Situated Expectancy-Value Theory (SEVT)** — expectancies and values are *situation-bound hierarchies*, not a single global "motivation score."

FOUNDER BELIEF under audit: MindCraft's identity thesis only converts to commercial retention / course-persistence if product reduces **cost** and raises **attainment + utility** *while* expectancy (FEI evidence) rises — not if we only make prettier explanations.

Claim we refuse as doctrine: "Story worlds automatically raise utility value." That is **SPECULATION**. Utility is a *student-authored* link to goals (Hulleman tradition), not a brand aesthetic.

XXXVII.2 The model in product language (not poster language)

Construct	Plain definition	MindCraft surface (hypothesis)	Failure if ignored
Expectancy	"Can I succeed *on the next relevant tasks*?"	Gap-scan → level gate; soft-wrong → retry; tutor witness; FEI micro-wins	High value + low expectancy → avoidance / crutch (Bastani Base risk, Part XXXIII)
Intrinsic value	Enjoyment / interest in the activity itself	Story-module engagement; challenge–skill (XXXI)	Fun without transfer = Duo-streak trap (XXXV)
Attainment value	Importance to *who I am* / identity	Identity re-storying (XXV); integral role in worlds (XXXVI)	Scoreboard status ≠ "math person"
Utility value	Usefulness for current/future goals	ACT roadmap honesty; parent-facing "why this concept"; student-written relevance prompts	Pure test-prep cynicism; or fake career lore

Cost	Effort, opportunity, emotional/psychological price	Anxiety load (XXIV); session length; shame of failure; time vs other activities	High cost kills choice even when expectancy/value look "ok"

FACT (classic split): Meece, Wigfield & Eccles (1990, *Journal of Educational Psychology*, doi:10.1037/0022-0663.82.1.60) — in young adolescents (N=250), **performance expectancies** predicted subsequent math grades; **value perceptions** predicted **course enrollment intentions**; math anxiety was predicted by ability/expectancy/value perceptions but did not independently drive grades/enrollment in their SEM.

Product translation: Instrument **both** outcome classes. `retry_120s` / transfer ≈ expectancy path. Intent-to-continue / challenge_accept / "take next ACT practice block" ≈ value path. Do not collapse both into a streak KPI (standing ban).

XXXVII.3 Subjective task value — four components, four commercial mistakes

A. Intrinsic value ≠ engagement theater

FACT: Intrinsic value = enjoyment of the task (Eccles-Parsons et al., 1983; Wigfield & Eccles, 2000, *CEP*, doi:10.1006/ceps.1999.1015).

HYPOTHESIS: Story can raise intrinsic value *if* it encodes structure and preserves integral roles (XXVIII / XXXVI). Seductive details that raise "interest" but crush learning (Harp & Mayer; Sundararajan & Adesope — carry) are **fake intrinsic** for MindCraft's purpose.

Kill: Marketing that equates "delightful world" with durable math choice without FEI + value instruments.

B. Attainment value is the identity wedge

FACT (theory refinement): Attainment value moved from "importance of doing well" toward linkage with personal/social identities (Eccles, 2005/2009 line; summarized in Eccles & Wigfield, 2024, *Educational Psychology Review*, doi:10.1007/s10648-024-09888-9).

FOUNDER BELIEF: "I am becoming someone who can think mathematically" is attainment value. Streaks are not.

Commercial implication: Parent and teen copy should sound like *identity competence*, not *gamified attendance*. Allowed: "Evidence you can trust yourself on the next hard problem." Banned: "Don't break your streak — that means you're a math person."

C. Utility value — the only mass-market EVT intervention with strong field RCTs

FACT (RCT): Hulleman & Harackiewicz (2009, *Science*, doi:10.1126/science.1177067) — ninth-grade science students randomly assigned to write how material connects to their lives (or a known other) vs summary control; **relevance writing raised interest and grades especially for students with low success expectations**.

FACT (lab + classroom): Hulleman, Godes, Hendricks & Harackiewicz (2010, *JEP*, doi:10.1037/a0019506) — utility writing raised perceived utility and interest, again strongest for lower performers/expectancies; utility perceptions mediated interest effects.

HYPOTHESIS for MindCraft: A 2–4 minute **student-authored** utility prompt *after* a micro-win (not before first attempt under high threat) beats brand-authored “math is useful for STEM careers” copy — especially for Maya with low expectancy.

Red Team wound: Effects are domain/context sensitive; science classroom ≠ ACT app session; dosage and timing matter (SEVT situating). Do not paste “Hulleman proved story worlds work.”

Kill: Shipping utility as *staff-written* career propaganda dumped pre-attempt (CLT + seductive + autonomy threat).

D. Cost — historically neglected, commercially decisive

FACT: Cost was theorized early (effort, opportunity, psychological cost of failure) but empirically under-measured for decades (Wigfield & Cambria reviews; Eccles & Wigfield, 2020 SEVT paper stresses cost elaboration).

FACT (measurement): Flake, Barron, Hulleman, McCoach & Welsh (2015, *CEP*, doi:10.1016/j.cedpsych.2015.03.002) — validated multi-dimensional cost (task effort, outside effort, loss of valued alternatives, emotional cost); cost is separable from expectancy/value and relates to student outcomes.

HYPOTHESIS: For anxious Maya, **emotional cost** and **effort cost** dominate early sessions; for busy juniors, **opportunity cost** (vs sleep/jobs/sports) dominates weekly return. Product that only raises “fun” while leaving shame-of-wrongness intact fails SEVT choice.

Product translation: Soft-wrong / hide-correctness diagnostic (C4) / tutor tone (Part XXX) are **cost interventions**, not just pedagogy niceties. Session design that demands 40 minutes of lore before first attempt raises opportunity + effort cost without expectancy gain (tie to EQ-3).

XXXVII.4 Hierarchies and situations (why one “motivation score” is a product bug)

FACT: SEVT emphasizes *hierarchies* of expectancies and values across competing activities and situations (Eccles & Wigfield, 2020; Wigfield & Eccles developmental reviews). Choosing math means math beats alternatives *this week*, not that motivation is globally high.

HYPOTHESIS: Duo wins short-horizon intrinsic/utility-for-streak; Khan wins low-cost access library; ChatGPT Base raises short-term utility (“get the answer”) while crushing expectancy-for-unaided performance (Part XXXIII/XXXV). MindCraft’s wedge is **attainment + expectancy evidence with managed cost** — not out-streaking Duo or out-librarying Khan.

SPECULATION (commercial): Parents purchase when they believe we move the *child’s hierarchy* toward math without catastrophic emotional cost — “they’ll open it again without a fight” is a cost/expectancy story sold as calm, not as dopamine.

XXXVII.5 Socializers — parents and tutors as EVT machinery

FACT (model structure): Socializers’ beliefs and behaviors shape children’s interpretations, affective memories, and thus expectancies/values (Eccles-Parsons et al., 1983; SEVT socialization arm).

Carry: Part XXXII (parent anxiety transmission) — parent math anxiety can raise the child's **emotional cost** and lower expectancy via homework help patterns (Maloney/Beilock line).

HYPOTHESIS: Parent dashboard copy that celebrates *process evidence* (retry, transfer) raises parent-transmitted attainment/utility messages; copy that only shows streak/percentile can raise performance pressure → cost.

Product implication: Parent WTP experiments (Part XXVII queue) should manipulate EVT framing: "identity/competence evidence" vs "minutes practiced" vs "AI explanations unlimited." Pre-register which frame shifts purchase intent without raising reported homework conflict.

XXXVII.6 Mapping MindCraft features → EVT levers (audit table)

Feature / bet	Primary lever	Secondary	Claim ladder max without new data
Gap-scan + level gating	Expectancy calibration	Cost ↓ (fewer impossible first items)	L1 product intent
Soft-wrong → retry_120s	Expectancy evidence	Emotional cost ↓	L1–L2 with FEI instrumentation
Story worlds (structure-encoding)	Intrinsic + attainment	Utility if student links	L0–L1; EQ/HIST required
Student utility writing (post-win)	Utility	Intrinsic (situational interest)	L1; EVT-1 experiment
ACT exam-mode roadmap	Utility (instrumental)	Expectancy if honest sequencing	L1 honesty; not "score guarantee"
Tutor human sessions	Expectancy + cost	Attainment (witnessed identity)	L1; do not claim 2-sigma
Solver without guards	Short-term utility (answer)	Expectancy ↓ (crutch)	Kill as hero feature
Streaks / leaderboards	Pseudo-attainment / extrinsic	Cost ↑ under threat	Standing ban as North Star

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XXXVII.7 Commercial implications (copy, positioning, growth)

Marketing language that survives

- **Allowed:** "Expectancy without value is empty drills. Value without expectancy is inspirational avoidance. We design for both — and we treat the emotional price of being wrong as a first-class design constraint."
- **Allowed:** "Utility is something the student *writes*, not a slogan we paste."
- **Allowed (parent):** "Fewer fights at the kitchen table" = cost reduction claim — keep qualitative until measured.
- **Banned:** "Our world makes math useful" (untested utility).

- **Banned:** “Fun = they’ll choose Precalculus.”
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity.

Positioning vs competitors

Competitor	EVT profile (audit, not insult)	MindCraft contrast
Khan	Low cost access; mixed expectancy; thin attainment	Session FEI + identity evidence
Duo	Intrinsic/habit; cost managed via brevity; attainment via streak fiction	Transfer + real expectancy
Brilliant	Intrinsic puzzle; utility unclear for ACT Maya	Exam-honest utility + anxiety cost
ChatGPT Base	Max short-term utility; expectancy poison	Guarded Solver + unaided transfer

Growth / North Star hygiene

Optimize a **pair**: expectancy evidence (`retry_120s`, transfer_pass) **and** a value/choice proxy (`challenge_accept`, weekly voluntary return, intent-to-take-next-block). If value proxy rises while transfer falls, you are Duo-pilling the product — kill the feature.

XXXVII.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
EVT-1	Post-win student utility writing vs staff utility blurb vs none	3-arm A/B after micro-win	situational interest + 7d return + transfer@7d	Utility arms lose transfer or raise time-to-next-attempt badly
EVT-2	ACT-utility framing vs identity-attainment framing in session CTA	2-arm	challenge_accept; parent/teen preference	Attainment wins preference but tanks attempt rate in anxious segment (segmented kill)
EVT-3	Cost questionnaire (Flake-style short form) pre/post first week	Observational → then intervene on top cost	which cost dimension predicts churn	If emotional cost predicts churn but product only ships “more fun,” roadmap invalid

EVT-4	Parent copy: competence-evidence vs minutes vs AI-unlimited	WTP / conflict items	purchase intent; reported homework conflict	Competence frame raises conflict without WTP lift
EVT-QUAL	10 Maya interviews: “why would you take another math course?”	Appendix B + EVT probes	coded expectancy /value/cost	Students cannot articulate any non-grade reason → attainment/utility product gap

Pre-reg (XXXIV): EVT-* identify levers on **in-app choice and short-horizon learning**. They do **not** identify long-term course enrollment or career STEM entry without follow-up designs. Do not sell “we increase AP Calculus enrollment” from EVT-1.

XXXVII.9 Confidence table

Claim	Label	Confidence
Choice ≈ f(expectancy, STV) in Eccles tradition	FACT (large corpus)	High
Expectancy→performance, value→enrollment intentions (adolescent math)	FACT (Meece et al., 1990 SEM)	Medium–High (sample/era limits)
STV has intrinsic / attainment / utility / cost structure	FACT (theory + measurement work)	High
Utility-value writing can lift interest/grades for low-expectancy students	FACT (Hulleman & Harackiewicz, 2009)	Medium–High (context limits)
Cost is multi-dimensional and outcome-relevant	FACT (Flake et al., 2015)	Medium–High
Story worlds raise utility by themselves	SPECULATION	Low
MindCraft should instrument E and V separately	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Soft-wrong + tutor tone primarily cut emotional cost	HYPOTHESIS	Medium
Streaks raise attainment value	SPECULATION / likely false for identity	Low (against)

XXXVII.10 What this chapter kills

1. **Kill:** Single “motivation” KPI as if E and V were the same dial.
2. **Kill:** Staff-authored “math is useful” dumps as a substitute for student-authored utility.
3. **Kill:** Treating story delight as proof of course-choice impact.
4. **Wound:** Pure ACT-utility positioning — survives for juniors near test date; wounds long-horizon identity brand if it becomes the only message.
5. **Survive (constrained):** Identity-transformation thesis — **if** product explicitly raises expectancy *and* attainment/utility *and* designs against cost, with EVT-1…4 on the roadmap.

Doctrine until data: Design and market the **E×V×Cost triad**. Expectancy evidence without value is a drill app. Value talk without expectancy is a poster. Ignoring cost is how anxious students churn while NPS looks fine among the already-confident.

Part XXXVIII — Goal Orientation: Mastery vs Performance Framing

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When MindCraft frames goals as *get better at this* vs *look smart / beat others / don’t look dumb*, which motivational pattern do we induce — and what must product + copy refuse?

Owners: Learning Science · Product · Growth / Positioning · Red Team

Commercial job: Replace empty “mastery learning” branding with a **goal-structure audit**: which surfaces cue mastery-approach, which cue appearance/performance, which cue avoidance — and which we kill even if they juice short-term engagement.

XXXVIII.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution / helplessness), XXXI (challenge–skill), and XXXVII (expectancy-value) answer *can I*, *why did I fail*, *is difficulty calibrated*, and *will I choose math*. They do not fully answer: **what success definition is the product teaching?**

A product that says “identity transformation” while UI, CTAs, and parent dashboards scream *percentile / streak / don’t mess up* is running a **performance-avoidance climate** with mastery marketing on top.

FOUNDER BELIEF under audit: MindCraft’s North Star metrics (`retry_120s`, `challenge_accept`, `transfer_pass`) only stay coherent if the **goal structure** of sessions is mastery-approach dominant. If recognition and evaluation structures are performance/appearance, those metrics will fight the UI.

Claim we refuse as doctrine: “Any competitive element raises motivation.” That collapses normative, appearance, and avoidance goals — meta-analyses show they are not the same animal.

XXXVIII.2 The constructs (product language, not poster language)

Goal	Plain definition	Typical student cue	MindCraft failure mode if dominant
Mastery-approach	Improve competence vs self/task standard	"Understand this; try a harder variant"	None if paired with transfer; risk = slow coverage if time unmanaged
Mastery-avoidance	Avoid failing to learn / avoid imperfect understanding	"Don't leave gaps; don't misunderstand"	Perfectionism, over-checking, refusal to attempt under uncertainty
Performance-approach (normative)	Outperform peers	Rank, leaderboard, "top of class"	Can raise grades for some; often raises threat for Maya; identity ≠ rank
Performance-approach (appearance)	Demonstrate talent / look smart	Public correctness, "prove you're smart"	Meta-analytically worse than normative; shame + shallow strategy
Performance-avoidance	Avoid looking incompetent	Hide wrongness; skip hard items; Solver crutch	Helpless pattern; FEI collapse; Part XXX / XXXIII risk

FACT (classic dichotomy + classroom structures): Ames (1992, *Journal of Educational Psychology*, doi:10.1037/0022-0663.84.3.261) — classroom **task, evaluation/recognition, and authority** structures make mastery vs performance goals salient and elicit qualitatively different motivational patterns; interventions must treat structures as interdependent, not one "growth mindset poster."

FACT (student perceptions → strategies): Ames & Archer (1988, *JEP*, doi:10.1037/0022-0663.80.3.260) — among 176 secondary students, perceived **mastery** classroom goals related to more effective strategies, preference for challenge, positive attitude, and effort→success beliefs; perceived **performance** salience related to ability focus, negative ability evaluation, and attributing failure to low ability.

FACT (implicit theories → goals → patterns): Dweck & Leggett (1988, *Psychological Review*, doi:10.1037/0033-295X.95.2.256) — entity vs incremental theories orient students toward performance vs learning goals, which set up helpless vs mastery-oriented response patterns after difficulty.

Product translation: Soft-wrong + retry is not only an expectancy/cost intervention (XXXVII) — it is a **goal-structure** intervention. Public scoreboards and "don't break the streak" are evaluation/recognition structures that cue performance/appearance.

XXXVIII.3 Approach–avoidance split (why "performance goals" is too coarse)

FACT (trichotomous model): Elliot & Church (1997, *JPSP*, doi:10.1037/0022-3514.72.1.218) — mastery, performance-approach, and performance-avoidance are separable; mastery facilitated intrinsic motivation; performance-approach enhanced graded performance; **performance-avoidance harmed both** intrinsic motivation and grades (college classroom path model).

FACT (2×2 framework): Elliot & McGregor (2001, *JPSP*, doi:10.1037/0022-3514.80.3.501) — adds **mastery-avoidance**; four goals show distinct antecedents/outcomes; mastery-avoidance sits between mastery-approach (more adaptive) and performance-avoidance (more maladaptive) on several indicators.

HYPOTHESIS for MindCraft: Anxious Maya often defaults to **performance-avoidance** ("don't look dumb") or **mastery-avoidance** ("I must understand perfectly before I try"). Product that only bans leaderboards but still frames every miss as identity threat will not fix the pattern.

Kill: Marketing that equates "mastery mode" with "zero mistakes allowed on the path." That is mastery-avoidance cosplay.

XXXVIII.4 Performance-approach is not one knob — measurement kills bad slogans

FACT (meta-analysis on measures): Hulleman, Schrager, Bodmann & Harackiewicz (2010, *Psychological Bulletin*, doi:10.1037/a0018947) — across 243 correlational studies (N≈91k), "performance-approach" scales that emphasize **normative outperformance** correlate *positively* with performance outcomes (r≈.14), while scales emphasizing **appearance / looking talented** correlate *negatively* (r≈−.14). Same label, different constructs.

FACT (extended outcomes meta): Senko & Dawson (2017, *Journal of Educational Psychology*, doi:10.1037/edu0000160) — the normative vs appearance split generalizes beyond grades to competence perceptions and self-regulation; age confounds with measure type were addressed.

FACT (theory status review): Senko, Hulleman & Harackiewicz (2011, *Educational Psychologist*, doi:10.1080/00461520.2011.538646) — achievement goal theory at a crossroads: multiple controversies (including when performance-approach helps) require precise operationalization before practice claims.

Commercial implication: If MindCraft ever ships social comparison, it must pre-register whether the cue is **normative skill evidence** (dangerous for threatened learners; still not identity) or **appearance theater** (ban). Default = **neither**. Do not cite "performance goals sometimes help grades" as license for public shame UX.

SPECULATION (positioning): "Beat your friends" copy will read as Duo/Brilliant hybrid; "prove you're smart" reads as SAT-cram Instagram. Neither is the identity-transformation wedge.

XXXVIII.5 Classroom / product goal *structures* (TARGET), not vibes

FACT (schoolwide / TARGET lineage): Maehr & Midgley (1991, *Educational Psychologist*, doi:10.1080/00461520.1991.9653140) — motivation change requires altering school/classroom goal structures, not only exhorting students; dimensions map to task design, authority, recognition, grouping, evaluation, time (TARGET family; Ames, 1992).

FACT (review): Meece, Anderman & Anderman (2006, *Annual Review of Psychology*, doi:10.1146/annurev.psych.56.091103.070258) — classroom goal structures predict personal goals and achievement-related patterns; mastery structures are the design target for adaptive motivation.

FACT (longitudinal TARGET → mastery goals): Lüftenegger et al. (2014, *Educational Psychology*, doi:10.1080/01443410.2013.814189) — TARGET as a latent structure positively predicted students' mastery goal orientations (N=1680 secondary; longitudinal).

MindCraft TARGET audit (hypothesis map):

TARGET dim	Mastery-leaning product move	Performance/appearance trap
Task	Varied challenge matched to skill (XXXI); format variety with transfer	Same-item grind that feels like "prove speed"
Authority	Student choice of next concept *after* honest diagnosis	Forced path that only optimizes for looking ahead of peers
Recognition	Process evidence: retry, transfer, bridge repair	Streak flames, public %ile, "genius" badges
Grouping	Tutor as witness; optional co-op without rank	Ranked classrooms / public wrongness
Evaluation	Soft-wrong diagnostic (C4); hide correctness when graph-corrupting	Instant public right/wrong as identity verdict
Time	Short sessions that still allow struggle → retry	Timed pressure that cues avoidance

FOUNDER BELIEF: PawHub "Practice the weak spot" is mastery-approach **if** copy is "repair this gap" and **not** "you're behind everyone." Gap language can flip to performance-avoidance under shame framing.

XXXVIII.6 Interaction with expectancy, anxiety, and AI crutches

FACT (goal pursuit × expectancy): Senko & Hulleman (2013, *JEP*, doi:10.1037/a0031136) — for mastery-approach *and* performance-approach, the harder a goal appears to attain, the less students pursue it (with costs to interest/achievement); mastery-approach goals were generally judged easier to attain than performance-approach; situational cues (material complexity) shift mastery-approach expectancies.

Carry: Part XXIV (anxiety) + XXXIII (AI overtrust) — under high threat, performance-avoidance + Solver-without-guards is the cheap escape from a hard-looking mastery or performance goal.

HYPOTHESIS: Level gating after gap-scan (easy→L3, kinda→L2, hard→L1) is a **goal-attainment-expectancy** design: it makes mastery-approach goals feel attainable. Removing gates to "challenge everyone" without expectancy supports will increase avoidance and Solver dependence.

Product implication: CTAs should make the next mastery goal *feel attainable* (Senko & Hulleman) without lying about transfer. Honest sequencing ≠ performance theater.

XXXVIII.7 Competitive product audit (goal climate lens)

Competitor	Dominant goal climate (audit)	What MindCraft must not copy
Duolingo	Habit + streak recognition; appearance of consistency	Streak-as-identity / streak-as-North-Star (standing ban)
Khan	Mastery language + energy points / badges mixed	"Mastery" label without transfer_pass
Brilliant	Puzzle competence; mild normative "smart people do this"	Appearance-smart branding without anxiety cost design
ChatGPT Base	Outcome utility ("get it right now") → avoidance of struggle	Solver as hero; unaided transfer optional

HYPOTHESIS: MindCraft's wedge is **mastery-approach structures with expectancy-managed tasks and anti-appearance evaluation** — not out-ranking Duo and not out-explaining Khan.

XXXVIII.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** "We design for improvement goals — understand, repair, transfer — not for looking smart in public."
- **Allowed:** "Wrong answers are evidence for the next attempt, not a verdict on who you are." (ties XXX / soft-wrong)
- **Allowed (parent):** "We measure whether they take the next hard problem, not whether they kept a streak."
- **Banned:** "Become a top 1% math student" as hero claim (appearance/normative mashup; undefended).
- **Banned:** "Never break your streak — that's how mastery works."
- **Banned:** Empty growth-mindset posters without TARGET structure changes (Dweck mis-applied as sticker).
- **Banned:** 2-sigma / tutoring-is-free absolutes.

Feature claim ladder (pre-reg hygiene, Part XXXIV)

Feature	Goal climate claim max without GO-* data
Soft-wrong → retry_120s	L1 intent: mastery-approach evaluation structure
`challenge_accept` CTA	L1: approach (need to verify mastery vs performance wording)
`transfer_pass`	L1–L2 anti-fake-mastery (competence standard ≠ appearance)
Leaderboards / public ranks	L0 default kill for Maya segment
Streak North Star	Kill (standing ban)

“Mastery graph” marketing	L1 only if transfer instrumented; else label is cosmetics

Growth / North Star hygiene

`challenge_accept` is **not automatically mastery-approach**. Instrument **why** they accept: repair gap (mastery) vs beat peer (normative) vs avoid looking behind (avoidance). If acceptance rises because of shame CTAs, kill the copy even if the metric looks green.

XXXVIII.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
GO-1	Mastery-approach CTA (“repair this gap”) vs normative (“catch up to peers”) vs appearance (“show you’re smart”)	3-arm on Practice launch	challenge_accept; retry_120s; 7d return; anxiety item	Normative/appearance win short accept but lose retry/transfer or raise anxiety
GO-2	Recognition: process badge (retry/transfer) vs streak badge vs none	3-arm	transfer@7d; identity item; self-reported goal	Streak badge wins engagement, loses transfer / raises avoidance
GO-3	Evaluation: soft-wrong first vs instant public correctness	2-arm on early session	next-attempt latency; avoidance skips; Solver opens	Instant correctness raises “finish” rate but kills retry_120s
GO-4	Goal-climate pulse (short PALS-like mastery/performance classroom items adapted to app) weekly	Observational → then redesign recognition	which climate predicts churn / Solver dependence	If product self-reports “mastery brand” but climate score is performance → roadmap invalid
GO-QUAL	10 Maya interviews: “What would ‘doing well’ here mean?”	Appendix B + goal probes	coded mastery / normative / appearance / avoidance	Majority answer only “not look dumb” → evaluation/recognition redesign mandatory

Pre-reg (XXXIV): GO-* identify **in-app goal climate → short-horizon FEI behaviors**. They do **not** identify long-term GPA or STEM major from CTA wording alone. Do not sell "we create mastery-oriented students for life" from GO-1.

XXXVIII.10 Confidence table

Claim	Label	Confidence
Mastery vs performance goal structures shape strategies/attitudes	FACT (Ames & Archer, 1988; Ames, 1992)	High
Implicit theories orient goals → helpless vs mastery patterns	FACT (Dweck & Leggett, 1988)	High
Performance-avoidance is broadly maladaptive for motivation/grades	FACT (Elliot & Church, 1997 lineage)	High
2×2 adds mastery-avoidance as distinct	FACT (Elliot & McGregor, 2001)	High
"Performance-approach" effects depend on normative vs appearance ops	FACT (Hulleman et al., 2010; Senko & Dawson, 2017)	High
TARGET-like structures can raise mastery orientations	FACT (Maehr & Midgley; Lüftenegger et al., 2014)	Medium–High (school ≠ app)
Soft-wrong + process recognition = mastery evaluation structure	HYPOTHESIS	Medium–High
Any leaderboard helps MindCraft's Maya segment	SPECULATION / likely false	Low (against)
Streaks instantiate mastery goals	SPECULATION / false for identity	Low (against)
MindCraft should ban appearance-goal UX by default	FOUNDER BELIEF / HYPOTHESIS	Medium–High

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XXXVIII.11 What this chapter kills

1. **Kill:** Treating "mastery" as a brand adjective without evaluation/recognition structures that cue improvement standards.
2. **Kill:** Streak / public rank / "look smart" as North Star or hero copy (appearance + standing streak ban).
3. **Kill:** Collapsing all performance goals into "competition is motivating."
4. **Wound:** Normative comparison as a growth lever — survives only as a pre-registered, segmented experiment (GO-1), never as default Maya UX.

5. **Survive (constrained):** Identity-transformation thesis — **if** product TARGET structures are mastery-approach dominant, avoidance/appearance cues are designed out, and GO-1…4/QUAL sit on the roadmap.

Doctrine until data: Market and build **mastery-approach goal structures** (task challenge + process recognition + non-shaming evaluation + attainable next goals). Performance-avoidance and appearance goals are product bugs dressed as engagement. Normative comparison is not the default growth loop.

Part XXXIX — Interleaving vs Blocking: When to Mix Problem Types

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When should MindCraft mix problem types (interleave) vs group same-type drills (block) — and what must product + copy refuse to claim?

Owners: Learning Science · Product (Practice sequencing) · Growth / Positioning · Red Team

Commercial job: Turn "we shuffle questions" from a cute engineering detail into a **defensible practice-design claim**: strategy *selection*, not just strategy *execution* — without selling interleaving as always-on magic or as engagement theater.

XXXIX.1 Why this chapter exists

Part XXIX (spacing / retrieval) already flags interleaving as a desirable difficulty and hypothesizes **Block** → **fade** → **interleave** → **spaced retrieval**. Part XXVI (CLT) warns that early overload kills learning. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance if framed as shame.

This chapter densifies the **mix vs block** decision for Practice sessions, ingredient cards, and ACT-style mixed review — because textbooks and most edtech default to **blocked practice** (a dozen slope problems after the slope lesson), and students experience that fluency as "I get it" while exams demand **choose the strategy from the problem itself**.

FOUNDER BELIEF under audit: Identity transformation requires competence evidence that survives *transfer*. Blocked fluency is often **illusory competence** — high session accuracy, weak delayed discrimination.

Claim we refuse as doctrine: "Always interleave everything from the first item." That ignores schema formation, CLT, and similarity moderators — and can spike threat for anxious Maya before `retry_120s` is healthy (XXIX.4).

XXXIX.2 Constructs (product language)

Schedule	Plain definition	What the student practices	Failure mode if misused

Blocked	Consecutive items share the same strategy / concept	*Execute* a known procedure	Fluency illusion; weak strategy choice on exams
Interleaved	Consecutive items require different strategies (no safe "same as last")	*Select* then execute	Early overload; frustration; avoidance if unscaffolded
Hybrid / faded	Short block for schema → rising mix	Execute → select	Best candidate for MindCraft default

Operational definition for MindCraft (HYPOTHESIS): Interleaving = consecutive Practice items do **not** share the same primary strategy class (concept + dominant ingredient / archetype family). Mere format shuffle (word → diagram) with the same strategy is **not** full interleaving — it is representation variety (useful; different claim).

XXXIX.3 Mechanism: why mixture changes the pedagogical demand

FACT (math laboratory / classroom line): Rohrer & Taylor (2007, *Instructional Science*, doi:10.1007/s11251-007-9015-8) — Experiment 2: after learning multiple problem types, **mixed** vs **blocked** practice; one-week delayed test vastly favored mixed practice. Experiment 1 separately showed spacing > massing for a single type. Shuffled textbooks alter **both** spacing and mixing; they are related but not identical levers.

FACT (strategy choice vs execution): Rohrer, Dedrick & Stershic (2015, *Journal of Educational Psychology*, doi:10.1037/edu0000001) — 126 seventh-graders, same problems over ~3 months, interleaved vs blocked arrangement; after review, unannounced tests favored interleaving at 1 day (d = 0.42) and 30 days (d = 0.79). Core mechanism claim: blocked practice lets students know the strategy *before reading the problem*; interleaved practice forces strategy choice from problem features — the skill exams actually require.

FACT (not only look-alike problems): Rohrer, Dedrick & Burgess (2014, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-014-0588-3) — interleaved benefit is **not** limited to superficially similar problem kinds; reordering changes pedagogical demand (choose vs execute) even when types do not look alike.

FACT (classroom-scale RCT): Rohrer, Dedrick, Hartwig & Cheung (2020, *JEP*, doi:10.1037/edu0000367; Online First 2019) — preregistered cluster RCT, 54 seventh-grade classes, mostly interleaved vs mostly blocked assignments over months; after interleaved review for both, one-month unannounced test: 61% vs 38%, d = 0.83. Teachers implemented without training; anonymous pre-results survey supported feasibility. Caveats remain (materials, population), but this is the strongest classroom efficacy signal in the math interleaving line.

Product translation: MindCraft's `transfer_pass` and mixed ACT practice are **closer to the interleaved demand** than a single-concept blocked grind. A session that is 12 items of `linear_equations` after "Learn linear equations" is textbook-default blocking — fine as *early schema practice*, dangerous as the *only* schedule if we market exam readiness.

XXXIX.4 Meta-analytic moderators (do not overclaim "interleaving always wins")

FACT (inductive learning meta-analysis): Brunmair & Richter (2019, *Psychological Bulletin*, doi:10.1037/bul0000209) — 59 studies / 238 effect sizes / 158 samples; overall interleaving vs blocking Hedges' g = 0.42. **Material matters:** paintings/visual high ($g \approx 0.67$); **mathematical tasks small but positive ($g \approx 0.34$)**; words favor **blocking** ($g \approx -0.39$); expository texts ambiguous. Moderators: stronger interleaving when **between-category similarity is high**, **within-category similarity is low**, and material is more complex — consistent with attentional bias / sequential attention accounts (Carvalho & Goldstone lineage).

FACT (attentional bias framework): Carvalho & Goldstone (2015, *Frontiers in Psychology*, doi:10.3389/fpsyg.2015.00505) — study sequence biases what is encoded: interleaving emphasizes **between-category differences**; blocking emphasizes **within-category similarities**. Prediction: when categories are hard to tell apart, interleave; when the job is to notice what members of one category share, block can help.

HYPOTHESIS for MindCraft: Interleave **confusable** strategy families (e.g., independent vs dependent probability; similar ACT archetypes that share surface cues but need different ingredients). Block when introducing a *new* ingredient representation until the student can execute with worked-example fade (XXVI) — then mix against near-miss foils.

Kill: Marketing that says "our AI interleaves everything like elite tutors" without specifying **confusable sets** and **when blocking is intentional**.

XXXIX.5 Fluency illusion, affect, and why students prefer blocking

FACT / TRADITION (desirable difficulties): Conditions that slow acquisition (spacing, interleaving, generation) often improve later retention/transfer (Bjork lineage; see XXIX.4). Interleaving feels harder *during* practice; blocked practice feels fluent.

HYPOTHESIS (Koriat/Bjork-style monitoring, applied): Students and parents misread blocked fluency as mastery. Parent dashboards that celebrate "12/12 on slope today" without a mixed delayed check sell **performance theater** (XXXVIII) dressed as learning.

Carry from XXIV / XXXVII / XXXVIII: Raising interleaving dose raises short-term error rate → for anxious Maya, that can cue performance-avoidance unless (a) soft-wrong / mastery evaluation, (b) expectancy-managed difficulty, (c) explicit frame: "this mix trains choosing, not proving you're dumb."

Product implication: Copy for interleaved sessions: **"Practice choosing the right move"** — not "hard mode for elites" and not streak pressure.

XXXIX.6 Sequencing doctrine (ties XXVI + XXIX)

FOUNDER BELIEF / HYPOTHESIS (sequenced schedule):

1. **Acquire (block-lite):** short same-strategy set + worked examples / cards until execution is stable enough that errors are mostly slips, not total confusion.
2. **Discriminate (interleave):** mix with **near-miss** types that share surface features (bridge gaps, format gaps, ACT archetype neighbors).

3. **Retain (space):** redistribute the same mix across days (XXIX) — interleaving *guarantees* some spacing of each type, but lag still needs explicit scheduling.
4. **Transfer check:** unannounced mixed set → `transfer_pass`; failure routes to diagnosis (forgot vs never-had; wrong strategy class vs execution bug).

Contradiction to hold: Some math/category studies report null or blocking advantages under specific similarity structures (noted in Brunmair & Richter's discussion of mixed math findings; e.g., certain representation-interleaving designs). MindCraft must treat **interleaving axis** (strategy class) as primary for exam readiness, and treat **representation mix** as a separate experiment track (format axis / C2–C3).

XXXIX.7 Competitive / product audit (schedule lens)

Surface	Typical schedule	What MindCraft should do
Textbook / workbook	Heavy blocking after lesson	Do not clone as "mission complete"
Khan-like mastery	Often skill-blocked until "mastered"	Mastery label without mixed delayed check = fluency risk
Duo-like	Habit + short items; mix varies by skill tree	Habit ≠ discrimination training
Brilliant	Puzzle variety; not always systematic near-miss mix	Variety ≠ controlled interleaving
ChatGPT Solver	Single-problem depth	Zero schedule pedagogy; cannot substitute mix training
MindCraft Practice today	Often concept-scoped sessions + bank shuffle	Shuffle within one concept ≠ full strategy interleaving; need **cross-type mix modes**

HYPOTHESIS: PawHub "Practice the weak spot" correctly targets a gap, but a pure blocked weak-spot grind can raise local accuracy while leaving **strategy selection** untrained. Pair weak-spot blocks with a **discrimination coda**: 3–5 interleaved near-misses including the weak type.

XXXIX.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** "Exams don't announce the chapter. We train choosing the strategy — not only repeating the last one."
- **Allowed:** "Blocked practice feels easy; mixed practice builds the skill tests actually measure."
- **Allowed (parent):** "We check whether they can pick the right approach on a mixed set days later — not just a perfect streak on one topic."
- **Banned:** "Interleaving always doubles scores" / absolute effect-size marketing (Rohrer classroom $d \approx 0.83$ is **one** RCT context; Brunmair math $g \approx 0.34$ is the broader inductive-math signal — do not collapse).

- **Banned:** “Always mix from day one.”
- **Banned:** Selling shuffle/random as science without near-miss design.
- **Banned:** 2-sigma / tutoring-is-free / streak-as-mastery.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without IL-* data
Within-concept item shuffle	L0–L1 “variety”; **not** full interleaving claim
Cross-strategy interleaved coda after weak-spot block	L1 intent: discrimination training
Delayed mixed `transfer_pass`	L1–L2 anti-fluency-illusion
“Interleaving improves ACT” hero claim	L0 until IL-1/IL-2 pre-registered

Growth / North Star hygiene

Session accuracy under **blocked** schedules will look better than under interleaved ones. **Do not** optimize North Star on blocked accuracy. Prefer: mixed delayed accuracy, strategy-selection error rate (wrong procedure class), `transfer_pass`, and `retry_120s` after mix-induced misses.

XXXIX.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
IL-1	Blocked-only weak-spot session vs block + interleaved near-miss coda	2-arm; same total items	Delayed (48h) mixed accuracy; strategy-class error rate	Coda hurts delayed mix **and** raises avoidance without transfer gain
IL-2	Early interleave (from item 1) vs fade-in interleave after short block	2-arm on new concept	Acquisition errors; `retry_120s`; 7d mixed test	Early-all-mix wins delayed score but kills return/anxiety for Maya segment
IL-3	Near-miss interleave (confusable archetypes) vs far mix (unrelated easy types)	2-arm	Discrimination errors on near-miss test	Far mix equals near-miss on discrimination → “any shuffle” doctrine dies

IL-4	Parent/student judgment: blocked fluency vs mixed score — which do they trust?	Survey + optional choice	Preference for blocked; willingness-to-pay framing	If parents only buy blocked dashboards, copy/education mandatory before UX ships mix-heavy
IL-QUAL	10 Maya interviews: “When did practice feel like the real test?”	Appendix B	coded choose-vs-execute moments	Nobody describes strategy choice → product isn’t creating the demand state

Pre-reg (XXXIV): IL-* identify **schedule → discrimination / delayed mix**. They do **not** identify long-term STEM identity from one coda. Do not sell “interleaving creates math people.”

Supersedes / densifies: XXIX INT-1 (interleave after fade vs blocked-only) → prefer IL-1/IL-2 as the operational pair.

XXXIX.10 Confidence table

Claim	Label	Confidence
Mixed/interleaved math practice often beats blocked on delayed tests	FACT (Rohrer & Taylor, 2007; Rohrer et al., 2015, 2020)	High (math classroom/lab line)
Mechanism includes forced strategy selection (not only spacing)	FACT / strong interpretation in Rohrer line	High
Benefit can appear for non-look-alike problem kinds	FACT (Rohrer et al., 2014)	High
Overall inductive interleaving effect moderated by material & similarity	FACT (Brunmair & Richter, 2019)	High
Math subset effect smaller than visual category learning	FACT (g≈0.34 vs higher for paintings)	High
Blocking can help when within-category similarity learning is the job	FACT (attentional bias framework) / HYPOTHESIS for early schema	Medium–High
Always-interleave-from-item-1 is optimal for anxious Maya	SPECULATION / likely false	Low (against)
Within-concept shuffle ≡ interleaving science claim	SPECULATION / false	Low (against)

Default MindCraft path = short block → near-miss interleave → spaced mix check	FOUNDER BELIEF / HYPOTHESIS	Medium–High

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XXXIX.11 What this chapter kills

1. **Kill:** Treating blocked session accuracy as proof of exam readiness or identity change.
2. **Kill:** “We randomly shuffle, therefore we interleave” as a marketing claim.
3. **Kill:** Absolute always-interleave / always-double-scores claims.
4. **Wound:** Pure weak-spot blocked grind as the only Practice mode — survives as *acquisition*, not as the full loop.
5. **Survive (constrained):** Desirable-difficulty interleaving for **strategy discrimination** — after schema foothold, with near-miss design, affect-safe framing, and delayed mixed `transfer_pass`.

Doctrine until data: Market and build **choose-the-strategy training** via faded interleaving of confusable types. Blocking is a scaffold, not the product. Fluency without mixed delayed evidence is a vanity metric.

Part XL — Self-Explanation Prompts: Chi/Renkl → Coach UX

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When do prompts that force the student to *generate* an explanation improve math learning — and what must MindCraft’s coach / cards / Solver refuse to do with “explain yourself” UX?

Owners: Learning Science · Product (coach, ingredient cards, Solver) · Growth / Positioning · Red Team

Commercial job: Turn “we explain everything” and “AI tutor talks at you” into a **defensible generation claim**: students build transferable understanding by explaining *why steps work* (and why common wrong moves fail) — without selling open-ended “justify your guess” as science, and without confusing fluent AI explanation *to* the student with self-explanation *by* the student.

XL.1 Why this chapter exists

Part XXVI (CLT / worked examples / fading) already puts examples and completion problems in the acquisition path. Part XXIX / XXXIX demand retrieval and strategy *selection*. Part XXXIII warns that fluent AI explanations can create overtrust. Experiment A (wizard coach) and soft-wrong already assume that *after* an error, something productive happens.

This chapter densifies the **prompt layer**: what the product asks the student to *produce* — principle for a step, why a misconception is wrong, anticipation of the next operator — vs what it merely *shows*.

FOUNDER BELIEF under audit: Identity as a mathematical thinker requires evidence that the student can *account for* a move, not only select A/B/C/D. Passive delivery without generation is Layer-1 scarcity relief, not identity transformation.

Claim we refuse as doctrine: "Any explain box after a wrong answer is self-explanation science." Untargeted prompts, explain-your-wrong-guess-without-feedback, and AI monologues are different mechanisms — some help, some hurt, some are theater.

XL.2 Constructs (product language)

Construct	Plain definition	Product surface	Failure mode
Self-explanation	Learner generates inferences beyond the given text/example *for themselves*	Prompted coach turns; card reflection; step-principle pick	Empty "why?" with no scaffold; AI does the explaining for them
Instructional explanation	Expert/AI/tutor explains *to* the learner	Solver narration; worked example text	Can reduce germane effort if overused (expertise / passivity)
Worked-example study	Study full solution before / instead of cold solving	Ingredient cards; fade path	Without SE prompts → shallow copy (Chi "poor" pattern)
Principle / glossary citation	Name the rule that justifies a step	Aleven-style menu; ingredient id pick	Menu-click without comprehension if too shallow
Misconception contrast	Explain why a common wrong path fails	Soft-wrong + "why not B"; bank distractors	Asking only "defend your wrong answer" before feedback

Operational definition for MindCraft (HYPOTHESIS): A self-explanation prompt is load-bearing only if (1) the to-be-explained content is tagged **correct or known-incorrect**, (2) the response format scaffolds principle / operator–goal / misconception contrast (not free essay only), and (3) success is measured by **transfer / delayed mix**, not by longer time-on-task or chat length.

XL.3 Classic effect: spontaneous quality, then elicitation

FACT (good vs poor example study): Chi, Bassok, Lewis, Reimann & Glaser (1989, *Cognitive Science*, doi:10.1207/s15516709cog1302_1) — talk-aloud while studying mechanics worked examples. "Good" students generated many explanations that refined conditions for actions and linked steps to principles; "poor" students under-explained, monitored poorly, and relied heavily on examples at solve time. Self-explanation quality predicted understanding and example-independence.

FACT (elicitation works): Chi, de Leeuw, Chiu & LaVancher (1994, *Cognitive Science*, doi:10.1016/0364-0213(94)90016-7) — eighth-graders prompted to self-explain after each line of a circulatory-system text outperformed controls who reread; high explainers built more accurate mental models. Prompts can *create* the behavior that spontaneous good students show.

FACT (quality ≠ mere time): Renkl (1997, *Cognitive Science*, doi:10.1016/S0364-0213(99)80017-2) — probability worked examples; time-on-task controlled. Learning gains still predicted by qualitative SE features: **principle-based** explanations, **operator–goal** explication, **anticipative** reasoning. Two effective profiles: anticipative reasoners and principle-based explainers.

Product translation: Coach UX should elicit *those* moves — “What rule licenses this step?” / “What is this step trying to achieve?” / “What happens next?” — not “Write anything about your feelings about math” and not “Hear the AI explain for 90 seconds.”

XL.4 Scaffolded prompts + fading (coach design spine)

FACT (fading + principle prompts): Atkinson, Renkl & Merrill (2003, *Journal of Educational Psychology*, doi:10.1037/0022-0663.95.4.774) — successively fading worked-out steps helps near transfer vs example–problem pairs, but far transfer is unreliable from fading alone. Combining fading with prompts to identify the **underlying principle** for each worked step produced medium-to-large effects on **near and far transfer without extra time on task**.

FACT (Cognitive Tutor classroom): Aleven & Koedinger (2002, *Cognitive Science*, doi:10.1016/S0364-0213(02)00061-7) — geometry Cognitive Tutor; students who explained steps by citing geometric rules (glossary) learned with greater understanding and better transfer than problem-solving-only peers. Interpretation: better-integrated declarative knowledge, less shallow procedural knowledge. Scales in classroom software — not only lab talk-alouds.

HYPOTHESIS for MindCraft: Ingredient `card_templates` + bridge cards are the worked-example substrate; coach prompts should (a) attach principle/ingredient labels to faded steps, (b) require a brief generation or structured pick *before* revealing the next full AI paragraph, (c) fade prompts as mastery rises (expertise reversal risk if we nag forever).

Tie to SAFE-CRAFT (XXVI): Micro-attempt / completion blank inside an example *plus* a principle prompt beats cold multi-step solve *or* passive read-only cards.

XL.5 Math meta-analysis: what we may claim (and must not)

FACT (math SE meta-analysis): Rittle-Johnson, Loehr & Durkin (2017, *ZDM*, doi:10.1007/s11858-017-0834-z) — prompted self-explanation in mathematics: small-to-moderate immediate gains — procedural knowledge ES ≈ .28, conceptual ES ≈ .33, procedural transfer ES ≈ .46. Effects **stronger when high-quality explanations are scaffolded** (training or structured responses). Time-on-task control did **not** erase the benefit (moderator). **Caveats they stress:** evidence for reliable classroom (esp. primary/secondary non-ITS) effects and for **retention over delay** is much thinner; delayed procedural/conceptual effects often weak/null while delayed transfer ES ≈ .32 in the subset that measured it; substantial heterogeneity (some null/negative studies).

Instructional recommendations from that paper (treat as design doctrine, not slogans):

1. Scaffold high-quality explanations (structure or train).
2. Design prompts so they do **not** steal attention from other critical content.
3. Prompt learners to explain **correct** information.

4. Prompt why **common misconceptions** are incorrect.

Kill: Marketing “self-explanation always raises grades” / absolute classroom-retention claims from lab immediate ES alone. Claim ladder: L1 “structured why-this-step prompts aim at transfer”; L2+ only after SE-* experiments with delayed mixed `transfer_pass`.

XL.6 Constraints: when prompts backfire

FACT (constraint review): Rittle-Johnson & Loehr (2017, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-016-1079-5) — four constraints: (1) SE favors principle-guided domains and transfer, can hurt detail/exception learning; (2) explaining **one’s own** (often wrong) ideas can entrench them — better to explain known-correct / known-incorrect content; (3) prompts focus attention — mis-aimed prompts tax the wrong content; (4) alternatives (instructional explanation, retrieval, unfamiliar problems) can match or beat SE in some conditions.

HYPOTHESIS for soft-wrong: Order matters. After a wrong MC choice: (A) brief correct-principle or contrastive “why B fails” prompt with scaffold → help; (B) “Explain why you picked C” *before* any correctness signal → risk of justifying the misconception (constraint 2). Prefer: soft-wrong frame → show contrastive target → generate/select why the foil fails → then optional deeper AI wrap (XXXIII: calibration > fluency).

Contradiction to hold: Open generation is the romantic “thinker” move; structured glossary/ingredient picks (Aleven) often deliver the measurable classroom gain. MindCraft should not shame structured picks as “not real thinking” — they are the scalable SE scaffold.

XL.7 Competitive / product audit (explanation lens)

Surface	Typical explanation pattern	MindCraft stance
Khan / video	Instructional explanation; optional reflection	Do not clone lecture-first as identity product
Duo	Rare deep SE; habit + bits	Habit ≠ principle generation
Brilliant	Puzzle insight; light justify	Insight ≠ scaffolded SE curriculum
ChatGPT Solver	High-fluency instructional explanation	**Not** self-explanation; overtrust risk (XXXIII)
Cognitive Tutor lineage	Step + rule citation	Closest evidence-backed software pattern
MindCraft today	Cards + Solver + coach drafts	Need **student generation gates** before AI walls of text; wire misconception bank into contrastive prompts

FOUNDER BELIEF: Solver that answers *for* Maya without a generation gate competes on Layer-1 (answers get cheap) and loses the identity thesis. Coach that forces a principle pick / misconception contrast competes on Layer 2–6.

XL.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** “We don’t just show the steps — we ask you to say *why* the step works.”
- **Allowed:** “Wrong answers become contrast: why that trap fails, not a shame spiral.”
- **Allowed (parent):** “We train explaining the rule, not only picking the letter — that’s what transfers.”
- **Banned:** “AI explains everything so tutoring is free.”
- **Banned:** “Any chat box = self-explanation science.”
- **Banned:** Absolute meta-analytic grade claims; 2-sigma; streak-as-understanding.
- **Banned:** Forcing students to justify unchecked wrong guesses as the default pedagogy.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without SE-* data
AI worked-example narration	L0 instructional help; **not** SE claim
Structured principle / ingredient pick on faded step	L1 SE scaffold (Aleven/Atkinson-aligned intent)
Contrastive “why this distractor fails” after soft-wrong	L1 misconception-SE intent
Open essay “explain your answer” every item	L0 until quality+affect validated; risk of load + avoidance
“SE raises ACT / identity” hero	L0 until SE-1/SE-2 delayed transfer

Growth / North Star hygiene

Chat turns, explanation word count, and Solver dwell time are **vanity** if uncoupled from `transfer_pass` and strategy-class errors. Prefer: rate of principle-consistent explanations (rubric or structured correct pick), misconception-contrast success, delayed mixed accuracy, `retry_120s` after contrastive miss — not tokens generated by the model.

XL.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
SE-1	Principle-prompt on faded card steps vs same cards with AI narration only	2-arm; matched time budget	48h near + far / mixed transfer; shallow-copy errors	Narration-only ≥ prompted on transfer **and** lower load/avoidance

SE-2	Soft-wrong → contrastive “why foil fails” vs soft-wrong → explain-own-choice first	2-arm	Misconception recurrence; `retry_120s`; delayed item isomorphic	Explain-own-first wins transfer → constraint-2 doctrine wounded
SE-3	Structured ingredient/glossary pick vs open text SE vs no SE	3-arm	Transfer; time; completion rate (Maya anxiety segment)	Open text wins transfer but kills return → ship structured default
SE-4	Prompt every step vs adaptive fade of SE prompts with mastery	2-arm	Transfer; prompt skip/nag irritation	Always-on prompts equal fade on transfer but raise churn → fade mandatory
SE-QUAL	10 Maya interviews: “When did an explanation feel like *yours* vs the app talking?”	Appendix B	coded ownership / passivity moments	Nobody distinguishes → UX isn’t creating generation

Pre-reg (XXXIV): SE-* identify **prompt regime → transfer / misconception recurrence**. They do **not** identify long-term identity from one session of explain boxes. Densifies Experiment A: coach content should be SE-constrained, not generic pep talk.

Supersedes / densifies: XXVI worked-example fading notes; Experiment D (explanation timing × anxiety) → prefer SE-2/SE-3 as operational pairs; XXXIII “pedagogy wrap” → wrap = student generation + calibration, not longer AI prose.

XL.10 Confidence table

Claim	Label	Confidence
Higher-quality spontaneous SE while studying examples predicts better learning	FACT (Chi et al., 1989)	High
Prompting SE can improve understanding vs reread / no prompt	FACT (Chi et al., 1994; math meta)	High
Qualitative SE features predict gains beyond time-on-task	FACT (Renkl, 1997)	High
Fading + principle prompts aid near & far transfer without extra time	FACT (Atkinson et al., 2003)	High

Step explanation via rule citation helps Cognitive Tutor geometry transfer	FACT (Aleven & Koedinger, 2002)	High
Math prompted-SE immediate effects small–moderate; scaffold helps	FACT (Rittle-Johnson et al., 2017 ZDM)	High
Classroom + delayed retention evidence thinner than immediate lab/ITS	FACT (same meta)	High
Explaining own wrong ideas can entrench; prefer correct/incorrect targets	FACT (constraint review)	High
AI instructional explanation ≡ self-explanation benefit	SPECULATION / false	Low (against)
Default MindCraft path = faded examples + structured principle/misconception prompts before AI wrap	FOUNDER BELIEF / HYPOTHESIS	Medium–High

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XL.11 What this chapter kills

1. **Kill:** Equating Solver/AI monologue with self-explanation pedagogy.
2. **Kill:** "Explain why you chose that" *before* correctness/contrast as the default after every wrong answer.
3. **Kill:** Absolute "SE always raises scores / always works in class / always sticks for months" marketing.
4. **Wound:** Open-ended explain boxes as the only SE UX — survives as optional depth, not as the scalable default.
5. **Survive (constrained):** Scaffolded prompts on **correct steps and known misconceptions**, paired with worked-example fading, measured by transfer — the coach spine for identity-relevant competence evidence.

Doctrine until data: Build and market **student-generated why** (structured first) on faded examples and contrastive wrongs. AI explains *after* or *around* that generation — never as a substitute for it. Fluency of the model is not mastery of the student.

Part XLI — Desirable Difficulties × Anxiety: When Struggle Helps vs When It Becomes Threat

Chapter status: Living evidence brief — Researcher tick 2026-07-30

Primary question: How should MindCraft reconcile Bjork-style desirable difficulties (spacing, interleaving, retrieval, generation) with Ashcraft-style math-anxiety / working-memory threat — so product + copy do not sell either "grind harder" or "never make it hard"?

Owners: Learning Science · Affect / Anxiety · Product (Practice sequencing + evaluation climate) · Growth / Positioning · Red Team

Commercial job: Resolve the I.4 bottleneck: ship faded desirable difficulties **without** converting anxious Maya's session into shame theater — and market the distinction between *productive struggle* and *threat* in language parents and students can trust.

XLI.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), and XL (self-explanation) all lean on **desirable difficulties**: conditions that slow short-term performance but improve durable access and transfer. Part XXIV (affect/anxiety) and Ashcraft's line show that math anxiety **already** loads working memory like a dual task. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance under performance threat.

v1.6 left an explicit bottleneck: *"Part XLI must resolve desirable difficulties × anxiety so interleaving/SE/spacing do not ship as shame."*

FOUNDER BELIEF under audit: Identity transformation requires competence evidence that survives delay and mix — which *requires* some desirable difficulty — but for threatened learners, unsequenced difficulty is not "elite training"; it is **threat + WM collapse + exit**.

Claim we refuse as doctrine: "Struggle is always good" / "always crank desirable difficulties" *and* its twin "anxious students should only get easy fluency drills forever." Both are commercially tempting; both fail Maya.

XLI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Desirable difficulty (DD)	Challenge that forces encoding/retrieval processes that support later access	Mixed review, spaced retries, generation before reveal	Feels worse today → students & parents quit if misframed
Undesirable difficulty	Challenge the learner cannot productively meet (knowledge, cues, or affect load)	Early full interleave; open SE under panic; high-stakes quiz theater	Errors without learning; avoidance; "I'm not a math person"
Math anxiety load	Worry/rumination consuming WM resources needed for multi-step math	Freeze, hint binge, rapid exit after first hard miss	Interpreted as "lazy" or "needs more struggle"
Stakes / evaluation climate	Whether the hard moment is framed as judgment vs training	Soft-wrong, mastery climate, no streak-as-worth	High stakes turn retrieval into threat (Hinze & Rapp)

Equipped success	Prior knowledge + cues sufficient to *respond* to the difficulty	Block-lite → fade → mix; worked examples; structured SE	DD without equipment = cruelty dressed as science

Operational definition (HYPOTHESIS): A MindCraft difficulty is *desirable* only when (a) the student has enough schema/cues to respond with non-chance success or productive partial success, **and** (b) evaluation climate keeps the moment in mastery/training frame, **and** (c) affect load is not already saturating WM. Else it is undesirable — even if the *same* schedule would be desirable for a calm, prepared peer.

XLI.3 Bjork line: difficulty that builds durable access

FACT (origin): Bjork (1994, in Metcalfe & Shimamura, *Metacognition: Knowing about knowing*, MIT Press, pp. 185–205) — conditions that slow acquisition (spacing, variation, testing/generation as learning events, contextual interference) can improve long-term retention and transfer relative to conditions that make performance look strong during practice.

FACT (desirable ≠ any difficulty): Bjork & Bjork (2011/2014 reviews; JARMAC 2020 comment, doi:10.1016/j.jarmac.2020.09.003) — many difficulties are undesirable forever; a difficulty is desirable only when the learner can respond successfully with the encoding/retrieval processes the difficulty is meant to trigger. *If the learner is not equipped, the difficulty becomes undesirable.*

FACT (learning vs performance illusion): Instructors and students systematically misread current fluency as learning (Bjork lineage; see also XXIX / XXXIX). Blocked / massed / reread conditions feel productive; spaced / interleaved / tested conditions feel worse *and often teach more*.

Product translation: MindCraft cannot sell "we make practice feel easy" as proof of learning, nor "we make practice feel hard" as proof of rigor. The commercial claim must be **durable choosing and retrieving under exam-like demand** — measured by delayed mix / `transfer_pass` / `solo_transfer_pass` — not by session comfort or session pain.

XLI.4 Ashcraft line: anxiety as a competing load

FACT (WM disruption): Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — high math-anxiety individuals show reduced WM spans (especially computation-based spans); dual-task math + memory load amplifies RT/errors. Anxiety acts as a **transitory disruption of working memory**, not merely as a stable "low ability" trait.

FACT (review / education bridge): Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, doi:10.3758/BF03194059) — math beyond simple retrieval depends on WM; high math anxiety functions like a resource-demanding secondary task (preoccupation with fear). Global consequence: avoidance of elective math, majors, and careers.

FACT (carry from XXIV): Association of math anxiety with performance is real but modest on average (meta ~r ≈ –0.17); mechanism candidates include WM competition + avoidance. Anxiety is **one bottleneck among many**, not destiny.

Conflict statement (HYPOTHESIS): Desirable difficulties often *increase* momentary retrieval/selection demand — exactly the demand anxiety already taxes. Unbuffered DD therefore risks converting a would-be desirable difficulty into an undesirable one for HMA students: same schedule, different learner state → different pedagogy.

XLI.5 Evidence that stakes and anxiety change whether "hard practice" helps

FACT (pressure can kill retrieval benefits): Hinze & Rapp (2014, *Applied Cognitive Psychology*, doi:10.1002/acp.3032) — high-stakes vs low-stakes quizzes: quiz performance similar, but **final test better after low-stakes**; only low-stakes quizzes beat reread. Authors propose **retrieval disruption**: pressure can impair the memory-formation benefits of retrieval even when online quiz scores look fine.

FACT (classroom retrieval often *lowers* reported test anxiety): Agarwal, D'Antonio, Roediger, McDermott & McDaniel (2014, *JARMAC*, doi:10.1016/j.jarmac.2014.07.002) — survey of 1,408 middle/high-school students in classrooms using retrieval practice; 72% reported feeling **less nervous** for unit tests/exams when retrieval practice was used; 92% said it helped learning. Self-report, not RCT — but directionally opposite to "tests always raise anxiety."

FACT (meta: practice tests tend to reduce test anxiety): Yang, Li, Zhao, Luo & Shanks (2023, *Educational Psychology Review*, doi:10.1007/s10648-023-09801-w) — 24 studies / 25 effects / N≈3,374; practice tests reduce test anxiety with Hedges' $g \approx -0.52$; easy practice tests more mitigating than difficult ones. **Caveat:** mostly classroom/quiz anxiety outcomes — not a free pass to ship brutal mixed ACT sims as day-one "practice tests."

FACT (anxiety can reshape *what* is learned, not only performance): Fioriti, Pizzie, Evans, Green & Lyons (2025, *JEP:LMC*, doi:10.1037/xlm0001453) — adults learning unfamiliar multiplication across days: high math anxiety associated with **impaired procedural learning** on unrepeated problems but **accelerated retrieval learning** on heavily repeated problems — consistent with effort allocation toward declarative retrieval routes and away from procedural efficiency gains.

HYPOTHESIS for MindCraft: Anxious Maya may *prefer* and even accelerate on repeated retrieval of the same items (looks like mastery, feels safer) while under-building procedural flexibility and strategy selection — exactly the exam skills interleaving/SE target. Product that only rewards repeated-item fluency can **serve anxiety's preference** while failing transfer.

XLI.6 Resolution doctrine (not a slogan)

FOUNDER BELIEF / HYPOTHESIS — SAFE-DD stack (ties FEI / mastery climate / IL / SE):

1. **Equip before you strain.** Short block-lite + worked-example / card fade until execution errors are mostly slips (XXVI, XXXIX). DD without schema is undesirable.
2. **Lower stakes while raising retrieval.** Soft-wrong, mastery climate, no public rank, no streak-as-worth (XXXVIII; Hinze & Rapp). Retrieval as *training*, not judgment.
3. **Dose difficulty to remaining WM.** Prefer structured SE picks over open essays under high affect (XL SE-3); prefer faded interleave over day-one full mix (IL-2); prefer spaced short sets over marathon mixed threat.
4. **Frame the feeling.** Copy: "This mix trains *choosing* — feeling slower is the workout, not proof you're behind." Never: "hard mode for elites" / "grind through the pain."

5. **Measure the right win.** Session accuracy under block is not the North Star. Prefer `retry_120s`, delayed mix / `transfer_pass`, `solo_transfer_pass`, and anxiety-segment retention — not streaks or hint-free speed alone.
6. **Do not abandon DD for the anxious segment.** Permanent easy fluency is identity colonization by avoidance (XXV). Graded exposure to equipped difficulty *is* the path — therapy-adjacent but not a therapy claim (queue id 58).

Kill the false dichotomy: Bjork vs Ashcraft is not "science wars." Bjork already says unequipped difficulty is undesirable; Ashcraft explains *why anxiety unequips* (WM). The product job is **re-equip + destake + sequence**, then apply DD.

XLI.7 Competitive / marketing implications

Competitor pattern	What it often does under anxiety	MindCraft wedge
Duolingo-like streaks	Turns miss into identity failure; pressures retrieval	Ban streak-as-learning; train without shame timers
Blocked textbook / many apps	Fluency comfort; weak strategy choice	Honest: early block OK; claim exam-ready only after mix
Brutal "exam sim" day one	High stakes + hard mix → Hinze-style disruption + exit	Soft sim → rise stakes after `retry_120s` healthy
ChatGPT tutor monologue	Soothes without generation; may reduce threat short-term	Student-generated why *before* AI wrap (XL); calm ≠ mastery
"Growth mindset" posters	Empty struggle praise without equipment	Struggle + scaffolds + mastery climate + transfer metrics

Marketing language that survives

- **Allowed:** "Hard practice only after you're equipped — then we mix so exams don't surprise you."
- **Allowed:** "Feeling slower on mixed sets is the training signal, not a verdict."
- **Allowed (parent):** "We separate *training retrieval* from *graded judgment* so practice builds memory instead of panic."
- **Banned:** "Struggle always equals learning."
- **Banned:** "We never make anxious students struggle."
- **Banned:** Streak / blocked fluency / AI comfort as proof of readiness.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without DD×ANX data
Soft-wrong + mastery copy on hard items	L1 affect-safe practice climate
Block → fade → interleave schedule	L1 sequenced DD (intent aligned XXXIX)
Low-stakes spaced retrieval before "exam mode"	L1 Hinze-aligned retrieval hygiene

“Our struggle pedagogy raises ACT for anxious students”	L0 until DD-1/DD-2
Always-on max mix / open SE under threat	L0 — likely undesirable difficulty

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XLI.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
DD-1	Faded DD (block→mix + soft-wrong) vs early full mix same items	2-arm; stratify high vs low math-anxiety pretest	7d `transfer_pass`; `retry_120s`; exit after first miss	Early-mix wins transfer **and** retry in HMA segment → fade doctrine wounded
DD-2	Low-stakes retrieval sets vs same items framed as scored “exam drill”	2-arm (Hinze-style stakes)	Delayed retention; state anxiety; return D+1	High-stakes ≥ low-stakes on delayed + no anxiety cost → stakes doctrine overstated
DD-3	Structured SE under mix vs open SE under mix for HMA segment	2-arm (densifies SE-3)	Completion; transfer; hint binge	Open SE wins without churn → structured-default less necessary
DD-4	Repeated-item fluency path vs varied procedural path (Fioriti-inspired)	2-arm within HMA	Procedural transfer / strategy-class errors vs item-repeat accuracy	Repeat-fluency equals varied on transfer → anxiety-shaped preference harmless
DD-QUAL	10 Maya interviews: “When did hard practice feel like training vs attack?”	Appendix B	coded stake/equipment moments	Nobody separates → UX framing invisible

Pre-reg (XXXIV): DD-* identify **schedule × stakes × anxiety → transfer/return**. They do **not** identify long-term identity from one hard session. Densifies Experiment D and IL-2/SE-3 as the anxiety-gated versions of desirable difficulty.

Supersedes / densifies: XXIX.4 “never introduce DD as shame”; XXXIX IL-2; XL SE-3; I.4 bottleneck → now has an operational SAFE-DD stack pending DD-1/2.

XLI.9 Confidence table

Claim	Label	Confidence
Spacing/interleaving/retrieval/generation can improve long-term learning vs fluent short-term conditions	FACT (Bjork lineage)	High
Difficulties are desirable only if the learner can respond successfully	FACT (Bjork & Bjork)	High
Math anxiety disrupts WM / acts like dual-task load	FACT (Ashcraft & Kirk, 2001; Ashcraft & Krause, 2007)	High
High stakes can erase retrieval-practice benefits on delayed tests	FACT (Hinze & Rapp, 2014)	High
Classroom retrieval programs often associate with *lower* reported exam nerves	FACT (Agarwal et al., 2014; self-report)	Medium–High
Practice testing meta-analytically reduces test anxiety (esp. easier quizzes)	FACT (Yang et al., 2023)	High
HMA may accelerate repeated-item retrieval learning while impairing procedural learning	FACT (Fioriti et al., 2025)	Medium–High (adult lab; arithmetic)
SAFE-DD (equip → destake → dose → frame → measure) is the right MindCraft default	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Struggle always” or “never struggle anxious kids” is good marketing	SPECULATION / false	Low (against)

XLI.10 What this chapter kills

1. **Kill:** “Desirable difficulties always help — turn difficulty up for everyone.” (Ignores Bjork’s own *equipped* clause + Ashcraft WM.)
2. **Kill:** “Anxious students should only get easy, blocked fluency forever.” (Serves avoidance; fails transfer; Fioriti-shaped trap.)
3. **Kill:** High-stakes “exam mode” as the default vehicle for retrieval practice. (Hinze & Rapp.)
4. **Wound:** Self-report that “quizzes reduce anxiety” as license for brutal day-one mixed sims — survives as *low-stakes, well-sequenced* retrieval, not as cruelty.
5. **Survive (constrained):** Faded, low-stakes, mastery-framed desirable difficulties that produce delayed mix evidence — the commercial spine for “exam-ready without panic theater.”

Doctrine until data: Ship **SAFE-DD**: equip → lower stakes → dose to WM → frame as training → measure delayed transfer and `retry_120s` in the anxiety segment. Market productive struggle only when those guards are visible. Never sell pain or comfort as mastery.

Part XLII — Social Comparison & Leaderboards: When Ranks Help vs Hurt Novices

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 3)

Primary question: When do social-comparison surfaces (especially leaderboards) motivate learning versus threaten identity, and what should MindCraft default for Maya — the anxious / novice / identity-fragile segment?

Owners: Motivation · Affect / Anxiety · Product (Practice / Community UX) · Growth / Positioning · Red Team

Commercial job: Kill "ranks = engagement" as a North Star; harden the I.4 TARGET rule (default-ban appearance UX) into a shippable comparison policy competitors can be audited against.

XLII.1 Why this chapter exists

Part XXXVIII (goal orientation) already bans **appearance** goal structures and streak-as-North-Star. Part XXIV / XLI show social-evaluative threat taxes working memory and can turn desirable difficulty into threat. Competitive products (Duolingo-like ranks, classroom Top-N boards, public classroom streaks) still sell comparison as "motivation."

FOUNDER BELIEF under audit: Identity transformation needs competence evidence — not a public ladder that tells bottom-half Maya she is behind *people* rather than behind a *skill criterion*.

Claim we refuse as doctrine: "Leaderboards motivate learners" (full stop) — and its twin "if we anonymize, any infinite rank is fine."

XLII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Social comparison	Evaluating self via others' standing (Festinger)	Any peer rank, percentile, "class average," public tutor callout	Turns mastery climate into appearance climate
Upward / downward / lateral	Compare to better / worse / similar others	Top-N board; "you beat 80%"; neighbor band	Extreme upward gap → give-up; downward → complacent ego boost without learning

Absolute vs relative board	All players ranked vs local neighbors / Top-N only	Infinite classroom board vs "near you" strip	Absolute + real names = max face threat for low ranks
Criterion / mastery feedback	Progress vs a task standard, not vs peers	Path progress, concept mastery bars, personal bests	Still can be gamed if criterion ≠ transfer
Normative / appearance feedback	Standing vs others; looking smart/fast	Public leaderboard, streak leagues	Identity threat; WM load; exit after first public miss

Operational definition (HYPOTHESIS): A MindCraft comparison surface is *safe* only when (a) the default is **self/criterion**, (b) peer comparison is **opt-in**, (c) comparison targets are **attainable** (local band, not unreachable Top-1), (d) metrics ranked are **effort/process-compatible** with mastery climate (not speed-only or streak-only), and (e) low standing is never **publicly named** for the anxiety / novice segment.

XLII.3 Festinger → competition: the comparison engine

FACT (origin): Festinger (1954, *Human Relations*, "A Theory of Social Comparison Processes") — when objective standards are ambiguous, people evaluate abilities/opinions by comparing with others; ability comparison carries a unidirectional drive upward (improve toward better standards).

FACT (selection + reaction meta): Gerber, Wheeler & Suls (2018, *Psychological Bulletin*, doi:10.1037/bul0000127) — across 60+ years: when choosing up vs down, strong preference for **upward** comparison under no threat; **contrast** (self-evaluation moves *away* from the target) dominates reactions — ability estimates and affect often decline after upward contact. People often choose upward comparisons *even though* the typical reaction is self-deflating contrast.

FACT (competition model): Garcia, Tor & Schiff (2013, *Perspectives on Psychological Science*, doi:10.1177/1745691613504114) — social comparison is a major source of competitive attitudes/behavior; situational amplifiers include **proximity to a standard** (near #1 vs far away) and **N** (few vs many competitors). Ranking UIs are engineered comparison machines.

Product translation: Shipping a leaderboard is not a neutral "engagement widget." It is a deliberate injection of comparison → contrast risk → competitive framing. For Maya, that often means **appearance climate** (XXXVIII), not mastery.

XLII.4 Leaderboards can move behavior — under narrow conditions

FACT (HE experiment): Landers & Landers (2014, *Simulation & Gaming*, doi:10.1177/1046878114563662) — random assignment to a course wiki **leaderboard** (not tied to grades) increased project interactions (~+29.6) vs control; time-on-task mediated academic performance. Supports: leaderboards *can* raise engagement via goal/conflict attributes **in that setting**.

Caveats (FACT / scope): Undergraduate / course project; not math-anxious high-school Maya; not public infinite classroom shame; achievement metric was interaction volume, not durable transfer. Landers' theory of gamified learning (2014, doi:10.1177/1046878114563660) treats game attributes as mediators/moderators of *instructional* quality — gamification does not replace pedagogy.

FACT (HE systematic review): Li, Liang, Fryer & Shum (2024, *Journal of Computer Assisted Learning*, doi:10.1111/jcal.13077) — 20 articles / 22 studies / 29 interventions (2014–2023), **all higher education**; ~half ≤1 hour. Leaderboards *can* benefit motivation/engagement/performance, but **effects hinge on design**. Authors recommend (among others): absolute boards for motivation *with* anonymity especially for low ranks; mix performance + engagement criteria; caution that HE short-duration findings do **not** generalize cleanly to K–12.

HYPOTHESIS: Landers-style gains are most likely when learners already have task competence + low identity threat + voluntary/low-stakes ranking. MindCraft's default user is the opposite segment on day one.

XLII.5 Math-specific: anxiety × infinite ranks

FACT (primary maths game, large N): Almo, Rocha, Brennan & Dondio (2024, *International Journal of Serious Games*, doi:10.17083/ijsg.v11i4.794) — N=1,389 Irish primary students, 6-week *Seven Spells* programme with an **infinite leaderboard**. Leaderboard enjoyment predicted by rank position **and** maths anxiety: **maths-anxious players disliked the leaderboard more than non-anxious peers even after controlling for position**. Preference for playing against others also predicted enjoyment. Small correlation between liking the board and liking the game.

Implication (HYPOTHESIS): For anxious learners, public infinite rank is not a free engagement boost — it is a trait-moderated threat surface. "They're low because they're bad at the game" does not exhaust the mechanism; anxiety predicts dislike **net of rank**.

FACT (negative feedback climate): Fong, Patall, Vasquez & Stautberg (2019, *Educational Psychology Review*, doi:10.1007/s10648-018-9446-6) — meta-analysis (78 studies): negative feedback reduces intrinsic motivation vs positive feedback; harm lessens when feedback includes instructional how-to-improve detail and uses **criterion-based** standards (vs purely normative). Public low rank is often pure normative negative feedback without instructional content.

Bridge to XLI / XXIV: Social-evaluative threat (public miss, public bottom third) is exactly the stakes Hinze & Rapp warn can disrupt retrieval benefits — and Ashcraft-style WM load. Infinite boards amplify that climate at scale.

XLII.6 Resolution doctrine — SAFE-COMPARE

FOUNDER BELIEF / HYPOTHESIS — SAFE-COMPARE stack (ties TARGET / SAFE-DD / FEI):

1. **Default = criterion, not peers.** Show progress vs path / mastery / personal best. Peers off by default.
2. **Opt-in competition only.** Never force infinite boards onto gap-scan novices or high-anxiety segment.
3. **If peers: local + attainable.** Prefer neighbor bands / similar-ability cohorts over Top-1 theater (Garcia proximity + Gerber contrast).
4. **Anonymize low standing.** No named public bottom ranks (Li et al. Rec. 2; Bai et al. cited therein).
5. **Rank what mastery climate allows.** Prefer process-compatible signals (retries completed, transfer attempts, concepts unlocked) over speed leagues / streak leagues. Never rank "hint-free seconds" as identity.

6. **Instructional wrap on any relative miss.** If relative feedback ships, attach *how to improve next* (Fong moderators) — not only "you're #47."
7. **Measure the right win.** `retry_120s`, `challenge_accept` (mastery-coded), `transfer_pass` / `solo_transfer_pass` — not `leaderboard_views` or rank delta as learning.

Kill the false dichotomy: "Competition is toxic" vs "competition is always motivating." Competition can raise time-on-task for some prepared learners (Landers). For Maya default, **unbuffered infinite peer rank is appearance UX** — banned until SC experiments say otherwise.

XLII.7 Competitive / marketing implications

Competitor pattern	Comparison mechanism	MindCraft wedge
Duolingo-like leagues / ranks	Normative weekly standing; streak adjacent	Mastery path progress; ban streak/rank as learning proof
Classroom Top-N boards	Extreme upward targets; face threat	Criterion bars; optional local bands
Khan / practice dashboards	Often self-paced; weaker public rank	Keep self/criterion; don't "gamify catch-up" with shame ranks
Brilliant prestige signals	Ability-appearance among peers	Sell *solo transfer* and *challenge-seeking why*, not percentile flex
ChatGPT tutor alone	No peer board — but social comparison still via school/parents	Own the *safe* comparison story: progress vs yesterday's self

Marketing language that survives

- **Allowed:** "Progress against *your* path — not a public ladder."
- **Allowed:** "Competition is optional; mastery climate is the default."
- **Allowed (parent):** "We don't put anxious kids on a named scoreboard to 'motivate' them."
- **Banned:** "Leaderboards keep students hooked" as a learning claim.
- **Banned:** Rank / league / streak as North Star or as proof of identity change.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free; empty mindset posters.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without SC data
Criterion progress / personal bests default	L1 mastery-climate alignment (XXXVIII)
Opt-in anonymized local band	L1 SAFE-COMPARE intent
"Our leaderboard raises ACT / identity for anxious Maya"	L0 until SC-1/SC-2

Forced infinite named classroom board	L0 — appearance UX; likely harm for HMA

XLII.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
SC-1	Criterion-only progress vs infinite named leaderboard (same practice items)	2-arm; stratify high vs low math-anxiety	`retry_120s`; exit after public miss; 7d `transfer_pass`	Infinite board wins transfer **and** retry in HMA → default-ban wounded
SC-2	Opt-in local anonymized band vs forced absolute board	2–3 arm	Enjoyment; state anxiety; return D+1; challenge_accept motive codes	Forced absolute ≥ opt-in local on retention without anxiety cost → anonymity/opt-in overstated
SC-3	Rank process metrics (retries, mix attempts) vs rank speed/streak	2-arm within volunteers	Strategy-class errors; `transfer_pass` vs session speed	Speed rank equals process rank on transfer → metric choice moot
SC-4	Criterion negative feedback + how-to-improve vs rank-only "you're #N"	2-arm (Fong-inspired)	Intrinsic motivation proxy; retry	Rank-only ≥ instructional criterion → Fong moderator fails in product
SC-QUAL	10 Maya interviews: "When did seeing others' scores help vs freeze you?"	Appendix B	coded comparison threat moments	Nobody distinguishes → copy invisible

Pre-reg (XXXIV): SC-* identify **comparison surface × anxiety → return/transfer**. They do **not** identify that "healthy competition builds character" from one league season.

Densifies: XXXVIII appearance ban; I.4 TARGET "leaderboard kill for Maya default"; XXIV social-evaluative threat hypothesis.

XLII.9 Confidence table

Claim	Label	Confidence
People use others to evaluate abilities when standards are ambiguous	FACT (Festinger, 1954)	High
Upward comparison often preferred; contrast often dominates self-eval/affect	FACT (Gerber et al., 2018)	High
Ranking contexts amplify competitive social comparison	FACT (Garcia et al., 2013)	High
Leaderboards can increase time-on-task / mediate performance in some HE settings	FACT (Landers & Landers, 2014)	High (scope-limited)
Leaderboard effects in education are design-dependent; HE evidence ≠ K–12	FACT (Li et al., 2024)	High
Maths-anxious players dislike infinite boards more, net of rank	FACT (Almo et al., 2024)	High
Normative negative feedback tends to hurt intrinsic motivation vs positive; criterion + how-to softens	FACT (Fong et al., 2019)	High
SAFE-COMPARE default (criterion, opt-in, local, anonymize low, process metrics) is right for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
"Leaderboards motivate everyone / raise learning for Maya"	SPECULATION / false as universal	Low (against)

XLII.10 What this chapter kills

1. **Kill:** Universal "leaderboards motivate students" as product doctrine or marketing. (Design- and trait-dependent; HE ≠ Maya.)
2. **Kill:** Infinite named public ranks as default engagement for novices / high math-anxiety. (Almo et al.; TARGET / appearance.)
3. **Kill:** Rank delta / league standing / streak leagues as North Star or as proof of identity transformation.
4. **Wound:** Absolute-board motivation gains from HE reviews as license for K–12 public boards — survives only as *design hypothesis*, not as Maya default.
5. **Survive (constrained):** Opt-in, anonymized, attainable, criterion-wrapped comparison for learners who already seek competition — never as the onboarding spine.

Doctrine until data: Ship **SAFE-COMPARE**: criterion default → peer opt-in → local/attainable → anonymize low standing → rank mastery-compatible process → instructional wrap → measure transfer/retry, not rank views. Market “progress without public ladders.” Never sell ranks as belonging.

Part XLIII — Habit Formation Science (Wood/Clear): Cue Design Without Identity Colonization

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 6 ≡ Red Team slot; ch43 never written → prefer Researcher per rotation)

Primary question: How should MindCraft use habit science (context cues, repetition, automaticity) to raise *return to practice* without making streaks / daily check-ins the student’s identity — or mistaking habit for learning?

Owners: Motivation · Product (Practice / notifications / recognition) · Growth / Positioning · Red Team

Commercial job: Harden Insight 5 / HID into a shippable **SAFE-HABIT** stack competitors (especially Duo-like streak products) can be audited against; kill “habit = mastery” and “21 days” folklore in copy and KPIs.

XLIII.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance / streak North Star ban), XLI (SAFE-DD), and XLII (SAFE-COMPARE) already treat streaks and ranks as threat surfaces when they become *worth*. Market gravity still pulls edtech toward Duolingo-shaped habit loops: push cue → micro-session → streak reward → loss aversion.

FOUNDER BELIEF under audit: Identity transformation needs *reliable return to difficulty* — so cue design is load-bearing — but the habit that forms must be “open practice and try the next hard thing,” not “protect a number that proves I am consistent.”

Claims we refuse as doctrine:

1. “Streaks = learning” / streak length as North Star.
2. “Habits take 21 days” as science.
3. Treating popular packaging (*Atomic Habits*) as peer-reviewed habit theory without Wood/Gardner/Lally grounding.
4. Habit loops that **colonize identity** — the student becomes “a streak-keeper” rather than “someone who can recover from wrong.”

XLIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Habit (scientific)	Learned context→response association in procedural memory; cue activates response with limited goal mediation (Wood)	Stable time/place/preceding-action cue for practice open	Shipping “motivation” when the real lever is *cue stability*
Automaticity	Behavior feels efficient / less deliberative	Self-report automaticity; shorter time-to-first-problem	Confusing app-open automaticity with *math skill* automaticity
Context cue	Physical setting, time, people, prior action in a sequence	After-school phone dock; tutor session end → 10-min solo; calendar block	Push spam as fake context; unstable cues never habitize
Implementation intention	If-then plan: when Y, do X (Gollwitzer)	Onboarding “After dinner, I open MindCraft Map”	Vague goals (“practice more”) with no situational trigger
Identity colonization (product sense)	Habit metric becomes who the student *is* / fears losing	Streak as self-worth; freeze culture; XP-as-self	HID failure — return without identity change; break → exit
SAFE-HABIT	Cue + small start + miss-forgiveness + learning metrics above habit metrics	See XLIII.6	Habit theater that still ranks streak / DAU as proof

Operational definition (HYPOTHESIS): A MindCraft habit surface is *safe* only when (a) cues target **opening equipped practice**, not protecting a counter, (b) miss recovery is cheap (Lally: one miss ≠ wipe), (c) recognition prefers process/identity markers over streak length, and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never streak days.

XLIII.3 Wood: habits are context–response, not vibes

FACT (review): Wood & Rünger (2016, *Annual Review of Psychology*, doi:10.1146/annurev-psych-122414-033417) — habits are learned associations between recurring **context cues** and responses, strengthened gradually via associative / reward learning; once formed, perception of the cue activates the response representation with limited need for executive control. Defining features of habit automaticity include cue activation and **relative insensitivity to short-term goal/outcome changes**.

FACT (habit–goal interface): Wood & Neal (2007, *Psychological Review*, “A New Look at Habits and the Habit–Goal Interface”) — goals help *start* repetition and exposure to cues; once habits form, contexts can trigger responses without a mediating goal. Habit change therefore often requires changing cues / environments, not only “trying harder.”

FACT (systems view): Wood, Mazar & Neal (2021, *Perspectives on Psychological Science*, habits vs goals as separate but interacting systems) — habitual responses can activate regardless of current goals; people may still inhibit them, but cue-driven activation is real. Product translation: a streak notification is a **Pavlovian/context cue** engineered by the app — not proof the student wants math.

Product translation: If MindCraft wants return, design **stable contexts** (same after-school window; end-of-tutor-session stack; device home-screen tile) more than pep talks. If MindCraft wants identity, do **not** make the strongest cue "don't break the number."

XLIII.4 Clear / Fogg packaging vs the evidence spine

FACT (popularization, not primary science): Clear (*Atomic Habits*, 2018) packages cue / craving / response / reward loops and "habit stacking" ("After [current habit], I will [new habit]") for a mass audience. Fogg (*Tiny Habits*) popularized tiny start + anchor habits. Neither replaces Wood's procedural-memory account.

FACT (what stacking actually is): Habit stacking's evidence spine is largely **implementation intentions** — Gollwitzer & Sheeran (2006, *Advances in Experimental Social Psychology*, doi:10.1016/S0065-2601(06)38002-1): across 94 tests, if-then plans show a medium-to-large effect on goal attainment ($d \approx 0.65$), via heightened cue accessibility and automated initiation.

HYPOTHESIS: Clear is usable *copy scaffolding* for onboarding ("After tutoring, open Map for one mission") **if** MindCraft cites the Gollwitzer mechanism and measures initiation → equipped practice, not Clear's compound-growth metaphors as science.

Kill: Marketing that says "built on Atomic Habits science" as if the book were a methods paper. Prefer: "cue-based practice plans (implementation intentions; habit research)."

XLIII.5 How long, how fragile, what to measure

FACT (timeline myth): Lally, van Jaarsveld, Potts & Wardle (2010, *European Journal of Social Psychology*, doi:10.1002/ejsp.674) — daily repetition in a consistent context; automaticity rises on an asymptotic curve; among good-fit participants, median time to ~95% of asymptote ≈ **66 days** (range **18–254**). Missing **one** opportunity did not materially derail formation. ~Half of motivated participants still failed to perform consistently enough to reach habit status in the study window.

Kill: "21 days to form a habit" (Maltz folklore, not this literature). Also kill product copy that implies a week-long streak *is* a habit.

FACT (definition hygiene): Gardner (2015, *Health Psychology Review*, doi:10.1080/17437199.2013.876238) — habit as a *process* whereby a stimulus generates an impulse to act via learned S–R association; impulse need not equal behavior. Most health-habit studies are correlational / self-report-heavy — confidence ceiling for edtech transfer claims.

FACT (measurement trap): Verplanken & Orbell (2003, *Journal of Applied Social Psychology*, doi:10.1111/j.1559-1816.2003.tb01951.x) — SRHI includes history of repetition, automaticity, **and expressing identity**. Self-report habit strength can smuggle identity language into a frequency metric.

Product implication (HYPOTHESIS): Do not treat SRHI-like “this is typically me” items as proof of *math identity transformation* (Part XXV / id 61 queue). Separate: (1) practice-open automaticity, (2) math self-efficacy / narrative identity scales, (3) transfer metrics.

XLIII.6 Duolingo-shaped products: habit works — and colonizes

FACT (industry, self-report): Duolingo publicly designs streaks using habit research framing; company blog claims learners who reach short streaks are far more likely to continue (e.g., 7-day streak correlational lifts — *not* causal RCTs published as independent learning science). Streak freezes / separation of streak vs daily goal are retention experiments on **engagement**, not transfer.

FACT (misuse qualitative): Hadi Mogavi et al. (2022, *Learning @ Scale*, doi:10.1145/3491140.3528274) — forum analysis + interviews on Duolingo: **gamification misuse** when users fixate on XP/streaks/leaderboards and get distracted from learning; reasons include competitiveness, playfulness overlearning, dark nudges, compulsion, herding. Ramifications touch learning efficiency, well-being, and ethics.

Bridge to XLII / XXXVIII: Streak leagues + infinite ranks are the same family as appearance climates — consistency theater. Wood predicts cues will keep firing even when the *learning* goal cooled; Hadi Mogavi shows the product can train the wrong response (protect streak / farm XP).

FOUNDER BELIEF: MindCraft may borrow **cue stability + small start + miss forgiveness**. MindCraft must not borrow **streak-as-self** or **loss-aversion-as-pedagogy**.

XLIII.7 Resolution doctrine — SAFE-HABIT

FOUNDER BELIEF / HYPOTHESIS — SAFE-HABIT stack (ties HID / TARGET / SAFE-DD / SAFE-COMPARE):

1. **Cue the practice, not the counter.** Default cues: calendar/time, tutor-session end stack, story-world return point — not “your streak dies at midnight” as the primary cue.
2. **Implementation intention onboarding.** One written if-then in gap-scan aftermath or first session (“After [stable cue], I open one mission”). Measure initiation, not slogan recall.
3. **Tiny equipped start.** First action ≤ friction of one problem with SAFE-DD equipment — not a multi-screen habit tutorial.
4. **Miss ≠ identity failure.** One miss does not reset belonging (Lally). Prefer gentle resume copy over shame / “you broke it.” Streak freezes are a *retention patch*, not a learning feature — if shipped, never as North Star.
5. **Habit metrics subordinate.** Track `D1/D7 return`, cue-honoring rate, time-to-first-problem as *diagnostics*. Learning North Stars remain retry / transfer / solo transfer / challenge_accept (mastery-coded).
6. **Identity markers elsewhere.** Map fill, story unlocks, tutor recognition of *recovery*, parent “they tried the hard one again” — not streak length as “who they are.”
7. **No identity colonization.** Ban copy that equates worth with unbroken days. Ban ranking streaks (XLII). Ban streak leagues as proof of transformation.

Kill the false dichotomy: “We refuse all habit design” vs “we must out-Duo Duo.” Refuse *colonizing* habit design; keep *cue science*.

XLIII.8 Competitive / marketing implications

Competitor pattern	Habit mechanism	MindCraft wedge
Duolingo streaks / freezes	Loss aversion + daily cue; retention-first	Cue practice return; miss-forgiving; never streak-as-learning
Duo leagues + XP	Habit + appearance compound	SAFE-COMPARE + SAFE-HABIT; sell solo transfer
Khan session streaks / energy	Mild habit / progress bars	Keep criterion progress; don't add shame timers
Brilliant daily puzzles	Delight + return habit	Return *into* recoverable struggle, not only delight
ChatGPT "study every day" tips	Advice without cue architecture	Own *if-then* + *equipped mission* as the plan

Marketing language that survives

- **Allowed:** "Same time, same cue — open one hard problem and recover."
- **Allowed:** "Habits get you to the desk; identity is what you do when it's wrong."
- **Allowed (parent):** "We build a practice cue without making a broken streak mean failure."
- **Banned:** "Streaks create math identity."
- **Banned:** "21-day habit challenge = mastery."
- **Banned:** Streak / DAU as North Star or as proof of transformation.
- **Banned:** Bloom 2-sigma; tutoring-is-free; empty mindset posters.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without HAB data
If-then practice cue + tiny equipped start	L1 initiation design (Gollwitzer-aligned)
Miss-forgiving resume without streak shame	L1 HID hygiene
"Our streak raises ACT / math identity"	L0 until HAB-1/HAB-2
Forced streak + public streak rank as default	L0 — colonization UX

XLIII.9 Experiments spawned

ID	Question	Design	Primary	Kill condition

HAB-1	If-then cue plan vs streak-primary onboarding (same missions)	2-arm; new users	D7 return; time-to-first-problem; 14d `transfer_pass`	Streak arm wins transfer **and** identity item → on-ramp-only doctrine wounded
HAB-2	Miss-shame copy (“you broke your streak”) vs miss-forgiving resume	2-arm after forced miss	Return D+1; state anxiety; `retry_120s`	Shame ≥ forgive on return without anxiety cost → forgive overstated
HAB-3	Stable calendar cue vs push-only reminders	2-arm	Cue-honoring rate; 4-week automaticity proxy	Push-only equals calendar on return → context-stability claim weak
HAB-4	Recognition: process/recovery badge vs streak badge vs none	3-arm (ties GO-2)	`challenge_accept` motives; identity item; streak obsession items	Streak badge wins engagement **and** transfer → HID weaker than feared
HAB-QUAL	10 Maya interviews: “What does your streak (elsewhere) make you feel when you miss?”	Appendix B	coded colonization vs helpful cue	Nobody reports threat → copy over-indexed

Pre-reg (XXXIV): HAB-* identify **cue/recognition × return/transfer**. They do **not** identify that “daily habit builds mathematicians” from DAU alone.

Densifies: Insight 5 / HID; I.4 TARGET streak ban; XXXVIII appearance; XLII streak-league kill.

XLIII.10 Confidence table

Claim	Label	Confidence
Habits are context–response associations; cue-triggered; relatively goal-insensitive once strong	FACT (Wood & Rünger, 2016; Wood & Neal, 2007)	High
Goals help form habits via repetition/exposure; changing cues aids habit change	FACT (Wood line)	High

Implementation intentions raise goal attainment (medium–large)	FACT (Gollwitzer & Sheeran, 2006)	High
Automaticity plateaus on the order of weeks–months; median ~66d in Lally sample; high variance; one miss often OK	FACT (Lally et al., 2010)	High
“21 days” is not the Lally result	FACT	High
Clear/Fogg are popular packaging; stacking ≈ if-then planning	HYPOTHESIS / FACT (mechanism mapping)	Medium–High
Gamification can distract from learning (Duo case)	FACT (Hadi Mogavi et al., 2022)	High (qualitative)
Streaks can raise return without identity / transfer	HYPOTHESIS (HID; industry)	Medium–High
SAFE-HABIT is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Streaks = learning / identity”	SPECULATION / false as universal	Low (against)

XLIII.11 What this chapter kills

1. **Kill:** Streak length / DAU / “habit streak” as North Star or as proof of identity transformation. (HID; standing ban densified with Wood + Duo misuse.)
2. **Kill:** “21-day habit” marketing as science. (Lally variance + asymptote.)
3. **Kill:** Equating *Atomic Habits* branding with peer-reviewed habit doctrine. (Use Wood/Gardner/Gollwitzer; Clear as optional UX copy only.)
4. **Kill:** Shame-on-miss and public streak ranks as default engagement for Maya. (Colonization; XLII.)
5. **Wound:** “Any daily reminder is habit science” — survives only if tied to **stable context + repeated equipped response**, not push spam.
6. **Survive (constrained):** Cue design, tiny starts, implementation intentions, and miss forgiveness as **on-ramps** to FEI / transfer — never as the product thesis.

Doctrine until data: Ship **SAFE-HABIT**: cue practice (not the counter) → if-then onboarding → tiny equipped start → miss ≠ identity failure → habit metrics subordinate → identity markers elsewhere → ban streak colonization. Market “we get you back to the hard problem — without making a broken streak mean you’re not a math person.”

Part XLIV — Intrinsic Motivation Killers (Deci): Controlling Rewards and the Caveats

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 9; hour%6≠0; 3 researcher entries since synthesizer v1.6 → Researcher, not Synthesizer)

Primary question: When do MindCraft rewards, XP, badges, streaks, and "do this for the prize" UX *undermine* the intrinsic interest and autonomy we claim to build — and when are they informational rather than controlling?

Owners: Motivation · Product (Practice / recognition / parent copy) · Growth / Positioning · Red Team

Commercial job: Turn the vague "extrinsic bad / Deci says so" slogan into a shippable **SAFE-REWARD** stack competitors can be audited against; kill both "all rewards kill love of math" absolutism and "gamify everything" as identity strategy.

XLIV.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance climates), XLII (SAFE-COMPARE), and XLIII (SAFE-HABIT) already treat streaks, ranks, and consistency theater as threat surfaces. Market gravity still pulls edtech toward *expected tangible / token rewards for doing the task*: XP for opening the app, coins for finishing any problem, badges for login streaks, parent dashboards that score "minutes practiced."

FOUNDER BELIEF under audit: Identity transformation needs *felt autonomy* and *competence information* — not the absence of all rewards. The killer is **controlling functional significance** (pressure to behave for the inducement), not the existence of feedback or unexpected acknowledgment.

Claims we refuse as doctrine:

1. "All rewards kill intrinsic motivation" (Cameron–Pierce counter-literature exists; CET is differentiated).
2. "Gamification is always fine if engagement rises" (engagement ≠ free-choice interest; Hadi Mogavi / Duo misuse).
3. "Verbal praise always helps kids" (Deci: controlling praise / "you should keep it up" can undermine).
4. Treating SDT as a poster ("autonomy / competence / relatedness") without contingency design.

XLIV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Intrinsic motivation	Doing the activity for inherent interest / satisfaction	Free-choice return to hard problems; curiosity asks; challenge_accept for mastery reasons	Confusing "opens app for XP" with interest
Extrinsic motivation (spectrum)	Behavior for separable outcome; can be more/less autonomous (external → introjected → identified → integrated)	Parent-pressure vs "I chose ACT path because…"	Treating all extrinsic as poison or all as fine

Controlling vs informational	Event felt as pressure to behave vs competence feedback (CET)	"Finish 10 for a coin" vs "you used the bridge correctly"	Same badge can be either — design of *how* it is framed
Engagement-contingent reward	Expected reward for *doing* the task (no quality bar)	XP for any attempt; login coins	Classic undermine path (Deci 1999)
Completion-contingent	Expected reward for finishing	"Complete mission → loot" regardless of strategy quality	Similar undermine; weak competence signal
Performance-contingent	Reward for meeting a standard	Badge for transfer pass; tutor note for recovery	Can inform competence *or* control — climate decides
SAFE-REWARD	Prefer informational feedback; avoid expected tangible for mere engagement; autonomy-supportive framing	See XLIV.7	Reward theater that still trains external PLOC

Operational definition (HYPOTHESIS): A MindCraft reward surface is *safe* only when (a) it primarily conveys **competence / strategy information**, (b) expected tangible / token rewards are **not** the reason to open practice, (c) framing is autonomy-supportive (choice, rationale, non-pressuring language), and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never XP velocity or badge count.

XLIV.3 SDT spine (needs, not posters)

FACT (overview): Ryan & Deci (2000, *American Psychologist*, 55(1), 68–78, doi:10.1037/0003-066X.55.1.68) — Self-Determination Theory posits innate needs for **competence, autonomy, and relatedness**; social contexts that support these needs facilitate intrinsic motivation and internalization; contexts that thwart them yield alienation and depleted motivation.

FACT (CET): Cognitive Evaluation Theory (Deci & Ryan line; summarized in Deci, Koestner & Ryan, 1999) — rewards and external events have **functional significance**: controlling aspects shift perceived locus of causality outward (undermine intrinsic motivation); informational aspects affirm competence (can enhance). Competence feedback without autonomy does not reliably enhance intrinsic motivation.

Product translation: MindCraft's brand voice ("you're becoming someone who recovers") must not be paired with UX that says "do this *or* lose the prize." Relatedness (tutor, story world, parent) is not a license for controlling rewards.

XLIV.4 The classic finding — and the differentiated meta

FACT (origin): Deci (1971, *Journal of Personality and Social Psychology*, 18(1), 105–115, doi:10.1037/h0030644) — expected extrinsic rewards for an interesting activity can reduce subsequent free-choice engagement (the "undermining" / overjustification pattern that launched the literature).

FACT (Deci–Koestner–Ryan 1999 meta): Deci, Koestner & Ryan (1999, *Psychological Bulletin*, 125(6), 627–668, doi:10.1037/0033-2909.125.6.627) — 128 studies. As predicted by CET:

- **Engagement-contingent** rewards undermine free-choice intrinsic motivation ($d \approx -0.40$).
- **Completion-contingent** ($d \approx -0.36$) and **performance-contingent** ($d \approx -0.28$) also undermine free-choice behavior on average.
- Engagement- and completion-contingent also hurt self-reported interest ($d \approx -0.15$ / -0.17).
- **Unexpected** tangible rewards: no average undermine on free-choice.
- **Positive feedback / verbal rewards:** enhance free-choice ($d \approx 0.33$) and interest ($d \approx 0.31$) *on average*.
- Tangible rewards tended to be **more detrimental for children** than college students; verbal rewards less enhancing for children than for college students.

FACT (education restatement): Deci, Koestner & Ryan (2001, *Review of Educational Research*, 71(1), 1–27, doi:10.3102/00346543071001001) — argue Cameron & Pierce (1994) understated undermining; conclude tangible rewards have a substantial undermining effect when CET contingencies are respected; defend keeping CET for educational practice.

Product implication (HYPOTHESIS): Expected XP/coins for *mere engagement* (open app, any answer, any completion) is the highest-risk MindCraft pattern for Maya-aged users. Informational soft-wrong + strategy feedback is the low-risk pattern. Performance-contingent badges *can* work if they mark real competence (transfer) and are not administered as pressure.

XLIV.5 Caveats — Cameron/Pierce, climate, and "not all rewards"

FACT (counter-meta): Cameron & Pierce (1994, *Review of Educational Research*, 64(3), 363–423, doi:10.3102/00346543064003363) — across 96 between-groups experiments, *overall* reward did not decrease intrinsic motivation; verbal praise increased it; the clearest negative effect was expected tangible rewards for *simply doing* a task (free-choice time after reward removal). They argued negative effects are circumscribed and avoidable.

FACT (debate): Ryan & Deci (1996, *RER*), Lepper et al. (1996), Kohn (1996) vs Cameron & Pierce (1996, *RER*) — methodological dispute over aggregation, moderators, and what "overall" means. **Not settled as "rewards are harmless."**

FOUNDER BELIEF / resolution for MindCraft: Do **not** cite "Deci proved all gamification is evil." Do **not** cite Cameron to greenlight engagement-contingent loot. Cite the **contingency**: expected tangible for mere doing is the kill zone; informational verbal/competence feedback is the build zone; performance-contingent is climate-sensitive.

FACT (controlling praise): Ryan (1982) and Ryan, Mims & Koestner (1983) — positive feedback administered with controlling language (e.g., "you *should* keep up the good work") reduces intrinsic motivation relative to informational administration. Deci 1999 summarizes interpersonal climate as a moderator of performance-contingent rewards.

Kill: "We praise a lot, so SDT is satisfied." Praise can be controlling.

FACT (choice as autonomy support): Patall, Cooper & Robinson (2008, *Psychological Bulletin*, 134(2), 270–300, doi:10.1037/0033-2909.134.2.270) — meta of 41 studies: providing choice enhances intrinsic motivation, effort, task performance, and perceived competence. Effects stronger for instructionally *irrelevant* choices, when ~2–4 successive choices are offered, when rewards are **not** stacked after the choice manipulation, and vs highly controlling controls; stronger for children than adults.

Product implication: Meaningful micro-choice (which story frame, which representation after mastery structure is set) can support autonomy — but **choice + immediately dangling reward** weakens the choice benefit. Do not "choose your path → earn 50 coins."

XLIV.6 Competitive patterns under CET audit

Competitor pattern	CET reading	MindCraft wedge
Duolingo XP / streaks / leagues	Expected engagement- and completion-contingent tokens + appearance climate (XLII/XLIII)	Informational recovery feedback; no XP-as-reason-to-open; SAFE-HABIT cues
Khan energy / points-ish progress	Mild completion signals; mostly mastery bars	Keep criterion progress; avoid coinization of minutes
Brilliant delight unlocks	Intrinsic interest scaffold; some completion loot	Keep delight *inside* the problem; don't pay for opening
ChatGPT "I'll quiz you if you want"	Low token reward; high fluency risk (XXXIII)	Autonomy of asking ≠ competence evidence; measure solo transfer
Parent dashboards: minutes / day streak	Parent-facing controlling contingency spills to student	Report `retry_120s`, transfer, challenge_accept — not "minutes bought with coins"

Bridge: XLIII says cue the practice, not the counter. XLIV says: if you must reward, reward **informative competence**, not engagement.

XLIV.7 Resolution doctrine — SAFE-REWARD

FOUNDER BELIEF / HYPOTHESIS — SAFE-REWARD stack (ties TARGET / SAFE-COMPARE / SAFE-HABIT / FEI):

1. **No expected tangible / token for mere engagement.** Ban default XP-for-open, coins-for-any-answer, loot-for-login. (Deci 1999 engagement-contingent.)
2. **Feedback > inducement.** Soft-wrong, strategy labels, bridge-gap naming, and "you recovered" tutor notes are the primary reinforcer. Prefer unexpected acknowledgment over announced prizes.
3. **If performance-contingent, mark real competence.** Badges/unlocks tied to `transfer_pass` / `solo_transfer_pass` / mastery-coded `challenge_accept` — not to streak days or raw volume. Administer without "you should" pressure.

4. **Autonomy-supportive framing.** Offer rationale; allow meaningful micro-choice (Patall); never “do this or lose the reward.”
5. **Don’t stack choice with dangling rewards.** Choice then immediate coin undermines the autonomy gain.
6. **Child-risk premium.** Maya-aged users are in the higher tangible-harm band of Deci 1999 — stricter default than adult SaaS gamification norms.
7. **Metrics hygiene.** Track XP/badge counts as *diagnostics of system behavior*, never as North Stars. Learning NS remain retry / transfer / solo transfer / identity.
8. **Parent copy:** Sell “she chose the harder one again” and competence evidence — not “she earned 400 XP.”

Kill the false dichotomy: “No rewards ever” vs “gamify everything.” Refuse *controlling engagement loot*; keep *informational competence signals* and story unlocks that are earned by transfer, not by attendance.

XLIV.8 Marketing language that survives

- **Allowed:** “We don’t pay kids to open the app — we show them what they figured out.”
- **Allowed:** “Unlocks follow real transfer, not login streaks.”
- **Allowed (parent):** “Progress you can trust: recovery and transfer — not points theater.”
- **Banned:** “Rewards create love of math.”
- **Banned:** “XP builds identity.”
- **Banned:** Absolute “all extrinsic motivation is bad” / “tutoring is free so motivation is free.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streak/XP as North Star.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without IM data
Informational soft-wrong + strategy feedback (no engagement loot)	L1 CET-aligned hygiene
Unexpected tutor/parent acknowledgment after recovery	L1 (unexpected tangible / social)
“Our XP system raises intrinsic math interest”	L0 until IM-1/IM-2
Expected coins for any completion as default	L0 — engagement-contingent kill zone

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XLIV.9 Experiments spawned

ID	Question	Design	Primary	Kill condition

IM-1	Engagement-loot (XP for any attempt) vs no-loot informational feedback (same missions)	2-arm; new users	Free-choice analog: voluntary hard-problem rate post-session; 14d interest item; `transfer_pass`	Loot arm wins free-choice **and** transfer → engagement-loot ban overstated
IM-2	Performance badge on `transfer_pass` (informational copy) vs same badge with controlling copy (“you should keep earning”)	2-arm	`challenge_accept` motives; interest; return	Controlling ≥ informational on interest → climate claim weak
IM-3	Micro-choice without reward vs choice + dangling coins	2-arm (Patall moderator)	Intrinsic motivation proxy; effort; autonomy need satisfaction	Choice+coins ≥ choice-only → “don’t stack” overstated
IM-4	Unexpected recovery acknowledgment vs announced prize for recovery	2-arm	Free-choice return; attribution coding	Announced ≥ unexpected on free-choice without control feelings → unexpected preference weaker
IM-QUAL	10 Maya interviews: “When a learning app gives points, what does that make you feel about the math?”	Appendix B	coded controlling vs informational	Nobody reports control → copy over-indexed

Pre-reg (XXXIV): IM-* identify **reward contingency × free-choice/interest/transfer**. They do **not** identify “gamification builds mathematicians” from XP velocity.

Densifies: TARGET appearance ban; SAFE-HABIT; SAFE-COMPARE; Insight on extrinsic traps; standing streak NS ban.

XLIV.10 Confidence table

Claim	Label	Confidence

Autonomy, competence, relatedness support intrinsic motivation / wellness	FACT (Ryan & Deci, 2000)	High
Expected engagement-/completion-contingent tangible rewards undermine free-choice IM on average	FACT (Deci, Koestner & Ryan, 1999)	High
Unexpected tangible rewards do not show average undermine	FACT (Deci et al., 1999)	High
Positive feedback enhances IM on average; controlling administration can reverse	FACT (Deci et al., 1999; Ryan line)	High
Tangible rewards more detrimental for children than college samples	FACT (Deci et al., 1999)	High
Cameron & Pierce find overall null with circumscribed negatives	FACT (1994); interpretation contested	Medium (dispute)
Choice enhances IM / effort / competence perceptions	FACT (Patall et al., 2008)	High
Stacking rewards after choice weakens choice benefit	FACT (Patall moderator)	Medium–High
SAFE-REWARD is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“All rewards kill intrinsic motivation” / “XP creates math love”	SPECULATION / false as universals	Low (against)

XLIV.11 What this chapter kills

1. **Kill:** Expected engagement-contingent XP/coins/loot as default motivation architecture for Maya. (Deci 1999 *d*≈–0.40 free-choice.)
2. **Kill:** Marketing that “rewards create love of math” or “XP = identity.” (External PLOC; HID.)
3. **Kill:** Absolute slogan “all extrinsic rewards are bad” used to ban informational feedback and earned transfer unlocks. (CET differentiation; Cameron caveat acknowledged.)
4. **Kill:** Controlling praise / “you should keep earning” as the voice of competence support.
5. **Wound:** “Performance badges are always safe” — survives only with informational climate + real competence criterion.
6. **Survive (constrained):** Unexpected acknowledgment, strategy-informative feedback, autonomy-supportive micro-choice, and transfer-gated unlocks under **SAFE-REWARD**.

Doctrine until data: Ship **SAFE-REWARD**: no expected tokens for mere engagement → feedback over inducement → performance marks only for real transfer/recovery → autonomy framing → don’t stack choice with dangling coins

→ child-risk premium → XP never North Star. Market "we don't pay kids to practice — we make the competence visible."

Part XLV — Mathematical Resilience: Growth Under Adversity Without Grit Theater

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 12 ≡ Red Team slot, but ch45 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: What is *mathematical resilience* as an actionable construct (Johnston-Wilder & Lee), how does it differ from grit posters and empty "bounce back" copy, and what product stack keeps Maya in the growth zone without dumping her into the danger zone?

Owners: Motivation / Affect · Product (Practice / coach / tutor briefing) · Positioning · Red Team

Commercial job: Replace "build grit" / "resilience mindset" marketing with a shippable **SAFE-RESILIENCE** stack competitors can be audited against — growth-zone design, support recruitment, struggle-as-normal, value without coercion — tied to FEI / SAFE-DD / SAFE-REWARD.

XLV.1 Why this chapter exists

Parts XXIV (anxiety), XXV (self-efficacy / habit≠identity), XXX (attribution / helplessness), XXXI (flow / challenge–skill), XLI (SAFE-DD), and XLIV (SAFE-REWARD) already treat threat, recovery, and controlling incentives. Market and school language still collapses everything into **grit** or generic **resilience** slogans: "just push through," "bounce back harder," streak-as-character, leaderboard-as-grit-proof.

FOUNDER BELIEF under audit: Identity transformation under difficulty needs *domain-specific* resilience — knowing struggle is normal in math, valuing the work, knowing how to work at it, and **recruiting support** before anxiety flips the lid — not a personality transplant labeled grit.

Claims we refuse as doctrine:

1. Duckworth-style grit as MindCraft's North Star or primary product claim (Credé et al. wound the construct).
2. "Desirable difficulty" = keep students in the danger/anxiety zone (conflicts SAFE-DD / Ashcraft).
3. Empty resilience posters without support architecture (tutor cards, peers, ontology scaffolds).
4. Treating mathematical resilience as synonymous with growth-mindset wall art.

XLV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Developmental resilience (Masten)	Ordinary adaptive systems enabling competence despite adversity — not rare "super kids"	Design that protects basic systems (safety, belonging, competence feedback)	Selling "extraordinary character" as the product
Mathematical resilience (JW/Lee)	Positive stance that lets learners engage despite affective barriers: growth belief, value, knowing how to work at math, knowing how to recruit support	Stay in growth zone; ask for help; recover after stuck; continue study path	Posterizing the four aspects without UX
Growth Zone Model	Comfort (easy / little growth) → Growth (challenge + support) → Danger/Anxiety (threat / thinking collapses)	Challenge calibrated + support on tap; exit danger fast	Framing red-zone panic as "grit training"
Grit (Duckworth)	Perseverance + consistency of interest toward long-term goals	Tempting NS copy	Near-conscientiousness; weak intervention leverage (Credé)
SAFE-RESILIENCE	Equip → name struggle → recruit support → destake danger → measure recovery/transfer — never grit KPI	See XLV.7	Resilience theater that still shames help-seeking

Operational definition (HYPOTHESIS): A MindCraft session builds mathematical resilience only when the student (a) experiences struggle as *expected* not as personal defect, (b) has a clear path to **support** (cards, tutor, peer, worked scaffold) before threat spikes, (c) returns to challenge after recovery (`retry_120s` / `challenge_accept`), and (d) can transfer without the model finishing the thought — not when streak or "grit score" rises.

XLV.3 Ordinary magic — resilience is not a unicorn trait

FACT: Masten (2001, *American Psychologist*, 56(3), 227–238, doi:10.1037/0003-066X.56.3.227) — resilience in development is often **ordinary magic**: competence despite adversity typically arises from normative adaptational systems (attachment, learning systems, self-regulation, community supports). Greatest threats are those that *compromise* these protective systems. Policy implication: protect and restore ordinary systems rather than hunt rare resilient personalities.

Product translation: MindCraft should not market "we forge resilient geniuses." Market "we keep the ordinary protective systems online while math gets hard" — safety, relatedness (tutor/world), competence information, recoverable miss. That is Masten-aligned and FEI-aligned.

Kill: "Resilience is a rare character trait we select for."

XLV.4 Mathematical resilience as a pragmatic construct

FACT (origin framing): Johnston-Wilder & Lee (2010) introduced **mathematical resilience** as a pragmatic positive construct opposite avoidance, anxiety, and helplessness — enabling learners to engage with mathematics despite affective barriers (cited as foundational in later scale and handbook work; e.g. Lee & Johnston-Wilder construct summaries; Johnston-Wilder & Lee handbook chapter, doi:10.4324/9781315740713-34).

FACT (four aspects — construct literature): Lee & Johnston-Wilder summarize mathematical resilience as four pragmatic aspects:

1. **Growth** — belief that mathematical capability can develop with effort and appropriate support (growth/incremental theory of learning).
2. **Value** — mathematics has personal / world value and the learner is valued in the learning community.
3. **Knowing how to work at mathematics** — struggle is expected; strategies exist for stuckness.
4. **Knowing how to recruit support** — peers, adults, resources; stuck is not permanent solitude.

FACT (Growth Zone Model): Johnston-Wilder et al. (2013 onward; e.g. pre-service teacher applications, Warwick WRAP) — three zones:

- **Comfort (green):** routine; little growth.
- **Growth (amber/yellow):** challenge requiring support, persistence, agency; productive struggle.
- **Danger / anxiety (red):** challenge perceived beyond capability even with support; survival response; repeated exposure → persistent anxiety / avoidance.

HYPOTHESIS (product): MindCraft's "soft-wrong + cards + tutor" stack is literally a **support-recruitment machine** for staying in the growth zone. Competitors that only escalate difficulty (or shame help) shove Maya into red.

Bridge to SAFE-DD (XLI): Desirable difficulties belong in the *growth* zone with equipment and destaking — not in the danger zone. "Struggle" ≠ "panic."

XLV.5 Measurement — MRS is attitudes, not magic scores

FACT: Kooken, Welsh, McCoach, Johnston-Wilder & Lee (2016, *Measurement and Evaluation in Counseling and Development*, 49(3), 217–242, doi:10.1177/0748175615596782) — developed and validated the **Mathematical Resilience Scale (MRS)** via EFA/CFA across three samples. Three correlated factors: **Value**, **Struggle**, and **Growth**. Positioned as a way to gauge likelihood of participation and persistence in mathematics.

Product implication (HYPOTHESIS): MRS-style items (value / struggle-as-normal / growth belief) can inform parent research and longitudinal identity panels — but they are **not** a North Star and not a gamified in-app grit meter. Self-report attitudes ≠ transfer. Pair with `retry_120s`, `challenge_accept`, `transfer_pass`, help-recruitment events (opened card / asked tutor / peer).

Kill: Shipping a "Resilience Score™" badge as proof of identity change (L0 → L3 leap; Part XXXIV hygiene).

XLV.6 Grit theater — what we refuse to sell

FACT (origin): Duckworth, Peterson, Matthews & Kelly (2007, *Journal of Personality and Social Psychology*, 92(6), 1087–1101, doi:10.1037/0022-3514.92.6.1087) — grit as perseverance and passion for long-term goals; predictive claims in selective samples fueled popular education discourse.

FACT (meta wound): Credé, Tynan & Harms (2017, *Journal of Personality and Social Psychology*, 113(3), 492–511, doi:10.1037/pspp0000102) — 584 effects, 88 samples, N≈66,807: higher-order grit structure not confirmed; grit **very strongly** overlaps conscientiousness; overall grit–performance/retention relations **modest**; perseverance-of-effort facet carries more criterion validity than consistency-of-interest; **interventions to enhance grit may have only weak effects**; construct validity questioned; primary utility may lie in perseverance facet alone.

FOUNDER BELIEF / resolution: Perseverance *after equipped struggle* remains commercially meaningful (`retry_120s`). Packaging that as Duckworth "grit" invites overclaim, personality fatalism, and equity harm (blame the child's character when curriculum/support failed). Prefer **mathematical resilience** language: growth zone + support recruitment + struggle-as-normal.

Standing Constitution already warned (core OS): grit/resilience hype is contested; structural inequality and curriculum quality matter more than character slogans. This chapter densifies that kill with Credé + JW/Lee specificity.

Kill: "MindCraft builds grit." **Survive (constrained):** "MindCraft keeps you in the growth zone — challenge with support, recovery without shame."

XLV.7 Competitive audit — who shoves students into the red zone?

Pattern	Zone reading	MindCraft wedge
Duolingo streak shame / league pressure	Danger + appearance climate (XLII/XLIII); "grit" via fear of break	Miss ≠ identity failure; cue practice not counter; no grit KPI
Khan mastery lock without affect design	Can strand in stuck without emotional exit tools	Soft-wrong + ingredient cards as support recruitment
Brilliant hard delight	Often growth-zone when curiosity high; weak for high-anxiety Maya	SAFE-DD equip/destake before delight escalation
Unguarded ChatGPT "just try harder / here's the answer"	Either skips struggle (comfort via answer) or fluent wrong confidence	PWC + solo transfer; support = pedagogy wrap not answer dump
School "resilience assembly"	Poster without recruit-support architecture	Operationalize support buttons in-session

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XLV.8 Resolution doctrine — SAFE-RESILIENCE

FOUNDER BELIEF / HYPOTHESIS — SAFE-RESILIENCE stack (ties FEI / SAFE-DD / TARGET / SAFE-REWARD / EVT value):

1. **Name the zones in product language.** Comfort / growth / danger — coach and tutor briefs use this, not "be gritty."
2. **Equip before escalate.** SAFE-DD first: representations, faded examples, destaked attempts — then raise challenge.
3. **Struggle is expected, not character failure.** Attribution copy (XXX): strategy and effort-with-strategy; never "you're not resilient enough."
4. **Support recruitment is a first-class action.** Cards, ask-tutor, peer, worked scaffold — log as positive resilience behavior, not as "cheating the streak."
5. **Exit the danger zone fast.** Anxiety spike → reduce stakes, shorten item, switch representation, offer support — do not double dose "desirable difficulty."
6. **Value without coercion.** Eccles utility/attainment (XXXVII) and story worlds — not "suffer because grit."
7. **Measure recovery and transfer, not grit scores.** `retry_120s`, help-then-solo sequences, `challenge_accept`, `transfer_pass` / `solo_transfer_pass`; optional MRS-like survey offline.
8. **Parent copy:** Sell "she got stuck, recruited help, and came back" — not "she's a gritty kid now."

Kill the false dichotomy: "Coddle forever in comfort" vs "forge character in the red zone." Refuse both. Ship **growth-zone challenge with support on tap**.

XLV.9 Marketing language that survives

- **Allowed:** "Hard math with a way out — support when you're stuck, challenge when you're ready."
- **Allowed:** "Stuck isn't failure. Stuck is when you recruit the next tool."
- **Allowed (parent):** "We train recovery and transfer — not grit slogans."
- **Banned:** "We build grit." / "Resilience score." / "Push through the panic."
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as proof of resilience; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without RES data
Growth-zone framing + support CTAs in coach	L1 hygiene aligned with JW/Lee
Soft-wrong + card recruitment → retry	L1 mechanism claim
"Raises mathematical resilience (MRS)" as product outcome	L0 until RES-1
"Builds grit / character"	L0 — banned packaging

XLV.10 Experiments spawned

ID	Question	Design	Primary	Kill condition

RES-1	Growth-zone coach copy + explicit support CTA vs grit/perseverance copy (same difficulty)	2-arm; anxiety-stratified	`retry_120s`; help-recruit rate; post affect; `transfer_pass`	Grit copy ≥ growth-zone on retry+transfer without raising danger affect → zone framing overstated
RES-2	Support-first after miss (card/tutor offer) vs no offer (same soft-wrong)	2-arm	Time-to-retry; danger-zone self-report; solo transfer later	No-offer ≥ support-first on retry without harm → recruitment less load-bearing
RES-3	Escalate difficulty after equip vs escalate without equip (SAFE-DD×RES)	2×2	Challenge_accept; WM/anxiety proxy; transfer	Unequipped escalate wins transfer in high-anxiety → SAFE-DD sequencing weak
RES-4	MRS Value/Struggle/Growth Δ over 4 weeks of SAFE-RESILIENCE UX vs control	Survey + behavior	MRS factors + behavioral NS	MRS rises, behavior flat → attitude theater
RES-QUAL	10 Maya interviews: “When you’re stuck in math, what does ‘being resilient’ mean?”	Appendix B	coded grit-vs-support meanings	Everyone already means support recruitment → copy shift lower priority

Pre-reg (XXXIV): RES-* identify **zone framing × support recruitment × recovery/transfer**. They do **not** identify “we create resilient mathematicians” from MRS alone.

Densifies: SAFE-DD; FEI; TARGET; standing grit-hype ban; help-seeking as virtue not streak cheat.

XLV.11 Confidence table

Claim	Label	Confidence
Developmental resilience often arises from ordinary protective systems	FACT (Masten, 2001)	High

Mathematical resilience = growth + value + how-to-work + recruit support	FACT (JW/Lee construct line)	High (construct); Medium (intervention effect sizes sparse)
Growth Zone Model: comfort / growth / danger	FACT (JW et al. model literature)	High as pedagogy model; Medium as RCT-proven UX
MRS measures Value, Struggle, Growth attitudes	FACT (Kooken et al., 2016)	High
Grit higher-order structure weak; overlaps conscientiousness; modest prediction; weak intervention leverage	FACT (Credé et al., 2017)	High
Danger-zone exposure as “training” harms anxious learners	HYPOTHESIS (Ashcraft line + SAFE-DD)	Medium–High
SAFE-RESILIENCE is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Grit posters raise transfer” / “Resilience Score™ = identity”	SPECULATION / false as doctrine	Low (against)

XLV.12 What this chapter kills

1. **Kill:** Grit / “build character” as MindCraft marketing North Star. (Credé 2017; Constitution grit warning.)
2. **Kill:** Treating panic / danger-zone as desirable difficulty. (Growth Zone Model + SAFE-DD.)
3. **Kill:** Resilience as rare trait branding (“extraordinary kids”). (Masten ordinary magic.)
4. **Kill:** In-app Resilience Score™ as proof of identity transformation. (MRS ≠ transfer; claim ladder.)
5. **Wound:** Growth-mindset language alone — survives only inside growth + value + struggle know-how + **support recruitment**, not as wall art.
6. **Survive (constrained):** Growth-zone challenge with support on tap; struggle-as-normal; recovery metrics; parent story of “stuck → help → return.”

Doctrine until data: Ship **SAFE-RESILIENCE**: name zones → equip before escalate → struggle expected → support recruitment first-class → exit danger fast → value without coercion → measure recovery/transfer not grit. Market “hard math with a way out.”

Part XLVI — Teacher/Tutor Expectancy: Pygmalion Without Magic, Golem Without Labels

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 15; hour%6≠0 → Researcher; 5 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: What do teacher/tutor expectancy effects actually justify for MindCraft — and what briefing ethics keep college tutors from installing a Golem on Maya?

Owners: Tutor ops / Product (briefing cards) · Equity · Positioning · Red Team

Commercial job: Replace "we believe in every student" poster copy and "Pygmalion IQ" folklore with a shippable **SAFE-EXPECTANCY** stack — evidence-based briefs, class-level high challenge + support, ban stigmatizing labels, measure mediation behaviors not vibes.

XLVI.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution), XXXVI (equity), XXXVII (EVT expectancy), and XLV (support recruitment) already treat *student-side* "can I?" and belonging. They do not fully answer: **what beliefs do human tutors carry into the session, how do those beliefs leak into climate/input/output/feedback, and how should MindCraft's pre-session briefing be designed so gap-scan weakness does not become a self-fulfilling low label?**

FOUNDER BELIEF under audit: A knowledge-graph + tutor hybrid only compounds identity if the *human* in the loop treats Maya as capable of the next hard step with support — not as a fixed "weak student" whose low mastery scores license colder climate, thinner input, and fewer chances to respond.

Claims we refuse as doctrine:

1. Marketing the classic Pygmalion IQ story as MindCraft's differentiator ("our tutors raise IQ by believing harder").
2. Briefing tutors with trait labels ("low ability," "slow," "not STEM material") drawn from gap-scan or weakness ranking.
3. Treating teacher-expectancy literature as proof that positive vibes alone move transfer.
4. Ignoring Golem / differential-treatment risks for stigmatized or "weakness-tagged" students.

XLVI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Induced expectancy (Pygmalion paradigm)	Authority given false "bloomers" info → treats them differently → outcomes shift	Early-session / cold-start briefing window	Selling IQ magic; ignoring replication limits
Accuracy vs prophecy	Expectations often predict because they track real prior performance	Gap-scan / graph as *task* evidence, not character	Confusing prediction with causation
Golem effect	Low expectations → colder / more dogmatic treatment → worse performance	Ban negative trait labels in tutor cards	"Honest" weakness dump that licenses withdrawal

Four-factor mediation	Climate, input, output opportunity, feedback carry expectancy into behavior	Audit tutor/coach on those four channels	Soft climate without hard input/output
High-expectation teacher (class-level)	High challenge + high support for *all*, not only nominated stars	Tutor playbook: everyone gets hard Qs + scaffolds	Favorites / "bloomers only" tracks
SAFE-EXPECTANCY	Evidence brief → high challenge-with-support → no trait labels → mediate via CIOF → measure behaviors	See XLVI.7	Belief posters; Golem briefs

Operational definition (HYPOTHESIS): A MindCraft tutor session is expectancy-*safe* only when (a) the brief states **next playable tasks and scaffolds**, not trait judgments, (b) the tutor offers **hard-enough** work with support on tap (XLV growth zone), (c) climate stays warm under struggle, and (d) the student gets **output opportunity** (speak/solve before tutor finishes) — not when the tutor "believes harder" while finishing the math.

XLVI.3 Origin story — real, contested, not marketing copy

FACT: Rosenthal & Jacobson (1968, *Pygmalion in the Classroom*; also *Urban Review* summary, doi:10.1007/BF02322211) — teachers told that randomly selected elementary students were poised for intellectual blooming; those students later showed larger IQ gains on average in the reported analyses, especially in early grades. The study launched classroom expectancy research and intense methodological critique (e.g. Thorndike and later replication debates).

FACT (meta, induction credibility): Raudenbush (1984, *Journal of Educational Psychology*, 76(1), 85–97, doi:10.1037/0022-0663.76.1.85) — synthesis of 18 Pygmalion-style experiments: expectancy effects on pupil IQ are larger when teachers have had **little prior contact** with pupils at induction (≤~2 weeks); effects shrink when teachers already know students well. Grade pattern: stronger early-grade effects in the corpus; design features of induction matter.

Product translation: MindCraft's **first tutor sessions and cold-start briefs** are the high-leverage expectancy window — analogous to early-year induction. Once a tutor has rich behavioral evidence, "random bloomers" folklore is the wrong metaphor; **accurate, task-level** expectations dominate.

Kill: "Pygmalion proves we raise IQ by believing." Sell mediation design in the first hours of contact, not IQ alchemy.

XLVI.4 What 35 years actually say (Jussim & Harber)

FACT (review): Jussim & Harber (2005, *Personality and Social Psychology Review*, 9(2), 131–155, doi:10.1207/s15327957pspr0902_3) — after ~35 years:

1. Self-fulfilling prophecies in classrooms **do occur**, but effects are **typically small**, do not accumulate greatly across perceivers/time, and may dissipate more than snowball.

2. **Powerful** effects may selectively hit students from **stigmatized** social groups.
3. Whether expectancy effects reliably change *intelligence*, and whether they do more harm than good in general, remains unclear.
4. Teacher expectations often predict outcomes **because they are accurate**, not only because they are self-fulfilling.

HYPOTHESIS for MindCraft: Gap-scan / mastery graphs will *accurately* predict struggle. Accuracy is not permission to install a Golem. The commercial risk is **translating accurate weakness into low-challenge, low-warmth treatment** — exactly the path that can still produce small-but-real prophecy costs, amplified for already-stereotyped students (ties Part XXXVI equity).

Wound: "Expectancy effects are huge and ubiquitous." Survives only as: small average effects; context/stigma moderators; accuracy dominates prediction.

XLVI.5 Mediation — climate, input, output, feedback

FACT (behavioral mediation): Brophy & Good (1970, *Journal of Educational Psychology*, doi:10.1037/h0029908) — observational study: teachers demanded better performance and praised it more for high-expectation students; accepted poor performance more readily from lows and were less likely to praise good performance from lows when it occurred.

FACT (four-factor meta): Harris & Rosenthal (1985, *Psychological Bulletin*, 97(3), 363–386, doi:10.1037/0033-2909.97.3.363) — 31 meta-analyses of mediating behaviors; Rosenthal's four-factor frame (**climate**, **feedback**, **input**, **output**/opportunity opportunity) organizes how expectancies become treatment. Expectancy→behavior and behavior→outcome links support multiple mediators (e.g. warmer climate, longer/richer interaction, questioning). Later summaries often note **feedback** as the weaker of the four average links, pushing affect/effort emphasis.

Product checklist (HYPOTHESIS — CIOF audit):

Channel	Tutor/coach do	Anti-pattern
Climate	Warmth under struggle; curiosity about wrong paths	Cold distance after "weak" brief
Input	Richer explanations, multiple representations (cards)	Thin "just try" for lows; lectures only for highs
Output	Wait time; let Maya finish; SE prompts (Part XL)	Tutor finishes every thought (Solver crutch)
Feedback	Criterion + strategy (XXX); soft-wrong	Praise only highs; criticism without how-to

Kill: Climate-only positivity ("you've got this!") without input/output redesign — that is poster Pygmalion.

XLVI.6 Golem risk — negative expectancy is the ops hazard

FACT: Babad, Inbar & Rosenthal (1982, *Journal of Educational Psychology*, 74(4), 459–474, doi:10.1037/0022-0663.74.4.459) — distinguished positive expectancy outcomes ("Galatea") from **negative** ones ("Golem"). Teachers high in susceptibility to biasing information treated low-potential nominees more negatively (notably dogmatic behavior); unbiased teachers treated groups more equitably. Differential negative treatment showed up in teacher behavior **and** student task performance.

FACT (student-perceived differentiation): Brattesani, Weinstein & Marshall (1984, *Journal of Educational Psychology*, 76(2), 236–247, doi:10.1037/0022-0663.76.2.236) — in classrooms where students perceive **high differential treatment**, teacher expectations contribute more to achievement than in low-PDT classrooms. Students can *see* the track.

Product translation: A tutor card that says "Maya is weak on linear equations / low confidence / hard gap-scan" is biasing information. High-bias tutors will leak Golem through climate and dogmatism. Low-PDT design = same high standards + differentiated *scaffolds*, not differentiated *respect*.

Kill: "Honest labels help tutors calibrate." Honest **task evidence** helps; honest **trait verdicts** harm.

XLVI.7 High-expectation teachers — class-level, not favorites

FACT: Rubie-Davies (2007, *British Journal of Educational Psychology*, doi:10.1348/000709906X101601) — contrasts high- vs low-expectation *teachers* (expectations for the class as a whole). High-expectation teachers spent more time teaching, oriented lessons, linked prior knowledge, explained carefully, asked mixed open/closed questions including **high-level questions of all students**, and managed more positively; lows were more dismissive of incorrect answers and relied on closed questioning.

HYPOTHESIS: MindCraft's competitive wedge vs "AI that flatters then finishes" and vs "tutor who only drills the labeled weak topic gently" is **Rubie-Davies-shaped ops**: every student gets high-level questioning + scaffolds; incorrect answers get learning support, not dismissal.

Bridge to SAFE-RESILIENCE / SAFE-DD: High expectation ≠ danger zone. High expectation = growth-zone challenge with support recruitment — not grit theater and not thin input for "lows."

XLVI.8 Competitive audit (expectancy lens)

Competitor pattern	Expectancy mechanism	MindCraft counter
Duo	Habit/streak as "you're the type who shows up"; little academic expectancy mediation	Transfer + CIOF; no streak-as-belief
Khan	Self-serve library; weak human expectancy channel	Human tutor brief + graph without Golem labels
Brilliant	Prestige challenge; can cue performance-approach for novices	Mastery structures + expectancy-managed gates

ChatGPT Base	Sycophantic high-warmth climate + answer dump (fake high expectancy, zero output demand)	Warm climate **plus** forced solo output / PWC
Unbriefed college tutor	Forms expectancies from accent, race, “seems lost,” or raw weakness dump	SAFE-EXPECTANCY brief + playbook

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XLVI.9 Resolution doctrine — SAFE-EXPECTANCY

FOUNDER BELIEF / HYPOTHESIS — SAFE-EXPECTANCY stack:

1. **Brief tasks, not traits.** Tutor card: current concept, recommended level, formats struggled, successful scaffolds last time, next playable question IDs — **never** “low ability / not a math person.”
2. **Protect the cold-start window.** First sessions: explicit high-challenge-with-support script (Raudenbush induction logic).
3. **Class-level high expectation.** Rubie-Davies: hard questions for everyone; wrong answers → teach, don’t dismiss.
4. **Mediate via CIOF.** Climate warmth + rich input (cards) + output opportunity (wait / SE) + strategy feedback — audit these, not “belief intensity.”
5. **Accuracy without Golem.** Graph weakness drives *scaffold choice*, not *respect withdrawal*.
6. **Stigma premium.** Extra review when briefs could activate stereotype (XXXVI); ban demographic proxies in tutor UI.
7. **AI coach inherits the same rules.** No “you’re behind your peers” cold open; no finishing her thought under “support.”
8. **Measure mediation + FEI, not IQ folklore.** Tutor wait-time, student talk ratio, help-then-solo, `retry_120s`, `transfer_pass` / `solo_transfer_pass`.

Kill the false dichotomy: “Ignore data, just believe” vs “dump the weakness label so tutors are realistic.” Refuse both. Ship **evidence-based high challenge with equitable treatment**.

XLVI.10 Marketing language that survives

- **Allowed:** “Tutors see what to teach next — not who she is forever.”
- **Allowed:** “High challenge for everyone, support on tap — no tracks of respect.”
- **Allowed (parent):** “We brief tutors on the next skill, not on a fixed ability story.”
- **Banned:** “Pygmalion IQ gains.” / “Believe and they bloom.” / “We tell tutors who’s weak so they know.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as expectancy proof; tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without EXP data

Trait-label ban + task-only tutor brief	L1 hygiene aligned with Golem literature
CIOF playbook / wait-time coaching	L1 mechanism claim
“Raises IQ via tutor expectancy”	L0 — banned
“Eliminates achievement gaps via Pygmalion”	L0 until EXP-* + equity design (Jussim caveats)

XLVI.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
EXP-1	Task-only brief vs trait/weakness -label brief (same content otherwise)	2-arm tutors; record session	Student talk ratio; wait-time; post affect; `transfer_pass`	Label brief ≥ task-only on transfer without colder climate → Golem risk overstated for our tutors
EXP-2	High-expectation playbook (hard Q + scaffold) vs “remediate gently only” after same gap-scan	2-arm	`challenge_accept`; solo transfer; help-recruit	Gentle-only wins transfer+return in high-anxiety → Rubie-Davies transfer fails for Maya
EXP-3	Cold-start script (first 2 sessions) vs unstructured tutor open	2-arm new matches	FEI events week 1; retention to session 3	Unstructured ≥ scripted → induction window less load-bearing
EXP-4	AI coach: warm+finish vs warm+output-demand (SE/wait)	2-arm	`solo_transfer_pass`; PWC	Warm+finish ≥ warm+output on solo transfer → climate-only expectancy myth revived
EXP-QUAL	10 tutor + 10 Maya: “What did the brief make you assume?”	Interview	coded trait vs task inferences	Tutors already ignore labels → UI priority shifts

Pre-reg (XXXIV): EXP-* identify **brief content** × **CIOF mediation** × **transfer**. They do **not** identify “we create bloomers” or IQ change.

Densifies: Equity (XXXVI); EVT expectancy; SAFE-RESILIENCE support; FEI; tutor ops queue item 68.

XLVI.12 Confidence table

Claim	Label	Confidence
Rosenthal & Jacobson launched expectancy research; original IQ story is contested	FACT	High
Induced expectancy effects larger when teacher knows student less (Raudenbush, 1984)	FACT	High
Average self-fulfilling effects small; accuracy often drives prediction; stigma can amplify (Jussim & Harber, 2005)	FACT	High
Differential treatment observable; highs get more demand/praise (Brophy & Good, 1970)	FACT	High
Four-factor mediation frame useful (Harris & Rosenthal, 1985)	FACT	High (framework); Medium (which channel dominates in tutoring apps)
Golem / high-bias negative treatment real (Babad et al., 1982)	FACT	High
High-expectation teachers differ in practice for whole class (Rubie-Davies, 2007)	FACT	High (observational/descriptive line); Medium (causal ops transfer)
SAFE-EXPECTANCY is right default for MindCraft tutor briefing	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Pygmalion IQ” / belief posters as product claim	SPECULATION / false as doctrine	Low (against)

XLVI.13 What this chapter kills

1. **Kill:** Marketing classic Pygmalion IQ blooming as MindCraft proof. (Raudenbush limits; Jussim accuracy; claim ladder.)
2. **Kill:** Tutor briefs that install trait/weakness identity labels. (Babad Golem; Weinstein PDT.)
3. **Kill:** Climate-only positivity without input/output demand. (Harris & Rosenthal mediation; Bastani/PWC line.)
4. **Kill:** “Just be realistic — tell tutors she’s weak” as ops hygiene.
5. **Wound:** Expectancy as a large, cumulative social-problem engine — survives only as small average effects with stigma/PDT moderators.
6. **Survive (constrained):** Cold-start high-challenge-with-support; task-only briefs; CIOF audit; Rubie-Davies whole-class high expectation; measure mediation + transfer.

Doctrine until data: Ship **SAFE-EXPECTANCY**: brief the next skill, not the fixed self; high challenge for everyone with scaffolds; warm climate under struggle; force student output; never weaponize gap-scan as a Golem tag. Market “tutors see what to teach next — not who she is forever.”

Part XLVII — Sleep, Stress, and Learning: Timing Practice Without All-Nighter Theater

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 18 ≡ Red Team slot, but ch47 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: How should MindCraft time practice and exam-week product behavior given sleep-dependent consolidation and stress/pressure drain on working memory — without becoming a sleep-coaching app or glorifying late-night grind?

Owners: Product (session timing / exam modes) · Parent messaging · Affect pipeline · Red Team

Commercial job: Ship a **SAFE-TIMING** stack — overnight spacing as consolidation infrastructure, exam-week load softening, ban streak/all-nighter heroism, measure delayed transfer not midnight minutes.

XLVII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXIV/XLI (anxiety × desirable difficulty), XXXI (flow/challenge–skill), and XLV (growth vs danger zone) already treat *within-session* difficulty and affect. They do not fully answer: **when** should Maya practice relative to sleep, and what should the product do in high-stakes stress weeks when encoding, attention, and WM are compromised?

FOUNDER BELIEF under audit: Identity transformation requires recoverable struggle *and* biological conditions that let struggle stick. A product that pushes “one more mission” at 1 a.m. during exam week can look engaged while sabotaging consolidation and installing danger-zone affect.

Claims we refuse as doctrine:

1. Marketing MindCraft as a sleep-hygiene / wellness brand (“we fix teen sleep”).
2. Celebrating all-nighters, midnight streaks, or “grind hours” as learning proof.
3. Treating acute exam pressure as desirable difficulty.
4. Absolute claims that “sleep always consolidates math” independent of encoding quality, task type, and sleep opportunity.

XLVII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Sleep-dependent consolidation	Offline reprocessing after encoding that stabilizes/reshapes memory	Space practice across nights; delay transfer checks	Sell “sleep = magic” without encoding first
Encoding under restriction	Multi-night short sleep impairs *new* learning capacity	Soften new-concept loads after chronic short sleep flags	Blame student “laziness” for restriction biology
Pressure / choke	High-stakes appraisal consumes WM needed for hard math	Destake practice; hide peer ranks in exam week	Treat choke as low ability
Math-anxiety dual-task	Worry acts like a secondary load on WM	SAFE-DD before hard items	Add timers/leaderboards under anxiety
SAFE-TIMING	Time practice for consolidation + protect WM under stress	See XLVII.8	Sleep-app feature creep; grind theater

Operational definition (HYPOTHESIS): A MindCraft practice plan is timing-*safe* when (a) hard encoding sessions are followed by a real sleep opportunity before high-stakes transfer claims, (b) multi-night sleep debt / high stress triggers **load softening** (fewer new concepts, more retrieval of known, support on tap), and (c) engagement metrics never reward post-midnight volume as mastery.

XLVII.3 Sleep consolidates — and evolves — memory

FACT (systems review): Diekelmann & Born (2010, *Nature Reviews Neuroscience*, 11, 114–126, doi:10.1038/nrn2762) — sleep optimizes consolidation depending on learning conditions and sleep timing; slow-wave sleep (SWS) and REM support complementary system- vs synaptic-consolidation roles; consolidation can change representations quantitatively *and* qualitatively.

FACT (broader offline processing): Stickgold (2005, *Nature*, 437, 1272–1278, doi:10.1038/nature04286) — converging evidence that offline memory reprocessing during sleep is a major component of how memories are formed and shaped (not a single all-or-none event).

FACT (integrative/schema angle): Walker & Stickgold (2010, *Nature Reviews Neuroscience*, 11, 218, doi:10.1038/nrn2762-c1) — beyond veridical preservation, sleep (often REM-linked in their framing) supports unitization, assimilation into remote networks, and abstraction of rules/schemas — relevant to MindCraft’s transfer goal, not only verbatim item memory.

FACT (mechanism exemplar): Gais & Born (2004, *PNAS*, 101(7), 2140–2144, doi:10.1073/pnas.0305404101) — elevating acetylcholine during SWS-rich sleep blocked declarative word-pair consolidation while sparing a nondeclarative skill task — evidence that the *neurochemical milieu* of SWS matters for declarative consolidation, not “rest” generically.

Product translation: Spacing missions across nights (Part XXIX) is not only spacing theory — it is a **consolidation window**. Claiming “exam-ready” from a same-day blocked accuracy binge without overnight delay is scientifically lazy (ties XXXIX readiness signal).

Kill: “Sleep replaces practice.” Consolidation operates on what was encoded. No encoding → nothing to consolidate.

XLVII.4 Adolescent sleep restriction is causal for learning damage

Maya is a high-schooler. Population sleep debt is not a vibe; experimental restriction shows classroom-relevant harm.

FACT (simulated classroom): Beebe, Rose & Amin (2010, *Journal of Adolescent Health*, 47(5), 523–525, doi:10.1016/j.jadohealth.2010.03.005) — within-subject multi-night restriction vs healthy duration: sleep-restricted adolescents showed poorer learning, more inattentive behaviors, and lower arousal in a simulated classroom (medium-to-large effects; small *n*, preliminary but experimental).

FACT (review): Beebe (2011, *Pediatric Clinics of North America*, doi:10.1016/j.pcl.2011.03.002) — synthesizes cognitive, behavioral, and functional consequences of inadequate sleep in children/adolescents; chronic restriction is common and not benign.

FACT (long-term factual retention): Cousins, Wong & Chee (2019, *Journal of Adolescent Health*, 65(4), 549–557, doi:10.1016/j.jadohealth.2019.04.030) — adolescents (15–18) with 5 h sleep opportunity for 5 nights vs 9 h controls: after four restriction nights, learning detailed factual material showed **more forgetting** at 30 min (~26%), Day 3 (~34%), and Day 42 (~65%) relative to controls. Vigilance also fell; memory impairment did not simply track vigilance correlation.

HYPOTHESIS for MindCraft: A “productive” ACT cram week of late missions on short sleep can **look** like engagement while destroying the encoding that gap-scan and practice are supposed to build. Parent-visible “minutes practiced” without sleep context is a vanity metric.

Wound: “Weekend catch-up fixes the week.” Literature flags incomplete restitution and cumulative cost (see Beebe 2019-line summaries); do not market weekend binge as full repair.

XLVII.5 Stress and pressure — WM theft, not character failure

Sleep loss and evaluative stress often co-occur in exam weeks. Separate the mechanisms.

FACT (math anxiety × WM): Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, 14, 243–248, doi:10.3758/BF03194059) — math performance depends on WM for beyond-retrieval arithmetic; high math anxiety behaves like a dual-task (worry consumes capacity); anxious students also avoid math pathways. Complements Ashcraft & Kirk (2001) line already in Part XXIV.

FACT (choking under pressure): Beilock & Carr (2005, *Psychological Science*, 16(2), 101–105, doi:10.1111/j.0956-7976.2005.00789.x) — performance pressure harmed **high**-WM individuals on the most WM-demanding math problems — pressure consumes the capacity their superior strategies rely on. Not “dumb under stress”; often “capable under load.”

Bridge to SAFE-DD / FEI: Exam-week timers, public ranks (XLII), and “prove you’re ready tonight” framing are pressure injectors. Desirable difficulty assumes encoding capacity; pressure + anxiety can erase the dose.

HYPOTHESIS: Affective check-in stress > 0.7 already softens `target_mastery` in `/recommend` — that is the correct *direction*. Extend it: under high stress / self-reported short sleep, prefer retrieval + support recruitment over new-concept flooding and appearance UX.

Kill: "Pressure makes diamonds" as practice design. Pressure makes WM poorer for the problems that need it most.

XLVII.6 Competitive audit (timing lens)

Competitor pattern	Timing / stress mechanism	MindCraft counter
Duo	Streaks + late-night nagging; habit over consolidation	SAFE-HABIT + SAFE-TIMING: miss≠identity; no midnight streak heroism
Khan	Infinite library; learner can cram to exhaustion	Spaced missions + delayed transfer as readiness
Brilliant	Delight / prestige sessions; can invite performance pressure	Mastery climate; destake under stress
ChatGPT Base	Always-on 2 a.m. answers; no consolidation pedagogy	Human/AI wrap that schedules *struggle then sleep*, not answer dumps
Tutor shops	Pack hours before exam Saturday	Brief tutors: encode early in week; retrieval near exam; never shame sleep

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XLVII.7 Exam weeks — special product mode, not harder grind

SPECULATION → product hypothesis: Default "more missions" during ACT week maximizes anxiety and restriction. Safer mode:

1. **Front-load encoding** earlier in the week when sleep opportunity is better.
2. **Shift to retrieval + mixed review** 48–72 h pre-exam (XXIX/XXXIX).
3. **Lower appearance cues** (no leaderboards, no streak threats).
4. **Keep growth-zone items** but exit danger fast (XLV); prefer scaffolds over new ontology leaps.
5. **Parent copy:** sleep is part of the study plan — not "rest is for quitters."

FOUNDER BELIEF: Selling calm competence under exam load beats selling heroic sleep debt.

XLVII.8 Resolution doctrine — SAFE-TIMING

FOUNDER BELIEF / HYPOTHESIS — SAFE-TIMING stack:

1. **Encode → sleep → transfer claim.** Treat overnight delay as part of readiness (with XXIX spacing), not optional wellness.
2. **Never reward midnight volume.** Ban streak/XP incentives that spike after local quiet hours; engagement ≠ consolidation.
3. **Restriction-aware load.** If multi-night short sleep or high stress is signaled (check-in, parent note, exam-week flag): fewer new concepts, more retrieval, CIOF support (XLVI), SAFE-DD destake.
4. **Pressure ≠ DD.** Timers, ranks, and “prove it tonight” are kill switches under high anxiety/stress.
5. **Exam-week mode.** Soften novelty; maximize recoverable retrieval; protect sleep opportunity messaging.
6. **Do not become a sleep clinic.** Nudge timing and load — do not diagnose disorders or sell melatonin theater.
7. **Tutor briefs include timing.** “Practice early evening when possible; protect sleep before transfer checks” — task ops, not character judgment.
8. **Measure delayed `transfer_pass` / `solo_transfer_pass`,** not minutes-after-midnight.

Kill the false dichotomy: “Ignore biology, just grind” vs “pivot the company into sleep coaching.” Refuse both. Ship **timing-aware learning load**.

XLVII.9 Marketing language that survives

- **Allowed:** “Hard practice sticks better when she sleeps on it.”
- **Allowed:** “We don’t measure mastery at 1 a.m.”
- **Allowed (parent):** “Exam week ≠ all-nighter week — we shift to review and protect sleep.”
- **Banned:** “Our app fixes teen sleep.” / “Grind while they sleep.” / Streak-as-study-ethic.
- **Banned:** Bloom 2-sigma; empty mindset; tutoring-is-free; pressure-as-pedagogy.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without TIM-* data
Overnight spacing before transfer checks	L1 mechanism aligned with consolidation reviews
Exam-week load softening + destake	L1 product hygiene under Ashcraft/Beilock
“Sleep mode raises ACT scores”	L0–L1 only with pre-reg; not L3 identity
Sleep-clinic / wellness brand claims	L0 — banned

XLVII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition

TIM-1	Same content: practice ending ≥1 h before habitual bedtime vs late-night block; transfer next evening	2-arm	next-day / 48h `transfer_pass`	Late ≥ early → consolidation timing overstated for our items
TIM-2	Exam-week mode (retrieval+ destake) vs "more new missions" in last 5 days pre-mock	2-arm	mock score; affect; sleep opportunity self-report	More-new wins mock+return → softening wrong default
TIM-3	High-stress check-in → auto soft path vs ignore stress	2-arm	`retry_120s`; WM-demand item accuracy; return	Ignore ≥ soft on transfer without affect cost → modifier useless
TIM-4	Parent message: "sleep is study" vs "push one more hour"	message A/B	next-night session timing; parent WTP/trust	Push-hour increases volume *and* next-day transfer → grind narrative survives
TIM-QUAL	10 Maya exam-week diaries: when practice helped vs harmed	interview	coded sleep debt × danger zone × cram	Students already self-time well → UI priority drops

Pre-reg (XXXIV): TIM-* identify **timing × load × delayed transfer**. They do **not** identify clinical sleep treatment or ACT score miracles from a bedtime tip.

Densifies: XXIX spacing; XLI SAFE-DD; XLII compare ban; XLIII habit without colonization; XLV danger exit; affect check-in; Part LXIII exam-pressure queue.

XLVII.11 Confidence table

Claim	Label	Confidence
Sleep supports declarative/procedural consolidation with stage-sensitive mechanisms (Diekelmann & Born, 2010; Stickgold, 2005)	FACT	High

Sleep can support schema/abstraction beyond verbatim storage (Walker & Stickgold, 2010)	FACT	Medium–High (active research debate on REM vs SWS mapping)
Low ACh during SWS matters for some declarative consolidation (Gais & Born, 2004)	FACT	High (specific manipulation); Medium (product generality)
Multi-night adolescent restriction impairs classroom learning/attention (Beebe et al., 2010; Beebe, 2011)	FACT	High (direction); Medium (effect sizes / sample limits)
Restriction impairs long-term factual retention in teens (Cousins et al., 2019)	FACT	High
Math anxiety consumes WM (Ashcraft & Krause, 2007)	FACT	High
Pressure can choke high-WM math performance (Beilock & Carr, 2005)	FACT	High
SAFE-TIMING is right default for MindCraft scheduling/exam mode	FOUNDER BELIEF / HYPOTHESIS	Medium–High
MindCraft-as-sleep-app is a good growth wedge	SPECULATION / false as doctrine	Low (against)

XLVII.12 What this chapter kills

1. **Kill:** All-nighter / midnight-streak heroism as learning culture. (Cousins retention; Beebe classroom; SAFE-HABIT.)
2. **Kill:** Treating exam pressure and timers as desirable difficulty. (Beilock choke; Ashcraft dual-task; SAFE-DD.)
3. **Kill:** Same-day cram accuracy as exam-ready proof without overnight/delayed transfer. (Diekelmann/Born; XXIX/XXXIX.)
4. **Kill:** Pivoting brand into clinical sleep wellness.
5. **Wound:** “Sleep always helps every math skill equally.” Survives only as: condition- and task-dependent consolidation; encoding first.
6. **Survive (constrained):** Encode→sleep→claim; restriction/stress-aware load; exam-week retrieval mode; parent “sleep is study”; measure delayed transfer.

Doctrine until data: Ship **SAFE-TIMING**: practice hard enough to encode, sleep enough to consolidate, soften novelty under debt and pressure, never sell grind-at-1-a.m. as identity. Market “hard practice sticks better when she sleeps on it.”

Part XLVIII — Transfer & Variation Theory: Far Transfer Design in the Ontology

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 21; hour%6≠0 → Researcher; 7 researcher entries since synthesizer v1.6 → not yet Synthesizer; this tick becomes the 8th)

Primary question: How should MindCraft design question sequences and “transfer” claims so Maya discerns *critical mathematical structure* — without selling story-world immersion or blocked accuracy as far transfer?

Owners: Product (questionBank / ontology / pathfinder) · Content ops · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-TRANSFER** stack — deliberate variation/invariance for discernment, hugged near transfer toward ACT formats, bridged high-road abstraction only where designed, measure delayed/format/solo transfer — kill “narrative = transfer for free.”

XLVIII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), XL (self-explanation), XLI (SAFE-DD), and XLVII (encode→sleep→claim) already constrain *when* and *how hard* practice should feel. They do not fully answer: **what pattern of examples** makes a concept transferable, and what marketing language about “transfer” survives evidence?

MindCraft’s ontology already encodes ingredients, bridges, combinations, formats, and archetypes. The risk is product language that treats any format change, story skin, or mixed set as “far transfer” — while competitors sell fluency theater as readiness.

FOUNDER BELIEF under audit: Identity as a mathematical thinker requires *flexible discernment* of structure across contexts — not one beautiful tutorial problem solved once. That flexibility is **designed**, not an automatic side-effect of lore.

Claims we refuse as doctrine:

1. Story worlds automatically produce far transfer.
2. Blocked same-day accuracy ≡ exam-ready / transferable skill.
3. “We teach for transfer” without specifying content×context dimensions (Barnett & Ceci).
4. Random interleaving or random story skins as a substitute for *principled* variation of critical aspects.

XLVIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Near / far transfer	Application of learning across some distance in content and/or context	Label every transfer claim with *which* dimensions moved	Vague “far transfer” marketing

Low-road / high-road	Automatic similarity trigger vs mindful abstraction	Hugging (ACT-like items) vs bridging (explicit compare/abstract prompts)	Expect far transfer from low-road drill alone
Object of learning	What is to be discerned (content + critical aspects)	Ingredient / bridge as the intended discernment target	Teach "one thing" without opening critical dimensions
Variation / invariance	What changes vs what stays fixed across a sequence	Example sets that separate critical aspects	Vary everything at once → noise; vary nothing → surface fixation
SAFE-TRANSFER	Transfer claims + sequences that survive evidence	See XLVIII.8	Story-skin theater; blocked-accuracy readiness

Operational definition (HYPOTHESIS): A MindCraft mission is transfer-*safe* when (a) the sequence makes at least one critical aspect *discernable* via planned contrast/generalization/separation/fusion, (b) "transfer" checks change a declared Barnett–Ceci dimension (format, surface story, temporal delay, or solo vs scaffolded), and (c) marketing never claims far transfer from immersion alone.

XLVIII.3 Far transfer is rare without design — name the dimensions

FACT (taxonomy): Barnett & Ceci (2002, *Psychological Bulletin*, 128(4), 612–637, doi:10.1037/0033-2909.128.4.612) — a century of "does far transfer occur?" debates compared apples to oranges; they propose a nine-dimension content×context framework. Estimating a *single* far-transfer effect size is misguided; evidence is substantial under some conditions and untested under others.

FACT (prosecution): Detterman (1993, in Detterman & Sternberg, Eds., *Transfer on Trial*, Ablex) — argues spontaneous far/general transfer is rare; apparent transfer often reflects cues/hints or other mechanisms; humans more often learn specific facts and patterns than free-floating abstractions.

FACT (analogical noticing failure): Gick & Holyoak (1980, *Cognitive Psychology*, 12, 306–355, doi:10.1016/0010-0285(80)90013-4) — subjects who read an analogous military "fortress" story often failed to spontaneously apply the convergence principle to Duncker's radiation problem; **hints** dramatically raised solution rates. Structural isomorphism ≠ noticed transfer.

Product translation: MindCraft must not market "she'll just see it on the ACT" after a kitchen-world linear-equations arc. Spontaneous far transfer is the *exception*. Design either **hugs** the exam context or **bridges** with explicit abstraction (compare two instances; name the invariant).

Kill: "Our stories create far transfer." Stories can raise utility/attainment (XXXVII) and belonging (XXXVI); they do not substitute for noticing structure.

XLVIII.4 Mechanisms that *do* teach for transfer — low road, high road, hug, bridge

FACT (mechanisms): Salomon & Perkins (1989, *Educational Psychologist*, 24(2), 113–142, doi:10.1207/s15326985ep2402_1) — **low-road** transfer: extensive, varied practice → automatic triggering in similar stimulus conditions; **high-road** transfer: mindful abstraction, forward- or backward-reaching search for connections. Findings of transfer/nontransfer reflect whether those conditions were met.

FACT (instructional strategies): Perkins & Salomon (1992, "Transfer of Learning," *International Encyclopedia of Education*, 2nd ed.; building on Perkins & Salomon, 1988, *Educational Leadership*, 46(1), 22–32) — **hugging** (practice that approximates target performance) vs **bridging** (prompt abstraction, metacognition, planned connections). Education gets transfer when designed for it — not by hoping.

HYPOTHESIS for MindCraft:

- **Hug:** format-tagged ACT/GCSE-like items, timed retrieval near exam, archetype-matched instances (Layer 2/3) after ingredient mastery.
- **Bridge:** coach prompts that force comparison ("what stayed the same?"), self-explanation of the invariant (XL), and bridge-gap cards when two concepts are known but unconnected.
- **Low-road alone** (Duo-like drill skins) will not deliver ACT problem-type novelty. **High-road alone** (chatty abstraction without varied instances) will not stick.

Wound: Absolute "transfer can't be taught" (hard Detterman reading). Survives only as: spontaneous far transfer is rare; *cued and designed* transfer is the product job.

XLVIII.5 Variation theory — discernment requires planned difference

FACT (foundations): Marton & Booth (1997, *Learning and Awareness*, Lawrence Erlbaum) — learning is experiencing an object of learning in a new way; discerning critical aspects requires experiencing patterns of **variation and invariance**.

FACT (math classroom design): Kullberg, Runesson Kempe & Marton (2017, *ZDM Mathematics Education*, 49, 559–569, doi:10.1007/s11858-017-0858-4) — variation theory as a design principle changes *what is made possible to learn*; teachers using VT systematically open critical aspects (e.g., relating concepts rather than teaching one-at-a-time isolation); example sets matter, and teaching must **draw attention** to the patterns (presence of variation ≠ experienced variation).

FACT (dimensions of possible variation): Watson & Mason (line; e.g., Watson & Mason, 2005/2006 math-education elaborations of Marton) — to learn a mathematical concept is to become aware of which aspects can vary while remaining an example, and over what **range of permissible change**. Teacher and student may stress different dimensions.

FACT (Chinese bianshi tradition): Gu, Huang & Marton (2004, in Fan et al., Eds., *How Chinese Learn Mathematics*, World Scientific, pp. 309–347, doi:10.1142/9789812562241_0012) — Chinese "teaching with variation" systematically juxtaposes conceptual and procedural variants so students deepen structure; indigenous practice, not a Western bolt-on.

FACT (curriculum variation problems): Sun (2011a, *Educational Studies in Mathematics*, 76(1), 65–85, doi:10.1007/s10649-010-9263-4) — Chinese textbook "variation problems" organize connectedness (one problem, multiple changes/solutions) to highlight invariants; contrasts with curricula that treat problems as isolated context decorations.

Product translation for the ontology:

1. Pick the **object of learning** (ingredient or bridge), not “finish five questions.”
2. Hold the critical structure **invariant**; vary one dimension at a time (numbers → representation → story skin → exam archetype).
3. After separation, **fuse** (vary jointly) only when WM/SAFE-DD allows (XXVI/XLI).
4. Prompt noticing (“what changed / what stayed the same?”) — ties XL self-explanation.
5. Interleaving (XXXIX) is related but not identical: IL trains discrimination among *categories*; VT trains discernment of *critical aspects within* an object of learning. Use both deliberately.

Kill: “More random story skins = better transfer.” Random surface change without invariant focus is entertainment variance, not Martonian variation.

XLVIII.6 Competitive audit (transfer / variation lens)

Competitor pattern	Transfer mechanism	MindCraft counter
Khan	Huge item pool; learner may block by topic; transfer left to chance	VT sequences + delayed mixed transfer checks
Duo	Low-road habit; skin-deep variety; streaks as proof	SAFE-HABIT + SAFE-TRANSFER: variety serves discernment, not XP
Brilliant	Elegant interactive instances; may under-specify exam hug	Hug ACT formats after insight; measure `transfer_pass`
ChatGPT Base	Fluent explanations; weak structured example contrast; sycophancy risk (XXXIII)	Deterministic card/path spine; SE + contrast prompts; no monologue≡mastery
Tutor shops	Ad-hoc worksheets; expert tutors often vary well by instinct	Codify variation patterns in bank + tutor briefs (CIOF-compatible, XLVI)

XLVIII.7 Ontology / bank design implications (MindCraft-specific)

HYPOTHESIS — variation layers already latent in the stack:

- **Ingredient card styles** (geometric / algebraic / procedural) = representation variation with concept held.
- **Format axis** (`word_problem`, `diagram`, `symbolic_expression`, …) = context/content-adjacent transfer dimensions — prefer matched then force a declared hop.
- **Bridges** = high-road objects of learning (connection itself is the critical aspect).
- **Archetypes (Layer 2)** = exam-pattern hug targets after near mastery.

- **Combinations** = fusion stage (multiple ingredients co-activated) — do not dump at cold start.

SPECULATION → product rule: Generation and bank curation should tag each practice *bundle* with: `invariant_aspect`, `varied_dimension`, `transfer_check_dimension`. Untagged bundles default to "drill," not "transfer mission."

FOUNDER BELIEF: Pathfinder recommendations that only maximize weakness severity without sequencing variation will grind the same surface until boredom — and still fail the ACT's novel wrap.

XLVIII.8 Resolution doctrine — SAFE-TRANSFER

FOUNDER BELIEF / HYPOTHESIS — SAFE-TRANSFER stack:

1. **Name the hop.** Every transfer claim states which Barnett–Ceci-style dimension moved (format, story surface, delay, scaffold removal, domain).
2. **Design variation, don't sprinkle it.** Sequences: contrast → separate critical aspect → generalize range → fuse when ready.
3. **Hug the exam; bridge the abstraction.** ACT-like items after structure is discernable; explicit compare/SE prompts for high-road.
4. **Never equate story immersion with transfer.** Narrative is motivation/belonging infrastructure (WORLD_VISION / XXXVI–XXXVII), not a transfer engine by itself.
5. **Blocked accuracy ≠ transferable skill.** Require delayed / mixed / format-shifted / solo checks (`transfer_pass`, `solo_transfer_pass`).
6. **Draw attention to the pattern.** Coach/tutor line: "What stayed the same?" is load-bearing UX, not flavor text.
7. **Respect load and anxiety.** Fusion + far hops only under SAFE-DD; exam week prefers hugged retrieval (XLVII).
8. **Measure what you sell.** Instrument transfer hops; kill missions that raise same-surface accuracy without hop gains.

Kill the false dichotomy: "Transfer is a myth, only drill" vs "Our world magically creates genius generalizers." Refuse both. Ship **designed discernment**.

XLVIII.9 Marketing language that survives

- **Allowed:** "We change one thing at a time until the structure is obvious — then we change the wrapping."
- **Allowed:** "Ready means she can do it in a new format tomorrow, not five identical items tonight."
- **Allowed (parent):** "Practice looks varied on purpose — same math backbone, different disguises."
- **Banned:** "Immersive stories create far transfer." / "AI explanations transfer automatically." / Blocked-streak-as-mastery.
- **Banned:** Bloom 2-sigma; empty mindset; tutoring-is-free; vague far-transfer heroics.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without TR-* data

VT-sequenced bundles + noticing prompts	L1 mechanism aligned with Marton/Kullberg/Gu
Format-hop / delayed transfer checks as readiness	L1 product metric hygiene (Barnett–Ceci + Gick/Holyoak)
“Raises ACT by teaching far transfer”	L0–L1 only with pre-reg estimand; not L3 identity
Story-world ≡ far transfer	L0 — banned

XLVIII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
TR-1	Same concept: VT sequence (one varied dimension) vs random surface skins	2-arm	next-day format-hop `transfer_pass`	Random ≥ VT → principled variation overstated
TR-2	Bridge object: explicit compare/SE vs monologue explanation then items	2-arm	bridge-gap item accuracy; `solo_transfer_pass`	Monologue ≥ compare → XL/bridge UX useless
TR-3	Hug-then-hop: ACT-archetype items after VT vs archetype-first cold	2-arm	archetype accuracy; affect; time	Cold ≥ VT-prep → hug sequencing wrong
TR-4	Transfer check: delayed mixed vs same-day blocked as “mission complete” gate	2-arm	48h transfer; return	Same-day gate predicts 48h as well → delay rule overstated
TR-QUAL	10 Maya think-alouds on “what changed?” prompts	interview	coded discernment of invariant	Students already name invariants → prompt priority drops

Pre-reg (XXXIV): TR-* identify **sequence design × declared transfer hop**. They do **not** identify “stories cause identity” or ACT score miracles from a single varied set.

Densifies: XXIX delayed readiness; XXXIX IL discrimination; XL SE; XLI SAFE-DD; format-weakness workstream; Layer 2/3 archetype bank; Part XLIX misconceptions queue.

XLVIII.11 Confidence table

Claim	Label	Confidence
Far-transfer debates need multi-dimension taxonomy (Barnett & Ceci, 2002)	FACT	High
Spontaneous far/general transfer is rare; cues matter (Detterman, 1993; Gick & Holyoak, 1980)	FACT	High
Low-road vs high-road + hug/bridge are usable design levers (Salomon & Perkins, 1989; Perkins & Salomon, 1992/1988)	FACT	High
Discernment requires variation/invariance patterns (Marton & Booth, 1997; Kullberg et al., 2017)	FACT	High
Chinese variation / bianshi is a systematic task-design tradition (Gu et al., 2004; Sun, 2011a)	FACT	High
SAFE-TRANSFER is right default for MindCraft bank/path/copy	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Story immersion alone produces far transfer	SPECULATION / false as doctrine	Low (against)

XLVIII.12 What this chapter kills

1. **Kill:** Marketing story worlds as automatic far-transfer engines. (Gick/Holyoak noticing; Detterman rarity.)
2. **Kill:** Blocked same-surface accuracy as transferable / exam-ready proof. (Barnett–Ceci; XXIX/XXXIX readiness.)
3. **Kill:** Random cosmetic variety as “variation theory.” (Kullberg: what is made possible to learn; attention to patterns.)
4. **Kill:** AI monologue explanations as sufficient for transfer. (Hint/schema need; XL SE; XXXIII trust.)
5. **Wound:** “Transfer cannot be taught.” Survives only as: spontaneous far transfer is rare; designed hug/bridge/VT can raise hop success.
6. **Survive (constrained):** Named transfer hops; VT sequences; hug ACT + bridge abstraction; noticing prompts; delayed/format/solo metrics.

Doctrine until data: Ship **SAFE-TRANSFER**: vary on purpose so the backbone becomes obvious, hug the exam when claiming readiness, bridge with explicit abstraction, never sell lore or identical-item streaks as transfer. Market “same math backbone, different disguises — then prove it in a new wrapping tomorrow.”

Part XLIX — Misconceptions as Productive: Diagnostic Wrong Answers & Soft-Wrong Science

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 00 ≡ Red Team slot, but ch49 never written → prefer Researcher per rotation; 8 researcher entries since synthesizer v1.6 → Synthesizer deferred to next eligible non-RT hour)

Primary question: How should MindCraft treat wrong answers — as shame tokens to erase, or as *diagnostic resources* that drive soft-wrong coaching, misconception routing, and honest mastery updates?

Owners: Product (Diagnostic / practice soft-wrong / Layer-4 misconception memory) · Engine (ingredient/misconception links) · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-MISCON** stack — elicit → classify (lack vs confident misconception) → springboard without public shame → route remediation → measure retry + recalibration — kill "instant red-X pedagogy" and "all wrongs are equal."

XLIX.1 Why this chapter exists

MindCraft already ships pieces of this thesis: soft-wrong UX, hide-correctness diagnostic (C4), Eedi misconception bank, Layer-1 `canonical_misconception_family` fields, and Experiment A (wizard coach on soft-wrong). What the Constitution lacked was an evidence spine that says *why* those features are load-bearing — and which competitor postures they kill.

Parts XXX (attribution), XXXVIII (mastery climate), XL (self-explanation), XLI (SAFE-DD), and XLVIII (SAFE-TRANSFER) constrain how misses feel and how sequences vary. They do not fully answer: **what is a productive wrong answer**, and when does revealing correctness help vs corrupt learning and the graph?

FOUNDER BELIEF under audit: Identity as a mathematical thinker grows when a miss becomes *information about structure* — not a grade, not a personality verdict, and not a fluent AI monologue that skips the student's idea.

Claims we refuse as doctrine:

1. Misconceptions are unitary "bugs" that instruction must confront-and-replace wholesale.
2. Instant correctness feedback = better learning for every miss.
3. Soft-wrong tone alone equals diagnostic teaching.
4. All incorrect selections update mastery the same way (guess ≡ confident wrong ≡ slip).
5. Marketing "we celebrate mistakes" without routing, classification, or retry instrumentation.

XLIX.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Misconception (classic)	Stable, theory-like wrong idea that interferes with learning	Named `misconception_id`; distractors that diagnose it	Treat every wrong as a deep theory
Knowledge-in-pieces / resources	Fragmentary, context-sensitive knowledge elements; flawed *and* productive	Route to refine/reorganize, not only "delete"	Romanticize every wrong as deep wisdom
Diagnostic wrong	Incorrect response that reveals *which* resource/misconception fired	Option → misconception map; confidence tier	Beauty distractors with no diagnostic map
Soft-wrong	Low-shame evaluation structure: miss is safe, informative, retry-inviting	Tone + no public rank + coach invite	Soft tone with no next move = empty kindness
Hide-correctness (C4)	Serve real items; record outcome; do **not** reveal key during diagnostic	Gap-scan / FEI probes	Revealing a wrong key as "gotcha" corrupts trust + can seed wrong keys into memory theater
SAFE-MISCON	Productive-error doctrine that survives evidence	See XLIX.8	Mistake-celebration posters; red-X gamification

Operational definition (HYPOTHESIS): A MindCraft miss is *productive* when (a) the response maps to a known misconception family or resource pattern, (b) the UX destakes identity threat (soft-wrong / mastery climate), (c) the next move uses the wrong as a springboard (compare, conflict, SE, bridge card) rather than only stating the key, and (d) telemetry separates lack-of-knowledge, slip, and high-confidence misconception for graph updates.

XLIX.3 Misconceptions reconceived — flawed *and* productive

FACT (constructivist critique): Smith, diSessa & Roschelle (1993/1994, *Journal of the Learning Sciences*, 3(2), 115–163, doi:10.1207/s15327809jls0302_1) — acknowledge empirical findings of misconceptions research but reject the accompanying view that students hold strongly held flawed ideas that instruction must *confront and replace*. That view overemphasizes discontinuity with experts and conflicts with constructivism: advanced knowledge is built from prior understandings. They sketch learning as refinement and reorganization within a complex-systems view; misconceptions are **both flawed and productive**.

FACT (knowledge in pieces): diSessa (1993, *Cognition and Instruction*, 10(2–3), 105–225, doi:10.1207/s1532690xci1002&3_2) — intuitive knowledge as weakly organized elements (p-prims / sense of mechanism), not a coherent wrong theory ready for wholesale replacement. Expertise densifies and reorganizes the system.

FACT (instructional lens): Hammer (1996, *Journal of the Learning Sciences*, 5(2), 97–127, doi:10.1207/s15327809jls0502_1) — “misconceptions” vs “p-prims” change what a teacher *sees* as the instructional task (replace a theory vs refine activation conditions of resources).

Product translation: MindCraft’s Layer-1 `failure_mode` / `canonical_misconception_family` and Eedi `mis_` IDs should drive **refinement routes** (when does this resource work? when does it fail?) — not only a shameful “you have Misconception X” badge. Coach copy: “That move works for this case — here’s where it breaks” beats “Wrong. Correct answer is B.”

Kill: Product language that treats Maya as carrying a disease-like misconception to be purged. Survive: diagnostic naming for *engine routing*, student-facing language as productive refinement.

XLIX.4 Errors as springboards — inquiry, not erasure

FACT (teaching experiment): Borasi (1994, *Journal for Research in Mathematics Education*, 25(2), 166–208, doi:10.5951/jresematheduc.25.2.0166) — secondary students can capitalize on errors as **springboards for inquiry**; variations of the strategy exist; the point is stimulating mathematical inquiry from the error, not only teacher-side diagnosis for remediation.

FACT (error analysis tradition): Radatz (1979, *Journal for Research in Mathematics Education*, 10(3), 163–172, doi:10.5951/jresematheduc.10.3.0163) — systematic error analysis has long diagnostic value for teachers; Borasi’s contribution is turning that value toward *student* inquiry.

FACT (diagnostic teaching / conflict): Bell (1993a, *Educational Studies in Mathematics*, 24, 5–34; 1993b, *Educational Studies in Mathematics*, 24, 115–137) and Swan’s diagnostic-teaching line (e.g., Swan, 2001, “Dealing with misconceptions in mathematics,” in *Issues in Mathematics Teaching*, Routledge) — long-term learning improves when lessons **expose** existing thinking, generate cognitive conflict across methods/answers, and resolve via reflective discussion — vs curricula that avoid analyzing errors or front-load explanations then graded practice.

HYPOTHESIS for MindCraft: Soft-wrong + wizard coach should implement a *compressed* springboard cycle: expose idea → contrast with invariant/conflict item → student-generated why (XL) → optional AI wrap under PWC — not Duo-style instant red/green with explanation dump.

Wound: Classroom conflict discussion does not transfer 1:1 to solo app UX. Survive as: design *micro-conflict* (two worked paths; “which claim survives?”) rather than claiming Socratic seminar in a phone UI.

XLIX.5 Soft-wrong science — safety is necessary, not sufficient

FACT (affect × attempt): Prior Constitution densification (XXIV anxiety/WM; XXXVIII mastery climate; XLI SAFE-DD destake) — threat appraisal suppresses attempt quality; evaluation structure that makes misses identity-safe raises willingness to engage difficulty.

FOUNDER BELIEF: Soft-wrong is mastery *evaluation structure*, not brand voice. Without diagnostic mapping + next move, soft-wrong is empty kindness.

HYPOTHESIS: Soft-wrong's commercial wedge vs Khan/Duo is not "nicer feedback" — it is **recoverable struggle with competence evidence** (retry_120s, challenge_accept, solo_transfer_pass). Competitors can copy warmer copy; they rarely ship miss → misconception route → bridge/ingredient card → delayed transfer check.

Contradicting pressure: Parents often want instant correctness theater ("show the answer"). Claim ladder: L0–L1 session metrics for soft-wrong; do not sell L3 identity from tone alone (XXXIV).

Kill: Soft-wrong as marketing slogan without Experiment A instrumentation. Survive: soft-wrong as FEI stage (informative miss) gated by SAFE-MISCON routing.

XLIX.6 Confidence, calibration, and hide-correctness

FACT (hypercorrection): Butterfield & Metcalfe (2001, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(6), 1491–1494, doi:10.1037//0278-7393.27.6.1491) — contrary to interference predictions, **high-confidence errors** were *more* likely to be corrected after feedback than low-confidence errors (hypercorrection).

FACT (mechanism sketch): Butterfield & Metcalfe (2006, *Memory & Cognition* / follow-on attentional work; see also Metcalfe & Finn line) — corrective feedback after surprising high-confidence errors captures more attention; later work notes latent knowledge / "knew it all along" contributions. Product implication: confidence on wrongs is diagnostically precious, not just a UI flourish.

FACT (authentic math diagnostics online): Foster, Woodhead, Barton & Clark-Wilson (2022 advance of 2021 online-first; *Educational Studies in Mathematics*, doi:10.1007/s10649-021-10084-7) — large-scale diagnostic quiz data: higher confidence on an incorrect response associated with higher later correction probability on analogous items (hypercorrection signal in authentic math learning), after controls discussed in-paper; confidence assessment can prompt reflection.

FACT (calibration hygiene): Confidence–accuracy work in diagnostics (e.g., overconfidence on alternative conceptions; gender patterns in under/overconfidence across studies) — poor calibration is common; separating *misinformed* (confident wrong) from *uninformed* (low confidence / blank) changes instruction.

Product translation — C4 hide-correctness: During gap-scan / identity-sensitive diagnostics, revealing keys mid-flow can (a) spike threat, (b) turn the session into answer-hunting, (c) risk encoding wrong "gotcha" narratives. Record outcomes + confidence; defer key reveal to a designed springboard moment (or never in pure diagnostic). **Do not** market hide-correctness as "we never teach the answer" — that is false and banned under tutoring-is-free style slogans.

HYPOTHESIS: Practice mode should often *use* hypercorrection: after a high-confidence miss, prioritize attended corrective feedback + isomorphic retest; after low-confidence miss, prioritize scaffold/hug before confrontation (SAFE-DD equip).

Kill: Treating every wrong as equal graph evidence. Survive: confidence-tiered update policy (aligns with Layer-4 evidence_update_policy intent: don't over-update from one item; separate can-do from can-recognize).

XLIX.7 What this means for the bank, ontology, and Solver

Bank / C5: Distractors should be designed misconceptions, not random wrongs. Eedi ingestion already mints `mis_` IDs — enrichment linking misconceptions → ingredients remains the annotation bottleneck (CLAUDE.md). Without links, soft-wrong cannot route.

Ontology: Prefer nested ingredient `failure_mode` + Layer-2 `common_misconception_ids` as the join surface for coach cards. Variation sequences (XLVIII) that juxtapose the misconception-triggering surface vs the invariant are VT done right.

Solver / AI: Under PWC (XXXIII), AI wrap after student springboard — not "fluent correct answer" that ignores the selected distractor. Sycophancy risk: agreeing with a wrong path. Guard: name the student's move, state where it applies, show the break.

Competitive wedge (XXXV): Duo celebrates streaks and instant binary feedback; ChatGPT delivers fluent keys; Khan explains. MindCraft sells **diagnostic wrong → recoverable path** with evidence the parent can see (retry, transfer) without leaderboard shame (SAFE-COMPARE).

XLIX.8 SAFE-MISCON stack (ship rule)

1. **Elicit** — Real items; prefer misconception-tagged distractors.
2. **Classify** — Lack / slip / high-confidence misconception (confidence + item family).
3. **Destake** — Soft-wrong mastery climate; no public shame; no streak-as-punishment for miss.
4. **Springboard** — Micro-conflict, compare, SE prompt, or Borasi-style inquiry move *using the wrong*.
5. **Route** — Ingredient / bridge / VT contrast card from misconception map.
6. **Reveal with purpose** — Feedback timing by confidence tier; hide-correctness in diagnostic; attended feedback after high-confidence miss.
7. **Measure** — `retry_120s`, recalibration Δ, misconception recurrence, delayed isomorphic accuracy — not "mistakes celebrated" vanity.

Marketing language that survives: "Wrong answers that teach," "find the pattern in the miss," "confidence that matches skill."

Marketing language that dies: "We never say you're wrong," "mistakes are magic," instant red-X as rigor, Soft-wrong™ as identity transformation (L3 from tone).

XLIX.9 Claim ladder

Claim	Max ladder without new data
Soft-wrong raises attempt/retry vs harsh red-X	L1 (Experiment A)
Misconception-tagged distractors improve routing vs random wrongs	L1
Confidence-tiered feedback beats uniform instant key	L1

Hide-correctness diagnostic yields cleaner weakness ranking	L1 (C1/C4 product bet)
Productive-error UX → identity as mathematical thinker	L3 — ban until transfer + identity items

XLIX.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
MIS-1	Soft-wrong + springboard coach vs soft-wrong + key-only	2-arm	`retry_120s`; next-item accuracy	Key-only ≥ springboard → inquiry UX overstated
MIS-2	Misconception-mapped distractors vs random wrongs	2-arm	route precision; recurrence of same mis-ID	Random ≥ mapped → bank annotation ROI fails
MIS-3	Confidence-tiered feedback (hypercorrection-aware) vs uniform instant	2-arm	high-conf miss correction @ 48h	Uniform ≥ tiered → tiering complexity unjustified
MIS-4	Hide-correctness diagnostic vs reveal-as-you-go for gap-scan	2-arm	completion; affect; later `worstWeakness` stability	Reveal ≥ hide on weakness quality + affect → C4 overstated
MIS-QUAL	10 Maya think-alouds after soft-wrong	interview	coded: shame vs curiosity; names own idea?	Students already springboard unaided → coach priority drops

Pre-reg (XXXIV): MIS-1…4 identify **error-handling design × classification**. They do **not** identify “celebrating mistakes causes identity.”

Densifies: Experiment A; C4 diagnostic; Layer-4 misconception memory; XL SE; XLVIII VT conflict pairs; Part L confidence calibration queue.

XLIX.11 Confidence table

Claim	Label	Confidence

Misconceptions are both flawed and productive resources (Smith et al., 1993/94)	FACT	High
Knowledge-in-pieces / p-prims reframes instructional task (diSessa, 1993; Hammer, 1996)	FACT	High
Errors can be springboards for inquiry (Borasi, 1994)	FACT	High
Diagnostic teaching / conflict discussion beats error-avoidant exposition for long-term learning (Bell/Swan line)	FACT	Medium–High (classroom evidence; app transfer open)
High-confidence errors are hypercorrected after feedback (Butterfield & Metcalfe, 2001; Foster et al. math diagnostics)	FACT	High
SAFE-MISCON is right default for MindCraft soft-wrong / diagnostic / bank	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Soft tone alone transforms identity	SPECULATION / false as doctrine	Low (against)

XLIX.12 What this chapter kills

1. **Kill:** Misconceptions-as-disease / confront-and-replace-only pedagogy as company doctrine. (Smith/diSessa/Roschelle; Hammer.)
2. **Kill:** Instant red-X + answer dump as the default "rigorous" product. (Borasi springboard; Bell/Swan diagnostic teaching.)
3. **Kill:** Soft-wrong as empty brand kindness without diagnostic route + retry metrics.
4. **Kill:** Equal graph updates for guess, slip, and confident misconception. (Hypercorrection / calibration literature.)
5. **Kill:** Hide-correctness marketed as "we never teach answers."
6. **Wound:** Classroom conflict discussion ≡ mobile UX. Survive as designed micro-conflict + SE, not seminar cosplay.
7. **Survive (constrained):** Misconception-mapped distractors; confidence-tiered feedback; soft-wrong as FEI informative miss; springboard-before-monologue; MIS-1…4.

Doctrine until data: Ship **SAFE-MISCON**: treat wrong answers as diagnostic fuel, destake identity threat, springboard from the student's idea, route to structure, measure retry and recalibration — never sell mistake-celebration posters or red-X theater as learning science.

Part L — Confidence Calibration: Over/Underconfidence, Hide-Correctness, and Honest Self-Knowledge

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 06 ≡ Red Team slot, but ch50 never written → prefer Researcher per rotation; 0 researcher entries since synthesizer v1.7)

Primary question: How should MindCraft treat student *confidence* — as a vibe to inflate, a score to gamify, or a *calibrated signal* that drives feedback, graph updates, and identity-safe diagnostics?

Owners: Product (Diagnostic C4 / soft-wrong / confidence UX) · Engine (evidence_update_policy / confidence tiers) · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-CALIB** stack — elicit confidence → classify miss type → destake → recalibrate without inflation theater → tier feedback → measure `confidence_miss_tier` — kill "raise confidence" as North Star and "how sure?" without routing.

L.1 Why this chapter exists

The research question of this Constitution is how humans become *confident mathematical thinkers*. That phrase is dangerous if "confident" means louder, happier, or more certain. The commercial and scientific load-bearing meaning is **calibrated competence**: knowing when you know, knowing when you do not, and updating both skill and self-appraisal under informative misses.

Parts XXV (self-efficacy), XXX (attribution), XXXIII (AI overtrust), XLIX (SAFE-MISCON / hypercorrection), and the v1.7 metric `confidence_miss_tier` already point here. What was missing was a chapter that (a) defines calibration rigorously, (b) separates over- from underconfidence product responses, (c) ties C4 hide-correctness to metacognitive hygiene, and (d) refuses confidence theater as marketing.

FOUNDER BELIEF under audit: Identity as a mathematical thinker includes *honest self-knowledge* — not perpetual certainty, and not chronic self-doubt dressed as humility.

Claims we refuse as doctrine:

1. Raise confidence = improve learning (always).
2. Confidence meters / "Belief Score™" as identity North Stars.
3. Asking "how sure?" without changing feedback, routing, or graph updates.
4. Hide-correctness = never teaching answers.
5. Dunning–Kruger as a meme that justifies shaming low performers.
6. Instant correctness reveal as the only path to calibration.

L.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Calibration	Match between stated confidence and empirical accuracy across items	Confidence × correctness curves; bias & absolute accuracy indices	One-item "I feel sure" = calibrated
Overconfidence	Confidence systematically exceeds accuracy	High-conf wrongs; skipped review; false mastery	Shame the student; ignore hypercorrection opportunity
Underconfidence	Accuracy exceeds confidence	Correct + low conf; stuck on secure content	Force "believe harder" posters; skip challenge progression
Confidence of response	Certainty that *this* answer is correct (item-level)	0–10 or low/med/high next to answer	Global "math confidence" survey only
Hide-correctness (C4)	Record outcome; defer or withhold key reveal in diagnostic	Gap-scan / FEI probes	Never-teach-answers slogan; corrupt key theater
SAFE-CALIB	Calibration doctrine that survives evidence	See L.8	Confidence gamification; inflation theater

Operational definition (HYPOTHESIS): MindCraft treats calibration as a *session and graph* property when (a) item-level confidence is elicited on diagnostic and soft-wrong paths, (b) high-conf misses trigger attended corrective / springboard sequences, (c) low-conf corrects unlock challenge progression rather than endless drill, (d) mastery updates weight miss type (guess ≠ confident misconception), and (e) marketing claims stay at L0–L1 until CAL experiments report.

L.3 Calibration science — what "well calibrated" means

FACT (definition): Lichtenstein & Fischhoff (1977, *Organizational Behavior and Human Performance*, 20, 159–183; classic line) and Fischhoff, Slovic & Lichtenstein (1977, *Journal of Experimental Psychology: Human Perception and Performance*, 3(4), 552–564, doi:10.1037/0096-1523.3.4.552) — a judge is well calibrated if, over the long run, for propositions assigned probability p, the proportion true ≈ p. Extreme certainty is routinely *inappropriate*: people are wrong too often when they feel certain.

FACT (bias pattern): Across general-knowledge and many educational tasks, the dominant bias is **overconfidence**, especially when accuracy is near chance; as accuracy rises, overconfidence often shrinks and underconfidence can appear at high accuracy (Lichtenstein & Fischhoff, 1977 summary pattern; replicated in educational monitoring work).

FACT (measurement plurality): Lingel, Lenhart & Schneider (2019, *ZDM*, metacognitive monitoring in mathematics — ERIC EJ1222655) — absolute accuracy, relative accuracy (e.g. Gamma), and diagnostic sensitivity/specificity are *not interchangeable*. Method choices change conclusions. Seventh-graders showed **pervasive overconfidence**; retrospective judgments and some scale formats improved monitoring precision vs prospective.

Product translation: MindCraft should log item-level confidence + correctness and report **bias** (over/under) and **discrimination** (does higher confidence predict correctness?) separately. Do not ship a single "Calibration %" vanity metric parents can misunderstand.

Kill: One glamorous calibration number as growth KPI. Survive: `confidence_miss_tier` + bias curves for product learning, not for public leaderboards (SAFE-COMPARE).

L.4 Overconfidence, underconfidence, and the dual commercial problem

FACT (school math overconfidence): García, Rodríguez, González-Castro, González-Pienda & Torrance (2016, *Metacognition and Learning*; ERIC EJ1105998) — elementary students solving math problems were poorly calibrated and tended toward **overconfidence**; inaccurate calibrators differed in reported metacognitive process use.

FACT (ability × miscalibration): Kruger & Dunning (1999, *Journal of Personality and Social Psychology*, 77(6), 1121–1134, doi:10.1037/0022-3514.77.6.1121) — bottom-quartile performers on humor/grammar/logic tasks grossly overestimated ability; incompetence can impair the metacognitive skill needed to recognize incompetence. Improving skill can improve self-appraisal.

Wound (popular DK): The meme version ("stupid people are always overconfident") overclaims, ignores regression-to-the-mean critiques, and invites product cruelty. Use as **FACT** about *metacognitive dual burden in low skill*, not as a brand insult toward Maya.

FACT (underconfidence is also costly): Foster's CA program (Foster, 2016, *Educational Studies in Mathematics*, 91(2), 271–288, doi:10.1007/s10649-015-9660-9; Foster, 2016, *Mathematics Teaching*, 251, 11–13; Centre for Mathematical Cognition summary) — underconfidence can reduce satisfaction, block progression to harder material, and trap students in repetitive secure practice; overconfidence can embed uncorrected errors and complacency. Pedagogical aim is **appropriate** confidence, not maximal confidence.

HYPOTHESIS for MindCraft: Overconfident Maya needs destaked confrontation + springboard (SAFE-MISCON + hypercorrection). Underconfident Maya needs earned-win evidence and challenge unlocks (FEI), not pep talks. One "confidence booster" feature cannot serve both.

Kill: "We build confidence" as undifferentiated slogan. Survive: "confidence that matches skill" / "know what you know."

L.5 Confidence assessment in classrooms — promise and null on attainment

FACT (CA mechanism): Confidence assessment asks students to rate certainty alongside each answer; scoring that rewards correct high-confidence and penalizes incorrect high-confidence (and also punishes under-reporting when correct) incentivizes *truthful* ratings (Foster, 2016; Gardner-Medwin line in HE). Aim includes discouraging guess-and-hope.

FACT (student acceptability): Foster (2016 ESM) — secondary students can make usable confidence judgments on procedures; generally positive about CA in short trials.

FACT (null on summative attainment — load-bearing contradiction): Foster (2021/2022, *International Journal of Science and Mathematics Education*, 20, 1411–1429, doi:10.1007/s10763-021-10207-9) — quasi-experimental CA across four secondary schools (N = 475), meta-analytic Cohen's $d \approx -0.02$ [95% CI –0.22, 0.19], Bayes factor favoring null on attainment gains vs controls. Conclusion: incorporating CA into low-stakes formative assessments does **not** appear detrimental — and also does **not** show a clear positive attainment effect from CA alone.

Commercial implication (critical): Shipping a “how sure?” widget is cheap and marketable. Evidence says **CA alone is not an attainment engine**. MindCraft’s wedge is not novelty CA — it is CA **wired** to soft-wrong routing, misconception maps, confidence-tiered feedback, and graph policy. Competitors can copy a confidence slider; they rarely ship the spine.

HYPOTHESIS: MindCraft CA improves *process metrics* (retry, misconception route hit, delayed isomorphic correction after high-conf miss) before it moves ACT scores. Claim ladder: process L1 first; attainment L2+ only after CAL + MIS experiments.

Kill: Marketing “confidence assessment raises grades.” Survive: CA as diagnostic fuel + metacognitive prompt, non-inferior on attainment per Foster 2021, valuable when coupled to FEI/SAFE-MISCON.

L.6 Hypercorrection, contradicting reasons, and feedback design

FACT (hypercorrection): Butterfield & Metcalfe (2001, *JEP:LMC*, 27(6), 1491–1494, doi:10.1037//0278-7393.27.6.1491) — high-confidence errors more often corrected after feedback than low-confidence errors.

FACT (math diagnostics): Foster, Woodhead, Barton & Clark-Wilson (*Educational Studies in Mathematics*, doi:10.1007/s10649-021-10084-7) — authentic online math diagnostic data show hypercorrection-consistent patterns; confidence assessment can prompt reflection.

FACT (debiasing via contradicting reasons): Koriat, Lichtenstein & Fischhoff (1980, *Journal of Experimental Psychology: Human Learning and Memory*, 6(2), 107–118) — listing reasons *against* the chosen answer improved appropriateness of confidence; supporting reasons alone did not. Confidence tracks strength of supporting evidence; people neglect disconfirming evidence unless prompted.

Product translation:

1. After high-conf miss → attended feedback + micro-conflict / “why might this be wrong?” (SE + Koriat) before fluent AI monologue (PWC / XXXIII).
2. After low-conf miss → scaffold/hug first (SAFE-DD), then lighter confrontation.
3. After low-conf correct → show competence evidence; invite slightly harder item (underconfidence unlock).
4. C4 hide-correctness during gap-scan: elicit confidence + answer; **defer** key reveal so the session stays diagnostic, not answer-hunting — then use purposeful reveal in practice springboards (XLIX).

Kill: Uniform instant red/green for every miss. Survive: confidence-tiered reveal + contradicting-reason prompt (aligns MIS-3).

L.7 Hide-correctness, AI trust, and graph integrity

FOUNDER BELIEF: C4 exists because revealing keys mid-diagnostic can spike threat, turn FEI into answer theater, and (if keys are wrong) poison trust. Recording outcomes without reveal preserves graph honesty for weakness ranking

(C1).

HYPOTHESIS: Calibration *improves* when students later reconcile confidence with delayed feedback in a destaked springboard — not when every item is a public gotcha.

Link to XXXIII: Fluent AI tutors create **overtrust** (calibration failure toward the *system*). Student self-appraisal and AI-skepticism are sibling problems.

Layer-4 alignment: Confidence tiers operationalize `evidence_update_policy` — high-conf wrong → stronger misconception evidence; low-conf wrong → weaker / "uninformed" path.

Kill: Equal mastery deltas for all wrongs; "we never teach the answer." Survive: purposeful reveal; confidence-weighted updates; AI wrap after student generation.

L.8 SAFE-CALIB stack (ship rule)

1. **Elicit** — Item-level confidence on diagnostic and on soft-wrong / challenge items (not only global surveys).
2. **Classify** — Correct/incorrect × low/med/high → miss classes (guess, slip, confident misconception, underconfident correct).
3. **Destake** — Soft-wrong / mastery climate; never public-shame high-conf wrongs; no streak punishment for honest low confidence.
4. **Recalibrate** — Show mismatch privately ("sure + wrong" or "unsure + right") as information, not identity verdict.
5. **Tier feedback** — Hypercorrection path for high-conf miss; hug-then-dose for low-conf miss; unlock challenge after underconfident correct.
6. **Prompt against** — One contradicting reason / "where does this break?" before answer dump (Koriat).
7. **Measure** — `confidence_miss_tier`, calibration bias Δ, high-conf miss correction @ delay, underconf→challenge_accept — not Belief Score™ or confidence streaks.

Marketing language that survives: "Know what you know," "confidence that matches skill," "wrong answers that teach," "honest diagnostics."

Marketing language that dies: "We raise confidence," "believe in yourself" as product, Confidence League ranks, "never say you're wrong," CA-raises-grades claims without data.

L.9 Competitive wedge (brief)

Duo (binary + streaks), Khan (explain-first), Brilliant (scaffold dopamine), and ChatGPT tutors (fluent certainty → overtrust) all fail *calibration* differently. MindCraft's L1-safe line: not louder confidence — **earned, calibrated** confidence that survives a format hop (solo transfer + confidence on unaided items + PWC).

L.10 Claim ladder

Claim	Max ladder without new data
Item-level CA is usable with teens and non-inferior on attainment vs BAU	L1 (Foster 2016; 2021 null)
High-conf misses deserve different feedback than low-conf misses	L1 (hypercorrection literature)
Contradicting-reason prompts improve confidence appropriateness	L1 (Koriat et al., 1980)
C4 hide-correctness improves weakness ranking quality + affect vs reveal-as-you-go	L1 product bet (MIS-4 / CAL-4)
Confidence-tiered graph updates beat equal-wrong updates	L1
CA alone raises summative math attainment	**Banned** pending contrary RCT (Foster 2021 against)
Calibrated confidence → identity as mathematical thinker	L3 — ban until transfer + identity instruments

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L.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
CAL-1	Confidence-tiered feedback vs uniform instant key	2-arm	high-conf miss correction @ 48h; `retry_120s`	Uniform ≥ tiered → tiering unjustified
CAL-2	Contradicting-reason prompt vs explain-correct-only after miss	2-arm	calibration bias Δ; next isomorphic accuracy	Explain-only ≥ prompt → Koriat UX overstated for app
CAL-3	Underconfident-correct → challenge unlock vs more secure drill	2-arm	`challenge_accept`; later accuracy	Drill ≥ unlock on growth → underconf policy wrong
CAL-4	Hide-correctness diagnostic vs reveal-as-you-go (shared with MIS-4)	2-arm	completion; affect; `worstWeakness` stability	Reveal wins on weakness + affect → C4 overstated
CAL-5	Confidence-weighted mastery updates vs equal-wrong updates	2-arm offline / sim + online	prediction of next-session accuracy	Equal ≥ weighted → engine complexity unjustified

CAL-QUAL	10 Maya interviews: meaning of “sure” / shame after high-conf miss	interview	coded: threat vs curiosity; wants reveal when?	Students refuse confidence UX → redesign elicit

Pre-reg (XXXIV): CAL-* identify **confidence × feedback / update design**. They do **not** identify “raising confidence causes identity.”

Densifies: MIS-3/4; C4; Layer-4 evidence policy; XXV self-efficacy; XXXIII AI overtrust; metric `confidence_miss_tier`.

L.12 Confidence table

Claim	Label	Confidence
Calibration = long-run match of confidence to accuracy (Fischhoff/Lichtenstein line)	FACT	High
Overconfidence is common; extreme certainty often unwarranted	FACT	High
School math students often poorly calibrated / overconfident (García et al.; Lingel et al.)	FACT	High
DK dual burden for low skill is real; meme uses are overstated	FACT / WOUND	Medium–High
CA usable + non-inferior attainment; not a proven attainment booster alone (Foster 2016; 2021)	FACT	High
Hypercorrection after high-conf errors (Butterfield & Metcalfe; Foster et al.)	FACT	High
Contradicting reasons improve confidence appropriateness (Koriat et al., 1980)	FACT	High
SAFE-CALIB is right default for MindCraft diagnostic / soft-wrong / graph	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Raise-confidence marketing transforms identity	SPECULATION / false as doctrine	Low (against)

L.13 What this chapter kills

1. **Kill:** "Raise confidence" as undifferentiated North Star or brand promise. (Foster: appropriate confidence; under- and over- both hurt.)
2. **Kill:** Confidence meters, Belief Score™, or confidence streaks as identity engines. (SAFE-COMPARE / SAFE-HABIT alignment.)
3. **Kill:** "How sure?" UI without tiered feedback, routing, or update policy. (Foster 2021 null on CA-alone attainment.)
4. **Kill:** CA / confidence assessment marketed as raising grades without new evidence.
5. **Kill:** Equal graph updates for guess vs high-confidence misconception.
6. **Kill:** Hide-correctness as "we never teach answers."
7. **Kill:** Dunning–Kruger shaming as brand voice.
8. **Wound:** Classroom CA ≡ app CA; HE medical CA ≡ HS math. Survive as: elicit + truthful incentive design + wired spine, not worksheet cosplay.
9. **Survive (constrained):** Item-level confidence; SAFE-CALIB stack; hypercorrection-aware feedback; underconfidence unlocks; CAL-1…5; coupling to SAFE-MISCON.

Doctrine until data: Ship **SAFE-CALIB**: treat confidence as a diagnostic signal for honest self-knowledge and differential pedagogy — never as a vibe to inflate, a league to rank, or a substitute for FEI evidence.

Part LI — Deliberate Practice in Tutoring Sessions: Ericsson vs Hambrick, and the 45-Minute Design

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 09; hour%6≠0; 1 researcher entry since synthesizer v1.7 → Researcher)

Primary question: What should a MindCraft tutoring / practice session *do* if "practice" is not volume, not streak minutes, and not answer delivery — and how far can Ericsson's deliberate-practice (DP) frame travel before Macnamara–Hambrick evidence kills overclaim?

Owners: Product (session architecture / tutor briefing) · Engine (weakness targeting / feedback loops) · Marketing claim ladder · Red Team

Commercial job: Ship a **SAFE-DP** session stack — diagnose → isolate edge skill → goal → effortful attempt → informative feedback → revise → brief rest → solo check — kill 10,000-hours theater, grind-volume North Stars, and "we do deliberate practice" as brand without a coach-designed edge task.

LI.1 Why this chapter exists

MindCraft sells sessions: college tutors, practice missions, Solver, Map. The Constitution's research question is identity transformation under difficulty. Deliberate practice is the popular science answer to "how experts get good." It is also a marketing trap — competitors and edtech decks cite Ericsson while shipping naive drill (more of the same), gamified volume (streaks/XP), or fluent AI monologue (pseudo-coaching).

Parts XXVI (CLT / fading), XXIX (spacing/retrieval), XXXI (challenge–skill), XXXIX (interleave), XL (self-explanation), XLI (SAFE-DD), and L (SAFE-CALIB) already constrain *what kind* of difficulty and feedback are safe. What was missing was a chapter that (a) defines DP vs purposeful vs naive practice, (b) faces the Macnamara–Hambrick variance numbers honestly, (c) translates math-education DP (Lehtinen line) into a **45-minute** tutor/product design, and (d) refuses talent-denial and grind-as-virtue as doctrine.

FOUNDER BELIEF under audit: A high-quality MindCraft session is closer to *coach-designed practice at the edge of ability with feedback* than to homework help, content library, or engagement loops.

Claims we refuse as doctrine:

1. 10,000 hours / "practice makes perfect" as causal story.
2. Accumulated minutes / streaks as proof of DP.
3. DP explains nearly all expert variance (popular overread).
4. Any hard drill = DP; AI chat without diagnosis = DP coach; talent-denial as brand.

LI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Naive practice	Repeating what you can already do; play / homework slog	High accuracy, low struggle, no edge target	"Completed 40 problems" vanity
Purposeful practice	Goal + focus + feedback, possibly self-directed (Ericsson & Pool, *Peak*, 2016)	Student sets a micro-goal; app gives feedback	Duo-like streaks dressed as purpose
Deliberate practice (strict)	Coach/teacher designs tasks beyond current reliable performance; immediate informative feedback; refinement; builds mental representations (Ericsson et al., 1993; Ericsson & Pool, 2016)	Tutor/engine picks weakness; task just beyond; feedback revises method	Labeling all structured homework "DP"
Structured practice (meta-analytic)	Broad "practice designed to improve" used in Macnamara et al. (2014) — often looser than Ericsson's DP	Hours logged in studies	Equating study-hours with DP

Mental representation	Organized internal model that lets experts monitor and improve (Ericsson line; Lehtinen et al., 2017)	Transfer / format hop / self-monitor	Fluency without representation
SAFE-DP	DP doctrine that survives Hambrick wound + anxiety/CLT gates	See LI.8	Grind theater; 10k-hours copy

Operational definition (HYPOTHESIS): A MindCraft block counts as *deliberate-practice-shaped* when (a) target is a diagnosed edge (concept, ingredient, bridge, format, or miss class), (b) task is beyond reliable solo performance but solvable with brief support, (c) feedback changes the *next attempt's method*, not only the answer key, (d) student generates at least one why / check before AI wrap (PWC / XL), and (e) session ends with a short unaided probe (`solo_transfer_pass` or isomorphic check) — not with "minutes practiced."

LI.3 Ericsson's core claim — what is actually said

FACT (foundational paper): Ericsson, Krampe & Tesch-Römer (1993, *Psychological Review*, 100(3), 363–406, doi:10.1037/0033-295X.100.3.363) — expert performance is strongly associated with accumulated *deliberate practice*: highly structured activities designed to improve specific aspects of performance, typically with teacher/coach guidance, clear goals, feedback, and effortful concentration. Mere experience / play / mindless repetition do not substitute.

FACT (effort constraint): Same paper and later summaries — effective daily DP duration is limited; early practice often ~1 hour or less; laboratory extended-practice protocols commonly ~1 hr sessions; pushing past optimal duration yields diminishing returns and fatigue. Rest is part of the training system, not laziness (aligns SAFE-TIMING).

FACT (Peak distinction): Ericsson & Pool (2016, *Peak*) — **purposeful practice** = well-defined goal, focus, feedback, get-out-of-comfort-zone; **deliberate practice** = purposeful practice *plus* a field with known training methods and a teacher who designs/adapts exercises. Popular "10,000 hours" readings flatten this distinction and often cite Gladwell's popularization rather than Ericsson's constraints.

Product translation: MindCraft's tutor + ontology spine can *aspire* to the coach role (diagnose → design edge task → feedback). Solo app loops without diagnosis are at best purposeful practice — and often naive practice with a nicer UI.

Kill: "10,000 hours of MindCraft." Survive: "coach-designed practice at your edge, in a session you can sustain."

LI.4 The Hambrick / Macnamara wound — variance is not destiny theater

FACT (meta-analysis): Macnamara, Hambrick & Oswald (2014, *Psychological Science*, 25(8), 1608–1618, doi:10.1177/0956797614535810) — across domains, deliberate-practice measures explained roughly **26%** of variance in games, **21%** music, **18%** sports, **4%** education, and **<1%** professions. Conclusion: DP is important, but **not as important as popular arguments claim**.

FACT (sports follow-up): Macnamara, Moreau & Hambrick (2016, *Perspectives on Psychological Science*, 11(3), 333–350, doi:10.1177/1745691616635591) — in sports, substantial unexplained variance remains; elite vs sub-elite comparisons weaken simple "more DP → elite" stories.

FACT (reply chain): Ericsson (2016, *Perspectives on Psychological Science*, doi:10.1177/1745691616635600) argues meta-analyses sum *heterogeneous practice hours* and dilute the construct; Macnamara & Hambrick (2016, doi:10.1177/1745691616635614) reply that Ericsson's criteria shift and that existing evidence does not support "largely accounted for by DP."

FACT (definition rescue attempt): Ericsson & Harwell (2019, *Frontiers in Psychology*, doi:10.3389/fpsyg.2019.02396) — re-emphasize original DP criteria; claim Macnamara et al. often measured **structured practice**, not strict DP; reanalyses with stricter inclusion raise explained variance. Independent methodologists still warn against treating any reanalysis as settling talent, opportunity, and starting-skill confounds.

Wound for MindCraft marketing: Education's **~4%** figure is a Red Team grenade against "science says deliberate practice is everything." Even if Ericsson is right that many education studies measured weak proxies, MindCraft cannot sell DP as a near-sufficient cause of identity or ACT gains.

HYPOTHESIS: MindCraft should treat DP as a **session design grammar** (how to spend 45 minutes), not as a **variance-explained brand claim**. Commercial strength is *individualized edge practice with feedback* — the part both camps agree matters more than mindless hours — without promising that practice volume explains Maya's future.

Kill: "Experts are made of deliberate practice — so we made you an expert factory." Survive: "Most homework isn't practice that changes you; we design the next hard thing on purpose."

LI.5 Mathematics education — drill vs deliberate practice

FACT (math DP review): Lehtinen, Hannula-Sormunen, McMullen & Gruber (2017, *ZDM Mathematics Education*, 49, 625–636, doi:10.1007/s11858-017-0856-6) — school math "practice" is often **drill-and-practice** that automatizes procedures but can yield *inert* routines rather than adaptive number knowledge; DP in expertise research involves thinking, problem solving, reflection, and dynamically retargeted training — not mechanical fluency alone. They summarize Ericsson (2016) features: coach-known skills, beyond-current-ability effort, specific goals, full attention, feedback + modification, building mental representations, improving prior skills with correct fundamentals.

FACT (study time ≠ improvement): Plant, Ericsson, Hill & Asberg (2005, *Contemporary Educational Psychology*, 30(1), 96–116) — in higher education, amount of study time did not significantly predict GPA once quality/concentration/goals entered; improvement tied to concentrated, goal-directed learning (cited in Lehtinen et al. as caution against raw hours).

HYPOTHESIS for MindCraft: Ontology ingredients + bridges are the *right decomposition* for DP-style targeting (isolate the failing subprocess, then rebuild), but only if sessions refuse "do 20 similar ACT items" as the default. Blocked accuracy is acquisition theater (XXXIX); DP needs the edge miss class (SAFE-MISCON / SAFE-CALIB), then varied probes (SAFE-TRANSFER).

Link to CLT / anxiety: Beyond-ability tasks without destake/equip violate SAFE-DD. DP's "near-maximal effort" is **not** permission for threat flooding. For Maya, the coach's first job is often making the edge task *informative*, not merely hard.

Kill: Back-to-basics grind marketed as Ericsson. Survive: adaptive practice that builds representations — Lehtinen's contrast as competitive copy vs worksheet mills.

LI.6 What a 45-minute MindCraft session should look like

FOUNDER BELIEF: Human tutor sessions (~45–60 min) and digital missions should share one DP-shaped spine so marketing demos, tutor briefs, and app UX do not diverge into "warm homework help" vs "streak game."

Proposed spine (HYPOTHESIS — SAFE-DP):

Minute block	Move	Maps to
0–5	Affect + goal: name *one* edge target (weakness / bridge / format / miss class)	FEI safety; E×V; SAFE-EXPECTANCY task-only language
5–12	Quick diagnose / warm isomorphic (hide-correctness OK if FEI)	C4; SAFE-CALIB elicit if used
12–28	Edge attempts: 1–3 items beyond reliable solo; student generation before explain	DP effort; PWC; SE (XL); SAFE-DD dose
28–36	Feedback that revises method: contradicting reason / springboard / ingredient card — not answer dump	Koriat; SAFE-MISCON; ingredient runtime
36–42	Near transfer / format hop / interleaved foil	SAFE-TRANSFER; XXXIX
42–45	Solo check + attribution ("what changed in your method?") + next cue	`solo_transfer_pass`; SAFE-HABIT if-then; no streak sermon

Effort constraint product rule: Prefer one sharp DP cycle over stuffing eight topics. Marathon "grind nights" violate SAFE-TIMING and Ericsson's own duration cautions.

Tutor ethics: Brief tutors on *task* targets from the graph — never trait labels (SAFE-EXPECTANCY). "Today we strengthen the functions→rate bridge" beats "she's weak and unmotivated."

AI role: Engine proposes the edge target and card order; LLM wraps language *after* student attempt (PWC). Unguarded Solver-as-session is not DP.

LI.7 Competitive wedge (brief)

Khan often explain-then-block; Duo sells streak volume; Brilliant scaffolds delight; ChatGPT tutors monologue without diagnosis; traditional tutoring delivers answers. MindCraft's L1-safe line: not "more practice" —

better-designed practice (one edge, method-changing feedback, solo proof). Minutes, XP, and fluent help are not DP.

LI.8 SAFE-DP stack (ship rule)

1. **Diagnose** — Pick one edge from graph / gap / tutor brief (concept, ingredient, bridge, format, or confidence-miss class).
2. **Isolate** — Decompose to the failing subprocess (ontology), not a random worksheet.
3. **Goal** — State a performance micro-goal for the block ("solve rate problems when the story hides the function").
4. **Stretch** — Tasks beyond reliable solo, inside SAFE-DD capacity (equip/destake first if anxious).
5. **Attempt** — Student generates; no AI monologue-first (PWC / XL).
6. **Feedback** — Informative, method-revising; confidence-tiered when elicited (SAFE-CALIB / MISCON).
7. **Revise** — Immediate retry or isomorphic variant that uses the new method.
8. **Prove** — Short unaided / format-hop check; log `solo_transfer_pass`, not minutes.
9. **Stop** — Respect effort constraint; schedule sleep/spacing (SAFE-TIMING / XXIX).
10. **Measure** — Edge-target hit rate, method-change coding, transfer — never 10k-hours progress bars.

Marketing language that survives: "Practice at your edge," "coach-designed sessions," "feedback that changes your next try," "show what you can do alone."

Marketing language that dies: "10,000 hours," "grind mindset," "practice makes perfect," "XP = mastery," "our AI is your deliberate-practice coach" without diagnosis spine, talent-denial ("anyone can be elite if they DP").

LI.9 Claim ladder

Claim	Max ladder without new data
Strict DP ≠ mere study hours / drill fluency	L1 (Ericsson 1993; Lehtinen 2017; Plant et al. 2005)
Popular "DP explains almost all expertise" overclaims; education variance small in Macnamara et al.	L1 (2014 meta; reply chain)
Coach/teacher-designed edge tasks + feedback beat naive repetition for skill change	L1 (expertise literature consensus on quality > raw hours)
MindCraft 45-min SAFE-DP spine improves solo transfer vs homework-help control	L1 product bet — needs DP experiments
DP volume alone creates identity as mathematical thinker	**Banned** (L3 without instruments; Hambrick wound)
Talent/opportunity irrelevant	**Banned** as brand

LI.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
DP-1	SAFE-DP spine vs homework-help / explain-first session (same tutor time)	2-arm	`solo_transfer_pass`; method-change code	Help ≥ DP spine → spine overstated
DP-2	One edge target vs multi-topic coverage in 45 min	2-arm	delayed isomorphic accuracy; cognitive load / affect	Coverage ≥ focus on learning → “one edge” wrong for ACT cram segment
DP-3	Engine-proposed ingredient target vs tutor-chosen topic without graph	2-arm	edge-hit; transfer	Tutor-only ≥ engine → graph targeting weak
DP-4	Feedback revises method (SE + springboard) vs answer-key-only after miss	2-arm	next-attempt success; `retry_120s`	Key-only ≥ method feedback → DP feedback claim weak
DP-5	Session cap (~45–60 focused min) vs encouraged grind (+30 min similar items)	2-arm	delayed accuracy; sleep/affect	Grind ≥ cap → effort-constraint messaging wrong for this population
DP-QUAL	10 Maya + 5 tutor interviews: “real practice” vs help	interview	coded: edge / threat / agency	Spine feels like cruelty → redesign destake

Pre-reg (XXXIV): DP-* identify **session architecture / targeting / feedback quality** — not “DP causes identity” or “explains elite careers.” Densifies FEI, PWC, SAFE-DD/TRANSFER/MISCON/CALIB/TIMING; prefer transfer metrics over minutes.

LI.11 Confidence table

Claim	Label	Confidence
Ericsson et al. (1993): DP = coach-guided, feedback-rich, effortful practice beyond current performance	FACT	High

Practice duration constrained; rest matters; purposeful ≠ deliberate without coach (*Peak*)	FACT	High
Macnamara et al. (2014): modest DP variance; ~4% education; definition dispute unsettled	FACT	High
Lehtinen et al.: math drill ≠ DP; Plant et al.: hours ≠ achievement when quality ignored	FACT	High
SAFE-DP is right default session grammar for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
10k-hours / talent-denial / grind NS transform learners	SPECULATION / false as doctrine	Low (against)

LI.12 What this chapter kills

1. **Kill:** 10,000-hours / "practice makes perfect" as causal brand.
2. **Kill:** Streak minutes, problem counts, or XP as DP proof.
3. **Kill:** "DP explains expertise — therefore our app creates experts" (education ~4% wound).
4. **Kill:** Hard drill ≡ DP; AI monologue ≡ DP coach; talent-denial; multi-topic coverage theater as default rigor.
5. **Wound:** Violin-lab DP ≠ anxious ACT novice tutoring — survive as quality grammar + SAFE-DD, not music cosplay.
6. **Survive:** SAFE-DP spine; ontology edge targeting; method-revising feedback; effort caps; DP-1…5; "better-designed practice" copy.

Doctrine until data: Ship **SAFE-DP**: one diagnosed edge, effortful attempt, feedback that changes the next try, short solo proof, stop before grind — never sell hours, streaks, or talent-denial as the learning engine.

Part LII — Community of Practice (Lave/Wenger): Legitimate Peripheral Participation — Careful Transfer

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 12 ≡ Red Team slot, but ch52 never written → prefer Researcher per rotation; 2 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What does *community of practice* / *legitimate peripheral participation* (LPP) actually claim — and how far can MindCraft transfer those ideas into product, tutoring, story worlds, and "Community" UX before the transfer becomes cosplay?

Owners: Product (social / tutor / world design) · Marketing claim ladder · Equity (whose community?) · Red Team

Commercial job: Ship a **SAFE-CoP** stack — participation trajectories with real old-timers and real practices, not Discord-as-community, leaderboard-as-belonging, or "we're a CoP" brand — while killing apprenticeship theater that ignores Boylan/Fuller wounds.

LII.1 Why this chapter exists

The Constitution's thesis is identity transformation under difficulty. Lave & Wenger's situated-learning line is the classic social account of how newcomers become members of a practice — midwives, tailors, quartermasters — not how students complete problem sets. Edtech decks love the phrase *community of practice* because it sounds like belonging, mentors, and identity without specifying *what practice* students are joining.

MindCraft already has social-adjacent surfaces: college tutors, story worlds, Community nav, parent/tutor roles, Map as shared ontology. Without this chapter, those surfaces drift toward (a) Duo-style leagues as belonging, (b) forum/Discord theater as CoP, or (c) AI chat as "apprenticeship" with no old-timer practice. Parts XXV (identity), XXXVI (equity of worlds), XLII (SAFE-COMPARE), XLVI (tutor expectancy), and LI (SAFE-DP coach role) constrain pieces; what was missing was an honest transfer of LPP into product law.

FOUNDER BELIEF under audit: Becoming a confident mathematical thinker is partly *joining a practice* (ways of posing, checking, arguing, recovering from wrong), not only acquiring items — but school/ACT prep is not a tailor shop, and naming the product a CoP does not make LPP true.

Claims we refuse as doctrine:

1. "MindCraft is a community of practice" as marketing without naming the practice and the peripheral→central trajectory.
2. Forums, streaks, or chat volume as proof of LPP.
3. AI monologue as old-timer apprenticeship.
4. School/ACT classrooms (or solo app sessions) as interchangeable with ethnographic apprenticeship sites.

LII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Community of practice (CoP)	People engaged over time in a joint enterprise with shared repertoire (Wenger, 1998)	Tutors + peers + world characters sharing *how math is done here*	Calling any group chat a CoP
Legitimate peripheral participation (LPP)	Newcomers participate in real practice at the edge; access and legitimacy grow toward fuller roles (Lave & Wenger, 1991)	Student contributes partial reasoning / checks; tutor/old-timer holds the full craft	Watching a video ≠ peripheral *participation*

Old-timer / newcomer	Asymmetric competence + responsibility in the practice	College tutor, veteran Maya peer, coach spine	Peer leaderboard as fake expertise hierarchy
Repertoire	Shared ways of talking, tools, stories, criteria of “good work”	Soft-wrong language; Map ingredients; story check-moves	Lore dump without practice criteria
Acquisition metaphor	Learning as gaining knowledge/skills as possessions (Sfard, 1998)	Mastery graph, item bank, transfer probes	Only metaphor → identity ignored
Participation metaphor	Learning as becoming a member of a practice (Sfard, 1998)	“How we recover from wrong here”	Only metaphor → contentless belonging
SAFE-CoP	CoP doctrine that survives Boylan/Fuller transfer wounds + Sfard pluralism	See LII.8	Apprenticeship cosplay; Discord theater

Operational definition (HYPOTHESIS): A MindCraft interaction counts as *LPP-shaped* when (a) the student does a real piece of the practice (attempt, explain, check, revise — not only consume), (b) an old-timer (human tutor or SE-constrained coach) models criteria of competence, (c) legitimacy is task/role based (you may try this edge), not trait based (you are a weak student — SAFE-EXPECTANCY), and (d) progress is visible as *changing participation* (more solo responsibility, wider repertoire) — not as XP.

LII.3 Lave & Wenger (1991) — what LPP actually says

FACT (foundational book): Lave & Wenger (1991, *Situated Learning: Legitimate Peripheral Participation*, Cambridge University Press, ISBN 9780521423748; doi:10.1017/CBO9780511815355) — learning viewed as situated activity has as its central defining characteristic **legitimate peripheral participation**: learners participate in communities of practitioners and move toward fuller participation in the sociocultural practices of the community. LPP names relations among newcomers and old-timers, activities, identities, artifacts, and communities of knowledge and practice — not a classroom seating chart.

FACT (cases): Their analytic cases are midwives, Yucatec midwives’ apprentices, Vai and Gola tailors, navy quartermasters, butchers, and Alcoholics Anonymous — craft and mutual-aid settings with durable joint enterprises, not standardized exam prep.

FACT (learning subsumed): They argue that becoming a participant *includes* skill learning; identity and membership are not add-ons after content delivery.

Product translation: MindCraft’s research question (“confident mathematical thinkers”) rhymes with *identity in practice*. The danger is importing the *label* while shipping acquisition-only UX (explain → pick A–D → streak).

Kill: “Situated learning = our story world backdrop.” Survive: “Participation means doing pieces of mathematical practice with growing responsibility.”

LII.4 Wenger (1998) — meaning, practice, community, identity

FACT: Wenger (1998, *Communities of Practice: Learning, Meaning, and Identity*, Cambridge University Press; doi:10.1017/CBO9780511803932) — social theory of learning with four interlocking components: **meaning** (experiencing the world as meaningful), **practice** (shared historical/social resources for mutual engagement), **community** (social configurations where competence is recognized), **identity** (learning changes who we are). CoPs are informal groupings that form as people pursue shared enterprises over time — primary unit of analysis is neither lone individual nor formal institution alone.

HYPOTHESIS for MindCraft: The product must specify the *enterprise* (e.g., recovering from productive wrongs on ACT-path concepts; telling-and-checking in a story world) and the *repertoire* (soft-wrong, SE prompts, Map language). Without enterprise + repertoire, "Community" nav is a social tab, not a CoP.

Wound: Corporate/edtech CoP literature often flattens Wenger into "knowledge-sharing groups." MindCraft must not sell community features as identity transformation (claim ladder L3) from engagement metrics (L0/L1).

LII.5 The transfer wound — school math is not a tailor shop

FACT (school mathematics critique): Boylan (2010, *Teaching and Teacher Education*, 26(1), 61–70, doi:10.1016/j.tate.2009.08.005) — LPP was developed from informal apprenticeship contexts; applicability to school classrooms is **problematic**, especially where teacher-centred, decontextualised procedural practices dominate "usual" school mathematics. He critiques LPP transfer on social practice, learning relationships, power, agency, and identity; proposes **ecologies of participation** to capture multidimensional participation (imposed/voluntary, engagement/disengagement, identity risk/maintain, marginal/peripheral, etc.). He notes that if LPP is not found in usual school math, treating it as a *universal* feature of situated learning is undermined — while still valuing Lave & Wenger as a contrastive model and intervention heuristic.

FACT (workplace reassessment): Fuller, Hodkinson, Hodkinson & Unwin (2005, *British Educational Research Journal*, 31(1), 49–68, doi:10.1080/0141192052000310029) — LPP remains insightful for workplace learning research, but has **significant limitations** in complex institutional settings (manufacturing, secondary schools): diverse patterns of participation; institutional environments configure opportunities and barriers that ethnographic craft cases under-specify.

FACT (metaphor pluralism): Sfard (1998, *Educational Researcher*, 27(2), 4–13, doi:10.3102/0013189X027002004) — acquisition and participation metaphors each entail desirable and harmful practices; **choosing just one** risks theoretical distortion and bad pedagogy. MindCraft's graph/mastery engine is acquisition-shaped; identity thesis is participation-shaped — doctrine needs both, not a CoP slogan that deletes assessment.

FACT (math classroom situated work — careful ally): Boaler (1999 AERA paper / related classroom studies; see also Boaler & Greeno identity/agency line summarized in math-education CoP literature) — classroom communities' constraints and affordances shape what students learn to *do*; formalized classrooms attune students to practices that may not transfer. Useful as evidence that *participation forms matter*, not as proof that MindCraft already is an apprenticeship CoP.

HYPOTHESIS: Treat LPP as a **design aspiration for tutor+practice sessions** (peripheral contribution → fuller solo responsibility) and a **diagnostic for fake community**, not as a claim that the app *is* a CoP.

Kill: Universal "students learn by LPP here." Survive: "Usual school math often blocks LPP; we design for legitimate attempts at the edge with old-timer criteria."

LII.6 What "practice" means for MindCraft (not vibes)

Align with Constitution practices already named:

1. **Diagnose** → **attempt** → **informative miss** → **revise** → **solo check** (FEI / SAFE-DP / SAFE-MISCON).
2. **Student-generated why before AI wrap** (PWC / XL).
3. **Named transfer hops** (SAFE-TRANSFER).
4. **Calibrated confidence talk** (SAFE-CALIB) — saying "I'm sure" as part of the repertoire, not a Belief Score™.
5. **Story worlds as wrappers of frozen math** (equity + meaning) — characters can model *peripheral roles* (watcher → helper → solver) without claiming ethnographic authenticity.

SPECULATION (to test): Peer "Community" features help only when they circulate the repertoire above (how we check; how we destake wrongs). Social chat about school stress without practice criteria may raise belonging feelings without changing mathematical participation (wound for growth loops).

LII.7 Competitive positioning

Competitor pattern	CoP-shaped risk	MindCraft wedge
Duo leagues / streaks	Belonging as rank theater (SAFE-COMPARE)	Criterion competence, not named infinite ranks
Khan comments / help forums	Peripheral *talk about* problems ≠ LPP in craft	Tutor/old-timer + attempt-first
Brilliant delight social	Shared puzzles without recovery repertoire	Soft-wrong + solo transfer as membership proof
ChatGPT "study buddy"	Fluency without legitimacy criteria	PWC + ontology spine; AI ≠ old-timer by default
Discord "study community"	CoP brand, acquisition optional	Enterprise + repertoire required for CoP copy

Commercial implication: Do not out-community Discord. Sell *becoming someone who participates differently in hard math* — with evidence (`solo_transfer_pass`, method-change after miss), not member counts.

LII.8 SAFE-CoP stack (product law)

1. **Name the practice** — joint enterprise in copy and UX (“how we recover and prove on this edge”), not “join our community.”
2. **Peripheral must be real work** — attempt, partial solution, check, SE prompt; watching/scrolling is not LPP.
3. **Old-timer required for CoP claims** — human tutor or SE-constrained coach with criteria; unguarded AI chat ≠ apprenticeship.
4. **Legitimacy without trait labels** — access to roles by task readiness; ban Golem scripts (SAFE-EXPECTANCY).
5. **Trajectory > tab** — measure changing participation (solo share up; help faded) not Community page visits.
6. **Plural metaphors** — keep acquisition instruments (graph, bank); never replace them with belonging KPIs (Sfard).
7. **Equity of membership** — whose history/stories define the repertoire (Part XXXVI); token “community” without belonging audit is a kill.
8. **Claim ladder** — CoP/LPP language max L1 product bet until CoP-* experiments; ban L3 “identity via community features” without instruments.

LII.9 Claim ladder

Claim	Max ladder without new data
LPP describes apprenticeship-style learning in craft/mutual-aid cases	L1 (Lave & Wenger, 1991)
Usual school math often fails LPP conditions; transfer is non-trivial	L1 (Boylan, 2010; Fuller et al., 2005)
Acquisition + participation metaphors both needed; one alone distorts	L1 (Sfard, 1998)
MindCraft tutor sessions can implement LPP-shaped trajectories	L1 product bet — needs CoP experiments
Forums / streaks / AI chat = CoP / LPP	**Banned**
Shipping Community features creates mathematical identity	**Banned** L3 without instruments

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LII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
CoP-1	LPP-shaped session (peripheral contribution + old-timer criteria + fade to solo) vs explain-first homework help	2-arm	`solo_transfer_pass`; coded participation moves	Help ≥ LPP spine → CoP framing overstated

CoP-2	Named repertoire onboarding ("how we check here") vs feature tour	2-arm	next-session method language; `retry_120s`	Tour ≥ repertoire → practice-naming useless
CoP-3	Human tutor old-timer vs unguarded AI buddy (same time)	2-arm	solo transfer; overtrust flags (XXXIII)	AI ≥ tutor on solo → old-timer claim weak for this segment
CoP-4	Community chat with repertoire prompts vs open social chat	2-arm	practice-talk rate; delayed accuracy	Open ≥ prompted → repertoire constraint wrong
CoP-5	Story peripheral roles (watcher→ helper→solver) vs flat problem list	2-arm	challenge_accept ; belonging items; transfer	Flat ≥ roles on learning → role trajectory theater
CoP-QUAL	10 Maya + 5 tutor: "who is in your math people?"	interview	coded: old-timer / repertoire / legitimacy	"Community" = friends only, no practice → social tab demote

Pre-reg (XXXIV): CoP-* identify **participation design / old-timer necessity / fake-community kill** — not "CoP causes identity" or "LPP is universal in apps."

LII.11 Confidence table

Claim	Label	Confidence
Lave & Wenger (1991): LPP as central descriptor of situated learning in apprenticeship cases	FACT	High
Wenger (1998): CoP as meaning –practice–community–identity	FACT	High
Boylan (2010): LPP transfer to usual school math is problematic; ecologies of participation	FACT	High
Fuller et al. (2005): LPP limited in complex institutions	FACT	High
Sfard (1998): do not choose only acquisition or only participation	FACT	High

SAFE-CoP is right default for MindCraft social/tutor design	FOUNDER BELIEF / HYPOTHESIS	Medium
Community features alone transform math identity	SPECULATION / false as doctrine	Low (against)

LII.12 What this chapter kills

1. **Kill:** “We are a community of practice” as brand without enterprise + repertoire + old-timer.
2. **Kill:** Forums, Discord, streaks, leagues, or chat volume as LPP proof.
3. **Kill:** AI monologue ≡ apprenticeship / old-timer.
4. **Kill:** Treating school/ACT/app sessions as ethnographically equivalent to Lave & Wenger’s craft cases.
5. **Wound:** Participation-only pedagogy that deletes mastery/transfer instruments (Sfard).
6. **Survive:** SAFE-CoP; LPP-shaped tutor spine; peripheral = real work; trajectory metrics; CoP-1…5; “join the practice of recovering and proving” copy — not “join the community.”

Doctrine until data: Ship **SAFE-CoP** — name the practice, require real peripheral work and old-timer criteria, measure participation trajectories under acquisition+participation pluralism — never sell social tabs or CoP labels as identity engines.

Part LIII — Religion / Ritual Light-Touch Design: Meaning Without Cult Dynamics

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 15; hour%6≠0; 3 researcher entries since synthesizer v1.7 → Researcher)

Primary question: How can MindCraft design *light-touch rituals* that mark transitions, regulate affect, and carry meaning — without drifting into cult dynamics, sacred-science branding, or high-arousal initiation theater?

Owners: Product (session / world / onboarding arcs) · Brand & copy · Equity / trust · Red Team

Commercial job: Ship a **SAFE-RITUAL** stack — brief, opt-in, secular, reversible meaning-markers tied to real practice — while killing “join the order,” streak liturgies, confession milieus, and unretracted ritual→grade overclaims.

LIII.1 Why this chapter exists

The Constitution’s thesis is identity transformation under difficulty. Humans mark identity shifts with *rituals*: greetings before hard work, closing circles after miss→revise, naming a first solo transfer. Edtech and consumer apps already smuggle ritual language — streaks as devotion, leagues as congregations, onboarding as initiation, founder myth as sacred science — usually without naming the load-bearing psychology or the failure modes.

MindCraft has natural ritual-shaped moments: gap-scan completion, first soft-wrong recovery, tutor session open/close, story-world threshold crossings, Map unlocks. Parts XXIV (affect), XXV (identity), XXXI (flow), XLIII (SAFE-HABIT), XLIV (SAFE-REWARD), XLV (resilience), LII (SAFE-CoP) constrain pieces. What was missing was an honest doctrine that (a) uses ritual science for *light* meaning, (b) borrows transition structure without religion cosplay, and (c) installs a cult-dynamics kill list so "meaning" never becomes totalism.

FOUNDER BELIEF under audit: Becoming a confident mathematical thinker needs *marked moments* (not only item completion) — but the company must never sell belonging-through-devotion, milieus that punish exit, or doctrine that overrides the student's evidence.

Claims we refuse as doctrine:

1. Rituals (or "sacred practice") as proven grade engines from unretracted lab theater.
2. Streak / league / XP liturgy as identity North Star.
3. High-arousal initiation, confession milieus, or "you're not one of us until…" gates.
4. Founder/product doctrine as sacred science that cannot be falsified (contradicts the lab itself).

LIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Ritual (psych)	Predefined, symbolic, often repetitive sequence that is *more than* its instrumental steps (Hobson et al., 2018)	Brief open/close; naming a miss class; candle-free "we begin"	Calling every UX animation a ritual
Routine / habit	Repeated behavior cue→response without required symbolic meaning (SAFE-HABIT)	If-then practice cue	Relabeling habit as sacred
Rite of passage	Transition marked by separation → liminal → reaggregation (van Gennep; Turner)	Gap-scan → liminal "not yet rated" → diagnosticCompleted + first path	Initiation hazing; permanent limbo
Imagistic mode	Infrequent, high-arousal rituals → vivid episodic memory; small exclusive groups (Whitehouse, 2004)	Rare world "boss" moments	Trauma theater; faction bonding over learning

Doctrinal mode	Frequent, low-arousal repetition → semantic/procedural memory; larger inclusive scale (Whitehouse, 2004)	Soft-wrong language; Map repertoire; session grammar	Indoctrination scripts; empty liturgy
Totalism / cult dynamics	Thought-reform pattern: milieu control, confession cult, sacred science, loaded language, doctrine-over-person, exit costs (Lifton)	Anti-patterns in copy, retention, community	"Family," purity tests, unfalsifiable brand
SAFE-RITUAL	Light-touch ritual doctrine that survives cult kills + retraction wounds	See LIII.8	Sacred streak; initiation cosplay

Operational definition (HYPOTHESIS): A MindCraft moment counts as *SAFE-RITUAL* when (a) it is brief and optional or easily skippable, (b) symbolism points at *practice repertoire* (recover, prove, calibrate) not brand devotion, (c) exit/skip has no status punishment, (d) no confession or purity demand, (e) claims stay on the ladder (affect/transition aid ≠ causal identity engine).

LIII.3 Psychology of ritual — regulatory functions (not magic grades)

FACT (integrative review): Hobson, Schroeder, Risen, Xygalatas & Inzlicht (2018, *Personality and Social Psychology Review*, 22(3), 260–284, doi:10.1177/1088868317734944) — organize empirical ritual work around three regulatory functions: **(a) emotion regulation**, **(b) performance goal states**, **(c) social connection**. Mechanisms include bottom-up features (repetition, formality, synchrony) and top-down meaning (belief that the sequence *is* a ritual / carries symbolic significance).

FACT (grief / control): Norton & Gino (2014, *Journal of Experimental Psychology: General*, 143(1), 266–272, doi:10.1037/a0031772) — rituals after losses (loved ones, relationships, lotteries) can reduce grief feelings and increase felt control relative to no-ritual controls. Useful as evidence that symbolic sequences can regulate emotion and agency — **not** as a math-learning RCT.

FACT (consumption / meaning frame): Vohs, Wang, Gino & Norton (2013, *Psychological Science*, 24(9), 1714–1721, doi:10.1177/0956797613478949) — ritualized actions can enhance consumption experience via greater involvement/savoring. Ally for "mark the moment" design; kill if used to justify gamified consumption of XP as learning.

WOUND (retraction — do not market): Brooks, Schroeder, Risen, Gino, Galinsky, Norton & Schweitzer (2016, *Organizational Behavior and Human Decision Processes*, 137, 71–85, doi:10.1016/j.obhdp.2016.07.004) — claimed pre-performance rituals reduce anxiety and improve performance (including high-stakes math-like tasks). **This article has been retracted.** MindCraft must **not** cite it as surviving evidence or sell "rituals raise exam scores." Treat as a Red Team kill on ritual→grade overclaim.

HYPOTHESIS for MindCraft: Prefer Hobson's *emotion regulation* and *goal-state* functions for session openers/closers and soft-wrong recovery markers; treat performance/grade claims as **banned** until unretracted, pre-registered evidence on MindCraft metrics (`retry_120s`, `solo_transfer_pass`, affect check-ins).

Kill: "Science proves rituals improve math performance." Survive: "Light symbolic sequences can regulate affect and mark transitions; grade claims need new evidence."

LIII.4 Modes of religiosity — why high-arousal initiation is the wrong default

FACT: Whitehouse (2004, *Modes of Religiosity: A Cognitive Theory of Religious Transmission*, AltaMira Press / Rowman & Littlefield, ISBN 978-0-7591-0615-4) — religions tend to coalesce around **imagistic** vs **doctrinal** poles. Imagistic: infrequent, high-arousal rituals → flashbulb-like episodic memory; small, exclusive, often intense group bonds. Doctrinal: frequent, low-arousal, routinized teaching/ritual → semantic and procedural memory; larger, more inclusive, centrally scaffolded communities.

FACT (extreme ritual ally — careful): Xygalatas, Mitkidis, Fischer, Reddish, Skewes, Geertz et al. (2013, *Psychological Science*, 24(8), 1602–1605, doi:10.1177/0956797612472910) — extreme rituals can increase prosociality in field settings. This is **not** a product recommendation for teens in ACT prep; it shows imagistic arousal can bond groups — which is exactly the risk surface for cult-adjacent design.

HYPOTHESIS: MindCraft's default should be **doctrinal-light**: frequent, low-arousal repertoire (session grammar, soft-wrong language, Map names) — not imagistic initiation (ordeal unlocks, shame fires, exclusive survivor factions). Rare story "threshold" moments may use mild imagistic *flavor* only if destaked, optional, and never required for belonging (SAFE-DD / equity).

Product translation: Gap-scan and first recovery are transitions, not hazing. World chapters can feel special without pain-as-proof.

Kill: Bootcamp/ordeal branding; "you haven't really suffered until…" Survive: Frequent low-arousal practice markers + rare optional celebration.

LIII.5 Rites of passage — structure without cult costume

FACT: van Gennep (1909; English trans. 1960, *The Rites of Passage*, University of Chicago Press) — life transitions commonly share three phases: **separation**, **margin/limen**, **aggregation** (reincorporation).

FACT: Turner (1969, *The Ritual Process*; see also "Liminality and Communitas") — elaborates the liminal phase as *betwixt and between*: ambiguous status, stripped markers, intense potential for *communitas* (anti-structure bonding). Powerful analytically; dangerous if product designers chase liminality as engagement hack (keep students permanently "in between," dependent on the brand for status).

HYPOTHESIS for MindCraft transitions:

Phase	Product move	Guardrail

Separation	Leave "I don't know where I stand" → enter gap-scan / session	No shame script; task language only (SAFE-EXPECTANCY)
Liminal	Hide-correctness diagnostic; soft-wrong; "not yet aggregated"	Time-bounded; no status mockery; skippable ceremony
Aggregation	`diagnosticCompleted`; first path; named repertoire ("how we recover here")	Grant competence access, not trait identity; no purity test

Link to SAFE-CoP: Aggregation should admit the student to *practice roles* (peripheral → fuller), not to a brand congregation.

Kill: Permanent liminality as retention. Survive: Clear exits into rights/obligations of the practice (you may attempt this edge solo).

LIII.6 Cult dynamics — the kill list (Lifton as anti-pattern checklist)

FACT (classic totalism criteria): Lifton (1961, *Thought Reform and the Psychology of Totalism*; later editions restate cult criteria) — thought-reform / totalistic environments characteristically include patterns such as: **milieu control**, **mystical manipulation**, **demand for purity**, **cult of confession**, **sacred science**, **loading the language**, **doctrine over person**, **dispensing of existence** (who counts as real/valid). Cult-like groups often combine charismatic leader-focus, thought-reform patterns, and exploitation of idealism.

FOUNDER BELIEF: MindCraft's research lab and product ethics require the *opposite* of sacred science: every doctrine chapter is falsifiable; Red Team is a feature; students may leave without metaphysical damnation.

Product anti-patterns (SPECULATION → treat as design law until wounded):

Lifton pattern	Edtech failure mode	MindCraft ban
Milieu control	App as only legitimate math space; shame offline help	Allow tutors, school, parents; no "only here" purity
Mystical manipulation	Streak miracles; "the algorithm chose you" destiny copy	Transparent recommend reasons; no fate talk
Demand for purity	Perfect streaks; zero-miss identity	Soft-wrong; SAFE-MISCON; miss as data
Cult of confession	Forced public failure theater / oversharing	Private by default; opt-in sharing
Sacred science	Unfalsifiable brand claims; ban critique	Claim ladder; lab kills; no 2-sigma marketing
Loading the language	In-group jargon that replaces thinking	Repertoire words OK; ban thought-stopping slogans
Doctrine over person	Override student evidence with brand story	Graph + outcomes over vibes; SAFE-CALIB

Dispensing of existence	Rank/league as who “counts”	SAFE-COMPARE; no exile UX

Kill: “We’re a movement / family / order.” Survive: “We’re a practice with optional meaning-markers and exit dignity.”

LIII.7 Competitive audit — ritual theater in the market

Duo streaks/leagues = doctrinal liturgy + mild imagistic competition (habit colonization; cue habits without sacred streak NS). Khan mastery energy = completion ritual (keep instruments; add recovery markers). Brilliant “aha” = imagistic spark (celebrate then probe transfer). GPT chat = pseudo-confession / endless liminality (SE-constrained close). Studygram hustle = purity + confession (ban grind-as-virtue).

Commercial implication: Position as **meaning without membership taxes** — mark *mathematical practice*, not brand devotion. Copy: “mark the recovery,” “open like a craft,” “close with what changed” — never “join the faithful.”

LIII.8 SAFE-RITUAL stack (product law)

1. **Light and short** — seconds to a minute; never a sermon that displaces attempts (SAFE-DP time budget).
2. **Optional / skippable** — no status loss for skip; no streak-as-morality.
3. **Point at practice** — symbolize recover / prove / calibrate / transfer — not founder, brand, or league.
4. **Doctrinal default** — frequent low-arousal repertoire; rare celebration only, never ordeal.
5. **Bounded liminality** — transitions resolve into clear rights (you may try X); no permanent betwixt for retention.
6. **Anti-totalism** — ban milieu monopoly, confession theater, sacred science, purity, exit shame.
7. **Affect ≠ grade claim** — may aim at calm / agency / belonging-to-practice; **no** unretracted ritual→score marketing.
8. **Equity** — no borrowed sacred forms that mock students’ real religions; secular by default (Part XXXVI).
9. **Claim ladder** — ritual language max L1 product bet until RIT-* experiments; ban L3 identity-via-ceremony without instruments.

LIII.9 Claim ladder

Claim	Max ladder without new data
Rituals can regulate emotion / goal states / connection (mechanism review)	L1 (Hobson et al., 2018)
Symbolic sequences can increase felt control after loss	L1 (Norton & Gino, 2014)
Imagistic vs doctrinal modes describe divergent ritual ecologies	L1 (Whitehouse, 2004)

Rites of passage often share separation–liminal–aggregation	L1 (van Gennep; Turner)
Lifton patterns are useful anti-design checks for edtech	L1 design heuristic (not a clinical diagnosis of competitors)
MindCraft light rituals improve `retry_120s` / affect / solo transfer	L1 product bet — needs RIT experiments
Rituals raise ACT scores / prove identity transformation	**Banned** without unretracted evidence
Streak liturgy / ordeal initiation / “family” milieus as identity engines	**Banned**

LIII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition
RIT-1	30s session open ritual (name edge + breath/pose optional) vs silent start	2-arm	pre-task affect; `challenge_accept`	Silent ≥ ritual on start+transfer → opener useless
RIT-2	Soft-wrong recovery marker (“name the miss class”) vs correct/incorrect only	2-arm	`retry_120s`; method-change code	Binary ≥ marker → recovery ritual theater
RIT-3	Gap-scan aggregation ceremony (brief) vs auto-redirect no mark	2-arm	next-session return; belonging-to-practice item	Auto ≥ ceremony → skip ceremony permanently
RIT-4	Optional celebration on first `solo_transfer_pass` vs none	2-arm	delayed transfer; skip rate	Forced > optional engagement but worse affect → force banned
RIT-5	Streak-framed cue vs practice-repertoire cue (same schedule)	2-arm	4-week return; identity items; anxiety	Streak ≥ repertoire on learning → still ban streak as NS; copy only
RIT-QUAL	10 Maya: which moments felt meaningful vs creepy/cultish?	interview	coded: light / pressure / exit fear	“Family/order” language → kill that copy

Pre-reg (XXXIV): RIT-* identify **transition/affect design and anti-totalism** — not "ritual causes identity" or "rituals raise scores."

LIII.11 Confidence table

Claim	Label	Confidence
Hobson et al. (2018): three regulatory functions of ritual	FACT	High
Norton & Gino (2014): rituals can ease grief / raise felt control	FACT	High
Brooks et al. (2016) ritual→performance paper	FACT it was published; **retracted** — do not use as evidence	High (against use)
Whitehouse (2004): imagistic vs doctrinal modes	FACT	High
van Gennep / Turner: passage / liminality structure	FACT	High
Lifton totalism criteria as edtech anti-patterns	HYPOTHESIS / design heuristic	Medium–High
SAFE-RITUAL is right default for MindCraft meaning UX	FOUNDER BELIEF / HYPOTHESIS	Medium
Ceremony alone transforms math identity	SPECULATION / false as doctrine	Low (against)

LIII.12 What this chapter kills

1. **Kill:** Ritual→exam-score marketing (especially via retracted Brooks et al., 2016).
2. **Kill:** Streak / league / XP as sacred liturgy or North Star.
3. **Kill:** High-arousal initiation, confession milieus, purity tests, exit shame.
4. **Kill:** "Movement / family / order" copy that loads Lifton patterns.
5. **Wound:** Permanent liminality-as-retention; imagistic faction bonding.
6. **Survive:** SAFE-RITUAL; doctrinal-light repertoire; bounded passages; anti-totalism checklist; RIT-1…5; "mark the recovery" copy — not "join the faithful."

Doctrine until data: Ship **SAFE-RITUAL** — brief, skippable, practice-pointing meaning-markers under an explicit cult-dynamics ban — never sell devotion, ordeal, or unretracted ritual-performance claims as the identity engine.

Part LIV — Military / Aviation Brief–Debrief: After-Action Reviews for Math Sessions

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 18 ≡ Red Team slot, but ch54 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable structure do military after-action reviews and aviation brief/debrief cycles offer for MindCraft tutoring and practice sessions — without militaristic cosplay, blame culture, or overclaiming effect sizes as ACT marketing?

Owners: Product (session spine / Notes close) · Tutor ops · Coach UX · Brand & copy · Red Team

Commercial job: Ship a **SAFE-AAR** stack — brief → attempt → structured debrief with objective traces — while killing bootcamp branding, lecture-as-debrief, trauma-debrief confusion, and "debriefs improve performance 25%" as an undifferentiated growth claim.

LIV.1 Why this chapter exists

Parts LI (SAFE-DP) and LIII (SAFE-RITUAL) already constrain *time budgets* and *light meaning markers*. What was missing is the **operational cycle** high-reliability domains use to convert experience into the next attempt: a short **brief** that sets intent and roles, an event under observation, and a **debrief / after-action review (AAR)** that compares expected vs actual performance without collapsing into critique-as-humiliation.

MindCraft already has the raw materials: session open (affect check-in), attempt traces, soft-wrong codes, outcomes posted to the graph, Notes/session summaries, and coach cards. Competitors either skip structured reflection (answer-delivery chat), fake it with streak liturgy (Duo), or replace it with instructor monologue (many tutoring products and AI wrap text). The commercial wedge is not "we're like the military." It is: **every hard attempt leaves a learning artifact the student helps interpret.**

FOUNDER BELIEF under audit: Identity transformation under difficulty requires *reviewed attempts*, not only more problems — but review must be non-punitive, student-participatory, and tied to the next edge (SAFE-DP), not to brand devotion (SAFE-RITUAL) or trait shame (SAFE-EXPECTANCY).

Claims we refuse as doctrine:

1. Militaristic / "warrior math" branding as belonging technology.
2. AAR ≡ critical-incident stress debriefing (trauma processing) — different animal.
3. Marketing "+25% performance" from debrief meta-analyses as if it were an ACT score guarantee.
4. Instructor/AI lecture that *calls itself* a debrief while blocking student self-analysis.

LIV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Brief	Short pre-event alignment on intent, success criteria, roles, likely traps (CRM / FAA markers)	Session open: target edge, level, “what counts as success,” optional coach role	10-min sermon; fate talk; trait labels
After-action review (AAR)	Systematic technique turning a recent event into learning via feedback, reflection, discussion (Keiser & Arthur)	Post-mission 4-question close; tutor end-of-session AAR	Critique lecture; blame; vibes-only
Debrief (training)	Structured reflection after simulation/performance; often facilitator-guided	Coach/tutor facilitation; self-led when traces exist	Endless chat liminality; AI monologue
Objective performance media	Records of what happened (video, logs, instruments) — not memory alone	Attempt list, soft-wrong codes, timing, outcomes JSON	Relying on “how did it feel?” only
Facilitation (vs lecture)	Elicit crew/student self-analysis; deepen participation (Dismukes et al.)	Advocacy–inquiry; student names frame first	Tutor/AI tells the moral of the story
Good judgment stance	Share evaluative judgment + curiosity about learner’s frames (Rudolph et al.)	Soft-wrong + “what were you seeing?”	Fake nonjudgmental praise; or blame
Alignment	Match participants ↔ focus ↔ measurement (team vs individual)	Solo practice AAR on solo metrics; tutor+student AAR on shared plan	Team vibes measured by individual XP
SAFE-AAR	Brief–attempt–debrief doctrine that survives militarism / blame / overclaim kills	See LIV.8	Bootcamp cosplay; 25%-as-marketing

Operational definition (HYPOTHESIS): A MindCraft close counts as *SAFE-AAR* when (a) a pre-brief named the intended edge and success criterion, (b) the debrief answers expected / actual / sustain / improve using **at least one objective trace**, (c) the student produces ≥1 self-analytic statement before any AI wrap, (d) the output is a next-attempt commitment (not a grade sermon), (e) no trait labels, no punishment for miss, no militaristic belonging tax.

LIV.3 Army AAR doctrine — four questions, not a roast

FACT (doctrine lineage): U.S. Army training circular **TC 25-20** (*A Leader's Guide to After Action Reviews*, 30 Sep 1993) and subsequent training doctrine institutionalize the AAR as a professional discussion focused on training objectives — not a humiliation ritual. Contemporary restatements (e.g., Center for Army Lessons Learned handbooks; USAID AAR guidance adapted from TC 25-20) center four questions:

1. What was **expected** to happen?
2. What **actually** happened?
3. What went **well**, and why?
4. What can be **improved**, and how?

FACT (process intent): Formal Army AAR guidance emphasizes open-ended questioning so participants discover gaps vs standards; facilitators must avoid critique-as-lecture (FM 25-101 Appendix G / CALL lineage). Focus: **results vs objectives**, sustain strengths, actionable fixes.

HYPOTHESIS for MindCraft: Map onto a 60–120s practice close and a 3–5 min tutor close — brief names expected edge; traces show actual; student names one sustain; one improve-how (SAFE-HABIT if-then or method change). Kill "AAR = public roast." Survive: standards-referenced discovery.

LIV.4 Meta-analytic evidence — real gains, careful transfer

FACT: Tannenbaum & Cerasoli (2013, *Human Factors*, 55(1), 231–245, doi:10.1177/0018720812448394) — meta-analysis of 46 samples (N = 2,136). Debriefs / AARs improve effectiveness vs control by about **25% on average** ($d \approx 0.67$). Effects similar for teams and individuals, simulated and real settings, medical and nonmedical samples. **Alignment** (participants, intent, measurement level) bolsters effects; facilitated and structured debriefs trend stronger (facilitation estimate based on fewer nonfacilitated studies).

FACT: Keiser & Arthur (2021, *Journal of Applied Psychology*, 106(7), 1007–1032; advance online 2020, doi:10.1037/apl0000821) — bare-bones meta-analysis of 61 studies (107 *ds*; 915 teams and 3,499 individuals). Overall AAR effect d ≈ **0.79** across multiple training criteria — larger than Tannenbaum & Cerasoli's task-performance-focused estimate. Two characteristics consistently help: **(a) alignment** to individual vs team, **(b) objective performance review media**. Facilitation interacts: self-led + team-aligned and self-led + objective media among stronger combinations; structure matters more in military than healthcare contexts.

HYPOTHESIS (transfer wound): These samples are largely military, healthcare simulation, and organizational training — **not** ACT prep for anxious Maya. Treat $d \approx 0.67$–0.79 as evidence that structured post-event reflection is a high-leverage training design pattern — **not** license to claim "MindCraft AARs raise SAT/ACT by 25%."

Commercial claim ladder:

- L1: Debriefs/AARs improve training outcomes in reviewed domains (cite metas).
- L1 product bet: SAFE-AAR improves MindCraft instruments (`retry_120s`, method-change codes, `solo_transfer_pass`).
- **Banned:** Undifferentiated "science says +25%" on landing pages without domain caveat and MindCraft pre-reg.

Kill: 2-sigma-style overclaim via borrowed meta d. Survive: "Structured brief–debrief with traces beats vibes-only close."

LIV.5 Aviation — brief as error management; debrief as facilitated self-analysis

FACT (CRM briefing as performance marker): FAA Advisory Circular **AC 120-51E** (*Crew Resource Management Training*) lists effective **briefings** among CRM behavioral markers: thorough, address coordination/planning/problems, establish open communication, set expectations for SOP deviations and automation duties. Briefings are error-management tools — they align the crew *before* the event, not after the crash.

FACT (CRM evolution): Helmreich, Merritt & Wilhelm (1999, *International Journal of Aviation Psychology*, 9(1), 19–32; see also Helmreich line of NASA/UT work) — CRM evolved toward error management: avoid, trap, mitigate. Briefing strategies and situation awareness modules are framed as countermeasures, not soft skills theater.

FACT (LOFT debrief facilitation): Dismukes, Jobe & McDonnell (1997, NASA TM-110442, *LOFT Debriefings: An Analysis of Instructor Techniques and Crew Participation*) — analyzed 36 LOFT debriefings at five U.S. airlines; developed the Debriefing Assessment Battery. **Facilitation** (vs instructor lecture) increases depth of crew participation and self-analysis of CRM performance; instructor skill varied dramatically; crews were responsive but rarely fully led their own debriefs. Companion: McDonnell, Jobe & Dismukes (1997, NASA TM-112192, *Facilitating LOS Debriefings: A Training Manual*).

HYPOTHESIS for MindCraft:

- **Brief** = gap-scan / mission card / tutor open: name the edge, success criterion, known misconception trap, role of coach (hints vs silence).
- **Debrief** = student-led analysis with tutor/coach as facilitator; AI may *prompt* advocacy–inquiry, must not deliver the sermon first (aligns with Part XL SE constraints and Part XXXIII AI trust).

Product translation: Tutor briefing ethics from SAFE-EXPECTANCY (task-only language) + CRM brief markers (coordination, traps, automation = "when does the coach speak?").

Kill: Captain-monologue onboarding. Survive: short interactive brief + facilitated close.

LIV.6 Healthcare simulation — good judgment, not fake nonjudgment

FACT: Rudolph, Simon, Dufresne & Raemer (2006, *Simulation in Healthcare*, 1(1), 49–55, doi:10.1097/01266021-200600110-00006; PubMed 19088574) — "There's no such thing as 'nonjudgmental' debriefing." **Debriefing with good judgment:** (1) frames drive actions → results; (2) stance of genuine curiosity (mistakes as puzzles); (3) **advocacy–inquiry** pairs (state observation + judgment, then ask what the learner's frame was). Helps instructors share evaluative judgment without destroying trust.

Link to SAFE-MISCON / SAFE-CALIB: Soft-wrong is not empty kindness; it is destaked judgment plus frame inquiry. Hide-correctness diagnostic is a special brief (no outcome reveal) — its "debrief" must not smuggle keys that corrupt the graph (C4).

HYPOTHESIS: MindCraft coach cards should default to advocacy–inquiry templates ("I notice you treated the inequality like an equation when you multiplied by −1 — what were you assuming?") rather than praise-only or scold-only.

Kill: "Nonjudgmental forever" (hides standards) and "brutal honesty" (hides curiosity). Survive: good-judgment debrief.

LIV.7 Distinctions and kill list (scope control)

Do not confuse training AARs with **critical-incident stress debriefing** (trauma processing — different ethics). Ban militaristic belonging ("warrior," bootcamp shame), streak-as-AAR, AI monologue labeled as debrief, vibes-only hot-wash when traces exist, public leaderboard "reviews" (SAFE-COMPARE), and "+25% smarter" ads from borrowed meta *d*.

Wound: Hot-wash alone under-captures learning — useful as phase 1, insufficient as the whole product. MindCraft default: immediate 60s close + optional next-day Notes deepening.

LIV.8 SAFE-AAR stack (product law)

1. **Brief before attempt** — name edge, success criterion, trap, coach role (CRM brief markers).
2. **Attempt under observation** — log objective media (outcomes, codes, timing).
3. **Four-question close** — expected / actual / sustain / improve (TC 25-20 lineage).
4. **Student speaks first** — ≥1 self-analytic statement before tutor/AI wrap (Dismukes facilitation).
5. **Good judgment** — advocacy–inquiry; standards visible; curiosity about frames (Rudolph).
6. **Align level** — solo metrics for solo practice; shared plan metrics for tutor sessions (Tannenbaum; Keiser).
7. **Next-attempt output** — one concrete improve-how tied to SAFE-DP time budget.
8. **Non-punitive climate** — miss = data; no trait labels (SAFE-EXPECTANCY); no militaristic shame.
9. **Claim ladder** — meta *d* stays L1 domain evidence; MindCraft ACT claims need AAR-* experiments.
10. **Anti-cosplay** — borrow the *cycle*, not the costume; secular practice language (SAFE-RITUAL).

LIV.9 Competitive audit — who actually closes the loop?

Khan tells correctness without student-led expected/actual. Duo substitutes streak liturgy for AAR. Brilliant celebrates aha but rarely closes sustain/improve. ChatGPT tutors lecture after chatty "goals." Human tutoring quality varies with facilitator skill (Dismukes variance). MindCraft already logs graph outcomes — formalize SAFE-AAR in practice close + Notes.

Commercial implication: Position as **reviewed struggle** — brief the edge, attempt with traces, close expected/actual — never "military-grade" or borrowed "+25%" miracles. Parent copy: expected vs happened → what changes next. Student copy: "name what you thought would happen."

LIV.10 Claim ladder

Claim	Max ladder without new data

AARs/debriefs improve training outcomes in meta-analyzed domains ($d \approx 0.67$–0.79)	L1 (Tannenbaum & Cerasoli, 2013; Keiser & Arthur, 2021)
Four-question AAR structure is institutionalized Army training practice	L1 (TC 25-20 lineage / CALL restatements)
Facilitation deepens self-analysis vs lecture in LOFT debriefs	L1 (Dismukes et al., 1997)
CRM briefings are performance markers for coordination / error management	L1 (FAA AC 120-51E; Helmreich line)
Good-judgment advocacy–inquiry is a validated simulation debrief stance	L1 method (Rudolph et al., 2006)
SAFE-AAR improves MindCraft `retry_120s` / method-change / solo transfer	L1 product bet — needs AAR experiments
"+25%" / meta *d* as ACT or identity guarantee	**Banned**
Bootcamp / warrior branding as belonging engine	**Banned**
AAR ≡ trauma CISD	**Banned** conflation

LIV.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
AAR-1	4-question practice close vs "nice job / try again" toast	2-arm	`retry_120s`; method-change code rate	Toast ≥ AAR on retry+transfer → close theater
AAR-2	Brief (edge+success+trap) vs no brief, same items	2-arm	`challenge_accept`; first-attempt process quality	No-brief ≥ brief → brief bloat; shorten or cut
AAR-3	Objective-trace AAR vs memory-only AAR	2-arm	accuracy of self-report; next-session transfer	Memory ≥ trace → still keep traces for graph; copy only
AAR-4	Student-first facilitation vs AI-monologue wrap	2-arm	SE quality rubric; `solo_transfer_pass`	Monologue ≥ student-first → contradicts XL; kill monologue default
AAR-5	Tutor trained on advocacy–inquiry vs business-as-usual close	2-arm cluster	session notes quality; student affect; 2-week transfer	BAU ≥ trained → training content fail

AAR-QUAL	10 Maya + 5 tutors: which closes felt useful vs military/shamey?	interview	coded: useful / cosplay / blame	“Bootcamp” language → kill that copy

Pre-reg (XXXIV): AAR-* identify **session architecture** (brief–debrief design) — not “military methods raise ACT” or identity-from-discipline myths.

LIV.12 Confidence table

Claim	Label	Confidence
Tannenbaum & Cerasoli (2013): debriefs $d \approx 0.67$	FACT	High
Keiser & Arthur (2021): AAR $d \approx 0.79$; alignment + objective media matter	FACT	High
TC 25-20 / Army AAR four-question core	FACT	High
Dismukes et al. (1997): facilitation → deeper crew self-analysis	FACT	High
Rudolph et al. (2006): good-judgment / advocacy–inquiry	FACT (method paper)	High
FAA CRM briefing markers	FACT (advisory doctrine)	High
Meta effects transfer cleanly to ACT prep for anxious teens	HYPOTHESIS / wounded	Low–Medium
SAFE-AAR is right default session spine for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Military cosplay improves belonging or scores	SPECULATION / false as doctrine	Low (against)

LIV.13 What this chapter kills

1. **Kill:** “Military-grade” / bootcamp / warrior branding as identity or efficacy claim.
2. **Kill:** Marketing meta-analytic *d* / “+25%” as undifferentiated ACT or IQ promise.
3. **Kill:** Conflating training AARs with trauma CISD.
4. **Kill:** Lecture or AI monologue that labels itself a debrief while blocking student self-analysis.
5. **Kill:** Vibes-only closes without objective traces when traces exist.

6. **Wound:** Hot-wash-only products; facilitator skill variance (train tutors; constrain AI).
7. **Survive:** SAFE-AAR; four-question close; CRM-style brief; facilitation + good judgment; AAR-1…5; copy as **reviewed struggle** — not costume cosplay.

Doctrine until data: Ship **SAFE-AAR** — brief the edge, attempt with traces, close with expected/actual/sustain/improve under non-punitive facilitation — and never sell discipline theater or borrowed percentage miracles as the identity engine.

Part LV — Sports Film-Study Pedagogy: Error Clips → Coach Cards

Chapter status: Living evidence brief — Researcher tick 2026-07-31 (UTC hour 21; hour%6≠0; 5 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable design does sports film-study / video-based feedback (VBF) offer for MindCraft practice review — turning attempt traces into sparse, student-controlled “clips” and coach cards — without sports-cosplay branding, feedback floods, or highlight-reel shame?

Owners: Product (Solver / Notes / coach cards) · Tutor ops · Coach UX · Brand & copy · Red Team

Commercial job: Ship a **SAFE-FILM** stack — objective attempt clips + less-is-more cueing + learner-timed review + self∪model pairing — while killing “we’re like a sports academy” marketing, monologue film rooms, and flood-of-tips as pedagogy.

LV.1 Why this chapter exists

Part LIV (SAFE-AAR) established the *cycle*: brief → attempt with traces → structured close. Sports film study is the **media layer** of that cycle — high-performance domains do not rely on memory alone; they rewatch selected moments, cue attention, and commit the next play. MindCraft already logs the analogue of game film: attempt lists, soft-wrong codes (`mis_` IDs), timing, format tags, outcomes JSON. Competitors either never replay process (answer-key chat), replay only correctness (Khan-style), or flood the learner with AI commentary that no one remembers a week later.

FOUNDER BELIEF under audit: Identity as a mathematical thinker grows when the learner can *see their own decision frames* on a specific attempt — not when a brand dresses practice as a locker room. Film-study transfer is about **selective objective media + facilitation**, not jersey cosplay.

Claims we refuse as doctrine:

1. Sports-academy / “elite athlete” branding as belonging or efficacy technology.
2. Coach monologue over 30 tips per “film room” (recall collapses).
3. Highlight-reel of only successes *or* only failures as identity theater.
4. Treating declarative recall of tips as if it were transfer to next-attempt performance.
5. Equating any video UI with film-study pedagogy.

LV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Film study / VBF	Structured review of performance via recorded moments + guided attention	Replay attempt; soft-wrong clip; coach card on *that* step	Passive highlight reel; shame montage
Performance analysis (PA)	Evaluate/describe performance with media (Pearson et al.)	Graph outcomes + process codes as “tape”	Stats without learner agency
Self-observation	Learner watches *own* prior attempt	“Replay your steps” before AI wrap	Narcissistic doom-scroll of errors
Skilled / expert model	Observation of a competent demonstration	Worked example / correct pathway card paired to same archetype	Model-only → “I can’t do that” gap
Self-modeling / feedforward	Edited view of self succeeding at a near-future challenge (Dowrick)	Show student’s own prior *successful* transfer on related edge	Fake montages that invent competence
Attentional cueing	Direct gaze/attention to the decision that mattered	Coach card highlights inequality sign flip / bridge gap	Cue everything = cue nothing
Self-controlled feedback	Learner chooses *when* to request video/KP review	“Review this attempt?” on demand vs forced after every item	Autonomy theater with no cue quality
SAFE-FILM	Film-study doctrine that survives sports cosplay / flood / shame kills	See LV.8	Academy branding; tip dumps

Operational definition (HYPOTHESIS): A MindCraft review counts as *SAFE-FILM* when (a) the clip is an **objective attempt trace** (not vibes), (b) ≤3 attentional cues / coaching points are attached, (c) the learner can request or skip replay (self-control), (d) when a miss is shown, a **paired skilled model or student’s own prior success** on a related edge is available (self∪model), (e) the output is a next-attempt commitment (SAFE-AAR improve-how), (f) no public shame reel, no sports-cosplay belonging tax.

LV.3 Applied models of observation — who / what / when / how

FACT: Ste-Marie, Law, Rymal, Jenny, Hall & McCullagh (2012, *International Review of Sport and Exercise Psychology*, 5(2), 145–176, doi:10.1080/1750984X.2012.665076) advance an **Applied Model for the Use of Observation (AMUO)**: practitioners should assess observer and task characteristics, then vary **who** is observed, **what** is observed (and instructional features), **when** observation occurs, and **how** information is delivered — not treat “show video" as a single intervention.

FACT: Ste-Marie et al. (2020, *Research Quarterly for Exercise and Sport*, doi:10.1080/02701367.2019.1693489) revisit AMUO across 2011–2018 applied-task articles; observation remains effective across a wider task base, but **guidelines still require intentional Who/What/When/How design** — effectiveness is not automatic from "video exists."

HYPOTHESIS for MindCraft: Map AMUO onto practice:

- **Who:** self (attempt replay) ± skilled model (worked pathway for same archetype) ± peer only when SAFE-COMPARE allows.
- **What:** the decision frame that produced the soft-wrong — not the whole worksheet.
- **When:** after attempt, before next isomorphic item; optional delayed Notes deepen (aligns SAFE-AAR hot-wash + later deepen).
- **How:** sparse cueing + student-first naming (facilitation), not AI monologue.

Kill: "We added a replay button" as film-study. Survive: designed observation with AMUO knobs.

LV.4 Video feedback as performance analysis — learning ≠ tip retention

FACT: Pearson, Webb, Milligan & Dicks (2023/2025 print, *International Review of Sport and Exercise Psychology*, 18(1), 417–439, doi:10.1080/1750984X.2023.2235700) — integrative review of video feedback as a facet of PA (16 studies from 5,838 screened; OSF registration noted). Delivery mode (coach-led classroom vs self-led), environment, and **scheduling** shape learning; scheduling often aids **declarative knowledge retention** more clearly than transfer to on-field performance. Collaborative coach–athlete practices may enhance engagement. Authors call for better frameworks and experimental variety — PA is common; *learning-optimized* PA is under-specified.

Wound (transfer): Elite sport PA classrooms ≠ Maya's 12-minute practice session. Treat Pearson et al. as a warning against equating "we reviewed the tape" with "performance transferred."

Commercial implication: MindCraft must instrument **next-attempt behavior** (`retry_120s`, method-change codes, `solo_transfer_pass`) — not tip-quiz scores after coach cards.

LV.5 Feedback floods kill recall — less is more

FACT: Mason, Farrow & Hattie (2020, *International Journal of Sports Science & Coaching*; doi:10.1177/1747954120951080) — exploratory study of one-to-one post-match video feedback in an AFL club (6 coach–player dyads). Coaches delivered on the order of **~30 feedback messages** per meeting. One week later, players recalled about **50% of summarised themes** but only about **6% of fine-grained feedback idea units**. Authors encourage a **"less is more"** approach.

HYPOTHESIS: AI tutors and "helpful" coach UIs recreate the AFL flood at machine speed — worse for anxious novices (SAFE-DD × anxiety; CLT). Cap coach-card points per clip at **1–3**. Prefer one bridge/misconception handle over a laundry list.

Product translation: Soft-wrong → single primary `mis_` / ingredient cue → one advocacy–inquiry prompt (Rudolph/SAFE-AAR) → one next-attempt tweak. Kill "here are eight tips from your film."

LV.6 Self-controlled video feedback — autonomy of *when*, not absence of coaching

FACT: van der Meer, van den Hoven, van der Kamp & Savelsbergh (2024, *Research Quarterly for Exercise and Sport*, 95(2), 537–545, doi:10.1080/02701367.2023.2275801; online 2023) — intermediate tennis players; self-controlled video feedback (request) vs yoked externally controlled schedule; coach provided video with attentional cueing and transitional tactical statements. Self-controlled group showed **larger tactical performance gains** vs pretest at posttest and **one-week retention**. Self-efficacy did not differ by group; gains were not predicted by self-efficacy / self-regulative questionnaire scores in this sample.

FACT (supporting line): Aiken, Fairbrother & Post (2012, *Frontiers in Psychology*, 3:338, doi:10.3389/fpsyg.2012.00338) — self-controlled video feedback benefits basketball set-shot learning relative to yoked schedules (motor-learning line often cited in later VBF work).

Wound: Meta-analytic debate exists on the strength of self-controlled feedback advantages (van der Meer et al. note McKay et al., 2022 vs Jimenez-Diaz et al., 2021). Do **not** market "science proves self-paced video raises ACT." Treat as design prior: **learner-timed review + quality cueing** beats forced tip dumps *or* no review.

HYPOTHESIS for MindCraft: After a miss (or optional after a hard success), offer "Review this attempt?" rather than auto-playing a 90s AI lecture. Forced review every item → fatigue and controlling climate (SAFE-REWARD / SAFE-DD).

LV.7 Self∪model pairing and feedforward — identity-safe media

FACT (pairing): Robertson, St. Germain & Ste-Marie (2018, *Journal of Motor Learning and Development*, 6(1), 18–34, doi:10.1123/jmld.2016-0027) — gymnasts learning two skills in a within-subject design: self-observation **paired with a skilled model** outperformed self-observation alone later in acquisition; error-identification **response sensitivity** was also higher for paired skills.

FACT (self-modeling / feedforward): Dowrick (1999, *Applied and Preventive Psychology*, 8(1), 23–39, doi:10.1016/S0962-1849(99)80009-2) reviews self-modeling; Dowrick (2012, *WIREs Cognitive Science*, 3(2), 215–230, doi:10.1002/wcs.1156) theorizes "learning from the future" via reconfigured component skills. Dowrick, Kim-Rupnow & Power (2006, *Journal of Special Education*, 39(4), 194–207, doi:10.1177/00224669060390040101) — video **feedforward** plus tutoring improved reading fluency; in 9/10 cases improvement rate was greatest during feedforward (small *N* — transfer carefully).

HYPOTHESIS: A miss clip should not stand alone. Pair with (1) a skilled worked pathway for the same archetype and/or (2) the student's own prior success on a related edge. Feedforward spirit without fake editing: **show competence already owned, then the gap.**

Kill: Failure-only shame reels; success-only dopamine montages that hide productive misconceptions (SAFE-MISCON). Survive: paired clip design.

LV.8 SAFE-FILM stack (product law)

1. **Clip = objective attempt media** — steps, codes, timing; not “how did it feel?” alone (extends Keiser objective-media finding via LIV).
2. **Less is more** — ≤3 cues per clip; prefer one primary soft-wrong handle (Mason et al.).
3. **AMUO design** — decide who/what/when/how before shipping replay UI (Ste-Marie 2012/2020).
4. **Self-controlled timing** — learner can request review; default is offer, not force-flood (van der Meer et al.; Aiken et al.).
5. **Self∪model pairing** — miss clip + skilled pathway and/or own prior success (Robertson/Ste-Marie line; Dowrick feedforward spirit).
6. **Student names the frame first** — then advocacy–inquiry coach card (SAFE-AAR / Rudolph).
7. **Transfer metric, not tip quiz** — instrument next attempt (Pearson wound).
8. **Anti-cosplay** — borrow the *media pedagogy*, not locker-room brand (SAFE-RITUAL / SAFE-AAR).
9. **No public error theater** — film study is private or tutor-dyad unless SAFE-COMPARE + consent (SAFE-CoP caution).
10. **Claim ladder** — sport VBF evidence stays L1 domain evidence; MindCraft ACT/identity claims need FILM-* experiments.

LV.9 Competitive audit — who runs film study?

ChatGPT-style tutors rarely replay *your* attempt structure and risk tip floods. Khan shows correctness with weak process pairing. Duo streak liturgy ≠ film study. Brilliant ahas rarely close a self∪model loop. Human “sports-style” tutoring inherits facilitator variance (Pearson). MindCraft already has attempt traces + soft-wrong + cards + Notes — **formalize SAFE-FILM**.

Commercial implication: Position as **reviewed decisions** — cue the one hinge, try the next play — never “train like the pros.” Parent copy: replay *how* they thought. Student copy: “pick a clip to review.”

LV.10 Claim ladder

Claim	Max ladder without new data
Observation interventions should vary who/what/when/how (AMUO)	L1 (Ste-Marie et al., 2012; 2020)
VBF-as-PA often improves declarative retention more clearly than transfer; design/delivery matter	L1 (Pearson et al., 2023)
Dense coach tip floods → poor week-later fine-grained recall (~6%); less-is-more recommended	L1 exploratory (Mason, Farrow & Hattie, 2020)
Self-controlled video feedback can outperform yoked schedules on tactical learning (tennis)	L1 single study (van der Meer et al., 2024)

Self∪skilled-model pairing can beat self-observation alone (gymnastics)	L1 (Robertson, St. Germain & Ste-Marie, 2018)
Video feedforward can accelerate skill acquisition in some educational samples	L1 small-N (Dowrick et al., 2006); theory (Dowrick, 2012)
SAFE-FILM improves MindCraft `retry_120s` / method-change / `solo_transfer_pass`	L1 product bet — needs FILM experiments
“Sports academy methods raise ACT / identity” undifferentiated	**Banned**
Highlight-shame reels / tip floods as pedagogy	**Banned**

LV.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
FILM-1	≤3-cue clip card vs uncapped AI tip flood after miss	2-arm	tip recall @24h; `retry_120s`; method-change	Flood ≥ sparse on transfer → revisit cue quality only
FILM-2	Self-controlled “Review?” vs forced replay every miss	2-arm	completion; affect; next-item transfer	Forced ≥ choice on transfer *and* better affect → keep forced for novices only
FILM-3	Miss-only clip vs miss+skilled model vs miss+own prior success	3-arm	error ID; `solo_transfer_pass`	Miss-only ≥ paired → pairing theater
FILM-4	AMUO-scripted coach card (who/what/when/how) vs generic praise/explain	2-arm	SE quality; transfer	Generic ≥ AMUO → card content fail
FILM-5	Film close + SAFE-AAR four questions vs film without AAR questions	2-arm	improve-how specificity; week-later transfer	No-AAR ≥ AAR → film without close still ok; keep AAR optional
FILM-QUAL	10 Maya: which clip UX felt useful vs shamey/sportsy?	interview	coded: useful / cosplay / flood / shame	“Athlete” language → kill that copy

Pre-reg (XXXIV): FILM-* identify **review media architecture** — not “sports methods raise scores” or identity-from-athleticism myths.

LV.12 Confidence table

Claim	Label	Confidence
Ste-Marie et al. AMUO who/what/when/how	FACT	High
Pearson et al. VBF-as-PA: delivery/scheduling; transfer caution	FACT	High
Mason et al.: ~30 tips; ~6% fine-grained week recall; less-is-more	FACT (exploratory, small *N*)	Medium–High
van der Meer et al.: self-controlled VBF > yoked on tennis tactics	FACT (single study)	Medium
Robertson, St. Germain & Ste-Marie: self∪model > self alone	FACT	Medium–High
Dowrick feedforward reading gains	FACT (small *N*)	Medium
Sport VBF transfers cleanly to ACT prep for anxious teens	HYPOTHESIS / wounded	Low–Medium
SAFE-FILM is right default review layer for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Sports-cosplay branding improves belonging or scores	SPECULATION / false as doctrine	Low (against)

LV.13 What this chapter kills

1. **Kill:** Sports-academy / pro-athlete / locker-room branding as efficacy or belonging claim.
2. **Kill:** Feedback floods (30 tips / uncapped AI wrap) labeled as film study.
3. **Kill:** Equating tip recall or “we watched the replay” with transfer / exam readiness.
4. **Kill:** Failure-only shame reels and success-only dopamine montages as identity engines.
5. **Kill:** Replay UI without AMUO who/what/when/how design.
6. **Wound:** Classroom PA and elite dyads under-transfer to short teen sessions; facilitator variance.
7. **Survive:** SAFE-FILM; less-is-more clips; self-controlled review timing; self∪model pairing; feedforward spirit via own prior success; FILM-1…5; copy as **reviewed decisions** — not sports cosplay.

Doctrine until data: Ship **SAFE-FILM** — objective attempt clips, sparse cues, learner-timed review, paired models, transfer metrics — stacked on SAFE-AAR's brief–debrief spine, and never sell athlete mythology or tip floods as the transformation engine.

Part LVI — Music Pedagogy Ladders: Scales → Repertoire → Recital Identity

Chapter status: Living evidence brief — Researcher tick 2026-08-01 (UTC hour 00 ≡ Red Team slot, but ch56 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable design does instrumental music pedagogy offer for MindCraft's learning ladder — technique drills → meaningful repertoire → public competence signal — without conservatory cosplay, minutes-practiced vanity, or recital theater as identity?

Owners: Product (Practice path / story quests / exam signal) · Coach UX · Brand & copy · Red Team

Commercial job: Ship a **SAFE-MUSIC** stack — scales-as-ingredient drills, repertoire-as-story/exam pieces with learner choice, recital-as-transfer proof not stage cosplay — while killing "practice like a virtuoso," hours-logged as mastery, and pass-floor exam identity.

LVI.1 Why this chapter exists

The core OS already sketches a music analogy (drill → quest → "recital" signal). Parts LI (SAFE-DP) and LV (SAFE-FILM) borrowed *session architecture* and *review media* from high-performance domains. Music pedagogy is the clearest **ladder** domain: beginners do not start with concertos; they build technique, then play repertoire they care about, then face graded exams or recitals that mark identity — *if* practice quality, motivation, and self-belief cohere.

MindCraft's product already has the three rungs in embryo: ingredient/concept drills (scales), story-wrapped and exam-tagged questions (repertoire), ACT/diagnostic public signals (recital). Competitors collapse the ladder: Duo gamifies minutes; Khan floods explanations; ChatGPT skips technique for fluent answers; Brilliant jumps to ahas without graded public proof.

FOUNDER BELIEF under audit: Mathematical identity grows when learners climb a **named ladder** — controlled technique → chosen meaningful pieces → evidence they can perform under evaluation — not when a brand dresses homework as a conservatory.

Claims we refuse as doctrine:

1. Conservatory / "virtuoso / prodigy" branding as belonging or efficacy.
2. Practice **minutes** or streaks as achievement (quantity without formal practice quality).
3. Scales-forever without repertoire *or* repertoire without technique scaffolds.
4. Recital/stage theater as identity without solo transfer evidence.
5. Treating a bare "pass" exam outcome as motivational success.

LVI.2 Constructs (product language)

Construct	Music meaning	MindCraft analogue	Failure mode if misused
Scales / technique	Isolated skill building with immediate feedback	Ingredient drills; format reps; bridge micro-skills	Endless drill without meaning → boredom / controlling motivation
Repertoire	Whole pieces worth playing; often student-chosen	Story quests; archetype instances; exam-tagged sets	Pretty narrative with no technique → theater
Recital / grade exam	Public or adjudicated performance identity marker	Diagnostic; timed mixed set; `solo_transfer_pass`	Stage cosplay; pass-floor as "done"
Formal practice	Goal-directed time with SRL + deliberate strategies	SAFE-DP session spine with goals + feedback	Naive play-through of worksheets
Ineffective strategies	Error-chasing without analysis; mindless repetition	Immediate correction spam; tip floods; blocked vanity accuracy	Hallam "ineffective strategies" factor
SAFE-MUSIC	Ladder doctrine that survives cosplay / quantity / theater kills	See LVI.8	Conservatory brand; minutes KPI

Operational definition (HYPOTHESIS): A MindCraft path counts as *SAFE-MUSIC* when (a) technique drills are **goal-tagged ingredients** with feedback (not infinite XP), (b) repertoire includes **learner-chosen or interest-aligned** problem sets when feasible, (c) public signals require **mixed / transfer** performance not blocked accuracy, (d) minutes practiced never outrank formal-practice quality metrics, (e) branding names the ladder (skills → pieces → proof) without virtuoso mythology.

LVI.3 Deliberate practice's home domain — music, not Gladwell hours

FACT: Ericsson, Krampe & Tesch-Römer (1993, *Psychological Review*, 100(3), 363–406, doi:10.1037/0033-295X.100.3.363) grounded deliberate practice in the training of instrumental musicians: effortful, teacher-guided activities with goals, immediate informative feedback, and repetition after reflection — distinct from playful performance or mere hours logged. Music academy students' structured solitary practice was the empirical prototype.

Wound (already in SAFE-DP / Part LI): Hambrick-line critiques and domain limits mean "10,000 hours" and undifferentiated practice time are **banned** marketing. Music supplies the *definition*, not a guarantee that ACT scores rise from branded "deliberate practice."

Product implication: Scales = short, goalful ingredient reps with feedback; never sell "log more hours this week" as the transformation engine (aligns Bonneville-Roussy below; SAFE-HABIT / SAFE-REWARD).

LVI.4 Strategy quality rises with expertise — not all practice develops

FACT: Hallam, Rinta, Varvarigou, Creech, Papageorgi, Gomes & Lanipekun (2012, *Psychology of Music*, 40(5), 652–680, doi:10.1177/0305735612443868) surveyed **N = 3,325** young instrumentalists across nine expertise/grade levels. Factor analysis yielded seven practice-related factors, including: systematic practice strategies; organization of practice; use of recordings/metronome for feedback; analytic strategies; **ineffective strategies**; concentration; immediate correction of errors. Statistically significant linear relationships with grade level appeared for systematic strategies, recordings/metronome use, ineffective strategies, and immediate error correction — but **not** for organization, analytic strategies, or concentration. Educational implication: expertise tracks *which* strategies learners adopt, not merely "they practiced."

HYPOTHESIS for MindCraft: Instrument `strategy_quality` proxies (sectioning hard items, requesting SAFE-FILM clip, naming soft-wrong before retry) — not `minutes_in_app`. Decline of enjoyment of practice with advancement in related Hallam work is a wound: advanced ladders need repertoire meaning, not more grind UI.

Kill: Progress bars powered only by time-on-task. Survive: strategy factors as product instrumentation.

LVI.5 Quantity is not enough — formal practice mediates time → achievement

FACT: Bonneville-Roussy & Bouffard (2015, *Psychology of Music*, 43(5), 686–704, doi:10.1177/0305735614534910; online 2014) propose an integrative framework: **formal practice** = goal-directed, focused practice that includes both self-regulation and deliberate practice strategies. In a 4-month prospective SEM study, their integrative model predicted musical achievement better than traditional time-only approaches; practice time predicted achievement **when associated with formal practice**, not as raw quantity.

Commercial implication: Parent dashboards that lead with "hours practiced" recreate the failed music predictor. Lead with formal-practice signals: goals set, feedback used, transfer passed. Copy: "focused reps on the weak bridge," never "she practiced 4 hours."

Aligns: SAFE-DP (goals/feedback/rep), SAFE-HABIT (cue ≠ colonization), streak ban as North Star.

LVI.6 Repertoire choice — interest upgrades practice behavior

FACT: Renwick & McPherson (2002, *British Journal of Music Education*, 19(2), 173–181, doi:10.1017/S0265051702000256) — case study of a beginning clarinettist ("Clarissa"). When practising **self-selected** repertoire vs teacher-assigned pieces, she engaged strategies typical of more advanced stages (silent fingering, silent thinking, singing), spent more time, and persevered through difficulty. Authors link choice/interest to higher-level cognitive engagement (consistent with broader motivation findings cited therein).

FACT / FOUNDER-adjacent: Evans (2015) applies Self-Determination Theory to music education: scales are often **not** intrinsically enjoyable; motivation quality depends on whether learners internalize the *value* of technique (identified regulation) vs pure external control (teacher said so / praise). External control of scale drills risks the Deci overjustification pattern already killed in SAFE-REWARD (Part XLIV).

HYPOTHESIS: MindCraft "repertoire" should mix (1) teacher/engine-assigned technique (scales) framed with **why this ingredient unlocks the next piece**, and (2) **choice among story arcs / problem sets** when mastery allows — choice upgrades strategy quality, not just delight metrics.

Kill: Forced syllabus-only grind with no choice forever. Kill: choice-only playground with no technique ladder. Survive: scales with identified value + repertoire choice windows.

LVI.7 Recital / grade exams — identity markers that can demotivate

FACT: Hallam, Papageorgi, Varvarigou & Creech (2020 print / 2019 online, *Psychology of Music*, doi:10.1177/0305735618816168) — **N = 2,131** musicians aged 6–19 across expertise and exam outcomes (fail → highly commended). Those awarded **merely a pass** tended to undertake the **least** practice and responded least positively on organization, recordings/metronome use, analytic strategies, social life, and self-belief in musical ability. Those who **failed** were most likely to adopt ineffective strategies and less likely to enjoy performing, playing, lessons, and practice. Motivation models in the paper foreground musical **identity**, self-belief, goals, attributions, and supportive environment — not exam theater alone.

Wound: Graded exams are identity technology *and* stress technology (parent/teacher reputation dynamics noted in related literature the authors cite). A MindCraft "recital" that is only a score reveal can create pass-floor demotivation or fail-shame — opposite of FEI.

HYPOTHESIS: Public signals must be (a) criterion-referenced transfer proofs (`solo_transfer_pass`, mixed formats), (b) mastery-climate framed (SAFE TARGET / Part XXXVIII), (c) never sold as "your recital proves you're a genius," (d) paired with SAFE-AAR debrief + SAFE-FILM clip on the weak bar, not curtain-call XP.

Kill: Recital cosplay / standing ovation UI as identity. Kill: celebrating bare pass as transformative. Survive: adjudicated *competence evidence* with recovery path.

LVI.8 SAFE-MUSIC stack (product law)

1. **Name the three rungs** — technique (scales) → repertoire (pieces) → proof (recital/transfer).
2. **Scales = goalful ingredient drills** with immediate feedback (Ericsson definition; SAFE-DP).
3. **Repertoire includes choice** when possible — interest upgrades strategy quality (Renwick & McPherson).
4. **Frame technique value** (identified regulation) — why this drill unlocks the piece (Evans / SDT); avoid pure controlling rewards (SAFE-REWARD).
5. **Formal practice > minutes** — instrument strategy quality; never North-Star hours (Bonneville-Roussy & Bouffard; Hallam 2012).
6. **Public signal = mixed/transfer proof** — not blocked accuracy or stage theater (Hallam 2019 wound + SAFE-TRANSFER).
7. **Pass-floor ≠ win** — design recovery for bare-pass and fail states; protect self-belief (Hallam exam outcomes).
8. **Anti-cosplay** — borrow the *ladder*, not conservatory/prodigy brand (SAFE-RITUAL / SAFE-AAR pattern).
9. **Stack with prior SAFE-*** — DP session spine, FILM review of the hard bar, AAR close, CALIB confidence tiers.

10. **Claim ladder** — music evidence stays L1 domain evidence; MindCraft ACT/identity claims need MUSIC-* experiments.

LVI.9 Competitive audit — who runs the ladder?

Competitor	Scales	Repertoire	Recital	Ladder failure
Khan	Skill practice	Weak story meaning	Unit mastery quizzes	Explanation flood; weak identity proof
Duo	Micro-drills	Limited choice depth	Leagues/streaks as "recital"	Quantity/streak liturgy (banned NS)
Brilliant	Insight puzzles	Delight-as-repertoire	Weak graded public ACT proof	Aha without adjudicated transfer
ChatGPT tutor	Skips scales	Answers as repertoire	No solo proof	Fluent wrongness; no ladder
Human music-style tutor	Often strong	Often assigned-only	Real juries	Variance; stress; syllabus boredom

Commercial implication: Position MindCraft as **skills** → **pieces** → **proof** — "drill the hinge, play the piece, show you can transfer" — never "train like Juilliard" or "practice minutes that compound." Parent copy: quality of practice and transfer proof. Student copy: "unlock the next piece" / "show what you can play cold."

LVI.10 Claim ladder

Claim	Max ladder without new data
Deliberate practice prototype is music training with goals, feedback, repetition	L1 (Ericsson et al., 1993)
Strategy factors shift with grade level in large youth sample; time alone insufficient framing	L1 (Hallam et al., 2012)
Practice time predicts achievement when tied to formal (SRL+DP) practice	L1 (Bonneville-Roussy & Bouffard, 2015)
Student-selected repertoire can upgrade practice strategies vs assigned pieces	L1 case study (Renwick & McPherson, 2002)
Scales often need internalized value, not pure intrinsic fun	L1 theoretical/applied SDT (Evans, 2015)
Bare-pass exam outcomes associate with weaker practice/motivation profiles	L1 observational (Hallam et al., 2019/2020)

SAFE-MUSIC improves MindCraft transfer / identity metrics vs minutes-led UX	L1 product bet — needs MUSIC experiments
“Conservatory methods raise ACT / identity” undifferentiated	**Banned**
Hours practiced / streak as mastery or recital	**Banned**

LVI.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
MUSIC-1	Named ladder UX (scales→piece→proof) vs flat question feed	2-arm	completion; `solo_transfer_pass`; perceived progress	Flat ≥ ladder on transfer → ladder is copy theater
MUSIC-2	Choice among 2–3 repertoire sets vs assigned-only	2-arm	strategy proxies; time-on-hard-items; affect	Assigned ≥ choice on transfer *and* affect → choice optional
MUSIC-3	Technique framed with “unlocks piece X” vs controlling “do your scales”	2-arm	persistence; identified reasons; next-piece start rate	Controlling ≥ identified → SDT frame fail
MUSIC-4	Parent dashboard: formal-practice metrics vs minutes-first	2-arm	parent WTP/comprehension; student pressure talk	Minutes-first better WTP without pressure → still wound educationally; don’t ship
MUSIC-5	Recital = mixed transfer set + AAR vs score-reveal only	2-arm	self-belief; retry; week-later transfer	Reveal-only ≥ AAR on belief *and* transfer → simplify
MUSIC-QUAL	10 Maya: does “scales/pieces/recital” language feel clarifying or elitist?	interview	coded: clarifying / cosplay / pressure	Conservatory words → kill that copy

Pre-reg (XXXIV): MUSIC-* identify **ladder architecture** — not “music methods raise scores” or prodigy-identity myths.

LVI.12 Confidence table

Claim	Label	Confidence
Ericsson et al.: DP rooted in music training structure	FACT	High
Hallam et al. 2012: large-N strategy factors × grade	FACT	High
Bonneville-Roussy & Bouffard: formal practice mediates time→achievement	FACT	High
Renwick & McPherson: choice upgrades strategies (case)	FACT (small *N*)	Medium
Evans SDT: scales need internalized value	FACT / applied theory	Medium–High
Hallam et al. 2019: pass-floor weakest practice/motivation profile	FACT (observational)	Medium–High
Music ladder transfers cleanly to ACT prep for anxious teens	HYPOTHESIS / wounded	Low–Medium
SAFE-MUSIC is right default path metaphor for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Conservatory cosplay / hours KPI improve belonging or scores	SPECULATION / false as doctrine	Low (against)

LVI.13 What this chapter kills

1. **Kill:** Conservatory / virtuoso / prodigy branding as efficacy or belonging claim.
2. **Kill:** Minutes practiced, streaks, or hours-logged as North Star mastery (quantity without formal practice).
3. **Kill:** Scales-only forever without repertoire meaning *or* repertoire theater without technique.
4. **Kill:** Recital/stage cosplay and bare-pass celebration as identity transformation.
5. **Kill:** Equating “we practiced a lot” with exam readiness or musical/math identity.
6. **Wound:** Graded public performance can demotivate (pass-floor, fail-shame, parent pressure); transfer carefully.
7. **Survive:** SAFE-MUSIC ladder; formal-practice metrics; choiceful repertoire; identified technique value; transfer-as-recital; MUSIC-1…5; copy as **skills** → **pieces** → **proof**.

Doctrine until data: Ship **SAFE-MUSIC** — goalful scales, meaningful chosen pieces, adjudicated transfer proof — stacked on SAFE-DP / SAFE-FILM / SAFE-AAR, and never sell conservatory mythology or practice minutes as the transformation engine.

Part LVII — Chess Annotation & Metacognition: Postmortem UX Without Grandmaster Cosplay

Chapter status: Living evidence brief — Researcher tick 2026-08-01 (UTC hour 03; hour%6≠0; 7 researcher entries since synthesizer v1.7 → Researcher)

Primary question: What transferable design does chess *annotation / postmortem* offer for MindCraft's after-attempt metacognition — student-generated explanation of critical positions, error classification into next drills, analysis of wins as well as losses — without grandmaster cosplay, engine-eval floods, or rating-as-identity?

Owners: Product (Practice debrief / Solver review / Notes) · Coach UX · ML soft-wrong → drill loop · Brand & copy · Red Team

Commercial job: Ship a **SAFE-ANNOT** stack — annotate before reveal, self-explain critical moves, classify → next drill, analyze successes not only failures — while killing "think like a GM," engine dump ≡ coaching, and games-played/rating as learning proof.

LVII.1 Why this chapter exists

Parts LIV–LVI borrowed *session close* (SAFE-AAR), *review media* (SAFE-FILM), and *ladder architecture* (SAFE-MUSIC) from adjacent high-performance domains. Chess is the classic cognitive-science laboratory for **move choice under uncertainty** and for **post-game annotation** as deliberate learning — the closest analogue to MindCraft's "attempt → review → next weak bridge" loop.

Competitors already fake the chess metaphor poorly: ChatGPT tutors monologue the "best line"; puzzle apps spam tactics without annotation; Duo/Khan celebrate streaks and accuracy without a student-authored postmortem. MindCraft's soft-wrong science (SAFE-MISCON), hide-correctness diagnostic (SAFE-CALIB), and coach self-explanation doctrine (Part XL / SAFE-SE) need a **postmortem UX law** that is evidence-shaped, not chess-club branding.

FOUNDER BELIEF under audit: Mathematical identity grows when learners *annotate their own critical attempts* — name what they saw, why they chose, what principle failed — then convert that into one next drill, not when an engine or LLM floods them with better answers.

Claims we refuse as doctrine:

1. Grandmaster / "think like a master" branding as efficacy or belonging.
2. Engine evaluation bars / tip floods labeled as "annotation."
3. Games played, rating, streaks, or puzzle counts as North Star mastery.
4. Loss-only shame review (never annotate wins / successful strategies).
5. Novice JOLs / vague "how well did you understand?" without forced restudy of critical moves.
6. AI postmortem as automatically closing skill gaps (self-selection wound).

LVII.2 Constructs (product language)

Construct	Chess meaning	MindCraft analogue	Failure mode if misused
Critical position	Branch point where evaluation swings	Soft-wrong item; bridge fail; timed blunder	Annotating every move → exhaustion
Annotation / postmortem	Written/verbal account of thought at critical nodes	Student-first debrief after attempt/session	Engine dump before student speech
Candidate moves	Alternatives considered before choice	Alternative strategies / formats considered	Fake multiple-choice theater
Self-explanation	Why this move / principle applies	Chi/Renkl-style why-prompts (Part XL)	"Explain" that restates the tip
Engine / AI feedback	Stockfish-style eval after human thought	Coach card / model solution *after* annotate	Reveal-first collapses generation
Template / chunk	Pattern recognition structures in LTM	Ingredient / archetype recognition	"Memorize more patterns" vanity
SAFE-ANNOT	Postmortem doctrine that survives cosplay / flood / rating kills	See LVII.9	Chess branding; eval bars as pedagogy

Operational definition (HYPOTHESIS): A MindCraft review counts as *SAFE-ANNOT* when (a) the learner generates explanation of **1–3 critical attempts before** correctness/model reveal, (b) annotation names a **principle or misconception family** (not vibes), (c) the loop ends in a **linked next drill** (motif → ingredient set), (d) wins/successes are eligible for annotation (not only failures), (e) engine/LLM text never substitutes for student generation, (f) branding borrows the *method*, not grandmaster mythology.

LVII.3 de Groot — verbal protocols are the annotation prototype

FACT: Adriaan de Groot's classic work (*Het denken van den Schaker*, 1946; English *Thought and Choice in Chess*, 1965/1978, Mouton / Amsterdam University Press lineage) collected **thinking-aloud protocols** from players of varying strength (including masters and world champions) who were asked to choose a move in unfamiliar tournament positions *as if in a real game*, verbalizing plans, calculations, and considerations. The object was the **choice-of-move problem**, not whole-game biography.

FACT (structure): de Groot described phases of chess thought commonly summarized as orientation → exploration → investigation → proof (see secondary summaries in the chess-programming / cognitive literature citing de Groot). Expertise showed up less as "search farther on every branch" in the original macrostructure story, and more as **better initial perception and selection of what to investigate**.

Wound (citation hygiene): Bilali■, McLeod & Gobet (2008, *Cognitive Science* line / "Sixty years of citing de Groot") document that textbooks often **misreport** de Groot as "experts and novices search identically." Later replications find additional search-structure differences. Product implication: do **not** market "masters don't calculate more — they just see" as a slogan that excuses skipping hard work *or* as a claim that MindCraft's annotation UI replicates grandmaster perception.

Product implication: Annotation UX = structured verbalization at critical nodes (what did you notice → candidates → why this choice → what would falsify it), not a vibes journal. Aligns SAFE-AAR question spine and Part XL self-explanation prompts.

LVII.4 Chunks and templates — recognition scaffolds search (not "genius")

FACT: Chase & Simon (1973) chunking theory — experts recall meaningful game positions far better than novices; advantage collapses on random boards — grounded expertise in **pattern memory**, not mystical IQ (classic result; see also Gobet & Simon extensions).

FACT: Gobet & Simon's **template theory** (e.g. Gobet & Simon, 1996; Gobet, 1998 *Cognition* comparison of expert-memory theories) proposes that recurring chunks evolve into richer templates with slots — unifying low-level chunks with higher-level schematic knowledge. Gobet (2005) notes deliberate practice matters because unsupervised "just play" can acquire **irrelevant** chunks that fail to improve (or even hinder) performance.

Commercial implication: MindCraft "annotation" should cue **ingredient/archetype recognition** ("which pattern is this?") before deep calculation monologues. Kill copy that says students must "think like Kasparov." Survive: pattern labels tied to ontology IDs after student attempt.

Kill: Genius branding. Kill: unstructured blitz-volume as chunk building. Survive: deliberate attention to the patterns that matter (SAFE-DP).

LVII.5 Self-explanation beats prediction-only for chess novices

FACT: de Bruin, Rikers & Schmidt (2007, *Contemporary Educational Psychology*, 32(2), 188–205, doi:10.1016/j.cedpsych.2006.01.001) — novices learning a King+Rook vs King endgame. Conditions: observe computer moves; predict next move; **predict + self-explain**. Self-explanation condition showed better principled understanding (predictions applying endgame principles) and more often checkmated in the test phase. Prediction alone did **not** beat observation.

HYPOTHESIS for MindCraft: Postmortem prompts must force **why/principle language**, not mere "what would you play next?" prediction clicks. This is the chess-domain replication of Chi/Renkl SE doctrine already in Part XL — use it to harden coach UX after attempts.

Kill: Silent engine scrubbing. Kill: "tap the best move" prediction theater without explanation. Survive: annotate-why before reveal.

LVII.6 Novice metacognition is fragile — JOLs can fail; forced restudy helps

FACT: de Bruin, Rikers & Schmidt (2005, *Applied Cognitive Psychology*, 19(2), 167–181, doi:10.1002/acp.1109) — novice chess endgame learning with judgments of learning (JOLs) and move selection for restudy. Forced selection of moves for restudy improved learning vs free selection (even controlling for number restudied). Providing JOLs showed better *self-regulatory behavior* on some measures but **no or negative** performance benefit vs no-JOL — authors argue JOLs may place high ineffective load on novice working memory.

FACT: de Bruin, Rikers & Schmidt (2007, *European Journal of Cognitive Psychology* / related metacomprehension skill-acquisition paper, doi:10.1080/09541440701326204) — experienced chess players showed higher metacomprehension accuracy and self-regulation than novices when learning an endgame; novice absolute accuracy near zero. Expertise affects monitoring quality in skill domains.

Product implication: Do not ship bare confidence sliders as "metacognition" for Maya-novices (also SAFE-CALIB wound). Prefer **forced critical-move restudy** + principle annotation. Gate fancy monitoring UI by expertise / after scaffolding.

Kill: "Rate your understanding 1–5" as the whole postmortem. Survive: select 1–3 critical items → explain → restudy linked drill.

LVII.7 Analyzing games beats mere volume — and wins matter

FACT (observational, large-N): Yiannakoulias (2026, *Simulation & Gaming*, doi:10.1177/10468781261443352) — ~2M online blitz games / ~2.1k active Lichess players (1600–1800 band). Computer-assisted game analysis positively associated with rating improvement; **analyzing wins** showed a clearer association than analyzing losses in main models; total games played only weakly associated. Authors stress measurement noise (analysis flag is blunt; biases toward null) and exploratory observational status — not an RCT.

HYPOTHESIS: MindCraft should prompt annotation on **successful hard items** and efficient solutions (Dowrick-adjacent feedforward / SAFE-FILM success clips), not only on failures — "study your brilliancies" as well as soft-wrongs. Loss-only review risks ostrich/shame dynamics the paper discusses via adjacent feedback literature.

Wound: Associational evidence ≠ "annotation raises ACT." Claim ladder stays L1 for chess rating deltas; MindCraft transfer needs ANNOT experiments.

Kill: "Play more puzzles" as improvement doctrine. Kill: failure-only shame reels. Survive: reflective analysis of critical successes and failures.

LVII.8 AI / engine feedback — self-selection and fluency wounds

FACT / high-relevance wound: Riedl & Bogert (2024/2026 update, arXiv:2409.18660) — 5+ years, ~52k individuals on an online chess platform with optional AI game analysis. Motivated and higher-skilled players **self-select** into AI feedback and use it more productively; apparent learning gains can be an **illusion of AI effectiveness** once endogenous motivation is accounted for; AI access can **widen skill gaps**; centralized AI feedback can reduce intellectual diversity (supported via platform natural experiments in the paper).

Aligns: Part XXXIII AI trust/sycophancy — fluent eval bars are not pedagogy; overtrust risk. SAFE-FILM less-is-more; SAFE-AAR facilitation not lecture.

Product implication: Optional "engine" / model solution is a **second pass** after student annotation. Do not market auto-analysis as closing equity gaps. Instrument who seeks postmortem help (selection bias).

Kill: "AI reviews every game so everyone improves equally." Kill: reveal-first Stockfish cosplay in Solver. Survive: human-first annotation → sparse model contrast → one drill.

LVII.9 SAFE-ANNOT stack (product law)

1. **Student annotates first** — 1–3 critical attempts before correctness / coach / model reveal.
2. **Prompt for principles** — why/candidates/what would change your mind (de Groot structure + de Bruin SE).
3. **Forced restudy selection** for novices — don't rely on raw JOLs (de Bruin 2005).
4. **Classify → next drill** — motif/misconception → ingredient set (close the loop; not insight theater).
5. **Annotate wins too** — successful strategies and hard-earned corrects (Yiannakoulias).
6. **Sparse AI second pass** — contrast, don't replace; watch self-selection (Riedl & Bogert).
7. **Pattern labels after attempt** — chunks/templates as ontology tags, not genius myths (Chase/Simon; Gobet).
8. **Anti-cosplay** — borrow postmortem method, not "grandmaster academy" brand (SAFE-RITUAL / SAFE-MUSIC pattern).
9. **Stack** — SAFE-AAR close, SAFE-FILM clip, SAFE-MISCON soft-wrong, SAFE-SE prompts, SAFE-CALIB tiers.
10. **Claim ladder** — chess evidence = L1 domain; MindCraft ACT/identity claims need ANNOT-* experiments.

LVII.10 Competitive audit — who runs a real postmortem?

Competitor	Annotation?	Student generation first?	Loop to next drill?	Failure mode
Khan	Hints/explanations	Often reveal/explain-first	Weak systematic soft-wrong→drill	Explanation flood
Duo	Minimal	No	Streak/XP loop	Quantity liturgy
Brilliant	Aha rationales	Partial	Weak exam transfer proof	Insight without postmortem habit
ChatGPT tutor	Fluent postmortem monologue	Usually model-first	Rare durable drill link	Sycophancy; fluent wrongness
Chess.com / Lichess	Engine analysis UX	Optional; often engine-first	Puzzle motifs if user closes loop	Eval bars; self-selection
Human tutor	Variable verbal postmortem	Often strong	Variable	Unscalable; shame risk

Commercial implication: Position MindCraft as **annotate → classify → drill** — "replay the critical move in your own words, then train the hinge" — never "AI grandmaster reviews your session." Parent copy: reflection quality and linked practice, not games played. Student copy: "name the mistake / name the win, then one next rep."

LVII.11 Claim ladder

Claim	Max ladder without new data
de Groot: verbal protocols reveal structured move-choice thought	L1 (classic methods)
Chase/Simon + Gobet/Simon: expertise tied to chunks/templates, not mysticism	L1
Self-explanation + prediction > prediction/observation for chess-novice principles	L1 (de Bruin et al., 2007 CEP)
Forced restudy selection helps novices; JOLs can fail/overload	L1 (de Bruin et al., 2005 ACP)
Metacomprehension accuracy rises with chess expertise in skill learning	L1 (de Bruin et al., 2007 EJCP)
Game analysis associated with rating gains; play volume weak; wins analysis salient	L1 observational (Yiannakoulias, 2026)
AI feedback gains partly selection; can widen gaps / reduce diversity	L1 observational + natural experiments (Riedl & Bogert)
SAFE-ANNOT improves MindCraft transfer / identity vs reveal-first UX	L1 product bet — needs ANNOT experiments
“Grandmaster methods” / engine dumps raise ACT undifferentiated	**Banned**
Rating / streak / games-played as mastery North Star	**Banned**

LVII.12 Experiments spawned

ID	Question	Design	Primary	Kill condition
ANNOT-1	Annotate-before-reveal vs reveal-first coach	2-arm	`retry_120s`; principle-coded annotation quality; week-later transfer	Reveal-first ≥ annotate on transfer → annotation is friction
ANNOT-2	SE-why prompts vs prediction-only taps	2-arm	principled language rate; next-drill completion	Prediction ≥ SE on transfer → simplify
ANNOT-3	Forced 2 critical restudies vs free JOL-only	2-arm	restudy adherence; transfer	JOL-only ≥ forced on transfer *and* affect → revisit

ANNOT-4	Annotate wins+losses vs losses-only	2-arm	return rate; self-belief; strategy reuse	Losses-only ≥ mixed on learning without shame cost → still prefer mixed ethically
ANNOT-5	AI second-pass on vs off after annotation	2-arm	overtrust incidents; transfer; time-to-drill	AI-on hurts calibration or transfer → default off for novices
ANNOT-QUAL	10 Maya: does “postmortem / annotate” language feel clarifying or chess-elitist?	interview	coded: clarifying / cosplay / shame	Chess words → plain “replay the hard step”

Pre-reg (XXXIV): ANNOT-* identify **postmortem architecture** — not “chess training raises ACT” or grandmaster-identity myths.

LVII.13 Confidence table

Claim	Label	Confidence
de Groot protocols established structured verbal move-choice analysis	FACT	High
Chunk/template accounts dominate expert chess memory explanations	FACT	High
de Bruin 2007: SE helps chess novices discover principles	FACT	High
de Bruin 2005: forced restudy > free; JOLs shaky for novices	FACT	High
Expertise improves metacomprehension in chess skill learning	FACT	Medium–High
Yiannakoulias: analysis↔rating; play volume weak (observational)	FACT (assoc.)	Medium
Riedl & Bogert: AI feedback self-selection / gap / diversity wounds	FACT (platform)	Medium–High

Chess postmortem transfers cleanly to anxious ACT teens' identity	HYPOTHESIS / wounded	Low–Medium
SAFE-ANNOT is right default review metaphor for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Grandmaster cosplay / engine-first dumps improve belonging or scores	SPECULATION / false as doctrine	Low (against)

LVII.14 What this chapter kills

1. **Kill:** Grandmaster / "think like a master" branding as efficacy or belonging.
2. **Kill:** Engine-eval floods and LLM monologues labeled as annotation or coaching.
3. **Kill:** Games played, rating, streaks, or puzzle counts as North Star mastery.
4. **Kill:** Loss-only shame review; never studying successful critical decisions.
5. **Kill:** Bare JOLs / confidence sliders as sufficient metacognition for novices.
6. **Kill:** "AI analyzes every attempt so gaps close equally" (selection + diversity wound).
7. **Wound:** Textbook oversimplification of de Groot search findings; don't sloganize perception.
8. **Survive:** SAFE-ANNOT; annotate-before-reveal; SE principles; forced critical restudy; classify→drill; annotate wins; sparse AI second pass; ANNOT-1…5.

Doctrine until data: Ship **SAFE-ANNOT** — student-generated postmortem on critical attempts, principle-tagged, linked to the next drill, stacked on SAFE-AAR / SAFE-FILM / SAFE-SE / SAFE-MISCON — and never sell chess mythology, engine bars, or volume metrics as the transformation engine.

Part LVIII — Therapy: Graded Exposure (Math Anxiety Ladders)

Chapter status: Living evidence brief — Researcher tick 2026-08-01

Primary question: How should MindCraft borrow *graded exposure / inhibitory learning* from anxiety treatment — without therapy cosplay, flooding, or "calm until zero" theater — so threatened learners approach math instead of avoiding it?

Owners: Affect / Anxiety · Learning Science · Product (Practice sequencing + stakes) · Growth / Positioning · Red Team

Commercial job: Turn math-anxiety avoidance into a **designed approach ladder** that parents can trust and students can finish — then market approach + expectancy violation, not "we cured anxiety" or day-one ACT sims as exposure therapy.

LVIII.1 Why this chapter exists

Parts XXIV (affect/anxiety), XLI (SAFE-DD), XLV (SAFE-RESILIENCE), and L (SAFE-CALIB) already say: anxiety taxes working memory, unbuffered desirable difficulty becomes threat, grit posters fail, and “raise confidence” is not a North Star. What they do **not** yet specify is the **approach schedule** when the threat *is the math itself* — timed items, word problems, graphs, exams, public board energy.

Competitors fake exposure poorly: Khan dumps harder items after a wrong answer; Duo streaks punish skip; Brilliant opens with delightful puzzles that never feel like the threat Maya avoids; ChatGPT tutors soothe (“you’ve got this”) while the student never approaches the feared format. Therapy apps and school worksheets sometimes cosplay CBT with breathing gifs and no expectancy violation.

FOUNDER BELIEF under audit: Identity transformation for high-math-anxiety (HMA) Maya requires **approaching feared math situations under destaked conditions until threat expectancies are violated** — not forever-easy fluency, not flooding, and not clinical cosplay.

Claims we refuse as doctrine:

1. Day-one full ACT / mixed timed exams labeled “exposure therapy.”
2. Habituation-only success (“stay until anxiety = 0”).
3. Clinical CBT / therapist branding without clinician oversight.
4. Permanent safety behaviors (hint binge, skip, parent rescue) as “support.”
5. Strict linear hierarchy forever (never vary order/context).
6. Anxiety meters / “Calm Score™” as North Star.
7. Exposure → ACT +2σ / Bloom marketing.

LVIII.2 Constructs (product language)

Construct	Therapy meaning	MindCraft analogue	Failure mode if misused
Fear hierarchy	Ranked approach steps from least → most feared	Format/stakes ladder (untimed soft → timed mix)	Hierarchy as forever easy-mode
Graded exposure	Approach feared CS while learning safety	Planned approach to avoided item types	Random harder = fake exposure
Expectancy violation	Feared outcome does not occur (or is tolerable)	Soft-wrong + survive + retry without catastrophe	Calm breathing without approach
Safety behavior	Acts that prevent learning (escape, ritual)	Instant full reveal; skip; streak protect	“Support” that blocks learning
Inhibitory learning	New CS–no US memory competes with fear memory	Multiple contexts/formats after approach	One safe session → exam panic
SAFE-EXPOSE	Exposure doctrine that survives flooding / cosplay / meter kills	See LVIII.9	Therapy branding; flood sims

Operational definition (HYPOTHESIS): A MindCraft session counts as *SAFE-EXPOSE* when (a) the learner **approaches** a previously avoided or high-SUDS math situation, (b) stakes are **destaked** enough that WM can engage (SAFE-DD), (c) a named **threat expectancy** is checked ("I'll freeze / blank / be stupid"), (d) outcome **violates or revises** that expectancy without catastrophe framing, (e) **safety behaviors are faded** (hints → cards → solo), (f) later practice **varies context/format** so learning is not locked to one cozy UI, (g) product never claims to *treat* clinical anxiety disorders.

LVIII.3 Why avoidance is the commercial enemy

FACT (construct + avoidance): Hembree (1990, *Journal for Research in Mathematics Education*, 21(1), 33–46, doi:10.2307/749455 / 10.5951/jresematheduc.21.1.0033) — meta-analysis of 151 studies: mathematics anxiety relates to poorer achievement, poorer attitudes, and is **bound directly to avoidance** of math; valid treatments that reduce anxiety tend to accompany improved performance.

FACT (WM disruption): Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — HMA individuals show reduced WM spans under math demand; anxiety acts as a **transitory dual-task load**, amplifying errors especially on carry/complex items (see also Ashcraft & Krause, 2007, *Psychonomic Bulletin & Review*).

Commercial implication: The product problem is not "Maya needs nicer explanations." It is **avoidance of the situations that would create competence evidence**. Marketing that promises comfort without approach sells temporary relief and loses the identity loop (FEI). Marketing that floods her with timed mixed ACT on day one sells rigor theater and churn.

Kill: "More content solves anxiety." Kill: "Just make it fun." Survive: designed approach to avoided formats under recoverable stakes.

LVIII.4 Classical graded exposure and math desensitization lineage

FACT (lineage): Systematic desensitization (Wolpe tradition) pairs gradual hierarchy ascent with incompatible responses (often relaxation). Education/counseling literature adapted this to math anxiety via hierarchy construction + gradual approach (see secondary reviews; e.g. IntechOpen synthesis citing Wolpe 1958 and school applications — cite cautiously as review, not primary RCT).

FACT (math-anxiety treatment comparison): Zettle (2003, *The Psychological Record*, 53, 197–215, doi:10.1007/BF03395440) — Acceptance and Commitment Therapy (ACT) vs systematic desensitization for mathematics anxiety; both approaches target anxiety-related avoidance, with different mechanisms (acceptance/values vs desensitization). Product takeaway: **approach + new relationship to anxious thoughts** matters more than branding one school of therapy.

FACT (case / protocol shape): Open-access childhood MA CBT case work (e.g. Coping Cat–inspired hierarchy + graded exposure; PMC8283868) shows the clinical shape — least→most feared math situations. **Wound:** N=1 ≠ product efficacy claim.

HYPOTHESIS for MindCraft: Borrow *hierarchy* + *graded approach* + *destake*; do **not** ship therapist roleplay or "we treat anxiety disorders."

Kill: Clinical cosplay. Survive: hierarchy of *math situations* (formats, timing, stakes), not feelings posters.

LVIII.5 Inhibitory learning upgrades habituation theater

FACT (mechanism shift): Craske, Treanor, Conway, Zbozinek & Vervliet (2014, *Behaviour Research and Therapy*, 58, 10–23, doi:10.1016/j.brat.2014.04.006; PMC4114726) — exposure optimized via **inhibitory learning**: goal is new inhibitory CS–no US associations and expectancy violation, **not** necessarily within-session fear reduction to zero. Strategies include expectancy violation, removal of safety signals, variability, multiple contexts, retrieval cues, affect labeling (among others).

FACT (clinical overview): Craske inhibitory-regulation line (doi:10.1159/000381574) — anxious individuals often show inhibitory learning/regulation deficits; strengthen inhibitory retrieval, not just "wait until calm." Variability-in-exposure reviews (e.g. PMC6884337) argue varying hierarchy order/intensity can aid expectancy violation and cross-context retrieval — school-product transfer remains empirical.

Product implication: Success metric is **approach + violated expectancy + later approach in new context** (`challenge_accept`, `retry_120s`, format hops), **not** "anxiety slider hit zero." Soft-wrong + survive is expectancy violation done right. Breathing-only without approach is cosplay.

Kill: Habituation-as-North-Star. Kill: "Stay in the hard problem until you feel calm." Survive: name the feared outcome → approach → check what happened → vary next.

LVIII.6 Extinction is fragile — renewal kills one-session miracles

FACT: Bouton (2002, *Biological Psychiatry*, 52(10), 976–986, doi:10.1016/S0006-3223(02)01546-9; PubMed 12437938) — extinction does **not** erase the original fear association; it creates new, **context-dependent** learning. Return of fear via **renewal** (ABA / ABC patterns) is expected when context changes after extinction.

FACT (learning & memory bridge): Bouton (2004, *Learning & Memory*, 11, 485–494) — contexts set the occasion for CS–US vs CS–no US; mere removal from the extinction context can renew responding.

Commercial implication: One cozy MindCraft session where Maya survived word problems does **not** license "anxiety cured" copy — exam hall / parental watching / timed ACT is a **new context**. Product must plan **multi-context approach** (home app → timed → mixed → low-stakes quiz climate) and retrieve inhibitory learning with cues (SAFE-AAR "what was I afraid would happen?").

Kill: One-and-done exposure marketing. Survive: spaced approach across formats/stakes (ties SAFE-TIMING / SAFE-TRANSFER).

LVIII.7 Brief cognitive tools that are *not* exposure — and must not replace it

FACT: Park, Ramirez & Beilock (2014, *Journal of Experimental Psychology: Applied*, doi:10.1037/xap0000013) — short **expressive writing** before a math exam reduced the HMA–LMA performance gap (N=80); among HMAs, anxiety/cause/insight language related positively to math performance. Related classroom writing finding: Ramirez & Beilock (2011, *Science*, doi:10.1126/science.1199427).

HYPOTHESIS: Writing / affect labeling can free WM *before* approach (Craske lists affect labeling among optimizers) — but writing **without** subsequent approach is incomplete for avoidance. Product: optional 60–90s worry-dump *then* hierarchy step — not a journaling app substitute for practice.

Kill: Mindfulness theater as the whole product. Survive: brief unload → approach → expectancy check.

LVIII.8 Crosswalk to SAFE-DD / FEI / soft-wrong

Exposure principle	MindCraft already-adjacent law	Product move
Equip before strain	SAFE-DD equip	Hierarchy starts below panic; cards available then faded
Destake while approaching	Soft-wrong; mastery climate	No public rank; no streak-as-worth during expose steps
Expectancy violation	FEI fear→evidence	Prompt: "What did you expect?" → "What happened?"
Remove safety signals	Hint fade / solo_transfer	Cap hint depth; require attempt before full reveal
Variability / multi-context	SAFE-TRANSFER / VT	After approach succeeds, hop format/timing
Do not flood	SAFE-RESILIENCE growth-zone	Ban day-one full timed mix for HMA segment

FOUNDER BELIEF: SAFE-EXPOSE is the **affective schedule** for SAFE-DD — difficulty dosing without approach design is incomplete for Maya who never opens the hard door.

LVIII.9 SAFE-EXPOSE stack (surviving product rules)

1. **Build a situation hierarchy, not a vibes hierarchy.** Steps = concrete math situations (untimed symbolic → word → diagram → timed soft → short mix). Prefer student-ranked SUDS-lite once, then product defaults.
2. **Approach before polish.** The session's job is approach + expectancy check, not explanation volume.
3. **Destake the approach.** Soft-wrong, mastery climate, no league, no streak identity (SAFE-COMPARE / HABIT / REWARD).
4. **Name expectancy → violate → label.** Short prompt triad; optional expressive writing only as WM unload.
5. **Fade safety behaviors.** Hints → structured cards → annotate (SAFE-ANNOT) → solo transfer. Permanent hint binge = failed exposure.
6. **Vary after success.** Do not lock learning to one cozy context (Bouton renewal).
7. **Measure approach, not calm meters.** Primary: `challenge_accept`, `retry_120s`, avoided-format approach rate, later `transfer_pass`. Anxiety self-report is covariate, not North Star.

8. **Anti-cosplay.** No "therapist mode," no diagnosis, no clinical claims. Plain language: "practice the scary step under safe rules."

LVIII.10 Competitive audit — who actually runs exposure?

Competitor	Hierarchy?	Destaked approach?	Expectancy check?	Failure mode
Khan	Weak (skill tree ≠ fear ladder)	Partial	Rare	Harder after miss ≈ flood/shame
Duo	No	No (streak threat)	No	Avoidance via skip; streak colonization
Brilliant	Puzzle delight ≠ feared exam format	Low stakes yes	Rare	Fun ≠ approach to avoided threat
ChatGPT tutor	Ad hoc	Emotional soothe	Often sycophantic reassurance	Soothe without approach; fluent wrongness
Human tutor	Often strong	Variable	Variable	Unscalable; trait-label risk (SAFE-EXPECTANCY)

Commercial implication: Position as **approach ladders for feared math situations** — scary step under safe rules, then prove in a new format — never "AI therapy" and never day-one elite timed drills.

LVIII.11 Claim ladder

Claim	Max ladder without new data
Math anxiety tied to avoidance; treatments that work often improve performance	L1 (Hembree, 1990)
Anxiety disrupts WM during math	L1 (Ashcraft & Kirk, 2001)
Exposure benefits from inhibitory learning / expectancy violation > calm-to-zero	L1 clinical translation (Craske et al., 2014+)
Extinction is context-bound; renewal expected	L1 (Bouton, 2002/2004)
Expressive writing can shrink HMA performance gap pre-test	L1 (Park/Ramirez/Beilock)
Zettle: SD and ACT both target math-anxiety avoidance pathways	L1 clinical trial lineage

SAFE-EXPOSE raises MindCraft `challenge_accept` / transfer for HMA segment	L1 product bet — needs EXP-O experiments
Flooding / day-one ACT sims / Calm Score™ / therapy cosplay	**Banned** as doctrine

LVIII.12 Experiments spawned

ID	Question	Design	Primary	Kill condition
EXP-O1	Hierarchy+destake vs random harder after miss	2-arm HMA segment	`challenge_accept`; exit; week transfer	Random ≥ hierarchy on transfer without churn cost → revisit
EXP-O2	Expectancy-violation prompts on vs off	2-arm	expectancy revision rate; `retry_120s`	Prompts add friction without learning → shorten/kill
EXP-O3	Safety-behavior fade vs permanent hints	2-arm	`solo_transfer_pass`; hint depth	Permanent hints ≥ fade on transfer → still prefer fade ethically; redesign fade
EXP-O4	Single-context approach vs multi-format renewal plan	2-arm	format-hop pass after delay	Single ≥ multi → still keep multi for exam ecology
EXP-O5	Worry-write then approach vs approach-only	2-arm HMA	accuracy under load; completion	Write hurts or delays without gain → optional/off
EXP-O-QUAL	10 Maya: “exposure ladder” language vs “practice the scary step”	interview	coded: clarifying / clinical / shame	Clinical words → plain language only

Pre-reg (XXXIV): EXP-O* identify **approach architecture for anxiety** — not “we treat anxiety disorders” or ACT score guarantees.

LVIII.13 Confidence table

Claim	Label	Confidence
Anxiety–avoidance–performance cluster is real at population level	FACT	High
WM dual-task account of online math-anxiety cost	FACT	High
Inhibitory learning reframes exposure away from habituation-only	FACT (clinical science)	High
Bouton renewal / context-bound extinction	FACT	High
Expressive writing helps some HMA test performance	FACT	Medium–High
Case/protocol SD for child MA generalizes to ACT teens in apps	HYPOTHESIS / wounded	Low–Medium
SAFE-EXPOSE is right default for MindCraft HMA segment	FOUNDER BELIEF / HYPOTHESIS	Medium
Therapy branding / flooding / calm meters improve belonging or scores	SPECULATION / false as doctrine	Low (against)

LVIII.14 What this chapter kills

1. **Kill:** Day-one full timed mixed ACT / flooding labeled as exposure therapy.
2. **Kill:** Habituation-only success (“anxiety must hit zero”).
3. **Kill:** Clinical CBT / therapist / diagnosis cosplay in product or ads.
4. **Kill:** Permanent hint binge, skip, or parent-rescue as “support.”
5. **Kill:** Anxiety meters / Calm Score™ / “we cured math anxiety” as North Star.
6. **Kill:** One cozy session ≡ exam-ready (renewal wound).
7. **Kill:** Bloom 2-sigma / exposure⟶score miracle marketing.
8. **Wound:** N=1 clinical cases and review chapters as efficacy proof.
9. **Survive:** SAFE-EXPOSE; hierarchy of situations; destaked approach; expectancy violation; fade safety behaviors; multi-context renewal plan; EXP-O1…5.

Doctrine until data: Ship **SAFE-EXPOSE** — graded approach to feared math situations under recoverable stakes, with expectancy checks and faded safety behaviors, stacked on SAFE-DD / FEI / SAFE-CALIB — and never sell therapy cosplay, flooding, or calm meters as the transformation engine.

Part LIX — Parent WTP Experiments Design

Chapter status: Living evidence + experiment-design brief — Researcher tick 2026-08-01

Primary question: How should MindCraft *measure* what parents will pay for — and which message frames move purchase intent — without confusing stated preference for cash, or selling score theater / vibe identity?

Owners: Growth / Positioning · Behavioral Econ · Product (parent surfaces) · Red Team

Commercial job: Ship a **pre-registered WTP / conjoint / message stack** that tells founders which attributes to build and which claims to kill — before pricing pages and parent dashboards ossify around the wrong North Star.

LIX.1 Why this chapter exists

Part XXVII mapped the post-AI WTP stack (answers → explanations → diagnosis → affect → accountability → trusted outcome signal) and spawned PAR-1 / COMP-1. Part XXXII flagged parent anxiety transmission; Part XXXVII's EVT-4 already asked competence-evidence vs minutes vs AI-unlimited. What the Constitution still lacked is the **methods contract**: how to run conjoint / discrete-choice and message tests so results can raise claim-ladder level (Part XXXIV L4 market causal) instead of producing inspirational survey quotes.

FOUNDER BELIEF under audit: Families will pay for *trusted diagnosis* + *transfer evidence* + *human accountability* more than for unlimited AI chat — **if** we prove it with trade-off designs, not founder intuition.

Claims we refuse as doctrine:

1. "Parents only buy score spikes" as settled fact without trade-off data.
2. "Identity transformation" as a purchasable SKU without FEI evidence parents can parse.
3. Likert "how much would you pay?" as serious WTP.
4. Stated WTP ≡ revenue (hypothetical bias ignored).
5. Streak / minutes / "AI unlimited" as the parent value prop by default.
6. One national average price as the pricing strategy.

LIX.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
WTP	Max price at which a package is chosen over alternatives / outside option	Monthly plan parents select in CBC	Treat survey WTP as sticker price
Attribute	Package feature with levels	Diagnosis depth; tutor hours; report type; price	Too many attributes → noise
CBC / DCE	Choice among multi-attribute profiles	Parent chooses Plan A vs B vs neither	Rank-all-features surveys
Part-worth	Utility of a level	Relative value of FEI report vs score-only	Over-interpret small segments

Message test	Framing → intent / comprehension	Landing / email / dashboard copy	Confuse click with retention
SAFE-WTP	Experiment doctrine that survives vibe pricing	See LIX.10	Sell identity without transfer proof

Operational definition (HYPOTHESIS): A MindCraft parent-pricing study counts as *SAFE-WTP* when (a) it uses **choice-based** trade-offs (or incentive-aligned follow-ups), not open "name a price," (b) attributes map to **shippable product surfaces**, (c) a **price** or clear outside option is present, (d) **hypothetical-bias mitigations** are pre-registered, (e) primary outcomes separate **purchase intent**, **comprehension of child's bottleneck**, and **homework-conflict risk**, (f) results update packaging — not vanity decks.

LIX.3 Market facts — demand exists; composition is uneven

FACT (US tutoring industry geography): Kim, Goodman & West (2024, CESifo Working Paper No. 11251 / IZA DP 17178) — U.S. private tutoring *centers* more than tripled ~1997–2022 (~3,000 → ~10,000); locations concentrate in high-income, high-parent-education areas; >half in top income quintile; residual association with Asian American population share conditional on income/education. School-market structure (private/charter/magnet) is only marginally predictive.

FACT (household spend snapshot, cited therein): Using Consumer Expenditure Survey calculations, authors report ~6–7% of U.S. families with children 6–17 paid for tutoring in the past year (2022), averaging about **$437 in months with a purchase**; top 10% / 5% of monthly spend near **$1,200 / $2,000**. (Treat as CES-derived descriptive, not MindCraft price ceiling.)

FACT (shadow-education framing): Bray (2010) and Bray & Lykins (2012, ADB) document private supplementary tutoring ("shadow education") as a global, fee-based parallel to schooling — useful for competitive context, not a US price guide.

Commercial implication: There is a real paid market adjacent to MindCraft — but it is **skewed**. Pricing and positioning that assume median-America "all parents pay $X for identity" will miss both the premium competitive segment and equity constraints. Segment conjoint (income × anxiety × exam stakes) is mandatory.

Kill: One national WTP number as strategy. Survive: segment-aware CBC + equity-aware packaging.

LIX.4 Why Likert WTP fails — use stated *choice*

FACT (method foundation): Louviere, Hensher & Swait (2000, *Stated Choice Methods*, Cambridge) — stated preference via controlled choice experiments recovers relative attribute utilities under random-utility assumptions; preferable to unstructured preference talk when products are multi-attribute.

FACT (RUM lineage): McFadden (1974) random-utility discrete choice — choices reveal trade-offs; with a monetary attribute, researchers can express part-worths as WTP for levels.

FACT (tutoring-adjacent DCE): Gambi & De Witte (2024, *Higher Education*, doi:10.1007/s10734-024-01358-z) — DCE + information experiment on HE students' tutoring preferences/consumption; high stated WTP coexists with low private uptake; SES demand vs ability-to-pay tension; behavioral frictions (status quo, image) matter. **Wound for K-12 parents:** student HE sample ≠ Maya's parent — method transfers, coefficients do not. Broader HE DCE literature (UK/EU/Polish university-attribute studies building on Louviere et al.) shows the same tool recovers monetary trade-offs for staff expertise, flexibility, and mode.

Product implication: Stop asking "Would you pay for an identity-transforming math world?" Ask: *Between these two monthly packages, which do you choose — or neither?*

LIX.5 Hypothetical bias — stated ≠ paid

FACT: Murphy, Allen, Stevens & Weatherhead (2005, *Environmental and Resource Economics*, 30(3), 313–325) — meta-analysis of hypothetical vs actual WTP; median hypothetical/actual ratio ~1.35 with severe positive skew; choice-based elicitation associated with lower bias than some contingent-valuation formats (model-sensitive).

FACT (mitigation lineage): Cummings & Taylor (1999, *American Economic Review*) — "cheap talk" scripts can reduce hypothetical overstatement (effects heterogeneous later). Schmidt & Bijmolt (2020, *JAMS*, doi:10.1007/s11747-019-00666-6) — consumer-goods meta: prefer indirect/choice over direct "what would you pay?"; incentive-aligned mechanisms help when feasible.

HYPOTHESIS for MindCraft: Parent CBC will still inflate absolute dollars; **rank-order of attributes** (FEI report vs streak dashboard vs AI chat hours vs human tutor minutes) is the decision-useful output. Validate top packages with a small **deposit / waitlist / paid pilot** before national pricing.

Kill: Survey WTP as P&L input. Survive: relative part-worths → packaging; cash tests → prices.

LIX.6 Attribute set for MindCraft parent CBC (shippable)

Design rule (FOUNDER BELIEF): Every attribute level must map to a screen or ops cost we can actually ship in 90 days.

Attribute	Levels (draft)	Ties to doctrine
Diagnosis depth	Vague "needs practice" / concept weakness list / bridge+format gap map	Ontology advantage; XXVII trust
Evidence report	Score/minutes only / FEI pack (`retry_120s`, challenge-seeking, transfer) / FEI + honest plateau note	PAR-1; anti-streak
Human accountability	None / async tutor note weekly / 2× live 25-min / 4× live	Scarcity layer

AI help policy	Unlimited chat / guarded pedagogy wrap / attempt-first then cards	XXXIII; SE constraints
Practice architecture	Streak gamification / mastery climate soft-wrong / SAFE-EXPOSE ladder language	XLII–XLIV; LVIII
Price	e.g. $49 / $99 / $179 / $299 per month (calibrate to segment)	Must include realistic span
Outside option	“Neither — keep free Khan/GPT/status quo”	Critical for WTP

HYPOTHESIS: Diagnosis depth + FEI report + modest human accountability will dominate unlimited AI and streak theater **in the anxious-middle segment**; score-chaser segment may overweight timed-test sims — measure, don’t assume.

Red Team: If FEI report wins stated choice but parents cannot *explain* the child’s bottleneck after viewing it, comprehension failed — packaging theater.

LIX.7 Message tests are not conjoint (run both)

Conjoint answers *what package*; messages answer *which story opens the wallet without raising homework conflict.*

Pre-registered message arms (builds EVT-4 / COMP-1):

Arm	Parent-facing frame	Risk
M1 Score relief	“Raise ACT math”	Overclaim; retention crash if flat
M2 Homework peace	“End nightly fights”	Soothing without competence
M3 Diagnosis trust	“See the exact bridge that’s stuck”	Jargon; need plain language
M4 Identity / confidence	“Become someone who doesn’t freeze”	Belief Score™ / raise-confidence kill (v1.7–v1.8)
M5 Hybrid	Diagnosis + transfer check + honest plateau	Longer copy; may win retention

Primary outcomes: click → trial start; 14-day activation; parent comprehension quiz (name bottleneck); reported homework conflict; **not** vanity “brand love.”

Kill: Identity-only ads that cannot point to a measurable FEI metric. Survive: hybrid that sells method, shows evidence.

LIX.8 Qualitative gate before fielding CBC

FACT / process: Part XXVII and QUAL-P (XXXII) already require ~10 parent interviews. CBC attributes invented in a founder Slack will mis-specify levels.

Protocol (HYPOTHESIS): Semi-structured interviews → code "what I thought I bought / what would make me cancel / words I trust / words that feel scammy" → freeze attribute glossary → then CBC. Do not reverse the order.

Segment quotas (SPECULATION until interviews): (1) HMA-Maya parents (anxiety/avoidance), (2) score-chaser / selective-college, (3) homework-peace exhausted, (4) price-sensitive remedial. Run latent-class or split models; do not average them into one "parent."

LIX.9 Equity and pricing ethics

FACT (concentration): Kim et al. (2024) — private center supply concentrates where income/education are high. Packaging only for that segment reproduces shadow-education inequality.

FOUNDER BELIEF: Charging for diagnosis+accountability is legitimate if transfer evidence is real (XXVII.7). Illegitimate if the paid tier is streak cosmetics + GPT wrap.

Guardrails:

- Paid claims ↔ measured FEI / learning metrics.
- No "guaranteed score jump" in any CBC level or ad arm.
- Consider a **honest free tier** that still shows diagnosis (trust acquisition) vs freemium that hides the bottleneck (dark pattern).
- Report segment WTP separately; do not market top-quintile WTP as universal.

LIX.10 SAFE-WTP stack (surviving product rules)

1. **Choice over chatter.** CBC/DCE or incentive-aligned choice; ban standalone "name your price."
2. **Ship-mapped attributes only.** No fictional features.
3. **Always include price + neither.** Otherwise WTP is fantasy.
4. **Bias protocol.** Cheap talk / certainty / consequentiality / small cash validation — pre-register which.
5. **Three outcomes.** Intent ≠ comprehension ≠ conflict. Optimize packaging on all three.
6. **Segment, don't average.** Latent class or pre-defined segments.
7. **Messages ≠ packages.** Run COMP/EVT message tests alongside CBC.
8. **Anti-theater.** FEI report that parents cannot interpret loses even if chosen.
9. **Anti-overclaim.** No 2σ, no guaranteed ACT points, no "AI tutor replaces human" as paid promise.
10. **Update doctrine with data.** Part-worths that kill founder favorites get logged as kills.

LIX.11 Claim ladder

Claim	Max ladder without new data

US private tutoring centers grew sharply; high-income concentration	L1 (Kim/Goodman/West 2024)
Shadow education is a real global fee market	L1 (Bray line)
Stated choice recovers relative utilities better than unstructured WTP talk	L1 (Louviere/Hensher/Swait; McFadden)
Hypothetical WTP inflates; choice formats help but do not erase bias	L1 (Murphy et al. 2005; Cummings & Taylor)
HE tutoring DCE: high WTP, low uptake, SES frictions	L1 adjacent (Gambi & De Witte 2024) — transfer wound to K-12 parents
MindCraft parents prefer FEI+diagnosis over unlimited AI / streaks	**L0–L1 product bet** — needs WTP-* experiments
Survey dollars = revenue forecast	**Banned**

LIX.12 Experiments spawned

ID	Question	Design	Primary	Kill condition
WTP-1	FEI report vs score/minutes vs streak dashboard in CBC	CBC, price + neither	part-worths; segment classes	Streak/minutes dominate FEI *and* later retention → revisit packaging (still educational wound)
WTP-2	Diagnosis depth (vague / concept / bridge+format) × price	CBC	WTP for bridge+format; comprehension	Depth raises intent but kills comprehension → simplify language
WTP-3	Human minutes vs AI-unlimited vs guarded wrap	CBC	part-worth human; churn proxy intent	AI-unlimited wins on intent *and* 30-day solo transfer → thesis wounded
WTP-4	Message arms M1–M5	RCT landing/email	trial start; conflict; comprehension	Identity arm wins clicks + raises conflict without retention → kill identity-only
WTP-5	Cheap-talk / certainty calibration on vs off	2× embedded in CBC	calibration gap	No gap → still keep cash pilot

WTP-6	Deposit / paid pilot of top CBC winner vs #2	Field	pay-through; 30-day retention	Stated winner loses cash → trust revealed preference
WTP-QUAL	10 parents: cancel triggers + trusted words	Interview	codebook → attribute freeze	Founder glossary ≠ parent glossary → rewrite levels

Pre-reg (XXXIV): WTP-* identify **parent package preference and message effects** — not "we proved identity sells" or national price optimality from N=200 survey.

Ties: EVT-4, PAR-1, COMP-1, MUSIC-4, TIM-4 become arms or covariates inside this family — do not fork duplicate surveys.

LIX.13 Confidence table

Claim	Label	Confidence
Paid tutoring demand exists and is income-skewed in the US center market	FACT	High
CES spend snapshots describe *purchasers*, not all parents' WTP	FACT	High
CBC/DCE is the right class of tool for multi-attribute tutoring packages	FACT (methods)	High
Absolute stated dollars will be biased upward	FACT / HYPOTHESIS	High
FEI+diagnosis beats unlimited AI for MindCraft's anxious-middle segment	HYPOTHESIS	Medium
Hybrid message (diagnosis + transfer + honest plateau) wins retention vs score-only	HYPOTHESIS	Medium
Identity-only parent ads are net-positive	SPECULATION / against doctrine until data	Low (against)
One price point fits all segments	SPECULATION / false as strategy	Low (against)

LIX.14 What this chapter kills

1. **Kill:** Likert / open-ended "how much would you pay?" as pricing research.

2. **Kill:** Treating survey WTP as revenue or national sticker price.
3. **Kill:** “Parents only buy scores” *or* “parents buy identity” as settled without CBC.
4. **Kill:** Streak / minutes / AI-unlimited as default parent value props without losing on part-worths.
5. **Kill:** Guaranteed ACT points / 2σ / Bloom in any package level or ad.
6. **Kill:** Averaging anxious-middle and score-chaser into one “parent persona.”
7. **Wound:** Importing HE or non-US tutoring conjoint coefficients as US ACT-parent truth.
8. **Survive:** SAFE-WTP; CBC with price+neither; ship-mapped attributes; bias protocol; message≠package; WTP-1…6 + QUAL; cash validation before scale.

Doctrine until data: Run **SAFE-WTP** — choice-based parent trade-offs on diagnosis, FEI evidence, human accountability, AI policy, and price — then package what wins *and* what parents can explain, not what flatters the identity thesis.

Part LX — Trust After AI Hallucination

Chapter status: Living evidence + repair-protocol brief — Researcher tick 2026-08-01

Primary question: After MindCraft’s AI is *caught* being fluently wrong (or quietly wrong and later exposed), how do we repair *calibrated* trust — and when must we escalate to a human — without apology theater, never-wrong marketing, or post-error abandonment?

Owners: AI / Product (Solver · coach) · Learning Science · Human Factors · Parent trust · Red Team

Commercial job: Ship a **SAFE-REPAIR** doctrine that turns inevitable model errors into a *positioning asset* (honest plateau + human accountability) rather than a churn event or a “AI that never fails” lie.

LX.1 Why this chapter exists

Part XXXIII owned *prevention and calibration*: pedagogy wrap, anti-sycophancy, Bastani unguarded-harm FACT, Lee & See appropriate reliance. Part L (SAFE-CALIB) owned *student* confidence after misses. What the Constitution still lacked is the **post-violation playbook**: what the product does in the *minutes after* a hallucination is detected — by checker, by student flag, by tutor review, or by exam-collapse discovery.

FOUNDER BELIEF under audit: Families and students will tolerate imperfect AI *if* the system (a) admits competence failure fast, (b) shows a concrete repair path tied to the Map/ontology, and (c) escalates to a human when stakes or repeated failure demand it — and will *punish* products that either hide errors or drown users in soft apologies without correction.

Claims we refuse as doctrine:

1. “Our AI doesn’t hallucinate” as marketing (false category; Bastani notes GPT Base hallucinations as mechanism risk).
2. Rote “Sorry!” as trust repair.
3. Empathy-only recovery that never corrects the math.
4. Permanent undertrust (“never trust the coach”) as the safety story.

5. Unlimited human escalation as the default UX (destroys unit economics and tutors).
6. Student thumbs-up / “was this helpful?” as proof the repair worked.

LX.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Hallucination / fluent wrongness	Confident false content	Wrong hint, invented step, affirming wrong setup	Treat only wild fabrications; ignore soft arithmetic slips
Trust violation	Event that breaks reliance calibration	Caught wrong coach move; silent wrong→exam miss	Every minor UI glitch as “trust crisis”
Competence vs integrity failure	Ability miss vs values/honesty miss (Kim et al.)	Bad algebra vs hiding known wrongness	Apologize for everything the same way
Trust repair	Actions that restore *appropriate* reliance	Admit → correct → verify → optional escalate	Maximize liking, not calibration
Misuse / disuse	Over-reliance / under-reliance (Parasuraman & Riley)	Copying Solver; abandoning coach after one miss	Chase either extreme as “engagement”
Escalation	Hand-off to human when AI insufficient	Tutor queue / session brief with error packet	Escalate everything or nothing
SAFE-REPAIR	Surviving post-error doctrine	See LX.10	Apology cosplay; never-wrong brand

Operational definition (HYPOTHESIS): A MindCraft post-error flow counts as *SAFE-REPAIR* when (a) the system **detects or accepts** the error signal, (b) labels it as a **competence** miss (not student shame), (c) issues an **explanatory** repair with a **deterministic correction path** (bank key / checker / Map ingredient), (d) invites a **solo re-attempt** before full solution dump, (e) offers **escalation** only under explicit triggers, and (f) measures repair by **re-engagement + verified next attempt**, not apology NPS.

LX.3 Prior art already in the Constitution (do not re-litigate)

FACT (high-school math field RCT): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — unguarded GPT access inflated practice then **harmed** solo exam (~–17% vs never-AI); guarded “GPT Tutor” largely neutralized harm but did **not** show positive learning vs control; Base hallucinations and crutch copying are named mechanisms. Perception ≠ learning.

FACT (novice verification failure): Li, Song, Sundaram & Karahalios (L@S 2025, doi:10.1145/3698205.3729550) — most learners failed to detect chatbot factual errors; undetected errors harmed outcomes and self-efficacy.

FACT (appropriate reliance): Lee & See (2004, *Human Factors*, doi:10.1518/hfes.46.1.50_30392) — design for calibration, resolution, specificity; misuse and disuse are both failures.

FACT (use/misuse/disuse): Parasuraman & Riley (1997, *Human Factors*) — operators over-rely, under-rely, or abuse automation; trust is a primary mediator.

FACT (trust layers): Hoff & Bashir (2015, *Human Factors*, doi:10.1177/0018720814547570) — dispositional, situational, and learned trust; post-error recovery is largely a **learned-trust** problem.

Product implication: Prevention (PWC / ontology / checkers) remains primary. This chapter owns what happens when prevention fails in the wild — which it will.

LX.4 Trust repair is a designed capability, not a tone setting

FACT (HMI agenda): de Visser, Pak & Shaw (2018, *Ergonomics*, doi:10.1080/00140139.2018.1457701; "From 'automation' to 'autonomy'") — as systems become more autonomous, **trust repair** after violations becomes a first-class design requirement; competence vs integrity framing and context/risk matter; human–human repair findings transfer only partially.

FACT (human–human baseline): Kim, Ferrin, Cooper & Dirks (2004, *Journal of Applied Psychology*, doi:10.1037/0021-9010.89.1.104) — apology repairs **competence** violations better than denial; for **integrity** violations, denial of culpability can outperform apology (when evidence supports innocence). Follow-on attribution work (Kim, Dirks, Cooper & Ferrin, 2006, *OBHDP*, doi:10.1016/j.obhdp.2005.07.002) shows internal vs external blame interacts with violation type.

HYPOTHESIS for MindCraft math errors: Nearly all Solver/coach hallucinations are **competence** failures. Correct response class = **apology + ownership + concrete fix**, not "the curriculum is wrong" externalization, and not integrity-theater ("we would never…"). If the product *knew* the key and still affirmed the student's wrong answer (sycophancy), treat that as closer to **integrity** — repair requires admitting the agreement failure, not only the math slip.

Wound (field realism): Liu, Du, Ma, Zhang & Yan (2024, *IEEE Trans. Human-Machine Systems*, doi:10.1109/THMS.2024.3434680) — after automated-driving failure in a test-track study (N=257), verbal apology+explanation+promise **did not restore damaged trust**; human-voice repair only partially mitigated other attitudes, mainly among less experienced drivers. **Transfer wound:** ADS safety failure ≠ wrong algebra hint — but the kill candidate survives: **do not assume apology scripts restore trust scores**.

Kill: Apology copy as the repair KPI. Survive: behavioral re-calibration after a verified correct path.

LX.5 What users prefer after LLM mistakes (preference ≠ recovery)

FACT (preregistered preference study): Zhang et al. (2025/26; *CHI* / ACM, doi:10.1145/3793679; arXiv:2507.02745) — N=162 Prolific; pairwise preferences among **rote**, **explanatory**, and **empathic** apologies after bias, fabrication (hallucination), and factual-error scenarios. **Explanatory** apologies generally preferred; **empathic** preferred in bias contexts; for hallucinations, users saw seriousness but showed **no clear apology-type preference** — uncertainty, not a settled recipe.

FACT (critical review): Harland et al. (2025, *Artificial Intelligence Review*, doi:10.1007/s10462-025-11305-8) — AI apology literature (2020–2023 synthesis) stresses that apology is only one element; identification, explanation, and **system behavior change** are required for meaningful repair. Purely discursive apology without corrective action is ethically and practically thin.

HYPOTHESIS: For Maya after a wrong coach step, the winning stack is **explanatory ownership + Map-linked correction + destaked re-attempt**, not empathic padding ("that must feel frustrating") alone. Empathy may reduce shame (Part XXIV) but does not restore calibration.

Contradiction / limit: Some GenAI service-recovery work finds gratitude/appreciation or humorous styles can raise *willingness to forgive* in consumer settings (e.g. Systems 2024 mixed methods on language style × gratitude/apology — domain = service UX, not math learning). Treat forgive-intent as **engagement theater** unless paired with verified next-attempt success.

Commercial implication: Marketing may say "when we're wrong, we show our work and hand you a human." Product must actually do the second half.

LX.6 Escalation to human — scarce, triggered, packetized

FACT (hybrid classroom deployment): Kazemitabaar et al. (2025, arXiv:2510.14457) — programming course N=82; 673 AI hints; students rated 146 (22%) unhelpful; of those, only **16 (11%)** escalated to instructors. Instructor replies on escalated cases were incorrect/insufficient ~half the time. Escalation is **underused** by students and **not automatically high-quality** for humans.

FACT (transparency shifts strategy, not accuracy): Nagashima, Hladký & Rief (arXiv:2606.03822; ECTEL 2026) — fallibility warning increased hint-seeking without changing identical ITS behavior or immediate performance (already in XXXIII).

HYPOTHESIS — MindCraft escalation triggers (product contract):

1. **Checker conflict:** model claim ≠ bank key / symbolic checker → auto-flag; do not wait for student detective work.
2. **Repeat competence miss:** ≥2 verified wrong coach moves in one mission on same concept/ingredient → offer human.
3. **Affect spike + stuck:** high self-reported threat / long idle after soft-wrong → soft escalate (tutor ping), not more AI monologue.
4. **Integrity-class event:** sycophantic affirm of wrong answer against known key → mandatory human-visible audit packet.
5. **Parent/exam stakes mode:** optional "human-reviewed session" SKU (ties SAFE-WTP human-minutes attribute) — not free infinite chat.

Escalation packet (HYPOTHESIS): Human receives: problem ID, student attempt trace, AI utterance, contradiction evidence, ontology IDs, student affect flag — not a blank "help?" thread. Aligns with SAFE-AAR / SAFE-ANNOT (trace before coach dump).

Wound: Human-in-the-loop is not magic (Kazemitabaar instructor error rate). Escalation must preserve **tutor attention hygiene** — packetized, not "AI failed, you debug the LLM live."

Press-adjacent note (SPECULATION until primary paper cited in-repo): Hybrid AI-draft + human-approve tutoring reports (e.g. Eedi/LearnLM coverage in education press) suggest low hallucination rates *under* human gatekeeping — useful as competitive *direction*, not as MindCraft proof. Prefer primary methods papers before marketing "near-zero hallucinations."

LX.7 Identity stakes — why math trust violations cut deeper

HYPOTHESIS: For Maya, a fluent wrong coach move is not only a UX bug; it reactivates the identity narrative "even the helper confirms I'm lost" *or* the opposite overtrust "the AI said I'm right" (Parts XXV, XXX, L). Repair must separate:

- **System competence failure** (external, unstable, specific — Weiner-friendly)
- **Student strategy miss** (controllable after correct feedback)

Never: "you're just not a math person" and never: "trust me completely next time."

FOUNDER BELIEF: Soft-wrong already destakes *student* error. Post-hallucination UX must destake *AI* error without teaching helplessness toward tools (Lee & See disuse).

LX.8 Competitive positioning

Competitor move	Failure mode	MindCraft counter (HYPOTHESIS)
ChatGPT raw	Hallucinate + apologize vaguely + continue	Checker + Map correction + solo re-attempt
"AI tutor never wrong" ads	Integrity violation when caught	Honest fallibility + repair SLA
Unlimited chat, no human	Disuse after salient miss; or silent misuse	Triggered human minutes as paid scarce resource
Thumbs-up quality loop	Bastani/Bo: perception fails	`repair_reattempt_pass`, `escalation_resolve_48h`

Copy that survives Red Team: "We catch wrong steps, show the fix, and bring a tutor when it matters."

Copy that dies: "Our AI never makes mistakes" / "Sorry you're frustrated!" without a corrected path.

LX.9 SAFE-REPAIR stack (surviving product rules)

1. **Detect** — prefer deterministic contradiction over student-reported vibes.
2. **Classify** — competence vs integrity-class (sycophancy-against-key).
3. **Own** — short explanatory apology; no blame-the-student; no vague empathy wall.
4. **Correct** — show Map/ingredient + checked next step; ban full solution until re-attempt window unless escalate.
5. **Re-attempt** — destaked item; measure success.

6. **Calibrate UI** — temporarily lower coach confidence badge on that skill grain (SAFE-CALIB cousin).
7. **Escalate** — only on triggers; send packet; SLA visible to parent/student.
8. **Learn** — log to evaluation set; do not only "prompt harder."
9. **Anti-cosplay** — no therapy script, no airline-captain apology theater, no never-wrong brand.

LX.10 Experiments (REPAIR family)

ID	Contrast	Primary outcome	Falsifier
REPAIR-1	Explanatory+correct path vs rote apology vs empathy-only after injected wrong hint	`repair_reattempt_pass` @10m; trust calibration item	Empathy-only wins reattempt → revisit stack
REPAIR-2	Auto-detect+own vs silent correct-and-continue	next-mission return; overtrust score	Silent fix retains more misuse → keep detect
REPAIR-3	Escalation offer after 1 vs 2 vs 3 verified AI misses	escalate rate; human resolve quality; cost/session	Instant escalate floods tutors without lift
REPAIR-4	Packetized tutor handoff vs blank "talk to tutor"	time-to-correct; tutor CSAT; reattempt	Packet no better → packet UX fail
REPAIR-5	Fallibility banner always-on vs post-error-only vs none	hint abuse; solo transfer	Always-on destroys use without calibration gain
REPAIR-6	Parent-visible "AI miss + repair" FEI note vs hidden	parent trust / WTP message arm	Hidden wins retention → wound transparency thesis
REPAIR-QUAL	10 Maya + 5 tutors: what "sorry" means after wrong hint	codebook → copy freeze	Founder apology ≠ user need

Pre-reg (XXXIV): REPAIR-* identify **post-error calibration and escalation efficiency** — not "apology increases ACT scores" or "human tutors eliminate hallucinations."

Ties: AIT-, *CAL*-, ANNOT-, *AAR*-, WTP human-minutes, EXP-O affect gates.

LX.11 Confidence table

Claim	Label	Confidence

Unguarded generative help can harm solo math learning; hallucinations contribute	FACT	High
Novices often fail to detect fluent errors	FACT	High
Appropriate reliance (not max trust) is the right design target	FACT (HF)	High
Apology scripts alone restore post-failure trust	SPECULATION / often false	Low (against) — Liu et al. wound
Explanatory repairs preferred over rote for many LLM errors	FACT (preference)	Medium–High
Preference among apology types for *hallucination* is unsettled	FACT	High
Competence-framed ownership + corrective action is right class for math slips	HYPOTHESIS	Medium–High
Triggered, packetized human escalation beats unlimited chat *and* no-human	HYPOTHESIS	Medium
Parent-visible repair notes raise WTP for hybrid SKUs	HYPOTHESIS	Medium
“Never hallucinates” is a viable brand claim	SPECULATION / false	Low (against)

LX.12 What this chapter kills

1. **Kill:** “Our AI doesn’t / won’t hallucinate” marketing.
2. **Kill:** Rote or empathy-only apology as the repair product.
3. **Kill:** Silent overwrite of wrong AI output without student-visible ownership when detection exists.
4. **Kill:** Thumbs-up / satisfaction as proof of trust repair.
5. **Kill:** Infinite free human escalation as default.
6. **Kill:** Permanent “never trust the AI” as brand (disuse).
7. **Wound:** Importing ADS trust-repair nulls or consumer GenAI “forgiveness” as ACT learning proof.
8. **Survive:** SAFE-REPAIR; competence ownership + checked correction + re-attempt; triggered packetized escalation; REPAIR-1…6 + QUAL.

Doctrine until data: After a hallucination, MindCraft sells **recoverable truth** — detect, own, correct, prove, escalate when scarce human attention is worth it — not warmth, not invincibility.

Part LXI — Measurement of Math Identity

Chapter status: Living evidence + measurement brief — Researcher tick 2026-08-01

Primary question: How should MindCraft *measure* mathematics identity — without mistaking a Likert bump for transformation, without inventing a proprietary Belief Score™, and without demand-effect theater?

Owners: Learning Science · Product analytics · Diagnostic / FEI · Parent messaging · Red Team

Commercial job: Ship a **SAFE-IDMEASURE** doctrine: validated subscale items + behavioral FEI corroboration + narrative probes — so marketing, experiments, and parent copy can claim *identity movement* only when measurement survives demand effects.

LXI.1 Why this chapter exists

The Constitution's research question is identity transformation. Parts XXV (narrative identity / habit≠identity), XXXVII (expectancy-value), L (SAFE-CALIB), and LIX (SAFE-WTP) already ban empty confidence and survey-dollars-as-revenue. What was still missing is the **instrument layer**: which published scales are usable for Maya-aged students, what they actually factor into, how demand and social desirability corrupt them, and how MindCraft should triangulate without building a vanity Identity Meter™.

FOUNDER BELIEF under audit: Identity change is real and commercially load-bearing — but only if we measure it with instruments that have validity evidence *and* refuse to treat a post-session "I am a math person" checkbox as product success.

Claims we refuse as doctrine:

1. A proprietary "Math Identity Score™" as North Star.
2. Two HSLS-style items alone as causal proof of FEI.
3. Pre→post Likert lift after identity-themed copy as marketing evidence.
4. Competence/self-efficacy survey ≡ mathematics identity.
5. Ignoring recognition (others' appraisals) because it is hard to productize.
6. Treating qualitative storytelling as "unscientific" *or* Likert as "the only science."

LXI.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Mathematics identity	How a learner sees self *in relation to math* (thickening self-perception)	"Math person" / story world belonging without cosplay	Brand slogan without instrument
Interest	Desire/curiosity to engage math	Challenge accept; story-module return	Fun NPS ≡ identity

Recognition	Self + perceived others as “math person”	Tutor/parent brief language; peer opt-in	Forced public praise / leaderboard recognition
Competence / performance beliefs	Beliefs about understanding / doing math	Mastery graph; calibrated confidence	Self-efficacy survey alone as identity
Narrative identity	Reifying, endorsable, significant stories (Sfard & Prusak)	Notes / AAR “what changed about me as a learner”	Essay theater without transfer
Demand / SDR	Response bias toward expected / socially approved answers	Post-mission identity survey after identity ads	Survey lift sold as FEI
SAFE-IDMEASURE	Surviving measurement doctrine	See LXI.9	Identity Score™ productization

Operational definition (HYPOTHESIS): MindCraft claims *identity movement* only when (a) a **validated multi-item subscale** (interest + recognition + competence/performance) moves with preregistered effect size, (b) at least one **behavioral FEI** marker co-moves (`challenge_accept`, `solo_transfer_pass`, or destaked retry), and (c) a light **narrative probe** shows actual→designated story shift — not when any single channel moves alone.

LXI.3 What the field actually measures (not slogans)

FACT (SEM on college calculus, large N): Cribbs, Hazari, Sonnert & Sadler (2015, *Child Development*, 86(4), 1048–1062, doi:10.1111/cdev.12363) — FICSMath; more than 9,000 U.S. college calculus students. SEM of mathematics identity: competence/performance beliefs act largely *indirectly* on “math person” identity via **interest** and **external recognition**. Competence alone is **not** sufficient for identity development.

Product kill: Shipping only “I feel smarter at math” / self-efficacy meters as identity proof. Survive: design and messaging that create *interest* and *recognition*, not competence theater alone.

FACT (K–12 instrument validation): Cribbs & Utley (2023/2024, *Mathematics Education Research Journal*, 36, 767–789, doi:10.1007/s13394-023-00474-w) — N=**1559** students grades **5–12** in two U.S. districts; adapted undergraduate items; expert content review + readability (Flesch–Kincaid ~grade 3); EFA/CFA support a **16-item** scale with three factors: **interest, recognition, competence/performance** (competence and performance again load together, echoing Cass et al., 2011). Fit indices in CFA: CFI≈0.97, RMSEA≈0.063, SRMR≈0.028. Negatively correlated with math anxiety (mAMAS) as concurrent validity.

Commercial implication: For Maya (high-school), the closest *validated* efficient instrument is this lineage — not an invented MindCraft rubric. Prefer licensed/adapted item sets with permission over DIY “identity KPIs.”

FACT (two-item national survey tradition): HSLS:09 and follow-ups include math-identity items of the form *I see myself as a math person* / *others see me as a math person* (Ingels et al., NCES HSLS documentation; Cronbach α reported ~0.84–0.88 in base/follow-up). Miller & Wang (2019, *Journal of Youth and Adolescence*, 48, 2038–2050, doi:10.1007/s10964-019-01115-x) use classroom-practice → motivational beliefs → mathematics identity mediation (N=525 sixth-graders); identity operationalized in that literature stream often collapses to short “math person” self/other items.

Wound: Cribbs & Utley (2023) explicitly note that several large studies using NAEP/HSLS-style items **did not report full validity evidence** for the measure as used. Two items are useful for **screening and national comparison**, weak as sole **product experiment** endpoints.

FACT (science-identity ancestor): Carlone & Johnson (2007, *Journal of Research in Science Teaching*, 44, 1187–1218, doi:10.1002/tea.20237) — recognition, competence, performance as analytic lens for science identity (women of color). Hazari et al. added **interest** when moving to physics/math identity measurement traditions that Cribbs inherits.

FACT (narrative operationalization): Sfard & Prusak (2005, *Educational Researcher*, 34(4), 14–22, doi:10.3102/0013189X034004014) — identity as collections of **reifying, significant, endorsable** stories; learning as closing the gap between **actual** and **designated** identity narratives. Not a Likert substitute — a complementary analytic.

LXI.4 Demand effects, social desirability, and identity advertising

FACT (response styles): Kreitchmann, Abad, Ponsoda, Nieto & Morillo (2019, *Frontiers in Psychology*, 10:2309, doi:10.3389/fpsyg.2019.02309) — social-desirability and acquiescence biases distort Likert scores; ignoring them worsens validity; modeling SDR or forced-choice designs can help; each strategy has tradeoffs (reliability vs criterion validity).

HYPOTHESIS (MindCraft-specific demand): After a session that *markets* identity (“become a math person”), immediately asking “I am a math person” maximizes **demand characteristics** — students infer the desired answer. Risk rises when coaches, tutors, or parents use the same phrase as praise (Part XLVI expectancy ethics).

HYPOTHESIS: Separating **measurement windows** from **identity marketing copy** (e.g., measure at week boundaries, not mission-complete splash) reduces artifactual lift.

Kill: Pre/post “math person” Likert on the same screen as identity-themed celebration UI, sold as FEI evidence.

Survive: Blinded or temporally separated administration; behavioral corroboration; optional forced-choice / IRT-aware scoring later if scale volume justifies it.

LXI.5 What must not be confused with identity

Nearby construct	Why it is not identity	Constitution home
Self-efficacy	Can rise without “math person” story or recognition	XXV, Bandura
Calibration / confidence	Match of certainty to accuracy — honesty, not belonging	L / SAFE-CALIB
Mastery graph score	Performance evidence; necessary but not sufficient (Cribbs 2015)	Engine / XXI

Streak / habit	Cue-driven return ≠ identity thickening	XLIII / SAFE-HABIT
Anxiety reduction	Correlated (Cribbs & Utley) but distinct	XXIV, LVIII
Parent WTP	Purchasing ≠ student identity	LIX / SAFE-WTP

FOUNDER BELIEF (constrained): FEI still holds — fear→evidence→identity — but **identity** here means the Cribbs-factor bundle + narrative shift, not a mood.

LXI.6 Equity and recognition measurement (do not flatten)

FACT / tradition: Recognition is not only teacher praise; Martin and equity-oriented identity work stress how race, gender, and institutional positioning shape who gets to be seen as mathematical (see Part XXXVI equity audit; Carlone & Johnson recognition as gate).

HYPOTHESIS: Product "recognition" that is only AI cheerleading ("Great job, math person!") will inflate Likert recognition subscales without changing peer/teacher/parent stories — a **false recognition** pathway.

Product rule: Prefer recognition that is (a) **earned** (tied to specific transferable performance), (b) **multi-source when possible** (tutor brief, parent note, optional peer — never forced leaderboard), (c) **non-trait** (task language per SAFE-EXPECTANCY).

LXI.7 Recommended MindCraft measurement stack

Tier A — Primary quantitative (product experiments): Adapt Cribbs & Utley (2023) **16-item / 3-factor** instrument (or a preregistered short form of the same factors) for 5–12; administer at baseline, 4 weeks, 8 weeks — **not** on every mission complete.

Tier B — Behavioral corroboration (always co-primary): `challenge_accept`, `solo_transfer_pass`, `retry_120s` after informative miss, destaked re-attempt after soft-wrong. Identity claim fails if Likert rises and these stay flat (XXV falsifier).

Tier C — Narrative light probe (qual, Appendix B adjacent): 2–3 prompts: actual story ("Right now, what kind of math learner are you?"), designated story ("Who are you becoming?"), recognition source ("Who has treated you as capable at math lately — and about what?"). Code for actual→designated gap closure (Sfard & Prusak), contamination→redemption themes (McAdams tradition in XXV).

Tier D — Screening only: HSLS-style two items for funnel diagnostics and parent dashboard *context* — never as sole RCT endpoint.

Explicit ban: No composite "Identity Score™" marketed to parents. Report **factor means + behavioral FEI**, or say nothing.

LXI.8 Experiments (IDM family)

ID	Contrast	Primary endpoints	Kill / survive rule
IDM-1	Cribbs 16-item at weeks 0/4/8 vs HSLS 2-item only	Factor α; sensitivity to FEI intervention	2-item moves, 16-item flat → kill 2-item-as-proof
IDM-2	Measure at week boundary vs mission-complete splash	Δ interest/recognition; demand proxy	Splash-only lift → demand wound
IDM-3	Interest+recognition-focused UX vs competence-meter UX	Identity factors; `challenge_accept`	Competence UX lifts efficacy not identity → confirms Cribbs SEM
IDM-4	Earned tutor/parent recognition note vs AI "math person" praise	Recognition subscale; transfer	AI praise lifts Likert only → kill false recognition
IDM-5	Likert-only claim vs Likert+behavioral co-primary in experiment report	Pre-reg analysis plan adherence	Likert-alone "win" rejected
IDM-QUAL	10 Maya interviews: what "math person" means; demand cues	Codebook → item wording freeze	Founder phrase ≠ student phrase

Pre-reg (XXXIV): IDM-* identify **measurement validity and demand**, not "MindCraft raises identity by X% forever."

Ties: EFF-, *EVT*-, CAL-*, WTP message arms (identity vs score packaging), HIST recognition equity.

LXI.9 SAFE-IDMEASURE stack (surviving doctrine)

1. **Use validated factors** — interest, recognition, competence/performance — not inventing a brand score.
2. **Triangulate** — subscale + FEI behavior + light narrative; one channel alone is insufficient.
3. **Separate measure from marketing** — no identity survey on identity celebration screens.
4. **Competence ≠ identity** — graph and efficacy gains without interest/recognition are incomplete.
5. **Recognition must be earned and multi-source-capable** — ban empty AI "math person" praise as the recognition product.
6. **Report factors, not a trademarked composite** — parent copy may say "students see themselves differently *and* take harder problems" only when both move.
7. **Guard demand/SDR** — preregister; consider timing, neutral framing, later forced-choice if needed.
8. **Longitudinal ownership** — single-session Δ is SPECULATION; 4–8 week windows are the minimum serious bar (feeds Part LXII).

LXI.10 Commercial implications (copy, product, positioning)

Marketing language that survives: "We track whether students *and* the people around them start treating them as someone who does hard math — and whether their practice behavior matches that story."

Marketing language that dies: "+32% Math Identity Score™"; "Become a math person in one session"; "Our AI recognizes you as a genius."

Positioning vs Khan / Duo / Brilliant / GPT tutors: Competitors can show accuracy, streaks, or fluency. MindCraft's differentiated *claim* is identity — which **raises the measurement bar**, not lowers it. If we cannot measure better than a streak app's mood survey, we do not get to use identity as the brand wedge.

Growth / instrumentation: Add IDM battery to research panel / experiment flags; do not ship Identity Meter on the default student HUD.

Parent WTP link (LIX): Identity packaging in conjoint is legitimate *only* if IDM-5-style evidence exists; otherwise sell FEI session proof and transfer, not identity poetry.

LXI.11 Confidence table

Claim	Label	Confidence
Competence/performance beliefs are insufficient alone for math identity; interest + recognition matter	FACT (SEM)	High
Validated 5–12 multi-item instrument exists (Cribbs & Utley 2023)	FACT	High
HSLS two-item forms are widely used but thin as sole experiment endpoints	FACT / HYPOTHESIS	High / Medium–High
Immediate post-mission identity Likert is demand-contaminated	HYPOTHESIS	Medium–High
AI praise alone produces false recognition subscale lift	HYPOTHESIS	Medium
Narrative probes add validity beyond Likert	HYPOTHESIS (theory-backed)	Medium
Proprietary Identity Score™ improves learning or WTP	SPECULATION / refuse	Low (against)
Single-session identity transformation is measurable and marketable	SPECULATION / refuse	Low (against)

LXI.12 What this chapter kills

1. **Kill:** Math Identity Score™ / Belief Score™-style composites as North Star.
2. **Kill:** Two-item “math person” pre/post as causal FEI proof.
3. **Kill:** Competence or confidence survey ≡ identity.
4. **Kill:** Identity survey glued to identity marketing splash screens.
5. **Kill:** Empty AI “you’re a math person” as recognition product.
6. **Kill:** Single-session identity transformation claims.
7. **Wound:** Importing undergraduate-only instruments without 5–12 validity evidence.
8. **Survive:** SAFE-IDMEASURE; Cribbs-factor triangulation; IDM-1…5 + QUAL; behavioral co-primary.

Doctrine until data: MindCraft may *aim* at identity — but may **claim** identity movement only when validated interest/recognition/competence–performance subscales move **with** FEI behavior (and preferably narrative gap closure), under demand-aware administration.

Part LXII — Longitudinal Identity Change Models

Chapter status: Living evidence + horizon brief — Researcher tick 2026-08-02

Primary question: What does *honest* math-identity change look like at **8 / 26 / 52 weeks** — and which commercial claims survive fade-out, stability, and dimensional-comparison evidence?

Owners: Learning Science · Product analytics · Growth / retention · Parent messaging · Red Team

Commercial job: Ship a **SAFE-LONGID** doctrine: staged claim horizons so MindCraft can sell *direction and persistence*, not “math person in eight weeks” theater, and so subscription / package design matches how identity actually moves.

LXII.1 Why this chapter exists

Part LXI (SAFE-IDMEASURE) fixed *how* to measure identity. This chapter fixes *when* a movement counts as change vs noise, demand, or temporary lift. Founders and growth teams instinctively want a short horizon (“show identity by week 8 for renewals”). Developmental evidence is slower, stickier, and often **differentiating away from math** through high school — not smoothly ascending toward “math person.”

FOUNDER BELIEF under audit: Identity transformation is the product thesis — but durable identity change is a **multi-horizon** process. Selling L3 identity claims on L1 session windows (Part XXXIV claim ladder) is already banned; selling them on an **8-week Likert bump without persistence** is the next failure mode.

Claims we refuse as doctrine:

1. Eight-week subscale lift alone ≡ transformation (marketable or scientific).
2. Linear “always becoming more of a math person” growth curves for Maya.
3. One-semester tutoring ≡ occupational / STEM-pipeline identity.
4. End-of-program effects that ignore fade-out after support ends.

5. Stability evidence as proof that identity *cannot* change (fatalism).
6. Multi-year school-department reforms as templates for a 45-minute app session.

LXII.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Horizon	Follow-up window for an estimand	8w / 26w / 52w experiment clocks	Selling 52w outcomes on 8w data
Stability	Rank-order / mean identity often sticky	Baseline IDM battery	"Nothing moves → kill product" *or* "must move weekly"
Differentiation	Math vs verbal self-concepts diverge over grades	Challenge accept in *math* not generic hustle	Generic confidence copy
Persistence vs fade-out	Treatment effects that hold vs shrink after end	Retention after plan pause / summer	Celebrate end-of-program only
Process signal	Early FEI / interest / approach markers	`challenge_accept`, destaked retry	Process ≡ identity
SAFE-LONGID	Surviving longitudinal claim doctrine	See LXII.9	Forever-identity packaging

Operational definition (HYPOTHESIS): MindCraft may claim *identity movement at horizon H* only when SAFE-IDMEASURE triangulation holds **at H**, and for H ≥ 26 weeks also shows **non-fade** relative to an earlier post-test (or preregistered maintenance criterion). Eight weeks may support *process / direction* claims, not L3 identity brand claims.

LXII.3 Developmental facts (not slogans)

FACT (dimensional comparison meta-analysis): Wan, Lauermann, Bailey & Eccles (2021, *Psychological Bulletin*, 147(9), 867–889, doi:10.1037/bul0000340) — 142 samples, ~211k participants, 426 effect sizes. Correlations between math- and language-related motivational beliefs *decline* across grades (from roughly r≈0.32 in Grades 1–4 toward near-zero / slight negative in Grades 9–12). Students increasingly treat domains as **either/or**.

Commercial implication: High-school Maya is in the window where "I'm a words person, not a math person" consolidates. Product copy that treats identity as a generic confidence dial fights the developmental grain.

FACT (person-type trajectories across K–12): Wan, Tian, Guo & Lauermann (2026, *Child Development*, doi:10.1093/chidev/aacag069) — four U.S. longitudinal samples (Ns ≈ 971–5,681; ages ~6–18). Annual classification into math person / language person / both-high / neither-high. **Both-high shrinks**; language-person increases; **math-person rises through middle school then plateaus in high school**; girls' shift toward language is steeper. In samples with adult follow-up, high-school math-person status associates with more math-intensive occupations.

Product kill: Marketing that implies continuous year-on-year math-identity ascent through HS. Survive: **sustain and deepen** math identity in the HS plateau window — especially for students at risk of language-only sorting.

FACT (expectancy–value over years, not weeks): Simpkins, Davis-Kean & Eccles (2006, *Developmental Psychology*, 42(1), 70–83, doi:10.1037/0012-1649.42.1.70) — N=227; 5th-grade out-of-school math/science activity → 6th/10th-grade expectancies–values → high-school course taking beyond grades. Belief→choice chains run on **multi-year** clocks.

FACT (related classroom→belief→choice chain): Simpkins, Davis-Kean & Eccles / follow-on EVT longitudinal work (e.g., classroom experiences in middle school predicting later courses and aspirations; see also Wang lineage cited in LXI via Miller & Wang 2019) — identity-adjacent motivational beliefs are **path-dependent**, not session-local.

FACT (attainment value / "science person" as persistence predictor): Andersen & Ward (2014, *Science Education*, 98(2), 216–242, doi:10.1002/sce.21092) — HSLS:09 high-ability ninth-graders; science attainment value ("science person"-type identification) predicts STEM persistence plans across Black, Hispanic, and White subgroups (with utility operating differently by group). Snapshot is grade 9, but the *outcome* is planned persistence — a longitudinal *intent*.

Wound / transfer caution: HSLS items are thin (LXI); use as directional evidence that **identity-like attainment value predicts plans**, not as MindCraft's instrument.

LXII.4 Stability is real — and so is change under the right ecology

FACT (college sample stability): Cribbs, Tassell, Hazari, Sadler & Sonnert (2022, *RIPEM*, 12(2), 68–91, doi:10.37001/ripem.v12i2.2923) — N=131 undergraduates; Wilcoxon signed-rank on global math-identity self-perceptions over ~three college years → **non-significant change** (identity reported as stable). Qualitative themes: formative experiences often involve *performing* and *teaching/sharing*; many believe one *can* become a math person while still harboring innate-ability beliefs.

Interpretation for MindCraft (HYPOTHESIS): Stability under business-as-usual schooling is the null. A tutoring product that expects weekly identity Likert jumps is fighting a sticky construct. Change requires **repeated recognition + interest + competence evidence** (Cribbs SEM 2015; LXI), not one delightful onboarding.

FACT (multi-year equitable departments): Boaler & Staples (2008, *Teachers College Record*, 110(3), 608–645, doi:10.1177/016146810811000302) — Railside longitudinal mixed-methods (~700 students across three CA high schools). Heterogeneous, reform-oriented department associated with larger achievement gains and narrowing of ethnic gaps by roughly the **end of year two** on several measures; students also reported more enjoyment / progression. Horn (2008, *Mathematical Thinking and Learning*, 10(3), 201–239, doi:10.1080/10986060802216177) — turnaround student identities emerge across **departmental worlds over years**, not a single classroom moment.

Kill: Citing Railside/Horn as proof that an edtech session architecture produces identity in weeks.

Survive: Identity ecology is **relational and sustained**; MindCraft's analogue is tutor+parent recognition + FEI practice cadence over months — not a splash screen.

FACT (short belief interventions ≠ identity doctrine): Boaler, Dieckmann, Pérez-Núñez, Sun & Williams (2018, *Frontiers in Education*, 3:26, doi:10.3389/feduc.2018.00026) — teacher-randomized MOOC (~six ~15-min sessions, Oct–Dec) shifted math *beliefs*/engagement and standardized scores vs control on a ~semester research clock. Useful as proof that **beliefs can move in months with math-specific messages** — but Constitution already bans empty mindset posters; do **not** rebrand MOOC belief lifts as Cribbs-factor identity or FEI.

LXII.5 Fade-out: why 8-week wins die without maintenance

FACT (fade-out is normal): Bailey, Duncan, Cunha, Foorman & Yeager (2020, *Psychological Science in the Public Interest*, 21(2), 55–110, doi:10.1177/1529100620915848) — fade-out of educational intervention effects is **widespread**, often coexists with some persistence, and is substantive (not only measurement artifact). Persistence depends on skill type, institutional constraints, and **complementarities** with later environments.

FACT (earlier geometric-decline pattern): Bailey, Duncan, Odgers & Yu (2017, *Journal of Research on Educational Effectiveness*, 10(1), 7–39, doi:10.1080/19345747.2016.1232459) — post-treatment impacts commonly shrink within 1–2 years; early skill gains are poor guarantees of long-run advantage without supporting environments.

HYPOTHESIS (MindCraft-specific): An 8-week FEI + interest lift that is followed by **disengagement, exam-only cramming, or AI-crutch Solver use** will fade on identity factors by 26–52 weeks — even if week-8 IDM looks green.

Product implication: Retention design is not a growth vanity metric; it is **identity infrastructure**. Pause/summer plans need light maintenance cues (SAFE-HABIT subordinate to FEI), not streak guilt.

LXII.6 Horizon model for MindCraft (8 / 26 / 52)

Horizon	What can honestly move	Allowed commercial speech	Forbidden speech
~8 weeks	Process: `challenge_accept`, destaked retry, soft-wrong return; possible early interest/recognition subscale *direction* under SAFE-IDMEASURE	“Students start approaching harder problems and sticking after misses”	“Became a math person”; “identity transformed”; Identity Score™
~26 weeks	Triangulated factor movement + FEI co-primary; narrative actual→designated shift; first fade check vs week 8	“Over a semester, self-view *and* challenge behavior moved together” (only if true)	Guaranteed ACT points; occupational STEM destiny
~52 weeks	Persistence of week-26 gains; course/intent signals where observable; maintenance under reduced support	Annual renewals tied to *sustained* approach + transfer, not mood	“Forever math person after one year with us” without follow-up

FOUNDER BELIEF (constrained): Package tiers and parent CBC (Part LIX) should map to horizons: diagnostic + 8w process proof; semester identity-direction report; annual persistence review — never sell the annual claim with only an 8w dashboard.

SPECULATION (explicit): Exact week cutoffs are product conventions informed by school semesters and subscription cycles, not natural laws. Preregister windows; do not retrofit.

LXII.7 Analytic models (what "longitudinal" means in practice)

MindCraft does not need a 10-year panel to be rigorous. Minimum serious designs:

1. **Repeated measures / growth:** IDM factors at 0/8/26/52; random slopes; report **individual heterogeneity** (not only means).
2. **Latent transition / person-types (inspired by Wan 2026):** math / language / both / neither-style states — watch **both-high erosion** and math-person **plateau** risk.
3. **Joint process–identity models:** FEI events as time-varying covariates of interest/recognition (tests FEI pathway).
4. **Fade contrast:** end-of-active-support vs +8–12 weeks post-pause.
5. **Demand control:** never collect IDM on identity-celebration UI (LXI).

Pre-reg (XXXIV): LONG-* experiments identify **horizon-valid estimands**, not "MindCraft creates math people by Δ=X."

LXII.8 Experiments (LONG family)

ID	Contrast	Primary endpoints	Kill / survive rule
LONG-1	8w IDM+FEI vs waitlist / active control	Interest/recognition Δ; `challenge_accept`	Factor↑ FEI flat → kill identity claim; FEI↑ factor flat → process-only speech
LONG-2	26w continuation vs 8w-then-pause	Factors at 26; fade from 8	Pause erases 8w lift → kill "8w locks identity"
LONG-3	Recognition-rich tutor/parent notes cadence vs AI praise cadence (extends IDM-4)	Recognition subscale; transfer; 26w hold	AI-only recognition fades faster → confirms false recognition
LONG-4	Maintenance micro-dose (1–2 FEI sessions/mo) vs full stop after 26w	52w factors + FEI	No maintenance → fade; informs annual package

LONG-5	Person-type transition coding (Wan-style) exploratory	Transition probs; gender split	Both-high collapse without math sustain → HS copy priority
LONG-QUAL	10 Maya + 10 parent interviews at 0/8/26: story change vs score obsession	Narrative codebook	Parent wants 8w identity certificate → WTP honesty constraint

Ties: IDM-, *EVT*-, FEI/SAFE-CRAFT, WTP package horizons, REPAIR trust persistence, HIST equity of who gets sustained recognition.

LXII.9 SAFE-LONGID stack (surviving doctrine)

1. **Stage claims by horizon** — 8w = process/direction; 26w = triangulated identity movement; 52w = persistence/maintenance.
2. **Expect stickiness** — null is stable identity under weak ecology (Cribbs 2022); design for repeated FEI + earned recognition.
3. **Respect differentiation** — HS students sort math vs language; sustain math identity, don't sell generic grit.
4. **Plan for fade-out** — end-of-program green lights are incomplete; measure post-pause.
5. **Do not launder mindset MOOCs or multi-year school reforms as app-session identity proof.**
6. **Align packages to clocks** — CBC attributes and renewals match measurable horizons (LIX).
7. **Heterogeneity first** — mean lifts can hide non-responders; report who moves.
8. **Keep SAFE-IDMEASURE** — same instruments, longer clocks; no new Identity Score™.

LXII.10 Commercial implications (copy, product, positioning, growth)

Marketing language that survives: "In the first weeks we look for *approach under safety*. Over a semester we look for whether students' *self-story and challenge behavior* move together. Over a year we look for whether that holds when life gets busy."

Marketing language that dies: "Math person in 8 weeks"; "Guaranteed identity transformation this semester"; "Join and permanently become STEM"; linear forever-up charts on landing pages.

Positioning vs Khan / Duo / Brilliant / GPT tutors: Competitors optimize session accuracy, streaks, or fluency — short clocks. MindCraft's wedge is **durable self-as-math-doer**. That only works if we **own the long clock** in measurement and packaging. If we market identity on Duo's time horizon, we lose the wedge and inherit their vanity metrics.

Growth: Instrument cohort retention *as* identity maintenance; annual plans should include LONG-4-style maintenance, not only denser content. Churn after week 6 is not merely CAC failure — it is **hypothesis failure** for L3 claims.

Vision / future: The company roadmap should treat 26–52 week panels as first-class research infrastructure (research panel flag), not a nice-to-have after feature shipping.

LXII.11 Confidence table

Claim	Label	Confidence
Math/verbal motivational beliefs differentiate across school years	FACT (meta)	High
Math-person rates plateau in HS; both-high shrinks; language-person rises	FACT (multi-sample longitudinal)	High
Global math identity can be stable over years under status-quo college experience	FACT (small N)	Medium–High
Multi-year equitable school departments can shift achievement/identity ecology	FACT (case / mixed methods)	Medium–High (external validity to edtech low)
Educational intervention effects commonly fade without supporting environments	FACT (reviews)	High
8w triangulated lift without 26w hold is insufficient for L3 identity marketing	HYPOTHESIS / doctrine	Medium–High
Staged 8/26/52 product clocks are the right commercial mapping	FOUNDER BELIEF / HYPOTHESIS	Medium
MindCraft will produce 52w identity persistence with maintenance micro-dosing	SPECULATION	Low–Medium (test via LONG-4)

LXII.12 What this chapter kills

1. **Kill:** “Math person in 8 weeks” / single-horizon identity transformation marketing.
2. **Kill:** Treating end-of-program identity Likert as durable without fade checks.
3. **Kill:** Linear always-rising math-identity curves for high school.
4. **Kill:** Equating short mindset/belief MOOC effects with Cribbs-factor identity + FEI.
5. **Kill:** Citing Railside/Horn multi-year worlds as proof of session-scale app identity.
6. **Kill:** Annual “forever STEM identity” packaging without 52w persistence evidence.
7. **Wound:** Using HSLS two-item snapshots as longitudinal product proof (re-anchor LXI).
8. **Survive:** SAFE-LONGID staged horizons; differentiation-aware HS positioning; fade-aware retention; LONG-1…5 + QUAL.

Doctrine until data: Identity is the North Star *target*, not an 8-week badge. Ship process truth early; claim identity movement only on triangulated **semester+** clocks; claim persistence only when fade-out is measured and survived.

Part LXIII — ACT / Exam Pressure Special Case

Chapter status: Living evidence + affect-separation brief — Researcher tick 2026-08-02

Primary question: How is **high-stakes exam affect** (ACT/SAT/state tests) different from **learning-session affect** — and what product, copy, and metric rules follow so MindCraft does not confuse destaked FEI practice with exam readiness (or flood Maya with day-one timed sims)?

Owners: Learning Science · Practice / diagnostic UX · Parent messaging · Growth · Red Team

Commercial job: Ship a **SAFE-EXAM** doctrine: separate *learn* clocks from *perform under pressure* clocks; sell honest prep architecture, not guaranteed points or calm-app theater.

LXIII.1 Why this chapter exists

MindCraft already owns math anxiety (XXIV), desirable difficulties under threat (XLI / SAFE-DD), sleep/stress timing (XLVII), and graded approach ladders (LVIII / SAFE-EXPOSE). Parents still buy **exam outcomes**. Tutors still schedule **timed mixed sets**. Product still risks one of two failures:

1. **Flooding:** treat every practice minute as a mini-ACT (performance climate; WM choke; churn).
2. **Under-transfer:** optimize cozy FEI accuracy and market "exam-ready" without ever dosing pressure.

FOUNDER BELIEF under audit: Identity transformation happens in *learning affect* (safety → recoverable miss → earned win). College-gatekeeping happens in *exam affect* (stakes, clock, evaluation threat). Conflating them either destroys learning or lies about readiness.

Claims we refuse as doctrine:

1. Day-one full timed ACT / mixed flood labeled "exposure therapy" or "rigor."
2. Low-stakes blocked accuracy ≡ ACT-ready.
3. Guaranteed ACT point lifts / 2σ / Bloom in any package (already SAFE-WTP / standing ban).
4. Calm Score™ / "zero anxiety before test day" as North Star (SAFE-EXPOSE).
5. Expressive-writing or one pre-test ritual as substitute for content + approach ladders.
6. Stereotype-threat lab effects as unlimited marketing for "our UI closes the ACT gap."
7. Cortisol / stress-bias evidence as "tests are invalid so skip prep."

LXIII.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused

Learning affect	Destaked / mastery-climate emotion during acquisition	FEI session; soft-wrong; hide-correctness diagnostic	Treat as identical to test day
Exam / evaluation affect	Worry + physiological arousal under evaluative stakes	Timed mixed; mock ACT week; parent-visible score	Flood every session
Math anxiety (MA)	Domain-specific threat of math	Concept avoidance; Solver over-reliance	Collapse into “nerves”
Test anxiety (TA)	Evaluation anxiety across subjects	Clock panic; blanking; rank fear	Only calming apps
Pressure dose	Intentional stakes/time/mix intensity	Laddered timed sets after equip	Hardness-as-virtue
Readiness signal	Delayed mixed under mild→moderate pressure	`transfer_pass` / timed mix after SAFE-DD	Blocked green checks
SAFE-EXAM	Surviving exam-pressure doctrine	See LXIII.9	Score guarantees; flood/cozy extremes

Operational definition (HYPOTHESIS): A MindCraft path is *exam-honest* when (a) acquisition stays under SAFE-DD / SAFE-EXPOSE destake rules, (b) pressure is **dosed after equip**, (c) readiness claims require **delayed mixed** performance under declared stakes (not blocked accuracy), (d) parent copy separates “building the skill” from “proving under the clock,” (e) no package promises fixed ACT points.

LXIII.3 Exam pressure is real — and not the same as math anxiety

FACT (test anxiety meta): Hembree (1988, *Review of Educational Research*, 58(1), 47–77, doi:10.3102/00346543058001047) — 562 studies: test anxiety relates to poorer performance; inversely to self-esteem; treatments that reduce TA often accompany better test/GPA outcomes. Classic baseline: evaluation threat is a performance construct, not “motivation fluff.”

FACT (math anxiety meta): Hembree (1990, *JRME*, 21(1), 33–46, doi:10.5951/jresematheduc.21.1.0033) — 151 studies: MA tied to achievement, attitudes, and **avoidance**. Prep products that only soothe without approach leave the avoidance loop intact (LVIII).

FACT (MA vs TA on math performance): Caviola, Toffalini, Giofrè, Mercader Ruiz, Sz■cs & Mammarella (2022, *Educational Psychology Review*, 34, 363–399, doi:10.1007/s10648-021-09618-5) — 177 studies / ~906k participants. MA–math achievement overall $r \approx -0.30$; TA–math achievement $r \approx -0.23$. Both matter; MA is the stronger domain hitch. WM mediation of anxiety→math is present but **weak**.

Commercial implication: Parent copy that says only “we reduce test nerves” undersells the math-specific threat. Copy that says only “we fix concepts” undersells evaluation affect on test week. Product needs **both rails**: concept FEI *and* laddered exam affect — sequenced, not mashed.

FACT (WM under math stress): Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224); Beilock (2008, *Current Directions in Psychological Science*, 17(5), 339–343, doi:10.1111/j.1467-8721.2008.00602.x); Beilock & Carr (2005, *Psychological Science*, 16(2), 101–105) — pressure/worry consumes WM; high-WM performers can choke when strategies rely on WM. Exam pressure is a **cognitive-resource** problem, not only a vibes problem.

LXIII.4 High-stakes weeks change bodies — scores are not pure learning meters

FACT (field cortisol × high-stakes tests): Heissel, Adam, Doleac, Figlio & Meer (2021, *Education Finance and Policy*, 16(2), 183–208, doi:10.1162/edfp_a_00306; NBER w25305) — low-income New Orleans students grades 3–8; saliva cortisol in testing weeks vs baseline weeks. Homeroom cortisol ~18% higher before high-stakes tests (boys ~35%). Large cortisol *increases or decreases* associated with ~0.40 SD lower-than-expected scores vs other academic performance — suggestive **stress bias** in score interpretation (correlational on the score link; still product-relevant).

Wound: Sample is K–8 high-stakes *school* tests, not ACT sitters. Mechanism transfers as caution; coefficients do not become "ACT −80 points for everyone."

Product implication: Do not treat a student's worst timed mock as pure "doesn't know content." Instrument **condition** (sleep, stakes frame, prior flooding) when diagnosing. Do not market "tests are unfair so skip practice" — parents buy access; MindCraft sells **honest preparation under known bias**, not nihilism.

LXIII.5 Pressure during *learning* can kill the testing effect

FACT (high-stakes quizzes blunt retrieval benefits): Hinze & Rapp (2014, *Applied Cognitive Psychology*, 28(4), 597–606, doi:10.1002/acp.3032) — high-stakes vs low-stakes quizzes: initial quiz accuracy similar, but **final retention better after low-stakes**; only low-stakes beat reread controls. Performance pressure during practice can disrupt encoding benefits of retrieval (retrieval-disruption account).

HYPOTHESIS (MindCraft): Running every practice card as a scored, parent-visible, timed ACT item will look "serious" and **underproduce learning** relative to destaked retrieval + later pressure doses.

Tie to SAFE-DD / SAFE-TIMING: Desirable difficulty ≠ maximum stakes. Pressure is a **transfer/context** variable after equip, not the acquisition default.

LXIII.6 Brief worry-offload helps some — it is not the product

FACT (expressive writing before exams): Ramirez & Beilock (2011, *Science*, 331(6014), 211–213, doi:10.1126/science.1199427) — brief pre-exam expressive writing about worries improved exam performance, especially for habitually test-anxious students (lab + classroom field experiments).

Kill: Rebranding MindCraft as a journaling app that "boosts ACT." Writing is a **pre-performance micro-tool**, optional in SAFE-EXPOSE / SAFE-AAR brief — never a substitute for ontology-backed practice, FEI, or pressure ladders.

Survive: Optional 2–3 minute worry dump before *timed* modules for high-TA segment; measure whether timed accuracy rises without reducing later `solo_transfer_pass`.

LXIII.7 Stereotype threat: real lab phenomenon, fragile marketing claim

FACT (women's math under threat): Spencer, Steele & Quinn (1999, *Journal of Experimental Social Psychology*, 35(1), 4–28, doi:10.1006/jesp.1998.1373) — describing a difficult math test as gender-diagnostic can depress women's performance; reducing threat can eliminate the gap in those designs.

FACT / wound (overclaim): Stoet & Geary (2012, *Review of General Psychology*, doi:10.1037/a0026617) — many purported replications are weak or confounded; enthusiasm that stereotype threat *explains* the gender math gap outruns the clean evidence.

Commercial rule: Design against stereotype-cueing copy and tokenism (XXXVI). Do **not** sell "our story world removes stereotype threat → +X ACT." Equity is a design constraint and HIST target, not a score guarantee.

LXIII.8 Two clocks: learn vs prove

Phase	Affect goal	Stakes / time	Allowed metrics	Forbidden claim
Acquire	Learning affect; FEI	Destaked; untimed or soft clock	`challenge_accept`, destaked retry, soft-wrong return	"Exam ready"
Consolidate	Mild evaluative spice	Short timed slices; criterion feedback	Near-mix accuracy; calibration	Leaderboard as NS
Prove (prep)	Exam affect dose	Declared timed mixed / mock	Delayed mixed under pressure; error taxonomy	Guaranteed composite points
Prove (real)	External gate	Official ACT/SAT	Post-hoc outcome study only	Causal brand claim without pre-reg

FOUNDER BELIEF (constrained): Gap-scan and hide-correctness diagnostic (C4 / SAFE-CALIB) stay in **Acquire** affect — they measure knowledge/confidence without turning onboarding into humiliation theater. Timed ACT sims belong **after** equip and approach ladders (SAFE-EXPOSE), not as day-one brand ritual.

SPECULATION: Optimal first full timed mixed set sits after ~2–4 weeks of concept FEI for a typical Maya — test via EXAM-1; do not hardcode without data.

LXIII.9 SAFE-EXAM stack (surviving doctrine)

1. **Separate affects** — learning affect ≠ exam affect; never optimize one metric for both.
2. **Destake to learn; dose to prove** — acquisition defaults low-stakes (Hinze & Rapp); pressure is scheduled.
3. **Ladder pressure** — time → mix → parent-visible mock; gate with SAFE-DD / SAFE-EXPOSE.
4. **Readiness ≠ blocked green** — claim exam-prep progress only with delayed mixed under declared stakes.
5. **Dual rail diagnosis** — miss under cozy ≠ miss under clock; log condition tags.
6. **Micro-tools subordinate** — expressive writing / brief AAR optional; never the hero feature.
7. **Anti-guarantee** — no ACT point promises; no 2σ; package "prep architecture," not outcomes.
8. **Anti-nihilism** — stress-bias evidence informs interpretation and equity, not "skip the exam."
9. **Stereotype-safe copy** — no threat-cueing; no threat-removal score claims.
10. **Parent speech hygiene** — "building under safety" vs "proving under the clock" in WTP messages (LIX).

LXIII.10 Experiments (EXAM family)

ID	Contrast	Primary endpoints	Kill / survive rule
EXAM-1	Early timed-flood vs destake-then-dose	4w delayed mixed; churn; `challenge_accept`	Flood ↑ early mock, ↓ delayed / ↑ churn → kill flood-default
EXAM-2	Parent-visible daily score vs private criterion	Learning retention; anxiety item; retention	Visible score ↑ TA, ↓ Hinze-style retention → kill public scoreboard default
EXAM-3	Optional expressive writing before timed module	Timed accuracy; transfer next session	Helps HTA only → segment; helps none → demote
EXAM-4	Pressure dose after SAFE-EXPOSE ladder vs same dose without ladder	Timed mix; approach; SUDS/expectancy	Ladder wins → keep; equal → simplify
EXAM-5	Readiness CTA: blocked accuracy vs delayed mixed under mild time	Parent trust; false-ready rate	Blocked CTA predicts mock fail → kill blocked-as-ready
EXAM-QUAL	10 Maya + 10 parent: "ready" language meanings	Codebook: score vs approach vs calm	Parents equate calm≡ready → copy correction

Ties: SAFE-DD, SAFE-EXPOSE, SAFE-CALIB, SAFE-TRANSFER, SAFE-WTP, SAFE-TIMING, LONG horizons for post-exam identity (don't claim identity from one ACT sit).

LXIII.11 Commercial implications (copy, product, positioning, growth)

Marketing language that survives: “We build skill under conditions where working memory can work — then we practice *proving* under the clock, on purpose.” / “Low-stakes struggle first; timed mixed when the ladder says so.” / “We won’t sell you a point guarantee; we will show you readiness signals that include pressure.”

Marketing language that dies: “ACT +4 guaranteed”; “Day-one real exam conditions”; “Become calm and the score follows”; “Our AI removes test anxiety”; “Practice accuracy means test-day ready.”

Positioning vs Khan / Duo / Brilliant / GPT tutors: Khan/Duo optimize volume and streaks under mostly low evaluative threat; ChatGPT optimizes fluent help (crutch risk); Brilliant optimizes delight. MindCraft’s wedge is **recoverable struggle** → **competence evidence** → **solo transfer**, then an explicit **pressure ladder** competitors either skip (too cozy) or flood (too harsh). Own the sequencing story.

Growth: Instrument `timed_mix_attempted`, `pressure_dose_level`, `mock_vs_expected`, and false-ready (blocked-green then mock-fail). Do not grow on raw timed-item counts. Exam-season campaigns sell **architecture** (ladder + diagnosis), not miracle composites.

Vision: Future ACT track is a first-class *affect curriculum* parallel to the concept ontology — pressure as a designed dimension, not an accidental byproduct of “serious mode.”

LXIII.12 Confidence table

Claim	Label	Confidence
TA and MA both negatively relate to math achievement (MA stronger in large meta)	FACT	High
Evaluative pressure can consume WM and induce choke	FACT	High
High-stakes weeks can shift cortisol; large swings associate with worse-than-expected scores	FACT (field; score link correlational)	Medium–High
High-stakes practice quizzes can erase retrieval-learning benefits despite equal quiz scores	FACT (lab)	High
Expressive writing can help some high-TA students on exams	FACT	Medium–High
Stereotype threat can depress lab math performance; gap-explanation overclaim is unsafe	FACT + wound	Medium
Destake-then-dose beats flood-default for MindCraft retention + delayed mixed	HYPOTHESIS	Medium–High

SAFE-EXAM sequencing is the right commercial mapping for ACT parents	FOUNDER BELIEF / HYPOTHESIS	Medium
Exact week of first full mock is universal	SPECULATION	Low (EXAM-1)

LXIII.13 What this chapter kills

1. **Kill:** Day-one timed ACT flood as default pedagogy or "exposure."
2. **Kill:** Blocked / cozy accuracy marketed as exam-ready.
3. **Kill:** Guaranteed ACT points / 2σ / Bloom in exam-season copy.
4. **Kill:** Calm-to-zero / Calm Score™ as exam North Star.
5. **Kill:** Journaling / expressive writing as the product.
6. **Kill:** "We eliminate stereotype threat → score lift" marketing.
7. **Kill:** Using stress-bias evidence to sell anti-prep nihilism.
8. **Wound:** Treating K–8 cortisol coefficients as ACT effect sizes.
9. **Survive:** Dual-rail MA+TA framing; destake-to-learn / dose-to-prove; EXAM-1…5; parent speech split; pressure as designed curriculum dimension.

Doctrine until data: Exam pressure is a **special case of affect**, not a synonym for math anxiety and not a synonym for desirable difficulty. Ship learning under safety; schedule proving under the clock; sell the architecture, never the composite.

Part LXIV — Multilingual / ELL Math Identity

Chapter status: Living evidence + language–math separation brief — Researcher tick 2026-08-02

Primary question: How do **language load** and **math load** come apart for English learners / multilingual students — and what product, copy, and metric rules stop MindCraft from misdiagnosing identity, dumbing down math, or selling translation theater?

Owners: Learning Science · Question bank / formats · Story worlds (equity) · Diagnostic UX · Parent messaging · Red Team

Commercial job: Ship a **SAFE-ELL** doctrine: measure and scaffold linguistic access without collapsing "math person" into English fluency; sell equitable *access to mathematical practice*, not simplified forever tracks or "we translate so identity follows."

LXIV.1 Why this chapter exists

MindCraft already owns equity of story worlds (XXXVI), identity measurement (LXI), exam pressure (LXIII), and format-tagged practice. A large and growing share of U.S. secondary students are classified as English learners (ELs) or are multilingual without the label. Product still risks three failures:

1. **Confound:** treat wrong answers on dense English word problems as proof of weak *math* identity / mastery.
2. **Deficit curriculum:** strip context and discourse until only procedures remain — then call it "ELL support."
3. **Access theater:** auto-translate, glossaries, or bilingual badges marketed as identity transformation.

FOUNDER BELIEF under audit: Mathematical identity is participation in *mathematical practices* (reasoning, representing, arguing) — not conversational English polish. Language can gate that participation without defining the student's mathematical self.

Claims we refuse as doctrine:

1. "Math is language-free, so EL gaps are ability gaps."
2. Forever-simplified / procedure-only tracks as equity.
3. Vocabulary flashcards or translation alone as math-identity product.
4. Auto-translate / dual-language UI as proof MindCraft "closes the EL gap."
5. Homogeneous "ELL" persona in copy (one level, one home language, one need).
6. Using EL status as a soft excuse to skip challenge, FEI, or pressure ladders.
7. Marketing "story worlds fix belonging for bilingual kids" without HIST / ELL experiments.

LXIV.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Language load	Linguistic complexity demanded to parse/produce the task	Dense word_problem stems; idioms; passive voice	Call it "math hardness"
Math load	Conceptual / procedural demand of the mathematics	Ingredient DAG; format; mix	Ignore language confound
BICS vs CALP	Conversational fluency vs academic language	Chatty UI vs problem register	Assume chatty English $\Rightarrow$ ready for ACT stems
Language as resource	Home language / multimodal resources support math	Optional L1 notes; diagrams; gestures analogue in UI	Token "hola" branding
Language as deficiency	Treat EL status as lack	Remedial forever track	Identity colonization as "catch-up English"
Differential item functioning (DIF)	Items unfairly harder for ELs at equal math proficiency	Wordy ACT-style items	Bank blindness

SAFE-ELL	Surviving multilingual doctrine	See LXIV.9	Translate-theater; deficit tracks

Operational definition (HYPOTHESIS): A MindCraft path is *language-honest* when (a) diagnostics and outcomes can separate language-parse fails from math-concept fails, (b) scaffolds reduce *unnecessary* linguistic complexity without removing mathematical discourse opportunities, (c) story wrap does not add stereotype-cueing or opaque cultural idioms as load, (d) identity claims never equate English fluency with being a "math person," (e) challenge and transfer standards stay intact (SAFE-TRANSFER / SAFE-EXAM).

LXIV.3 The EL–math score gap is real — and partly linguistic access

FACT (NAEP EL gaps): NCES Condition of Education — Mathematics Performance (using 2022 NAEP): grade 4 EL average 216 vs non-EL 239 (**23-point** gap); grade 8 EL 241 vs non-EL 277 (**36-point** gap). Gaps are persistent across assessment years since disaggregation began (1996 for grades 4/8). Grade 12 (2019): EL 111 vs non-EL 152 (**41-point** gap). Source: NCES COE indicator CNC / Digest table 222.12.

Wound: NAEP gaps confound language classification, opportunity to learn, SES, and math proficiency. They are **market and equity facts**, not proof that ELs "aren't math people," and not MindCraft effect sizes.

Commercial implication: Parent and school copy can name the access problem honestly. Do not sell "we close the 36-point NAEP gap." Sell architecture that reduces *construct-irrelevant* language load while keeping math demand.

LXIV.4 Linguistic complexity changes who the item measures

FACT (linguistic modification helps): Abedi & Lord (2001, *Applied Measurement in Education*, 14(3), 219–234, doi:10.1207/S15324818AME1403_2) — 1,174 grade-8 students; NAEP math items vs parallel **linguistically modified** items. ELs scored lower than proficient English speakers overall; modification produced small but significant gains, with larger benefits for ELs, low-SES, and lower-track math students. Preference interviews favored revised wording.

FACT (DIF on word problems): Martiniello (2008, *Harvard Educational Review*, 78(2), 333–368, doi:10.17763/haer.78.2.70783570r1111t32) — linguistic complexity of MCAS grade-4 math word problems associated with DIF against ELs relative to non-ELs of **comparable math proficiency**; think-alouds show reading-comprehension traps (syntax, unfamiliar vocabulary, dense noun phrases) that are not the math construct.

HYPOTHESIS (MindCraft): A wrong answer on a high-language-load `word_problem` after a correct `symbolic_expression` / `diagram` item on the same ingredient is more often a **language-gate** than a concept hole — and treating it as identity evidence ("I'm bad at math") is product malpractice.

Product implication: Prefer format triangulation (already in bank formats) before mastery or identity updates. Flag high linguistic-complexity stems in authoring. Offer a **language-simplified parallel** for practice (same math, lower incidental English) without making it the only track.

LXIV.5 Conversation fluency ≠ academic math language

FACT (BICS/CALP distinction): Cummins (1979, *Working Papers on Bilingualism*, No. 19, 121–129) introduced **BICS** (basic interpersonal communicative skills) vs **CALP** (cognitive/academic language proficiency) to warn that conversational L2 fluency is not academic register command. Restated in Cummins (2008, in Hornberger, *Encyclopedia of Language and Education*, doi:10.1007/978-0-387-30424-3_36).

Commercial kill: Onboarding that hears fluent chat English and assumes ACT word-problem readiness. Conversely: withholding hard math because a student is still building CALP. Support **register access** (glosses of academic terms, multimodal stems) while keeping cognitive demand.

LXIV.6 Sociocultural turn: participation and resources, not vocab lists

FACT (situated bilingual math): Moschkovich (2002, *Mathematical Thinking and Learning*, 4(2–3), 189–212, doi:10.1207/s15327833mtl04023_5) — moves beyond vocabulary-acquisition views toward participation in mathematical practices; competence includes multimodal and bilingual resources, not only formal English lexicon.

FACT (equitable practice principles): Moschkovich (2013, *Journal of Urban Mathematics Education*, 6(1), doi:10.21423/jume-v6i1a204) — principles/guidelines: treat language as **resource not deficiency**; emphasize academic achievement (not only English learning); support discourse about math ideas (not low-level language drills as the main diet); draw on multiple classroom resources (objects, drawings, graphs, gestures) and home languages.

FACT (field synthesis): de Araujo, Roberts, Willey & Zahner (2018, *Review of Educational Research*, 88(6), 879–919, doi:10.3102/0034654318798093) — review of 75 empirical K–12 EL×math studies (2000–2015) through a sociocultural lens; documents persistent marginalization patterns and the need for cumulative, not only case-study, research.

FACT (task selection wound): de Araujo (2017, *Curriculum Inquiry*, doi:10.1080/03626784.2017.1368351) — secondary teachers accommodating ELLs often selected **repetitive, procedure-focused, context-stripped** tasks, driven by beliefs that ELLs as a homogeneous group cannot handle richer work.

Kill: MindCraft "ELL mode" that only serves blocked procedures and never story/discourse/transfer. That is deficit curriculum dressed as care — and it starves identity (LXI recognition + competence evidence).

Survive: Multimodal + optional L1 support + full mathematical practice (FEI, soft-wrong, transfer).

LXIV.7 Story worlds and identity: equity constraint, not EL miracle

Tie to XXXVI / SAFE-IDMEASURE: Story wrap can support belonging or add idiom load and tokenism. Bilingual characters without mathematical agency are costume. Empty AI praise in simplified English ("you are a math person") remains banned (LXI).

HYPOTHESIS: For multilingual Maya, recognition that counts is **being heard as a mathematical thinker** (even with imperfect English or code-switching notes), not celebrating English progress as math progress.

SPECULATION: Optional student L1 annotation on Solver traces may raise `challenge_accept` for mid-proficiency ELs without hurting delayed transfer — test in ELL-3; do not ship as brand without data.

LXIV.8 Two loads: diagnose separately

Signal	Likely load	Product move	Forbidden interpretation
Fails wordy stem; passes symbolic/diagram twin	Language	Offer simplified parallel; gloss; do not slash mastery hard	“Not a math person”
Fails all formats on ingredient	Math	FEI / SAFE-MISCON route	“Just a language issue — skip concept”
Fluent chat; fails academic stems	CALP gap	Register scaffolds; keep math demand	Assume ready for ACT flood
Avoids word problems only	Threat + language	SAFE-EXPOSE ladder on language-load situations	Forever avoid word problems
Parent wants “English tutoring branded as math”	Market pull	Separate message: math practice with language access	Rebrand as ESL app

FOUNDER BELIEF (constrained): Hide-correctness diagnostic (C4) must not silently treat language-gated misses as calibrated underconfidence in *math*. Prefer format-balanced probes for EL-classified or self-identified multilingual students.

LXIV.9 SAFE-ELL stack (surviving doctrine)

1. **Separate loads** — language load ≠ math load; instrument both when possible.
2. **Language as resource** — optional L1 / multimodal supports; never deficiency branding.
3. **Modify incidental English, not math** — Abedi-style linguistic simplification for practice parallels; keep discourse opportunities (Moschkovich).
4. **No deficit track** — ban procedure-only forever “ELL path” (de Araujo task-selection wound).
5. **Format triangulation before identity/mastery verdicts** — especially on `word_problem`.
6. **Heterogeneity** — never one ELL persona; ask language history / preference lightly.
7. **Anti-theater** — translation/glossary ≠ identity; ≠ NAEP-gap closure claim.
8. **Anti-excuse** — EL status does not waive FEI, transfer, or SAFE-EXAM dosing — sequence scaffolds, don’t skip.
9. **Story equity** — no idiom-heavy unexplained slang as default; no token bilingual costume (XXXVI).
10. **Parent speech hygiene** — “access to show what they know” vs “easier math forever.”

LXIV.10 Experiments (ELL family)

ID	Contrast	Primary endpoints	Kill / survive rule
ELL-1	High-language stem vs linguistically modified twin (same math)	Accuracy delta; time; self-report “stuck on words”	Large EL-only delta → keep twins; no delta → deprioritize authoring cost
ELL-2	Format-balanced diagnostic vs word-problem-heavy diagnostic	Mastery misclassify rate; identity item demand	Word-heavy ↑ false weak → require balance for multilingual
ELL-3	Optional L1 note / gloss on vs off	`challenge_accept`; delayed `solo_transfer_pass`	Helps approach, hurts transfer → redesign; helps both → ship optional
ELL-4	Procedure-only “support” track vs full multimodal practice	4w transfer; belonging/recognition items	Procedure track ↓ transfer / identity → kill deficit default
ELL-5	Auto-translate banner vs linguistic simplify + diagram	Trust; accuracy; reattempt	Translate-only wins UX but loses transfer → kill as sole solution
ELL-QUAL	10 multilingual Maya (+ parents): “math person” vs “English hard”	Codebook: load attribution	Students blame self-as-dumb for word traps → copy + UI fix

Ties: SAFE-TRANSFER, SAFE-MISCON, SAFE-IDMEASURE, SAFE-EXAM (ACT stems are high CALP), SAFE-EXPOSE (language-load as situation tier), XXXVI HIST-EQ.

LXIV.11 Commercial implications (copy, product, positioning, growth)

Marketing language that survives: “We separate *reading the problem* from *doing the math* — so language never masquerades as destiny.” / “Support for multilingual learners means access to full mathematical practice, not a forever-easy track.” / “Same math, clearer language when needed — then prove it in more than one format.”

Marketing language that dies: “Math is universal, language doesn’t matter”; “ELL mode = easier problems”; “We translate, so the gap closes”; “Fluent English means ACT-ready”; “Bilingual avatar = belonging solved.”

Positioning vs Khan / Duo / Brilliant / GPT tutors: Khan/Duo often add language demand without diagnosing it; GPT tutors can drown ELs in fluent English monologue (SE risk, XL) or over-simplify into procedures; Brilliant’s delight still assumes English UI literacy. MindCraft’s wedge: **format + linguistic-access scaffolding inside identity-safe struggle** — recoverable misses that are honestly classified.

Growth: Instrument `language_simplify_used`, `format_mismatch_flag`, `self_report_word_stuck`, and EL/multilingual opt-in (privacy-sensitive). Do not grow on “ELL engagement minutes.” School sales: lead with **fairer**

evidence of what students know, not miracle gap closure.

Vision: Question bank and story modules treat linguistic complexity as a first-class authoring dimension beside format and concept — parallel to pressure as a designed affect dimension (LXIII).

LXIV.12 Confidence table

Claim	Label	Confidence
Persistent NAEP EL vs non-EL math score gaps	FACT	High
Linguistic modification can improve math-item performance for ELs / lower-track students	FACT	High
Linguistic complexity can create DIF against ELs at comparable math proficiency	FACT	High
Conversational fluency ≠ academic language proficiency (BICS/CALP)	FACT (framework; widely used)	High
Equitable EL math instruction emphasizes resource view + discourse, not vocab-only	FACT (design principles / review)	High
Teachers often over-accommodate with procedure-stripped tasks	FACT (small qualitative)	Medium
Format triangulation reduces false “weak math” labels for multilingual users	HYPOTHESIS	Medium–High
SAFE-ELL is the right commercial mapping for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Optional L1 notes raise approach without hurting transfer	SPECULATION	Low (ELL-3)

LXIV.13 What this chapter kills

1. **Kill:** Math-is-language-free → EL gap = ability/identity verdict.
2. **Kill:** Forever procedure-only / context-stripped “ELL support” tracks.
3. **Kill:** Vocabulary drills or auto-translate as the math-identity product.
4. **Kill:** “We close the NAEP EL gap” / guaranteed equity lift marketing.
5. **Kill:** Homogeneous ELL persona and stereotype-cueing bilingual costume.

6. **Kill:** Using EL status to waive challenge, FEI, transfer, or exam dosing.
7. **Wound:** Treating NAEP gaps as pure language or pure math — they are entangled.
8. **Survive:** Load separation; linguistic modify without math dilute; format triangulation; Moschkovich resource/discourse stance; ELL-1…5; parent speech split.

Doctrine until data: Multilingual math identity work is **access to mathematical practice under honest load diagnosis** — not English tutoring in a math skin, and not easier math forever. Scaffold the language gate; keep the math climb.

Part LXV — Gender & Math Stereotypes Update

Chapter status: Living evidence + stereotype/identity brief — Researcher tick 2026-08-02

Primary question: What does *current* evidence (not 1990s posters) say about gender, math performance, stereotypes, and identity — and what copy, product, and metric rules stop MindCraft from selling innate-gap myths, threat-removal score miracles, or pink-STEM cosplay as transformation?

Owners: Learning Science · Story worlds (equity) · Identity measurement · Parent messaging · Diagnostic / exam UX · Red Team

Commercial job: Ship a **SAFE-GENDER** doctrine: treat mean performance parity and stereotype/interest gaps as separable facts; design belonging without essentialism; never sell "we fixed stereotype threat so scores rose."

LXV.1 Why this chapter exists

MindCraft already owns belonging/anxiety (XXIV), expectancy-value (XXXVII), equity of story worlds (XXXVI), identity measurement (LXI), exam pressure (LXIII), and ELL load separation (LXIV). Gender stereotypes remain a marketing landmine and a product design constraint:

1. **Ability myth:** parents and culture still narrate "boys are better at math" even when mean school performance is near parity.
2. **Threat theater:** early stereotype-threat (ST) findings get oversold as a turnkey score fix.
3. **Cosplay equity:** pink branding, "girl power" badges, or token women-in-STEM stickers sold as identity transformation.

FOUNDER BELIEF under audit: Mathematical identity for Maya is earned participation + recognition + competence evidence (LXI) — not gendered packaging. Stereotypes can gate *interest, belonging, and course choice* even when average test scores look equal.

Claims we refuse as doctrine:

1. Innate male math superiority as product premise or parent copy.
2. "Stereotype threat explains the gender STEM gap" as a settled marketing claim.
3. ST-removal / affirmation as guaranteed ACT points.
4. Girl-washed UI or "for her" tracks as the identity product.
5. Homogeneous "girls in STEM" persona (ignores race, class, ELL, interest).

6. Using gender to soften challenge, FEI, or transfer standards.
7. "We close the gender STEM gap" without measured mechanisms and fade checks.

LXV.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Mean performance gap	Average score difference by gender	Bank accuracy; ACT practice	Inflate tiny ds into destiny
Variance / tail gap	Male excess at extremes (debated)	High-end challenge path	Summers-style innate marketing
Stereotype (explicit)	Stated "math is for boys" beliefs	Parent/student surveys	Treat as only signal
Stereotype (implicit)	IAT-style gender–science associations	Not productized	Fake IAT gamification
Stereotype threat	Cue → underperformance under evaluative pressure	Exam-pressure UX (LXIII)	Miracle intervention brand
Cultural field stereotypes	Who belongs in CS/eng culture	Story characters; tutor tone	Token costume without practice
Relative strength / interest	Intra-individual math vs verbal preference	Course / path choice	Essentialize preferences as biology
SAFE-GENDER	Surviving gender×math doctrine	See LXV.9	Pink STEM; ST→score; innate gap

Operational definition (HYPOTHESIS): A MindCraft experience is *gender-honest* when (a) it does not encode ability stereotypes in copy or character casting, (b) it separates performance evidence from interest/identity evidence, (c) it designs belonging via recognition of mathematical practice (not gendered aesthetics), (d) any threat/belonging intervention is pre-registered and not sold as score insurance, (e) challenge and transfer standards stay intact across gender.

LXV.3 Mean math performance: parity is the update

FACT (gender similarities — broad): Hyde (2005, *American Psychologist*, 60(6), 581–592, doi:10.1037/0003-066X.60.6.581) reviewed 46 meta-analyses and advanced the **gender similarities hypothesis**: males and females are similar on most psychological variables; overstated difference claims carry costs.

FACT (school math scores near zero d): Hyde, Lindberg, Linn, Ellis & Williams (2008, *Science*, 321(5888), 494–495, doi:10.1126/science.1160364) analyzed U.S. state assessments covering **>7 million** students; weighted mean gender effect size ≈ **d = 0.0065** (grades 2–11) — consistent with no meaningful mean difference. NAEP hard / complex items at grade 12 averaged about **d ≈ 0.07** (trivial).

FACT (meta update): Lindberg, Hyde, Petersen & Linn (2010, *Psychological Bulletin*, 136(6), 1123–1135, doi:10.1037/a0021276) — overall gender difference in math performance ≈ **0**; small male advantages appear in some high-school / high-performing slices (e.g., reported ds on the order of ~0.2–0.4 in those slices — still not "boys own math").

Commercial implication: Parent copy can say **school math scores no longer support a boys-are-better story**. Do not sell MindCraft as closing a huge mean gap that largely is not there. Sell the remaining problem: **interest, confidence, recognition, course choice, and stereotype climate**.

Wound: Parity on means does not erase STEM major / career segregation, anxiety gaps, or high-tail debates. Parity ≠ identity solved.

LXV.4 Affect and attitudes still gendered — update the 1990s carefully

FACT (attitudes/affect meta — classic still cited): Hyde, Fennema, Ryan, Frost & Hopp (1990, *Psychology of Women Quarterly*, 14(3), 299–324, doi:10.1111/j.1471-6402.1990.tb00022.x) found gender differences in math attitudes/affect often larger than in performance — stereotypes, confidence, and anxiety channels.

HYPOTHESIS (MindCraft, tying XXXVII / LXI): For many Mayas, the binding constraint is **expectancy + attainment value + recognition**, not a missing lecture. Stereotype climate can tax those without moving mean item accuracy much in low-stakes practice.

Kill: "Girls underperform on our bank, so we need easier girl mode." Prefer diagnose anxiety, format, language load, and mastery climate first.

LXV.5 Implicit stereotypes predict gaps — not a MindCraft IAT product

FACT (nation-level IAT ↔ TIMSS gaps): Nosek et al. (2009, *PNAS*, 106(26), 10593–10597, doi:10.1073/pnas.0809921106) — ~70% of >500k gender–science IATs across 34 countries showed science=male associations; **nation-level implicit stereotypes predicted** 8th-grade TIMSS science/math sex differences (science: r ≈ 0.60); explicit self-report stereotypes did not add predictive validity.

Wound: Ecological / correlational; IAT samples are volunteer web-takers; does not identify MindCraft causal paths. Do **not** ship a "stereotype IAT score" as identity metric.

Survive: Cultural stereotype climate is real enough to design against in story casting, tutor language, and parent speech — without psychologizing students via IAT gamification.

LXV.6 Stereotype threat: origin real, school-age literature wounded

FACT (classic college ST): Spencer, Steele & Quinn (1999, *Journal of Experimental Social Psychology*, 35(1), 4–28, doi:10.1006/jesp.1998.1373) — women underperformed men when a math test was described as showing gender differences; performance equalized under gender-fair framing (highly identified college samples).

FACT (children/adolescents meta — small + bias risk): Flore & Wicherts (2015, *Journal of School Psychology*, 53(1), 25–44, doi:10.1016/j.jsp.2014.10.002) — 47 effect sizes; mean effect ≈ **g = –0.22**; **none of tested moderators significant**; multiple signs of **publication bias**; called for large pre-registered replications.

FACT (large null registered report): Flore, Mulder & Wicherts (2018, *Comprehensive Results in Social Psychology*, 3(1), doi:10.1080/23743603.2018.1559647) — Dutch high schools, **N = 2,064**, ages ~13–14; **no overall ST effect** on girls' math scores; no moderated ST effects for domain/gender ID, math anxiety, or difficulty in that sample.

FACT (U.S. school-age nulls): Ganley, Mingle, Ryan, Ryan, Vasilyeva & Perry (2013, *Developmental Psychology*, 49(10), 1886–1897, doi:10.1037/a0031412) — three studies, **N = 931**; **no evidence** school-age girls' math performance was impacted by ST activation methods from implicit to explicit; discuss publication-bias risk in the child ST literature.

Doyle & Voyer (2016, *Learning and Individual Differences*, 47, 103–116, doi:10.1016/j.lindif.2015.12.018) — meta-analysis of stereotype *manipulations* on math/spatial tests; useful as additional ST literature, not as MindCraft score insurance.

Commercial kill (already foreshadowed in LXIII standing bans): Do not market "we remove stereotype threat → +ACT points." Destake-to-learn / dose-to-prove remains the exam doctrine; ST cues are one pressure channel among many, not a magic lever.

Survive: Avoid **gratuitous stereotype cues** in product (gendered ability jokes, "even for girls," male-only genius tropes). That is hygiene, not an efficacy claim.

LXV.7 Cultural stereotypes as gatekeepers (interest ≠ ability)

FACT (CS/eng culture stereotypes): Cheryan, Master & Meltzoff (2015, *Frontiers in Psychology*, 6:49, doi:10.3389/fpsyg.2015.00049) — stereotypes of computer science/engineering as male, isolated, machinery-focused, "brilliance"-coded gatekeep girls' interest; **diversifying representations of people/work/environments** can raise belonging and interest without necessarily changing ability beliefs overnight.

Tie to XXXVI / SAFE-IDMEASURE: Story worlds and tutor casting are identity technology. Token "add a girl character" without mathematical agency is costume. Empty AI praise ("you're a math girl!") is banned recognition theater (LXI).

HYPOTHESIS: For MindCraft, the highest-leverage gender-equity move is **recognition of real mathematical practice** (annotations, soft-wrong springboards, transfer proof) inside non-stereotyped worlds — not a separate pink product SKU.

LXV.8 Gender-equality paradox: segregation ≠ mean inability

FACT (PISA / STEM graduation pattern): Stoet & Geary (2018, *Psychological Science*, 29(4), 581–593, doi:10.1177/0956797617741719) — PISA science/math/reading **N = 472,242**; girls similar or better in science in 2/3 of countries; yet sex differences in *relative* academic strengths and STEM degree pursuit **increase with national gender equality** (the “gender-equality paradox”).

FACT (stereotype mediation account): Breda, Jouini, Napp & Thebault (2020, *PNAS*, 117(49), 31063–31069, doi:10.1073/pnas.2008704117) — ~300k 15-year-olds, 64 countries; stereotype that **“math is not for girls”** is *stronger* in more egalitarian/developed countries and can **account for** the paradox linking development to math-intensive field segregation (vs pure “preferences free to express” story alone).

Wound: Cross-country observational; contested commentary exists (e.g., Richardson et al. 2020 on Stoet & Geary). Do not import as U.S. ACT causal law.

Commercial implication: U.S. parent markets are high-equality contexts where **horizontal gendering of interests** can be strong even when scores look equal. Selling “girls can’t do math so buy us” is false; selling “interest and stereotype climate still steer courses” is honest. Neither justifies pink cosplay nor challenge-softening.

LXV.9 SAFE-GENDER stack (surviving doctrine)

1. **Parity hygiene** — do not premise product on a large mean ability gap; cite similarities when talking scores.
2. **Separate channels** — performance ≠ interest ≠ identity ≠ stereotype climate; instrument separately (EVT + IDM).
3. **No innate-gap marketing** — ban Summers-style or “boys brains” copy forever.
4. **No ST→score guarantee** — Flore/Ganley wounds; ST is not ACT insurance (ties SAFE-EXAM).
5. **Cue hygiene** — no gendered ability jokes, “even you,” male-genius tropes in stems/stories/AI.
6. **Belonging via practice** — Cheryan-style diversify *who does math here*; recognition of work > aesthetics.
7. **Anti-cosplay** — no girl-only easy track; no pink SKU as equity.
8. **Anti-excuse** — gender does not waive FEI, transfer, or exam dosing — sequence support, don’t skip.
9. **Intersectionality light** — gender × race × ELL × class; one “STEM girl” persona is a kill (ties XXXVI / ELL).
10. **Parent speech** — “scores are close; confidence and course choice still get gendered” vs “we’ll make her a mathematician in 8 weeks.”

LXV.10 Experiments (GEND family)

ID	Contrast	Primary endpoints	Kill / survive rule
GEND-1	Stereotype-cued copy vs cue-clean copy (same math)	Accuracy; challenge_accept; belonging/recognition items	Cue hurts approach → enforce hygiene; no accuracy delta → still keep hygiene as ethics
GEND-2	Gendered “for her” skin vs neutral craft UI	4w transfer; identity items; reattempt	Skin ↑ vanity engagement, ↓ transfer → kill cosplay

GEND-3	Diverse mathematician agency in story vs token cameo	Belonging; challenge_accept; HIST-EQ codes	Token ≤ baseline → require agency casting
GEND-4	Recognition-of-practice feedback vs “girl power” praise	IDM factors; solo_transfer_pass	Praise-only ↑ Likert, flat transfer → kill praise track
GEND-5	Destaked practice → later gendered-stakes probe (exam ladder)	Accuracy under stakes; skip rate	Destake helps approach without transfer → redesign; both → keep SAFE-EXAM sequencing
GEND-QUAL	10 Maya interviews: “when did ‘math person’ feel gendered?”	Codebook: cue sources (peers/parents/media/UI)	Feeds copy bans + story brief

Ties: SAFE-EXAM, SAFE-IDMEASURE, SAFE-LONGID, SAFE-EXPECTANCY, XXXVI HIST-EQ, XXXVII EVT.

LXV.11 Commercial implications (copy, product, positioning, growth)

Marketing language that survives: “Average school math scores don’t justify ‘boys are better’ — what still gets gendered is confidence, belonging, and who keeps going.” / “We design so stereotypes never become the curriculum.” / “Recognition for real mathematical work — not girl-washed packaging.”

Marketing language that dies: “Fix her math brain”; “stereotype-threat vaccine → +ACT”; “built for girls” easy mode; “we close the STEM gender gap”; male-genius lore; empty “math girl” badges.

Positioning vs Khan / Duo / Brilliant / GPT tutors: Competitors can accidentally amplify stereotypes (male-coded “genius grind,” pink gamification, or fluent AI that mirrors parent stereotypes). MindCraft’s wedge: **identity-safe struggle + competence evidence** under cue hygiene — same FEI spine for every student, with story equity that diversifies *agency*, not difficulty.

Growth: Instrument `stereotype_cue_flag` (authoring), belonging/recognition items (IDM), course-intent proxies — not “% female DAU” as learning NS. School/parent sales: honesty about interest/identity channels; never innate-gap fear marketing.

Vision: Gender enters MindCraft as **climate and recognition design**, co-equal with language load (ELL) and exam affect (EXAM) — first-class authoring constraints, not a seasonal campaign.

LXV.12 Confidence table

Claim	Label	Confidence

U.S. mean K–12 math gender gaps are trivial in large assessments	FACT	High
Overall meta gender math performance difference ≈ 0 (with small HS/high-tail caveats)	FACT	High
Math attitudes/affect often show larger gender differences than performance	FACT (classic meta; still directionally relevant)	Medium–High
Nation-level implicit gender–science stereotypes correlate with TIMSS gaps	FACT	Medium–High (correlational)
School-age ST average effect small and publication-bias-vulnerable; large RR null exists	FACT	High
Cultural STEM stereotypes gate interest/belonging and are malleable via representation	FACT / HYPOTHESIS (lab⟶product transfer)	Medium–High
Gender-equality paradox pattern in PISA/STEM degrees	FACT (pattern); mechanisms contested	Medium
Essentialist math-gender stereotypes help explain that paradox	HYPOTHESIS (Breda et al. account)	Medium
SAFE-GENDER is the right commercial mapping for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium
Cue-clean + practice recognition beats pink cosplay on transfer	HYPOTHESIS	Medium (GEND-2/4)

LXV.13 What this chapter kills

1. **Kill:** Innate male math superiority as marketing or product premise.
2. **Kill:** Large mean ability-gap narrative contradicted by Hyde/Lindberg line.
3. **Kill:** Stereotype-threat removal as ACT/score guarantee (Flore meta bias + Flore RR + Ganley nulls).
4. **Kill:** Pink / “for her” / girl-power packaging as identity transformation.
5. **Kill:** Token STEM women without mathematical agency; empty “math girl” AI praise.
6. **Kill:** “We close the gender STEM gap” / 8-week mathematician claims.
7. **Kill:** Softening FEI/transfer because the student is a girl.
8. **Wound:** Treating preferences in high-equality countries as pure biology (Stoet/Geary vs Breda tension).
9. **Survive:** Mean parity hygiene; cue hygiene; Cheryan belonging-via-representation; EVT/identity channel separation; GEND-1…5; parent speech that names confidence/choice without innate myths.

Doctrine until data: Gender×math work at MindCraft is **stereotype-climate hygiene + earned recognition inside equitable practice** — not ability mythology, not ST theater, and not pink cosplay.

Part LXVI — Socioeconomic Constraint on "Grit"

Chapter status: Living evidence + equity/positioning brief — Researcher tick 2026-08-02

Primary question: When schools, apps, and marketers sell *grit* as the lever for math success, what does evidence say about construct validity — and how do socioeconomic structure, cognitive bandwidth, and opportunity-to-learn constrain (or fake) "perseverance" as a product story?

Owners: Equity · Parent messaging / WTP · Motivation (SAFE-RESILIENCE) · Product (session load, access) · Red Team

Commercial job: Ship a **SAFE-STRUCTURE** doctrine: structure and scaffolds before slogans; never sell low-SES Maya's stuckness as a character defect; never ship a Grit Score™ as North Star.

LXVI.1 Why this chapter exists

Part XLV already kills grit *theater* as a resilience brand. Part LIX documents tutoring markets skewed by income. Parts LXIV–LXV separate language load and gender climate from ability myths. What remains unfinished is the **class channel**: parents and schools still hear "build grit" as the fix for achievement gaps that Reardon-scale income gradients and poverty-related cognitive load make structural.

FOUNDER BELIEF under audit: MindCraft can help identity under difficulty only if it treats SES as a **constraint on available bandwidth, time, tutoring dollars, and sense of control** — not as proof that some kids lack passion-and-perseverance DNA.

Claims we refuse as doctrine:

1. Grit as MindCraft's primary product claim or North Star metric.
2. "Poor kids just need more grit" / character lectures as equity.
3. Grit Score™, streak-as-grit, or leaderboard-as-character as learning proof.
4. Ignoring tutoring-market SES skew when pricing or packaging (ties SAFE-WTP).
5. Treating low session completion under scarcity as moral failure.
6. Importing Alan-style classroom grit RCTs as U.S. ACT guarantees.
7. Softening FEI/transfer standards "because SES" instead of removing cognitive taxes and adding support.

LXVI.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused

Grit (Duckworth)	Perseverance + consistency of interest for long-term goals	Tempting brand copy	Near-conscientiousness ; slogan equity
Perseverance of effort (PE)	Facet with more criterion validity	`retry_120s`, challenge_accept	Relabel PE as total grit
Consistency of interest (CI)	Facet weaker for academics	Path stickiness	Shame topic switching
Conscientiousness	Big Five trait overlapping grit	Study habits UX	Jangle fallacy product
Sense of control	Belief one can shape outcomes (SES-linked)	Destaked agency; clear next step	“Just believe harder”
Cognitive bandwidth / scarcity load	Poverty-related concerns consume WM/attention	Reduce UI tax; short sessions	Blame “lazy focus”
Income achievement gap	90–10 family-income score gaps	Market + equity fact	Sell grit as gap closer
SAFE-STRUCTURE	Structure > slogans doctrine	See LXVI.9	Character lectures; Grit Score™

Operational definition (HYPOTHESIS): A MindCraft experience is *structure-honest* when (a) it never diagnoses low engagement under scarcity as low grit, (b) it minimizes cognitive taxes (forms, clutter, timed floods) especially for bandwidth-taxed households, (c) it measures approach/recovery/transfer behavior not trait self-report, (d) pricing/access experiments acknowledge ability-to-pay (SAFE-WTP), and (e) challenge standards stay intact while support density rises.

LXVI.3 What grit claimed — and what the meta wound

FACT (origin construct): Duckworth, Peterson, Matthews & Kelly (2007, *Journal of Personality and Social Psychology*, 92(6), 1087–1101, doi:10.1037/0022-3514.92.6.1087) defined grit as perseverance and passion for long-term goals; across adult attainment, Ivy GPA, West Point retention, and Spelling Bee samples, grit accounted for **~4%** of variance in success outcomes on average, correlated highly with conscientiousness, and showed some incremental prediction beyond IQ/conscientiousness in those samples.

FACT (short scale): Duckworth & Quinn (2009, *Journal of Personality Assessment*, 91(2), 166–174, doi:10.1080/00223890802634290) — Grit-S (8 items; PE + CI facets).

FACT (meta wound): Credé, Tynan & Harms (2017, *Journal of Personality and Social Psychology*, 113(3), 492–511, doi:10.1037/pspp0000102) — **584** effect sizes, **88** samples, **N = 66,807**. Higher-order grit structure not confirmed; grit only **moderately** related to performance/retention; grit **very strongly** related to conscientiousness; **PE ■ CI** for criterion validity; interventions to “enhance grit” likely **weak** on performance; construct validity questioned; utility may lie mainly in the perseverance facet.

FACT (policy critique): Credé (2018, *Educational Researcher*, 47(9), doi:10.3102/0013189X18801322) — no empirical reason to fuse passion+perseverance into one construct; little support that grit is a *particularly* strong educational predictor or that it is readily intervention-responsive as marketed.

Commercial kill: Do not sell MindCraft as a grit factory. Do not ship Grit Score™. If any perseverance language survives, it must mean **designed recovery after soft-wrong** (SAFE-RESILIENCE / SAFE-DD), not trait branding.

Survive: Domain-specific persistence behaviors (`retry_120s`, help-recruitment, transfer) remain product signals — subordinate to FEI, not a personality product line.

LXVI.4 SES is not a grit deficit — structure shows up in scores and markets

FACT (income achievement gradient): Reardon (2011), in Duncan & Murnane (Eds.), *Whither Opportunity?* (Russell Sage) — the U.S. **90–10 income achievement gap** grew roughly **30–40%** for cohorts born ~2001 vs ~25 years earlier; income gaps became large relative to historical black–white gaps; gap is already large at school entry. (Later work debates trend magnitude; the *level* of SES–achievement association remains a central market fact.)

FACT (paid tutoring skew): Kim, Goodman & West (2024, CESifo WP 11251 / IZA DP 17178) — U.S. tutoring-center expansion and high-income concentration (also summarized in Part LIX). Bray (2010) / Bray & Lykins (2012, ADB) — shadow education is stratified.

Commercial implication: Parents who can buy "more grit" narratives often already buy **structure** (tutors, quiet time, test prep). Positioning that shames lower-resource families for "not pushing hard enough" is false and brand-toxic. Equity copy must name **access, time, and bandwidth**, not character.

Wound: Income gaps confound OTL, school quality, health, stress, language, and prior knowledge. They are not MindCraft effect sizes and not proof that grit interventions close Reardon gaps.

LXVI.5 Poverty taxes cognition — "not trying" is often load

FACT (scarcity → cognitive performance): Mani, Mullainathan, Shafir & Zhao (2013, *Science*, 341(6149), 976–980, doi:10.1126/science.1238041) — financial concerns experimentally reduce cognitive performance among lower-income mall participants (not well-off); the same Indian sugarcane farmers perform worse **pre-harvest** (scarce) than **post-harvest** (less scarce), not explained by nutrition/work-time alone; authors calibrate lab effects on the order of **~13 IQ points** / comparable to a night of sleep loss on Raven's-style tasks. Claim is about **context of poverty**, not fixed traits of "poor people."

HYPOTHESIS (MindCraft): Exam-week fees, noisy shared devices, after-job homework slots, and parent bill stress are **dual-task taxes** on math WM — same family as anxiety load (Ashcraft) and exam pressure (SAFE-EXAM). Product that adds cognitive taxes (dense UI, day-one timed ACT flood, endless forms) will look like "low grit" in analytics while actually measuring scarcity.

Product rules: Prefer short sharp DP cycles (SAFE-DP); offline-friendly / resume drafts; low-chrome practice; destake-to-learn before stakes; never moralize incomplete missions.

Kill: "They bounced off because they lack grit." Prefer diagnose load, anxiety, language, format, and access first.

LXVI.6 Sociology of grit: SES → sense of control → perseverance (not pure will)

FACT (SES–control–grit path): Kwon (2021, *Socius*, 7, doi:10.1177/23780231211005216) — U.S. and South Korea survey data; higher SES partially translates into grit (especially **perseverance of effort** in Korea) **via stronger sense of control**; education sometimes **negatively** associates with consistency-of-interest, suggesting stick-to-it-iveness can be a *supplementary* resource when other resources are scarce — not a simple "rich = gritty" cartoon.

HYPOTHESIS: Sense of control in MindCraft is engineered by **clear diagnosis** → **next ingredient** → **recoverable miss**, not by posters. Low-control contexts need *more* scaffold clarity, not louder character talk.

Tie to EVT (XXXVII): Expectancy collapses when structure fails. Selling passion without expectancy is inspirational avoidance.

LXVI.7 Can grit be taught? Real RCT — narrow transfer, not ACT insurance

FACT (classroom intervention RCT): Alan, Boneva & Ertac (2019, *Quarterly Journal of Economics*, 134(3), 1121–1162, doi:10.1093/qje/qjz006) — randomized educational intervention in elementary samples; treated students more likely to exert effort to build task-specific ability; follow-up ~**2.5 years** later ≈ **0.2 SD** on a standardized math test.

Wound / kill for marketing: This is **not** a license to claim MindCraft "installs grit → +ACT points." Sample/setting ≠ U.S. high-stakes ACT; effect is modest; Credé line still wounds trait-level marketing and easy malleability claims. Borrow the *design idea* (normalize failure → try again with strategy) already owned by SAFE-RESILIENCE / soft-wrong — do not import the brand name as North Star.

SPECULATION: MindCraft's closest analogue is **attempt** → **soft-wrong** → **support recruitment** → **retry**, which trains perseverance *behavior* without claiming personality change.

LXVI.8 Intersection with identity, ELL, gender, and exam pressure

SES rarely travels alone. Language load (SAFE-ELL), stereotype climate (SAFE-GENDER), and high-stakes affect (SAFE-EXAM) compound bandwidth taxes. Homogeneous "gritty hustle" personas erase that stack.

FOUNDER BELIEF: Recognition of mathematical practice (SAFE-IDMEASURE) must be available without requiring paid co-curricular polish that only high-income students get. Otherwise identity becomes another SES luxury good.

LXVI.9 SAFE-STRUCTURE stack (surviving doctrine)

1. **Structure before slogans** — scaffolds, diagnosis, cards, tutor briefs beat grit posters.
2. **No Grit Score™ / streak-as-character** — already banned as NS; reaffirm under SES equity.
3. **Bandwidth hygiene** — minimize cognitive taxes; short sessions; resume without shame.

4. **Behavior ≠ trait** — instrument `retry_120s`, help-recruit, transfer — not Duckworth Likert as product KPI.
5. **PE not CI worship** — allow healthy topic/path revision; don't shame interest shifts.
6. **Access honesty** — pricing/scholarship experiments acknowledge ability-to-pay (SAFE-WTP); don't equate unpaid with ungritty.
7. **No character-blame equity** — never explain SES gaps as missing passion.
8. **No ACT-from-grit guarantee** — Alan ≠ exam insurance; SAFE-EXAM dosing stays.
9. **Support density under scarcity** — more on-ramps, not softer math standards.
10. **Parent speech** — "we remove friction and build recoverable struggle" vs "we build grit."

LXVI.10 Experiments (STRUCT family)

ID	Contrast	Primary endpoints	Kill / survive rule
STRUCT-1	Low-chrome / resume-friendly practice vs high-tax clutter UI	Session completion; retry_120s; transfer	Clutter ↓ completion more for self-reported scarcity → enforce bandwidth hygiene
STRUCT-2	Soft-wrong + support recruitment vs grit-poster interstitial (same items)	challenge_accept; help-recruit; delayed accuracy	Poster ≥ scaffold on transfer → unexpected; else kill poster track
STRUCT-3	Grit-S / PE Likert vs behavioral perseverance suite as predictors	Incremental R^2 on transfer_pass	Likert adds nothing → ban trait KPI
STRUCT-4	Scholarship / access messaging vs "build character" messaging (parent CBC arm)	WTP; brand trust; offense codes	Character-blame ↓ trust in lower-income segment → kill
STRUCT-5	Exam-week destake ladder vs day-one timed flood under self-reported stress/scarcity	Skip rate; accuracy under later stakes	Flood harms scarce/stressed more → SAFE-EXAM+STRUCTURE
STRUCT-QUAL	10 Maya + 10 parent interviews: time, devices, work, quiet, money stress	Codebook of structural barriers	Feeds copy bans + session defaults

Ties: SAFE-RESILIENCE, SAFE-WTP, SAFE-EXAM, SAFE-DD, SAFE-DP, SAFE-ELL, SAFE-GENDER, SAFE-IDMEASURE.

LXVI.11 Commercial implications (copy, product, positioning, growth)

Marketing language that survives: “Struggle isn’t a character test — it’s a design problem.” / “We build recoverable practice and clear next steps, not grit lectures.” / “When life is loud, the product should get quieter — short sessions, saved progress, support on tap.”

Marketing language that dies: “We forge grit”; “poor performers lack passion”; Grit Score™; streak-as-moral-proof; “just push through” as equity; guaranteed ACT from character curriculum.

Positioning vs Khan / Duo / Brilliant / GPT tutors: Duo-style streaks and hustle aesthetics can read as character judgment when life constraints dominate. ChatGPT fluency doesn’t remove bandwidth taxes. MindCraft’s wedge: **ontology diagnosis + soft-wrong support + transfer proof** that respects structural load — structure as product, not sermon.

Growth: Segment analytics by access proxies (device class, session time-of-day, scholarship flag) without building a shady “SES score.” Instrument structural barriers in QUAL; never optimize shame. School/district sales: lead with OTL and scaffolding, not grit PD.

Vision: Class enters MindCraft as **bandwidth, access, and sense-of-control design** — co-equal with ELL load and gender climate — so identity transformation is not a luxury good.

LXVI.12 Confidence table

Claim	Label	Confidence
Duckworth grit predicts some success outcomes (~4% avg variance in origin paper)	FACT	High
Credé meta: strong conscientiousness overlap; weak higher-order grit; PE > CI; weak intervention expectation	FACT	High
Credé 2018: grit not a uniquely strong / easily trained edu lever as marketed	FACT	High
U.S. income achievement gaps are large and historically salient (Reardon line)	FACT	High (trend debates Medium)
Tutoring spend / centers concentrate among higher-income families	FACT	High
Poverty-related concerns can impair cognitive performance (~sleep-loss / ~13 IQ calibration in Mani et al.)	FACT	High (context transfer to HS homework Medium)
SES → sense of control → grit facets (Kwon)	FACT (observational, cross-cultural)	Medium–High

Alan et al. classroom grit intervention → ~0.2 SD math at 2.5y	FACT	High (external validity to MindCraft/ACT Low–Medium)
SAFE-STRUCTURE is the right commercial mapping	FOUNDER BELIEF / HYPOTHESIS	Medium
Bandwidth hygiene + soft-wrong beats grit posters on transfer under scarcity	HYPOTHESIS	Medium (STRUCT-1/2)

LXVI.13 What this chapter kills

1. **Kill:** Grit as MindCraft North Star, primary ad claim, or Grit Score™.
2. **Kill:** “Low-SES students lack grit” / character-blame equity.
3. **Kill:** Streaks or leaderboards as proof of character.
4. **Kill:** Importing Alan RCT as ACT-point insurance.
5. **Kill:** Treating scarcity-driven dropout as moral failure.
6. **Kill:** Softening math standards instead of removing cognitive taxes.
7. **Wound:** Treating all perseverance language as forbidden — behavioral recovery metrics still survive under SAFE-RESILIENCE.
8. **Survive:** Credé construct hygiene; Mani bandwidth humility; Reardon/WTP structural honesty; Kwon control pathway; STRUCT-1…5; structure-before-slogans parent speech.

Doctrine until data: Socioeconomic constraint means MindCraft sells **scaffolds, bandwidth hygiene, and recoverable struggle** — never grit sermons as the fix for structural gaps.

Part LXVII — Ontology as Diagnosis Moat

Chapter status: Living evidence + positioning brief — Researcher tick 2026-08-02

Primary question: When does a structured knowledge representation (concept/ingredient graph + persistent student state) beat fluent chat tutoring for *diagnosis* — and what, exactly, can MindCraft claim as a commercial moat without cosplaying ALEKS or inventing “AI tutor superiority”?

Owners: Engine (ontology / pathfinder) · Product (Map / Solver / Notes) · Positioning vs Khan/Duo/Brilliant/GPT · Red Team

Commercial job: Ship a **SAFE-ONTOLOGY** doctrine: diagnosis is a stateful graph problem first; language models are bookends; the moat is longitudinal, privacy-respecting student state joined to a canonical ontology under FEI — not a JSON file and not chat fluency.

LXVII.1 Why this chapter exists

Parts XXXIII / XXXV / LX already kill unguarded Solver-as-hero, Bastani-Tutor≡MindCraft category errors, and apology≡repair. What remains unfinished is the **engine wedge**: founders believe the standardized ontology + mastery graph is the product moat. Competitors can wrap GPT tomorrow. Parents hear "AI tutor" as interchangeable. Sales needs a defendable sentence for *why not just ChatGPT*.

FOUNDER BELIEF under audit: Knowledge graphs beat chat when the job is *which gap, in what order, with what evidence* — not when the job is *sound fluent*.

Claims we refuse as doctrine:

1. "We have a knowledge graph" ≡ durable competitive moat by itself.
2. RAG-wrapped ChatGPT ≡ MindCraft diagnosis (category error with Bastani-Tutor≡ontology).
3. Chat fluency / token volume as proof of diagnosis quality.
4. Ontology coverage theater (node count as learning).
5. Blocked accuracy on graph-sequenced drills ≡ exam-ready / identity change.
6. Absolute "ITS beat humans / 2-sigma" Bloom marketing.
7. Selling the graph without transfer / solo proof (PWC still binds).

LXVII.2 Constructs (product language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Student model	Latent mastery / knowledge state over skills	Concept + ingredient mastery; bridge gaps	Treat chat memory as student model
Knowledge tracing (BKT lineage)	Bayesian update of skill mastery from attempts	`/record-outcomes`, decay, strength scoring	Black-box "AI knows you" without inspectable state
Knowledge space / learning space	Combinatorial feasible states; outer fringe	Pathfinder trim; "ready next"	ALEKS cosplay; assessment-only brand
Model tracing / cognitive model	Step-level comparison to expert production rules	Ingredient cards + soft-wrong classify	Claim step-model without traces
Ontology / concept graph	Directed prereqs, bridges, combinations	Layer-1 ontology; Map	Node-count vanity; stale alias maps
Diagnosis	Infer *what is broken* and *what to teach next*	Gap-scan → `/recommend` → worstWeakness	Fluency-as-diagnosis
Pedagogy wrap (PWC)	Guards + spine; solo transfer ≥ no-AI	Solver under Map	Unguarded chat hero
SAFE-ONTOLOGY	Graph-first diagnosis doctrine	See LXVII.9	Graph file as moat slogan

Operational definition (HYPOTHESIS): A MindCraft experience is *diagnosis-honest* when (a) every practice/session outcome updates an inspectable concept/ingredient/bridge state, (b) next-action is constrained by ontology edges not free chat wander, (c) LLM text cannot invent mastery or hide wrongness, (d) parent/tutor can see *why this next*, and (e) success is still `solo_transfer_pass` / delayed mix — not chat satisfaction.

LXVII.3 What structured tutors actually earned (and did not)

FACT (relative tutoring effectiveness): VanLehn (2011, *Educational Psychologist*, 46(4), 197–221, doi:10.1080/00461520.2011.611369) — human tutoring effect ≈ **d = 0.79** vs no tutoring (far below Bloom's mythic 2σ); step-based ITS ≈ **d = 0.76**. Interaction *grain* matters more than "AI" branding; answer-only CAI sits weaker than step-based tutors.

FACT (K–12 math ITS meta): Steenbergen-Hu & Cooper (2013, *Journal of Educational Psychology*, 105(4), 970–987, doi:10.1037/a0032447) — 26 reports / 34 samples; overall ITS vs classroom often **near zero to small positive** (g ≈ 0.01–0.09); shorter interventions looked stronger than year-long; general-population samples outperformed low-achiever samples on average. **Wound:** "We have ITS architecture" is not an ACT guarantee and not equity magic.

FACT (college ITS meta): Steenbergen-Hu & Cooper (2014, *Journal of Educational Psychology*, 106(2), 331–347, doi:10.1037/a0034752) — moderate positive overall (g ≈ **0.32–0.37**); ITS beat many non-human alternatives; **worse than human tutoring** where compared (negative mean in that thin subset). Pedagogy and teachers still matter.

Commercial implication: Sell *inspectable diagnosis + recoverable struggle*, not "our AI is as good as a tutor" or Bloom 2σ. VanLehn's anti-2σ fact stays standing ban.

LXVII.4 Persistent skill models — the job chat still fails

FACT (knowledge tracing): Corbett & Anderson (1994/1995, *User Modeling and User-Adapted Interaction*, 4, 253–278, doi:10.1007/BF01099821) — Bayesian *knowledge tracing* over production-rule skills in an ACT-based programming tutor; mastery-driven exercise selection; model predicts posttest from component mastery estimates.

FACT (cognitive tutors as method, not costume): Anderson, Corbett, Koedinger & Pelletier (1995, *Journal of the Learning Sciences*, 4(2), 167–207, doi:10.1207/s15327809jls0402_2) — decade of ACT-R Cognitive Tutor lessons: explicit cognitive model, model-tracing assistance, knowledge-tracing mastery sequencing. Achievement gains tied to *component mastery design*, not chatbot persona.

FACT (knowledge spaces): Doignon & Falmagne (1985, *International Journal of Man-Machine Studies*, 23(2), 175–196, doi:10.1016/S0020-7373(85)80031-6) introduced knowledge-space combinatorics; Falmagne, Koppen, Villano, Doignon & Johannesen (1990, *Psychological Review*, 97(2), 201–224, doi:10.1037/0033-295X.97.2.201) — build/test/search knowledge spaces; practical "what student can do / ready to learn" lists. ALEKS commercializes this lineage (Falmagne & Doignon *Learning Spaces*, 2011; Matayoshi et al. JMP 2021 practical KST/ALEKS perspective) — millions of users, assessment→fringe learning.

HYPOTHESIS (chat wound): Stateless or weakly stateful LLM tutors optimize for *plausible next utterance*. They do not reliably maintain a falsifiable mastery vector across weeks, bridge gaps, or format axes (Shetye 2024 /

conversational-tutor reviews in Part XXXIII line; systematic KT×LLM reviews note diagnosis as open problem). Bastani et al. (2025, *PNAS*, doi:10.1073/pnas.2422633122) shows unguarded GPT Base ↑ practice then ↓ solo exam (~17%) — fluency without learning architecture.

Kill: “ChatGPT remembers our conversation” ≡ knowledge tracing. Memory ≠ mastery model.

Survive: Hybrid design already in MindCraft doctrine — deterministic spine (ontology, outcomes, recommend) with LLM as language bookends (classify/render), not the adjudicator of truth.

LXVII.5 When graphs beat chat (and when they do not)

HYPOTHESIS — graphs win when:

1. **Prerequisite / bridge failure** — student “knows” A and B but fails A→B (MindCraft bridge-gap severity). Chat often re-explains A.
2. **Longitudinal planning** — exam-mode path, decay, spacing need durable state (SAFE-TIMING / SAFE-EXAM).
3. **Comparable severity ranking** — C1 worstWeakness across concept + format gaps needs shared scales, not vibes.
4. **Tutor/parent accountability** — “why this next” must be auditable (SAFE-EXPECTANCY task briefs).
5. **Anti-hallucination ground** — Map-checked correct after repair (SAFE-REPAIR).

HYPOTHESIS — chat wins (or ties) when:

1. Open-ended *wording* help after diagnosis is already known.
2. Affective micro-coaching language (still SE-constrained; Parts XL / A).
3. Rapid authoring of surface variants *after* essence/keys verified — not as the diagnostic authority.

Wound (ALEKS already exists): Knowledge-space assessment is not a greenfield moat. Differentiation cannot be “we have states.” Differentiation is **ontology grain (ingredients/bridges) × FEI identity loop × human tutor witnessing × transfer proof** under PWC.

SPECULATION: A pure chat competitor can fake Map UI screenshots; they cannot fake longitudinal `eventCount`, bridge evidence, and `solo_transfer_pass` without building the spine — *if* MindCraft instruments and publishes those metrics honestly.

LXVII.6 Ontology is infrastructure, not the North Star

FOUNDER BELIEF refined: The ontology is the *spine* that makes diagnosis, cards, and recommendations non-hallucinated. It is not the emotional product Maya buys and not the parent headline.

FACT-adjacent commercial pattern: Steenbergen-Hu metas + VanLehn show structured tutors can help *and* can underwhelm vs classroom/human baselines depending on implementation. Selling “42 concepts” or “179 ingredients” as the hero metric invites coverage theater (same failure mode as hours-practiced / node-count).

Product rules:

- Map shows *evidence-backed* weakness and next route — not a trophy wall of nodes.
- Gap-scan seeds state; practice appends; never let Solver overwrite mastery from fluent dialogue alone.
- Soft-wrong + misconception tags update ingredient/bridge state (SAFE-MISCON).
- Generation fills coverage holes; generation does not replace ontology joins (C5).

Kill: Homepage that leads with "powered by our knowledge graph" without a human-need sentence (fear → evidence → identity).

LXVII.7 Positioning vs Khan / Duo / Brilliant / GPT

Competitor pattern	What they optimize	MindCraft counter under SAFE-ONTOLOGY
Khan	Content breadth + mastery dashes	Diagnosis of *your* bridge/format gap + recoverable miss — not another video library
Duo	Streak / habit loops	Habit subordinate to FEI; graph state ≠ streak (SAFE-HABIT)
Brilliant	Delightful interactive puzzles	Delight without longitudinal diagnosis is entertainment; we keep Map memory
ChatGPT tutors	Fluent explanation / homework finish	Fluency is cheap; **inspectable next-skill + solo transfer** is scarce (PWC)

Marketing language that survives: "We don't just explain the problem — we keep a living map of what broke and what unlocks next." / "Chat can talk; diagnosis needs memory you can trust." / "Your tutor sees the same map the AI must obey."

Marketing language that dies: "Our AI is smarter than ChatGPT"; "knowledge graph = guaranteed ACT"; "2-sigma Cognitive Tutor"; "we're ALEKS but cooler"; node-count bragging; Solver as the product hero.

LXVII.8 Moat economics (what compounds)

HYPOTHESIS (data moat): Privacy-respecting longitudinal events (outcomes, soft-wrongs, bridge evidence, format profile) joined to a *shared* canonical ID contract compound; prompt libraries and chat logs commoditize faster.

HYPOTHESIS (workflow moat): Tutor ops that brief from Map (Part LXVIII queued) + student Map literacy create switching costs chat tabs lack — *if* QA and FEI training are real (not Discord theater; SAFE-CoP).

Wound: Without instrumentation, "ontology moat" is founder poetry. Without privacy ethics (queue id 69), event density becomes liability.

Claim ladder: L1 = A/B on recommend quality / time-to-useful-next; L2 = tutor agreement with system diagnosis; L3+ identity still needs SAFE-IDMEASURE / LONGID — graph Δ alone ≠ "became a math person."

LXVII.9 SAFE-ONTOLOGY stack (surviving doctrine)

1. **Diagnosis before dialogue** — state update and next-action from ontology/outcomes; chat cannot set mastery.
2. **Inspectable student model** — concept/ingredient/bridge/format fields a tutor can audit.
3. **LLM = bookends** — classify/render under PWC; never adjudicate ground truth alone.
4. **Bridges & formats first-class** — not only node mastery (C1 severity).
5. **No graph-file moat slogan** — moat = state × ontology × FEI × transfer proof.
6. **No RAG≡MindCraft** — retrieval ≠ knowledge tracing.
7. **No fluency-as-diagnosis / token KPIs** — ban chat satisfaction as learning proof.
8. **No Bloom/ITS-2σ marketing** — VanLehn standing ban.
9. **Coverage fills gaps; coverage ≠ product** — generation serves bank holes under verify.
10. **Parent/tutor speech** — “living map of gaps” not “smarter chatbot.”

LXVII.10 Experiments (ONTO family)

ID	Contrast	Primary endpoints	Kill / survive rule
ONTO-1	Ontology-constrained next (`/recommend`) vs free chat “what should I do next?” (same student state hidden from chat)	Time-to-attempt; topic relevance coded; transfer@7d	Chat ≥ ontology on transfer → wound moat claim
ONTO-2	Bridge-gap card route vs re-explain prerequisite concept only	Bridge item accuracy; severity Δ	Re-explain wins → bridge detector theater
ONTO-3	Tutor shown Map diagnosis vs tutor chat-only brief	Tutor–system agreement; session FEI events	Map no lift → briefing UX fail not ontology fail
ONTO-4	Outcomes-appended state vs chat-memory “profile” summary only	Delayed mix; calibration of next difficulty	Summary≈state → simplify; else keep event spine
ONTO-5	Solver under Map guards vs unguarded Solver (replication of PWC/AIT)	`solo_transfer_pass`	Unguarded ≥ guarded → revisit PWC
ONTO-QUAL	10 parent + 10 tutor: “why not ChatGPT?” codebook	Message resonance; confusion codes	Feeds SAFE-WTP CBC attribute “living diagnosis map”

Ties: PWC, SAFE-REPAIR, SAFE-MISCON, SAFE-TRANSFER, SAFE-CALIB, SAFE-DP, SAFE-WTP.

LXVII.11 Commercial implications (copy, product, growth, vision)

Copy: Lead with the *job* — find the real gap, practice the unlock, prove it alone — not with graph jargon. Use “map,” “gap,” “next unlock,” “your tutor sees the same map.”

Product: Keep PawHub / Practice launches driven by `/recommend` + playable questions; never let Solver become the homepage. Instrument diagnosis quality (ONTO-1…4) before scaling ontology marketing.

Growth: School/district pitch = shared diagnostic language across tutors + audit trail; consumer pitch = “ChatGPT finishes homework; we build a map you still own when the chat ends.”

Vision: Thirty-year bet stays identity transformation — ontology is the **memory substrate** that makes earned competence evidence cumulative rather than session-amnesiac.

LXVII.12 Confidence table

Claim	Label	Confidence
Human tutoring ≈ d=0.79; step ITS ≈ d=0.76 vs no tutoring (VanLehn 2011); Bloom 2σ not baseline	FACT	High
K–12 math ITS meta often small vs classroom (Steenbergen-Hu & Cooper 2013)	FACT	High
College ITS moderate positive; not reliably ≥ human tutors (2014 meta)	FACT	High
BKT / Cognitive Tutor lineage: explicit skill model + mastery sequencing predicts/posttests	FACT	High
Knowledge spaces / ALEKS: combinatorial states → “can do / ready to learn”	FACT	High
Unguarded GPT can harm solo exam performance (Bastani et al. 2025 PNAS)	FACT	High
LLM chat alone lacks reliable longitudinal diagnostic student models	HYPOTHESIS (strongly supported)	Medium–High

MindCraft moat = state×ontology×FEI×transfer, not graph file or chat	FOUNDER BELIEF / HYPOTHESIS	Medium
Ontology-constrained next beats free chat on delayed transfer (ONTO-1)	HYPOTHESIS	Medium
SAFE-ONTOLOGY is right commercial mapping	FOUNDER BELIEF	Medium

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LXVII.13 What this chapter kills

1. **Kill:** Knowledge-graph file / node count as the moat slogan.
2. **Kill:** RAG or chat memory ≡ knowledge tracing / MindCraft diagnosis.
3. **Kill:** Fluency, thumbs-up, or Solver tokens as diagnosis quality.
4. **Kill:** Bloom 2σ / "ITS = human tutor" absolute marketing.
5. **Kill:** ALEKS-cosplay assessment brand without FEI identity loop.
6. **Kill:** "Smarter than ChatGPT" comparative AI brag.
7. **Wound:** Treating chat language help as useless — it survives as *bookend* under PWC.
8. **Survive:** VanLehn grain lesson; Corbett/Anderson student models; KST fringe idea; Bastani guardrails; ONTO-1…5; diagnosis-before-dialogue parent speech.

Doctrine until data: Ontology beats chat when the scarce job is **trusted, longitudinal diagnosis under FEI** — and only then. Chat remains language infrastructure, never the adjudicator of what Maya knows.

Part LXVIII — Human-in-the-Loop Tutor Ops

Chapter status: Living evidence + ops/product brief — Researcher tick 2026-08-03

Primary question: What does *quality* human tutoring look like as an operating system — playbooks, QA, FEI training, escalation — when MindCraft already has Map diagnosis and AI bookends?

Owners: Tutor ops · Product (tutor brief / session tools) · Engine (Map as shared truth) · Positioning vs "AI replaces tutors" / "warm humans = quality"

Commercial job: Ship a **SAFE-HITL** doctrine: humans are not a vibe layer on chat; they are trained operators who brief from inspectable state, prompt more than lecture, refuse polite false mastery, escalate when AI fails, and are coached at a cadence that survives scale — or the tutor brand is theater.

LXVIII.1 Why this chapter exists

Part LXVII argues diagnosis is a stateful graph problem. Part XLVI bans trait-label expectancy. Part LX kills apology≡repair and requires triggered human escalate. Part LI wants deliberate-practice session spines. Part LIV wants brief/debrief. None of that ships unless **tutor operations** exist: who briefs how, what “good session” means, how QA works, how new tutors learn FEI without Discord cosplay.

FOUNDER BELIEF under audit: College tutors + Map + AI bookends beat AI-alone *and* beat warm untrained humans — but only if ops encode pedagogy, not “hire nice people.”

Claims we refuse as doctrine:

1. Warm human presence ≡ tutoring quality.
2. Session hours / booked minutes as quality KPI.
3. “Do you understand?” / student smile as mastery check.
4. Slack/Discord chat ≡ CoP or tutor QA (SAFE-CoP already wounded forum≡LPP).
5. AI coach monologue as tutor training.
6. Empathy scripts without Map task brief (SAFE-EXPECTANCY).
7. Infinite free human escalation (SAFE-REPAIR).
8. “Our tutors are Ivy” as substitute for playbook + transfer proof.

LXVIII.2 Constructs (ops language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Naturalistic tutoring	Typical unskilled peer/near-peer dialogue	Default college tutor without playbook	Assume “1:1” automatically deep
Scaffold / prompt	Elicit student construction vs deliver explanation	Coach cards; SE prompts; AAR student-first	Lecture-first “help”
Comprehension-gauging Q	“Got it?” checks	End-of-explain polls	False mastery theater
Politeness tax	Face-saving softens needed correction	Soft-wrong avoided; wrong keys unchallenged	Rapport > diagnosis
Teacher–AI complementarity	Humans for what software cannot; shared orchestration	Tutor + Map + Solver under PWC	Glasses cosplay / dashboard vanity
Coaching (PD)	Sustained, observation-linked development	Tutor QA cycles; FEI rubrics	One-off onboarding deck
SAFE-HITL	Ops doctrine for human loop	See LXVIII.9	Human-as-marketing-prop

Operational definition (HYPOTHESIS): A MindCraft tutor session is *HITL-honest* when (a) pre-brief is Map/task-only (no trait labels), (b) talk ratio favors student construction over tutor monologue, (c) soft-wrongs are elicited not papered over, (d) AI outputs are challenged against Map ground truth, (e) close uses short AAR tied to next practice, and (f) QA samples sessions against a FEI rubric — not star ratings alone.

LXVIII.3 What naturalistic tutors actually do (and miss)

FACT (collaborative dialogue, not expert strategy): Graesser, Person & Magliano (1995, *Applied Cognitive Psychology*, 9(6), 495–522, doi:10.1002/acp.2350090604) — naturalistic tutoring with *normal unskilled* tutors (grad→undergrad research methods; HS→7th-grade algebra) is dominated by collaborative problem solving, question answering, and explanation on specific examples. Sophisticated strategies common in ITS aspiration (deep error diagnosis systems, ideal Socratic method, affect tactics) were largely **absent or underdeveloped**.

FACT (question quality > question count): Graesser & Person (1994, *American Educational Research Journal*, 31(1), 104–137, doi:10.3102/00028312031001104) — student questions ~240× more frequent in tutoring than classroom; after experience, **quality** of student questions correlated with achievement; **frequency** did not. Tutors need training to improve question-asking — volume is not the metric.

FACT ("Do you understand?" misleads): Person, Graesser, Magliano & Kreuz (1994, *Learning and Individual Differences*, 6(2), 205–229, doi:10.1016/1041-6080(94)90010-8) — quality of student *answers* is the reliable signal of understanding; student questions are weak; answers to tutor comprehension-gauging questions are **very misleading**.

FACT (politeness can inhibit pedagogy): Person, Kreuz, Zwaan & Graesser (1995, *Cognition and Instruction*, 13(2), 161–188, doi:10.1207/s1532690xci1302_1) — Gricean rules and Brown & Levinson politeness strategies appear throughout tutoring dialogue; they can support rapport **and** inhibit effective tutoring via softened/indirect feedback — especially in less constrained domains.

Commercial implication: Marketing "real tutors" without ops is selling the *setting* Graesser already showed works *somewhat* even when unskilled — while missing the failure modes that destroy FEI (false mastery checks, polite non-correction, lecture-heavy help). Ops must train against those defaults.

LXVIII.4 Prompting beats pouring — training content, not personality

FACT (interactive scaffolding ≈ didactic when prompts replace explanations): Chi, Siler, Jeong, Yamauchi & Hausmann (2001, *Cognitive Science*, 25(4), 471–533, doi:10.1207/s15516709cog2504_1) — when tutors were **suppressed from giving explanations and feedback** and pushed to prompt/scaffold, students learned **as effectively** as with traditional didactic tutoring; gains tied to deeper/more scaffolding episodes and greater student control (with reading-ability limits).

HYPOTHESIS (MindCraft mapping): Tutor training should default to *prompt → student try → soft-wrong diagnose → Map-aligned next*, not *explain until nod*. AI coach cards that monologue violate Chi's interactive lesson and Part XL SE doctrine.

Cross-link: SAFE-DP (45-min spine), SAFE-SE / Part XL, SAFE-MISCON (productive wrongs), SAFE-AAR (student-first close).

Kill: "Our tutors explain better than AI" as the brand. Explanation is cheap (Parts XXXIII/XXXV). Scarce skill = **eliciting construction under diagnosis**.

LXVIII.5 Human–AI complementarity is an ops design, not a slogan

FACT (teacher–AI co-orchestration can lift learning): Holstein, McLaren & Aleven (2020, *AI Magazine*, 41(2), doi:10.1002/aaai.12058) — field evidence that students learn more when teachers and AI tutors work *together* with real-time analytics support (Lumilo line) than in AI-supported classrooms without that teacher orchestration aid. Design premise: AI personalizes content/pace; humans handle situations software is ill-suited for; conflicts arise if AI and human plans are uncoordinated.

FACT (co-design of orchestration tools): Holstein, McLaren & Aleven (2019, *Journal of Learning Analytics*, 6(2), 27–52, doi:10.18608/jla.2019.62.3) — iterative co-design of real-time teacher awareness tools for AI-enhanced K–12; analytics must match practitioner needs, not dump model internals.

HYPOTHESIS (1:1 transfer): MindCraft's analogue is not smart glasses — it is **pre-session Map brief + live soft-wrong/bridge alerts + Solver escalate packet** (SAFE-REPAIR). Tutor without Map is Holstein's "AI classroom without orchestration support." Map without trained tutor is dashboard vanity.

Wound: Classroom orchestration ≠ remote 1:1 college tutor. Do not market Lumilo effect sizes as ACT guarantees. Borrow *complementarity principle*, not costume.

Kill: "AI replaces tutors" *and* "humans don't need the graph." HITL means shared truth + divided labor.

LXVIII.6 Coaching tutors beats hiring theater — and scale is the wound

FACT (coaching meta): Kraft, Blazar & Hogan (2018, *Review of Educational Research*, 88(4), 547–588, doi:10.3102/0034654318759268) — 60 causal-design studies; pooled coaching effects ≈ **0.49 SD** on instructional practice and ≈ **0.18 SD** on student achievement. Much evidence from U.S. literacy coaching in early grades. **Scale wound:** larger effectiveness trials show only a **fraction** of small efficacy-trial effects — coaching is hard to take to scale while keeping quality.

HYPOTHESIS: MindCraft tutor ops that treat onboarding as a PDF + one shadow session will reproduce the scale wound. Surviving design = small-cohort coaching cycles with session clips / traces (SAFE-FILM sparseness; SAFE-AAR), FEI rubric, and Map-brief fidelity checks — not "hire 200 tutors" vanity.

Cross-link SAFE-EXPECTANCY: Briefs are task/concept/next-move only; ban "she's weak / anxious type" labels that become Golem scripts.

Kill: Headcount as product quality. Kill star-rating-only QA (smile ≠ transfer).

LXVIII.7 Playbook skeleton (product-facing)

HYPOTHESIS — minimum viable tutor playbook (pre→during→post):

1. **Pre (≤3 min):** Open Map diagnosis — target concept/bridge/format, recommended level, last soft-wrongs; task brief only (SAFE-EXPECTANCY).
2. **Open:** One destaked warm item or prior miss — not day-one ACT flood (SAFE-EXAM).
3. **Core:** Prompt-heavy cycle; student writes/tries before tutor speech; treat wrong answers as diagnostic (SAFE-MISCON); no permanent hint leash (SAFE-EXPOSE).
4. **AI rule:** Solver/coach text is draft; Map-checked ground truth; on hallucination run SAFE-REPAIR escalate packet — tutor does not improvise integrity denial.
5. **Close (SAFE-AAR light):** What was the goal? What evidence of progress? What broke? What is the next solo attempt?
6. **QA sample:** Rubric on talk ratio, Map fidelity, soft-wrong use, no trait labels, escalate correctness — not "energy."

SPECULATION: Parents will pay for *auditable* tutor quality (session brief they can see + solo transfer trend) more than for "elite tutor" adjectives — test under SAFE-WTP CBC attribute "tutor follows your Map."

LXVIII.8 Positioning

Competitor pattern	What they optimize	MindCraft HITL counter
ChatGPT / AI tutors	Fluent homework finish	Human + Map when AI fails; solo transfer still NS
Marketplace tutors	Hours / credentials	Playbook + Map brief + FEI QA
Khan / content	Video + practice	Diagnosis-informed human time, not another explainer
Duo streaks	Habit	Tutor does not become streak police (SAFE-HABIT)

Marketing that survives: "Your tutor works from the same living map the practice engine uses." / "We train tutors to make you think — not to nod along." / "When the AI is wrong, a human takes the packet — with a repair path, not a shrug."

Marketing that dies: "Real humans, zero AI"; "AI so good you don't need tutors"; "Ivy League tutors"; hours-booked vanity; Discord community as quality assurance.

LXVIII.9 SAFE-HITL stack (surviving doctrine)

1. **Map-briefed humans** — every session opens from inspectable diagnosis (SAFE-ONTOLOGY).
2. **Prompt > pour** — Chi-interactive default; lecture is the exception.
3. **Answer evidence > "got it?"** — ban comprehension-gauging as mastery.
4. **Polite ≠ correct** — train direct-but-kind correction; soft-wrong welcome.
5. **Complementarity** — AI for pace/draft language; human for diagnosis conflict, affect, escalate.

6. **Coaching cycles** — sustained QA; resist scale-without-fidelity (Kraft wound).
7. **FEI training** — tutors scored on recoverable struggle + next solo plan, not vibes.
8. **Escalate with packets** — SAFE-REPAIR; no infinite free human chat.
9. **No trait briefs** — SAFE-EXPECTANCY.
10. **No Discord≡QA / hours≡quality / Ivy≡playbook.**

LXVIII.10 Experiments (HITL family)

ID	Contrast	Primary endpoints	Kill / survive rule
HITL-1	Map-briefed tutor vs chat-summary brief only	Tutor–system agreement; FEI event rate; student talk ratio	Chat brief ≈ Map → briefing UX fail
HITL-2	Prompt-first playbook vs explain-first default	Soft-wrong rate; `retry_120s`; delayed mix	Explain-first wins transfer → revisit Chi mapping
HITL-3	Coached tutors (biweekly clip QA) vs onboarding-only	Rubric fidelity @30d; `solo_transfer_pass`	No lift → coaching theater (Kraft scale wound)
HITL-4	Tutor+Map+guarded Solver vs AI-alone practice	Persistence; transfer; escalate appropriateness	AI-alone ≥ HITL on transfer → human ops not earning keep
HITL-5	Ban “got it?” + require worked evidence vs free probes	False-mastery incidents coded	Free probes no worse → training cost unjustified
HITL-QUAL	10 tutor + 10 parent interviews on brief/QA trust	Message codebook	Feeds WTP attribute “Map-aligned tutor”

Ties: ONTO-3, EXP-, *REPAIR*-, AAR-, *DP*-, SE / Part XL, SAFE-EXPECTANCY.

LXVIII.11 Commercial implications

Copy: Sell *trained operators on a shared map*, not “humans vs robots.” Lead with what the tutor *does differently* in the first three minutes (opens your gap, not a generic worksheet).

Product: Tutor console = Map brief + soft-wrong feed + escalate packet + 4-question close. Student-facing: same Map the tutor sees (trust). Instrument HITL-1…4 before scaling marketplace headcount.

Growth: B2B/school pitch = consistent tutor quality via playbook + audit trail; consumer pitch = “not a random tutor hour — a session aimed at your next unlock.”

Vision: Thirty-year identity bet needs *witnesses* who can recognize competence honestly (Part LXI recognition factor) without colonizing with trait labels — HITL is how recognition stays human and falsifiable.

LXVIII.12 Confidence table

Claim	Label	Confidence
Naturalistic unskilled tutoring is collaborative Q&A/examples; deep ideal strategies rare (Graesser et al. 1995)	FACT	High
Student question *quality* (not count) links to achievement after experience (Graesser & Person 1994)	FACT	High
"Do you understand?" answers mislead; answer quality is better signal (Person et al. 1994)	FACT	High
Politeness strategies can inhibit effective tutoring (Person et al. 1995)	FACT	High
Prompt/scaffold-suppressed-explanation tutoring ≈ didactic on learning (Chi et al. 2001)	FACT	High
Teacher–AI co-orchestration with real-time awareness can improve learning vs AI-classroom without it (Holstein line)	FACT	Medium–High (classroom context)
Coaching improves instruction (~0.49 SD) and achievement (~0.18 SD); scale shrinks effects (Kraft et al. 2018)	FACT	High
Map-briefed + coached MindCraft tutors beat unbriefed warmth on FEI/transfer	HYPOTHESIS	Medium
SAFE-HITL is right commercial mapping	FOUNDER BELIEF	Medium

LXVIII.13 What this chapter kills

1. **Kill:** Warm human presence / Ivy credentials as quality proof.

2. **Kill:** Session hours or booking volume as North Star.
3. **Kill:** “Got it?” / smile / star-rating as mastery or QA.
4. **Kill:** Discord/Slack community ≡ tutor training or QA.
5. **Kill:** Explain-first tutor brand (“we explain better than ChatGPT”).
6. **Kill:** AI-replaces-tutors *and* humans-don’t-need-Map dual fantasies.
7. **Kill:** One-off onboarding without coaching cycles (ignore Kraft scale wound).
8. **Wound:** Overclaiming Lumilo classroom effect sizes as MindCraft ACT insurance.
9. **Survive:** Graesser naturalistic baseline + training targets; Chi prompt default; Holstein complementarity; Kraft coaching-with-fidelity; HITL-1…5; Map-briefed FEI ops.

Doctrine until data: Human tutors are a **trainable operating system** on top of diagnosis — not a marketing prop and not an unexamined 1:1 myth. Without playbooks, QA, and FEI training, “human-in-the-loop” is just another expensive chatbot with a face.

Part LXIX — Privacy & Student Affect Data

Chapter status: Living evidence + product/ethics brief — Researcher tick 2026-08-03

Primary question: What affect data may MindCraft collect, infer, store, and act on — especially anxiety/stress telemetry — without turning identity transformation into emotional surveillance?

Owners: Product (check-in / recommend modifiers) · Engine (affective_state reads) · Legal/trust · Parent marketing · Tutor ops (brief hygiene)

Commercial job: Ship a **SAFE-PRIVACY** doctrine: voluntary, purpose-bound, short-lived, non-biometric affect *signals* that soften pedagogy — never face/voice emotion AI, never Anxiety Score™, never “FERPA compliant” as ethics proof, never trait labels for tutors or ads.

LXIX.1 Why this chapter exists

Parts XXIV / XLI / XLVII / LVIII / LXIII already treat affect as load-bearing for learning. The live stack already uses a **pre-session text check-in** (short free text → structured fields → `affective_state/{student_id}/latest`) so `/recommend` can soften `target_mastery` when stress is high and inject self-reported struggles. That is a product bet: affect informs *dose and destake*, not a mood dashboard brand.

The commercial temptation is the opposite: webcam “engagement,” voice stress, continuous Anxiety Score™, tutor heatmaps of “frustrated,” parent dashboards of raw emotion, or selling “empathic AI” against Khan/Duo. Emotion AI edtech markets that pitch. The research and child-rights literature say that pitch is scientifically fragile and ethically hot.

FOUNDER BELIEF under audit: MindCraft can use *self-authored* affect for FEI safety without becoming an emotional surveillance company — *if* collection is narrow, reversible, and never treated as ground-truth emotion.

Claims we refuse as doctrine:

1. Facial / vocal emotion AI ≡ personalization or empathy.
2. Anxiety Score™ / continuous mood meter as North Star or parent report.
3. “FERPA compliant” / school-official exception ≡ ethical clearance.
4. Opt-out checkboxes as meaningful consent for minors under compulsion.
5. Inferred affect as durable trait in tutor briefs (SAFE-EXPECTANCY collision).
6. Affect data as ad targeting, insurer, or third-party enrichment feedstock.
7. Soften-challenge forever because a model “detected stress.”

LXIX.2 Constructs (privacy language for product)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Affective computing	Systems that sense/respond to affect (Picard)	Soften recommend; expose ladders	Webcam “empathy” cosplay
Emotion recognition tech (ERT)	Claim to read emotion from face/voice/biometrics	**Default ban**	Barrett wound + McStay/Katirai harms
Self-report check-in	Student-authored state in words/scales	Pre-session 2–3 sentences → structured fields	Coercion; trait storage; no expiry
Contextual integrity	Privacy = appropriate flows for context norms	Tutoring context ≠ workplace emotion monitoring	“Parents already know stress” excuse
Purpose limitation	Use only for stated education/pedagogy purpose	Soften dose / inject struggle / destake	Scope creep to marketing/scoring
Emotional surveillance	Continuous monitoring that shapes behavior under watch	Live face/voice dashboards	Panopticon anxiety; performative calm
SAFE-PRIVACY	Doctrine for affect data	See LXIX.9	Privacy theater via policy PDF

Operational definition (HYPOTHESIS): An affect data practice is *MindCraft-honest* when (a) the student (or parent for young minors) knowingly authors the signal, (b) purpose is limited to session pedagogy modifiers + optional tutor *task* briefing, (c) retention is short and deletable, (d) no biometric ERT, (e) no durable emotion trait on the student card, and (f) refusal to check in still yields a usable (less customized) path.

LXIX.3 Affective computing ≠ license to watch faces

FACT (field founding + early ethics): Picard (1997, *Affective Computing*, MIT Press) framed computing that relates to, arises from, or deliberately influences emotion — and devoted explicit attention to moral/social risks (privacy, manipulation, deceptive machines). Picard (2003, *International Journal of Human-Computer Studies*, 59(1–2), 55–64, doi:10.1016/S1071-5819(03)00052-1) restates the privacy criticism head-on: emotions are "ultimately personal and private"; detection/recognition/manipulation can be the "ultimate breach" if done without acceptable norms — while also arguing ethical HCI uses exist when transparent and user-benefiting.

Commercial implication: Citing Picard to sell classroom cameras is selective quotation. The founding literature already flags privacy as first-class. MindCraft's lawful heir is **user-controlled self-report → respectful response**, not covert sensing.

LXIX.4 Emotion-from-face science does not support product ERT

FACT (expressions are not reliable emotion meters): Barrett, Adolphs, Marsella, Martinez & Pollak (2019, *Psychological Science in the Public Interest*, 20(1), 1–68, doi:10.1177/1529100619832930) — the common view that a person's emotional state can be readily inferred from facial movements is **not** supported as a strong, general rule. Expressions vary across cultures, situations, and people; similar face configurations map to multiple categories; scowls often communicate something *other* than the expected emotion. They urge caution for AI, education, and security consumers of emotion research.

FACT (edtech emotional AI risk): McStay (2020, *Learning, Media and Technology*, 45(3), 270–283, doi:10.1080/17439884.2020.1686016) — evaluates facial-coding emotional AI marketed into education/SEL quantification; finds serious method risks (material effects on students) plus ethical/legal clash of private EdTech interests with public education goods; concludes **significant risks** in classroom deployment.

FACT (ethics review of ERT): Katirai (2023, *AI and Ethics*, doi:10.1007/s43681-023-00307-3) — structured review (43 articles): biased/unfair outcomes from faulty premises; sensitivity of emotion data; harm risk in consequential settings including **education**; guidelines stress defined scope, fairness, privacy — and the review finds issues may be "potentially insurmountable" even as commercial ERT expands.

FACT (soft biometrics / weak privacy consensus): McStay (2020, *Big Data & Society*, doi:10.1177/2053951720904386) — stakeholder interviews + UK survey: only a *weak* consensus on privacy for emotional AI / soft biometrics; limited window to set stronger norms before deployment normalizes.

Kill: Webcam engagement meters, "we can see when you're stuck," voice-stress ACT coach, any vendor demo that maps smile→mastery.

Wound: Text sentiment models over check-in prose are *not* Barrett-safe either — they are inferences. Prefer structured student choices ("stressed / okay / ready") plus free text stored as evidence of *self-report*, not as ground-truth emotion labels sold upstream.

LXIX.5 Student privacy law is procedural — ethics still required

FACT (FERPA's limits in big-data edtech): Zeide (2016, *Drexel Law Review*, 8, 339–480; SSRN:2821837) — FERPA's FIPPs-flavored access-control model **delegates** most privacy decisions to institutions via broad exceptions (notably the school-official exception); transparency/oversight/accountability are thin; notice-and-consent is a poor fit under compulsory schooling and credential pressure.

FACT (purpose-limitation theater): Zeide (2017, *University of Miami Law Review*, 71(2), 494–533) — "education purpose" limitations in FERPA/state reforms are often **procedural** (on behalf of / authorized by schools), not substantive student-interest protections; data-driven tools change what student information *is* and how decisions are made; ethical discourse must go beyond checkbox purpose.

FACT (contextual integrity): Nissenbaum (2010, *Privacy in Context*, Stanford University Press; see also 2004, *Washington Law Review*, 79, 119) — privacy violations are inappropriate information *flows* relative to context norms (actors, attributes, transmission principles), not merely "data left the device."

HYPOTHESIS (MindCraft mapping): A stress check-in that only the recommend engine and (optionally) a Map-briefed tutor see for *this session*, with short TTL, can respect tutoring-context norms. Piping the same signal into parent weekly "Anxiety Score," marketing cohorts, or tutor trait cards **breaks** contextual integrity even if a lawyer stamps FERPA.

Kill: Marketing "bank-grade / FERPA compliant" as proof we may collect facial emotion or forever-retain affect. Compliance ≠ appropriateness.

LXIX.6 What MindCraft should collect (and refuse)

FOUNDER BELIEF → product rule (provisional SAFE-PRIVACY):

Allowed (with hygiene)	Banned by default
Voluntary pre-session text / short Likert stress	Webcam, mic emotion, wearables for "engagement"
Explicit struggle tags student selects	Auto-inferred personality / "fragile" traits
Session-scoped modifiers (soften mastery target; destake)	Permanent Anxiety Score™ on profile / parent SKU
Aggregate, de-identified product analytics	Sale/enrichment of affect to third parties
Tutor brief: *task* implications only ("high stress → destake first item")	Tutor brief: emotion labels as person descriptors
Student delete / skip check-in without product exile	Dark-pattern mandatory mood gate every open

Cross-links: SAFE-TIMING (stress softens load); SAFE-DD / SAFE-EXPOSE (destake before dose); SAFE-EXPECTANCY (no trait labels); SAFE-EXAM (learn vs prove rails); SAFE-HITL (tutors see Map tasks, not psych dossiers); SAFE-STRUCTURE (scarce bandwidth ≠ moral failure).

HYPOTHESIS: Soften-on-high-stress survives only if we measure that softened paths still produce `retry_120s` and delayed mix — not merely calmer NPS. Permanent under-challenge from stale stress flags is a privacy *and* learning failure.

LXIX.7 Commercial positioning

Vs Emotion-AI EdTech: Do not compete on “empathic cameras.” Compete on **honest self-report → inspectable pedagogy change** (Map + FEI). That is a trust wedge for parents who fear surveillance schools and for districts whose procurement now asks biometric questions.

Vs ChatGPT tutors: Generic chat may *perform* empathy in text while logging prompts indefinitely. MindCraft can claim *narrower* affect handling: purpose-bound fields, short retention, no biometric ERT — but only if true in the privacy policy *and* the schema.

Parent copy (SAFE-WTP compatible): “We ask how today’s session feels — so practice doesn’t pile on when stress is high. We don’t watch faces or sell mood data.” Do not promise clinical mental-health care.

Growth: Treat privacy review packets (data map, retention, ERT ban, delete path) as a **sales asset** for school pilots — Zeide implies schools outsource tools; buyers need substantive answers, not FERPA slogans.

LXIX.8 Experiments (PRIV family)

ID	Contrast	Primary endpoints	Kill / survive rule
PRIV-1	Check-in on (session TTL) vs no check-in	`retry_120s`; delayed mix; skip rate	Check-in harms transfer or >X% feel coerced → redesign/remove
PRIV-2	Soften-on-high-stress vs ignore stress	WM-demand item accuracy; return; transfer	Ignore ≥ soften on learning without affect cost → modifier useless
PRIV-3	Task-only tutor affect note vs emotion-label brief	Student trust items; expectancy language in tutor talk	Label brief ↑ trait talk / ↓ challenge → ban labels (EXPECTANCY)
PRIV-4	24h TTL vs 30d retain affect fields	Modifier usefulness; parent/student deletion requests	Long retain no lift → shorten; creep detected → kill retain
PRIV-5	Explicit ERT mock (face) vs text check-in preference	Forced choice; WTP attribute in CBC	Face preferred *and* lifts FEI → reopen only with ethics board (unlikely)
PRIV-QUAL	10 Maya + 10 parent interviews on affect data comfort	Codebook: surveillance vs care	Feeds WTP “no webcam emotion” attribute

Ties: TIM-3, EXP-O-, *EXAM*-, HITL-1, WTP-*, SAFE-EXPECTANCY.

LXIX.9 SAFE-PRIVACY doctrine (provisional)

1. **Self-authored first.** Prefer student (or parent-mediated) report over inferred biometrics.
2. **Purpose-bound.** Affect → pedagogy modifiers and task briefs only; never ads, scores-as-SKU, or trait archives.
3. **Short TTL + delete.** Session-to-days, not forever; visible delete/skip.
4. **ERT ban.** No face/voice/wearable emotion recognition in product roadmap without extraordinary evidence *and* rights review (Barrett + McStay bar is high).
5. **Contextual integrity.** Flows must match tutoring norms; parent summaries = actions taken ("we destaked"), not raw emotion dumps.
6. **Refusal-safe.** Skip check-in still gets a full product.
7. **Compliance ≠ ethics.** FERPA/COPPA/state student-privacy laws are floors; Nissenbaum appropriateness is the product bar.
8. **Measure learning, not calm.** Soften flags judged by FEI metrics, not Anxiety Score™ decline.

LXIX.10 Confidence table

Claim	Label	Confidence
Picard founding work includes privacy/manipulation concerns (1997; 2003 IJHCS)	FACT	High
Facial movements are unreliable general emotion meters (Barrett et al. 2019)	FACT	High
Classroom facial emotional AI carries significant method + ethics risks (McStay 2020 LMT)	FACT	High
ERT ethics literature flags bias, sensitive data, education harms (Katirai 2023)	FACT	High
FERPA/school-official + purpose limits are weak substantive privacy for edtech (Zeide 2016/2017)	FACT	High
Contextual integrity is the right evaluation frame for affect flows (Nissenbaum)	FACT (framework)	High
Text check-in soften → better FEI than no affect use	HYPOTHESIS	Medium
SAFE-PRIVACY is right commercial mapping for MindCraft	FOUNDER BELIEF	Medium–High

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LXIX.11 What this chapter kills

1. **Kill:** Facial/vocal/wearable emotion AI as MindCraft personalization.
2. **Kill:** Anxiety Score™ / continuous mood dashboard as NS or parent SKU.
3. **Kill:** "FERPA compliant" as substitute for substantive affect ethics.
4. **Kill:** Opt-out theater / mandatory mood gates for minors.
5. **Kill:** Emotion labels as durable tutor-brief traits.
6. **Kill:** Affect data sale, ad cohorts, or insurer-style enrichment.
7. **Kill:** Empathy-camera marketing vs Khan/Duo.
8. **Wound:** Treating LLM sentiment over journals as "measured anxiety."
9. **Survive:** Voluntary short-TTL self-report → soften/destake; PRIV-1…5; Barrett/McStay/Zeide/Nissenbaum stack; ERT ban as trust wedge.

Doctrine until data: Affect is for **permission and dosing**, not for watching. If the company needs a camera to know a student is scared, it has already lost the FEI plot — and the parent trust market.

Part LXX — Constitution Meta: How the Lab Updates Beliefs

Chapter status: Living evidence + process brief — Researcher tick 2026-08-03

Primary question: How should MindCraft revise Constitution claims so the company gets *truer* product law — not a longer PDF, not a shrine of founder priors, and not "science-backed" marketing from unfalsified prose?

Owners: Research Lab · Founders · Product/analytics (experiment falsifiers) · Brand (claim ladder hygiene)

Commercial job: Ship a **SAFE-LABMETA** doctrine: label → check → kill/wound/survive → demote marketing until FEI evidence; Bayesian *practice* means model-check and revise, not vibes-in-Bayes-clothing.

LXX.1 Why this chapter exists

The Constitution is already large. Without a belief-update OS, volume becomes cargo cult: agents append chapters, confidence tables fill with "Medium," and marketing quietly promotes FOUNDER BELIEF as FACT because the PDF looks thick.

FOUNDER BELIEF under audit: A living research Constitution can be a *competitive* asset — sharper copy, killed bad metrics, defensible trust packets — *if and only if* it behaves like a research programme with falsifiers, not a manifesto that only grows.

Claims we refuse as doctrine:

1. Chapter count / page count / "150–300 pages" as progress North Star.
2. "Bayesian" branding without prior, likelihood, or model check — vibes posterior.
3. Cite-washing: DOI lists that do not change a product rule or kill a slogan.
4. Repetition of FOUNDER BELIEF until it is treated as FACT.

5. Synthesizer as marketing refresh that preserves weak claims.
6. "Science-backed" / "research Constitution" in ads without named falsifiers and claim ladder L0–L4.
7. Fake numeric confidence (e.g. "87% sure") without evidence protocol.

LXX.2 Constructs (lab language for company law)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Falsifiability	Scientific claims risk refutation by observation (Popper)	Every SAFE-* row needs a kill condition	Unfalsifiable identity slogans
Research programme	Hard core + protective belt of auxiliaries (Lakatos)	FEI hard core; SAFE-* as adjustable belt	Hardening every slogan as core
Progressive vs degenerating	Novel predictions confirmed vs endless ad-hoc patches	Experiments that bite product	Adding caveats forever, never killing
Bayesian practice	Build → infer → *check model against data* → revise (Gelman/Shalizi)	Tick: write → Red Team → experiment → synthesizer	Prior worship / no checks
Researcher degrees of freedom	Flexible analysis inflates false positives (Simmons et al.)	Post-hoc metric shopping after A/B	"It worked on engagement"
Pre-registration / disclosure	Timestamp plans; reduce publishability bias (Nosek et al.)	Part IX families; Appendix D sketches	Secret endpoint swaps
Replication reality	Many published effects shrink or vanish (OSC; Ioannidis frame)	Distrust one study → ACT guarantee	Overclaim from single paper

Operational definition (HYPOTHESIS): A Constitution update is *honest* when (a) claims keep FACT/HYPOTHESIS/FOUNDER BELIEF/SPECULATION labels, (b) Red Team can name what would kill the claim, (c) commercial copy cannot climb the claim ladder without matching evidence, and (d) synthesizer *removes* as often as it merges.

LXX.3 Falsifiability is the lab's anti-marketing vaccine

FACT (demarcation / method): Popper (1959 English ed. of *Logik der Forschung*; Routledge reprints) argues scientific theories are distinguished by *falsifiability*: universal claims are never proven by accumulation of confirming instances, but can be contradicted by reproducible effects. Single stray anomalies do not auto-refute; *reproducible* refuting effects do.

Commercial implication: MindCraft marketing that cannot state a disconfirming user outcome ("if `solo_transfer_pass` does not beat unguarded chat, we retire Solver-as-hero") is not "science-backed" — it is branding with footnotes.

Kill: "We transform identity" without behavioral co-primaries (SAFE-IDMEASURE / LONGID).

Survive: Soft-wrong, Map diagnosis, PWC — each already carries falsifiers in Part IX families.

LXX.4 Treat the Constitution as a Lakatosian programme — not a Kuhn shrine

FACT (research programmes): Lakatos (1970, in Lakatos & Musgrave, *Criticism and the Growth of Knowledge*; developed in *The Methodology of Scientific Research Programmes*) models science as sequences of theories with a **hard core** protected by a **protective belt** of auxiliaries. The negative heuristic forbids directing every anomaly at the core; the positive heuristic guides how to revise the belt. Programmes are judged progressive vs degenerating by whether they yield novel confirmed content rather than endless patchwork.

HYPOTHESIS (MindCraft mapping):

- **Hard core (methodologically sticky):** Fear → Evidence → Identity (FEI); identity ≠ answer delivery; explanations-are-cheap market hypothesis; claim ladder L0–L4.
- **Protective belt (must take hits):** Individual SAFE-* product rules, UX defaults, copy bans, experiment families.
- **Degenerating smell:** New chapters that only add caveats around a failing metric (e.g. streaks) without killing the metric.

FOUNDER BELIEF: FEI may stay core while many SAFE-* rows die. That is health, not betrayal.

Kill: Treating every Red Team kill as optional flavor text while shipping the banned feature anyway.

Wound: Over-hardening FEI itself — if transfer never moves and threat never drops, the core must face a crisis review (quarterly "what we believed that was wrong" memo in Part XXIII).

LXX.5 "Bayesian" without model checking is cosplay

FACT (practice ≠ inductivist myth): Gelman & Shalizi (2013, *British Journal of Mathematical and Statistical Psychology*, 66(1), 8–38, doi:10.1111/j.2044-8317.2011.02037.x) argue that successful Bayesian *data analysis* fits hypothetico-deductive practice: models are built, posteriors computed, then **checked** (e.g. posterior predictive checks) and **revised**. Model checking sits outside naive "Bayes = automatic rationality / pure induction." Priors are part of the model, not sacred beliefs immune to data.

Commercial implication: Lab agents must not write "we Bayesian-updated to High confidence" after citing three friendly papers. The analogue of a posterior predictive check is: **did the experiment family falsify the product rule?** If experiments are unrun, confidence stays capped (typically Medium for HYPOTHESIS; FOUNDER BELIEF never becomes FACT by eloquence).

Lab update rule (HYPOTHESIS → doctrine candidate):

1. **Prior:** label + stated confidence + commercial temptation.

2. **Likelihood:** citations with contradicting evidence mandatory.
3. **Check:** Red Team tick + named falsifier in Part IX.
4. **Revise:** kill / wound / survive; synthesizer promotes only survivors to I.4.
5. **Demote copy:** marketing may use only surviving I.4 rows at the claim-ladder level earned.

Kill: Bayesian Score™ / "posterior confidence" widgets in investor decks.

Survive: Epistemic labels + experiment falsifiers + synthesizer deletions.

LXX.6 Why published ed/psych findings are a hostile prior

FACT (positive predictive value frame): Ioannidis (2005, *PLoS Medicine*, 2(8), e124, doi:10.1371/journal.pmed.0020124) argues many published research findings are likely false when studies are small, effects small, designs flexible, bias high, or pre-study odds poor — claimed findings can track prevailing bias.

FACT (replication estimate): Open Science Collaboration (2015, *Science*, 349(6251), aac4716, doi:10.1126/science.aac4716) — 100 psychology replications: original significant rate ~97%; replication significant rate **36%**; replication effect sizes roughly **half** originals on average.

FACT (flexibility): Simmons, Nelson & Simonsohn (2011, *Psychological Science*, 22(11), 1359–1366, doi:10.1177/0956797611417632) show undisclosed flexibility in collection/analysis ("researcher degrees of freedom") can make almost any hypothesis appear significant; propose disclosure requirements (and relatedly motivate pre-registration culture).

FACT (incentive repair): Nosek, Spies & Motyl (2012, *Perspectives on Psychological Science*, 7(6), 615–631, doi:10.1177/1745691612459058) — incentives favor novelty/publishability over truth; propose restructuring practices toward transparency (including open workflow tools such as OSF) so accuracy can compete with publishability.

Commercial implication: MindCraft must not turn a single encouraging tutoring or mindset paper into ACT-point packaging. External literature sets **priors and ban lists**; company truth requires **MindCraft experiments** (FEI endpoints), not literature cosplay.

Kill: "Meta-analysis says tutoring works → our app raises ACT 3 points."

Survive: Literature → mechanism hypothesis → pre-registered family → ship only what survives.

LXX.7 How a tick is supposed to change belief (not page count)

Map the existing cadence (skill + Part XXIII) onto Gelman/Shalizi + Lakatos:

Tick	Epistemic job	Degenerate failure
Researcher	Add labeled claims + citations + commercial implication + experiment IDs	Cite-wash; no falsifier; fluff length

Red Team	Attempt kill of weakest claim; update confidence/contradictions	Soft applause; no log of kill/wound
Synthesizer	Merge survivors into I.4; *delete* duplicates/weak args; refresh Exec Summary	Accumulate rows; never demote

HYPOTHESIS (stopping rule): If ≥8 researcher ticks land without synthesizer, provisional SAFE- *rows are* not* company law for ads — they are candidates. Marketing that quotes provisional doctrine as settled FACT violates claim ladder.

SPECULATION: Unattended agents drift toward agreeable synthesis; Red Team hours (UTC % 6 == 0) exist to counter that drift — unless the next chapter was never written (prefer Researcher so the queue does not starve).

LXX.8 Commercial positioning

Vs competitors' "science-backed" pages: Do not out-footnote Khan/Duo/Brilliant. Out-*kill*: publish (internally, then selectively externally) what metrics and slogans you retired. Trust is a progressive research programme signal.

Parent/district copy: "We pre-register what 'better' means — retry, transfer, solo prove — and we drop features that only juice streaks." Never: "Our 200-page Constitution proves identity transformation."

Growth: Experiment instrumentation (Part XXI) is GTM infrastructure: schools and serious parents buy *inspectable* claims. SAFE-LABMETA makes SAFE-WTP and district privacy packets coherent — same honesty stack.

Investor hygiene: Deck claims must cite I.4 survivors + claim ladder level, not chapter count since last board meeting.

LXX.9 Experiments (LABMETA family)

ID	Contrast	Primary endpoints	Kill / survive rule
LABMETA-1	Pre-registered FEI endpoints vs post-hoc engagement primary	Pre-reg adherence; false-positive feature ships	Post-hoc primary wins shipping → process fail; restore pre-reg gate
LABMETA-2	Red Team mandatory before I.4 promote vs promote-on-write	Kill rate; later reverse-kills	Zero kills over N ticks → RT theater; change RT incentives
LABMETA-3	Marketing copy lint vs I.4+ladder vs unrestricted "science-backed"	Claim-ladder violations caught; parent trust items	Lint catches nothing while ads overclaim → strengthen lint

LABMETA-4	Synthesizer deletion quota (must demote ≥1 weak row) vs merge-only	Rows removed; Exec Summary length	Merge-only → synthesizer fail; enforce demote
LABMETA-5	External skeptical reader quarterly vs internal-only	Claims wounded/killed by outsider	No external wounds ever → capture; require outsider
LABMETA-QUAL	5 staff write “belief we retired” memos	Codebook of real vs performative kills	Feeds quarterly wrong-belief memo

Ties: DAG- *(estimands), Part XXI metrics, claim ladder L0–L4, all SAFE-* falsifiers.

LXX.10 SAFE-LABMETA doctrine (provisional)

1. **Labels are load-bearing.** FACT/HYPOTHESIS/FOUNDER BELIEF/SPECULATION stay visible; repetition does not promote labels.
2. **Falsifier or it does not ship as law.** No SAFE-* without kill condition and experiment family hook.
3. **Check, then update.** Citations without Red Team / model-check analogue are bibliography, not belief change (Gelman & Shalizi).
4. **Hard core is small.** FEI + claim ladder + cheap-explanations market bet; everything else is belt.
5. **Progressive or cut.** Auxiliaries that only patch failing North Stars (streaks, calm meters, costume pedagogy) get killed, not footnoted.
6. **Hostile prior on single papers.** OSC/Ioannidis/Simmons frame: external lit informs; MindCraft FEI data decides shipping.
7. **Disclosure over cleverness.** Pre-reg endpoints; no silent metric swaps (Nosek; Simmons).
8. **Synthesizer must delete.** Merge without demotion is document inflation.
9. **Page count is vanity.** Truth density and killed claims are the lab KPIs.
10. **Marketing ≤ evidence.** “Science-backed” only for I.4 survivors at earned ladder level.

LXX.11 Confidence table

Claim	Label	Confidence
Popperian falsifiability as demarcation/method ideal (1959 LoSD)	FACT (philosophy)	High
Lakatos hard core / protective belt / progressive vs degenerating (1970)	FACT (framework)	High
Bayesian practice centers model checking + revision (Gelman & Shalizi 2013)	FACT	High

Many published findings have poor PPV / replication attrition (Ioannidis 2005; OSC 2015)	FACT	High
Analysis flexibility inflates false positives; disclosure helps (Simmons et al. 2011)	FACT	High
Incentives skew publishability; transparency reforms help (Nosek et al. 2012)	FACT	High
SAFE-LABMETA maps correctly onto MindCraft commercial risk	FOUNDER BELIEF	Medium–High
Enforcing deletion quotas + copy lint raises parent trust	HYPOTHESIS	Medium

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LXX.12 What this chapter kills

1. **Kill:** Page/chapter count as lab or company North Star.
2. **Kill:** “Bayesian update” theater without checks, experiments, or label discipline.
3. **Kill:** Cite-washing that never changes I.4 or ads.
4. **Kill:** Promoting FOUNDER BELIEF → FACT by repetition or PDF girth.
5. **Kill:** Synthesizer-as-brochure that never demotes weak doctrine.
6. **Kill:** “Science-backed Constitution” marketing without falsifiers + claim ladder.
7. **Kill:** Fake precision confidence numbers as persuasion.
8. **Wound:** Treating FEI hard core as unchallengeable forever without quarterly crisis review.
9. **Survive:** Label → Red Team → experiment → synthesizer demote/promote; Gelman/Shalizi check ethos; Lakatos belt discipline; hostile prior on single-study overclaim; LABMETA-1…5.

Doctrine until data: The lab’s product is **revised belief that constrains shipping and copy** — not pages. If the Constitution cannot kill a beloved feature, it is a mood board with DOIs.

Part LXXI — Peer vs Near-Peer vs Expert Tutoring

Chapter status: Living evidence + hire/ops brief — Researcher tick 2026-08-03

Primary question: At what *grain of tutor expertise* does MindCraft get learning and identity gains — same-age peer, near-peer (college→HS), or expert (teacher/pro) — and what hire bar survives evidence rather than Ivy theater?

Owners: Tutor ops · Hiring · Product (tutor matching) · Positioning vs marketplace “elite tutors” / peer-only Discord help

Commercial job: Ship a **SAFE-TUTORGRAIN** doctrine: MindCraft's default human is a *trained near-peer operator on the Map* — not "any peer is fine," not "PhD/Ivy ≡ better FEI," and not Bloom 2σ expert cosplay. Hire for trainable Map fidelity + prompt>pour; measure grain with transfer, not résumé.

LXXI.1 Why this chapter exists

Part LXVIII (SAFE-HITL) specifies *how* tutors operate once hired. This chapter specifies *who* to hire and what "expertise" is allowed to mean in copy. MindCraft's production model is college tutors with high-school students — **near-peer / cross-age** in Topping's typology — while competitors sell either peer Discord vibes or "expert" marketplace credentials. The wrong grain kills unit economics *or* pedagogy: untrained same-age peers default to knowledge-telling; credentialed experts still fail diagnosis and can over-pour; untrained near-peers reproduce Graesser's naturalistic baseline without FEI.

FOUNDER BELIEF under audit: Near-peer + Map + HITL playbook beats (a) expert-without-Map lecture and (b) peer-without-structure chat — on solo transfer and recoverable struggle — at a cost parents will pay.

Claims we refuse as doctrine:

1. Expert / Ivy / teacher-credential ≡ better FEI outcomes without Map+playbook.
2. Any peer help (Discord, classmate) ≡ MindCraft tutoring.
3. Bloom "2-sigma human tutors" as marketing (standing ban; VanLehn wound).
4. Domain-expert monologue ≡ tutoring quality (Chi interactive kill already).
5. Tutor learning as a substitute for *student* transfer (Roscoe/Chi bias).
6. Hire bar = GPA / school brand without fidelity QA.

LXXI.2 Constructs (ops language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Same-age peer	Tutor ≈ tutee grade/age; often reciprocal	Optional student↔student practice (not default paid tutor)	Unstructured Discord ≡ product
Near-peer / cross-age	Tutor older/further along same trajectory	College tutor ↔ HS Maya	"Big sibling vibes" without playbook
Expert tutor	Teacher / experienced / high domain+pedagogy	Rare escalate / coach-of-tutors	Ivy/PhD SKU as quality
Knowledge-telling	Deliver summary; low monitoring/elaboration	Explain-first college tutor default	Fluency without FEI
Knowledge-building	Monitor, integrate, elaborate (tutor *and* tutee)	Prompt>pour; soft-wrong springboard	Treat as automatic from "being smart"

Expertise grain	How much credential/experience changes outcomes	Hire bar + matching policy	Resume theater
SAFE-TUTORGRAIN	Doctrine for who tutors	See LXXI.9	Peer-chaos or expert-cosplay

Operational definition (HYPOTHESIS): A MindCraft human tutor is *grain-honest* when they are (a) near-peer by default (college→HS), (b) screened for Map-brief fidelity and prompt-first moves, not school brand, (c) coached under SAFE-HITL, and (d) escalated to expert grain only for packetized hard cases (SAFE-REPAIR / clinical-affect out of scope) — never marketed as "experts on every call."

LXXI.3 Peer tutoring works — and structure is the moderator

FACT (classic school meta): Cohen, Kulik & Kulik (1982, *American Educational Research Journal*, 19(2), 237–248, doi:10.3102/00028312019002237) — 65 tutoring-program evaluations; tutored students outperform controls on exams and show more positive subject attitudes; *tutors* also gain understanding/attitudes; little/no self-esteem effect. Peer/near-peer school tutoring is not a null intervention.

FACT (updated peer meta + moderators): Leung (2015, *Journal of Educational Psychology*, 107(2), 558–579, doi:10.1037/a0037698) — comprehensive peer-tutoring meta; positive impact on achievement; moderators include role structure (same-age reciprocal vs nonreciprocal vs cross-age), educational level, and program parameters grounded in role theory / interdependent contingencies — i.e. **design beats "put two kids together."**

FACT (HE typology): Topping (1996, *Higher Education*, 32(3), 321–345, doi:10.1007/BF00138870) — peer tutoring in further/higher education has many formats; cross-year small-group, PSI, and supplemental models are among those with substantial evidence and wider dissemination merit. MindCraft's college→HS pairing is closer to **cross-age / cross-year** than to same-year reciprocal undergrad SI — do not cite Topping as "Discord study groups work."

Commercial implication: Peer/near-peer is *evidence-eligible* as a delivery grain. Unstructured peer chat is not. Sell structured near-peer sessions, not "community help."

LXXI.4 Expert humans are not 2σ magic — and diagnosis is weak

FACT (human tutoring ES vs folklore): VanLehn (2011, *Educational Psychologist*, 46(4), 197–221, doi:10.1080/00461520.2011.611369) — review finds human tutoring effect size ≈ **d = 0.79** vs no-tutoring same content — far below Bloom's folklore ~2.0; step-based ITS ≈ **d = 0.76**, nearly matching human. Interaction-plateau: finer-than-step grain does not reliably beat step grain.

FACT (tutors mis-model students): Chi, Siler & Jeong (2004, *Cognition and Instruction*, 22(3), 363–386, doi:10.1207/s1532690xci2203_4) — novice-but-domain-knowledgeable tutors monitor student understanding poorly; miss most student-initiated knowledge deficits; everyday tutors are usually novices (Fitz-Gibbon cited therein) yet still produce gains (ties Cohen et al.).

Cross-link: Graesser naturalistic tutors lack deep ideal strategies (Part LXVIII); Putnam-style curricular scripts appear even in experienced tutors (cited in tutoring/ITS literature). Expertise ≠ accurate student model.

Kill: "Our expert tutors close the 2-sigma gap." Kill resume≡diagnosis. Wound: treating VanLehn's human≈ITS parity as "AI replaces humans" — MindCraft's bet is FEI + Map complementarity (SAFE-HITL / SAFE-ONTOLOGY), not ITS cosplay.

LXXI.5 Peer tutors default to knowledge-telling

FACT (tutor learning review): Roscoe & Chi (2007, *Review of Educational Research*, 77(4), 534–574, doi:10.3102/0034654307309920) — peer tutors *can* learn from tutoring via reflective knowledge-building (monitor, integrate, elaborate), but show a pervasive **knowledge-telling bias** — summarizing/delivering with little monitoring — even when trained; tutor-learning potential is often unrealized.

FACT (process study): Roscoe & Chi (2008, *Instructional Science*, 36(4), 321–350, doi:10.1007/s11251-007-9034-5) — untrained peer tutors learn most when explaining/responding incorporates reflective knowledge-building; knowledge-telling dominates; inferential tutee questions more often elicit elaborative tutor responses.

HYPOTHESIS (MindCraft): College near-peers without HITL training will **pour** (tell) under parent pressure to "explain until she gets it," reproducing the bias Roscoe/Chi document — and colliding with Chi et al. (2001) prompt≈didactic lesson (SAFE-HITL). Training must suppress telling-as-default, not celebrate "natural explainers."

Kill: Hire for "great at explaining." Kill tutor-learning / "tutors grow too" as the student-facing value prop (nice externality; not the NS).

LXXI.6 Near-peer is the default grain — expert is escalate, not SKU

HYPOTHESIS (grain map):

Grain	When it earns keep	When it fails
Same-age peer	Optional structured practice pairs; CoP peripheral work with old-timer (SAFE-CoP)	Paid product; unmoderated forums; peer ranks (SAFE-COMPARE)
Near-peer (college→HS)	Default paid human; Map-briefed; prompt-first; FEI QA	Vibes-only; knowledge-telling; trait expectancy briefs
Expert (teacher/coach)	Coach-of-tutors; hard escalate packets; rare clinical/pedagogy consult	Every-session expert SKU; 2σ marketing; lecture-first "master teacher"

FOUNDER BELIEF: Near-peer *rapport + recent struggle memory* can support belonging/recognition (Part LXI) without expert costume — *if* Map prevents "I remember how hard this was" from becoming soft-wrong avoidance (politeness tax, Part LXVIII).

SPECULATION: Parents may *say* they want experts in surveys and *choose* near-peer+Map+transfer report in CBC (test under SAFE-WTP; do not treat Likert "prefer certified teacher" as WTP).

Wound: Classroom peer-tutoring metas ≠ remote 1:1 ACT prep. Borrow structure moderators (Leung), not K–8 effect sizes as ACT guarantees.

LXXI.7 Hire bar (product-facing)

HYPOTHESIS — minimum viable tutor screen (ordered):

1. **Content floor:** Can work target ACT-band items at recommended level without Solver crutches (spot-check).
2. **Map fidelity drill:** Given a fake brief, names concept/bridge/format and next move — no trait labels (SAFE-EXPECTANCY).
3. **Prompt>pour drill:** On a soft-wrong, asks before explains; refuses "got it?" as close (SAFE-HITL).
4. **Repair drill:** On planted AI wrongness, owns + Map-checks + destaked reattempt path (SAFE-REPAIR).
5. **School brand / GPA:** Tie-break only — never primary.

Ops: Expert grain = internal coach + sampled session QA (Kraft coaching fidelity wound already in LXVIII), not a parent-facing "Expert Tutor" upsell that implies every minute is master-teacher.

LXXI.8 Positioning

Competitor pattern	What they optimize	MindCraft TUTORGRAIN counter
Marketplace "elite tutors"	Credentials / hourly	Near-peer + Map + fidelity QA; publish grain honesty
Peer Discord / homework help	Availability / vibe	Structured session ≠ chat; CoP ban without repertoire
AI-only tutors	Fluency / cost	Human near-peer for escalate + recognition; solo transfer NS
School peer programs	Coverage / cost	Cross-age trained operators, not same-age homework swap

Marketing that survives: "College tutors trained on your living Map — not random homework help." / "We hire for how they coach struggle, not the logo on their hoodie." / "Expert coaches train our tutors; your session is aimed near-peer time, not a TED talk."

Marketing that dies: "PhD/Ivy tutors"; "2-sigma expert instruction"; "peers teaching peers" as unpaid Discord; "any human is better than AI" without grain+playbook.

LXXI.9 SAFE-TUTORGRAIN stack (surviving doctrine)

1. **Default grain = trained near-peer** (college→HS), not same-age peer product and not expert-every-call.
2. **Structure > pairing** — Leung/Topping: role design and training beat “put humans together.”
3. **No 2σ / expert-magic claims** — VanLehn human≈0.79; Bloom folklore banned.
4. **Expertise ≠ student model** — Chi/Siler/Jeong monitoring wound; Map carries diagnosis.
5. **Suppress knowledge-telling** — Roscoe/Chi; prompt>pour hire+QA.
6. **Hire bar = fidelity drills**, not school brand.
7. **Expert = escalate + coach-of-tutors**, not default SKU.
8. **Tutor learning ≠ student NS** — externality only.
9. **Ties:** SAFE-HITL, SAFE-ONTOLOGY, SAFE-EXPECTANCY, SAFE-REPAIR, SAFE-CoP, SAFE-WTP.

LXXI.10 Experiments (GRAIN family)

ID	Contrast	Primary endpoints	Kill / survive rule
GRAIN-1	Trained near-peer vs untrained near-peer (same Map)	Soft-wrong rate; talk ratio; `solo_transfer_pass`	Untrained ≈ trained → training theater
GRAIN-2	Near-peer+Map vs expert-credential tutor without Map brief	Map agreement; FEI events; transfer	Expert-no-Map ≥ near-peer+Map → revisit grain bet
GRAIN-3	Near-peer+HITL vs same-age structured peer pair	Transfer; `challenge_accept`; safety codes	Peer-pair ≥ near-peer on transfer → paid grain unjustified
GRAIN-4	Hire-by-fidelity-drills vs hire-by-school-brand	30d rubric fidelity; escalate appropriateness	Brand hires win fidelity → drills wrong
GRAIN-5	Parent CBC: near-peer+Map report vs “certified expert” attribute	Choice share; WTP	Expert attribute dominates at equal price → message/package work (not auto-hire experts)
GRAIN-QUAL	10 tutor + 10 parent interviews on “who should tutor Maya”	Codebook: credential vs process trust	Feeds copy + GRAIN-5

Ties: HITL-, *ONTO*-, EXP-, *REPAIR*-, WTP-, *CoP*-.

LXXI.11 Commercial implications

Copy: Lead with *trained near-peer on your Map*. Never lead with Ivy. Never imply every tutor is a master teacher. Never call Discord “tutoring.”

Product: Matching defaults to near-peer; expert badge only on coach-of-tutors / escalate path. Instrument GRAIN-1/2 before raising expert SKUs. Tutor application = drills in LXXI.7.

Growth: B2B story = consistent cross-age operators under one playbook (cheaper than expert marketplace, more accountable than peer programs). Consumer story = "someone who recently walked this path — with a diagnosis, not vibes."

Vision: Identity transformation needs honest witnesses (LXI). Near-peers can be witnesses without colonizing as gurus — if grain and playbook stay honest for thirty years of hire scale (ties id 75 workforce drift).

LXXI.12 Confidence table

Claim	Label	Confidence
School tutoring programs help tutees (and often tutors) on achievement/attitudes (Cohen et al. 1982)	FACT	High
Peer tutoring positive on achievement; structure/role moderators matter (Leung 2015)	FACT	High
HE peer tutoring has evidenced formats; cross-year among stronger lines (Topping 1996)	FACT	Medium–High (review, heterogeneous)
Human tutoring ~d=0.79; ITS ~d=0.76; not Bloom 2.0 (VanLehn 2011)	FACT	High
Novice tutors monitor understanding poorly (Chi, Siler & Jeong 2004)	FACT	High
Peer tutors show knowledge-telling bias; knowledge-building rarer (Roscoe & Chi 2007/2008)	FACT	High
Trained Map-briefed near-peers beat expert-no-Map and peer-no-structure on FEI/transfer	HYPOTHESIS	Medium
SAFE-TUTORGRAIN is right commercial mapping	FOUNDER BELIEF	Medium

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LXXI.13 What this chapter kills

1. **Kill:** Ivy / PhD / "expert tutor" résumé as quality proof or default SKU.

2. **Kill:** Bloom 2-sigma human-tutor marketing (reinforce standing ban via VanLehn).
3. **Kill:** Unstructured peer Discord / classmate help ≡ MindCraft tutoring.
4. **Kill:** “Great explainers” hire bar (knowledge-telling bias).
5. **Kill:** Expertise ≡ accurate diagnosis (Chi monitoring wound; Map required).
6. **Kill:** Tutor-learning / “tutors grow too” as student North Star.
7. **Wound:** Importing K–8 peer-meta effect sizes as ACT score guarantees.
8. **Wound:** Reading VanLehn ITS≈human as license for AI-only brand without FEI.
9. **Survive:** Cohen/Leung/Topping structure-eligible peer/near-peer; VanLehn anti-2σ; Chi monitoring + Roscoe/Chi telling bias; GRAIN-1…5; trained near-peer default + expert-as-escalate.

Doctrine until data: MindCraft hires **trainable near-peer operators**, not peer chaos and not expert cosplay. Expertise grain is a product decision with falsifiers — résumé theater is not a research conclusion.

Part LXXII — Generated Question Validity & Key Risk

Chapter status: Living evidence + bank/ops brief — Researcher tick 2026-08-03

Primary question: When may MindCraft *generate* practice/diagnostic items (LLM or template), and what validity / answer-key failure modes make generation a product hazard rather than a coverage moat?

Owners: Engine (generation / verify) · Product (Practice / diagnostic bank) · Trust & Safety · Positioning vs “AI made infinite questions” competitors

Commercial job: Ship a **SAFE-GENQ** doctrine: generation is a *coverage accelerator under a verify loop*, never a substitute for keyed truth; bad keys poison the mastery graph and identity evidence; sell *bank trust*, not item count.

LXXII.1 Why this chapter exists

Parts XXXIII / LX / LXVII already kill fluent wrongness as tutoring, apology-as-repair, and chat-as-diagnosis. The remaining bank-specific hazard is quieter: **MindCraft’s own generation path can invent stems that look ACT-ready while shipping wrong keys.** Internal verify already dropped ~30% of a batch (104 kept / 45 dropped) — too high to scale into the live bank. Diagnostic hide-correctness (C4) and `/record-outcomes` make a wrong key *especially* dangerous: the product records “failure” against a false ground truth and rewrites Maya’s Map.

FOUNDER BELIEF under audit: Ontology-joined generation + deterministic key verify can fill uncovered concepts/formats faster than human authoring — *if and only if* shipping gates treat key validity as a hard fail, not a soft QA note.

Claims we refuse as doctrine:

1. LLM-generated items ≡ ready-to-ship practice without independent key verify.
2. Item-count / coverage % as North Star or marketing hero.
3. Unverified keys as diagnostic or mastery ground truth.
4. “AI wrote a thousand questions” as trust or ACT readiness.

5. Isomorphic clones / surface variants ≡ far transfer or exam readiness.
6. Fluency of stem / explanation ≡ arithmetic or keyed correctness.
7. Verify-off generation to “move fast” on identity or FEI claims.

LXXII.2 Constructs (bank language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Automatic item generation (AIG)	Model/template-driven item production with cognitive + psychometric intent	`ml/generation/` essence → LLM → format-tagged `Question`	Free-form LLM dump without item model
Item model / template	Spec of which features may vary	Layer-3 joins + essence prompts; C5 `Question` schema	Prompt-only “models” with no constrained slots
Key / scoring inference	Map observed response → scored outcome	`correctAnswer` / options; bank-key verify (AIT-6)	Wrong key → poisoned events
Isomorph / clone	Surface variants intended as psychometrically similar	Format/surface rewrites of a parent	Clone flood ≡ learning
Verification / review loop	Content+logic check before operational use	`--verify` pass; drop list; human spot-check	Soft “looks fine” gate
Interpretation/use argument (IUA)	Kane: validate *uses* of scores, not “the test”	Gap-scan / practice outcomes → Map / FEI claims	Ship items → claim identity without key evidence
SAFE-GENQ	Generation under verify doctrine	See LXXII.9	Coverage theater

Operational definition (HYPOTHESIS): A generated item is *shippable* only when (a) it matches C5 schema and ontology `conceptId`/`format`, (b) the keyed answer survives an independent check (symbolic/arithmetic and/or SME), (c) distractors are not merely fluent nonsense, (d) it is barred from hide-correctness diagnostic and mastery writes until (b) passes, and (e) batch drop rate is below an explicit gate (target \<~10% before scale; current ~30% = **do not scale**).

LXXII.3 What classical AIG actually earned

FACT (item models as AIG spine): Gierl & Lai (2012, *International Journal of Testing*, 12(3), 273–298, doi:10.1080/15305058.2011.635830) — AIG is a two-step practice: specialists build *item models* (templates specifying manipulable features), then algorithms instantiate many items. Benefits include volume and potential parameter prediction — not “LLM fluency.”

FACT (review process for generated items): Gierl & Lai (2016, *Educational Measurement: Issues and Practice*, 35(4), 6–20, doi:10.1111/emip.12129) — generated items still need a structured review focused on the *content and logic in the generation procedure*, not vibes after the fact. Illustration domains include mathematics (CCSS-linked) and surgical education.

FACT (quality vs traditional items — distractor wound): Gierl, Lai & Turner (2012, *Medical Education*, 46(8), 757–765, doi:10.1111/j.1365-2923.2012.04289.x) and follow-on quality comparisons (e.g. Gierl et al., *Medical Education*, 2013/quality indicators line) — AIG items can approach traditional quality on many indicators; **distractors** are a recurring weak spot (implausible options more often than hand-authored items). Experts cannot reliably "eyeball which is AIG."

FACT (feasibility / Kane framing in medical AIG): Reviews of AIG in medical assessment (e.g. Falcão et al., 2022, *Advances in Health Sciences Education*, doi:10.1007/s10459-022-10092-z) map validity work onto Kane-style inferences (scoring → generalization → extrapolation → implication). Scoring/construction clarity and psychometric spread can support use; **implication** (high-stakes decisions) is not free.

Commercial implication: MindCraft may borrow *strong-theory AIG discipline* (constrained models + review of the generator, not only the stem). It may **not** borrow "we generated thousands, therefore bank is valid."

LXXII.4 Validity is about *use* — Kane binds the bank

FACT (argument-based validity): Kane (2013, *Journal of Educational Measurement*, 50(1), 1–73, doi:10.1111/jedm.12000) — what is validated are *interpretations and uses* of scores, not the instrument as a fetish object. More ambitious claims need more evidence. Negative consequences can kill a *use* even if a thinner interpretation survives.

Applied to MindCraft (HYPOTHESIS):

Use	Ambition	Key-risk if wrong
Low-stakes practice with reveal + repair	Low–medium	Trust ding; recoverable via SAFE-REPAIR
Hide-correctness diagnostic → `/seed-assessment` / outcomes	High	Silent Map corruption; false weakness
Parent "diagnosis report" / FEI evidence	High	WTP and trust burn (SAFE-WTP / SAFE-REPAIR)
ACT-readiness or identity claims	L3+ (banned from L1 item A/Bs)	Claim-ladder violation (XXXIV)

Kill: "Items exist in the bank" ≡ validated for diagnostic or identity use.

Survive: Separate *coverage use* (practice under reveal) from *evidence use* (diagnostic / mastery writes) until key gates pass.

LXXII.5 LLM generation is not classical AIG (key failure rates)

Classical AIG manipulates constrained slots inside an item model. LLM generation samples fluent text that *looks* like an item. That difference matters for keys.

FACT (ungrounded GPT math error rates): Bastani et al. (2025, *PNAS*, doi:10.1073/pnas.2422633122) — GPT Base correct only ~**51%** when repeatedly asked for answers on practice problems; logical errors ~42%, arithmetic ~8%; problem-specific heterogeneity. Students often fail to detect or refuse to check. Unguarded use ↑ concurrent practice then ↓ solo exam.

FACT (classroom ChatGPT wrongness): Primary-school robot/ChatGPT study (IEEE AIRC 2024, doi:10.1109/airc61399.2024.10672220) — LLM inaccurate on ~**13/29** curriculum maths prompts (~55% accuracy); students' ability to spot errors tracked *their own* correctness — prior knowledge gates detection.

FACT / REVIEW (LLM math learning concerns): Findings-EMNLP analyses (e.g. "Three Questions…" 2023, doi:10.18653/v1/2023.findings-emnlp.201) — LLMs can mis-grade human work, invent wrong intermediates, and need tool/augmentation discipline; fluent rationale ≠ valid math.

FACT (internal engineering audit): MindCraft generation `--verify` pass retained 104 / dropped 45 (~**30%** bad-key rate). Lab standing order already: drop rate must fall before `--tested --formats all` scale and before `syncGeneratedQuestions` into the live bank.

Kill: Prompt-scale generation without verify as "coverage solution."

Wound: Treating Bastani's *tutor* wrongness as identical to *item-authoring* wrongness — related mechanism (fluent false math), different product surface (bank key vs live chat).

Survive: AIT-6 bank-key verify as load-bearing; generation inherits PWC — LLM is bookend, deterministic check is spine.

LXXII.6 Clones, coverage, and the fake-mastery path

FACT (isomorph intent vs clone pejorative): AIG literature distinguishes isomorphs (psychometrically similar variants) from free variants (Gierl & Lai ITEMS module / Gierl & Haladyna AIG line; Bejar item-modeling tradition). Specialists often disparage shallow 1-layer "clones" as detectable and pedagogically thin.

HYPOTHESIS (MindCraft): Format-tagged generation that only swaps surface nouns while freezing the same arithmetic skeleton inflates `questionCount` without buying SAFE-TRANSFER / interleaving discrimination. Clone floods can also teach answer patterns rather than strategy selection.

Ties: SAFE-TRANSFER (story≠hop; blocked≠ready), SAFE-MISCON (distractors must be diagnostic), SAFE-ONTOLOGY (coverage fills holes; coverage ≠ product), SAFE-CALIB (hide-correctness only on keyed truth).

Kill: "1,500 questions" / "AI-generated bank" as the marketing hero metric.

Survive: Publish *verify rate*, *key-fail taxonomy*, and *concepts still unverified* more loudly than raw count when talking to serious buyers.

LXXII.7 Failure taxonomy (ops)

Failure class	Example	Student harm	Graph harm	Gate
Arithmetic key fail	Stem OK; keyed option wrong	Learns false fact; trust burn	False miss/hit events	Hard fail
Logical key fail	Wrong method encoded as correct	Misconception installed	Bridge/concept mis-update	Hard fail
Ambiguous stem	Multiple defensibly correct	Frustration / “gotcha” affect	Noisy outcomes	Soft fail / rewrite
Implausible distractors	Options no Maya would choose	Easy gaming; shallow practice	Overconfident mastery	Soft fail
Format lie	Tagged `diagram` but text-only / unreadable	Format-axis diagnosis false	Format-gap severity wrong	Hard fail for format claims
Ontology misjoin	Wrong `conceptId` / ingredient	Wrong path / cards	Poisoned recommendations	Hard fail
Clone-only batch	N surface twins	Illusion of variety	Inflated exposure counts	Soft fail / cap

FOUNDER BELIEF: Arithmetic/logical key fails are *product-stopping* for diagnostic and mastery writes; ambiguous/implausible are *batch-quality* issues that still block “science-backed bank” copy.

LXXII.8 Positioning

Competitor pattern	What they optimize	MindCraft GENQ counter
“Infinite AI questions”	Volume / novelty	Verify gate; publish drop reasons
Unguarded ChatGPT homework	Fluency / speed	Bank keys + Map; no live key from chat
Static textbook banks	Stability	Generate only into holes; human/ACT seeds remain gold
Clone drill apps	Throughput	Cap isomorphs; measure delayed mix not count

Marketing that survives: “Every practice key is checked before it can rewrite your Map.” / “We generate to fill holes — we ship only what survives verify.” / “Coverage without corrupted diagnosis.”

Marketing that dies: “AI-generated question bank” as hero; “unlimited unique ACT questions”; “our AI never gets the answer wrong”; using generation volume to imply exam readiness or identity change.

LXXII.9 SAFE-GENQ stack (surviving doctrine)

1. **Verify-before-ship** — no live-bank sync without independent key check.
2. **Key fails are hard fails** — arithmetic/logical wrongness blocks diagnostic + `/record-outcomes` paths.
3. **IUA honesty (Kane)** — bank existence ≠ validated diagnostic/identity *use*.
4. **LLM ≠ classical AIG** — constrain with essence/ontology joins; never prompt-and-pray.
5. **Drop-rate gate** — do not scale while verify drop ■ ~10%; current ~30% = pause scale.
6. **Coverage ≠ North Star** — fill ontology holes; measure transfer / FEI, not `questionCount`.
7. **Clone discipline** — isomorph caps; format tags must be true.
8. **Repair continuity** — if a shipped key is later found wrong: SAFE-REPAIR + event quarantine playbook.
9. **Ties:** SAFE-REPAIR, SAFE-ONTOLOGY, SAFE-CALIB (C4), SAFE-TRANSFER, SAFE-MISCON, PWC/AIT-6, SAFE-LABMETA (no cite-wash of "AI bank").

LXXII.10 Experiments (GENQ family)

ID	Contrast	Primary endpoints	Kill / survive rule
GENQ-1	Verify-on vs verify-off generated items in *revealed* practice	Key-error rate spotted by students; `retry_120s`; trust items	Verify-off ≈ verify-on on trust → verify theater
GENQ-2	Verified generated vs human/ACT-seed items (same concept/format)	Delayed mix / `solo_transfer_pass`	Generated ■ seed → generation only for holes, not replacement
GENQ-3	Diagnostic eligibility: verified-only vs mixed bank under C4 hide-correctness	Post-scan Map agreement with tutor gold; false-weakness rate	Mixed bank ↑ false weakness → keep diagnostic sealed
GENQ-4	Prompt-hardened arithmetic vs baseline generator	Verify drop rate; residual fail taxonomy	Drop still ≥20% after harden → architecture change (tool-calc / template-first)
GENQ-5	Parent CBC: "verified bank" attribute vs "AI-generated unlimited Qs"	Choice share; WTP	Unlimited-AI attribute wins → message work, not ship unverified
GENQ-QUAL	10 tutor reviews of dropped vs kept items	Codebook: fail classes; SME disagreement rate	Feeds GENQ-4 prompt/tool design

Ties: AIT-6, REPAIR-, *ONTO*-, CAL-, *TR*-, WTP-, *LABMETA*-.

LXXII.11 Commercial implications

Copy: Lead with *checked keys* and *honest coverage*. Never lead with generation volume. Never imply generated ≡ ACT-official. Never claim diagnostic science on unverified items.

Product: Keep generated JSON inert until drop rate clears gate; seal C4 diagnostic + mastery writes to verified keys only; instrument key-fail reports from tutors; quarantine path when a live key is revoked.

Growth: District/parent trust packet (ties id 73) should show verify rates and refusal to ship bad keys — same posture as SAFE-PRIVACY honesty. Competitors selling infinite AI drills are the foil: *we refuse to poison the graph*.

Vision: Thirty-year identity product needs a bank Maya can *believe*. A single wrong key in a hide-correctness diagnostic teaches the company the wrong student. Generation remains a tool for holes under ontology joins — never the adjudicator of truth.

LXXII.12 Confidence table

Claim	Label	Confidence
AIG rests on item models + algorithmic instantiation (Gierl & Lai 2012)	FACT	High
Generated items require procedure-focused review (Gierl & Lai 2016 EMIP)	FACT	High
AIG distractors often weaker than traditional (Gierl/Lai/Turner medical AIG line)	FACT	Medium–High
Validity concerns interpretations/uses; ambitious uses need more evidence (Kane 2013)	FACT	High
Unguarded GPT Base ~51% correct on Bastani practice answer queries; harms solo exam	FACT	High
ChatGPT/robot maths answers ~55% accurate in cited primary classroom study	FACT	Medium (single study)
MindCraft verify drop ~30% on audited batch	FACT (internal audit)	High for that batch; medium as ongoing rate
Prompt hardening + tool-calc can push drop \<10% without abandoning LLM stems	HYPOTHESIS	Medium
SAFE-GENQ is the right commercial mapping	FOUNDER BELIEF	Medium–High

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LXXII.13 What this chapter kills

1. **Kill:** LLM-generated items ≡ ready-to-ship bank without independent key verify.
2. **Kill:** Item-count / “AI wrote N questions” as hero metric or ACT/identity proof.
3. **Kill:** Unverified keys as diagnostic (C4) or `/record-outcomes` ground truth.
4. **Kill:** Fluency of stem/explanation ≡ keyed correctness.
5. **Kill:** Clone/isomorph floods ≡ transfer or exam readiness.
6. **Kill:** Scaling generation while verify drop remains ~30%.
7. **Wound:** Equating classical constrained AIG validity evidence with unconstrained LLM item dumps.
8. **Wound:** Using Bastani chat-tutor error rates as a precise forecast of *authoring* error without batch audits.
9. **Survive:** Gierl/Lai item-model discipline; Kane IUA for bank *uses*; Bastani + classroom LLM wrongness as hazard priors; internal verify audit; GENQ-1…5; verify-before-ship + diagnostic seal.

Doctrine until data: MindCraft generates to **fill ontology holes**, ships only **verified keys**, and treats key failure as a **graph integrity** incident — not a content backlog inconvenience.

Part LXXIII — District Procurement & Privacy Review as GTM

Chapter status: Living evidence + GTM/ops brief — Researcher tick 2026-08-03

Primary question: What does a U.S. K–12 *district* actually require before MindCraft can sell into schools — and how should that privacy/procurement gate reshape packaging, copy, and roadmap?

Owners: GTM / partnerships · Legal/trust · Product (data inventory) · Parent B2C (contrast) · Engine (affect / PII surfaces)

Commercial job: Ship a **SAFE-PROCURE** doctrine: district revenue is gated by a *trust packet* (DPA + data map + deletion/breach + no-biometric / no-ad use), not by Bloom theater or “FERPA compliant” badges; privacy review *is* the sales process.

LXXIII.1 Why this chapter exists

Parts XXVII / LIX sell parents. Part LXIX constrains affect telemetry. Part LX kills “never hallucinates.” Part LXXII seals the bank. None of that closes a **district** deal. District buyers run a different machine: legal review, data-privacy agreements, security questionnaires, sometimes state-specific overlays — often *before* a pilot classroom ever opens.

FOUNDER BELIEF under audit: MindCraft’s identity / FEI story only reaches school budgets if procurement friction is treated as a *product surface* (shippable packet + NDPA readiness), not a lawyer afterthought once a champion teacher clicks “I agree.”

Claims we refuse as doctrine:

1. “FERPA compliant” / school-official badge ≡ district-ready or ethical clearance.

2. Teacher click-wrap ToS as a substitute for LEA-controlled contracts.
3. Biometric / face-voice “empathy” as a school differentiator.
4. Free teacher tools bypass the same privacy scrutiny as paid tools.
5. Marketing-first RFP chase before a data inventory + DPA pack exists.
6. Parent B2C WTP copy pasted into district RFPs as the trust story.
7. Anxiety Score™ / sell-affect / ad targeting of pupil records (SAFE-PRIVACY collision).

LXXIII.2 Constructs (procurement language)

Construct	Research / policy meaning	MindCraft analogue	Failure mode if misused
LEA / district buyer	Local educational agency that owns procurement + policy	School/district SKU path	Selling only to a teacher’s card
DPA / NDPA	Data privacy agreement; SDPC National Data Privacy Agreement template	Signed DPA + Exhibit A data map	One-off legal chaos per district
School official (FERPA)	Exception allowing disclosure under direct control for legitimate educational interest	Contractual school-official posture	Exception ≡ ethics / marketing stamp
Click-wrap ToS	Teacher-accepted provider terms	Banned as primary school path	Binds LEA to hostile terms
Trust packet	Pre-assembled answers buyers ask	DPA + inventory + retention/deletion + breach + subprocessors + ERT ban	PDF theater without product truth
State overlay	SOPPA, CA EDC 49073.1, etc.	Exhibit G / state terms readiness	Assume federal floor is enough
SAFE-PROCURE	Doctrine: privacy review = GTM gate	See LXXIII.9	Compliance cosplay

Operational definition (HYPOTHESIS): MindCraft is *district-sellable* when (a) a standard NDPA-compatible DPA is ready, (b) Exhibit A lists every student-data category actually collected (including check-in affect and mastery events), (c) purpose limitation + no targeted advertising + deletion-on-exit are product-true, (d) biometric ERT is contractually and productively banned, and (e) free classroom pilots use the same approval path as paid — not a ToS side door.

LXXIII.3 Federal floor: FERPA / PPRA / COPPA bind the *school*, not your badge

FACT (ED PTAC — online services guidance): U.S. Department of Education Privacy Technical Assistance Center (PTAC), *Protecting Student Privacy While Using Online Educational Services: Requirements and Best Practices* (Feb 2014) — when schools use third-party online educational services as part of school activity, FERPA (and often PPRA/COPPA) still apply; best practice is written agreements covering collection, use, retention, disclosure, destruction, security, breach, and access — not informal app installs.

FACT (ED PTAC — Model Terms of Service): PTAC, *Protecting Student Privacy While Using Online Educational Services: Model Terms of Service* (Mar 2016; ERIC ED585306) — click-wrap TOS that a teacher accepts can govern what a provider may collect, use, and share; those terms may conflict with FERPA/PPRA/best practice. PTAC supplies a checklist of "GOOD!" provisions (purpose limitation, security, modification notice/consent, breach response, etc.) for evaluating provider agreements.

FACT (CoSN toolkit — partnering / click-wrap risk): Consortium for School Networking (CoSN), *Protecting Privacy in Connected Learning* toolkit Part 2 (*Partnering with Service Providers*, updated ~2023) — flowchart for FERPA/PPRA/COPPA decisions; warns that teacher click-wrap can bind the school system to terms that misalign with district policy and law; ED recommends free online educational services go through the *same or similar* approval process as paid services; FTC best practice: COPPA appropriateness decisions at district/admin level, not delegated to individual teachers.

Commercial implication: District GTM is blocked or delayed by *agreement shape*, not by whether MindCraft's landing page says "privacy-first." A champion teacher who click-wraps MindCraft can create legal risk for the LEA — and create a *bad* sales motion for us (unsigned, non-compliant, later ripped out).

Kill: "Teachers love us / free classroom → privacy doesn't apply."

Survive: Same approval rigor for free pilots as for paid SKUs.

LXXIII.4 Standardization: SDPC / NDPA as the industry contracting rail

FACT (SDPC National DPA): Student Data Privacy Consortium (SDPC / Access 4 Learning Community) publishes the **National Data Privacy Agreement (NDPA)** (v2 announced Apr 2024; subsequent point releases) to set common expectations between LEAs and providers and reduce one-off contracts across ~13,000 U.S. districts. NDPA explicitly frames Provider as **School Official** with legitimate educational interest under FERPA when LEA policy permits, under LEA direct control; DPA terms take precedence over conflicting ToS/privacy policies for Student Data treatment; cites FERPA (20 U.S.C. § 1232g / 34 C.F.R. Part 99), PPRA (20 U.S.C. § 1232h), COPPA (15 U.S.C. §§ 6501–6506).

FACT (registry scale — SDPC claims): SDPC states that since 2016, hundreds of thousands of standard DPAs have been executed/subscribed (public materials cite 275,000+ and a large Resource Registry of signed agreements). Treat the *existence of a shared rail* as FACT for GTM planning; treat exact dollar-savings marketing math as vendor/community estimate (not MindCraft-validated).

HYPOTHESIS (MindCraft): Refusing NDPA-compatible contracting while chasing district RFPs maximizes legal cost and cycle time. The moat is not "we invented a unique privacy PDF" — it is product truth that maps cleanly into Exhibit A (data categories) and Exhibit B (schedule of data) without lying about affect check-ins, LLM subprocessors, or tutoring session artifacts.

Kill: Custom 40-page privacy essay as the only offer.

Survive: NDPA-ready pack + honest Exhibit A as table stakes; state exhibits as overlays.

LXXIII.5 State overlays are not optional footnotes

FACT (California contract mandates): California Education Code § 49073.1 (AB 1584) — LEA contracts for digital storage of pupil records and/or digital educational software that accesses/stores/uses pupil records must include specified clauses, including: LEA ownership/control of pupil records; pupil-generated content retention options; prohibition on use beyond contract purposes; parent/eligible-pupil review procedures; security/confidentiality actions (training); unauthorized-disclosure notification procedures; certification of non-retention after contract end (with enforcement description); joint FERPA compliance description; prohibition on using PII in pupil records for targeted advertising. Contracts missing required terms are vulnerable to cure/void dynamics under the statute's structure.

FACT (Illinois SOPPA): Illinois Student Online Personal Protection Act (105 ILCS 85/) — regulates covered information collected by operators for K–12 school purposes; requires written agreements covering data practices between schools and operators; defines covered information broadly (educational record fields, biometric information, evaluations, identifiers, voice recordings, geolocation, etc.); operator prohibitions include targeted advertising based on acquired information, amassing profiles except for K–12 school purposes, and selling/renting student information (with acquisition exceptions). Illinois districts commonly execute DPAs via SDPC rails under SOPPA.

SPECULATION (biometric tightening): Proposed Illinois school-code bills (e.g. SB 1239 line) seek to ban district acquisition of biometric systems for students. Direction of travel alone makes face/voice emotion AI a district GTM poison pill — consistent with SAFE-PRIVACY science kills.

Commercial implication: A "national" privacy page that ignores CA 49073.1 ad bans or SOPPA covered-info breadth fails the first serious CA/IL review. Build the packet for the *hard* states; lighter states ride along.

LXXIII.6 FERPA badge ≠ ethics, ≠ sales readiness (Zeide wound)

FACT (FERPA's institutional delegation): Zeide (2016, *Drexel Law Review*, 8, 339–480; SSRN:2821837) — FERPA's access-control model delegates most privacy decisions to institutions via broad exceptions (notably school-official); transparency/oversight/accountability are thin; notice-and-consent fits poorly under compulsory schooling.

FACT (purpose-limitation theater): Zeide (2017, *University of Miami Law Review*, 71(2), 494–533) — "education purpose" limits are often procedural (authorized by schools), not substantive student-interest protections.

FACT (contextual integrity): Nissenbaum (2010, *Privacy in Context*, Stanford University Press; also 2004, *Washington Law Review*, 79, 119) — privacy = appropriate information flows for context norms, not merely "stayed inside FERPA."

Ties to LXIX: Even a signed DPA does not make Anxiety Score™, emotion-trait tutor cards, or ad cohorts appropriate. Procurement clears *permission*; doctrine still constrains *use*.

Kill: Homepage "FERPA compliant" as the district hero claim.

Survive: “We sign NDPA-class DPAs; LEA owns pupil records; no targeted ads; no biometrics; deletion on exit; here’s the data map.”

LXXIII.7 What buyers actually ask (trust packet contents)

HYPOTHESIS (MindCraft trust packet = GTM SKU): Ship one downloadable + SE-backed artifact covering:

1. **DPA / NDPA stance** — standard agreement + state exhibits.
2. **Data inventory (Exhibit A)** — account PII; practice/diagnostic events; Map state; session artifacts; optional affect check-in; subprocessors; explicit non-collection (face/voice biometrics, continuous mood).
3. **Purpose limitation** — pedagogy / mastery / FEI ops — not ads or enrichment.
4. **Retention / deletion** — TTLs (affect); deletion-on-exit certification.
5. **Security + breach** — access control, encryption-in-transit, ML auth, notify timeline.
6. **AI honesty** — SAFE-REPAIR + SAFE-GENQ; no “AI never wrong.”
7. **Rights path** — LEA/parent review via district intermediary (PTAC preference).
8. **Pilot ≠ click-wrap** — written approval for free trials (CoSN/ED).

FOUNDER BELIEF: The packet is more load-bearing for district ARR than another hero video. Parent WTP (SAFE-WTP) is a parallel motion — do not conflate CBC tests with LEA legal review.

LXXIII.8 Positioning vs competitors

Competitor pattern	MindCraft PROCURE counter
Free teacher viral + click-wrap	Same approval as paid; DPA-first pilots
“FERPA compliant” badge	NDPA + Exhibit A + no-ad / no-bio clauses
Emotion-AI engagement	Contractual + product ERT ban (SAFE-PRIVACY)
Infinite AI tutor / bank	Repair + verified keys in trust packet

Marketing that survives: “Built for district DPAs.” / “LEA owns the records.” / “No biometrics. No targeted ads on pupil data.” / “Affect check-ins optional, short-lived, never sold.”

Marketing that dies: “FERPA compliant AI tutor”; “teachers just click to start”; “cameras see struggle”; “unlimited free classroom without IT.”

LXXIII.9 SAFE-PROCURE stack (surviving doctrine)

1. **Privacy review = GTM gate** — no district revenue theater without a trust packet.
2. **NDPA-ready contracting** — prefer standard DPA rails over one-off essays.
3. **Click-wrap is not the school path** — written LEA-controlled agreements; free ≠ unvetted.

4. **Exhibit A honesty** — inventory includes mastery events, session artifacts, optional affect; no hidden LLM training on covered student content without contractual allowance.
5. **No targeted advertising on pupil records** — product + contract (CA 49073.1 posture nationally).
6. **Biometric / ERT ban** — procurement poison + science kill (LXIX).
7. **FERPA badge ≠ ethics** — Zeide/Nissenbaum; compliance floor under SAFE-PRIVACY.
8. **Separate motions** — parent B2C WTP ≠ district legal review; both needed.
9. **Ties:** SAFE-PRIVACY, SAFE-REPAIR, SAFE-GENQ, SAFE-HITL, SAFE-WTP, SAFE-LABMETA (no "science-backed" without falsifiers in RFP claims).

LXXIII.10 Experiments (PROCURE family)

ID	Contrast	Primary endpoints	Kill / survive rule
PROCURE-1	Trust-packet-first outreach vs feature-demo-first to district IT/privacy	Time-to-security-questionnaire complete; DPA signature rate	Demo-first wins cycle time *and* DPA rate → packet overbuilt
PROCURE-2	NDPA-standard offer vs custom long-form DPA only	Legal cycle days; counsel revision count	Custom always faster → revisit NDPA hypothesis
PROCURE-3	Free pilot requiring LEA approval vs teacher click-wrap	Pilot survival at 90d; forced-removal rate; champion NPS	Click-wrap ↑ short adoption but ↑ removals → keep approval gate
PROCURE-4	Parent B2C message vs district trust-packet message on same landing	Qualified district lead rate vs parent trial rate	One message wins both → still keep packets separate operationally
PROCURE-5	Explicit "no biometric / no pupil-data ads" attribute in district CBC-style choice	Choice share among admins	Attribute flat → still keep as risk insurance, not hero
PROCURE-QUAL	10 interviews: district privacy officers / tech directors	Codebook: blockers, Exhibit A surprises, AI red lines	Feeds packet v1 contents

Ties: PRIV-, *WTP*-, REPAIR-, *GENQ*-, HITL-*.

LXXIII.11 Commercial implications

Copy: Lead district pages with *contractable trust* (DPA, ownership, no ads, no biometrics, deletion). Never lead with Bloom 2-sigma, "AI that never errs," or emotion cameras. Keep parent score/WTP language off the procurement packet.

Product: Maintain a living data inventory tied to real collections; short-TTL affect; refuse ERT features that would fail SOPPA/biometric reviews; document subprocessors; make deletion/export paths operable, not aspirational.

Growth: Treat trust-packet completion as a pipeline stage (like SOC questionnaire). Instrument days-in-legal. Prefer SDPC/NDPA participation over inventing a boutique privacy brand.

Vision: Thirty-year identity company that cannot clear a district privacy desk does not get the students who most need structured help. Procurement hygiene is not anti-vision — it is how the vision enters schools without becoming surveillance edtech.

LXXIII.12 Confidence table

Claim	Label	Confidence
PTAC 2014/2016: written agreements + Model ToS checklist; click-wrap risk	FACT	High
CoSN toolkit: free apps need similar approval; teacher click-wrap risk; COPPA admin-level decisions	FACT	High
SDPC NDPA exists as national DPA rail; school-official framing; DPA > conflicting ToS for Student Data	FACT	High
SDPC scale/savings marketing numbers	FACT (org claim) / treat cautiously	Medium
CA EDC § 49073.1 mandatory contract clauses incl. no targeted ads	FACT	High
IL SOPPA regulates operators + written agreements; broad covered information	FACT	High
Zeide 2016/2017 FERPA limits; Nissenbaum contextual integrity	FACT	High
Pending IL biometric school bans will pass everywhere soon	SPECULATION	Low–Medium
Trust-packet-first shortens district cycle vs demo-first	HYPOTHESIS	Medium
SAFE-PROCURE is the right commercial mapping	FOUNDER BELIEF	Medium–High

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LXXIII.13 What this chapter kills

1. **Kill:** “FERPA compliant” badge ≡ district-ready or ethical clearance.
2. **Kill:** Teacher click-wrap / free viral classroom as the primary school GTM path.
3. **Kill:** Biometric / face-voice empathy features as school differentiators.
4. **Kill:** Marketing-first RFP chase without DPA + Exhibit A inventory.
5. **Kill:** Parent B2C WTP copy as the district trust story.
6. **Kill:** Targeted advertising / sell-affect on pupil records.
7. **Wound:** Treating NDPA signature alone as proof of contextual integrity (still need SAFE-PRIVACY use limits).
8. **Wound:** Assuming federal floor covers CA/IL overlays without Exhibit G work.
9. **Survive:** PTAC Model ToS discipline; CoSN free=paid approval; SDPC/NDPA rail; CA 49073.1 / SOPPA overlays; Zeide/Nissenbaum wound on badge ethics; PROCURE-1…5; trust packet as GTM gate.

Doctrine until data: MindCraft sells into districts by **shipping contractable truth** — NDPA-ready DPA, honest data map, no biometrics, no pupil-data ads, deletion on exit — and treats privacy review as the **front door**, not the cleanup crew.

Part LXXIV — Spaced Retrieval Schedules in Product UX

Chapter status: Living evidence + product/UX brief — Researcher tick 2026-08-04

Primary question: How should MindCraft schedule *returns* to concepts (lags, expanding vs equal spacing, exam-calendar anchors) so competence evidence persists — without shipping streak theater or “optimal brain algorithm” marketing?

Owners: Product · Engine (path / practice) · Growth/copy · Red Team

Commercial job: Ship a **SAFE-SCHED** doctrine: schedule retrieval to the *retention horizon* (homework days vs ACT weeks), prefer delayed first retrieval + equal-ish lags over expanding-SRS cosplay, and sell durable returns — never “our AI found your perfect interval.”

Builds on: Part XXIX (MoC / spacing substrate), XXXIX (interleaving), XLI (desirable difficulties × anxiety), XLVII (timing/sleep), LXIII (exam dual rail).

LXXIV.1 Why a second spacing chapter

Part XXIX established spacing and retrieval as *identity substrate* (Memory of Competence). This tick answers the *product* question XXIX left open: **what schedule ships in UX** when Maya has an ACT date, a homework week, and a mastery graph that currently clears concepts more than it *returns* them?

FOUNDER BELIEF under audit: MindCraft can out-position Khan/Duo/Brilliant/ChatGPT on durable competence if the practice surface schedules scientifically honest returns — not if we only mass missions until a local mastery mark.

Claims we refuse as doctrine:

1. Expanding-SRS / “optimal interval AI” as hero marketing.
2. Streaks / daily-open as the spacing engine (SAFE-HABIT collision).
3. Cram week ≡ ACT-ready (SAFE-EXAM / blocked-accuracy kill).
4. Overlearning tonight (40 more of the same) as retention strategy.
5. Restudy / tip re-read as “review mode” without generation.
6. Shuffle / random mix ≡ spaced retrieval science.
7. Guaranteed ACT points from any spacing schedule.

LXXIV.2 Constructs (schedule language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
ISI / lag	Gap between study/practice episodes	Days between concept returns	One-night massing sold as mastery
Retention interval (RI)	Delay to criterial test	Homework due / ACT date / 28d MoC check	Optimize for tonight’s score only
Massed vs spaced	Same total practice, different distribution	Mission tonight vs split across sessions	“Finish the pad” as North Star
Expanding schedule	Growing ISIs (e.g. 1-5-9)	Naive SRS / Duo-like ease	Short-term glow; long-term weaker than equal
Equal / fixed lag	Similar ISIs (e.g. 5-5-5)	Calendar returns matched to RI	Feels “less smart” than expanding UI
Retrieval practice	Generate from memory, not restudy	Soft-wrong reconstruct; spaced quiz	Hint dump / worked-example re-watch
SAFE-SCHED	Doctrine: horizon-matched returns	See LXXIV.9	Algorithm cosplay

Operational definition (HYPOTHESIS): After `transfer_pass` (or a solid concept exposure), MindCraft schedules at least one **retrieval return** at an ISI scaled to the student’s next use horizon (≈ homework days; ≈ 10–20% of RI for multi-month ACT prep per Cepeda ridgeline intuition), with the *first* return delayed enough to be effortful — not an immediate re-quiz while the worked example is still in working memory.

LXXIV.3 Spacing is real; optimal ISI scales with how long you need it

FACT (meta-synthesis): Cepeda, Pashler, Vul, Wixted & Rohrer (2006, *Psychological Bulletin*, 132(3), 354–380, doi:10.1037/0033-2909.132.3.354) — 839 assessments / 317 experiments / 184 articles on distributed practice in verbal recall; spacing and lag effects are robust; the ISI that maximizes final-test retention **increases as the retention interval increases**.

FACT (temporal ridgeline): Cepeda, Vul, Rohrer, Wixted & Pashler (2008, *Psychological Science*, 19(11), 1095–1102, doi:10.1111/j.1467-9280.2008.02209.x) — >1,350 learners; gaps up to ~3.5 months; final tests up to ~1 year; at a given RI, performance rises then falls as ISI grows; optimal gap *increases* with RI, but as a *proportion* of RI the optimal gap *shrinks* (roughly ~20–40% of a 1-week RI → ~5–10% of a 1-year RI). Educational practices that ignore this interaction are inefficient.

FACT (math massing vs distributing): Rohrer & Taylor (2006, *Applied Cognitive Psychology*, 20(9), 1209–1224, doi:10.1002/acp.1266) — college students; distributing 10 math problems across two sessions (1-week gap) vs massing 10 in one session: spacing benefit **nil at 1-week test**, **large at 4-week test**; extra same-session problems (overlearning) did **not** help 1- or 4-week retention. Textbook massing + overlearning is the wrong default format.

Commercial implication: Parent/student copy that celebrates "cleared the topic tonight" without a scheduled return is selling a 1-week illusion. Product must show *return appointments* (Map / Practice path), not only first-clear fireworks.

Kill: Overlearn-tonight / finish-the-pad as retention strategy.

Survive: Same problem budget, split across sessions matched to RI.

LXXIV.4 Expanding retrieval is not the long-horizon winner

FACT (tradition): Landauer & Bjork (1978) proposed expanding retrieval as a superior long-term schedule — still the folk intuition behind many SRS UIs.

FACT (lab reverse at delay): Karpicke & Roediger (2007, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 33(4), 704–719, doi:10.1037/0278-7393.33.4.704) — vocabulary pairs; expanding (e.g. 1-5-9) beat equal spacing at a **10-min** final test (replicating Landauer & Bjork short delays); **equal spacing (e.g. 5-5-5) beat expanding at 2 days**, with or without feedback. Experiment 3: delaying the *first* retrieval improved long-term retention; expanding the later intervals added little once first-test delay was controlled.

FACT (review nuance): Roediger & Karpicke line (incl. 2010 *Memory & Cognition* text materials; 2011 spaced-retrieval review) — expanding advantages are mixed; longer delays often favor equal spacing or null; confounds of first-test position are load-bearing.

HYPOTHESIS (MindCraft): Shipping an "expanding SRS brain" as brand differentiates on *algorithm mystique*, not on durable math competence. For ACT horizons (weeks–months), prefer **horizon-matched equal-ish lags** + delayed first return after acquisition; use short expanding only for *stabilizing* a fragile new skill in-session or next-day — never as the hero story.

Kill: "Our AI found your perfect expanding interval."

Survive: Calendar honesty: "Come back in N days because that gap matches your exam/use horizon."

LXXIV.5 Retrieval vs restudy — and the math caveat

FACT (testing effect — prose / education): Roediger & Karpicke (2006a, *Psychological Science*, 17(3), 249–255, doi:10.1111/j.1467-9280.2006.01693.x) — repeated testing beats repeated studying on delayed retention; restudy can inflate immediate confidence without durable gain. Roediger & Karpicke (2006b, *Perspectives on Psychological Science*, 1(3), 181–210, doi:10.1111/j.1745-6916.2006.00012.x) — selective review of testing as learning.

FACT (classroom math spacing × amount): Lyle, Bego, Hopkins, Hieb & Ralston (2020, *Educational Psychology Review*, 32, 277–295, doi:10.1007/s10648-019-09489-x) — engineering precalculus; within-subjects amount × spacing of quiz retrieval; **within-semester** retention benefited from more practice *and* more spacing; **across-semester** (~4 weeks into calculus) benefited **exclusively from increased spacing**. Spacing beats piling more same-window quizzes for transfer-to-next-course durability.

FACT / contradicting nuance (math meta 2025): Murray, Horner & Göbel (2025, *Educational Psychology Review*, 37, article 75, doi:10.1007/s10648-025-10035-1) — spaced vs massed in math: overall $g \approx 0.28$ (27 studies / 53 ES; robust small–medium); isolated-material subset larger ($g \approx 0.43$) than course-embedded ($g \approx 0.24$). Testing vs restudy in math: $g \approx 0.18$ across 7 studies / 32 ES, but **95% CI crossed zero** — math retrieval-vs-restudy evidence is thinner/less conclusive than spacing-vs-massing. Effects may be **smaller than verbal-domain folklore**.

Red Team wound: Do not advertise "science-backed testing effect in math" as if g were large and settled. Spacing-vs-massing is the stronger commercial claim; retrieval-vs-passive-review remains **HYPOTHESIS-strength for math procedures** pending MindCraft FEI data — still prefer generation over tip-restudy as *design default* (aligned with XXIX / XL), but measure it.

Kill: "Retrieval practice always +0.8σ in math — trust us."

Survive: Spaced returns beat cram nights; instrument generation vs restudy inside SCHED experiments.

LXXIV.6 ACT calendar vs cram week (product dual rail)

FACT (from Cepeda ridgeline + Rohrer 4-week): When RI is exam-scale (weeks–months), massed "ACT bootcamp week" optimizes the wrong end of the lag×RI surface. Rohrer's 1-week null / 4-week large gap is the cautionary tale for last-week cramming sold as readiness.

Ties to LXIII (SAFE-EXAM): Destake-learn / dose-prove. Spacing belongs on the *learn* rail; timed mixed sets belong on the *prove* rail near the exam — not day-one flood.

HYPOTHESIS — Exam Horizon Scheduler:

1. Onboarding / gap-scan captures (or estimates) exam date → RI.
2. Engine sets default ISI ≈ f(RI) using Cepeda-style proportions (conservative band; A/B later) for concepts on the ACT path.
3. After `transfer_pass`, enqueue a **Return** card on Practice/Map (not a streak nag).
4. Fail return → classify forgot vs never-had (XXIX.6); never punish with Calm Score™ / grit meter.
5. Final 7–14 days: increase *mixed / timed prove* density; do **not** abandon spaced review of weak bridges.

SPECULATION: Parents will pay for "shows up again before the test" if the dashboard shows *scheduled returns completed* and *28d retention probes* — more than for "day streak." Validate via WTP CBC (LIX), not founder vibes.

LXXIV.7 Anxiety and desirable difficulty (dose the lag)

FACT / tradition: Spacing and delayed retrieval are desirable difficulties (Bjork line) — they raise short-term failure rates.

Ties to XLI (SAFE-DD) / LVIII (SAFE-EXPOSE): Equip → destake → dose. Do not drop a cold 3-week lag on a high-anxiety novice's first success.

HYPOTHESIS — Sequenced Scheduling:

1. Acquisition block: short lags OK (same session / next day) until `retry_120s` healthy.
2. Then stretch ISI toward RI-matched equal spacing.
3. Soften targets under high affective stress (LXIX modifiers) — shorter lag, not zero challenge forever.
4. Never frame lag failure as character ("you broke your streak").

LXXIV.8 Competitive positioning (what not to out-copy)

Competitor pattern	Mechanism	MindCraft response
Duo-like expanding SRS + streak	Habit + short-lag expanding	Equal-ish horizon lags; streak ≠ learning (XXI / XLIII)
Khan mastery clear	Local accuracy	Return appointments + MoC 7/28d
Brilliant delight / puzzle binge	Engagement massing	Delight *inside* spaced sessions, not instead of them
ChatGPT cram chat	Fluency tonight	Retrieval generation + scheduled re-ask of decisive step
ACT cram packs	Massed week	Horizon scheduler + prove rail late

Competitive wedge (FOUNDER BELIEF): Sell "your Map remembers to bring it back" — inspectable returns — vs "our model optimized your brain."

LXXIV.9 Doctrine — SAFE-SCHED (provisional)

1. **Horizon-match ISI to RI** (homework vs ACT); do not use one universal "daily" as science.
2. **Delay first retrieval** after acquisition; prefer equal-ish multi-day lags for long RI over expanding-SRS hero UI.
3. **Generation > restudy** in Review/Return modes (measure; don't overclaim math testing-effect size).
4. **Overlearning tonight ≠ retention**; redistribute problem budget across sessions.
5. **Returns are product surface** (Practice/Map cards), not push-nag streaks.
6. **Exam dual rail:** spaced learn early; timed mixed prove late — never day-one flood.
7. **Anxiety-aware lag stretch** (SAFE-DD); lag failure → diagnose forgot/never-had.
8. **Copy ban:** no "perfect interval AI," no streak-as-spacing, no guaranteed ACT points from schedule.

Confidence: High that spacing-vs-massing is real (incl. math). Medium that equal > expanding at ACT lags (lab verbal evidence strong; MindCraft math FEI pending). Medium-low that math testing-vs-restudy effect size is large (Murray et al. 2025 wound). Medium that horizon-scheduler UX beats streak UX on retention *and* WTP.

LXXIV.10 Experiments

ID	Question	Design	Primary
SCHED-1	Horizon-matched equal lag vs massed clear-after-mastery	A/B after transfer_pass	retention @ 7d / 28d; solo_transfer_pass
SCHED-2	Expanding UI schedule vs equal-lag returns (same total retrievals)	A/B	14d / 28d retention; first-return success
SCHED-3	Delayed first return (≈1–3d) vs immediate same-session re-quiz	A/B	7d retention; retry_120s
SCHED-4	Generation Return (decisive step) vs restudy tip/example	A/B	retention; transfer_pass
SCHED-5	ACT-date scheduler on vs off (path concepts)	A/B / cohort	mixed delayed accuracy; exam prove set
SCHED-QUAL	10 Maya + 10 parent interviews: return-card language vs streak language	Qual	message resonance; WTP attributes

Falsifier: Challenge-seeking / streak / session minutes rise while 28-day MoC collapses → schedule theater; demote SAFE-SCHED claims and kill streak-as-spacing copy.

Pre-register: SCHED-* before shipping “SRS / smart review” marketing (SAFE-LABMETA).

LXXIV.11 So what for MindCraft commercially

- **Copy:** “We bring topics back before they fade” / “returns matched to your test date” — not “AI-optimized spaced repetition.”
- **Product:** Return queue on Practice + Map; exam-date field drives ISI; Review mode must force generation.
- **Metric:** `return_due`, `return_completed`, retention@7/28 — co-primary with FEI; streak demoted.
- **Kill list:** expanding-SRS brand, cram≡ready, overlearn-tonight, restudy-as-review, shuffle≡spacing.
- **Growth:** Parent dashboard shows scheduled returns + retention probes (feeds LIX CBC).
- **Vision:** Identity needs autobiographical *durable* evidence; scheduling is how the product keeps the evidence from evaporating between sessions.

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Part LXXV — Tutor Workforce Pipeline & Quality Drift

Chapter status: Living evidence + ops/GTM brief — Researcher tick 2026-08-04

Primary question: How does MindCraft keep *FEI fidelity* across hire → onboard → tenure → churn when college near-peers burn out, relapse into knowledge-telling, or scale faster than coaching capacity?

Owners: Tutor ops · People/hiring · Product (tutor console + QA) · Finance (unit economics) · Positioning vs marketplace “we hired N tutors” vanity

Commercial job: Ship a **SAFE-WORKFORCE** doctrine: the product is not a warm bench of humans — it is a *pipeline that preserves Map-briefed prompt>pour fidelity over tenure*. Headcount, tenure months, and “wellness” theater are not North Stars; fidelity@30/90d + solo transfer under coached tutors are.

LXXV.1 Why this chapter exists

Parts LXVIII (SAFE-HITL) and LXXI (SAFE-TUTORGRAIN) answer *how sessions run* and *who to hire*. Neither answers the scale failure mode every tutoring company hits: **quality drifts after the photo of the hire cohort**. New tutors pass drills, then under parent pressure and fatigue they pour explanations, skip Map briefs, soft-pedal soft-wrongs, and quit — forcing rehire that resets the Kraft scale wound. Competitors sell marketplace volume ("1,000 expert tutors") or peer Discord availability. MindCraft's bet dies if pipeline ops treat onboarding as a one-time PDF.

FOUNDER BELIEF under audit: A smaller coached near-peer bench with measurable fidelity over tenure beats a larger uncoached bench on `solo_transfer_pass` and parent-auditable quality — at lower long-run CAC of *quality sessions*, even if raw tutor count looks worse in a pitch deck.

Claims we refuse as doctrine:

1. Tutor headcount / hours-booked as quality or growth NS.
2. Tenure months ≡ pedagogical improvement (seniority theater).
3. One-off onboarding ≡ coaching (Kraft scale wound already in LXVIII).
4. Wellness posters / yoga stipends as substitute for workload + QA design.
5. "We retain tutors" without fidelity instrumentation.
6. Burnout as moral failure of individual tutors (structure > slogans; ties SAFE-STRUCTURE).
7. Marketplace "elite tutor" SKU as pipeline strategy.
8. Bloom 2σ / Ivy hire as attrition insurance.

LXXV.2 Constructs (ops language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Pipeline	Hire → train → deploy → coach → retain/exit	Funnel with fidelity gates at each stage	Resume pile ≡ pipeline
Quality drift	Practice fidelity decays vs playbook after deploy	Rubric score ↓ @30/90d without coaching	Blame "bad hires" only
Burnout	Exhaustion, depersonalization, reduced accomplishment	Tutor quit intent + session softness	Wellness theater without load redesign
Dosage / tutor type	Who delivers how often (Nickow framework)	Near-peer session cadence + Map support	Import K–12 school-day ES as ACT guarantee
Coaching at scale	Sustained observation-linked PD	Clip/trace QA cycles (HITL-3)	Efficacy-trial cosplay at marketplace scale
SAFE-WORKFORCE	Doctrine for durable human quality	See LXXV.9	Headcount vanity

Operational definition (HYPOTHESIS): A MindCraft tutor workforce is *pipeline-honest* when (a) hire screens use fidelity drills (SAFE-TUTORGRAIN), (b) deploy requires Map-brief open before first explain, (c) coaching cycles continue past onboarding with rubric samples, (d) burnout/load signals trigger *caseload or schedule redesign* not slogans, (e) exit interviews + fidelity trends feed hire-bar revision, and (f) marketing never equates bench size with learning quality.

LXXV.3 Tutoring works — and *who* + *how often* moderates (not "any human")

FACT (field-experiment meta): Nickow, Oreopoulos & Quan (2024, *American Educational Research Journal*, 61(1), 74–107, doi:10.3102/00028312231208687) — systematic review/meta of tutoring RCTs; overall pooled effect ≈ **0.288 SD**. Effects tend to be largest for programs using **teachers or paraprofessionals**, earlier grades, ≥**3 days/week**, and **during school**. Nonprofessional/parent lines are weaker on average in the program typology.

Wound (mandatory): This is PreK–12 *program* evidence, often school-embedded high-dosage models — not remote college→HS ACT prep marketplaces. Borrow the *moderator logic* (tutor accountability, frequency, structure), not the pooled ES as MindCraft ACT insurance. Near-peer college tutors are closer to trained non-teachers than to certified teachers — which is why SAFE-HITL playbooks + Map must carry what credential would otherwise proxy.

Cross-link: VanLehn human≈0.79 / ITS≈0.76 (LXXI); Graesser naturalistic under-strategy baseline (LXVIII). Pipeline must manufacture structure the meta says matters — it will not appear from "hire nice college kids."

Commercial implication: Sell *structured*, *frequent-enough*, *Map-accountable sessions*, not "access to humans." Kill pitch slides that lead with tutor count.

LXXV.4 Coaching lifts practice — and collapses when scaled without fidelity

FACT (coaching meta): Kraft, Blazar & Hogan (2018, *Review of Educational Research*, 88(4), 547–588, doi:10.3102/0034654318759268) — 60 causal-design studies; pooled coaching ≈ **0.49 SD** on instructional practice and ≈ **0.18 SD** on achievement. **Scale wound:** larger effectiveness trials show only a **fraction** of small efficacy-trial effects — taking coaching to scale while keeping quality is the hard problem.

HYPOTHESIS (MindCraft mapping): Quality drift is what the Kraft scale wound *looks like from inside the company*: early cohort gets founder/coach attention (efficacy); hire wave two gets a Loom + checklist (effectiveness collapse). HITL-3 (coached vs onboarding-only) is the product falsifier; WORK-* experiments extend it across tenure.

Kill: "We trained them once." Kill headcount OKRs that outrun coach capacity (coaches-per-tutor ratio as a hard constraint).

LXXV.5 Burnout predicts quit — and harms the interaction students feel

FACT (attrition meta): Madigan & Kim (2021, *Teaching and Teacher Education*, 105, 103425, doi:10.1016/j.tate.2021.103425) — teacher burnout dimensions relate positively to intentions to quit (exhaustion r+ ≈ **.41**, depersonalization ≈ **.32**, reduced accomplishment ≈ **.21**); job satisfaction relates negatively (r+ ≈ **–.40**); burnout +

satisfaction together ≈ **27%** of quit-intent variance, with burnout carrying more relative weight than satisfaction confers protection; relationships appear to have strengthened over time in their moderation checks.

FACT (exhaustion → instructional quality → students): Klusmann, Aldrup, Roloff, Lüdtke & Hamre (2022, *Journal of Educational Psychology*, 114(6), 1442–1460, doi:10.1037/edu0000703) — large multilevel sample; teachers' emotional exhaustion associated with lower student self-concept, interest, and (in German) achievement; exhaustion linked to less emotional support and classroom organization; instructional-quality paths partially mediate exhaustion→student outcomes.

HYPOTHESIS: Fatigued MindCraft tutors will (a) skip Map briefs, (b) relapse into knowledge-telling (Roscoe/Chi), (c) overuse politeness soft-pedaling (Person et al. 1995), and (d) quit — each is a *fidelity* failure, not only an HR failure. Wellness Slack channels without caseload caps do not fix Klusmann's mediation path.

Wound: Teacher classroom metas ≠ gig college tutors. Use burnout→quit and burnout→interaction-quality as *mechanisms to instrument*, not K–12 turnover rates as MindCraft forecasts.

Kill: "Tutors just need grit." Kill Burnout Score™ as student-facing or parent SKU. Kill treating retention % as proof of quality.

LXXV.6 Drift mechanisms inside MindCraft's model

HYPOTHESIS — five drift channels (ordered by ops leverage):

1. **Brief skip** — under time pressure, tutor opens chat not Map (`map_brief_open` ↓).
2. **Pour relapse** — parent "just explain it" + fatigue → knowledge-telling (SAFE-TUTORGRAIN).
3. **False-mastery probes** — "got it?" returns as session closer (Person et al. 1994).
4. **Soft-wrong avoidance** — politeness tax rises when exhausted (rapport over diagnosis).
5. **QA starvation** — hire rate > coach sample rate → Kraft effectiveness collapse.

SPECULATION: College tutors' academic calendars create *batch churn* (semesters) — pipeline must assume partial cohort replacement yearly; design for re-certify drills, not permanent tenure mythology.

FOUNDER BELIEF: Instrumenting drift channels early is cheaper than brand repair after parents notice "tutor #4 this month who just lectures."

LXXV.7 Pipeline stages (product-facing)

HYPOTHESIS — minimum viable workforce pipeline:

Stage	Gate	Fail action
Source	Near-peer grain; content floor	Reject expert-SKU theater as default
Screen	Fidelity drills (LXXI.7)	School brand never primary

Onboard	Shadow + scored Map-brief session	No deploy on deck-only
Deploy	`map_brief_open` required in console	Soft-block explain-first
Coach	Biweekly sample @30/90d rubric	Caseload pause if fidelity↓
Load	Cap concurrent FEI-heavy students	Redesign schedule before pep talk
Exit	Fidelity trend + reason codes	Feed hire/coach revision

Ties: SAFE-HITL playbook; SAFE-EXPECTANCY (no trait briefs even when tired); SAFE-REPAIR escalate packets must not become infinite unpaid emotional labor.

LXXV.8 Positioning

Competitor pattern	What they optimize	MindCraft WORKFORCE counter
Marketplace volume	Tutor count / credentials	Fidelity@tenure + Map audit trail
Peer Discord help	Availability	Structured pipeline ≠ chat bench
AI-only	Cost / 24-7	Humans kept only if WORK-* fidelity holds
“Ivy tutors” brand	Status	Hire-by-drill; coach-of-tutors internal

Marketing that survives: “Every tutor is coached against the same Map your practice uses.” / “We measure whether tutors still make students think at day 90 — not how many we hired.” / “Smaller bench, higher fidelity.”

Marketing that dies: “1,000 tutors ready now”; tenure-as-quality; wellness-as-pedagogy; retention-% hero metrics; expert marketplace cosplay.

LXXV.9 SAFE-WORKFORCE stack (surviving doctrine)

1. **Fidelity over tenure > headcount** — instrument rubric@30/90d before scaling bench.
2. **Coaching is continuous** — onboarding ≠ PD (Kraft scale wound).
3. **Coach capacity gates hiring** — no hire wave without sample-QA bandwidth.
4. **Burnout → redesign load**, not grit posters (Madigan/Klusmann mechanisms).
5. **Drift channels named** — brief skip, pour relapse, got-it?, soft-wrong avoid, QA starve.
6. **Grain stays near-peer default** — expert = escalate/coach (SAFE-TUTORGRAIN).
7. **Retention without fidelity is vanity** — keep rate subordinate to FEI QA.

8. **No Burnout Score™ / wellness theater as product.**
9. **Ties:** SAFE-HITL, TUTORGRAIN, EXPECTANCY, REPAIR, STRUCTURE, WTP (auditable quality attribute).

LXXV.10 Experiments (WORK family)

ID	Contrast	Primary endpoints	Kill / survive rule
WORK-1	Continuous coaching vs onboarding-only (extend HITL-3 to 90d)	Rubric fidelity@30/90d; `solo_transfer_pass`	No fidelity lift → coaching theater
WORK-2	Hire-rate capped by coach capacity vs uncapped surge hire	Fidelity variance; escalate appropriateness	Surge ≥ capped on fidelity → capacity gate wrong
WORK-3	Caseload cap + schedule redesign vs wellness-module add-on	Quit intent proxy; brief-skip rate; pour rate	Wellness ≥ cap on drift → load redesign unjustified
WORK-4	Console hard-require Map-brief vs optional brief	`map_brief_open`; soft-wrong rate; talk ratio	Optional ≈ required → UX nag enough
WORK-5	Parent CBC: "coached fidelity report" vs "more tutors / Ivy" attributes	Choice share; WTP	Volume/Ivy dominates → packaging work (not auto-abandon fidelity)
WORK-QUAL	10 tutor + 10 parent interviews on drift/churn trust	Codebook: what "quality held" means	Feeds copy + WORK-5

Ties: HITL-, *GRAIN*-, EXP-, *WTP*-, REPAIR-*.

LXXV.11 Commercial implications

Copy: Lead with *coached operators whose quality is checked over time*. Never lead with bench size. Never imply seniority months = better FEI.

Product: Tutor console enforces Map-brief; ops dashboard shows fidelity@tenure and coach utilization; pause hiring when QA bandwidth saturated. Instrument WORK-1…4 before marketplace expansion narrative.

Growth: B2B story = consistent quality under turnover (schools hate random substitutes). Consumer story = "your tutor is still working from your Map at week 12 — we check." Unit economics: count *fidelity-weighted session capacity*, not raw tutor FTEs.

Vision: Thirty-year identity witnesses (LXI) require a workforce that does not quietly become lecturebots. Pipeline honesty is how human recognition stays falsifiable as the company scales.

LXXV.12 Confidence table

Claim	Label	Confidence
Tutoring RCTs pool ~0.288 SD; stronger for teacher/para, ≥3×/week, during school (Nickow et al. 2024)	FACT	High
Coaching ~0.49 SD instruction / ~0.18 SD achievement; scale shrinks effects (Kraft et al. 2018)	FACT	High
Burnout dimensions predict quit intent; burnout outweighs satisfaction as risk (Madigan & Kim 2021)	FACT	High
Exhaustion linked to worse instructional support/organization and student outcomes via quality paths (Klusmann et al. 2022)	FACT	High (classroom context)
MindCraft drift channels (brief skip / pour / got-it / soft-wrong / QA starve) drive FEI loss	HYPOTHESIS	Medium
Coach-capacity-gated hiring + continuous QA beats surge headcount on transfer	HYPOTHESIS	Medium
SAFE-WORKFORCE is right commercial mapping	FOUNDER BELIEF	Medium
Semester batch churn dominates college-tutor attrition	SPECULATION	Low–Medium

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LXXV.13 What this chapter kills

1. **Kill:** Tutor headcount / hours-booked / retention-% as North Stars.
2. **Kill:** Tenure months ≡ quality improvement.
3. **Kill:** One-off onboarding ≡ coaching (reinforce Kraft wound).
4. **Kill:** Wellness theater / grit posters as burnout strategy.
5. **Kill:** Burnout Score™ or emotion-wellness SKU for tutors-as-product.
6. **Kill:** Marketplace volume / Ivy bench as pipeline strategy.

7. **Wound:** Importing Nickow pooled ES or K–12 dosage as ACT score guarantees for remote near-peer.
8. **Wound:** Treating Madigan/Klusmann teacher samples as exact quit-rate forecasts for gig tutors.
9. **Survive:** Nickow moderator logic (structure/accountability/frequency); Kraft coaching-with-scale-humility; Madigan burnout→quit; Klusmann exhaustion→interaction quality; WORK-1…5; fidelity-over-tenure pipeline.

Doctrine until data: MindCraft's human layer is a **coached fidelity pipeline**, not a headcount metric. If quality drifts after hire, the Constitution's HITL/TUTORGRAIN rows were theater — and parents will notice before the pitch deck does.

Part LXXVI — Forgetting Curves as Product Honesty

Chapter status: Living evidence + product/GTM brief — Researcher tick 2026-08-04

Primary question: How should MindCraft show *decay risk and delayed competence* so parents and students trust retention claims — without shipping mastery fireworks, fake Ebbinghaus branding, or "we cured forgetting" theater?

Owners: Product (Map / Practice / parent dashboard) · Growth/copy · Engine (mastery decay honesty) · Red Team

Commercial job: Ship a **SAFE-FORGET** doctrine: treat forgetting as expected physics of memory, surface retention probes and scheduled returns as the product truth, and refuse copy that equates tonight's clear with durable identity evidence.

Builds on: Part XXIX (Memory of Competence), L (SAFE-CALIB), LXII (SAFE-LONGID), LXIII (SAFE-EXAM), LXXIV (SAFE-SCHED).

LXXVI.1 Why this chapter exists

SAFE-SCHED answered *how to space returns*. This tick answers the *honesty* problem: edtech UIs celebrate local mastery ("topic cleared," "100% tonight," green graph) while human memory predictably fades — and learners' own forecasts systematically ignore that fade. Competitors sell fireworks. MindCraft's identity thesis dies if the dashboard lies about time.

FOUNDER BELIEF under audit: Parents will pay more for a product that *admits forgetting and schedules against it* (inspectable 7d/28d probes + returns) than for a product that looks permanently mastered after one good session — if the honesty is framed as competence protection, not as "you're failing."

Claims we refuse as doctrine:

1. Mastery fireworks / "cleared forever" after a blocked session.
2. Personal Ebbinghaus curve / "forgetting score" as hero metric or shame meter.
3. Streaks as proof memory held (SAFE-HABIT / SAFE-SCHED collision).
4. Cram-week green boards ≡ ACT-ready (SAFE-EXAM).
5. Student self-rating of "I still know it" without delayed retrieval probe.
6. Guaranteed retention % or ACT points from any decay UI.
7. Fake neuroscience ("we reverse the forgetting curve").

8. Absolute “tutoring freezes memory” / Bloom 2σ retention theater.

LXXVI.2 Constructs (honesty language)

Construct	Research meaning	MindCraft analogue	Failure mode if misused
Forgetting curve	Retention declines with delay (savings / recall)	Expected fade between returns	Brand cosplay; shame curve
Retention interval (RI)	Delay to criterial test	Homework / ACT / MoC probe	Optimize only for tonight
Judgment of learning (JOL)	Predicted future recall	Confidence / “got it?” / parent belief	Fluency-as-mastery
Foresight bias	Study-time cues inflate JOLs	Clear-screen glow after hints	Sell glow as durable learning
Learning vs performance	Current fluency ≠ later access	Blocked accuracy vs delayed mix	Leaderboard of tonight
SAFE-FORGET	Doctrine: show decay risk honestly	See LXXVI.9	Mastery-fireworks NS

Operational definition (HYPOTHESIS): MindCraft is *forgetting-honest* when (a) concept “cleared” states expire or downgrade without a delayed retrieval return, (b) parent/student surfaces show last-proof age + next return due, (c) JOLs/confidence are never the sole retention claim, (d) copy frames returns as protecting competence (not character), and (e) marketing never implies permanent mastery from one session.

LXXVI.3 Forgetting is real — and not a marketing metaphor

FACT (classic replication): Murre & Dros (2015, *PLoS ONE*, 10(7), e0120644, doi:10.1371/journal.pone.0120644) — successful replication of Ebbinghaus’s savings forgetting curve (1880/1885); one subject, ~70 hours of list learning/relearning across delays from 20 min to 31 days; results similar to the original; curve is robust and not perfectly smooth (likely upward jump near the 24-hour point).

FACT (form of forgetting): Wixted (2004, *Annual Review of Psychology*, 55, 235–269, doi:10.1146/annurev.psych.55.090902.141555) — review of psychology/neuroscience of forgetting; candidate functions (power, logarithmic) share decelerating proportional decay; exponential is a poor description of typical empirical forgetting functions. **Wound:** Do not ship a branded “your personal power-law curve” — the science supports *fade with time*, not a student-facing physics widget.

Cross-link (spacing as intervention): Cepeda et al. (2006) and Rohrer & Taylor (2006) — already SAFE-SCHED: distributing practice beats massing for longer RIs; overlearning tonight does not buy 4-week math retention. Forgetting honesty without scheduled returns is merely a sad dashboard.

Commercial implication: The Map must age evidence. A green node with no delayed probe is a *performance snapshot*, not a competence claim.

Kill: “Once cleared, always mastered.”

Survive: Time-stamped last proof + expected fade until return.

LXXVI.4 Learners (and products) ignore the fade they “know about”

FACT (JOLs ignore retention interval): Koriat, Bjork, Sheffer & Bar (2004, *Journal of Experimental Psychology: General*, 133(4), 643–656, doi:10.1037/0096-3445.133.4.643) — when JOLs are experience-based (fluency at study), item-by-item and aggregate predictions are largely **insensitive to anticipated retention interval**, while actual recall shows steep forgetting; people can access theory-based forgetting knowledge mainly when RI is made salient (within-subject RI manipulation or framing in terms of *forgetting* rather than remembering). Classic dissociation: predicted week-later recall can stay high while actual recall collapses.

FACT (foresight / illusion of competence): Koriat & Bjork (2005, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 31(2), 187–194, doi:10.1037/0278-7393.31.2.187) — study-time JOLs suffer an inherent study–test mismatch (answer present at study, absent at test); a posteriori associations inflate competence judgments relative to later cued recall.

FACT (remediation path): Koriat & Bjork (2006, *Memory & Cognition*, 34(5), 959–972, doi:10.3758/BF03193244) — foresight bias can be reduced by study–test experience (especially *test* experience) and by delayed JOLs; overconfidence has behavioral costs for study-time allocation.

HYPOTHESIS (MindCraft mapping): Session-end “How confident?” prompts and parent dashboards that show tonight’s percent correct reproduce Koriat’s experience-based JOL trap unless the UI forces theory-based forgetting salience (last-proof age, days-to-exam RI, scheduled return) and delayed retrieval probes.

Kill: Confidence slider / thumbs-up as retention KPI.

Survive: Delayed generation Returns as the honesty engine (ties SAFE-CALIB + SAFE-SCHED).

LXXVI.5 Performance glow is the competitor’s product — and our trap

FACT / tradition (desirable difficulties): Bjork & Bjork (2011) — “Making things hard on yourself, but in a good way,” in *Psychology and the real world*; conditions that raise current performance (massing, blocking, rereading fluency) often undermine long-term retention/transfer; fluency and familiarity create illusions of comprehension; desirable difficulties (spacing, interleaving, testing) feel worse early and work better later.

FACT (math massing wound — reuse): Rohrer & Taylor (2006, *Applied Cognitive Psychology*, 20(9), 1209–1224) — spacing benefit nil at 1-week test, large at 4-week; same-session overlearning did not help delayed retention.

HYPOTHESIS: Khan-style clear / Duo-style streak / Brilliant binge / ChatGPT cram-chat all sell *performance glow*. If MindCraft’s Map fireworks the same glow without aging nodes and MoC probes, we are not differentiated — we are cosplaying competitors with a prettier ontology.

FOUNDER BELIEF: The commercial wedge is *inspectable honesty*: “Last proved 18 days ago — return due before your ACT” beats “Mastered!” on trust once parents understand the stakes (validate with WTP CBC, LIX).

LXXVI.6 Product surface — forgetting-honest UX

HYPOTHESIS — minimum honest surfaces:

Surface	Must show	Must not show as NS
Concept node	Last proof timestamp; proof type (blocked vs delayed mix / transfer)	Permanent green from one session
Practice Return	Due date matched to RI; generation task	Streak nag / shame fade meter
Parent dashboard	Returns completed; retention@7/28 probe rates	"Your child never forgets"
Post-mission	"Held tonight — still need a return"	Confetti ≡ durable mastery
Exam rail	Dual rail: learn spaced / prove timed (LXIII)	Cram board as readiness

SPECULATION: Framing returns as "protecting what you earned" outperforms "you forgot" on anxiety-sensitive Mayas — test in FORGET-QUAL; never soften standards into permanent hint crutches (SAFE-EXPOSE / SAFE-DD).

Engine note (HYPOTHESIS): Temporal decay toward prior in the ML mastery model is aligned in spirit with SAFE-FORGET — but product UI must not expose raw Beta posteriors as a "Forgetting Score™." Show behavioral probe outcomes, not spooky brain meters.

LXXVI.7 Competitive positioning

Competitor pattern	What they sell	MindCraft FORGET counter
Mastery clear fireworks	Tonight accuracy	Aged evidence + delayed mix
Streak / XP	Habit continuity	Returns ≠ streaks
Expanding SRS mystique	Algorithm brain	Horizon-matched honesty (SAFE-SCHED)
Cram packs	Short-RI performance	RI-matched returns + prove rail
Chat fluency	Feeling of knowing	Generation probe after delay

Marketing that survives: "We bring it back before it fades." / "Cleared tonight ≠ ready on test day — here's your return." / "Parents see last proof age, not just progress bars."

Marketing that dies: Personal Ebbinghaus brand; cured-forgetting claims; mastery confetti as North Star; retention guarantees; neuroscience theater.

LXXVI.8 Parent WTP and identity clocks

Ties: SAFE-WTP (CBC attributes), SAFE-LONGID (8/26/52w fade checks), SAFE-IDMEASURE (Likert ≠ durable identity).

HYPOTHESIS: A parent CBC attribute "shows when skills last proved + schedules returns" should beat "shows topics mastered this week" and "daily streak" for high-stakes ACT segments — without implying score guarantees.

FOUNDER BELIEF: Identity transformation requires *autobiographical competence that survives delay*. Forgetting honesty is how the product refuses to mint false identity evidence.

Wound: Do not claim parents "only care about honesty" — some buy cram. Offer dual rail; measure choice share; do not moralize buyers who pick cram attributes (still refuse to *market* cram as durable identity).

LXXVI.9 SAFE-FORGET stack (surviving doctrine)

1. **Fade is expected** — design for decay; do not brand-shame it.
2. **Age the evidence** — last-proof timestamps; clear states expire without delayed return.
3. **JOLs ≠ retention** — never sell confidence/fluency as held knowledge (Koriat line).
4. **Delayed probes are the truth surface** — MoC 7d/28d + generation Returns.
5. **Frame returns as competence protection**, not character failure.
6. **No Forgetting Score™ / personal Ebbinghaus hero UI.**
7. **Ties:** SAFE-SCHED (how to schedule), SAFE-CALIB (how to elicit confidence), SAFE-EXAM (dual rail), SAFE-LONGID (fade checks), SAFE-WTP (package honesty).
8. **Copy ban:** cured forgetting; cleared forever; streak-as-memory; guaranteed retention %.

LXXVI.10 Experiments (FORGET family)

ID	Contrast	Primary endpoints	Kill / survive rule
FORGET-1	Aged-node UI (last proof + due return) vs permanent-clear fireworks	retention@7/28; return_completed; false-mastery quit	Fireworks ≥ aged on retention → honesty UX not load-bearing
FORGET-2	Post-mission "held tonight / return needed" vs confetti-only	7d return success; student JOL calibration error	Confetti ≥ honest on calibration → copy work
FORGET-3	Parent dashboard: proof-age + returns vs topics-mastered-this-week	CBC choice share; WTP (with LIX)	Mastered-week dominates → packaging, not auto-abandon honesty

FORGET-4	Theory-framed forgetting salience (days-to-exam RI shown) vs fluency-only confidence prompt	JOL–actual gap at delayed probe	Salience null → need stronger probe design
FORGET-5	Clear-state expiry without return vs sticky green	delayed mix accuracy; solo_transfer_pass	Sticky ≥ expiry → expiry may be too harsh / needs soft downgrade
FORGET-QUAL	10 Maya + 10 parent interviews: “protect what you earned” vs “you forgot” language	Codebook: threat vs agency	Feeds copy; anxiety-aware framing

Ties: SCHED-, *CAL*-, WTP-, *LONG*-, EXAM-*.

Falsifier: Honest decay UI raises anxiety / dropout while 28d MoC stays flat → wound framing; do not kill the decay science, redesign threat language (SAFE-DD).

LXXVI.11 Commercial implications

Copy: Lead with *time-honest competence* — returns before fade, proof age visible. Never lead with permanent mastery fireworks or neuroscience curves.

Product: Map ages nodes; Practice Return queue is first-class; parent view shows returns + retention probes; post-mission copy separates tonight’s performance from delayed readiness.

Metric: `last_proof_age_days`, `return_due`, `return_completed`, retention@7/28 co-primary with FEI; demote session accuracy and streak as learning claims.

Growth: Differentiate vs Khan/Duo/Brilliant/ChatGPT on *inspectable durability*, not content breadth or habit theater. District story: evidence that does not silently expire into false readiness.

Vision: Thirty-year identity witnesses need memories that survive delay. Forgetting honesty is how MindCraft refuses to sell a green lie.

LXXVI.12 Confidence table

Claim	Label	Confidence
Ebbinghaus-style savings forgetting replicates (Murre & Dros 2015)	FACT	High

Empirical forgetting often decelerating (power/log-like); not a student UI mandate (Wixted 2004)	FACT	High
Experience-based JOLs largely ignore anticipated RI; theory framing can restore sensitivity (Koriat et al. 2004)	FACT	High
Study-time foresight bias inflates competence; test experience / delayed JOLs help (Koriat & Bjork 2005/2006)	FACT	High
Fluency/massing create performance–learning traps (Bjork & Bjork 2011; Rohrer & Taylor 2006)	FACT	High
Aged-node + return UX beats fireworks on 28d retention and parent WTP	HYPOTHESIS	Medium
“Protect what you earned” framing beats shame-fade for anxious Mayas	HYPOTHESIS	Medium
SAFE-FORGET is right commercial mapping	FOUNDER BELIEF	Medium
Raw engine decay posteriors should stay off student-facing meters	SPECULATION	Medium

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LXXVI.13 What this chapter kills

1. **Kill:** Mastery fireworks / cleared-forever as durable competence claims.
2. **Kill:** Personal Ebbinghaus / Forgetting Score™ hero metrics.
3. **Kill:** Streak / tonight-accuracy as memory-held proof.
4. **Kill:** Confidence / “got it?” as retention KPI.
5. **Kill:** Cured-forgetting / reverse-the-curve neuroscience ads.
6. **Kill:** Guaranteed retention % or ACT points from decay UI.
7. **Wound:** Importing nonsense-syllable savings percentages as ACT readiness numbers.
8. **Wound:** Shame-framed fade meters that spike anxiety without lifting MoC.
9. **Survive:** Murre/Dros robustness of forgetting; Koriat JOL–RI dissociation; Bjork learning≠performance; Rohrer delayed math spacing; FORGET-1…5; aged evidence + scheduled returns as honesty stack (with SAFE-SCHED).

Doctrine until data: MindCraft sells **time-honest competence evidence**, not permanent green. If the Map cannot age a claim, the claim is marketing — and the Constitution’s identity thesis is theater.

Part LXXVII — Adaptive Spacing Algorithms vs Fixed Calendars

Chapter status: Living evidence + product/engine brief — Researcher tick 2026-08-04

Primary question: When (if ever) should MindCraft replace horizon-matched *fixed* return calendars with *adaptive* lag models (SM-2 / FSRS-style / half-life regression), and how do we do that without shipping black-box "perfect brain interval" brand cosplay?

Owners: Engine (return scheduler) · Product (Practice/Map Returns) · Growth/copy · Red Team

Commercial job: Ship a **SAFE-ADAPT** doctrine: keep SAFE-SCHED's inspectable horizon calendar as the *default*; allow adaptive *modulation inside declared bands*; refuse SM-2/FSRS/Duo-HLR as hero marketing or as a substitute for transfer probes.

Builds on: Part XXIX (MoC), LXXIV (SAFE-SCHED), LXXVI (SAFE-FORGET), LXIII (SAFE-EXAM), LXVII (SAFE-ONTOLOGY).

LXXVII.1 Why a third spacing chapter

SAFE-SCHED answered *what schedule shape* (ISI×RI, delayed first return, equal-ish lags). SAFE-FORGET answered *how to show decay honestly*. Competitors and eng blogs answer with a third move: **personalize the lag with an opaque model** — SM-2 ease factors, FSRS stability/difficulty, Duolingo half-life regression — and sell that as the science.

FOUNDER BELIEF under audit: Parents will pay for *inspectable returns matched to an exam/homework horizon* more than for "our AI found your perfect interval"; adaptive models may still *win retention efficiency* if they stay subordinate, auditable, and measured on delayed math transfer — not flashcard MAE.

Claims we refuse as doctrine:

1. SM-2 / Anki / FSRS as the MindCraft brand story.
2. Black-box "optimal interval AI" with no parent/student-visible due date rationale.
3. Flashcard recall-prediction gains ≡ ACT / solo_transfer readiness.
4. Engagement / DAU lift from adaptive SRS ≡ learning or identity (Settles wound).
5. Expanding adaptive schedules as default for multi-month RI (SAFE-SCHED kill retained).
6. Streaks / daily open as the adaptive engine.
7. Guaranteed retention % or ACT points from any scheduler.

LXXVII.2 Constructs (algorithm language)

Construct	Research / eng meaning	MindCraft analogue	Failure mode if misused

Fixed / calendar lag	Same ISI rule for a cohort (or RI band)	Exam-horizon Return queue	One-size misses item difficulty
Adaptive scheduler	Next ISI from learner×item history	Optional modulator on due date	Opaque brain-AI cosplay
SM-2	Heuristic ease-factor expanding schedule (Wo■niak / SuperMemo 2)	Anti-pattern as hero	Ease hell; vocab-shaped UX
FSRS / DSR	Difficulty–Stability–Retrievability models; predict recall prob	Possible *engine* candidate later	Flashcard MAE sold as math science
Half-life regression (HLR)	Trainable decay half-life (Settles & Meeder)	Lexeme-memory, not FEI	Engagement-as-proof
Personalized review	Select *what* to review given history (Lindsey et al.)	Priority among due Returns	Random “smart” without lag honesty
SAFE-ADAPT	Doctrine: calendar first; adapt inside bands	See LXXVII.9	Black-box SRS brand

Operational definition (HYPOTHESIS): MindCraft may adapt a concept’s next return ISI only inside a **declared band** derived from RI (e.g. ±30% of the SAFE-SCHED horizon lag), using inspectable inputs (last proof age, fail/pass on generation Returns, bridge-gap severity) — never a silent SM-2/FSRS clone that rewrites due dates without showing *why* and *when*.

LXXVII.3 Adaptive scheduling can beat fixed spacing — in the domains studied

FACT (lab optimization): Pavlik & Anderson (2008, *Journal of Experimental Psychology: Applied*, 14(2), 101–117, doi:10.1037/1076-898X.14.2.101) — ACT-R–based scheduler that chooses practice timing to maximize long-term gain per unit practice time outperformed comparison schedules on recall accuracy and latency for paired-associate / fact learning; optimal intervals *lengthen* as items stabilize.

FACT (classroom personalized review): Lindsey, Shroyer, Pashler & Mozer (2014, *Psychological Science*, 25(3), 639–647, doi:10.1177/0956797613504302) — middle-school foreign-language course; semester-long retrieval software; on a cumulative exam ~1 month after semester end, **personalized** review beat **massed** (~+16.5% retention) and beat **one-size-fits-all spaced** (~+10.0%). Personalization here selected *which* material to review given inferred memory strength — not merely expanding flashcard ease cosplay.

FACT (trainable production SRS): Settles & Meeder (2016, *ACL*, 1848–1858, doi:10.18653/v1/P16-1174) — half-life regression on Duolingo data reduced recall-prediction error >45% vs several baselines; an operational study reported **~12% higher daily student engagement**. HLR generalizes Leitner/Pimsleur-style ladders with fitted features.

Commercial implication: “Adapt > fixed” is a *real* claim in vocab / flashcard / lexeme settings. MindCraft cannot Red-Team it out of existence. The fight is **domain transfer + brand honesty**, not denial of personalization.

Kill: “All adaptive SRS is pseudoscience.”

Survive: Personalized review can beat generic spacing for item memory — measure whether that transfers to MindCraft’s FEI endpoints.

LXXVII.4 SM-2 and FSRS are engineering lineages — not math-identity science

FACT (SM-2 origin): Wo■niak’s SuperMemo SM-2 algorithm (documented 1987; SuperMemo 2 era) sets early intervals (classically 1 day then 6) then multiplies by an ease factor updated from 0–5 quality ratings — a hand-tuned expanding heuristic that powered decades of SRS apps (incl. Anki’s legacy default). It is **not** a peer-reviewed math-education RCT.

FACT (modern flashcard ML): Ye, Su & Cao (2022, *KDD ’22*, 4381–4390, doi:10.1145/3534678.3539081) — MaiMemo-scale memory model + stochastic shortest-path scheduler; reported ~12.6% improvement over SOTA methods on review-cost / memorization efficiency for language learning; deployed in production. Open FSRS builds on related DSR/DHP memory-state ideas and is widely used to **predict flashcard recall probability** and reduce review load vs SM-2 on Anki-style logs (engineering benchmarks; recall calibration ≠ transfer).

HYPOTHESIS (domain mismatch): Math FEI depends on *ingredient/bridge transfer under variation*, not binary “card recalled?” grades. Mapping a concept node to an SM-2 “card” collapses diagnosis (SAFE-ONTOLOGY), soft-wrong science (SAFE-MISCON), and solo transfer into a flashcard grade — the wrong grain.

Wound: Cite FSRS/Anki MAE wins as *engine inspiration* for predicting return urgency; do **not** cite them as evidence MindCraft will raise ACT or `solo_transfer_pass`.

Kill: “We use FSRS / SM-2 — therefore science-backed ACT prep.”

Survive: Optional recall-probability models as *priority signals* among already-due Returns.

LXXVII.5 Engagement and prediction accuracy are not identity outcomes

FACT / Red Team wound (Settles): The production win highlighted for HLR includes **daily engagement +12%**. Engagement is not FEI. Duo-shaped products can improve revisit rates while still selling streak theater and short-horizon expanding schedules that SAFE-SCHED already rejects for long RI.

Cross-link (SAFE-FORGET): Koriat & Bjork foresight bias — learners (and UIs) confuse fluent recent success with future recall. An adaptive model that *schedules* well can still be wrapped in mastery fireworks that *communicate* poorly.

Cross-link (SAFE-SCHED / Karpicke & Roediger 2007): Expanding schedules that look “smart” can lose to equal spacing at longer delays. Adaptive ≠ expanding; if the learner model quietly expands first intervals for dopamine, it collides with equal-ish long-RI doctrine.

Commercial implication: Never advertise “more daily opens from our scheduler” as learning proof. Instrument `retention_probe_7d/28d`, `solo_transfer_pass`, and `return_completed` — co-primary with any ADAPT A/B.

LXXVII.6 Fixed calendars remain the right *product* default

FACT (reuse — ISI×RI): Cepeda et al. (2006; 2008) — optimal lag scales with retention interval; educational systems that ignore RI are inefficient. A **visible calendar** tied to homework/ACT date is the UX that makes ISI×RI *legible* to parents (SAFE-WTP / SAFE-FORGET proof-age).

HYPOTHESIS — Calendar-first stack:

1. **Layer A (default):** Horizon-matched equal-ish lags from exam/homework RI (SAFE-SCHED Exam Horizon Scheduler).
2. **Layer B (optional adapt):** Within ± band, pull forward Returns with weak last proof / bridge-gap / failed generation; push back only after strong delayed probe — never past RI honesty.
3. **Layer C (selection):** When many Returns are due, personalized *priority* (Lindsey-style) picks which concept to surface first — still showing due dates.
4. **Hard ban:** Silent SM-2 ease hell; student-facing "easiness 2.1"; FSRS Score™; streak as scheduler.

SPECULATION: Inspectable Layer B will beat black-box Layer-B-as-brand on parent WTP even if flashcard MAE is slightly worse — test via CBC (LIX), not founder vibes.

LXXVII.7 Anxiety, load, and adaptive aggression

Ties to SAFE-DD / SAFE-EXPOSE / SAFE-PRIVACY: Adaptive systems that schedule reviews at low predicted retrievability maximize desirable difficulty — and can spike threat for high-anxiety Maya.

HYPOTHESIS: Cap adaptive "pull forward" under high self-authored stress modifiers (shorten challenge density / soften stakes, not erase Returns). Never use affect to *delete* horizon honesty.

Kill: Anxiety Score™ that auto-masses practice "for comfort."

Survive: Affect → lag *aggression* caps inside SAFE-SCHED bands.

LXXVII.8 Competitive positioning

Competitor pattern	Mechanism	MindCraft response
Anki / FSRS mystique	Recall-prob optimization	Engine-only; no SRS brand
Duo HLR + streak	Engagement + lexeme half-life	Returns + MoC probes; streak demoted
Khan clear-and-forget	Local mastery	Aged evidence + calendar Returns
"AI perfect interval" edtech ads	Authority cosplay	Inspectable due + why chip
ACT cram packs	Massed week	Horizon calendar + late prove rail

Competitive wedge (FOUNDER BELIEF): Sell "your Map shows the next return and why" — not "smarter than Anki."

LXXVII.9 Doctrine — SAFE-ADAPT (provisional)

1. **Calendar-first:** SAFE-SCHED horizon lags are the default product surface.
2. **Adapt inside bands:** Personalization may modulate ISI only within declared RI-derived bounds; show due date + reason.
3. **Selection ≠ mystique:** Prefer Lindsey-style *what to review among due* over silent SM-2 *rewrite of science*.
4. **Flashcard lineage ≠ FEI proof:** SM-2 / FSRS / HLR evidence is vocab/item-memory; do not claim ACT/transfer from it.
5. **Engagement ≠ learning:** Ban DAU/streak lifts as scheduler success.
6. **No black-box brand:** No "perfect interval AI," FSRS Score™, or easiness theater.
7. **Anxiety caps:** Stress modifiers limit adaptive aggression; never erase returns.
8. **Measure FEI:** ADAPT-* wins only if delayed probes / solo transfer hold or rise vs fixed calendar.

Confidence: High that personalized review can beat massed and generic spaced for *item* memory (Lindsey; Pavlik). Medium that MindCraft Layer-B banding helps retention without hurting trust. Low–medium that off-the-shelf FSRS on concept nodes helps `solo_transfer_pass` (domain mismatch). High that SRS-as-brand collides with SAFE-SCHED / SAFE-FORGET / competitive wedge.

LXXVII.10 Experiments

ID	Question	Design	Primary
ADAPT-1	Fixed horizon calendar vs calendar + within-band lag modulation	A/B after transfer_pass	retention@7/28; solo_transfer_pass
ADAPT-2	Due-set priority (personalized selection) vs FIFO due queue (same lags)	A/B	28d retention; return_completed
ADAPT-3	Inspectable "due + why" UI vs opaque "smart review" queue	A/B + parent CBC	trust / WTP; retention (non-inferiority)
ADAPT-4	FSRS-like urgency score as *priority only* vs full ISI rewrite	A/B	28d retention; complaint/confusion rate
ADAPT-5	Adaptive pull-forward under high stress: capped vs uncapped	A/B	anxiety state; retry_120s; retention

ADAPT-QUAL	10 Maya + 10 parent: “smart algorithm” vs “returns on your calendar” copy	Qual	message resonance; SRS-brand allergy

Falsifier: Adaptive arm raises opens/streaks while `retention_probe_28d` or `solo_transfer_pass` falls → demote adaptive claims; keep calendar-first.

Pre-register: ADAPT-* before any “AI spaced repetition” marketing (SAFE-LABMETA).

LXXVII.11 So what for MindCraft commercially

- **Copy:** “Returns on your calendar — adjusted when evidence says so” — never “FSRS-powered brain optimization.”
- **Product:** Ship Layer A Returns first; Layer B/C behind flags; every due chip shows horizon basis.
- **Engine:** Decay + proof-age already exist — wire them to *visible* due dates before importing Anki math.
- **Metric:** Co-primary retention/transfer; demote scheduler engagement vanity.
- **Kill list:** SM-2/FSRS hero, black-box perfect-interval AI, engagement≡learning, flashcard MAE≡ACT.
- **Growth:** Parent dashboard = next returns + why — feeds WTP without SRS cosplay.
- **Vision:** Identity needs durable competence evidence; algorithms may *protect* evidence efficiently, but the product promise is honesty and transfer — not scheduler mystique.

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Part LXXVIII — Cram Products vs Durable-Identity Positioning

Chapter status: Living evidence + GTM/product brief — Researcher tick 2026-08-04

Primary question: How should MindCraft compete with massed "cram pack / bootcamp week / guaranteed points" test-prep without either (a) becoming another cram SKU or (b) moralizing parents who buy short-horizon score relief?

Owners: Growth/copy · Product (Practice / Returns / exam dual rail) · Parent WTP · Red Team

Commercial job: Ship a **SAFE-DURABLE** doctrine: dual-rail GTM (durable learn rail + late prove rail); refuse cram≡ready and package ACT guarantees; price and message *returns of competence* while still selling honest timed practice when the calendar demands it.

Builds on: Parts XXVII (markets), XXIX (MoC), LIX (SAFE-WTP), LXIII (SAFE-EXAM), LXXIV (SAFE-SCHED), LXXVI (SAFE-FORGET), LXXVII (SAFE-ADAPT).

LXXVIII.1 Why this chapter exists

The ACT/SAT prep market already sells what parents can feel in a weekend: massed drills, strategy tips, score-lift ads, and package tiers with point targets. MindCraft's thesis is identity transformation via durable competence evidence. Those two value props collide in the same checkout moment.

FOUNDER BELIEF under audit: Parents will pay a durable-identity premium *if* the product makes delayed competence legible (returns, aged proof, solo transfer) *and* still offers an honest late-stage "prove under stakes" rail — not if we only preach spacing while competitors own the anxiety week.

Claims we refuse as doctrine:

1. Cram-week / bootcamp massing ≡ ACT-ready or "math person."
2. Guaranteed ACT/SAT point packages as MindCraft SKUs.
3. Overlearn-tonight / marathon session as retention science.
4. Score-only parent messaging as the default brand (SAFE-WTP wound retained).
5. Shame-parenting ("if you bought Kaplan you failed your kid").
6. Pure anti-exam idealism that deletes timed practice (SAFE-EXAM dual rail retained).
7. Streak / DAU / tonight-accuracy as proof of durable positioning.

LXXVIII.2 Constructs (market language)

Construct	Research / market meaning	MindCraft analogue	Failure mode if misused
Massed / cram	Practice compressed into short RI before test	Anti-pattern as *learn* rail	Fluency illusion; fast fade
Distributed / spaced	Same effort across ISIs matched to RI	Returns / SAFE-SCHED	Sold as moral purity, not product
Overlearning	Extra same-session reps after criterion	Blocked drill after clear	Wastes time vs spacing (Rohrer)
Cram product	Commercial SKU selling massed score relief	Competitor default	Owns anxiety week
Durable-identity offer	Pay for lasting competence + self-story	FEI + Returns + Map	Soft brand with no exam rail
Dual rail	Destake-to-learn / dose-to-prove	SAFE-EXAM	Learn rail only → lost GTM
SAFE-DURABLE	Positioning doctrine (this chapter)	See LXXVIII.9	Cram cosplay or exam denial

Operational definition (HYPOTHESIS): A MindCraft commercial offer is *durable-positioned* iff (1) default Practice schedule is horizon-matched spaced Returns, (2) parent-facing proof includes aged evidence / delayed probes, (3) any exam-week intensity is labeled *prove rail* not *mastery*, and (4) no package promises a point delta.

LXXVIII.3 Spacing beats massing for durable retention — including math

FACT (meta, verbal recall): Cepeda, Pashler, Vul, Wixted & Rohrer (2006, *Psychological Bulletin*, 132(3), 354–380, doi:10.1037/0033-2909.132.3.354) — quantitative synthesis of distributed practice; spacing > massing across retention intervals; lag effects depend on RI (ISI×RI).

FACT (optimal lag): Cepeda, Vul, Rohrer, Wixted & Pashler (2008, *Psychological Science*, 19(11), 1095–1102, doi:10.1111/j.1467-9280.2008.02209.x) — temporal ridgeline: optimal gap scales with desired retention interval (~10–20% of RI as a rough educational heuristic in the spacing literature).

FACT (math practice): Rohrer & Taylor (2006, *Applied Cognitive Psychology*, 20(9), 1209–1224, doi:10.1002/acp.1266) — college students learning a math procedure; distributed practise across sessions beat massing at a **4-week** test (large benefit), with little/no spacing advantage at a 1-week test in Exp. 1; Exp. 2: tripling same-session problems (**overlearning**) did **not** raise 1- or 4-week scores.

FACT (math meta): Murray, Horner & Göbel (2025, *Educational Psychology Review*, 37, article 75, doi:10.1007/s10648-025-10035-1) — spaced vs massed practice in mathematics: robust small-to-medium overall effect (**g = 0.28**; 27 studies); larger in isolated-learning designs (g = 0.43) than course-embedded (g = 0.24). Testing-vs-restudy effect in math was small and not robust (g = 0.18; CI crossed zero).

FACT (utility rating): Dunlosky, Rawson, Marsh, Nathan & Willingham (2013, *Psychological Science in the Public Interest*, 14(1), 4–58, doi:10.1177/1529100612453266) — distributed practice rated **high utility**; many popular student strategies (rereading, highlighting) rated low.

Commercial implication: Durable retention is the scientifically stronger long-horizon claim. Effect sizes in *math* are real but often modest — do **not** advertise "2× retention" or Bloom-scale miracles from spacing alone.

Kill: Cram ≡ science-backed durable ACT readiness.

Survive: Spacing/distribution as the default *learn* schedule; measure delayed math outcomes, not vibes.

LXXVIII.4 Learners (and buyers) prefer the wrong schedule

FACT (metacognitive illusion): Kornell (2009, *Applied Cognitive Psychology*, 23(9), 1297–1317, doi:10.1002/acp.1537) — GRE-type flashcard study; spacing beat massing and beat last-day cramming; spacing helped ~90% of participants, yet after the first session **~72% believed massing had been more effective**.

Cross-link (SAFE-FORGET): Koriat & Bjork foresight bias — fluent recent performance feels like lasting knowledge. Cram products sell that feeling on purpose.

HYPOTHESIS (parent market): Parent WTP for cram packs is partly purchase of *felt readiness* and social proof ("we did a program"), not Bayesian belief about 4-week retention. CBC must separate "weekend intensive" vs "returns on a calendar" vs "timed prove week" attributes (SAFE-WTP).

Commercial implication: Do not win by shaming the illusion. Win by making delayed competence *visible* (proof age, return chips, retention probes) so the durable offer has a felt artifact competitors lack.

Kill: "Parents only buy scores, so we must become a cram brand."

Wound: Score anxiety is real demand — dual-rail, don't deny.

Survive: Legible MoC > moral lecture.

LXXVIII.5 Test-prep market reality (without becoming it)

FACT (industry shape): College-admissions test prep remains a large, multi-billion-dollar category with formal prep (courses, online, 1:1 tutoring) and informal prep (books, practice tests). Brookings' synthesis (Reber, Roller & Sanderson, 2025, "Exam ready") notes paid prep is uneven by income; individual tutoring/coaching is often more associated with gains than lighter formats; effects of commercial prep are generally **modest** relative to aggressive advertising; rigorous evidence for high-priced "guaranteed score" packages is weak/absent.

FACT (ACT retest prep activities): Moore, Sanchez & San Pedro (ACT Research Report R1710, 2018) — among students who retested, hours with a **private tutor/consultant** related to composite retest gains in propensity-adjusted models; many other listed prep activities were not statistically related. Absolute adjusted differences were **small** (sub-point on the ACT composite scale in the reported contrasts) — not "+4 guaranteed."

SPECULATION / market observation: Cram packs and bootcamps will keep winning anxiety-week SEO and school-hallway word-of-mouth. MindCraft should not spend brand capital trying to out-Kaplan Kaplan on massed volume.

Commercial implication: Position against *mechanism* (massed fluency theater), not against *parents*. Offer: diagnosis + spaced returns + Map-briefed humans + late prove rail. Package what can be instrumented (returns completed, retention probes, solo transfer) — never point guarantees.

LXXVIII.6 Dual-rail GTM (durable + prove)

Reuse (SAFE-EXAM / SAFE-SCHED): Destake-to-learn early; dose-to-prove late; readiness = delayed mixed under stakes — not blocked tonight accuracy.

HYPOTHESIS — SAFE-DURABLE product stack:

1. **Learn rail (default SKU story):** Horizon-matched Returns; aged evidence; bridge-gap honesty; FEI session spine.
2. **Prove rail (calendar-triggered):** Timed mixed sets, stakes ladder, exam affect handling — labeled as pressure practice, not “you’re mastered.”
3. **Cram-intercept UX:** If user requests “only this week,” show tradeoff: massing can lift short-RI performance while risking 28d fade — then offer denser Returns inside the remaining horizon rather than silent overlearn-tonight.
4. **Parent surface:** Next returns + last proof age + what “ready” means (delayed mix) — CBC-tested vs score-guarantee competitor copy.

Kill: Exam-week massing sold as the MindCraft method.

Survive: Honest prove rail subordinate to durable learn rail.

LXXVIII.7 Competitive positioning

Competitor pattern	Mechanism sold	MindCraft response
Kaplan / Princeton Review packs	Massed content + strategies + score ads	Dual rail; no point guarantees
Local bootcamp week	Anxiety-week intensity	Cram-intercept + denser Returns
Duo-style daily streak prep	Habit / engagement	Demote streak; Returns NS
Khan clear-and-move-on	Local mastery fireworks	Aged evidence + generation Returns
ChatGPT cram chat	Fluent coverage theater	PWC + solo_transfer; no monologue≡ready
“AI perfect interval” apps	Scheduler mystique	SAFE-SCHED / SAFE-ADAPT calendar-first

Competitive wedge (FOUNDER BELIEF): Sell "still sharp when it counts — because we returned" — not "more problems this weekend."

LXXVIII.8 Equity and structure (do not moralize scarcity)

Cross-link (SAFE-STRUCTURE / SAFE-WTP): Higher-income families buy more formal prep (Brookings; ACT ecological briefs). Mocking cram purchases shames constrained families who may only afford a short intensive.

HYPOTHESIS: Durable positioning must include *access-honest* paths (shorter calendars with honest RI matching; free/LEA rails under SAFE-PROCURE) — not "only year-long subscriptions are ethical."

Kill: "Real parents buy identity programs; cram buyers are unserious."

Survive: Structure and calendar honesty over character judgment.

LXXVIII.9 Doctrine — SAFE-DURABLE (provisional)

1. **Default = durable learn rail:** Horizon-matched spaced Returns; generation probes; Map diagnosis.
2. **Exam intensity = prove rail:** Timed/stakes practice labeled honestly; never rebranded as mastery fireworks.
3. **No point-guarantee packages:** Ban ACT/SAT +N SKUs and "bootcamp ready in 7 days" as brand claims.
4. **Cram-intercept, don't cram-cosplay:** If massing is requested, show RI tradeoff and densify Returns — do not ship overlearn-tonight as science.
5. **Legible MoC for WTP:** Parent value prop = proof age + returns + delayed mix — CBC before score-only default.
6. **Modest math effect honesty:** Cite spacing as real (Murray g~0.28 class) — never 2-sigma or cured-forgetting.
7. **No shame GTM:** Compete with products, not with parents' anxiety purchases.
8. **Measure FEI + retention, not bootcamp NPS:** CRAM-* wins only if delayed probes / solo transfer hold vs massed-pack messaging arms.

Confidence: High that spacing/distribution beats massing for long-RI retention (Cepeda; Rohrer & Taylor; Murray et al.). High that learners mis-prefer massing (Kornell). Medium that parent CBC will pay a premium for legible returns vs cram when price is matched. Low–medium that dual-rail alone beats entrenched score ads without distribution/partner channels. High that point-guarantee packaging is a kill under SAFE-WTP / SAFE-EXAM / claim ladder.

LXXVIII.10 Experiments

ID	Question	Design	Primary
CRAM-1	Durable-returns parent CBC vs cram-pack / +points CBC attributes	Parent CBC (SAFE-WTP style)	WTP; choose-durable rate

CRAM-2	Cram-intercept UX (tradeoff + denser Returns) vs silent massed week mode	A/B near exam date	retention_probe_28d; solo_transfer_pass; session load
CRAM-3	Dual-rail labeled prove week vs unlabeled “final mastery blitz”	A/B	calibration; anxiety state; delayed mix
CRAM-4	Score-lift ad creative vs returns/proof-age creative (same offer)	Message test	CTR/qualified lead; complaint; misexpectation
CRAM-5	Massed equal-time week vs horizon Returns (same total minutes)	A/B after transfer_pass	retention@7/28; blocked vs mixed
CRAM-QUAL	10 parents: “weekend intensive” vs “calendar returns + prove week” stories	Qual	resonance; shame allergy; price anchors

Falsifier: Durable messaging wins surveys but massed arm wins retention@28d and solo_transfer → revisit schedule fidelity before brand claims.

Falsifier: Cram-intercept raises short-term satisfaction while 28d retention collapses → demote intercept; strengthen defaults.

Pre-register: CRAM-* before any anti-Kaplan attack ads or “we don’t cram” hero (SAFE-LABMETA).

LXXVIII.11 So what for MindCraft commercially

- **Copy:** “Built for what you still know later — plus timed practice when the date is close.” Never “+4 ACT points” or “crush it in a weekend.”
- **Product:** Keep SAFE-SCHED Returns as default; ship cram-intercept + prove-rail labeling; surface proof age on parent dashboard.
- **Pricing:** Packages = horizon length + tutor/Map depth — not score deltas.
- **Metric:** Co-primary `retention_probe_28d`, `solo_transfer_pass`, `return_completed`; demote bootcamp NPS / tonight accuracy.
- **Kill list:** Cram≡ready; point guarantees; overlearn-tonight; shame-parent GTM; streak-as-durable.
- **Growth:** Compete on legible durability in a market that sells fluency theater.
- **Vision:** Identity needs competence that survives the test date — and a company that can say so without lying about anxiety week.

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Part LXXIX — Feedback Timing: Immediate vs Delayed

Chapter status: Living evidence + product UX brief — Researcher tick 2026-08-04

Primary question: When should MindCraft reveal correctness, coach on a soft-wrong, or withhold judgment — and what timing claims are scientifically indefensible in marketing?

Owners: Product (Practice / Solver / Diagnostic) · Coach UX · Tutor ops · Brand & copy · Red Team

Commercial job: Ship a **SAFE-FBTIME** doctrine: mode-conditional timing (diagnostic hide / learn micro-delay + soft-wrong / prove-rail KR); sell *task-focused informative feedback*, never "instant AI feedback" or "science says always delay" as the brand.

Builds on: Parts XXVI (CLT/tutoring), XL (SAFE-SE), XLI (SAFE-DD), XLIX (SAFE-MISCON), L (SAFE-CALIB), LI (SAFE-DP), LVIII (SAFE-EXPOSE), LXIII (SAFE-EXAM), LXXVIII (SAFE-DURABLE).

LXXIX.1 Why this chapter exists

Edtech defaults to a green check the instant a student taps an answer. Tutors default to explaining before the student finishes thinking. Competitors market "instant personalized feedback" as the product. MindCraft already needs the opposite in places (hide-correctness diagnostic; attempt-first SE; soft-wrong springboards) — but without a timing law, those become inconsistent vibes.

FOUNDER BELIEF under audit: A short, intentional delay before correctness reveal (or coach pour) raises constructive retry and transfer *for learn-rail practice*, especially after soft-wrongs — while instant KR can be right for novices acquiring a procedure and for labeled prove-rail calibration. Timing is a *mode switch*, not a universal moral.

Claims we refuse as doctrine:

1. "Instant feedback" as the MindCraft hero claim.
2. "Always delay feedback" as desirable-difficulty cosplay.
3. Instant right/wrong ≡ mastery or identity progress.
4. Feedback ≡ learning (ignoring Kluger's one-third harm rate).
5. Soft-wrong delay as Calm Score™ / therapy theater.
6. Explanation-first pour timed as "support."
7. Streak / speed-to-green-check as North Star.

LXXIX.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Immediate feedback	KR/KCR/EF right after a response	Instant check / Solver reveal	Dependency; shallow processing
Delayed feedback	After items, minutes, or sessions	End-of-set review; proof age	Frustration; abandoned retry
Micro-delay	Seconds of forced attempt / SE before reveal	Soft-wrong coach gate	Feels like lag / broken UX
Hide-correctness	No right/wrong reveal in-session	Diagnostic C4	Corrupts Map if keys wrong
Self-focused FI	Praise, ego, comparative rank	"You're a math person!" toast	Attention off task (FIT)
Task/process FI	How to revise the strategy	Soft-wrong → ingredient card	Monologue without generation
SAFE-FBTIME	Mode-conditional timing law	This chapter	Timing silver-bullet ads

Operational definition (HYPOTHESIS): Timing policy is *SAFE-FBTIME-honest* iff (1) diagnostic never reveals correctness in-session, (2) learn-rail soft-wrongs require attempt + brief SE/micro-delay before coach, (3) prove-rail may use immediate KR under explicit stakes label, (4) feedback content is task/process-focused, and (5) no marketing claims a universal optimal delay.

LXXIX.3 Feedback is powerful — and often harmful

FACT (meta): Kluger & DeNisi (1996, *Psychological Bulletin*, 119(2), 254–284, doi:10.1037/0033-2909.119.2.254) — 607 effect sizes; average FI effect **d ≈ 0.41**, but **over one-third of interventions decreased performance**. Feedback Intervention Theory (FIT): effectiveness falls as attention moves from task learning toward self / meta-task processes.

FACT (education review): Hattie & Timperley (2007, *Review of Educational Research*, 77(1), 81–112, doi:10.3102/003465430298487) — feedback is among the strongest influences on achievement *when* it answers where am I going / how am I going / where to next; timing is a thorny moderator, not a free lunch. Self-level feedback is typically least useful for learning.

FACT (formative guidelines): Shute (2008, *Review of Educational Research*, 78(1), 153–189, doi:10.3102/0034654307313795) — formative feedback should be nonevaluative, supportive, timely, and specific; timing interacts with learner and task (immediate often better for difficult new material / low prior knowledge; delayed can support transfer and avoid over-reliance).

Commercial implication: Do not sell “more feedback faster.” Sell *task-focused next move*. Instant ego toasts and fluency theater violate FIT.

Kill: Feedback volume or speed as product quality.

Survive: Task/process feedback with mode-aware timing.

LXXIX.4 Timing is not a silver bullet (classic split + modern digital)

FACT (classic meta): Kulik & Kulik (1988, *Review of Educational Research*, 58(1), 79–97, doi:10.3102/00346543058001079) — 53 studies; **applied classroom** quizzes usually favored **immediate** feedback; **lab / test-content acquisition** studies often favored **delayed**. Timing effects are context-bound.

FACT (test learning): Butler, Karpicke & Roediger (2007, *Journal of Experimental Psychology: Applied*, 13(4), 273–281, doi:10.1037/1076-898X.13.4.273) — after MC practice on prose passages, **delayed** feedback beat immediate on later cued recall; authors attribute benefit partly to **spaced re-presentation** of information.

FACT (digital feedback meta): Brummer, de Boer, Mouw & Strijbos (2024, *Learning Environments Research*, 27(3), 453–476, doi:10.1007/s10984-024-09501-4) — 116 interventions; overall **g ≈ 0.41** vs controls; both immediate and delayed timing showed strong positive effects vs no-feedback baselines; authors argue **clarity/consistency of the timing regime** matters more than a universal clock; meta-regression favored **feedback focus** (process) over timing as the practical lever.

FACT (emerging CAL timing meta — preprint): Kandemir, Esposito, Gurgand & Ramus (preprint, HAL:hal-05546645; 51 studies / 160 ES, 1988–2024) — average immediate-vs-delayed contrast **g = 0.03**, 95% CI [–0.08, 0.13], *ns*; substantial heterogeneity (level, domain, response-time constraints). Cite as preprint, not journal-of-record; use as *warning against timing absolutism*, not as a kill of all delay effects.

Commercial implication: “Our AI gives instant feedback” and “we delay because science” are both overclaims. Position on **mode + focus + generation**, not milliseconds.

Kill: Universal immediate-or-delayed brand claim.

Survive: Consistent, labeled policies per rail (diagnostic / learn / prove).

LXXIX.5 Math-specific: prior knowledge flips the sign

FACT: Fyfe & Rittle-Johnson (2016a, *Journal of Educational Psychology*, 108(1), 82–97, doi:10.1037/edu0000053) — elementary math; when strategy knowledge was induced, **immediate verification feedback hurt** relative to no feedback; without induced knowledge, feedback helped.

FACT: Fyfe & Rittle-Johnson (2016b, *Journal of Experimental Child Psychology*, 147, 140–151, doi:10.1016/j.jecp.2016.03.009) — after strategy instruction, computer feedback (immediate or summative) beat no feedback for low prior knowledge on next-day posttest; **immediate** was particularly effective for mastery across knowledge levels in that design (correct-answer feedback without explicit right/wrong judgment).

FACT (CBLE synthesis): van der Kleij, Feskens & Eggen (2015, *Review of Educational Research*, 85(4), 475–511, doi:10.3102/0034654314564881) — elaborated feedback (EF) produced larger effects than KR/KCR especially for higher-order outcomes; **delayed timing** associated with smaller effects in their models — again: content richness and outcome grain matter more than a slogan about clocks.

HYPOTHESIS (MindCraft): Gap-scan "hard" / low event-count concepts → lean immediate task-focused KR after attempt; higher-knowledge / soft-wrong discrimination items → micro-delay + SE prompt before reveal; never pour explanation-first (Experiment D / SAFE-SE).

Kill: One timing default for all Map nodes.

Survive: Confidence/prior-knowledge tiered timing (ties SAFE-CALIB).

LXXIX.6 Soft-wrong delay vs green-check addiction

Reuse (SAFE-MISCON / SAFE-SE / SAFE-CALIB): Soft-wrongs are diagnostic springboards; student-generated why beats AI monologue; hide-correctness protects the graph when keys are sacred.

HYPOTHESIS — product stack:

1. **Diagnostic rail:** No correctness reveal; record outcome silently; never sell "instant grade."
2. **Learn rail:** Submit → 3–8s SE / "what went wrong?" gate → soft-wrong coach card (task/process) → optional Map link.
3. **Prove rail (SAFE-EXAM / SAFE-DURABLE):** Immediate KR allowed; label as pressure practice; no mastery fireworks.
4. **Tutor HITL:** Pause before pour; Map brief; prompt>explain (SAFE-HITL).
5. **Parent surface:** Show *quality of next move* (retry, transfer), not "feedback latency SLA."

Kill: Green-check dopamine as engagement KPI.

Wound: Some students hate micro-delay — instrument abandon rate; adapt length by prior knowledge, don't delete the gate.

Survive: Attempt-first + task feedback as brandable difference vs ChatGPT instant essay.

LXXIX.7 Competitive positioning

Pattern	Timing sold	MindCraft response
Duo / game apps	Instant KR + XP	Demote green-check; FEI metrics
Khan hints	Fast reveal / explain	Soft-wrong SE before hint binge
Brilliant	Puzzle delight, often immediate	Keep delight; add transfer delay probes
ChatGPT tutors	Instant fluent critique	PWC + micro-delay generation; no monologue≡feedback
Human tutors (untrained)	Explain-first “help”	HITL pause + Map
Cram packs	Instant score theater	SAFE-DURABLE dual rail

Competitive wedge (FOUNDER BELIEF): “We wait just long enough for *your* reason — then we diagnose” beats “instant AI feedback.”

LXXIX.8 Doctrine — SAFE-FBTIME (provisional)

1. **Mode-conditional clocks:** Diagnostic hide; learn micro-delay + SE; prove immediate KR labeled.
2. **Focus > timing:** Task/process feedback; ban self-level ego toasts as learning feedback.
3. **Content > clock:** Prefer informative next-step / soft-wrong contrast over bare KR when the goal is transfer.
4. **Prior-knowledge switch:** Novice acquisition may need faster KR; higher knowledge needs delay/generation.
5. **No timing silver-bullet ads:** Ban “instant feedback AI” and “always delayed = science.”
6. **Kluger honesty:** Instrument harm (abandon, worse delayed mix) — feedback can hurt.
7. **Keys first:** Never delay-or-reveal theater on unverified items (SAFE-GENQ).
8. **Measure FEI not latency NPS:** `retry_120s`, `solo_transfer_pass`, soft-wrong→drill; demote ms-to-green-check.

Confidence: High that FIs are heterogeneous and can harm (Kluger & DeNisi). High that timing effects are context-dependent (Kulik & Kulik; Brummer et al.; Kandemir preprint null average). High that delayed feedback can aid retention via spacing in test settings (Butler et al.). Medium that MindCraft micro-delay+SE lifts transfer vs instant KR on soft-wrongs (needs FB-*). Medium–high that prior knowledge moderates sign (Fyfe line). High that instant-feedback brand claims are a kill.

LXXIX.9 Experiments

ID	Question	Design	Primary

FB-1	Learn-rail micro-delay+SE vs instant KR after soft-wrong	A/B within concept	`retry_120s`; `solo_transfer_pass`; abandon
FB-2	Prior-knowledge tiered timing vs one global delay	Stratify by gap-scan / eventCount	Delayed mix; mastery Δ; frustration
FB-3	Diagnostic hide vs immediate KR diagnostic	A/B (C4 integrity)	Post-scan Map agreement; false weakness
FB-4	Prove-rail immediate KR vs delayed end-of-set KR	A/B near exam	Calibration; anxiety; delayed mix under stakes
FB-5	Process soft-wrong card vs self-praise toast (same timing)	A/B	Transfer; talk/attention proxies
FB-QUAL	10 Maya: “wait for my reason” vs “instant check” preference	Qual	Allergy to lag; trust; identity language

Falsifier: Micro-delay raises survey “thoughtfulness” but lowers transfer and raises abandon → shorten gate or restrict to high-knowledge tier.

Falsifier: Instant KR wins delayed mix for all segments → demote delay doctrine; keep only diagnostic hide + SE prompts without clock.

Pre-register: FB-* before any “instant AI feedback” or “we scientifically delay” campaign (SAFE-LABMETA).

LXXIX.10 So what for MindCraft commercially

- **Copy:** “Feedback that waits for your thinking — then diagnoses the bridge.” Never “instant personalized feedback” as hero.
- **Product:** Ship mode switches in Practice/Diagnostic/Prove; soft-wrong SE gate; tutor pause cue.
- **Positioning:** Against ChatGPT/Duo instant-KR theater; for attempt-first competence evidence.
- **Metric:** Co-primary `retry_120s`, `solo_transfer_pass`, soft-wrong→card; demote latency NPS / green-check rate.
- **Kill list:** Instant-feedback brand; always-delay cosplay; feedback≡learning; ego toasts as FI; green-check NS.
- **Growth:** Differentiate on *quality of struggle recovery*, not feedback speed SLA.
- **Vision:** Identity forms when students author the next move — clocks serve that authorship.

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Part LXXX — Bridge-Gap Product Narrative

Chapter status: Living evidence + Map/copy brief — Researcher tick 2026-08-04

Primary question: When should MindCraft narrate failure as a *connection* problem (knows A, knows B, fails A→B) rather than "weak concept," and what bridge claims survive competitive honesty?

Owners: Product (Map / Practice / Solver) · Engine (bridges / pathfinder) · Brand & copy · Tutor ops · Red Team

Commercial job: Ship a **SAFE-BRIDGE** doctrine: name and treat directed bridges as first-class diagnosis objects; sell *connection failures* as the honest weak-spot story; never sell bridge-count, chat re-explain, or "we connect everything" as the brand.

Builds on: Parts XXXV (competitive audits), XXXIX (interleaving), XL (SAFE-SE), XLVIII (SAFE-TRANSFER), XLIX (SAFE-MISCON), LXVII (SAFE-ONTOLOGY), LXVIII (SAFE-HITL), LXXIX (SAFE-FBTIME).

LXXX.1 Why this chapter exists

Edtech and tutors default to a *node* story: "She's weak at quadratic equations." Parents buy node stories. ChatGPT re-explains the node. Progress dashboards light up nodes. MindCraft's engine already encodes something sharper — **bridges** (directed enabling relations) and bridge-gap severity when two sides look mastered but the join fails — yet Map/copy still risk collapsing that into "another weak topic."

FOUNDER BELIEF under audit: Many high-friction ACT and classroom failures are *relational*: the student can execute each side in isolation and still cannot map, compare, or carry structure across the join. Naming that failure is commercially load-bearing — it is the honest difference from content libraries and fluent chat — *if* we instrument transfer across the bridge and refuse costume claims.

Claims we refuse as doctrine:

1. Bridge-count / ontology edge-count ≡ product quality or moat.
2. "We fix connections" / "no more gaps between topics" as absolute marketing.
3. Re-explaining concept A (or B) ≡ bridging A⟶B.
4. Prerequisite absence ≡ bridge gap (different diagnosis; different CTA).
5. Story-skin change ≡ bridge practice.
6. Bridge-gap severity as Anxiety Score™ / shame Map theater.
7. Chat memory of "we covered both" ≡ knowledge integration.

LXXX.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Fragmented knowledge	Ideas present but weakly linked	High node mastery, low join performance	Celebrate node greens
Bridge (product)	Directed enabling relation A⟶B	Ontology bridges; 1.5× priority in runtime	Treat as another node
Bridge gap	Can do A and B separately; fails tasks needing the join	`isBridgeGap` / severity on Map & `/recommend`	Mislabel as "weak A"
High-road / bridging instruction	Mindful abstraction + search for connections	Compare/SE prompts; bridge cards	Abstraction monologue without instances
Hugging	Practice that approximates target performance	ACT-like items on the join	Only hug one side
Relational comparison	Aligned mapping source⟷target	Side-by-side contrast cards; tutor gestures-as-UX	Analogy without cognitive supports
SAFE-BRIDGE	Connection-first narrative + measurement	This chapter	Bridge cosplay ads

Operational definition (HYPOTHESIS): A failure is a *bridge gap* iff (1) recent evidence shows non-trivial competence on both endpoint concepts/ingredients, (2) items that require the directed join fail at elevated rate relative to endpoint-matched controls, and (3) remediation targets the *relation* (compare, map, carry invariant) rather than re-teaching an endpoint in isolation. Marketing may claim bridge diagnosis only when (1)–(3) are instrumented.

LXXX.3 Knowing both sides ≠ using the join

FACT (spontaneous transfer failure): Gick & Holyoak (1980, *Cognitive Psychology*, 12(3), 306–355, doi:10.1016/0010-0285(80)90013-4) — learners who possessed an analogous “fortress” story often failed to apply the convergence principle to Duncker’s radiation problem until a **hint** made the connection available. Structural isomorphism does not guarantee noticed transfer.

FACT (transfer mechanisms): Salomon & Perkins (1989, *Educational Psychologist*, 24(2), 113–142, doi:10.1207/s15326985ep2402_1) — **low-road** transfer (automatic triggering after varied practice) vs **high-road** transfer (mindful abstraction and deliberate search for connections). Findings of transfer/nontransfer track whether those conditions were met — not whether “the topics were covered.”

FACT (instructional tactics): Perkins & Salomon (1988, *Educational Leadership*, 46(1), 22–32; summarized in Perkins & Salomon, 1992, “Transfer of Learning,” *International Encyclopedia of Education*) — **hugging** (approximate the target) and **bridging** (prompt abstraction, metacognition, planned connections). Education gets transfer when designed for it.

FACT (taxonomy honesty): Barnett & Ceci (2002, *Psychological Bulletin*, 128(4), 612–637, doi:10.1037/0033-2909.128.4.612) — “far transfer” is not one effect size; content×context dimensions must be named. Bridge-gap claims should specify *which* dimension moved (representation, problem type, temporal delay, solo vs scaffolded).

Commercial implication: “She’s done both units” is not readiness. Sell *join evidence*. Chat that re-explains each unit is the wrong remediator for a join failure.

Kill: Coverage ≡ connection.

Survive: Hinted / designed bridging as the product job (ties SAFE-TRANSFER).

LXXX.4 Knowledge integration — links are the learning object

FACT (science-education synthesis): Eylon & Linn (1988, *Review of Educational Research*, 58(3), 251–301, doi:10.3102/00346543058003251) — perspectives on concept learning and problem solving converge on mechanisms of change in knowledge organization; fleeting coverage of many topics underperforms deeper integration work.

FACT (KI framework): Linn & Eylon (2006, in Alexander & Winne, Eds., *Handbook of Educational Psychology*, 2nd ed., pp. 511–544) — knowledge integration as eliciting ideas, adding normative ideas, **distinguishing**, and reflecting toward a more coherent network. Fragmented “pieces” can coexist without productive links.

HYPOTHESIS (MindCraft): Soft-wrongs that fire when both endpoints look “green” are high-value KI events: elicit the student’s mapping, add the bridge card, force distinguish (“what stays the same across these two?”), then reflect into Map as bridge evidence — not as a third unrelated node.

Kill: More topics completed as integration proof.

Survive: Distinguish + reflect on the *relation* as the object of learning (Marton-style discernment in XLVIII applied to bridges).

LXXX.5 Analogies without supports are theater

FACT (classroom video): Richland, Zur & Holyoak (2007, *Science*, 316(5828), 1128–1129, doi:10.1126/science.1142103) — TIMSS 1999 video sample: U.S., Hong Kong, and Japanese eighth-grade lessons all used relational comparisons frequently, but **U.S. teachers provided fewer cognitive supports** (familiar source, visual source kept visible, spatial cues, gestures, imagery) that help students notice the mapping. Analogy *mention* ≠ analogy *learning*.

FACT (instructional agenda): Richland, Stigler & Holyoak (2012, *Educational Psychologist*, 47(3), 189–203, doi:10.1080/00461520.2012.667065) — many post–K–12 students still treat math as memorized (often incorrect) procedures rather than a relational system; refining instruction so students draw connections through **relational comparisons** is a research-backed path to flexible knowledge; U.S. classrooms under-use the supports that make comparisons work.

Product translation:

1. Bridge cards must **co-present** both sides (source visible while targeting the join) — not sequential monologues.
2. Prompt alignment (“point to what maps to what”) before pouring explanation.
3. Tutor HITL briefs: gesture/spatial UX analogues (highlight shared structure), not “explain quadratic again.”
4. Interleave near-miss types that share surface features but differ on the bridge (XXXIX) — discrimination of the join.

Kill: “We use analogies” as pedagogy claim without co-presence + student mapping.

Wound: Some students find dual displays overwhelming (CLT) — dose with SAFE-DD / prior-knowledge tiers.

Survive: Supported relational comparison as Map→Practice CTA copy.

LXXX.6 Engine reality vs UI honesty

Reuse (SAFE-ONTOLOGY): Diagnosis-before-dialogue; bridges prioritized over isolated nodes in ingredient runtime; chat optimizes next utterance, not falsifiable join state.

HYPOTHESIS — product stack:

1. **Map:** Render bridge gaps as *edges* (or edge callouts), not only as red nodes. Copy: “You can do X and Y — the hard part is connecting them.”
2. `/recommend` / **PawHub Practice:** Prefer playable bridge-gap missions when severity dominates plain `1−mastery` (ties C1 worstWeakness).
3. **Practice:** Object of learning = the bridge; VT: hold join invariant, vary one endpoint surface at a time; measure `solo_transfer_pass` on join items after delay.

4. **Solver / coach:** Soft-wrong → "which connection broke?" SE before endpoint re-teach (SAFE-SE / SAFE-FBTIME).
5. **Tutor ops:** Map brief must list top bridge gaps; prompt>pour on the relation (SAFE-HITL).
6. **Parent surface:** "Connection practice" / proof on joins — not "another weak chapter" shame.

SPECULATION: Competitors can screenshot a graph; they cannot ship longitudinal bridge evidence + solo join transfer without the spine — *if* MindCraft instruments and shows those metrics honestly (same data-moat logic as LXVII).

Kill: Node-only weakness narrative when join evidence exists.

Survive: Connection-first CTA when endpoints are non-naive.

LXXX.7 Competitive positioning

Pattern	Failure story sold	MindCraft response
Khan topic paths	Weak skill node	Name bridge when both skills green
Duo skill tree	Isolated skill XP	Ban XP-as-connection; join probes
Brilliant puzzles	Insight inside one puzzle	Keep delight; add explicit compare across puzzles
ChatGPT tutors	Re-explain whichever topic was named	Refuse re-teach-as-bridge; force mapping
Tutor shops	"Review chapter 5 again"	Brief on A→B join; TR-2 style compare
Cram packs	Massed node blitz	SAFE-DURABLE: join returns beat massed re-cover

Competitive wedge (FOUNDER BELIEF): "We find the connection you think you already have" beats "we explain every topic again."

LXXX.8 Doctrine — SAFE-BRIDGE (provisional)

1. **Connection > isolated node** when endpoint evidence is non-trivial and join items fail.
2. **Name the edge in copy** — Map, Practice CTA, tutor brief, parent note.
3. **Remediate the relation** — co-present compare, SE mapping, bridge cards; ban endpoint re-teach as default.
4. **Supports required** — visible source+target, alignment prompts; analogy mention ≠ pedagogy.
5. **Measure join transfer** — delayed / format-hop / solo items on the bridge; demote node greens as readiness.
6. **Distinguish diagnoses** — missing prerequisite ≠ bridge gap ≠ format gap (Part LXXXII queued).
7. **No bridge-count moat ads** — edges are substrate; FEI + `solo_transfer_pass` on joins are the claim.

8. **Severity with dignity** — no shame heatmaps; competence-protection framing (SAFE-FORGET honesty applies to aged bridge evidence too).

Confidence: High that spontaneous far/join transfer is rare without cues (Gick & Holyoak; Salomon & Perkins). High that hugging/bridging are designable tactics (Perkins & Salomon). High that U.S.-style analogy use often lacks cognitive supports (Richland et al. 2007). High that knowledge integration is about links, not coverage (Eylon & Linn; Linn & Eylon). Medium that MindCraft's bridge-gap detector matches the operational definition above (needs BRIDGE-* calibration against human raters). Medium that connection-first copy lifts WTP vs node-weakness copy (needs parent CBC). High that re-explain≡bridge and bridge-count≡moat are kills.

LXXX.9 Experiments

ID	Question	Design	Primary
BRIDGE-1	Bridge-card compare/SE vs endpoint re-teach after detected gap	A/B within bridge id	Join-item accuracy; `solo_transfer_pass`; time-to-remediate
BRIDGE-2	Map edge callout CTA vs node-weakness CTA	A/B	Mission start; completion; post join probe
BRIDGE-3	Detector vs human tutor labels on same sessions	Agreement study	Precision/recall of `isBridgeGap`; false "weak A"
BRIDGE-4	Co-present dual display vs sequential explain	A/B	Mapping SE quality; transfer; CLT abandon
BRIDGE-5	Parent CBC: "connection gap" vs "weak topic" message	CBC	WTP; trust; stigma
BRIDGE-QUAL	10 Maya: "I knew both but…" language	Qual	Identity language; allergy to Map edges

Falsifier: Endpoint re-teach matches or beats bridge remediation on delayed join items → demote connection-first default; keep bridges as secondary tags only.

Falsifier: Detector disagreement with tutors > threshold → freeze severity-driven PawHub routing until calibrated.

Falsifier: Connection copy raises shame/abandon without transfer lift → rewrite dignity framing; reduce edge heat.

Pre-register: BRIDGE-* before any "we fix the connections textbooks miss" campaign (SAFE-LABMETA).

LXXX.10 So what for MindCraft commercially

- **Copy:** "The gap isn't the chapter — it's the connection." Never "master every topic" as the hero.

• **Product:** Edge-aware Map; bridge missions in Practice; Solver SE on joins; tutor briefs list bridges first when severity wins.

• **Positioning:** Against chat re-explain and skill-tree XP; for inspectable join diagnosis (with SAFE-ONTOLOGY).

• **Metric:** Co-primary join `solo_transfer_pass`, bridge remediation success, detector↔tutor agreement; demote node greens / bridge-count.

• **Kill list:** Bridge-count moat; re-explain≡bridge; coverage≡connection; absolute “we connect everything”; shame-edge theater.

• **Growth:** Parent WTP for *named connection practice* after both sides look fine — a story Khan/Duo/ChatGPT do not own.

• **Vision:** Identity as a mathematical thinker is relational competence — seeing structure *between* ideas, not collecting lit nodes.

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Part LXXXI — Cold-Start Without Fake Mastery

Chapter status: Living evidence + onboarding/copy brief — Researcher tick 2026-08-05

Primary question: How should MindCraft initialize a student graph on day one so recommendations feel personal *without* manufacturing mastery fireworks, inflated greens, or “we already know you” theater?

Owners: Product (gap-scan / PawHub / Map) · Engine (seed-assessment / priors / event sources) · Brand & copy · Parent trust · Red Team

Commercial job: Ship a **SAFE-COLD** doctrine: humble priors + labeled seed evidence + hide-correctness probes; sell *honest mapping*, never day-one mastery or confidence≡mastery.

Builds on: Parts L (SAFE-CALIB), LXVII (SAFE-ONTOLOGY), LXXII (SAFE-GENQ), LXXVI (SAFE-FORGET), LXXIX (SAFE-FBTIME), LXXX (SAFE-BRIDGE). Product seam: `/seed-assessment`, C4 hide-correctness diagnostic, `population_priors`, `eventCount === 0` → Learn-next.

LXXXI.1 Why this chapter exists

Every adaptive product faces the same commercial temptation at cold start: empty state feels broken, so the UI invents competence — green trees, "You're already strong at…," personalized paths from zero observations. Parents buy the fireworks. Students learn the app rewards confident self-report and hint-skipping. The graph then *lies* for weeks.

MindCraft's gap-scan already does something sharper: confidence ratings → synthetic `assessment` events (source=`onboarding_assessment`) that *replace* prior seed, plus optional hide-correctness probes. That is the right *family* of move — **if** we refuse to market the seed as mastery, refuse easy-default skips, and keep Map language in "uncertain / untouched / early signal" until delayed proof arrives (SAFE-FORGET).

FOUNDER BELIEF under audit: The first session's job is *diagnosis hygiene*, not delight-as-competence. Personalization that survives is "we asked hard questions and stayed honest about uncertainty," not "your Map lit up on minute five."

Claims we refuse as doctrine:

1. Day-one green Map / mastery fireworks ≡ personalization.
2. Self-rated "easy" ≡ mastery or ACT readiness.
3. Population prior alone ≡ student-specific diagnosis worth selling.
4. Skip-scan / "I'll explore" defaults that write easy/high mastery.
5. Onboarding chat fluency or AI pep ≡ calibrated prior.
6. Cold-start "Identity Score™" or "math person already" splash.
7. Unverified generated keys as first diagnostic ground truth (SAFE-GENQ).

LXXXI.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Cold start	No / little learner history for KT or recs	New `users/{uid}`; empty graph	Fake greens to fill void
Prior (P(L0))	Initial P(know skill) before observations	Ontology `population_priors`; Beta seeds	Sell prior as "your level"
Seed evidence	Labeled synthetic events from onboarding	`/seed-assessment` confidence map	Treat seed as FEI mastery
Probe evidence	Observed correctness without reveal theater	C4 hide-correctness items	Reveal keys → answer-hunting

Calibration	Confidence matches accuracy long-run	SAFE-CALIB; hard/kinda/easy	Confidence≡truth
Fake mastery	UI/state claims competence without durable proof	Fireworks; green untouched nodes	Trust debt; wrong path
SAFE-COLD	Humble init + labeled seed + probes	This chapter	Personalization cosplay

Operational definition (HYPOTHESIS): A cold-start is *honest* iff (1) every non-prior belief is tagged by source (`onboarding_assessment` / `practice` / session), (2) Map/copy distinguish untouched, seed-only, and delayed-proof states, (3) no mastery confetti or "you've mastered X" until criteria beyond seed confidence, and (4) skip paths do not write high mastery. Marketing may claim "personalized from day one" only as *personalized uncertainty + next probe*, not personalized mastery.

LXXXI.3 Knowledge tracing needs a prior — priors are not biographies

FACT (BKT spine): Corbett & Anderson (1995, *User Modeling and User-Adapted Interaction*, 4, 253–278, doi:10.1007/BF01099821) — knowledge tracing maintains P(learned) per production rule from observed attempts (prior, learn, guess, slip). Mastery-based sequencing depends on those estimates; the model is only as honest as its updates and its **initial prior**.

FACT (individualizing P(L0)): Pardos & Heffernan (2010, in *UMAP 2010*, LNCS 6075, pp. 255–266, doi:10.1007/978-3-642-13470-8_24) — standard deployed BKT often used skill-specific but not student-specific initial knowledge; individualizing the prior (especially pooling information across skills for each student) improved prediction on ASSISTments problem sets (lower error on 33/42 sets in their evaluation). Cold start is a *parameter* problem, not a license to invent greens.

FACT (prior sensitivity): Students enter with heterogeneous P(L0); a single global "everyone starts at 0.1" or "everyone starts mastered" both mis-sequence practice (BKT robustness / prior-variation literature; Pardos & Heffernan individualization agenda).

HYPOTHESIS (MindCraft): Ontology `population_priors` are the right *default* P(L0) layer — inspectable, shared, not personalized theater. Student-specific movement must come from labeled seed + probes + practice, with decay toward prior (align copy with engine honesty).

Kill: Population prior sold as "we know Maya."

Survive: Prior as humble shared baseline; personalization = updates with provenance.

LXXXI.4 Self-report is necessary and systematically biased

Gap-scan asks hard / kinda / easy. That is commercially necessary (coverage across ~29 ACT concepts without a three-hour exam) and epistemically dangerous.

FACT (overconfidence): Fischhoff, Slovic & Lichtenstein (1977, *Journal of Experimental Psychology: Human Perception and Performance*, 3(4), 552–564) — people are often wrong when "certain"; calibration curves show hit rates below stated confidence.

FACT (reasons bias): Koriat, Lichtenstein & Fischhoff (1980, *Journal of Experimental Psychology: Human Learning and Memory*, 6(2), 107–118) — confidence is inflated by evidence *for* the chosen answer; prompting contradicting reasons improves appropriateness of confidence.

FACT (unskilled–unaware pattern): Kruger & Dunning (1999, *Journal of Personality and Social Psychology*, 77(6), 1121–1134, doi:10.1037/0022-3514.77.6.1121) — bottom-quartile performers grossly overestimated ability. (Directional product risk — not a student-facing label.)

FACT (math metacognition): Erickson & Heit (2015, *Frontiers in Psychology*, 6, 742, doi:10.3389/fpsyg.2015.00742) — students overpredicted math test scores alongside math anxiety; not simply "low confidence everywhere."

FACT (item-level blindness): Lindsey & Nagel (2015, *Physical Review Special Topics — Physics Education Research*, 11, 020103, doi:10.1103/PhysRevSTPER.11.020103) — at item level, students of all abilities struggle to foresee which questions they will miss — self-assessment is a blunt prior, not a key.

Commercial implication: Map "easy" → L3 gating is coherent; Map "easy" → green mastery badge is a lie. Pair confidence seed with hide-correctness probes (SAFE-CALIB / C4); never reveal keys during diagnostic (SAFE-GENQ).

Kill: Confidence≡mastery; "raise onboarding confidence" as success metric.

Survive: Confidence as *routing prior* + miss-tier signal; probes as corrective evidence.

LXXXI.5 Gaming the onboarding is fake mastery's twin

FACT (gaming + harm): Baker, Corbett, Koedinger & Wagner (2004, *CHI '04*, pp. 383–390) — "gaming the system" = succeeding by exploiting system properties rather than learning; gaming frequency correlated negatively with learning, distinct from other off-task behavior.

HYPOTHESIS (product): Cold-start analogues include tapping "easy" on everything, skipping gap-scan, or hint-binging first Practice to farm greens. Designs that reward scan-completion over scan-honesty invite this.

Product translation:

1. No streak/XP for "finished diagnostic."
2. Skip-scan must not write optimistic mastery; prefer untouched + population prior.
3. Retake gap-scan *replaces* onboarding seed — do not stack optimistic layers.
4. PawHub Practice/Learn-next read `eventCount` honesty — do not celebrate seed as conquered islands.

Kill: Diagnostic completion % as North Star.

Wound: Anxious under-raters ("hard" everywhere) — SAFE-CALIB underconfidence unlock still applies.

Survive: Friction that makes honest mapping cheaper than fake competence.

LXXXI.6 Adaptive testing envy — shorter ≠ fireworks

FACT (CAT premise): Adaptive testing selects informative items given current estimates and can shorten tests when banks/models are calibrated (CAT survey/primer tradition). Early steps still start from a prior; stopping rules are about *measurement error*, not celebration.

HYPOTHESIS: Borrow CAT's *information* ethic (ask what reduces uncertainty) without placement trophies or "you placed into Gold" skins. Gap-scan + sparse probes ≈ poor-man's adaptive placement; sell "where to start," not a belt.

Kill: Placement-league / belt / IQ-cosplay cold start.

Survive: Uncertainty-reducing item selection as the onboarding craft.

LXXXI.7 Engine + UI stack (SAFE-COLD)

Reuse: SAFE-ONTOLOGY (inspectable state); SAFE-CALIB + C4 hide-correctness; SAFE-FORGET (age evidence; no fireworks≡durable); SAFE-FBTIME (diagnostic hide); SAFE-GENQ (verified keys only for probes that write mastery).

HYPOTHESIS — product stack:

1. **Prior layer:** Population / ontology priors visible as *shared* baselines — never "your mastery."
2. **Seed layer:** `/seed-assessment` writes labeled synthetic events; UI copy: "your confidence map" / "starting signals," not "mastered."
3. **Probe layer:** Short hide-correctness set across concepts/formats; record outcomes without reveal; optional contradicting-reason micro-prompt on high-confidence picks (Koriat et al. mechanism, light-touch).
4. **Map:** Untouched ≠ red failure; seed-only ≠ green mastery; reserve strong chrome for delayed/generation proof.
5. **PawHub:** Learn-next = first `eventCount === 0` playable on exam path; Practice = worst playable weakness — both read honesty, not confetti.
6. **Parent:** "We started with an honest map" CBC arm vs "personalized genius path from minute one."

SPECULATION: Competitors can ship prettier empty-state trees; MindCraft's wedge is *provenance* — every early claim has a source tag parents and tutors can understand.

LXXXI.8 Competitive positioning

Pattern	Cold-start move	MindCraft response
Duo skill tree	Early greens / crowns	Untouched honesty; ban XP-as-mastery
Khan mastery	Energy + % bars from thin attempts	Seed ≠ mastery bar; delayed proof
Brilliant	Delight puzzle → "you're clever"	Delight ok; no cleverness≡graph

ChatGPT tutors	Instant "I know your level" from chat	Chat ≠ prior; refuse fluency prior
Bootcamp quizzes	Placement score theater	Uncertainty map; no belt
Adaptive apps	"AI personalized path" from 3 clicks	Label seed; show what was never probed

Competitive wedge (FOUNDER BELIEF): "We refuse to fake knowing you" is sellable trust — especially to parents burned by apps that looked personal and taught nothing.

LXXXI.9 Doctrine — SAFE-COLD (provisional)

1. **Humble prior first** — population/ontology priors initialize; they are not biography.
2. **Labeled seed only** — confidence → synthetic events with source tag; replace-on-retake, don't stack lies.
3. **Probes without reveal** — C4 hide-correctness; verified keys only; no answer-hunting diagnostic.
4. **Confidence ≠ mastery** — hard/kinda/easy routes level + struggle prior; never crowns.
5. **Map chrome matches evidence age** — untouched / seed-only / proved (SAFE-FORGET).
6. **No skip-to-green** — skip leaves untouched + prior; completion is not a learning KPI.
7. **Anti-gaming** — no streak/XP for scan finish; watch uniform-easy patterns as QA.
8. **Copy:** "Honest starting map" / "where to begin" — never "you've already mastered" / Identity Score™.

Confidence: High that BKT-style systems need explicit priors and that individualizing P(L0) matters (Corbett & Anderson; Pardos & Heffernan). High that confidence is miscalibrated and especially risky for low performers (Fischhoff et al.; Kruger & Dunning; Erickson & Heit). High that gaming harms learning when systems can be exploited (Baker et al.). Medium that MindCraft's current seed→event map is well-calibrated to later practice (needs COLD-*). Medium that honest-cold-start copy wins parent WTP vs fireworks (needs CBC). High that day-one mastery fireworks and confidence≡mastery are kills.

LXXXI.10 Experiments

ID	Question	Design	Primary
COLD-1	Seed-only vs seed+hide-correctness probes	A/B new users	7d `/recommend` stability; probe–seed disagreement rate; abandon
COLD-2	Map chrome: seed-as-mastery greens vs honest untouched/seed labels	A/B	Trust items; wrong-path starts; parent CBC
COLD-3	Uniform-easy / skip patterns → QA flag vs ignore	Observational + intervene	Later miss rate; gaming-like behavior

COLD-4	Contradicting-reason micro-prompt on “easy” ratings	A/B	Calibration vs probes; time cost
COLD-5	Parent CBC: “honest map” vs “personalized from minute one”	CBC	WTP; trust; stigma
COLD-QUAL	10 Maya: what felt fake vs useful in first session	Qual	Language for copy; fireworks allergy

Falsifier: Seed-as-mastery greens raise 7d retention *and* delayed transfer vs honest chrome → revisit chrome (unlikely; if retention-only, treat as engagement trap).

Falsifier: Probes add abandon without improving recommend quality → shorten probe set; keep seed.

Falsifier: Honest-map copy loses WTP with no trust gain → reframe dignity without fireworks; do not restore fake greens.

Pre-register: COLD-* before any “personalized genius path from day one” campaign (SAFE-LABMETA).

LXXXI.11 So what for MindCraft commercially

- **Copy:** “An honest starting map — not a fake report card.” Never day-one “mastered” / belts / Identity Score™.
- **Product:** Prior → labeled seed → hide-correctness probes → provenance on Map; PawHub reads `eventCount` and seed humility.
- **Positioning:** Against Duo greens, chat “I know you,” and placement theater; for inspectable cold-start (with SAFE-ONTOLOGY).
- **Metric:** Probe–seed disagreement, recommend stability 7d, skip/uniform-easy rate, parent trust; demote diagnostic completion % and early green counts.
- **Kill list:** Fake mastery fireworks; confidence≡mastery; skip-to-green; population-prior-as-biography; unverified diagnostic keys.
- **Growth:** Trust compounds when week-two practice matches week-one honesty — fireworks create churn when the lie shows.
- **Vision:** Identity transformation starts with *accurate self-location*, not premature crowning.

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Part LXXXII — Format-Axis Product Narrative

Chapter status: Living evidence + Map/Practice/copy brief — Researcher tick 2026-08-05

Primary question: When should MindCraft narrate weakness as a *format* failure (can do the idea symbolically, fails the diagram — or vice versa) rather than "weak concept," and what FormatId claims survive competitive honesty?

Owners: Product (Practice / Map / gap-scan) · Engine (format gaps / `getQuestions` format param) · Bank & generation · Brand & copy · Red Team

Commercial job: Ship a **SAFE-FORMAT** doctrine: treat input representation as a first-class diagnosis axis; sell *format-flexible competence* with conversion evidence; never sell format-count, decorative diagrams, or "we cover every format" as the brand.

Builds on: Parts XXVI (CLT), XXXIX (interleaving), XL (SAFE-SE), XLVIII (SAFE-TRANSFER / VT), LXIV (SAFE-ELL language vs math load), LXVII (SAFE-ONTOLOGY), LXXII (SAFE-GENQ), LXXX (SAFE-BRIDGE), LXXXI (SAFE-COLD). Product seams: `FormatId` / C2–C3, format-tagged bank, format-aware CTA, Layer-4 `representation_profile`.

LXXXII.1 Why this chapter exists

Edtech and tutors default to a *topic* story: "She's weak at functions." Parents buy topics. Progress dashboards light up topics. MindCraft's bank and recommend stack already encode something sharper — **formats** (`word_problem`, `diagram`, `number_line`, `symbolic_expression`, `coordinate_graph`, …) and format-preferring practice — yet Map/copy still risk collapsing "missed the graph" into "weak functions."

FOUNDER BELIEF under audit: A large share of ACT/classroom fails are *representational*: the student can execute the idea in one vessel and still cannot read, translate, or produce it in another. Naming that failure is commercially load-bearing — it is an honest difference from symbolic drill packs and fluent chat that only ever sees text — *if* we instrument conversion and refuse costume claims.

Claims we refuse as doctrine:

1. Format-count / FormatId vocabulary size ≡ product quality or moat.
2. “We teach every format” / “visual learning for all” as absolute marketing.
3. Pretty diagram / story art ≡ FormatId coverage or learning.
4. Concept mastery on one format ≡ format-flexible readiness.
5. Format shuffle / random vessel ≡ science without conversion practice.
6. Format Personality / learning-style quiz as North Star.
7. Unverified generated figures as diagnostic ground truth (SAFE-GENQ).

LXXXII.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
External representation	Visible encoding of math ideas	Item stem vessel; Map format tags	Treat as decoration
Format / vessel	Product vocab for input representation	`FormatId`; Layer-4 `representation_profile`	Collapse into topic only
Treatment	Transform within one register	Symbolic algebra within symbols	Call treatment “transfer”
Conversion	Translate across registers	Diagram ↔ equation practice	Re-teach topic instead
Representational competence	Knowing how a vessel encodes content	Format fluency + conversion prompts	Assume vessels are transparent
Format gap	Strong in format A, weak in B for same concept	Format severity in `/recommend` / worstWeakness	Mislabel as “weak concept”
SAFE-FORMAT	Format-first narrative + conversion measurement	This chapter	Format cosplay ads

Operational definition (HYPOTHESIS): A failure is a *format gap* iff (1) recent evidence shows non-trivial competence on the same concept in at least one other FormatId, (2) items in the weak FormatId fail at elevated rate relative to format-matched controls on that concept, and (3) remediation targets *reading/producing/converting* that vessel — not only re-explaining the concept in the student’s strong format. Marketing may claim format diagnosis only when (1)–(3) are instrumented.

LXXXII.3 Registers differ — conversion is the hard cognitive cut

FACT (semiotic registers): Duval (2006, *Educational Studies in Mathematics*, 61(1–2), 103–131) — mathematical activity requires coordinating semiotic representations; **treatment** (within one register) and **conversion** (across registers) are distinct cognitive processes; conversion is often the deciding factor for comprehension failures even when within-register work looks fine.

FACT (computational efficiency ≠ equivalence): Larkin & Simon (1987, *Cognitive Science*, 11(1), 65–100, doi:10.1111/j.1551-6708.1987.tb00863.x) — diagrammatic and sentential representations can be informationally equivalent yet differ sharply in search cost, pattern recognition, and which inferences are "free"; a diagram is sometimes worth ten thousand words *for a given operator set* — not as a marketing slogan that "visuals always help."

Commercial implication: "She knows linear equations" is incomplete. Sell *which vessels she can use* and whether she can convert. Chat that only ever explains in symbols cannot remediate a diagram-read failure.

Kill: Single-format accuracy ≡ concept mastery.

Survive: Conversion as a named learning object (ties SAFE-TRANSFER / SAFE-BRIDGE when the join is also a register join).

LXXXII.4 Multiple representations help only under design + competence conditions

FACT (functional taxonomy): Ainsworth (1999, *Computers & Education*, 33(2–3), 131–152, doi:10.1016/S0360-1315(99)00029-9) — MERs serve distinct functions: **complement**, **constrain**, and **construct**; conflicting evaluation results track unrecognized function mix and unsupported translation across representations.

FACT (DeFT): Ainsworth (2006, *Learning and Instruction*, 16(3), 183–198, doi:10.1016/j.learninstruc.2006.03.001) — effectiveness depends on design parameters, pedagogical functions, and the cognitive **tasks** of understanding each representation *and relating them*; learners often treat representations in isolation and fail to integrate.

FACT (representation dilemma): Rau (2017, *Educational Psychology Review*, 29(4), 717–761, doi:10.1007/s10648-016-9365-3) — students often must learn content they do not yet understand from visuals they do not yet understand; benefit from multiple visuals requires **representational competencies**; more visuals ≠ automatically better — conditions matter (T+SV vs T+MV hypotheses).

FACT (support moderator): Rexigel, Kuhn, Becker & Malone (2024, *Educational Psychology Review*, 36, article 124, doi:10.1007/s10648-024-09958-y) — meta-analysis (132 effect sizes): learning with more than two external representations can show small performance and cognitive-load advantages vs two, but benefits **primarily depend on appropriate support** — stacking formats without scaffolds is not a product strategy.

HYPOTHESIS (MindCraft): Format-axis Practice should (a) diagnose per FormatId, (b) remediate with co-present conversion supports (source vessel visible while targeting the weak one), and (c) measure delayed format-hop transfer — not dump three diagrams beside a stem for "engagement."

Kill: "More formats = better learning" without support + conversion tasks.

Wound: Dual/multi displays can overload (CLT / XXVI) — dose by prior knowledge and anxiety (SAFE-DD).

Survive: Supported conversion as Map→Practice CTA.

LXXXII.5 Coherence: decoration is not FormatId coverage

FACT (extraneous material hurts): Mayer, Heiser & Lonn (2001, *Journal of Educational Psychology*, 93(1), 187–198, doi:10.1037/0022-0663.93.1.187) — adding interesting but extraneous material can reduce understanding (coherence / cognitive constraints on multimedia).

HYPOTHESIS: Story-world chrome that does not encode the mathematical structure is *not* a diagram FormatId item. Bank tags must distinguish decorative illustration from structurally necessary diagrams/graphs/number lines. Generation that invents pretty figures without keyed structure fails SAFE-GENQ and SAFE-FORMAT together.

Kill: Story art ≡ visual-learning claim.

Survive: Structural FormatId tagging + verified figures only when they carry the math.

LXXXII.6 Engine + UI stack (SAFE-FORMAT)

Reuse: SAFE-ONTOLOGY (inspectable state); SAFE-TRANSFER / VT (vary representation, hold invariant); SAFE-BRIDGE (when join = register join); SAFE-ELL (language load ≠ math load; format triangulation); SAFE-GENQ (verified keys/figures); SAFE-COLD (format probes in hide-correctness without fireworks); C1 severity comparable across concept/bridge/format gaps.

HYPOTHESIS — product stack:

1. **Bank:** Every practice/diagnostic item carries FormatId; prefer format-matched draws with concept-pool fallback (C3).
2. **State:** Maintain `representation_profile` (or equivalent per-format evidence), not only concept mastery.
3. **Map /** `/recommend`**:** Surface format gaps when severity dominates plain `1−mastery`; copy: “You can do this symbolically — the hard part is the graph” (or reverse).
4. **PawHub / Practice CTA:** Format-aware mission labels; object of learning = vessel + conversion, not only topic name.
5. **Cards / tutor HITL:** Co-present source and target formats; SE prompt on what maps to what (SAFE-SE); never explain-only in the strong format.
6. **Parent:** “Format flexibility” CBC arm vs “more visuals” / learning-style quiz.

SPECULATION: Competitors can ship prettier diagram packs; MindCraft’s wedge is *falsifiable format state* — which vessels fail, with conversion missions, not a VARK costume.

LXXXII.7 Competitive positioning

Pattern	Format move	MindCraft response
Khan / drill apps	Mostly symbolic or text stems	Format profile + conversion missions
Duo	Skill tree greens ignore vessel	Ban format-blind mastery crowns
Brilliant	Delight visuals as brand	Delight ok; tag structure vs decoration

ChatGPT tutors	Text-only explanations	Text fluency ≠ diagram competence
Learning-style apps	“You’re a visual learner”	Kill style quiz; sell conversion evidence
ACT cram PDFs	Mixed vessels unlabeled	Label FormatId; dual-rail prove vs learn

Competitive wedge (FOUNDER BELIEF): “We find when the idea fails in the vessel” is sellable diagnosis — especially vs chat that never sees the graph the student cannot read.

LXXXII.8 Doctrine — SAFE-FORMAT (provisional)

1. **Format is first-class evidence** — tag items; update per-FormatId competence separately from concept greens.
2. **Conversion is the object** — remediate with co-present vessels + mapping prompts; measure format-hop transfer.
3. **Concept ≠ format-flexible** — single-vessel success never markets as full mastery.
4. **Support before stack** — more FormatIds without scaffolds is load, not pedagogy (Ainsworth / Rau / Rexigel et al.).
5. **Decoration ≠ diagram** — structural encodings only count as FormatId coverage.
6. **No Format Personality™** — ban learning-style quiz as North Star.
7. **Severity honesty** — format gaps compete in worstWeakness with concept/bridge gaps (C1).
8. **Copy:** “Where the idea breaks in the graph / word problem / number line” — never “we teach every format” / format-count moat.

Confidence: High that treatment≠conversion and conversion drives many math comprehension failures (Duval). High that MERs are conditional and translation is hard (Ainsworth 1999/2006; Rau 2017). High that unsupported multi-rep stacking is not automatic gain (Rexigel et al. 2024). High that diagram vs sentential efficiency differs by operators (Larkin & Simon). Medium that MindCraft’s current FormatId bank density is enough for reliable per-student format profiles (needs FORMAT-* + coverage audit). Medium that format-gap copy wins parent WTP vs “more visuals” (needs CBC). High that format-count moat, style quizzes, and decoration≡coverage are kills.

LXXXII.9 Experiments

ID	Question	Design	Primary
FORMAT-1	Format-gap CTA vs concept-only CTA when severity says format	A/B	Format-hop `solo_transfer_pass`; abandon; hint binge
FORMAT-2	Co-present conversion card vs strong-format re-explain	A/B within concept	Delayed weak-format accuracy

FORMAT-3	Structural diagram items vs decorative-art distractors	Bank A/B or audit	Error patterns; false “diagram” tags
FORMAT-4	Multi-format stack with supports vs unsupported triple display	A/B	Load proxies; transfer; time
FORMAT-5	Parent CBC: “format flexibility” vs “visual learning / every format”	CBC	WTP; trust; stigma
FORMAT-QUAL	10 Maya: when did a graph/word problem feel like a different subject?	Qual	Copy language; vessel shame

Falsifier: Format-gap CTAs raise engagement but not delayed weak-format transfer → demote copy; keep tagging for research only.

Falsifier: Co-present conversion increases load/abandon without transfer gain → dose/scaffold; do not restore explain-in-strong-format as default.

Falsifier: “Every format” marketing wins WTP with no trust cost → still ban as doctrine (overclaim); test quieter “graph vs symbols” specificity.

Pre-register: FORMAT-* before any “visual learning / every format / Format Score™” campaign (SAFE-LABMETA).

LXXXII.10 So what for MindCraft commercially

- **Copy:** “We find where the idea breaks in the format — graph, symbols, words — and practice the conversion.” Never format-count, VARK, or “we teach every format.”
- **Product:** FormatId on bank + `representation_profile` + format-aware `/recommend` CTA; co-present conversion cards; Map language for format gaps alongside bridges.
- **Positioning:** Against symbolic-only drills, chat text fluency, and learning-style quizzes; for inspectable format diagnosis (with SAFE-ONTOLOGY).
- **Metric:** Format-hop transfer, format-gap mission starts, false decorative tags, parent CBC; demote format-count and style-quiz completion.
- **Kill list:** Format-count moat; decoration≡coverage; concept≡format-flexible; shuffle≡science; Format Personality™.
- **Growth:** ACT parents already feel “she freezes on word problems / graphs” — name it, instrument it, sell the remediator.
- **Vision:** Identity as a math thinker includes *moving ideas across vessels*, not only green topic nodes.

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Part LXXXIII — Parent Dashboard Honesty UX

Chapter status: Living evidence + parent-surface brief — Researcher tick 2026-08-05

Primary question: How should MindCraft's parent-facing surface show *proof age, returns, and competence evidence* so families can act — without shame heatmaps, tonight-% fireworks, or surveillance cosplay?

Owners: Product (ParentDashboard / MoC surfaces) · Growth/copy · Brand · Privacy · Red Team

Commercial job: Ship a **SAFE-PDASH** doctrine: treat the parent dashboard as a *belief-correction + action* instrument (accurate effort/competence beliefs → autonomy-supportive next step), never as a gradebook clone, streak theater, or shame engine.

Builds on: Parts XXVII (parent anxiety), LIX (SAFE-WTP), LXIII (SAFE-EXAM), LXVI (SAFE-STRUCTURE), LXIX (SAFE-PRIVACY), LXXVI (SAFE-FORGET), LXXVIII (SAFE-DURABLE), LXXXI (SAFE-COLD), LXXXII (SAFE-FORMAT). Product seam: parent proof-age / returns / MoC without selling affect or raw wrong-answer feeds.

LXXXIII.1 Why this chapter exists

SAFE-FORGET defined *what* honesty means (age evidence; JOLs≠retention). SAFE-WTP defined *how to price* legible MoC. This tick answers the *parent UX* problem: edtech parent portals default to (a) real-time greens and missing-assignment panic, (b) gamification screenshots for adults, or (c) AI-written "supportive" digests that flatten diagnosis. MindCraft's identity thesis dies if parents only see tonight's glow — and dies differently if the dashboard arms controlling homework battles.

FOUNDER BELIEF under audit: Parents will pay and stay for a dashboard that *corrects optimistic misbeliefs with dated competence evidence* and offers one autonomy-supportive action ("return due before ACT — here's the mission"), outperforming both "100% mastered!" theater and red-alert shame — if copy never implies character failure or guaranteed points.

Claims we refuse as doctrine:

1. Tonight-% / streak / XP / minutes as parent North Star.
2. Mastery fireworks / "cleared forever" on the parent surface.
3. Shame heatmaps, public ranks, or "your child is behind classmates" as default chrome (SAFE-COMPARE collision).
4. Real-time wrong-answer surveillance / live session stalking as "involvement."
5. Anxiety Score™ / mood telemetry on parent portal (SAFE-PRIVACY).
6. Gradebook clone / missing-assignment spam as MindCraft differentiation.
7. AI weekly letter ≡ diagnosis (fluency cosplay).
8. Guaranteed ACT points or "never forgets" from any parent widget.
9. Controlling homework nag prompts as the product's primary parent loop.

LXXXIII.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Information friction	Parents hold biased beliefs about effort/progress	Optimistic "she's fine" vs dated Map	Leave bias untouched OR flood noise
Belief correction	Accurate cumulative facts update parent priors	Last-proof age; returns due; probe rates	Relative-rank shame as "correction"
Actionable message	Specific, improvable next step	One mission CTA / return due	Vague "encourage more math"
Autonomy support	Parent involvement that supports student agency	Share return + invite student ownership	Controlling nag / do-it-for-them
MoC surface	Memory of Competence made legible	Proof age + delayed probe pass	Tonight accuracy as MoC
SAFE-PDASH	Parent UX doctrine for honest MoC	This chapter	Surveillance / fireworks portal

Operational definition (HYPOTHESIS): A parent dashboard is *honesty-complete* when (a) every competence claim shows last-proof age and proof type (blocked vs delayed/transfer), (b) at least one next action is a scheduled return or format/bridge mission — not "open the app for XP," (c) comparative classmate ranks are off by default, (d) affect/biometric data never appear, (e) copy frames returns as protecting earned competence (SAFE-FORGET), and (f) the surface never claims permanent mastery or guaranteed scores.

LXXXIII.3 Parents systematically misbelieve — correction is load-bearing

FACT (effort/grade bias): Bergman (2021, *Journal of Political Economy*, 129(1), 286–322, doi:10.1086/711410) — field experiment providing biweekly detailed information on missed assignments and grades; parents hold **upwardly biased** beliefs about child effort; information attenuates bias and raises achievement via more accurate beliefs and lower monitoring costs (persuasion-game framing).

FACT (high-frequency automated alerts): Bergman & Chan (2021, *Journal of Human Resources*, 56(1), 125–158, doi:10.3368/jhr.56.1.1118-9837R1) — weekly automated texts on missed assignments, grades, and absences reduced course failures (~27%) and raised attendance (~12%) at very low variable cost; larger effects for below-median GPA and high-school students; **no** significant state-test lift — honesty about process metrics ≠ automatic standardized-score miracle.

FACT (cumulative absences misbelief): Rogers & Feller (2018, *Nature Human Behaviour*, 2, 335–342, doi:10.1038/s41562-018-0328-1) — personalized reports of *total accumulated* absences reduced chronic absenteeism (~10%+ in best arms) partly by correcting parents' underestimates; relative classmate-comparison framing did **not** add impact in their design — social-comparison "you're worse than peers" is not the active ingredient.

Commercial implication: Sell *dated, specific competence facts* (proof age, return due, which gap) — not relative shame leagues. Do not market parent alerts as ACT-point engines (SAFE-EXAM / SAFE-DURABLE).

Kill: Classmate rank as default parent "truth."

Survive: Cumulative, personalized, actionable honesty (last proof; next return).

LXXXIII.4 Improvement-framed, light-touch messages beat praise theater

FACT (teacher→parent weekly notes): Kraft & Rogers (2015, *Economics of Education Review*, 47, 49–63, doi:10.1016/j.econedurev.2015.04.001) — weekly individualized teacher messages raised credit-earning probability (~6.5 pp; ~41% reduction in failing to earn credit in their credit-recovery setting); effects driven largely by preventing dropout; **improvement-focused** messages showed the largest point estimates vs positive-continue messages (study underpowered to fully separate arms).

HYPOTHESIS (MindCraft mapping): Parent push copy should prefer one improvable object ("Return due in 3 days: linear equations — delayed mix") over ego praise ("Amazing streak!") or vague warmth. Improvement ≠ shame: name the skill/vessel/bridge, not the child's character.

Kill: Streak/XP push as parent engagement.

Wound: Pure positivity digests that leave beliefs untouched.

Survive: Single improvement CTA tied to Map evidence.

LXXXIII.5 Involvement type matters — control is not support

FACT (homework involvement meta): Xu, Guo, Feng, Ma, Zhang, Núñez & Fan (2024, *Psicothema*, 36(1), 1–14, doi:10.7334/psicothema2023.92) — three-level meta-analysis (28 studies, 252 effects, N≈378k): overall parental homework involvement weakly **negatively** related to achievement (r ≈ −0.064); **autonomy support** positively related; content support, parental control, frequency, and mixed dimensions largely unrelated or not helpful.

HYPOTHESIS: A dashboard that invites "sit beside and correct every item" or nag-frequency maximization recreates controlling involvement. Prefer surfaces that help parents *structure time and witness returns* while the student owns the attempt (ties SAFE-STRUCTURE bandwidth hygiene; SAFE-REWARD anti-controlling rewards).

Kill: Parent-as-second-tutor homework police UX.

Survive: Autonomy-supportive prompts ("Ask what felt hard on the return — don't solve it for them").

LXXXIII.6 Tech-mediated engagement is not magic — design the channel

FACT (systematic review caution): See, Gorard, Lu, Dong & Siddiqui (2021, *Educational Research and Evaluation*, doi:10.1080/13803611.2021.1924791) — promising evidence concentrates on **phone/text/email** school–parent communication that is two-way, personalized, and positive/constructive; evidence that generic online portals/devices alone improve outcomes is weak or inconclusive; research quality often low.

Wound for MindCraft: Shipping a pretty ParentDashboard SPA is not, by itself, an achievement intervention. The active ingredients are belief-correcting facts + actionable improvement framing + low friction — not dashboard feature count.

SPECULATION: MindCraft's wedge vs school gradebooks is MoC (proof age / delayed probes / format-bridge gaps), not faster missing-homework alerts. Competing on gradebook UX is a trap.

LXXXIII.7 Product surface — SAFE-PDASH stack

Reuse: SAFE-FORGET (age nodes; no fireworks); SAFE-DURABLE (legible MoC for WTP); SAFE-WTP (CBC arms); SAFE-COLD (no day-one greens); SAFE-FORMAT / SAFE-BRIDGE (name the real gap); SAFE-PRIVACY (no affect on parent view); SAFE-EXAM (dual rail labels); SAFE-COMPARE (ranks off).

HYPOTHESIS — minimum honest parent chrome:

Element	Must show	Must not
Proof strip	Last proof age; proof type; concept/format/bridge label	Permanent green / Identity Score™
Return agenda	Next 1–3 returns with due+why	Streak calendar as learning
One CTA	Open student's due mission / share talk prompt	"Buy more AI minutes" as primary
Weekly digest	Δ proof age; returns completed; one improvement sentence	AI essay of empty praise
Exam rail	Prove vs learn labeled intensity	Cram board as readiness

Privacy	Progress + MoC only	Mood, voice, face, raw chat dumps

FOUNDER BELIEF: Parent WTP rises when the dashboard makes *decay risk and the return plan* legible without humiliating Maya — validate with PDASH-5 CBC vs fireworks portal and vs shame-rank portal.

LXXXIII.8 Competitive positioning

Pattern	Parent move	MindCraft response
School SIS portals	Grades / missing work	MoC + returns; don't clone SIS
Duo / gamified apps	Streak screenshots to parents	Ban streak-as-learning for adults
AI tutor dashboards	Tonight mastery + chat summary	Aged evidence; verify claims
Test-prep portals	Score predictors / point guarantees	Dual-rail; no guarantees
Surveillance SKUs	Live wrong-answer feed	Aggregate MoC; student dignity

Competitive wedge (FOUNDER BELIEF): "See what still holds — and what's due back" is sellable trust language against both fireworks and panic portals.

LXXXIII.9 Doctrine — SAFE-PDASH (provisional)

1. **Beliefs before chrome** — correct optimistic misbeliefs with dated competence facts, not relative shame.
2. **Proof age is the hero metric on the parent surface** — tonight-% is subordinate.
3. **One improvement CTA** — return / format / bridge mission, Kraft-style specificity.
4. **Autonomy support > control** — never optimize nag frequency or parent-as-solver.
5. **No fireworks / no Ebbinghaus Score™ / no Anxiety Score™** on parent view.
6. **Ranks off by default** — Rogers–Feller: totals beat relative framing for absences; don't import league toxicity.
7. **Not a gradebook** — differentiate on MoC and diagnosis, not SIS feature parity.
8. **Copy:** "Last proved 18 days ago — return due before the exam." Never "mastered forever" / "behind her class" / guaranteed points.

Confidence: High that parents hold upwardly biased effort/progress beliefs and that information can move outcomes (Bergman 2021; Bergman & Chan 2021; Rogers & Feller 2018). High that improvement-framed light-touch messages can matter (Kraft & Rogers 2015). High that autonomy-supportive involvement beats controlling homework help on average (Xu et al. 2024). Medium that MindCraft's MoC framing wins parent WTP vs fireworks and vs shame (needs PDASH-* / ties WTP). High that streak/XP/surveillance/affect on parent portal are kills. Medium that portal-alone engagement without push facts underperforms (See et al. review caution).

LXXXIII.10 Experiments

ID	Question	Design	Primary
PDASH-1	Proof-age strip + return CTA vs tonight-% fireworks parent home	A/B	Parent open→student return start; 7d probe; cancel
PDASH-2	Improvement digest vs positivity-only digest	A/B	Return completion; parent–child talk prompt use
PDASH-3	Absolute proof-age facts vs classmate rank chrome	A/B	Trust; stigma; Maya anxiety proxy; retention
PDASH-4	Autonomy-support talk card vs “check their work tonight” nag	A/B	Controlling-involvement self-report; abandon
PDASH-5	Parent CBC: honest MoC portal vs fireworks vs shame-rank	CBC	WTP; trust; stigma (SAFE-WTP)
PDASH-QUAL	10 parents: what would make you trust a math app report?	Qual	Copy; shame triggers; desired actions

Falsifier: Proof-age honesty raises trust surveys but lowers renewal vs fireworks → keep honesty in product doctrine; change *framing/dose*, do not restore permanent-mastery lies.

Falsifier: Improvement digests increase conflict/controlling homework without return gains → soften tone; keep student-owned CTA; do not add live wrong-answer feeds.

Falsifier: Rank chrome wins short-run opens with trust cost → ban as default (SAFE-COMPARE).

Pre-register: PDASH-* before any “parent portal / live progress / Family Leaderboard” campaign (SAFE-LABMETA).

LXXXIII.11 So what for MindCraft commercially

- **Copy:** “See what still holds — and what’s due back.” Never streak-for-parents, forever-mastered, classmate shame, or point guarantees.
- **Product:** Parent home = proof strip + return agenda + one improvement CTA; weekly digest optional; ranks/affect off.
- **Positioning:** Against SIS clones, Duo screenshot culture, AI mastery letters, and panic surveillance SKUs; for inspectable MoC.
- **Metric:** Parent open→return start, returns completed after digest, proof-age CBC WTP; demote parent DAU on fireworks and nag frequency.
- **Kill list:** Tonight-% NS; shame ranks; live wrong-answer stalk; Anxiety Score™; gradebook cosplay; controlling nag loop.

- **Growth:** ACT parents already fear silent fade — name proof age, sell the return plan (SAFE-FORGET × SAFE-WTP).
- **Vision:** Family trust is part of identity transformation — adults witness competence honestly without colonizing the struggle.

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Part LXXXIV — Tutor Talk-Ratio Instrumentation

Chapter status: Living evidence + HITL telemetry brief — Researcher tick 2026-08-05

Primary question: How should MindCraft instrument *who talks, who constructs, and who waits* in human (and AI-assisted) tutoring so prompt>pour becomes measurable ops fidelity — without turning silence into a vanity North Star?

Owners: Tutor ops / HITL QA · Product (session telemetry) · Workforce coaching · Brand/copy · Red Team

Commercial job: Ship a **SAFE-TALK** doctrine: treat talk-ratio as a *fidelity signal for student construction* (scaffolding density, wait time, student why-before-pour), never as a Talk Ratio Score™, equal-time dogma, or "our tutors are quiet" marketing costume.

Builds on: Parts XL (SAFE-SE), LI (SAFE-DP), LIV (SAFE-AAR), LXVIII (SAFE-HITL), LXXI (SAFE-TUTORGRAIN), LXXV (SAFE-WORKFORCE), LXXIX (SAFE-FBTIME). Product seam: HITL QA telemetry that makes "prompt>pour" auditable without surveilling affect.

LXXXIV.1 Why this chapter exists

SAFE-HITL already requires talk ratio favoring student construction over tutor monologue — but the ops stack still lacks a *shippable instrumentation contract*. Without one, coaches grade vibes (“warm energy”), tutors lecture to fill silence anxiety, and marketing invents “Socratic AI” claims nobody can audit. Workforce fidelity (SAFE-WORKFORCE) also needs observable session features beyond tenure and star ratings.

FOUNDER BELIEF under audit: Parents and districts will trust MindCraft’s human loop more when we can show *prompt density + student constructive talk + wait discipline* correlating with solo transfer — not when we advertise “50/50 talk balance” or tutor silence as virtue.

Claims we refuse as doctrine:

1. Raw tutor-talk-% or student-talk-% as North Star / Talk Ratio Score™.
2. Equal 50/50 airtime as pedagogical law (context and ICAP mode matter).
3. Silence theater (long pauses with no elicitation) as “Socratic.”
4. AI monologue length dressed as “dialogue” via fake student-token counts.
5. “Got it?” check-understanding as student talk (Graesser frame step 5 ≠ construction).
6. Warmth / energy / Ivy brand as substitutes for talk instrumentation (SAFE-HITL / GRAIN kills).
7. Affect/voice emotion AI layered on talk metrics (SAFE-PRIVACY).
8. Bloom 2-sigma marketing from “more student talk.”

LXXXIV.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
IRF / IRE	Teacher Initiation–Response–Feedback/Evaluation triad	Default pour loop if F = lecture	Closed IRF forever
5-step tutoring frame	Tutor Q → answer → short FB → collaborative improve → check	Session skeleton	Skip improve; pour explain
Wait time 1 / 2	Pause after ask / after student turn (~3s+)	Coachable micro-behavior	Sub-1s snatch-back
ICAP mode	Passive < Active < Constructive < Interactive	Student why / soft-wrong / join	Words ≠ constructive
Prompt>pour	Scaffold/elicit before explanation dump	HITL playbook law	Explain-first brand
SAFE-TALK	Instrumentation doctrine for construction fidelity	This chapter	Ratio vanity / silence cosplay

Operational definition (HYPOTHESIS): A session is *talk-instrumented* when MindCraft can report, per session sample: (a) tutor vs student floor time and turn counts, (b) **prompt density** (eliciting moves / tutor turns), (c) **student constructive share** (why/self-explain/soft-wrong articulations — not yes/no or “got it”), (d) median wait time 1 and 2,

(e) Map-brief opened before first explain (`map_brief_open`), and (f) at least one solo attempt evidence after scaffolding — *without* claiming any single ratio proves mastery.

LXXXIV.3 Classroom IRF shows why “teacher talks most” is the default failure

FACT (IRF as classroom grammar): Sinclair & Coulthard (1975) described the Initiation–Response–Feedback exchange as a fundamental unit of classroom discourse — teacher-led, often evaluative, structurally favoring adult control of the floor.

FACT (talk imbalance persists in IRF-heavy rooms): Contemporary discourse analyses continue to find teacher-dominated floor time under IRF (often ~60%+ teacher share in coded secondary lessons) — illustrative of the pattern, not a universal constant.

Commercial implication: If MindCraft tutors recreate classroom IRF with longer lectures, the “human advantage” collapses into expensive school-talk. Instrument *pour-IRF* (short student token → explanation dump) vs *scaffold-IRF* (extended student construction → brief feedback → next prompt).

Kill: Marketing “conversation-based tutoring” without measuring construction.

Survive: IRF as descriptive grammar; redesign the Feedback move toward elicit-and-verify.

LXXXIV.4 One-to-one tutoring is collaborative — but tutors still drive the frame

FACT (naturalistic tutoring dialogue): Graesser, Person & Magliano (1995, *Applied Cognitive Psychology*, 9(6), 495–522, doi:10.1002/acp.2350090604) — normal (unskilled) tutors show collaborative multi-turn problem solving; a pervasive **5-step frame**: tutor asks → student answers → short feedback → collaborative improvement of the answer → tutor checks understanding. Median turns per tutor-initiated question were far above one (multi-turn collaborative answers). Deep diagnosis of idiosyncratic knowledge deficits was often underdeveloped.

FACT (questioning density vs classroom): Graesser & Person (1994, *American Educational Research Journal*, 31(1), 104–137, doi:10.3102/00028312031001104) — students ask far more questions in tutoring than in classrooms (order-of-magnitude higher rates in their corpora), yet tutors still control agenda via curriculum scripts and dialogue frames.

HYPOTHESIS (MindCraft mapping): Success is not “maximize student words.” It is *keep the collaborative improve step alive* (step 4) and refuse to skip from short feedback into monologue. Check-understanding (“got it?”) is the weakest step for identity/FEI — prefer answer evidence / soft-wrong / Map probe (SAFE-HITL).

Kill: “Got it?” as mastery talk.

Wound: Counting all student tokens equally (procedural yes ≈ constructive why).

Survive: Multi-turn collaborative improvement as the auditable unit.

LXXXIV.5 Construction beats tutor explanation volume — Chi's interaction test

FACT (prompt-constrained tutoring): Chi, Siler, Jeong, Yamauchi & Hausmann (2001, *Cognitive Science*, 25(4), 471–533, doi:10.1207/s15516709cog2504_1) — analyses of human tutoring support tutor-, student-, and interaction-centered contributions; critically, when tutors were **suppressed from giving explanations and feedback** and restricted to prompting, students learned **as effectively** as in freer tutoring, attributed to deeper/more scaffolding episodes and greater student control (with reading-ability limits on what self-study recovered).

FACT (ICAP hierarchy): Chi & Wylie (2014, *Educational Psychologist*, 49(4), 219–243, doi:10.1080/00461520.2014.965823) — engagement modes Passive < Active < Constructive < Interactive predict increasing learning; overt behaviors must map to knowledge-change processes — "active" clicking/speaking is not automatically constructive.

Commercial implication: SAFE-HITL's prompt>pour is not a vibe — it is Chi-compatible. **Instrument prompts that elicit construction**, not tutor silence. An AI that "talks less" while leaving the student passive fails ICAP; a near-peer who prompts why-before-reveal can win without Ivy monologue (SAFE-TUTORGRAIN).

Kill: Tutor-talk-% minimization as the product KPI.

Survive: Prompt density × constructive student share as fidelity features feeding `tutor_fidelity_*`.

LXXXIV.6 Wait time is cheap, coachable, and repeatedly evidenced

FACT (baseline snatch-back): Rowe (1974, *Journal of Research in Science Teaching*, 11(2), 81–94, doi:10.1002/tea.3660110202; see also Rowe 1986, *Journal of Teacher Education*, 37(1), 43–50, doi:10.1177/002248718603700110) — teachers typically wait **<1 second** after questions and after student responses; extending wait time 1 and wait time 2 toward ~3+ seconds associates with longer, more complex, more speculative student responses and broader participation.

FACT (review): Tobin (1987, *Review of Educational Research*, 57(1), 69–95, doi:10.3102/00346543057001069) — synthesizing classroom wait-time studies, longer pauses support higher-quality discourse when implemented with accountable responding norms (not isolated stopwatch theater).

HYPOTHESIS: Median WT1/WT2 on sampled sessions is a **coachable leading indicator** of pour-vs-prompt culture — especially for anxious Mayas who need think-space before soft-wrong (ties SAFE-DD / SAFE-EXPOSE / SAFE-FBTIME).

Kill: Stopwatch cosplay without elicitation quality.

Survive: Banded wait targets (e.g., WT1/WT2 medians ≥3s on open prompts) inside FEI QA rubrics.

LXXXIV.7 Dialogic teaching ≠ talk-count theater

FACT (large dialogic-teaching trial): Alexander (2018, *Research Papers in Education*; see also Alexander's dialogic-teaching programme documentation) — developing dialogic teaching is a multi-principle pedagogy (collective, reciprocal, supportive, cumulative, purposeful), not a single talk-ratio knob. Implementation requires repertoire and epistemology, not merely “more student talk.”

Wound for MindCraft: Importing “dialogic” as brand language without Map-grounded cumulation (SAFE-ONTOLOGY / SAFE-BRIDGE) recreates supportive chat that never remediates joins.

SPECULATION: Online one-to-one math corpora will keep finding tutor-dominant initiation with sparse conceptual talk unless products force scaffold ladders — Map brief + soft-wrong gates should move sequences more than a talk-% dashboard alone.

LXXXIV.8 Product surface — SAFE-TALK instrumentation stack

Reuse: SAFE-HITL (prompt>pour; Map brief; FEI QA); SAFE-WORKFORCE (fidelity@tenure); SAFE-SE (student-generated why); SAFE-FBTIME (micro-delay before reveal); SAFE-AAR (improve-how close); SAFE-PRIVACY (no emotion AI on voice); SAFE-GRAIN (hire by fidelity not Ivy pour).

HYPOTHESIS — minimum honest session telemetry (sampled + consented):

Signal	Definition	Use	Must not
`tutor_floor_share`	Tutor speaking time / (tutor+student)	Drift monitor	Sole NS
`student_constructive_share`	Why/SE/soft-wrong turns / student turns	Primary fidelity	Count “yeah/got it”
`prompt_density`	Eliciting prompts / tutor turns	Coach target	Reward vague “what do you think?” spam
`wait_ms_p50`	Median WT1 and WT2	Coach micro-skill	Shame tutors live mid-session
`pour_streak_max`	Longest consecutive tutor explain turns	Detect monologue	Ban all explanation
`map_brief_open`	Brief before first explain	HITL hard gate	Trait-label briefs
`annotate_before_reveal`	Why before key/coach	SE×FBTIME	AI monologue first

Ops rule (FOUNDER BELIEF): Coaches review **clips where pour_streak_max is high and constructive share is low**, not leaderboards of who talked least. Parent/district packets may show *aggregate fidelity bands* (“sessions coached for prompt-first practice”) — never per-child Talk Ratio Score™.

AI sessions: Apply the same schema to Solver/coach turns — fluent AI floor share with low student constructive share is a **sycophancy/pour risk** (Part XXXIII), not “personalized dialogue.”

LXXXIV.9 Competitive positioning

Pattern	Talk move	MindCraft response
Classroom / Zoom school help	IRF pour	Instrument scaffold improve-step
ChatGPT tutor	Fluent monologue	Ban fluency≡dialogue; require student why gates
Marketplace “Ivy explainers”	Expert pour as quality	Hire/coach on prompt density (GRAIN×HITL)
Duo / app streaks	Engagement without discourse	Irrelevant; don’t import streak as talk proxy
“Socratic AI” ads	Brand claim	Show prompt density + constructive share or stay quiet

Competitive wedge (FOUNDER BELIEF): “We coach tutors (and AI) to make students construct — and we can show the tape metrics” beats explain-first marketplaces and unmeasured Socratic cosplay.

LXXXIV.10 Doctrine — SAFE-TALK (provisional)

1. **Construction > airtime** — student constructive share and prompt density outrank raw talk-%.
2. **Prompt>pour is measurable** — elicit before explain; cap pour streaks in QA.
3. **Wait time is a micro-skill** — band WT1/WT2; coach snatch-back.
4. **“Got it?” ≠ evidence** — prefer soft-wrong, annotated why, Map probe.
5. **No Talk Ratio Score™ / silence NS** — fidelity feature for coaching, not vanity KPI.
6. **No emotion AI on talk** — acoustic floor time ok; affect inference banned (SAFE-PRIVACY).
7. **Same schema for human and AI** — fluent pour fails either channel.
8. **Copy:** “Prompt first — then prove.” Never “our tutors talk half the time” / Bloom-from-talk / “Socratic guaranteed.”

Confidence: High that IRF defaults to teacher control (Sinclair–Coulthard) and that naturalistic tutoring uses multi-turn collaborative frames under tutor agenda control (Graesser et al. 1995; Graesser & Person 1994). High that student construction can carry learning when tutor explanation is constrained (Chi et al. 2001) and that ICAP ranks engagement modes (Chi & Wylie 2014). High that wait-time extension changes discourse quality (Rowe; Tobin). Medium that MindCraft’s telemetry bundle predicts solo transfer (needs TALK-*). High that talk-% NS / silence theater / got-it checks are kills. Medium that dialogic-brand copy without Map cumulation underperforms (Alexander).

LXXXIV.11 Experiments

ID	Question	Design	Primary
TALK-1	Prompt-first playbook + wait coaching vs explain-first business-as-usual	Tutor RCT / stepped-wedge	`student_constructive_share`; `solo_transfer_pass`; `pour_streak_max`

TALK-2	Live talk dashboard for coaches vs weekly clip QA only	Ops A/B	`tutor_fidelity_30d`; coach time; tutor reactance
TALK-3	Soft-wrong + annotate-before-reveal gate vs free talk	Session A/B	SE quality; transfer; time-on-task
TALK-4	AI coach with prompt-density cap vs unconstrained monologue	Product A/B	Constructive share; hallucination repair need; abandon
TALK-5	Parent CBC: “coached prompt-first tutors” vs “Ivy explainers” vs “Socratic AI”	CBC	WTP; trust (ties SAFE-WTP / GRAIN)
TALK-QUAL	10 tutors: what makes silence feel unsafe?	Qual	Snatch-back drivers; playbook friction

Falsifier: Raising constructive share / wait time without transfer lift → keep as process hygiene; do not market talk metrics as learning proof; retarget prompts to Map gaps.

Falsifier: Talk dashboards increase tutor reactance / fake short student turns → remove live shame UI; keep sampled clip QA.

Falsifier: AI prompt-caps raise abandon without transfer gain → soften cap; keep why-before-reveal on soft-wrong paths only.

Pre-register: TALK-* before any “Socratic / talk-balanced tutoring” campaign (SAFE-LABMETA).

LXXXIV.12 So what for MindCraft commercially

- **Copy:** “Prompt first — then prove.” Never Talk Ratio Score™, 50/50 dogma, silence theater, or Bloom-from-talk.
- **Product:** Sampled session telemetry — constructive share, prompt density, wait bands, pour-streak — wired into HITL QA and workforce fidelity.
- **Positioning:** Against Ivy pour marketplaces and unmeasured Socratic AI; for auditable construction.
- **Metric:** `student_constructive_share`, `prompt_density`, `wait_ms_p50`, `pour_streak_max` feeding `tutor_fidelity_*`; demote raw talk-% and energy ratings.
- **Kill list:** Talk-% NS; got-it≡mastery; emotion-AI talk scoring; AI monologue-as-dialogue ads.
- **Growth:** District/parent trust packets can cite coaching-for-construction without biometric theater (SAFE-PROCURE × SAFE-HITL).
- **Vision:** Identity transformation requires the student to *speak the math into being* — instrumentation exists to protect that, not to gamify silence.

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Part LXXXV — Story-World Cognitive Load vs Identity Payoff

Chapter status: Living evidence + narrative-wrap design brief — Researcher tick 2026-08-05

Primary question: When does a story world *pay for its working-memory tax* with identity/belonging/transfer — and when is it seductive load that hurts learning while looking branded?

Owners: Story/content ops · Product (Solver / chapter / practice stems) · Learning science · Brand/copy · Equity (Part XXXVI) · Red Team

Commercial job: Ship a **SAFE-STORYLOAD** doctrine: narrative wrap is allowed only when it encodes the math structure, keeps language load banded, and graduates via learning+belonging experiments — never as franchise lore, immersion theater, or “story = engagement = learning.”

Builds on: Parts XXVI (CLT), XXVIII (invention-story / HIST), XXXVI (equity of worlds), XLVIII (SAFE-TRANSFER; story≠far-transfer), LXIV (SAFE-ELL language vs math load), LXXXII (SAFE-FORMAT). Product seam: story-module / stem reskin must freeze the math and wrap it — WORLD_VISION — under load budgets.

LXXXV.1 Why this chapter exists

WORLD_VISION and brand thesis bet that math-inside-a-world beats naked drill for Maya’s identity. Parts XXVIII and XXXVI already constrain *whose* story and whether invention lore is seductive trivia. They do not settle the **cognitive-load bargain**: every extra character beat, lore paragraph, and decorative cutscene consumes working memory that might have built the schema.

FOUNDER BELIEF under audit: Story-first is a defensible differentiator *if and only if* the wrap reduces threat / raises situational interest / encodes structure — without exceeding WM for novices and anxious Mayas.

Claims we refuse as doctrine:

1. Immersion / lore depth as North Star or Story Engagement Score™.
2. "Story always helps learning" / narrative ≡ transfer (SAFE-TRANSFER kill).
3. Seductive franchise trivia as "world building."
4. Decorative art or soundtrack ≡ FormatId / pedagogy (SAFE-FORMAT).
5. Long pre-attempt lore dumps before first micro-win.
6. ELL/language-heavy wraps that conflate reading load with math ability (SAFE-ELL).
7. Token skin-tone diversity as load justification (XXXVI).
8. Story feature shipping without HIST / EQ / STORYLOAD experiment plan.

LXXXV.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Intrinsic load	Element interactivity of the math itself	Ontology ingredient / problem structure	Cannot "story away" hard joins
Extraneous load	Presentation that does not aid schema	Lore dump, split lore/diagram, BGM	Brand chrome tax
Seductive details	Interesting but irrelevant add-ons	Side quests, cute facts, unrelated plot	Interest without learning
Coherence principle	Exclude extraneous words/pictures/sounds	Tight wrap; silence optional	Multimedia clutter
Situational → individual interest	Triggered then maintained interest	World re-entry + role agency	Trigger without maintain
Structure-encoding wrap	Narrative makes the invariant memorable	Story-first question design	Scene then unrelated textbook ask
SAFE-STORYLOAD	Load-budgeted identity wrap	This chapter	Immersion NS / lore hero

Operational definition (HYPOTHESIS): A story module is *load-honest* when MindCraft can show: (a) the mathematical invariant the wrap encodes (structure test), (b) pre-attempt narrative word/screen budget within a band by level and language status, (c) time-to-first-attempt under a cap, (d) learning outcomes on retention/transfer not worse than matched plain stem (non-inferiority), and (e) optional belonging/interest lift — *without* claiming immersion proves identity.

LXXXV.3 CLT is the physics; story is optional mass

FACT (CLT core): Sweller (1988, *Cognitive Science*, 12(2), 257–285, doi:10.1207/s15516709cog1202_4) — problem-solving search can impose heavy load that impedes schema acquisition; instructional design must respect limited working memory.

FACT (intrinsic vs extraneous tradition): Chandler & Sweller and later CLT reviews distinguish load from the material's element interactivity vs load from how material is organized/presented (see also Sweller, van Merriënboer & Paas, 1998, *Educational Psychology Review*, 10, 251–296). Extraneous load is the designer's fault; intrinsic load is managed by sequencing and prior knowledge — not by prettier lore.

Commercial implication: MindCraft cannot sell "epic world" as pedagogy. The world is a *presentation choice* that must clear an extraneous-load budget. If the wrap does not redirect attention to schema-relevant relations, it is cost.

Kill: Story immersion as learning science.

Survive: CLT as hard constraint on narrative chrome.

LXXXV.4 Seductive details and coherence — the default kill for franchise lore

FACT (seductive details experiments): Harp & Mayer (1998, *Journal of Educational Psychology*, 90(3), 414–434, doi:10.1037/0022-0663.90.3.414) — undergraduates reading science text *with* interesting but irrelevant details recalled fewer main ideas and generated fewer transfer solutions than those without; diversion toward inappropriate organizing schemas is a leading account (not mere "fun distraction").

FACT (meta-analyses): Rey (2012, *Educational Research Review*, 7(3), 216–237, doi:10.1016/j.edurev.2012.05.003) — across ~39 experimental effects, seductive details show small-to-medium negative effects on retention and medium negative effects on transfer; moderators exist (time pressure, detail type, domain) but mean direction is caution. Sundararajan & Adesope (2020, *Educational Psychology Review*, 32, 707–734, doi:10.1007/s10648-020-09522-4) — later meta likewise supports a negative seductive-details pattern with coherence-relevant moderators.

FACT (coherence principle): Mayer's multimedia coherence principle — people learn more deeply when extraneous words, pictures, and sounds are excluded (handbook statements; e.g., Moreno & Mayer, 2000, *Journal of Educational Psychology*, 92(1), 117–125, doi:10.1037/0022-0663.92.1.117 on irrelevant sounds). Cute BGM, floating stickers, and lore sidebars are not free.

HYPOTHESIS (MindCraft mapping): Pre-question character backstory that does not encode the invariant is a textbook seductive detail. Structure-encoding narrative (the plot *is* the proportion / the journey *is* the function composition) is not "irrelevant" — it is germane packaging — **if** it stays short and co-located with the ask (no split-attention lore panel vs equation).

Kill: Lore-first chapter opens; museum trivia; soundtrack-as-engagement.

Wound: "Interestingness" as A/B primary without retention/transfer.

Survive: Coherence-first wrap; seductive-detail gate in content ops (aligns XXXVI structure test).

LXXXV.5 Interest development — the identity-side payoff (conditional)

FACT (four-phase interest): Hidi & Renninger (2006, *Educational Psychologist*, 41(2), 111–127, doi:10.1207/s15326985ep4102_4) — interest develops from triggered situational → maintained situational → emerging individual → well-developed individual interest; supports that mismatch phase can impede.

FACT (interest as context, not magic): Renninger, Ewen & Lasher (2002, *Learning and Instruction*, 12(4), 467–490, doi:10.1016/S0959-4752(01)00012-3) — inserting well-developed individual-interest contexts into text/word problems can scaffold meaning-focus, but can also *mask* difficulties or feel harder; teacher/product support still matters. Students often respond more to difficulty than to context alone.

FACT (personalization mixed): Walkington (2013, *Journal of Educational Psychology*, 105(4), 932–945, doi:10.1037/a0031882) — interest-personalized algebra contexts can improve performance/learning relative to generic problems in adaptive tutors (depth of connection matters). Van de Weijer-Bergsma & Van der Ven (2021, *Learning and Individual Differences*, 87, 101982, doi:10.1016/j.lindif.2021.101982) — two primary-school personalization studies found **no** effect of personalization on performance, enjoyment, or cognitive load (though enjoyment and load predicted performance). Personalization is not an automatic win.

HYPOTHESIS: MindCraft's identity payoff is less "pick Maya's favorite fandom" and more *maintained situational interest via agency in a coherent world* (role, re-entry, recognition of competence) tied to FEI attempts — not surface name-swap personalization.

Kill: Surface personalization / fandom skin as product hero.

Survive: Interest as a *possible* mediator of persistence (`retry_120s`) when wrap encodes structure and stays in load band; always co-primary with transfer.

LXXXV.6 Language load, anxiety, and format — compound taxes

FACT / doctrine reuse: SAFE-ELL — separate language load from math load; heavy prose can false-flag ability. SAFE-DD / Ashcraft line — anxiety consumes WM; extraneous narrative tax hits threatened Mayas hardest. SAFE-FORMAT — decorative story art ≠ diagram FormatId.

HYPOTHESIS: For high-anxiety / ELL / low-prior students, pre-attempt narrative budgets must be *shorter*, not richer. Offer plain-stem toggle or "skip to math" without shame; never gate belonging on lore completion.

Commercial implication: Parent copy cannot say "immersive story worlds" without "math stays clear — story carries the idea, not the obstacle." District reviewers will ask about reading demand (SAFE-PROCURE packet honesty).

LXXXV.7 Product surface — SAFE-STORYLOAD design contract

Reuse: XXVI attempt-grain; XXVIII HIST structure; XXXVI equity rubric; XLVIII name-the-hop (story world ≠ transfer world); LXIV linguistic modify without math dilute.

HYPOTHESIS — merge gates for story modules / themed stems:

Gate	Pass criterion	Fail →

Structure encode	Writer names the invariant the narrative makes memorable	Cut or rewrite
Ask binding	Numbers/relations in scene match the frozen item (no scene→unrelated textbook ask)	Reskin fail
Pre-attempt budget	Words/screens/time-to-first-attempt within level×language band	Compress; offer skip
Coherence	No BGM/stickers/side lore required to solve	Delete extraneous
Non-inferiority learning	Retention/transfer ≥ plain stem within pre-reg margin (STORYLOAD-1)	Demote to optional flavor
Belonging optional lift	EQ/HIST metrics only after learning gate	Do not ship on vibes
Equity scan	XXXVI stereotype / plurality checks	Rewrite

UX rules (FOUNDER BELIEF):

1. First micro-attempt comes fast — lore after or interleaved as completion prompts, not walls of text.
2. “Skip story” is always available; never shame.
3. World reveal scales with mastery (WORLD_VISION Horizons) — novices see *less* lore, not more.
4. Exam dual-rail (SAFE-EXAM): prove mode uses exam-like stems; story wrap is learn-rail default, not timed ACT costume.

LXXXV.8 Competitive positioning

Pattern	Story move	MindCraft response
Naked drill apps	Zero wrap	Wrap only if structure+load gates pass
Edutainment lore	Franchise immersion	Coherence + seductive-detail kill
Generic word-problem skins	Surface personalization	Depth/agency > name-swap; null results possible
Khan culture-thin library	Minimal narrative	Narrative risk accepted *with* instrumentation
Brilliant delight puzzles	Aesthetic engagement	Delight subordinate to transfer; no immersion NS

Competitive wedge (FOUNDER BELIEF): “Stories that carry the math — without drowning working memory” beats both sterile banks and lore-bloated games.

LXXXV.9 Doctrine — SAFE-STORYLOAD (provisional)

1. **CLT budgets narrative** — extraneous lore is a tax; intrinsic math difficulty is not cured by plot.
2. **Structure-encoding or cut** — wrap must make the invariant memorable; else seductive detail.
3. **Coherence > immersion** — exclude irrelevant words/pictures/sounds; no Story Engagement Score™.
4. **Interest is conditional** — situational interest may aid persistence; personalization is mixed evidence; never primary KPI.
5. **Novices get less lore** — mastery unlocks world depth (Horizons), not day-one encyclopedias.
6. **Language ≠ math** — band prose; skip-story; ELL hygiene (SAFE-ELL).
7. **Story ≠ transfer** — still name hops; measure `transfer_pass` / `solo_transfer_pass` (SAFE-TRANSFER).
8. **Copy:** “The story carries the idea — the math stays clear.” Never immersion hero / lore-as-learning / guaranteed identity-from-world.

Confidence: High that seductive details and coherence violations can impair retention/transfer (Harp & Mayer; Rey; Sundararajan & Adesope; Moreno & Mayer). High that WM limits constrain instructional search (Sweller). Medium that MindCraft structure-encoding wraps will clear non-inferiority (needs STORYLOAD-*). Medium–low that wrap lifts belonging* and* learning jointly (HIST-EQ / EQ still open). High that immersion NS / lore dumps / surface fandom skins are kills. Medium that skip-story + mastery-gated reveal protects anxious/ELL segments.

LXXXV.10 Experiments

ID	Question	Design	Primary
STORYLOAD-1	Structure-encoding wrap vs plain stem (same items)	A/B	Retention; `transfer_pass`; time-to-first-attempt; self-load rating
STORYLOAD-2	Pre-attempt lore long vs short budget	A/B	Learning; abandon; `retry_120s`
STORYLOAD-3	Skip-story available vs forced wrap	A/B	Use-rate; learning by anxiety/ELL segment
STORYLOAD-4	Mastery-gated world reveal vs day-one full lore	A/B	Learning; belonging; session length
STORYLOAD-5	Parent CBC: “clear story that carries the math” vs “immersive world” vs “no story / more practice”	CBC	WTP; trust (ties SAFE-WTP)
STORYLOAD-QUAL	10 Mayas: when does story feel like help vs homework reading?	Qual	Friction themes; skip motives

Falsifier: Wrap loses to plain stem on transfer beyond margin → demote story to optional cosmetic; keep brand elsewhere.

Falsifier: Short budget kills belonging with no learning gain → redesign agency/role, not add lore.

Falsifier: Skip-story users dominate and convert better → default plain; story opt-in.

Falsifier: Immersion CBC wins WTP but STORYLOAD-1 fails learning → do not ship immersion ads; dual-rail honest packaging only.

Pre-register: STORYLOAD-* / continue HIST-EQ before any “immersive story worlds improve learning” campaign (SAFE-LABMETA).

LXXXV.11 So what for MindCraft commercially

- **Copy:** “The story carries the idea — the math stays clear.” Never immersion NS, lore-as-pedagogy, or identity-from-world guarantees.
- **Product:** Structure-encode + ask-bind + pre-attempt budget + skip-story + mastery-gated reveal as merge gates for story modules/stems.
- **Positioning:** Against sterile banks *and* edutainment lore bloat; for load-honest narrative wrap.
- **Metric:** Non-inferiority on transfer vs plain; time-to-first-attempt; optional belonging; demote Story Engagement / lore completion.
- **Kill list:** Franchise trivia dumps; forced lore walls; BGM/sticker coherence violations; surface fandom personalization hero; story≡transfer ads.
- **Growth:** Parent/district packets show learning gates before world screenshots; equity rubric visible as craft not costume (XXXVI).
- **Vision:** Identity transformation uses story as *meaning technology under CLT* — worlds reveal as competence grows, they do not replace schema construction.

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Part LXXXVI — Competence Evidence as Social Proof

Chapter status: Living evidence + growth/copy brief — Researcher tick 2026-08-05

Primary question: What *witnessable competence artifacts* (solo-transfer / delayed proof) should MindCraft use as social proof for parents, tutors, and Maya — instead of vanity counts, vague praise, or mastery fireworks?

Owners: Growth/copy · Product (shareable MoC / portfolio surfaces) · Parent UX · Brand · Red Team

Commercial job: Ship a **SAFE-PROOF** doctrine: social proof must be *dated, specific, solo-capable competence evidence* others can inspect — never crowd size, star ratings, Identity Score™, or “she’s a math person now” testimonials without a named hop.

Builds on: Parts XXV (self-efficacy), XXVII/LIX (parent WTP), XLII (SAFE-COMPARE), XLVIII (SAFE-TRANSFER), LXI (SAFE-IDMEASURE), LXXVI (SAFE-FORGET), LXXVIII (SAFE-DURABLE), LXXXIII (SAFE-PDASH). Product seam: shareable `solo_transfer_pass` / proof-age artifacts that correct buyer uncertainty without shame leagues or guaranteed-points case studies.

LXXXVI.1 Why this chapter exists

SAFE-PDASH told parents *what to show* on a dashboard (proof age, returns, one autonomy CTA). SAFE-WTP told us to price legible MoC. Growth still defaults to the edtech social-proof stack: “Trusted by 50k families,” five-star walls, Ivy-tutor logos, AI-written “Maya feels confident!” quotes. That stack sells *affiliation and affect*, not *competence*. MindCraft’s identity thesis needs a different proof object: a third party (parent, tutor, skeptic buyer) can *witness* that the student solved a delayed / format-hop / join item **without** Solver open — and the claim names the concept/bridge/format and proof age.

FOUNDER BELIEF under audit: Parents and Maya buy (and stay) when social proof looks like *inspectable competence*, not herd size — if artifacts stay private-by-default, autonomy-supportive, and never imply guaranteed ACT points.

Claims we refuse as doctrine:

1. User-count / DAU / “#1 app” / star-average as primary learning social proof.
2. Vague testimonials (“amazing!” / “so confident!”) without a named competence object.

3. Mastery fireworks / tonight-% screenshots as shareable proof (SAFE-FORGET / SAFE-COLD).
4. Classmate ranks / Family Leaderboard as social proof (SAFE-COMPARE).
5. Identity Score™ / two-item HSLS pre-post as marketing proof (SAFE-IDMEASURE).
6. Guaranteed ACT-point case studies or "+N points in 8 weeks" packages (SAFE-EXAM / SAFE-DURABLE).
7. Empty AI praise / "math person" recognition without repertoire evidence (SAFE-IDMEASURE kill).
8. Forced public portfolios or viral share as North Star.

LXXXVI.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Social proof (persuasion)	Under uncertainty, people look to others' actions/opinions	Landing / parent CBC / tutor sales	Herd vanity without competence
Descriptive / provincial norms	"People like you / in this situation do X"	Similar-Maya case cards	Boomerang if framed as rank
Enactive mastery	Own successful performance → strongest efficacy source	`solo_transfer_pass` events	Scaffolded greens sold as mastery
Vicarious experience	Similar others succeed → "I can too"	Peer/near-peer artifacts; tutor film clips	Dissimilar Ivy hero ≠ Maya model
Witnessable artifact	Third party can inspect the competence claim	Shareable proof card: item type, hop, age, solo flag	Screenshot of streak / XP
SAFE-PROOF	Competence-first social proof doctrine	This chapter	Vanity / affect / points theater

Operational definition (HYPOTHESIS): A MindCraft social-proof unit is *SAFE-PROOF complete* when it (a) names a competence object (concept / bridge / FormatId), (b) states proof type (blocked vs delayed / transfer / solo), (c) shows proof age or date, (d) comes from a *similar* role (parent-of-anxious-Maya, near-peer tutor, not celebrity), (e) never claims guaranteed scores or permanent mastery, and (f) is share-opt-in with student dignity defaults (SAFE-PRIVACY / SAFE-PDASH).

LXXXVI.3 Social proof is real — and dangerous when it is only a crowd

FACT (principle + field norms): Cialdini's social-proof principle — under uncertainty, people use others' behavior as evidence of correct action — is the commercial spine of testimonials and "most guests…" signs (Cialdini, 2007/2009, *Influence*; field validation in Goldstein, Cialdini & Griskevicius, 2008, *Journal of Consumer Research*, 35(3), 472–482, doi:10.1086/586910). Descriptive-norm appeals beat generic exhortation; **provincial** norms (closest situational match) outperform broad "most people" claims.

FACT (boomerang risk): Schultz, Nolan, Cialdini, Goldstein & Griskevicius (2007, *Psychological Science*, 18(5), 429–434, doi:10.1111/j.1467-9280.2007.01917.x) — descriptive norms alone can *raise* undesired behavior among already-good performers (boomerang); adding an injunctive cue (approval of the desired behavior) reconstructs the message.

Commercial implication: "Most students on MindCraft practice daily" is weak and can shame or normalize the wrong thing. Prefer *similar-situation competence norms* ("Parents of ACT-anxious juniors who saw a delayed solo pass on linear systems…") plus injunctive frame that values the *return / solo attempt*, not volume. Never use "you're behind the average" as social proof (SAFE-COMPARE / SAFE-PDASH).

Kill: Crowd-size hero; classmate-average shame as proof.

Survive: Provincial, role-matched, competence-named norms; injunctive respect for honest returns.

LXXXVI.4 Mastery beats persuasion — artifacts must be enactive, not verbal

FACT (efficacy sources): Bandura (1977, *Psychological Review*, 84(2), 191–215, doi:10.1037/0033-295X.84.2.191) — efficacy expectations arise from performance accomplishments, vicarious experience, verbal persuasion, and physiological states; **performance accomplishments** are the most dependable.

FACT (school review): Usher & Pajares (2008, *Review of Educational Research*, 78(4), 751–796, doi:10.3102/0034654308321456) — across academic contexts, mastery experience is typically the most influential self-efficacy source; source strength varies by gender, ethnicity, ability, and domain; measurement of "sources" is often messy — product must not invent a Source Score™.

HYPOTHESIS (MindCraft mapping): Parent/tutor *verbal* persuasion ("you're so smart") and AI praise are the weakest durable channel. The growth surface should show **enactive** evidence: a dated solo delayed pass. Vicarious channel = similar Maya's artifact or a near-peer tutor film of *student construction* (SAFE-TALK / SAFE-FILM), not an Ivy explainer reel.

Kill: Praise-wall social proof; "AI says she's a math person."

Survive: Shareable mastery events tied to FEI metrics (`solo_transfer_pass`, `format_hop_solo_transfer_pass`, `join_solo_transfer_pass`).

LXXXVI.5 Portfolios are the right metaphor — if student-owned and reflective

FACT (portfolio definition): Paulson, Paulson & Meyer (1991, *Educational Leadership*, 48(5), 60–63) — a portfolio is a *purposeful* collection exhibiting efforts, progress, and achievements; it must include student participation in selection, criteria, and **self-reflection** — "done *by* the student, not *to* the student."

HYPOTHESIS: MindCraft's shareable proof card / Field Journal export is a *thin portfolio unit*: one (or few) items with reflection ("what I can do alone now") beats a dump of every green check. Parents witness competence; Maya owns selection — autonomy support, not surveillance scrapbook (SAFE-PDASH).

Wound: Math portfolio literature's outcome claims are thinner than rhetoric (see critiques of large-scale portfolio programs). Do not market "portfolios raise ACT scores." Market *inspectability* and belief correction.

Kill: Auto-publishing every session to parents/social.

Survive: Student-selected competence exhibits with age + solo flag.

LXXXVI.6 Honesty beats one-sided hype — two-sided proof builds credibility

FACT (two-sided ads): Eisend (2006, *International Journal of Research in Marketing*, 23(2), 187–198, doi:10.1016/j.ijresmar.2005.11.001) — meta-analysis: two-sided advertising (acknowledging a limitation) can enhance credibility and persuasion under conditions (e.g., when the negative is refutable or less important than gains) versus purely one-sided puffery.

HYPOTHESIS: Parent-facing case cards should admit dual-rail truth: "Blocked accuracy rose in week 2; delayed solo transfer on systems took until week 6 — returns still due." That is SAFE-FORGET honesty as *marketing craft*, not a confession booth. Guaranteed-points one-sided case studies are both banned and less credible to skeptical buyers.

Kill: "+4 ACT points guaranteed" social proof.

Survive: Dated competence + remaining gap + next return CTA.

LXXXVI.7 Product surface — SAFE-PROOF artifact contract

Reuse: SAFE-TRANSFER (name the hop); SAFE-FORGET (age evidence); SAFE-PDASH (autonomy; no shame ranks); SAFE-WTP (CBC arms); SAFE-IDMEASURE (no Identity Score™ splash); SAFE-DURABLE (no cram-as-ready); SAFE-COMPARE (ranks off); SAFE-PRIVACY (opt-in share; no affect).

HYPOTHESIS — minimum shareable proof card fields:

Field	Required	Banned substitute
Competence object	Concept / bridge / FormatId name	"Math improved!"
Proof type	Delayed / transfer / solo flag	Tonight blocked %
Proof age	Last-proof date or days since	"Mastered forever"
Similarity frame	Role match (anxious ACT junior / ELL / etc.)	Celebrity / Ivy-only
Student voice	Optional short reflection	AI-written confidence blurb
Share control	Opt-in; revoke; no public default	Forced feed / viral NS
Claim hygiene	No score guarantee; no permanent mastery	+points package language

Surfaces: Parent digests (one card); tutor briefing (film + artifact); marketing landing (anonymized provincial cases); in-app MoC share sheet — not App Store star walls as the learning claim.

LXXXVI.8 Competitive positioning

Competitor pattern	Social-proof move	MindCraft response
Duo / streak apps	League ranks, streak screenshots	Competence cards; ranks off
ChatGPT tutor ads	Fluency demos, “feels like a tutor”	Solo-pass bar; PWC
Cram/bootcamp	+points testimonials	Dual-rail dated MoC; no guarantees
Khan	Library scale / “free for all”	Diagnosis + witnessable hops, not item-count
Generic edtech	Star walls / DAU	Provincial competence norms

Competitive wedge (FOUNDER BELIEF): “Proof you can *see* — a solo pass on the real gap, with an age stamp” beats both herd vanity and guaranteed-points theater.

LXXXVI.9 Doctrine — SAFE-PROOF (provisional)

1. **Competence > crowd** — social proof names a hop and solo/delayed evidence; demote user-count / stars as learning claims.
2. **Enactive > verbal** — prefer mastery artifacts over praise walls and AI “math person” lines (Bandura; Usher & Pajares).
3. **Provincial similarity** — match role/situation; avoid broad “everyone” and classmate shame (Goldstein et al.; Schultz et al.).
4. **Age the claim** — proof age required; no permanent-mastery fireworks (SAFE-FORGET).
5. **Student-owned exhibit** — thin portfolio: select + reflect; not done *to* the student (Paulson et al.).
6. **Two-sided honesty** — admit returns due / remaining gaps; ban +points guarantees (Eisend; SAFE-DURABLE).
7. **Opt-in dignity** — share is voluntary; no viral NS; no affect telemetry on cards (SAFE-PRIVACY).
8. **Copy:** “See what she can do alone — dated, on the real gap.” Never star-hero / Identity Score™ / guaranteed ACT case study.

Confidence: High that social proof / descriptive norms move behavior under uncertainty (Cialdini tradition; Goldstein et al. 2008) and that descriptive-only norms can boomerang (Schultz et al. 2007). High that mastery experiences dominate verbal persuasion for efficacy (Bandura 1977; Usher & Pajares 2008). Medium that MindCraft proof cards beat star walls on parent WTP (needs PROOF-* / ties WTP/PDASH). High that vanity counts, Identity Score™, +points testimonials, and forced public portfolios are kills. Medium that student-selected thin portfolios raise retention without becoming homework theater.

LXXXVI.10 Experiments

ID	Question	Design	Primary
PROOF-1	Competence-card landing vs star-wall / user-count landing	A/B	Signup; trust; time-on-proof
PROOF-2	Provincial similar-Maya case vs generic testimonial	A/B	Conversion; perceived relevance
PROOF-3	Two-sided dated MoC case vs one-sided +points claim	A/B / CBC	WTP; credibility; regret
PROOF-4	Opt-in share of solo-transfer card vs auto parent push of greens	A/B	Share rate; student stigma; parent action quality
PROOF-5	Parent CBC: “dated solo proof” vs “streak screenshots” vs “ACT points guarantee”	CBC	WTP; trust (SAFE-WTP)
PROOF-QUAL	10 parents + 10 Mayas: what feels like real proof vs hype?	Qual	Artifact fields; shame triggers

Falsifier: Star-wall wins conversion and 90-day retention equally → still ban stars as *learning* claim; segregate storefront chrome from MoC doctrine.

Falsifier: Proof cards raise signup but increase controlling parent nag → redesign share copy toward autonomy (SAFE-PDASH).

Falsifier: Two-sided honesty loses CBC badly with no credibility gain → keep ban on guarantees; soften wording, do not restore +points ads.

Falsifier: Students refuse all sharing → keep private MoC; marketing uses anonymized aggregates only.

Pre-register: PROOF-* before any “trusted by N families / +points stories” campaign (SAFE-LABMETA).

LXXXVI.11 So what for MindCraft commercially

- **Copy:** “See what she can do alone — dated, on the real gap.” Never crowd-size learning claims, vague confidence quotes, or guaranteed ACT case studies.
- **Product:** Shareable proof-card schema (object, hop type, age, solo flag, opt-in); thin portfolio select+reflect; parent digest = one card + one return CTA.
- **Positioning:** Against Duo screenshot culture, bootcamp +points testimonials, and ChatGPT fluency demos; for witnessable solo competence.

- **Metric:** Proof-card view→action; share opt-in rate; parent CBC preference; co-primary `solo_transfer_pass` — demote star NPS / DAU-as-proof.
- **Kill list:** User-count hero; star walls as pedagogy; Identity Score™ splash; Family Leaderboard proof; AI praise testimonials; forced public portfolios.
- **Growth:** Landing and sales decks lead with provincial competence artifacts; district packets can show anonymized MoC samples without biometric theater (SAFE-PROCURE).
- **Vision:** Identity transformation becomes *socially legible* when competence can be witnessed — recognition of practice, not applause for streaks.

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Part LXXXVII — Bank Coverage Honesty in Marketing

Chapter status: Living evidence + GTM/copy brief — Researcher tick 2026-08-05

Primary question: How should MindCraft *advertise* question-bank coverage (ACT track, concept×level×format) without item-count heroics, “complete ACT bank” puffery, or GENQ volume theater — while still selling diagnosis trust?

Owners: Growth/copy · Product (Practice / Admin coverage) · Engine (bank audit / GENQ) · District GTM · Brand · Red Team

Commercial job: Ship a **SAFE-COVER** doctrine: coverage claims are *blueprint-honest matrices + use-tier labels*, never raw `questionCount`, never implied ACT-official completeness, never unverified generated volume as readiness.

Builds on: Parts XXXIV (claim ladder), XXXV (competitive audits), LIX (SAFE-WTP), LX (SAFE-REPAIR), LXVII (SAFE-ONTOLOGY), LXXII (SAFE-GENQ), LXXIII (SAFE-PROCURE), LXXXII (SAFE-FORMAT), LXXXVI (SAFE-PROOF). Product seams: `actOntologyCoverage.json`, Admin Testing coverage table, `questionBank` merge, GENQ verify gate, C4 diagnostic seal.

LXXXVII.1 Why this chapter exists

SAFE-GENQ already kills "AI wrote N questions" and unverified keys as diagnostic ground truth. Growth still faces a quieter hazard: **coverage theater** — leading landing pages, parent decks, and district packets with "1,500+ ACT questions," "covers every ACT topic," or "full ontology bank" while the live audit shows *partial* level coverage, ontology-only ACT-tested slots with zero static items, GCSE-tagged Eedi volume that must not be sold as ACT inventory, and generated JSON that is **inert until verify clears**. Item count is a *warehouse metric*; identity products sell *usable, keyed, blueprint-aligned practice for a named use*.

FOUNDER BELIEF under audit: Honest coverage tables (what exists, at which level/format, for which *use*) convert serious parents and LEAs better than inflated totals — and protect Map integrity when Maya's gap-scan only draws from sealed keys.

Claims we refuse as doctrine:

1. Item-count / "N questions" as marketing hero or North Star.
2. "Complete ACT coverage" / "every concept" without concept×level (and format) honesty.
3. Treating GCSE/Eedi/other-examTag inventory as ACT-prep proof.
4. Counting unverified / unsynced generated items in marketed coverage.
5. Coverage % ≡ exam readiness, identity change, or FEI proof.
6. Ontology node count / Layer-1 breadth ≡ playable practice bank.
7. "AI fills any hole instantly" as coverage substantiation (SAFE-GENQ).
8. Stale coverage audits sold as current product truth.

LXXXVII.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Domain representation	Content samples the intended domain	Ontology×bank join; ACT-tested set	Node list without items
Alignment	Agreement of assessment with expectations	Concept/level/format vs ACT blueprint intent	Count without DOK/range
Use / IUA	Kane: validate *interpretations and uses*	Practice reveal vs C4 diagnostic vs parent report	Bank exists ≡ ACT-ready
Coverage table	Public matrix of what is shippable where	Admin `actOntologyCoverage` + marketing cut	Totals without gaps

Use-tier label	Which claims a slot may support	Practice / diagnostic-sealed / not playable	One number for all uses
SAFE-COVER	Blueprint-honest coverage GTM	This chapter	Item-count hero

Operational definition (HYPOTHESIS): A MindCraft coverage claim is *SAFE-COVER complete* when it (a) names the exam/track and ontology join, (b) reports **playable** counts by concept×level (and format when claimed), (c) lists **gaps** as first-class (partial / ontology-only / verify-blocked), (d) separates practice-eligible from diagnostic-sealed inventory, (e) excludes unverified GENQ and wrong-examTag inflation, and (f) never implies official ACT endorsement or guaranteed score lift from bank size.

LXXXVII.3 Validity is about uses — coverage is not a fetish object

FACT (argument-based validity): Kane (2013, *Journal of Educational Measurement*, 50(1), 1–73, doi:10.1111/jedm.12000) — what is validated are *interpretations and uses* of scores, not "the test" as a thing. More ambitious claims need more evidence; negative consequences can kill a use even if a thinner interpretation survives.

FACT (Standards framing): AERA, APA, & NCME (2014). *Standards for Educational and Psychological Testing*. Washington, DC: AERA. — Validity is the degree to which evidence and theory support interpretations for proposed uses; statements should not claim unqualified "validity of the test." Content-related evidence must speak to the intended construct and use.

Applied (HYPOTHESIS):

Marketed claim	Ambition	Substantiation bar
"Practice items exist for topic X at L2"	Low	Playable keyed items in bank for that slot
"Gap-scan covers ACT-tested concepts"	High	Diagnostic-sealed keys + honest missing slots
"Full ACT Math coverage"	Extreme	Blueprint alignment + range/balance evidence — **do not claim** while partial/ontology-only gaps remain
"Bank proves identity / +ACT points"	Banned	Claim-ladder / SAFE-EXAM / SAFE-DURABLE

Kill: "Questions in the bank" ≡ validated ACT-readiness or identity use.

Survive: Use-tier coverage language tied to Kane IUA (ties SAFE-GENQ).

LXXXVII.4 Content evidence is representation — not warehouse volume

FACT (content-validity elements): Sireci (1998b, *Social Indicators Research*, 45(1–3), 83–117, doi:10.1023/A:1006985528729) — content validity concerns domain definition, **domain representation**, domain relevance, and appropriateness of development procedures — not how many items sit in a folder.

FACT (alignment as content evidence): Martone & Sireci (2009, *Review of Educational Research*, 79(4), 1332–1361, doi:10.3102/0034654309341375) — alignment studies evaluate congruence among curriculum, instruction, and assessment; content approaches (including SME congruence ratings) are a primary method family for gathering content-related evidence.

FACT (Webb criteria): Webb (2007, *Applied Measurement in Education*, 20(1), 7–25, doi:10.1080/08957340709336728) — alignment judgments use categorical concurrence, depth-of-knowledge consistency, range-of-knowledge correspondence, and balance of representation; "good enough" thresholds are pragmatic and partly subjective — volume alone never settles alignment.

Commercial implication: Marketing may show a **matrix** (concepts × levels × formats) and name missing cells. It may **not** treat "1,943 static questions" (current audit-scale warehouse) as proof the ACT blueprint is represented in depth, range, and balance. Sparse L3 on an ACT-frequent topic is a *coverage honesty* issue, not a footnote.

Kill: Item-count hero; ontology breadth ≡ bank representation.

Survive: Publish representation gaps; regenerate `actOntologyCoverage.json` before any coverage campaign.

LXXXVII.5 Alignment is hard — raters disagree; do not overclaim

FACT (rater variability): Herman, Webb & Zuniga (2007, *Applied Measurement in Education*, 20(1), 101–126, doi:10.1080/08957340709336732) — alignment pictures can change with which raters are used; agreement on topic/DOK is nontrivial. Large-scale programs still treat third-party alignment evidence as ethically and regulatorily important (WebbAlign / peer-review practice summarizing Webb 1997→; AERA/APA/NCME 2014 content evidence).

HYPOTHESIS: MindCraft's internal conceptId join is a *lightweight congruence audit*, not a full Webb study. Selling "aligned to ACT" as if a DOK/range panel ran is cite-wash (SAFE-LABMETA). Prefer: "Mapped to our ACT-tested ontology concepts; here is the playable matrix and holes."

Wound: Equating bank-alias joins with formal standards–assessment alignment studies.

Survive: Honest internal coverage tables as *minimum* GTM; commission real alignment only if making high-ambition district claims.

LXXXVII.6 Advertising law: prior substantiation for objective coverage claims

FACT (FTC prior substantiation): Federal Trade Commission (1984). *FTC Policy Statement Regarding Advertising Substantiation* (appended to *Thompson Medical Co.*, 104 F.T.C. 648, 839) — advertisers must have a **reasonable basis** for objective claims *before* dissemination; express/implied claims that suggest a level of proof require at least that level of support; failure to possess prior substantiation can violate Section 5 of the FTC Act.

FACT (consumer guidance): FTC Advertising FAQs (business guidance) — ads must be truthful and non-deceptive; advertisers need evidence for express *and* implied claims reasonable consumers take away; “tests prove”-class claims need the advertised level of proof.

Applied (HYPOTHESIS): “Complete ACT question bank,” “covers every tested topic,” and “1,500+ ACT questions” (when inventory mixes GCSE tags, partial levels, or unsynced GENQ) are **objective** coverage claims. Implied takeaway for parents often includes “enough for ACT prep / diagnosis.” Substantiation = dated coverage matrix + examTag discipline + verify seal — not a blog total.

Kill: Running coverage ads from stale audits or warehouse sums.

Survive: Pre-claim packet: regenerate audit → marketing cut of full/partial/ontology-only → use-tier footnotes.

LXXXVII.7 MindCraft bank reality (internal FACT for GTM)

FACT (product audit snapshot — regenerate before campaigns): `app/src/data/actOntologyCoverage.json` (audit summary fields) has reported on the order of ~29 ACT-tested concepts, mixed **full / partial / ontology_only** statuses, and nonzero **gapsNeedingContent** (e.g. partial level holes; ontology-only slots such as strategy/representation_translation/basic_equations with zero static bank match). Static warehouse totals can look large while L1/L2/L3 playable cells remain empty. CLAUDE.md / bank merge notes: multi-source bank (~static + ACT master + Eedi GCSE-tagged + generated stubs); several advanced concepts still zero-coverage; generation verify historically ~30% drop — **do not count unsynced GENQ**.

HYPOTHESIS: Parent- and LEA-facing honesty that *names* partial cells increases trust relative to “1,500+ questions” hero — testable via COVER- */ ties WTP-*.

Kill: Marketing the warehouse total without the gap list.

Survive: Admin Testing coverage table as the source of truth for copy; “we show the holes” as brand.

LXXXVII.8 Product surface — SAFE-COVER claim contract

Claim surface	Required fields	Banned substitute
Landing / ads	Track name; playable concept count; link or footnote to gaps	Bare “N questions”
Parent deck	Full / partial / missing summary; diagnostic seal note	“Complete ACT bank”
District packet	Dated audit; examTag policy; GENQ verify rate (SAFE-GENQ/PROCURE)	FERPA badge as coverage
In-app Learn/Practice	Playable gate (`hasPlayableQuestions`); no fake full path	Path nodes without items

Competitive foil	Diagnosis + sealed keys + honest matrix	Out-Khan on library size

Reuse: SAFE-GENQ (verify-before-count); SAFE-FORMAT (format claims need true FormatId stock); SAFE-ONTOLOGY (graph ≠ bank); SAFE-COLD (gap-scan only sealed keys); SAFE-PROOF (competence artifacts > library vanity); SAFE-WTP (CBC arms: honest matrix vs item-count hero).

LXXXVII.9 Competitive positioning

Competitor pattern	Coverage move	MindCraft response
Khan-scale library	Breadth / free forever	Diagnosis + playable honesty; do not out-count
Infinite AI drills	Volume / novelty	Verify gate; refuse poisoned Map (SAFE-GENQ)
Cram packs	“X tests worth of items”	Dual-rail; no completeness theater (SAFE-DURABLE)
ChatGPT homework	Unlimited stems	Keyed bank + seal; fluency ≠ coverage
Static publishers	Stable catalogs	Show ontology holes we still owe

Competitive wedge (FOUNDER BELIEF): “We publish the coverage matrix — including empty cells — and only sealed keys rewrite the Map” beats both library vanity and infinite-AI theater.

LXXXVII.10 Doctrine — SAFE-COVER (provisional)

1. **Matrix > total** — market concept×level (×format when claimed); demote item-count hero.
2. **Gaps are features** — partial and ontology-only cells appear in GTM, not only in eng docs.
3. **Use-tier labels** — practice vs diagnostic-sealed vs not playable (Kane IUA).
4. **ExamTag honesty** — do not sell GCSE/other inventory as ACT coverage.
5. **GENQ discipline** — unverified/unsynced generated items never enter marketed counts (SAFE-GENQ).
6. **Prior substantiation** — regenerate audit before coverage ads (FTC reasonable-basis posture).
7. **No ACT-official / complete-bank puffery** while holes remain.
8. **Copy:** “Here’s what you can practice now — and what we’re still filling.” Never “1,500+ ACT questions” as the learning claim.

Confidence: High that Kane/Standards bind *uses* not warehouses (Kane 2013; AERA/APA/NCME 2014). High that content evidence cares about representation/alignment, not raw N (Sireci 1998b; Martone & Sireci 2009; Webb 2007). High that FTC expects prior substantiation for objective ads (FTC 1984). Medium that honest matrices beat item-count heroes on parent/LEA conversion (needs COVER-*). High that counting unverified GENQ or wrong-examTag stock is a kill. Medium that lightweight ontology joins suffice for early GTM without full Webb panels.

LXXXVII.11 Experiments

ID	Question	Design	Primary
COVER-1	Honest matrix landing vs item-count hero landing	A/B	Signup; trust; bounce on gaps
COVER-2	Parent CBC: “shows holes + sealed keys” vs “1,500+ ACT questions” vs “AI unlimited”	CBC	WTP; credibility (SAFE-WTP)
COVER-3	District packet with dated coverage audit vs brochure totals	Sales A/B / qual	Advance-to-DPA; objection themes
COVER-4	In-app gap callout (“L3 not ready”) vs silent fail / dynamic-gen promise	A/B	Session start; trust; support tickets
COVER-5	Regenerate-audit cadence before campaigns vs stale JSON	Ops	Claim–product mismatch incidents
COVER-QUAL	10 parents + 5 tutors: what “covers ACT” means to them	Qual	Implied-claim codebook for FTC-style review

Falsifier: Item-count hero wins conversion *and* 90-day retention equally → still ban count as *learning/ACT completeness* claim; segregate storefront vanity from coverage doctrine.

Falsifier: Publishing gaps tanks signup with no trust gain → redesign framing (roadmap + sealed keys), do not restore “complete bank” lies.

Falsifier: Parents ignore matrices → keep doctrine for LEA/trust; simplify consumer cut to three numbers (full / partial / missing) + one CTA.

Pre-register: COVER-* before any “complete ACT / N questions / AI fills all holes” campaign (SAFE-LABMETA).

LXXXVII.12 So what for MindCraft commercially

- **Copy:** “Here’s what you can practice now — and what we’re still filling.” Lead with matrix + sealed keys. Never item-count / complete-ACT / infinite-AI coverage heroes.
- **Product:** Keep Admin coverage table + `actOntologyCoverage.json` as GTM source of truth; playable gates; use-tier labels on Learn/Practice; regenerate audit in release checklist before coverage claims.

- **Positioning:** Against Khan library vanity and AI-infinite drills; for blueprint-honest, verify-sealed practice that cannot silently poison diagnosis.
- **Metric:** Coverage-claim incidents (mismatched ads); matrix CTR→session; parent CBC preference; co-primary playable-slot fill rate — demote `questionCount` / generation volume.
- **Kill list:** Item-count hero; complete-ACT puffery; GCSE-as-ACT; counting unsynced GENQ; ontology-nodes≡bank; stale-audit campaigns.
- **Growth:** District trust packets include dated coverage + verify rates (SAFE-PROCURE × SAFE-GENQ); parent decks show holes as integrity, not weakness.
- **Vision:** A thirty-year identity company earns belief by refusing warehouse theater — Maya's Map only moves on keys that exist and are true.

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Part LXXXVIII — Worked-Example Fading in Solver UX

Chapter status: Living evidence + Solver/Practice UX brief — Researcher tick 2026-08-06

Primary question: How should MindCraft fade guidance from full worked examples → completion steps → solo solve in Solver (and Practice help rails) without permanent solution-dump, expertise-reversal tax, or "instant full answer" as the product brand?

Owners: Product (Solver / Practice) · Engine (ingredient cards / coach) · HITL tutors · Brand · Red Team

Commercial job: Ship a **SAFE-FADE** doctrine: guidance level is a *diagnosed state*, not a permanent chrome; attempt grain before reveal; completion problems as the default bridge; solo transfer as the proof that fading worked.

Builds on: Parts XXVI (CLT / fading), XL (SAFE-SE), LI (SAFE-DP), LXXIX (SAFE-FBTIME), LXXXIV (SAFE-TALK), LXXXV (SAFE-STORYLOAD), LXXXVI (SAFE-PROOF). Product seams: Solver help, ingredient cards, soft-wrong / SE gates, `solo_transfer_pass`, hint binge guards.

LXXXVIII.1 Why this chapter exists

Homework Help / Solver is the surface where competitors and ChatGPT win on *fluency*: paste problem → full worked solution → green check feeling. MindCraft already bans unguarded Solver hero (PWC) and AI-monologue≡SE. The remaining product hazard is quieter: **permanent full worked examples** (or always-on step dumps) that help absolute novices once, then become redundant load — and train help-seeking that never fades.

FOUNDER BELIEF under audit: A Solver that *fades* (full example → missing last steps → missing more → full problem) converts "I needed help" into "I can finish the join myself" — the identity move — better than unlimited perfect solutions. That claim is commercial only if we instrument attempt grain and solo transfer, not solution views.

Claims we refuse as doctrine:

1. Always-full-worked / solution-first Solver as learning North Star.
2. Never-fade step scaffolding ("we always show every step").
3. Completion-of-viewing ≡ mastery (watching steps = doing math).
4. Instant full answer as brand differentiator vs ChatGPT.
5. Expertise-blind help (same dump for L1 and near-fluent).
6. Hint binge / reveal streak without fade-up after success.
7. Fade Score™ / guidance-% as vanity NS.
8. Forward-only fade as dogma without local FADE-* check (backward often wins in lab).

LXXXVIII.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Worked example	Full problem + solution steps for study	Full coach card / Solver reveal	Permanent dump
Completion problem	Partial solution; learner finishes	Missing-step Solver; last-step blank	Fake blanks with answer peek
Fading	Successive removal of worked steps	Guidance ladder per concept/format	One-shot jump to solo
Expertise reversal	Novice-helpful guidance hurts experts	Same full example for advanced	Redundant load / boredom
Attempt grain	Learner generates before full reveal	SE / soft-wrong / blank-step gate	Watch-only "learning"
SAFE-FADE	Diagnosed fade ladder + solo proof	This chapter	Solution theater

Operational definition (HYPOTHESIS): A Solver/help path is *SAFE-FADE complete* when it (a) defaults to **attempt or completion** before full worked reveal on learn rail, (b) fades **backward** (omit final steps first) unless

FADE- *prefers otherwise for that schema, (c) raises guidance only on diagnosed struggle (not permanent chrome), (d) fades* up* after consecutive solo successes, (e) never counts solution-view as mastery, and (f) co-primary proves with `solo_transfer_pass` / delayed mixed accuracy — not hint satisfaction NPS.

LXXXVIII.3 Novices need worked examples — search is the tax

FACT (worked-example effect): Sweller & Cooper (1985, *Cognition and Instruction*, 2(1), 59–89, doi:10.1207/s1532690xci0201_3) — for algebra novices, studying worked examples as a substitute for conventional problem solving reduced time and errors on subsequent similar problems; conventional search imposed heavy load that retarded schema acquisition.

FACT (CLT framing): Sweller (1988, *Cognitive Science*, 12(2), 257–285, doi:10.1207/s15516709cog1202_4) — problem-solving search can consume working memory needed for schema construction; reducing extraneous search supports learning of complex material.

Applied (HYPOTHESIS): Day-one Solver that *only* withholds all guidance (pure struggle theater) violates CLT for true novices. SAFE-FADE is not "never help" — it is **help that starts complete enough to build schema, then steps down**.

Kill: Discovery-only / no-example Solver for first acquisition.

Survive: Full worked study *as a stage*, not as the product forever.

LXXXVIII.4 Expertise reverse — permanent examples become clutter

FACT (expertise reversal): Kalyuga, Ayres, Chandler & Sweller (2003, *Educational Psychologist*, 38(1), 23–31, doi:10.1207/s15326985ep3801_4) — instructional methods highly effective for inexperienced learners can lose effectiveness or become detrimental as expertise grows, because guidance becomes redundant and loads working memory with unnecessary cross-checking.

FACT (when solving beats examples): Kalyuga, Chandler, Tuovinen & Sweller (2001, *Journal of Educational Psychology*, 93(3), 579–588, doi:10.1037/0022-0663.93.3.579) — as learners gain expertise in a domain, problem solving can become superior to studying worked examples (classic reverse pattern under CLT).

Commercial implication: Marketing "we show every step, always" is a **novice costume** that taxes Maya once she has partial schemas — and looks identical to ChatGPT solution dump. Map/level/format evidence must *lower* default guidance after demonstrated competence.

Kill: Expertise-blind always-full-worked brand.

Survive: Guidance that tracks diagnosed grain (ties SAFE-COLD / SAFE-FORMAT / SAFE-ONTOLOGY).

LXXXVIII.5 Completion problems bridge example → solve

FACT (completion as bridge): van Merriënboer and colleagues introduced *completion problems* — given state, goal, and partial solution the learner must finish — as intermediates between fully worked examples and conventional problems; completion strategies emphasize completing increasingly larger parts of incomplete solutions (see van Merriënboer, 1990; van Merriënboer & de Croock, 1992, in programming; reviewed in Paas & van Merriënboer instructional-control work).

FACT (mental effort / transfer): Paas (1992, *Journal of Educational Psychology*, 84(4), 429–434, doi:10.1037/0022-0663.84.4.429) — in statistics problem solving, training with completion problems or worked examples yielded better transfer with lower test-phase mental effort than conventional problem training (completion ≈ worked examples on effort during training; both beat conventional on transfer efficiency).

Applied (HYPOTHESIS): Solver's default "help" should often be a **blank last step / blank middle join**, not a monologue. Ingredient cards already suggest multi-representation scaffolds — fade means those scaffolds lose steps over successes, not that more cards appear forever.

Kill: Help ≡ paste full solution.

Survive: Help ≡ completion problem with named missing grain (procedure, bridge, format).

LXXXVIII.6 Smooth fading beats static example–problem pairs

FACT (fading procedure): Renkl, Atkinson, Maier & Staley (2002, *Journal of Experimental Education*, 70(4), 293–315, doi:10.1080/00220970209599510) — successive integration of problem-solving elements into example study (complete example → increasingly incomplete → to-be-solved) fostered near-transfer learning vs traditional example–problem pairs; errors during learning mediated the effect; **backward fading** (omit last solution steps first) outperformed forward fading.

FACT (CLT transition argument): Renkl & Atkinson (2003, *Educational Psychologist*, 38(1), 15–22, doi:10.1207/S15326985EP3801_3) — as intrinsic load falls with schema growth, problem-solving demands can increase without overload; fading structures that transition; later stages change which activities are germane vs extraneous.

Applied (FOUNDER BELIEF → testable): MindCraft Solver ladder:

Stage	UX	Gate to next
E0 Full example	Study + SE prompt on a marked step	SE / principle tag
E1 Backward fade	Last step blank	Correct completion
E2 Deeper fade	Last two / critical join blank	Correct + no peek
E3 Full problem	Solo attempt	`solo_transfer_pass` delayed

Raise guidance only after miss class / pour streak — never as default forever.

Kill: Static example then unrelated dump; jump from E0 to E3 without completions.

Survive: Backward-first fade ladder with local FADE-* override rights.

LXXXVIII.7 Attempt grain — watching is not doing

FACT (passive example risk): Worked-example literature repeatedly notes that *passive* viewing weakens the effect; prompting self-explanation and active study of solutions strengthens learning (ties Chi / Renkl SE line — Part XL). MIT TLL synthesis of Sweller & Cooper (1985) and later work: insufficient examples and passive demos neutralize the worked-example effect.

HYPOTHESIS: Solver telemetry that optimizes `solution_reveal_rate` or time-on-help will invent help abuse (next chapter 89). Co-primary must be **student-generated steps before reveal** + delayed solo.

Reuse: SAFE-SE (why before wrap); SAFE-FBTIME (micro-delay on learn rail); SAFE-TALK (construction > pour); SAFE-PROOF (solo artifact).

Kill: View-complete ≡ session success.

Survive: `fade_step_attempt` events before unlock of next worked line.

LXXXVIII.8 Product surface — SAFE-FADE claim contract

Surface	Required behavior	Banned substitute
Solver first open (learn)	Attempt or completion before full dump	Instant ChatGPT-style full solution
After consecutive solos	Fade *up* (less guidance)	Keep dumping “to be helpful”
After struggle	Fade *down* one stage; name the missing grain	Permanent max scaffolding
Ingredient / coach cards	Steps can blank; SE on blank	Always-complete card deck
Marketing	“We fade help as you prove the join”	“Unlimited full solutions”
Prove / exam rail	Minimal worked reveal; KR timing (SAFE-FBTIME)	Learn-rail full examples under stakes

Competitive foil: Khan/Brilliant often lean example+practice; ChatGPT leans full solution fluency. MindCraft differentiates on **diagnosed fade + solo transfer**, not on prettier monologues.

LXXXVIII.9 Competitive positioning

Competitor pattern	Guidance move	MindCraft response
ChatGPT homework	Full fluent solution	Completion + fade + Map-check (SAFE-REPAIR)
Instant-feedback AI	Always show steps	Mode-conditional; attempt grain

Cram packs	Worked keys as content	Dual-rail; fade on learn, prove alone
“Socratic” bots	Endless questions *or* dump	Construction metrics; not talk-% (SAFE-TALK)
Static textbooks	Fixed example then exercises	Adaptive fade from evidence state

Competitive wedge (FOUNDER BELIEF): “Help that disappears when you can carry the step” beats both abandonment struggle and infinite solution theater.

LXXXVIII.10 Doctrine — SAFE-FADE (provisional)

1. **Stage, don’t costume** — full worked examples are an *acquisition stage*, not the brand forever.
2. **Completion is the default bridge** — blank critical / final steps before full reveal.
3. **Backward fade first** — omit last steps first unless FADE-* shows forward better for that schema.
4. **Expertise-aware** — raise guidance on struggle; lower after consecutive solos (expertise reversal).
5. **Attempt grain** — student-generated step / SE before unlocking more worked lines.
6. **Proof = solo transfer** — not solution views, Fade Score™, or help NPS.
7. **Copy:** “We fade the steps as you prove them.” Never “unlimited perfect solutions” or “we always show every step.”
8. **Tie rails** — learn rail may use E0–E2; prove/exam rail stays near E3 (SAFE-EXAM / SAFE-DURABLE).

Confidence: High that worked examples help novices vs conventional search (Sweller & Cooper 1985; Sweller 1988). High that expertise reverses example benefits (Kalyuga et al. 2003; Kalyuga et al. 2001). High that fading/completion structures transition better than static pairs (Renkl et al. 2002; Renkl & Atkinson 2003; Paas 1992). Medium that backward fade is the right default for MindCraft algebra/bridge schemas (needs FADE-*). Medium that parent WTP prefers fade story over unlimited solutions (needs WTP/FADE CBC). High that permanent full-dump Solver is commercially indistinguishable from ChatGPT and anti-identity.

LXXXVIII.11 Experiments

ID	Question	Design	Primary
FADE-1	Backward completion ladder vs always-full-worked Solver	A/B within concept	Near transfer; `solo_transfer_pass`; hint binge
FADE-2	Backward vs forward fade order	A/B	Transfer; errors during learning
FADE-3	Expertise-aware fade-up after 2 solos vs fixed E1 forever	A/B	Time-on-task; transfer; boredom/skip

FADE-4	SE-on-blank-step vs passive completion	A/B	Transfer; SE quality
FADE-5	Parent CBC: “fades help as you prove” vs “unlimited full solutions” vs “never shows answers”	CBC	WTP; trust (SAFE-WTP)
FADE-QUAL	10 Maya: when does a shown step feel like rescue vs theft of the win?	Qual	Fade UX codebook

Falsifier: Always-full-worked wins transfer *and* 26w identity equally → still ban as *brand* vs ChatGPT; segregate acquisition-stage examples from marketing hero.

Falsifier: Aggressive fade tanks novice retention with no solo gain → hold longer at E0/E1; do not restore permanent dump.

Falsifier: Forward fade beats backward on MindCraft bank → update ladder default; keep completion doctrine.

Pre-register: FADE-* before any “AI shows every step / unlimited solutions / never-fade scaffold” campaign (SAFE-LABMETA).

LXXXVIII.12 So what for MindCraft commercially

- **Copy:** “We fade the steps as you prove them.” Lead with completion → solo, not unlimited solutions.
- **Product:** Solver/help ladder E0→E3; blank-step attempt events; fade-up after solos; prove rail near E3; ingredient cards lose steps over success.
- **Positioning:** Against ChatGPT solution dump and struggle-theater apps; for diagnosed guidance that earns identity via solo carry.
- **Metric:** `fade_stage`, `fade_step_attempt`, `hint_binge`, co-primary `solo_transfer_pass` — demote `solution_reveal_rate` as success.
- **Kill list:** Always-full-worked hero; never-fade scaffolding; view≡mastery; expertise-blind dump; Fade Score™ NS; unlimited-solutions ads.
- **Growth:** Parent decks sell *earned independence of steps*; tutors QA to completion prompts not monologues (SAFE-HITL × SAFE-TALK).
- **Vision:** Thirty-year identity company teaches Maya to need the worked line less — help that graduates her, not a solution vending machine.

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Part LXXXIX — Help-Seeking vs Help Abuse

Chapter status: Living evidence + Solver/Practice/HITL brief — Researcher tick 2026-08-06

Primary question: When should MindCraft treat help use as adaptive self-regulation versus executive help abuse (or help avoidance), and what product/copy rules follow without turning “unlimited hints” or “never help” into the brand?

Owners: Product (Solver / Practice / coach) · Engine (hint ladder / telemetry) · HITL tutors · Brand · Red Team

Commercial job: Ship a **SAFE-HELP** doctrine: instrumental help that graduates into solo transfer; detect and discourage executive binge / bottom-out racing; invite help when avoidance is the failure; prove with deliberate help metrics + `solo_transfer_pass`, not hint satisfaction NPS.

Builds on: Parts XXVI (fading), XL (SAFE-SE), XLV (SAFE-RESILIENCE), LI (SAFE-DP), LVIII (SAFE-EXPOSE), LXXIX (SAFE-FBTIME), LXXXIV (SAFE-TALK), LXXXVIII (SAFE-FADE). Product seams: `hint_binge`, `ai_reveal_rate`, `help_recruit_then_solo`, fade ladder E0–E3, Solver reveal gates. Next sibling: id 91 hint economy (cost of peek).

LXXXIX.1 Why this chapter exists

SAFE-FADE answered *what* guidance looks like as expertise grows. This chapter answers *when students ask for it* — and when that ask is learning versus gaming.

FOUNDER BELIEF under audit: A product that makes help *always cheap and bottom-out-complete* trains Maya to recruit the answer, not the join. A product that shames all help trains avoidance and freeze. MindCraft’s identity claim requires **instrumental** help-seeking (help to understand the next step) and refuses **executive** help-seeking (help to finish without thinking) as success theater.

Claims we refuse as doctrine:

1. Unlimited hints / always-open bottom-out as learning North Star.
2. Hint volume or help NPS as proof of pedagogy quality.
3. Help avoidance as grit (struggle theater that bans support).
4. Metacognitive nag alone as guaranteed domain gains (Help Tutor caution).
5. Gaming detectors as shame dashboards for parents.
6. “AI always knows when you need help” black-box contingent brand.
7. Help Score™ / Talk-to-hint ratio vanity NS.
8. Permanent tutor dependency sold as “supportive.”

LXXXIX.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Instrumental help	Help aimed at learning how to proceed	Principle hint / completion blank / SE gate	Rare without design
Executive help	Help aimed at getting the answer done	Bottom-out dump / ChatGPT paste	Common default
Help abuse	Maladaptive overuse (e.g. hint racing)	Rapid reveal chain; `hint_binge`	Fake fluency
Help avoidance	Under-use when help would help	Stuck loops; refuse coach	Freeze / shame
Try-step abuse	Hasty guessing without sense-making	Spam submit / click farm	Gaming
Contingent help	Scaffold level tracks need	Fade down/up from evidence	Fixed forever chrome
SAFE-HELP	Instrumental + contingent + solo proof	This chapter	Hint vending

Operational definition (HYPOTHESIS): A help path is *SAFE-HELP complete* when it (a) distinguishes instrumental vs executive use in telemetry, (b) makes bottom-out / full reveal costly in *effort or stage* (not humiliation), (c) invites help after repeated miss without pouring the answer first, (d) co-primary proves with `help_recruit_then_solo` and `solo_transfer_pass`, and (e) never markets unlimited answer-complete help as identity transformation.

LXXXIX.3 Instrumental vs executive — the commercial fork

FACT (classroom help typology): Classic help-seeking research distinguishes help that supports learning how to solve from help that merely completes the task — often labeled instrumental vs executive (Nelson-Le Gall, 1981; Karabenick & Knapp, 1991; summarized in Roll, Aleven, McLaren & Koedinger, 2011, *Learning and Instruction*, 21(2), 267–280, doi:10.1016/j.learninstruc.2010.07.004).

Applied (HYPOTHESIS): ChatGPT-style Solver and "unlimited hints" ads optimize for *executive* completion feeling. Parent WTP for "my kid got unstuck" must be re-aimed at *instrumental* unstuck — named next grain + student-generated step — or MindCraft collapses into homework completion SaaS.

Kill: Help ≡ finish the worksheet.

Survive: Help ≡ recruit a scaffold that still requires Maya's next move (ties SAFE-FADE / SAFE-SE).

LXXXIX.4 Help abuse is frequent — and often anti-learning

FACT (early Cognitive Tutor logs): Aleven & Koedinger (2000/2001; reviewed in Aleven, McLaren, Roll & Koedinger, 2016, *IJAIED*, 26(1), 205–223, doi:10.1007/s40593-015-0089-1) — students often raced hint levels to the bottom-out answer (many pre-bottom levels viewed <1s) and under-asked after errors; on-demand help frequency correlated negatively with learning gains (selection + misuse both plausible).

FACT (model taxonomy): Aleven, McLaren, Roll & Koedinger (2006, *IJAIED*; also Aleven et al., 2004, ITS) — executable model with Help Abuse, Help Avoidance, and Try-Step Abuse; early offline eval classified a large share of actions as unproductive (Aleven et al., 2004 ~72% under a strict model — model-dependent, not a marketing statistic).

FACT (gaming): Baker and colleagues treat rapid help abuse / exploiting tutor regularities as primary "gaming the system," associated with reduced learning (Baker, Corbett & Koedinger line; Roll et al., 2011; Aleven et al., 2016).

Applied (FOUNDER BELIEF → testable): MindCraft's existing `hint_binge` (≥3 hints without independent solve) and `ai_reveal_rate` are the right *family* of detectors — but must not become parent shame ranks (SAFE-PDASH). Use them for coach fade, tutor QA, and product gates.

Kill: Hint clicks as engagement success.

Survive: Deliberate hint reading time + fewer levels when help is taken (Help Tutor behavioral signature).

LXXXIX.5 Help avoidance is the other failure — not grit

FACT: The same Aleven/Roll line documents frequent help avoidance — students who need help most often seek it least or seek it poorly (Aleven et al., 2003 overview; Wood & Wood, 1999; Nelson-Le Gall et al., 1990 on poorer self-assessment among lower prior-knowledge students — cited throughout the Help Tutor program).

FACT (contingent tutoring): Wood & Wood (1999, *Computers & Education*, 33(2–3), 153–169) — contingent tutoring adjusts help level with success/failure; process measures of help-seeking relate to prior knowledge and outcomes; help seeking is part of constructing a workable zone of support, not a character flaw.

Applied (HYPOTHESIS): After repeated soft-wrong / pour risk, MindCraft should *offer* instrumental help (completion blank, principle card) — not celebrate silent freeze as resilience. SAFE-RESILIENCE already recruits support; SAFE-HELP specifies the *kind*.

Kill: "Real math kids never ask for hints."

Survive: Help invite after struggle; still require attempt grain before bottom-out.

LXXXIX.6 Metacognitive feedback improves help skill — not always domain gain

FACT (Help Tutor studies): Roll et al. (2006 ITS; 2011, *Learning and Instruction*, 21(2), 267–280, doi:10.1016/j.learninstruc.2010.07.004) — metacognitive feedback made help use more deliberate (more time per level; fewer levels), sometimes persisting after feedback ended — **but domain learning gains did not reliably improve** (Aleven et al., 2016: "help helps, but only so much").

FACT (theoretical update): Aleven et al. (2016) — principle-based hints help when students sense-make; bottom-out can act like a worked example *if* self-explained (Shih, Koedinger & Scheines, 2008); otherwise executive racing yields thin learning.

Commercial implication: Do not sell "we tutor metacognition so scores rise." Sell **instrumental help + SE/fade gates + solo proof**. Metacognitive nags are support tools, not the FEI product.

Kill: Help-Tutor cosplay as ACT-point engine.

Survive: Deliberate-help UX + SAFE-SE on hints + FADE ladder; HELP-* before metacognition brand campaigns.

LXXXIX.7 Contingent scaffolding — system share of the load

FACT: Contingent / adaptive hint level (Wood & Wood, 1999; Luckin & du Boulay, 1999 — cited in Roll et al., 2011) reduces opportunities for some maladaptive patterns by matching help grain to estimated need — a design cousin of Cognitive Tutor delays between rapid hint requests.

Applied (HYPOTHESIS): MindCraft should prefer **contingent stage** (SAFE-FADE E0–E3 driven by struggle/solo evidence) over unlimited student-chosen bottom-out. Student agency remains: they can request the next instrumental grain; the system refuses instant executive completion on learn rail without attempt.

SPECULATION: Fully automatic "AI chooses every hint" without inspectable why will read as surveillance pedagogy to parents and as black-box to tutors — prefer named stage + reason chip (ties SAFE-ADAPT inspectability pattern).

Kill: Unlimited student-controlled answer dump as personalization.

Survive: Contingent ladder + requestable next grain + visible stage.

LXXXIX.8 Product surface — SAFE-HELP claim contract

Surface	Required behavior	Banned substitute
Solver learn rail	Instrumental first; bottom-out behind attempt/SE	Instant full answer
Practice hints	Levelled hints; binge gate; fade-up after solos	Hint spam = progress

After 2–3 misses	Offer help / completion; don't shame ask	Silent freeze as virtue
Tutor HITL	Prompt recruit of student why before pour	Tutor as answer key
Parent view	Aggregate deliberate-help / solo — not hint shame	Live hint-stalk dashboard
Marketing	"Help that teaches the next step"	"Unlimited hints / never stuck"
Prove / exam rail	Minimal executive help; KR timing (SAFE-FBTIME)	Learn-rail binge under stakes

Competitive foil: ChatGPT wins executive completion. Struggle-only apps win avoidance theater. Duo-like products win engagement loops. MindCraft differentiates on **instrumental, contingent help that still demands Maya's construction** — then disappears into solo transfer.

LXXXIX.9 Doctrine — SAFE-HELP (provisional)

1. **Instrumental > executive** — help must leave a student-generated step; bottom-out is a last grain, not the brand.
2. **Abuse and avoidance are both failures** — binge racing and freeze both fail FEI; design for both detectors.
3. **Contingent stage, not unlimited menu** — help level tracks evidence (SAFE-FADE); student requests next grain, not full dump by default.
4. **Sense-making gate** — hints without SE/attempt are incomplete (SAFE-SE × Aleven 2016).
5. **Metacognition is not score magic** — deliberate-help skill can improve without domain gains; co-primary stays transfer/solo.
6. **Dignity telemetry** — `hint_binge` / reveal rate for product + tutor QA; never parent shame ranks.
7. **Copy:** "Ask for the next step — not the finished answer." Never "unlimited hints so you're never stuck."
8. **Tie rails** — learn rail allows instrumental ladders; prove rail stays near solo (SAFE-EXAM / SAFE-DURABLE).

Confidence: High that executive hint racing/gaming is common and often anti-learning (Aleven/Koedinger; Baker; Aleven et al., 2016). High that instrumental vs executive is the commercial fork (Nelson-Le Gall / Karabenick via Roll et al., 2011). High that contingent tutoring beats free bottom-out as design prior (Wood & Wood, 1999). High that Help Tutor–style feedback can change help behavior without guaranteed domain gains (Roll et al., 2011; Aleven et al., 2016) — medium that SE+fade closes that gap (HELP-*). Medium parent CBC preference for instrumental copy (HELP-5). High that unlimited-hints brand ≈ ChatGPT completion.

LXXXIX.10 Experiments

ID	Question	Design	Primary
HELP-1	Contingent instrumental ladder vs unlimited bottom-out Solver	A/B within concept	`solo_transfer_pass`; `hint_binge`; near transfer

HELP-2	SE-required before next hint level vs free drill-down	A/B	Deliberate hint time; transfer; binge
HELP-3	Help-invite after 2 misses vs no invite (avoidance arm)	A/B	Help recruit; eventual solo; anxiety item
HELP-4	Metacognitive binge message vs silent gate vs neither	A/B/C	Help behavior; domain transfer (expect possible null on domain)
HELP-5	Parent CBC: “next-step help” vs “unlimited hints” vs “we never show answers”	CBC	WTP; trust (SAFE-WTP)
HELP-QUAL	10 Maya: when does a hint feel like a tool vs a theft of the win?	Qual	Instrumental/executive codebook

Falsifier: Unlimited bottom-out wins delayed solo transfer *and* 26w identity equally → still ban as *brand* vs ChatGPT; segregate rare emergency reveal from marketing.

Falsifier: Aggressive binge gates raise avoidance and drop retention with no solo gain → soften gate; keep instrumental-first doctrine.

Falsifier: Metacognitive messages alone raise ACT-relevant transfer → allow limited HELP-Tutor-style copy; still co-primary solo, not Help Score™.

Pre-register: HELP-* before any “unlimited hints / never stuck / AI knows exactly when to help” campaign (SAFE-LABMETA). Sibling id 91 (hint economy) owns peek-cost pricing experiments — do not duplicate as HELP-$.

LXXXIX.11 So what for MindCraft commercially

- **Copy:** “Ask for the next step — not the finished answer.” Lead with instrumental help + solo proof.
- **Product:** Contingent hint/fade ladder; SE/attempt before bottom-out; help-invite on avoidance; binge gates without shame UX.
- **Positioning:** Against ChatGPT executive completion and struggle-theater grit; for self-regulated help that graduates into independence.
- **Metric:** `hint_binge`, `ai_reveal_rate`, `help_recruit_then_solo`, deliberate hint time, co-primary `solo_transfer_pass` — demote hint count / help NPS as success.
- **Kill list:** Unlimited-hints hero; help NPS NS; avoidance-as-grit; Help Score™; parent hint-stalk; metacognition≡ACT points.
- **Growth:** Parent decks sell *earned independence of help*; tutors QA to instrumental prompts (SAFE-HITL × SAFE-TALK).
- **Vision:** Thirty-year identity company teaches Maya *when and how* to recruit support — then to need less of it — not a hint vending machine or a silence cult.

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Part XC — Product Analytics for FEI North Stars

Chapter status: Living evidence + instrumentation brief — Researcher tick 2026-08-06

Primary question: How should MindCraft instrument, govern, and commercially use FEI North Star events (`retry_120s`, `challenge_accept`, `transfer_pass`, `solo_transfer_pass`) without recreating vanity-metric theater or Goodhart-corrupted dashboards?

Owners: Product analytics · Engineering · Brand · HITL ops · Red Team · Founders

Commercial job: Ship a **SAFE-INSTRUMENT** doctrine: one FEI-aligned scoreboard that predicts durable competence and identity movement; demote DAU/streak/XP/accuracy as North Stars; validate leading events before marketing them; protect against metric gaming by students *and* by the company.

Builds on: Parts I.3 / VIII / IX Experiment C / XXI (Metrics Dictionary), XXXIV (claim ladder), XLIII (SAFE-HABIT), LXX (SAFE-LABMETA), LXXXVI (SAFE-PROOF). Immediate product ask in `NEXT_LAB.md`: align analytics with Part XXI. Sibling queue: 91 hint economy (peek cost once help metrics exist).

XC.1 Why this chapter exists

The Constitution already *named* the right events. Production still mostly optimizes what is easy to count: opens, minutes, streaks, tonight-% correct, tokens, "felt helpful." That gap is not a documentation problem — it is a **commercial operating system** problem.

FOUNDER BELIEF under audit: If MindCraft cannot see Fear → Evidence → Identity in the telemetry, the company will sell the competitors' story by accident: engagement theater (Duo-like), answer fluency (ChatGPT-like), or blocked accuracy (cram-pack-like). Identity transformation without instrumented FEI is a speech, not a product.

Claims we refuse as doctrine:

1. DAU / streak / XP / time-on-app as learning North Stars.
2. Blocked-session accuracy or Map green-% as readiness.
3. Coach thumbs-up / help NPS / "felt helpful" without `solo_transfer_pass`.
4. Shipping Experiment A–D without pre-registered primary events.
5. A single composite FEI Score™ sold to parents as proof.
6. Optimizing `retry_120s` alone until students click-farm retries.
7. Marketing "science-backed metrics" without Kane-style interpretation/use arguments.
8. Analytics as surveillance shame for parents or tutors (SAFE-PDASH / SAFE-HITL dignity).

XC.2 Constructs

Construct	Research meaning	MindCraft analogue	Failure mode
Vanity metric	Looks good; does not change decisions (Croll & Yoskovitz)	Raw DAU, downloads, streak length	Fake growth
OMTM / NSM	Focus metric for stage vs enduring success metric	Stage OMTM = `retry_120s`; NSM stack = challenge + transfer	Metric soup
Goodhart / Campbell	Target corrupts measure and process	Optimize accuracy → gaming	Corrupted FEI
Gaming the system	Exploit software regularities without learning (Baker)	Hint race, spam-retry, soft-wrong farm	Fake persistence
IUA (Kane)	Claims about *uses* of scores need evidence proportional to ambition	Parent deck claiming "identity" from 7-day retry	Overclaim
SAFE-INSTRUMENT	FEI events as governed product law	This chapter	Dashboard cosplay

Operational definition (HYPOTHESIS): Analytics are *SAFE-INSTRUMENT complete* when (a) the four FEI core events fire reliably in production, (b) Experiment C–style lagged prediction ranks them above streak/DAU for 8-week challenge-seeking, (c) anti-gaming companions (`hint_binge`, `ai_reveal_rate`, motive tags) are co-primary gates, (d) marketing uses only IUA-backed interpretations (L0–L2 until L3 evidence), and (e) no composite FEI Score™ is sold as a product hero.

XC.3 The four events — what they measure

FACT (internal doctrine spine): Part XXI already defines:

Event	Definition (abridged)	FEI role
`retry_120s`	New attempt within 120s of soft-wrong	Persistence under safety
`challenge_accept`	Chose harder level when offered (+ motive code)	Challenge-seeking
`transfer_pass`	Correct on varied item after mastery mark	Anti-fake-mastery
`solo_transfer_pass`	Transfer with AI/Solver closed or denied	Anti-crutch

HYPOTHESIS (working North Star composite, not a Score™): Challenge-seeking under safety *with* transfer — mastery-motive `challenge_accept`, return after miss (`retry_120s`), delayed mixed / solo transfer — predicts identity movement better than streaks (I.3; Experiment C prior).

Applied: Ship these four first. Secondary Part XXI events (`coach_shown`, `write_exit_to_retry`, `hint_binge`, …) are *guards and mediators*, not substitute North Stars.

Kill: “We measure learning” while only shipping session accuracy + DAU.

Survive: Named FEI events with motive tags and transfer co-primaries.

XC.4 Vanity metrics lose — on purpose

FACT (product analytics practice): Lean Analytics distinguishes vanity metrics (impressive, non-actionable totals) from actionable metrics that change behavior; it further advocates focusing on a stage-appropriate One Metric That Matters rather than “data puking” dashboards (Croll & Yoskovitz, *Lean Analytics*, O’Reilly; OMTM essays at leananalyticsbook.com / O’Reilly).

Applied (FOUNDER BELIEF → testable): MindCraft’s *enduring* North Star is not a single SaaS OMTM forever — identity is lagging — but early-stage focus should pick **one FEI leading indicator at a time** (today: `retry_120s` under coach experiments) while keeping the **NSM stack** (`challenge_accept + transfer_pass / solo_transfer_pass`) as the company scoreboard that OMTMs must ladder into.

Commercial implication: Investor and parent decks may show growth oxygen (activation, retention) in an appendix. The *hero* learning claim must name FEI events or dated solo-transfer artifacts (SAFE-PROOF) — never streak length.

Kill: Streak / DAU / XP velocity as learning proof.

Survive: FEI NSM stack + stage OMTM discipline.

XC.5 When measures become targets — Campbell & Goodhart

FACT: Campbell's law — the more a quantitative social indicator is used for decision-making, the more subject it is to corruption pressures and the more apt it is to distort the processes it monitors (Campbell, 1979, *Evaluation and Program Planning*, 2(1), 67–90, doi:10.1016/0149-7189(79)90048-X).

FACT: Goodhart's law (and Strathern's popular paraphrase) — when a measure becomes a target, it ceases to be a good measure — is the economics twin of the same corruption dynamic (Goodhart, 1975 monetary-policy statement; Strathern, 1997 paraphrase widely cited in accountability debates).

Applied (HYPOTHESIS): If tutors are paid on `retry_120s`, students will be coached to tap retry without sense-making. If marketing celebrates rising `challenge_accept` without motive codes, appearance-climate CTA wins (SAFE / TARGET / Part XXXVIII). If `transfer_pass` is gamed with isomorphic clones, fake mastery returns.

Design rule: Never attach high-stakes pay, shame ranks, or public heroboards to a *single* FEI event. Use **co-primary gates**: raise retry *without* raising `hint_binge` / `ai_reveal_rate`; raise challenge_accept *with* mastery-motive tag; raise transfer *with* declared hop + delayed mix (XXI.4).

Kill: FEI Score™ / single-KPI tutor commissions / parent shame ranks on retry.

Survive: Multi-metric decision rule + qualitative spot-checks (SAFE-LABMETA).

XC.6 Student gaming is also company risk

FACT: In intelligent tutoring systems, "gaming the system" — succeeding by exploiting software regularities (hint abuse, systematic guessing) rather than learning — is common and associated with poorer learning (Baker, Corbett, Koedinger & Wagner, 2004 line; Baker et al., 2009, AIED — tutor features explained a large share of gaming variance; Cocea, Hershkovitz & Baker — harmful gaming wastes learning opportunities).

FACT: Help-abuse / executive racing is already Constitution law under SAFE-HELP (Aleven/Koedinger; Roll et al., 2011; Aleven et al., 2016) — analytics must treat `hint_binge` and `help_executive_race` as *first-class companions* to FEI wins, not afterthoughts.

Applied: Instrument anti-gaming detectors *in the same ship* as FEI events. A week where `retry_120s` rises and `hint_binge` rises is **not** a FEI win — it is Campbell corruption in progress.

Kill: Celebrate raw retry/challenge lifts without gaming companions.

Survive: XXI.4 decision rule as shipping gate.

XC.7 Validity of uses — Kane before decks

FACT: Kane's argument-based approach holds that what is validated are proposed *interpretations and uses* of scores, not the numbers themselves; more ambitious claims require more evidence; evaluating score *uses* requires evaluating consequences (Kane, 2013, *Journal of Educational Measurement*, 50(1), 1–73, doi:10.1111/jedm.12000; see also Kane, 1992, *Psychological Bulletin*).

Applied (HYPOTHESIS): Emitting `retry_120s` is an L0/L1 engineering fact. Saying "FEI is working" from a 2-week A/B is L1. Saying "we build math identity" from those events without 26w triangulation is an L3 claim without warrant (SAFE-LONGID / claim ladder XXXIV). Parent-facing "proof" should prefer dated solo-transfer artifacts (SAFE-PROOF) over composite meters.

Commercial implication: Marketing and sales get an **IUA card** per claim: event → interpretation → allowed use → required evidence → banned overclaim. No card → no ad.

Kill: "Science-backed identity growth" from unvalidated dashboards.

Survive: Claim-laddered FEI reporting + Experiment C lagged prediction.

XC.8 Product surface — instrumentation contract

Surface	Required event / behavior	Banned substitute
Soft-wrong path	`retry_120s`, `coach_shown`, binge guards	Thumbs-up only
Level offer	`challenge_accept` + motive enum (mastery/appearance/norm)	Harder-click without why
Mastery mark	Delayed `transfer_pass` probe scheduled	Instant blocked green
Solver / AI	`solo_transfer_pass` / `ai_reveal_rate`	"Felt helpful" chat NPS
Experiment A	Pre-reg primary = `retry_120s`	Engagement-only success
Parent deck	Proof-age + named hop artifact	Streak / tonight-% hero
Tutor QA	FEI sample + talk/help guards	Hours booked NS
Board OMTM	One FEI leading metric per stage	Metric soup / DAU hero

Competitive foil: Duo wins streak oxygen. ChatGPT wins executive completion counts. Cram packs win blocked accuracy. MindCraft differentiates by publishing (internally first, externally carefully) **persistence under safety + challenge with motive + delayed solo transfer** — the FEI scoreboard competitors cannot honestly claim without changing their products.

XC.9 Doctrine — SAFE-INSTRUMENT (provisional)

1. **Ship the four** — `retry_120s`, `challenge_accept` (+motive), `transfer_pass`, `solo_transfer_pass` before any new vanity KPI.
2. **NSM stack, stage OMTM** — enduring scoreboard is FEI+transfer; current sprint focuses one leading event.
3. **Co-primary gates** — XXI.4: no FEI "win" if binge/reveal rises or transfer falls.
4. **Anti-Goodhart** — no single-event pay, shame, or public heroboard; rotate checks; keep qualitative audits.
5. **Anti-gaming companions** — `hint_binge`, `help_executive_race`, `ai_reveal_rate` ship with FEI.
6. **Kane IUA for uses** — decks/ads need interpretation/use cards; ambition ≤ evidence (L0–L4).
7. **Dignity** — analytics for product learning and tutor coaching, not parent/student humiliation.
8. **Copy:** "We measure return after a miss, choosing harder for mastery reasons, and solving later without the model." Never "we're #1 for engagement."

Confidence: High that vanity engagement metrics are weak durable-learning proxies (Lean Analytics practice + Constitution HID / SAFE-HABIT). High that Campbell/Goodhart corruption applies once FEI events become high-stakes targets. High that ITS gaming literature predicts students will exploit naive persistence metrics (Baker line). High that Kane IUA is the right hygiene for marketing uses. Medium that Experiment C will rank FEI events above streaks for 8-week challenge-seeking (pre-registered prior, not yet run). Medium on exact 120s window / motive enum reliability until instrumentation QA.

XC.10 Experiments

ID	Question	Design	Primary
INSTR-1	Do `retry_120s` / mastery-motive `challenge_accept` / `transfer_pass` beat streak & DAU predicting 8w challenge-seeking?	Lagged observational (Experiment C densify)	Ranked predictive validity; calibration plots
INSTR-2	Does publishing `retry_120s` as tutor KPI without co-gates raise binge/spam-retry?	Ops A/B (gate vs single KPI)	`hint_binge`; true retry quality sample
INSTR-3	Soft-wrong → coach instrumentation completeness vs dark funnel	Audit + fix ship	Event fire rate; missingness by surface
INSTR-4	Motive-coded `challenge_accept` vs uncoded harder-click	A/B CTA	Mastery-motive share; later transfer
INSTR-5	Parent CBC: FEI report card vs streak/accuracy hero deck	CBC (SAFE-WTP)	WTP; trust; comprehension

INSTR-QUAL	10 Maya + 5 tutors: which dashboard numbers feel honest vs gamable?	Qual	Corruption codebook

Falsifier: Streak/DAU predict 8w challenge-seeking and 26w identity as well as FEI events → still ban as *marketing* North Stars (habit≠identity); allow as oxygen metrics only.

Falsifier: Co-primary gates stall shipping for >1 quarter with no better learning outcomes → simplify to two-event gate (`retry_120s` + `solo_transfer_pass`) temporarily; keep anti-gaming.

Falsifier: Motive tags unreliable / demand-laden → drop parent-facing motive; keep internal coding research-only.

Pre-register: INSTR-* / Experiment C before any "our North Star proves identity" campaign (SAFE-LABMETA). Do not invent FEI Score™ packaging experiments that recreate Belief Score™ failures (SAFE-CALIB).

XC.11 So what for MindCraft commercially

- **Copy:** Lead with "return after a miss," "choose harder to learn," "prove it later without the model." Ban engagement/streak/accuracy hero claims as learning proof.
- **Product:** Instrument Part XXI FEI core + gaming companions this quarter; wire Experiment A primary to `retry_120s`.
- **Positioning:** Against Duo streak oxygen, GPT completion counts, and cram accuracy — for a FEI scoreboard tied to solo transfer.
- **Metric law:** NSM = challenge-seeking under safety with transfer; OMTM rotates among FEI leadings; vanity demoted.
- **Kill list:** DAU/streak/XP NS; blocked accuracy NS; FEI Score™; single-KPI tutor pay; thumbs-up≡learning; science-backed identity ads without IUA.
- **Growth:** Parent/LEA decks sell dated proof + FEI process honesty (SAFE-PROOF × SAFE-PDASH); sales enablement uses IUA cards.
- **Vision:** A thirty-year identity company measures the loop that builds identity — fear metabolized into evidence — not the vanity that only looks like school.

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Part XCI — Hint Economy & Contingent Scaffolding

Chapter status: Living evidence + Solver/Practice hint-ladder brief — Researcher tick 2026-08-06

Primary question: How should MindCraft price the *cost of a peek* (effort, stage, and timing — not humiliation) and make scaffolding *contingent* (up after miss, down after success) so soft principle hints beat free bottom-out racing — without renaming ChatGPT dump as "adaptive help"?

Owners: Product (Solver / Practice hints) · Engine (fade×hint state) · HITL tutors · Brand · Red Team

Commercial job: Ship a **SAFE-HINT** doctrine: soft before hard; contingency before menu; peek costs construction; fade×hint as one state machine; prove with solo transfer, not hint-click engagement.

Builds on: Parts XXVI (fading), XL (SAFE-SE), LXXXVIII (SAFE-FADE), LXXXIX (SAFE-HELP), XC (SAFE-INSTRUMENT). Product seams: hint levels, `hint_binge`, `help_deliberate_ms`, `help_executive_race`, `fade_stage`, SE gates. Sibling: id 92 explanation length (token tax on monologue hints).

XCI.1 Why this chapter exists

SAFE-HELP answered *when* help is instrumental vs abusive. SAFE-FADE answered *what stage* of worked guidance belongs on the learn rail. This chapter answers the *microeconomics of the hint button*: peek cost, soft vs hard levels, and contingency (Wood) + fading (van de Pol) so Maya constructs instead of shopping for answers.

FOUNDER BELIEF under audit: Free, instant, hard peeks train executive completion. Making the next *principle* cheap and the *answer* expensive — in effort, not shame — converts Solver from a dump into an identity machine, if parents hear "help that still asks you to think" and students still reach solo transfer.

Claims we refuse as doctrine:

1. Costless unlimited peeks / free bottom-out as learning North Star.
2. Soft-vs-hard confusion: treating static "hard scaffolds" (Saye & Brush) as the same as hard (=answer) hints.
3. Hint clicks, Hint Score™, or XP-for-hints as engagement success.
4. Non-contingent fixed chrome (same hard dump forever, or never any hint).

5. Fade ladder and hint ladder as two disconnected products.
6. Humiliation economy (shame timers, public peek counts) as "cost."
7. "AI always picks the perfect hint" black-box brand without inspectable stage.
8. Points / streaks unlocked by peeking.

XCI.2 Constructs (keep the two "soft/hard" axes separate)

Construct	Research meaning	MindCraft analogue	Failure mode
Scaffolding (classic)	Temporary support that enables success beyond unassisted ability	Coach / hint / tutor prompt	Permanent crutch
Contingency	Increase control after failure; decrease after success	Fade up/down + hint level	Fixed menu
Fading / transfer of responsibility	Support shrinks as competence grows	SAFE-FADE E0–E3	Never fade / never help
Soft hint (ITS)	Principle / goal / critical-feature prompt; not the answer	Principle card; "which join?"	Vague pep talk
Hard hint (ITS)	Near-answer / bottom-out	Full step reveal / expression dump	Executive race
Hard scaffold (Saye & Brush)	Static, preplanned support in the environment	Ingredient card templates; completion blanks	Mistaken for "hard hint"
Soft scaffold (Saye & Brush)	Dynamic human/peer contingent aid	Tutor HITL; adaptive coach	Mistaken for "soft hint"
Peek cost	Friction that preserves sense-making before answer	SE gate; dwell; stage lock	Shame timer
SAFE-HINT	Contingent soft→hard ladder with priced peeks	This chapter	Free dump

Operational definition (HYPOTHESIS): A hint path is *SAFE-HINT complete* when it (a) opens at a soft/principle grain unless diagnosis already requires harder support, (b) advances hard grain only after miss or deliberate request *with* attempt/SE, (c) decreases control after consecutive solo successes (contingency × fade), (d) makes bottom-out costly in *construction effort* (not humiliation), (e) never awards XP/streak for peeks, and (f) co-primary proves with `solo_transfer_pass` and lower `help_executive_race`.

XCI.3 Scaffolding is contingent by definition — not a hint vending machine

FACT (origin metaphor): Wood, Bruner & Ross (1976, *Journal of Child Psychology and Psychiatry*, 17(2), 89–100) — tutoring as scaffolding: recruitment, reduction of degrees of freedom, direction maintenance, marking critical features, frustration control, and demonstration when the learner can recognize the solution.

FACT (three characteristics): van de Pol, Volman & Beishuizen (2010, *Educational Psychology Review*, 22, 271–296) — scaffolding reviews converge on **contingency**, **fading**, and **transfer of responsibility**. Contingency (Wood, Wood & Middleton, 1978 line, carried forward): increase degree of control after failure; decrease after success.

FACT (computer contingent tutoring): Wood & Wood (1999, *Computers & Education*, 33(2–3), 153–169) — QUADRATIC algebra tutor implementing contingent support; help-seeking process measures predict learning beyond pretest; prior knowledge moderates how help use relates to outcomes.

Applied (HYPOTHESIS): MindCraft's hint button must encode *contingency*, not a flat shelf of equally cheap peeks. After two soft-wrongs, offer the next soft grain (mark critical feature / reduce degrees of freedom). After a clean solo, *lower* default control on the next isomorphic item. That is the same state machine as SAFE-FADE — not a second product.

Kill: Unlimited student-chosen answer menu labeled "scaffolding."

Survive: Contingent control that moves with evidence.

XCI.4 Soft before hard — Cognitive Tutor ladder evidence

FACT (hint sequences): Cognitive Tutors typically expose multi-level on-demand hints from general/goal reminders → increasingly specific → **bottom-out** near-answer (Aleven & Koedinger, 2000/2001; reviewed in Aleven, McLaren, Roll & Koedinger, 2016, *IJAIED*, 26(1), 205–223, doi:10.1007/s40593-015-0089-1).

FACT (abuse pattern): Students often race to bottom-out; large shares of pre-bottom levels viewed under ~1s; when help is requested, students frequently advance through all levels to the answer (Aleven & Koedinger, 2000/2001; Aleven et al., 2016). Help abuse correlates negatively with learning in multiple analyses; Baker's gaming line treats rapid help abuse as a primary gaming category.

FACT (caveat): Shih, Koedinger & Scheines (2008) — time spent with bottom-out can correlate positively with learning when treated as a worked-example opportunity (self-explanation), not as copy-paste (cited in Aleven et al., 2016). Soft-before-hard still holds; hard-without-sense-making fails.

Naming hygiene (FACT about literature collision): Saye & Brush (2002) call *hard scaffolds* the static, preplanned supports and *soft scaffolds* the dynamic teacher/peer supports. That axis is orthogonal to ITS soft/hard *hint specificity*. Product copy must not say "hard scaffolding" when it means "answer dump," or parents will hear cruelty; tutors must not hear "soft scaffolding" and think "vague AI pep talk."

Applied (FOUNDER BELIEF → testable): Default open = soft hint (critical feature / principle / "name the bridge"). Hard hint = gated. Bottom-out = last grain + SE/attempt tax (SAFE-HELP × SAFE-SE).

Kill: Soft and hard as brand poetry without ladder semantics.

Survive: Named levels: Goal → Principle → Join cue → Step sketch → Bottom-out.

XCI.5 The cost of a peek — effort economy, not shame economy

HYPOTHESIS (hint economy): If hard peeks are free in time, clicks, and social cost, rational students (and anxious ones) buy the answer. Learning products that "remove friction from help" optimize completion SaaS. Identity products must put friction on *executive* peeks while keeping *instrumental* peeks low-cost.

Legitimate costs (product priors):

1. **Construction cost** — write a why / fill a blank / name the FormatId or bridge before next hard level (SAFE-SE / SAFE-FADE).
2. **Stage cost** — hard levels unavailable until soft grain shown *or* diagnosed struggle (contingency).
3. **Dwell cost** — minimum readable time before next level advances (anti-race; Help Tutor behavioral signature).
4. **Budget cost (optional)** — limited hard peeks per session that refresh with solo successes (not daily streak theater).

Illegitimate costs (kills): Public peek shaming; parent live stalk of hint count; countdown humiliation; XP penalties that feel like punishment for asking; Anxiety Score™ gated on peeks.

FACT (support): Roll et al. (2011, *Learning and Instruction*, 21(2), 267–280, doi:10.1016/j.learninstruc.2010.07.004) — metacognitive feedback increased deliberate help (more time, fewer levels) without reliable domain-gain magic (Aleven et al., 2016). Cost that changes *behavior* is not automatically an ACT-point engine — co-primary stays transfer/solo (SAFE-INSTRUMENT).

Kill: Free hard peek as brand kindness.

Survive: Cheap soft peek + expensive hard peek in *effort*.

XCI.6 Fade × hint — one state machine

HYPOTHESIS: Running SAFE-FADE (E0 full example → E3 solo) and a separate always-on hard-hint menu recreates ChatGPT inside the "faded" product. The hint ladder *is* the local control knob inside a fade stage:

Fade stage	Default hint open	Hard peek policy
E0 (full example study)	Soft marks on the example	Bottom-out redundant — ban dump-over-example
E1 (completion)	Soft → principle on blank	Hard only after miss + SE on blank
E2 (sparse scaffold)	Soft first	One hard grain / attempt budget
E3 (near solo)	Soft invite after 2 misses	Hard rare; prove rail near zero

SPECULATION: Inspectable chip ("Stage E1 · next soft: name the join") beats black-box "AI chose your hint" for parent trust and tutor QA (SAFE-ADAPT inspectability pattern).

Kill: Fade theater with free hard menu underneath.

Survive: Coupled fade×hint state; tutor briefs read the same chip.

XCI.7 Product surface — SAFE-HINT claim contract

Surface	Required behavior	Banned substitute
Solver learn rail	Soft-first ladder; hard behind construction cost	Instant bottom-out
Practice hints	Contingent up/down; dwell anti-race	Hint spam = XP
Prove / exam rail	Soft only or none; no hard peeks as default	Learn-rail dump under stakes
Tutor HITL	Soft prompt before pour; match stage chip	Tutor as hard-hint vending
Parent view	Aggregate deliberate-help / solo — not peek ranks	Live “hints used” shame
Marketing	“Next-step hints that still ask you to think”	“Unlimited free hints / never stuck”
Analytics	`help_deliberate_ms`, `help_executive_race`, peek-budget use	Hint Score™ / peeks-as-DAU

Competitive foil: ChatGPT = costless hard peek. Struggle-only apps = infinite soft silence. Duo-like loops = XP for peeks. MindCraft = priced peeks under contingency that graduate into solo transfer.

XCI.8 Doctrine — SAFE-HINT (provisional)

1. **Soft before hard** — principle / critical-feature before bottom-out; name levels in UX.
2. **Contingency is law** — control up after failure, down after success (Wood; van de Pol).
3. **Peek cost = construction** — SE/blank/dwell/stage; never humiliation or parent stalk.
4. **Fade × hint = one machine** — no free hard menu under a “faded” costume.
5. **Two soft/hard vocabularies stay separate** — ITS hint specificity ≠ Saye & Brush hard/soft scaffolds.
6. **No reward for peeking** — no XP, streak, or Hint Score™ for reveals.
7. **Inspectable stage** — chip states why this grain; ban perfect-AI-hint mystique.
8. **Proof** — `solo_transfer_pass` + lower executive race; deliberate soft use OK.

Confidence: High that contingency/fading/transfer-of-responsibility define scaffolding (van de Pol et al., 2010; Wood et al., 1976). High that free bottom-out racing is common and often anti-learning (Aleven/Koedinger; Aleven et al., 2016; Baker). High that soft-before-hard is the right default. Medium that specific peek-budget numbers generalize — run HINT-*. High that shame economies violate SAFE-PDASH. Medium parent CBC for “next-step cost” copy (HINT-5). High that fade×hint decoupling recreates ChatGPT.

XCI.9 Experiments

ID	Question	Design	Primary
HINT-1	Soft-first contingent ladder vs free hard menu	A/B within concept	`solo_transfer_pass`; `help_executive_race`; near transfer
HINT-2	SE/blank cost before hard vs dwell-only vs neither	A/B/C	Deliberate ms; binge; transfer
HINT-3	Coupled fade×hint vs faded stage + free hard submenu	A/B	Solo transfer; hard-peek rate
HINT-4	Peek budget (N hard/session, refill on solo) vs unlimited hard	A/B	Executive race; avoidance; retention
HINT-5	Parent CBC: "next-step hints" vs "unlimited free hints" vs "we never hint"	CBC	WTP; trust (SAFE-WTP)
HINT-QUAL	10 Maya: when does a peek feel earned vs stolen?	Qual	Soft/hard phenomenology codebook

Falsifier: Free hard menu wins delayed solo transfer *and* 26w identity equally → still ban as *brand* vs ChatGPT; allow rare emergency reveal off-learn-rail.

Falsifier: Peek budgets spike avoidance and drop retention with no solo gain → drop budgets; keep soft-first + SE cost.

Falsifier: Soft-first harms true novices vs E0 full example → keep E0 stage; do not open with hard silence.

Pre-register: HINT-* before any "unlimited free hints / AI perfect peeks / Hint Score™" campaign (SAFE-LABMETA). Do not duplicate HELP-1 pricing arms — HINT owns peek-cost; HELP owns instrumental vs executive typology.

XCI.10 So what for MindCraft commercially

- **Copy:** "Next-step hints that still ask you to think." Lead with soft-first + solo proof. Never "unlimited free hints so you're never stuck."
- **Product:** Contingent soft→hard ladder coupled to fade stage; construction cost on hard peeks; inspectable stage chip.
- **Positioning:** Against ChatGPT costless dump and struggle-theater silence; for an *economy of peeks* that transfers responsibility.
- **Metric:** `help_deliberate_ms`, `help_executive_race`, hard-peek rate by stage, co-primary `solo_transfer_pass` — demote raw hint count / Hint Score™.

- **Kill list:** Free hard peeks as kindness brand; Hint Score™ / XP-for-hints; shame timers; black-box perfect-hint AI; fade costume with free dump underneath.
- **Growth:** Parent decks sell earned independence of peeks; tutors QA soft-before-pour (SAFE-HITL × SAFE-TALK).
- **Vision:** Teach Maya to recruit the smallest scaffold that works — then to need less of it.

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Part XCII — Explanation Length vs Germane Load

Chapter status: Living evidence + coach/Solver length brief — Researcher tick 2026-08-06

Primary question: When does a coach or Solver *explanation* create germane processing (schema-building attention to the join) versus a **token tax** — extraneous words that crowd working memory, suppress self-explanation, and cosplay as "thorough teaching"?

Owners: Product (coach copy / Solver wrap) · Engine (prompt budgets) · HITL tutors · Brand · Red Team

Commercial job: Ship a **SAFE-EXPLAIN** doctrine: short principle-first coach beats monologue; length follows expertise and element interactivity; student-generated why beats provided essay; prove with transfer/solo, not word count or "AI explained fully" satisfaction.

Builds on: Parts XXVI (CLT / fading), XL (SAFE-SE), LXXXV (SAFE-STORYLOAD), LXXXVIII (SAFE-FADE), LXXXIX (SAFE-HELP), XCI (SAFE-HINT). Sibling queue: id 93 retrieval failure modes. Product seams: coach card max length, soft-hint grain, SE-before-wrap, monologue-cap telemetry.

XCII.1 Why this chapter exists

SAFE-HINT priced the *peek*. SAFE-FADE staged the *worked example*. SAFE-SE required the *student why*. This chapter prices the *words after the why* (and the words that replace it): how long may a coach speak before thoroughness becomes overload?

Default AI tutor UX optimizes for fluent completeness — multi-paragraph solutions that feel caring. Default parent anxiety asks for "more explanation." Default LLM systems reward tokens. MindCraft's identity claim needs the opposite default: **the shortest explanation that marks the critical feature and leaves room for construction**.

FOUNDER BELIEF under audit: Verbose AI wrap trains passive receipt. Principle-length coach + SE + fade stage trains transferable competence — if parents hear "clear, not endless," and students still reach solo transfer.

Claims we refuse as doctrine:

1. Longer explanation ≡ better teaching / more learning.
2. Token count, "full solution essay," or Explanation Score™ as quality North Stars.
3. Always-on monologue wrap that displaces self-explanation (Kill #8 / SAFE-SE).
4. Same explanation length for novice and near-expert (expertise reversal ignore).
5. Seductive lore / pep talk / franchise trivia inside the coach card (SAFE-STORYLOAD).
6. "AI explained everything" as marketing proof of pedagogy.
7. Unlimited free hard peeks *plus* essay dumps (SAFE-HINT × this chapter).
8. Streaks/XP for reading long explanations.

XCII.2 Constructs (keep load types honest)

Construct	Research meaning	MindCraft analogue	Failure mode
Intrinsic load	Element interactivity of the math itself	Concept + bridge + FormatId difficulty	Soften the math instead of the wrap
Extraneous load	Processing that does not build the schema	Redundant essay, seductive asides, duplicate modalities	"Helpful" fluff
Germane processing	WM devoted to intrinsic structure (modern CLT: redistribute, not a third additive pile)	Naming the join; SE; marking critical features	Calling any effort "germane"

Token tax	Product term: words that consume WM without advancing schema	Multi-paragraph Solver dump	Fluency-as-care
Instructional explanation	Provided why/principle/operator text	Coach card; soft hint text	Essay replacing attempt
Self-explanation (SE)	Learner-generated why	SAFE-SE prompts	AI monologue labeled SE
SAFE-EXPLAIN	Length-contingent, principle-first, SE-preserving coach	This chapter	Monologue costume

Operational definition (HYPOTHESIS): An explanation path is *SAFE-EXPLAIN complete* when it (a) defaults to principle / critical-feature length (≈1–3 sentences or one marked step), (b) expands only after miss, demand, or diagnosed high element interactivity, (c) never replaces a required SE with a provided essay, (d) shortens as fade stage / expertise rises, (e) strips seductive and redundant material, and (f) co-primaries prove with `solo_transfer_pass` / near transfer — not dwell-on-essay or thumbs-up on "thoroughness."

XCII.3 Germane load is not "more words"

FACT (modern CLT): Sweller, van Merriënboer & Paas (2019, *Educational Psychology Review*, 31, 261–292, doi:10.1007/s10648-019-09465-5) reconceptualize germane load: it is working-memory resources devoted to *dealing with intrinsic load*, not a third additive load that rises when you add "learning activities." Reducing extraneous load frees capacity for intrinsic processing; germane is redistributive.

FACT (element interactivity): Sweller (2010, *Educational Psychology Review*, 22, 123–138, doi:10.1007/s10648-010-9128-5) — intrinsic load tracks element interactivity relative to expertise; instructional design should manage extraneous load and optimize germane *processing of* that intrinsic structure.

Applied (HYPOTHESIS): A 600-token coach monologue that re-derives every algebraic step the student already executed is mostly **extraneous** for a mid-fade learner — even if every sentence is "about math." Germane would be: mark the failed join, name the FormatId shift, invite one SE. Length is not a virtue metric.

Kill: "We maximized germane load by adding more explanation."

Survive: Cut extraneous wrap so WM can work the intrinsic join.

XCII.4 Provided explanations: small effects; SE often wins

FACT (meta-analysis): Wittwer & Renkl (2010, *Educational Psychology Review*, 22, 393–409, doi:10.1007/s10648-010-9136-5) — 21 studies / 28 comparisons of worked examples *with* vs *without* instructional explanations. Overall weighted mean effect **small** (d = 0.16). Benefits stronger for **conceptual knowledge** (d ≈ 0.36) than for transfer/problem-solving measures (often ns). When controls were **prompted to self-explain**, instructional explanations showed **no advantage** (subset $d \approx -0.01$).

FACT (withholding can beat providing): Richey & Nokes-Malach (2013, *Learning and Instruction*, 25, 22–34, doi:10.1016/j.learninstruc.2012.11.006) — across three experiments, students who had stepwise instructional explanations *withheld* showed **greater conceptual learning** than students who received them; achievement goals predicted learning more under withholding (constructive effort was volitional).

FACT (framework): Wittwer & Renkl (2008, *Educational Psychologist*, 43(1), 49–64) — instructional explanations often fail when mistimed, misleveled, or unused; design must fit generation *and* use conditions — not assume “explain more.”

Applied (FOUNDER BELIEF → testable): MindCraft’s coach should prefer **SE prompts + short principle marks** over default essay wraps. Long provided explanations are a *last* grain (like hard hints), not the brand voice.

Kill: AI monologue ≡ instructional science; “we explain thoroughly” as FEI proof.

Survive: Minimal instructional text that enables construction; expand only when SE stalls.

XCII.5 Coherence, redundancy, and the token tax

FACT (coherence): Mayer’s coherence principle — people learn better when extraneous material is excluded (supported across many multimedia tests; median ES often large in Mayer handbook summaries). Seductive details (interesting but irrelevant words/images/sounds) hurt retention/transfer (Harp & Mayer line; Moreno & Mayer, 2000, *Journal of Educational Psychology*, 92, 117–125 on minimizing irrelevant sounds).

FACT (redundancy): Chandler & Sweller (1991) / Sweller line — presenting the same information in unnecessary duplicate forms increases extraneous load. Mayer restricted redundancy (narration + on-screen text with animation) similarly taxes capacity (Mayer, Heiser & Lonn / Moreno & Mayer series; median ES ~0.7 in “nine ways” summaries).

Applied (HYPOTHESIS): Solver surfaces that show (1) full worked steps, (2) a paragraph restating those steps, and (3) a chatty pep layer create **content redundancy + seductive fluff**. Prefer one representation at a time: sparse step marks *or* short principle — not both plus lore.

Product prior: Soft hint = principle sentence. Hard peek = one revealed step. Bottom-out = minimal completed blank — not a TED talk. Story wrap stays outside the coach card budget (SAFE-STORYLOAD).

Kill: Duplicate text+voice essays; “engaging” asides inside the critical coach moment.

Survive: One clear signal; cut the rest.

XCII.6 Expertise reversal: length must fade

FACT: Kalyuga, Ayres, Chandler & Sweller (2003, *Educational Psychologist*, 38(1), 23–31) — instructional supports that help novices can **hurt** relative experts (expertise reversal). Detailed guidance becomes redundant; experts waste WM reconciling external text with already-formed schemas.

FACT (design implication): Kalyuga (2007, *Educational Psychology Review*, 19, 509–539) — tailor and fade guidance as expertise rises; fixed high-guidance designs are not "always safer."

Applied (HYPOTHESIS): Couple explanation *length* to SAFE-FADE stage and SAFE-HINT contingency:

Stage / expertise	Default coach length	Expand when
E0 novice example study	Short principle + marked critical feature on the example	Student SE fails / high anxiety + miss
E1 completion	One cue for the blank	Second miss
E2 sparse	Principle only	Demand + dwell
E3 near solo / prove rail	Invite SE; near-zero provided essay	Rare emergency reveal

Kill: Same multi-paragraph wrap for every student forever.

Survive: Length fades with responsibility transfer.

XCII.7 Assistance dilemma — not "more help"

FACT: Koedinger & Aleven (2007, *Educational Psychology Review*, 19, 239–264) — the **assistance dilemma**: how much information to provide vs withhold so that learners still do the constructive work that builds robust knowledge.

Bridge to product: SAFE-HELP / SAFE-HINT already reject unlimited executive dumps. SAFE-EXPLAIN rejects *verbose instrumental* dumps — explanations that look helpful, consume tokens, and still steal the generative work Richey & Nokes-Malach show matters for conceptual gain.

SPECULATION: LLM tutors systematically bias toward over-assistance because fluency and completeness are RLHF-shaped. Without an explicit length budget and SE gate, MindCraft will drift ChatGPT-ward even with "principle" system prompts.

XCII.8 Product surface — SAFE-EXPLAIN claim contract

Surface	Required behavior	Banned substitute
Soft coach / soft hint	≤ principle grain; mark join/FormatId	Multi-paragraph lecture
Soft-wrong path	SE before wrap; wrap ≤ short correction	Essay before student why
Solver learn rail	Length tied to fade stage	Always-full verbal restatement of steps
Prove / exam rail	Minimal KR; no lecture	Post-item monologue
Tutor HITL	Prompt > pour; pour stays short (SAFE-TALK)	Tutor as essay machine

Parent view	"Clear next step" copy — not word-count care	"AI explained for 10 minutes" proof
Marketing	"Short coaches that leave room to think"	"Unlimited thorough AI explanations"
Analytics	`coach_tokens`, `se_before_wrap`, wrap-expand rate	Explanation Score™ / tokens-as-quality

Competitive foil: ChatGPT = costless long dump. Khan video = long lecture outside attempt loop. Duo = short but often shallow XP copy. MindCraft = **length-contingent principle coach** under FEI.

XCII.9 Doctrine — SAFE-EXPLAIN (provisional)

1. **Short before long** — default principle / critical-feature length; expand only on evidence of need.
2. **Germane ≠ more text** — free WM for intrinsic joins; do not advertise "added germane load."
3. **SE before wrap** — provided essay never replaces student why (SAFE-SE).
4. **Length fades with expertise** — couple to fade×hint state (SAFE-FADE / SAFE-HINT).
5. **Coherence > charm** — strip seductive and redundant coach material.
6. **No token North Star** — ban Explanation Score™ / tokens-read / "thoroughness" thumbs as learning KPIs.
7. **Assistance dilemma honesty** — withhold when construction is the learning act.
8. **Proof** — `solo_transfer_pass` / conceptual items / lower wrap-expand binge — not essay dwell.

Confidence: High that modern CLT treats germane as processing of intrinsic structure, not additive word count (Sweller et al., 2019; Sweller, 2010). High that instructional explanations' average gain is small and often loses to SE prompts (Wittwer & Renkl, 2010). High that withholding stepwise explanations can improve conceptual outcomes (Richey & Nokes-Malach, 2013). High that coherence/redundancy and expertise reversal caution against always-longer guidance (Mayer line; Kalyuga et al., 2003). Medium on exact token caps / sentence budgets — run EXPLAIN-*. High that LLM defaults drift verbose without hard budgets.

XCII.10 Experiments

ID	Question	Design	Primary
EXPLAIN-1	Principle-short coach vs multi-paragraph monologue (same facts)	A/B within concept	`solo_transfer_pass`; conceptual item; wrap dwell
EXPLAIN-2	SE-before-wrap vs wrap-first essay	A/B	SE completeness; transfer; binge-expand
EXPLAIN-3	Length coupled to fade stage vs fixed long wrap	A/B	Solo transfer; expertise×length interaction

EXPLAIN-4	Soft-wrong: short mark vs long re-teach	A/B	Retry_120s; near transfer; anxiety self-report (secondary)
EXPLAIN-5	Parent CBC: “clear short coach” vs “thorough AI essays” vs “we never explain”	CBC	WTP; trust (SAFE-WTP)
EXPLAIN-QUAL	10 Maya: when does an explanation feel clarifying vs drowning?	Qual	Length phenomenology codebook

Falsifier: Fixed long monologue wins delayed solo transfer *and* 26w identity equally → still ban as *brand* vs ChatGPT; allow expandable “Show more” off default.

Falsifier: Ultra-short coach harms true novices on high-interactivity first exposure → keep E0 slightly richer marks; do not open with silence theater.

Falsifier: Parents refuse short-coach CBC entirely → redesign copy (“clear next step”) before surrendering to essay brand.

Pre-register: EXPLAIN- *before any “unlimited thorough AI explanations / Explanation Score™ / longest tutor wins” campaign (SAFE-LABMETA). Do not duplicate HINT hard-peek arms — EXPLAIN owns* provided-text length*; HINT owns soft→hard specificity.

XCII.11 So what for MindCraft commercially

- **Copy:** “Short coaches that leave room to think.” Never “the most thorough AI tutor.”
- **Product:** Default principle grain; SE-before-wrap; length tied to fade×hint; hard token/sentence budgets on coach prompts.
- **Positioning:** Against ChatGPT essay dumps and lecture-as-care; for *assistance-dilemma-honest* brevity that transfers.
- **Metric:** `coach_tokens`, `se_before_wrap`, wrap-expand rate — demote tokens / thoroughness NPS / Explanation Score™.
- **Kill list:** Longer≡better; monologue≡SE; token North Star; seductive coach fluff; fixed long wrap across expertise.
- **Growth:** Parent decks sell clarity and independence, not word count; tutors QA pour length (SAFE-TALK).
- **Vision:** Teach Maya to need *fewer* words from the coach — and to supply her own.

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Part XCIII — Retrieval Failure Modes in Practice UX

Chapter status: Living evidence + practice-session brief — Researcher tick 2026-08-06

Primary question: When a learner fails to retrieve mid-practice, which failure mode is occurring — tip-of-tongue / partial access, productive struggle, freeze/blanking, or avoidance — and what product response preserves construction without cosplaying "struggle theater" or dumping answers?

Owners: Product (Practice / Solver timers & hints) · Engine (spacing / soft-wrong) · HITL tutors · Brand · Red Team

Commercial job: Ship a **SAFE-RETRIEVE** doctrine: name the failure mode before treating it; wait + partial cue for TOT-class stalls; contingent soft scaffold for blank/freeze; unsuccessful attempt + feedback can potentiate learning; prove with retry/transfer — never Struggle Score™, blank-time NS, or "always stew / never stuck."

Builds on: Parts XXIX (spacing/retrieval), XLI (desirable difficulties × anxiety), XLIX (productive misconceptions), LXXXVIII–XCII (fade / help / instrument / hint / explain). Sibling queue: id 94 productive failure sequencing. Product seams: dwell timers, tip-cue UI, freeze detection, soft-wrong path, hint contingency.

XCIII.1 Why this chapter exists

Practice UX collapses many stalls into one button: "Need a hint?" Parents hear "struggle" and either panic or romanticize it. Competitors either (a) intervene instantly (ChatGPT dump), (b) gamify stuckness (streaks, pep), or (c) preach productive struggle without instruments.

MindCraft's FEI loop needs a finer grain: **retrieval failure is not one event.** A tip-of-tongue state that motivates continued search is not the same as working-memory blanking under anxiety, which is not the same as Kapur-style generation before instruction, which is not the same as help-avoidance.

FOUNDER BELIEF under audit: Distinguishing failure modes lets MindCraft keep desirable difficulty without flooding Maya after a freeze — and without selling "we never interrupt struggle" as brand.

Claims we refuse as doctrine:

1. All struggle ≡ productive struggle / desirable difficulty.
2. Blank-time, dwell-seconds, or Struggle Score™ as learning North Stars.
3. Always-wait / never-intervene (silence theater ≡ pedagogy).
4. Instant hard peek / full solution on first stall (executive dump).
5. Unsuccessful retrieval ≡ wasted / harmful (so skip attempt and show answer).
6. "Never stuck" marketing that legitimizes answer delivery.
7. Tip-of-tongue pep copy without a retrieval path (affirmation without cue).
8. Streaks/XP for enduring blank screens.

XCIII.2 Constructs (keep modes distinct)

Construct	Research meaning	MindCraft analogue	Failure mode
Retrieval attempt	Effort to access stored knowledge	First try; soft-wrong retry	Skip attempt → restudy tip
TOT (tip-of-the-tongue)	Feeling that an unretrieved item is known / recoverable; metacognitive state dissociable from access	Partial recall; "I almost have it"; letter/structure cue hunger	Treat as total ignorance
Omission / blanking	No candidate produced; freeze	Empty response; rapid panic peek	Force endless wait
Commission error	Wrong candidate produced	Soft-wrong path	Shame / instant dump
Productive struggle	Effort to make sense of math that is not immediately apparent (Hiebert & Grouws)	Sense-making dwell with progress markers	Romanticize despair
Productive failure (PF)	Generation/exploration before consolidation (Kapur) — design, not any fail	Optional struggle-first sequences (id 94)	Label every miss "PF"
Anxiety–WM tax	Anxiety consumes WM needed for math (Ashcraft line)	Freeze under timed/exam chrome	"Try harder" as fix

SAFE-RETRIEVE	Mode-contingent response to retrieval failure	This chapter	One-size struggle UX

Operational definition (HYPOTHESIS): A stall is *SAFE-RETRIEVE handled* when the product (a) attempts mode classification (TOT/partial vs blank/freeze vs productive progress vs avoidance), (b) defaults to wait + micro-cue for TOT-class, (c) offers contingent soft instrumental scaffold for blank/freeze after a bounded wait (SAFE-HINT), (d) treats commission errors via soft-wrong + SE (not dump), (e) never scores blank-seconds as virtue, and (f) co-primaries prove with `retry_120s` / `solo_transfer_pass` — not Struggle Score™.

XCIII.3 TOT is metacognitive — not "they don't know it"

FACT: Brown & McNeill (1966) established laboratory elicitation of tip-of-the-tongue states as distinct from simple forgetting — partial phonological/orthographic information often available when the full target is not.

FACT (review): Schwartz & Metcalfe (2011, *Memory & Cognition*, 39, 737–749, doi:10.3758/s13421-010-0066-8) synthesize direct-access and heuristic–metacognitive accounts: TOTs accompany some failed or slow retrievals; the *feeling* is informed by cues (partial info, related info, cue familiarity) and can change retrieval *behavior* (persist in search). TOT experience and retrieval success are dissociable.

Applied (HYPOTHESIS): In Practice, a student who can name "quadratic formula starts with negative b…" or sketch a triangle but cannot finish is often in a **TOT-class stall**, not a cold start. Correct product move: **bounded wait + partial cue** (first letter of step label, FormatId reminder, "name the bridge") — not full worked example, not "you don't know this — here's the lecture."

Kill: Instant dump on first hesitation; "blank = never learned."

Survive: Treat recoverable inaccessibility as a search problem; cue, don't replace.

XCIII.4 Unsuccessful retrieval can potentiate later learning

FACT: Kornell, Hays & Bjork (2009, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 35(4), 989–998) — unsuccessful retrieval attempts followed by study of the correct answer enhanced subsequent learning relative to study-only controls across multiple experiments (failed tests were not mere harm).

FACT: Richland, Kornell & Kao (2009, *Journal of Experimental Psychology: Applied*, 15(3), 243–257) — pretesting (often unsuccessful) on prose material improved later retention of those facts vs non-pretested content (pretesting / failed-test potentiation).

FACT (synthesis): Vaughn, Hausman & Kornell / Kornell & Vaughn line (e.g. Kornell & Vaughn, 2016, *Psychology of Learning and Motivation*, Vol. 65) — retrieval practice benefits appear even when the attempt fails, provided feedback/study follows; a two-stage view (attempt → process answer) undercuts "only successful retrieval counts."

Applied (FOUNDER BELIEF → testable): MindCraft should **require an attempt** before heavy help (already SAFE-HELP / SAFE-HINT). Marketing must not imply "wrong first tries damage the brain." Parent copy: "Trying before the coach — even when stuck — sets up the learning that follows."

Contradicting pressure (FACT/HYPOTHESIS): Anxiety–WM literature (Ashcraft & Kirk, 2001; Ashcraft, 2002) — under high anxiety, forced struggle without support can collapse performance. Mode matters: potentiation assumes a *search* is happening, not a panic freeze. Hence SAFE-RETRIEVE's freeze branch (soft scaffold after bound), not endless blank.

Kill: Skip-attempt restudy as default; "errors always scar memory" fear marketing.

Survive: Attempt → feedback; shorten wait when freeze signals dominate.

XCIII.5 Productive struggle ≠ every stuck screen

FACT: Hiebert & Grouws (2007) define productive struggle as intellectual effort to make sense of mathematics that is not immediately apparent — sense-making, not despair.

FACT: Warshauer (2015, *Journal of Mathematics Teacher Education*, 18, 375–400, doi:10.1007/s10857-014-9286-3) — 186 classroom struggle episodes; taxonomy of struggle types (get started, carry out process, explain, misconception/error) and teacher responses (telling → affordance). Productivity judged by maintaining cognitive demand, addressing the struggle, and building on student thinking — not by duration alone.

FACT (PF design): Kapur (2014, *Cognitive Science*, 38, 1008–1022) — productive *failure* is sequenced generation then consolidation, not any unsuccessful miss. Full PF sequencing is queue id 94; this chapter bans mislabeling every miss as "productive failure."

Applied (HYPOTHESIS): Detect **progress markers** during dwell (partial work, SE draft, diagram marks). Progress + TOT → wait/cue. No progress + peek urgency / affect → soft scaffold. Freeze → earlier help (optional SAFE-EXPOSE), not Struggle Score™.

Kill: "We maximize productive struggle" measured as blank minutes.

Survive: Sense-making effort with demand maintained; telling as last grain (Warshauer continuum ≈ SAFE-HINT soft→hard).

XCIII.6 Anxiety blanking — different medicine than TOT

FACT: Ashcraft (2002, *Current Directions in Psychological Science*, 11(5), 181–185) — math anxiety as tension/fear that interferes with performance; associated educational and cognitive consequences.

FACT: Ashcraft & Kirk (2001, *Journal of Experimental Psychology: General*, 130(2), 224–237) — math anxiety relates to reduced working-memory capacity available for math; dual-task-like tax.

Bridge to Part XLI: Desirable difficulties that raise challenge without WM support can become *undesirable* under anxiety. Timed chrome, shame clocks, and public blank timers are anti-patterns (also SAFE-EXAM / SAFE-PDASH).

Applied (HYPOTHESIS):

Signal (product proxies)	Likely mode	Default response

Partial work / self-report "almost" / slow edits	TOT / partial	Wait band + micro-cue; no full dump
Empty + rapid rage-click hint / affect spike	Freeze / blank	Soft instrumental after short bound; lower timer salience
Wrong worked path	Commission	Soft-wrong + SE before wrap (SAFE-EXPLAIN)
Idle + no interaction + skip	Avoidance	Help-invite (SAFE-HELP); do not score as grit
Generation phase before teach (designed)	PF (id 94)	Separate experiment arm — not default Practice miss

Kill: One dwell timer policy for all stalls; shame countdown.

Survive: Mode-contingent clocks and scaffolds.

XCIII.7 Product surface — SAFE-RETRIEVE claim contract

Surface	Required behavior	Banned substitute
Practice first stall	Classify; TOT → cue; freeze → soft soon	Instant hard peek
Soft-wrong	Attempt counted as retrieval work	Treat miss as moral failure
Hint ladder	Soft before hard; SE cost (SAFE-HINT)	Free bottom-out menu
Timers	Bound waits; no public blank shame	Struggle Score™ / blank-seconds XP
Coach copy	"Keep searching" only with a cue path	Empty pep / "embrace struggle" poster
Exam / prove rail	KR timing per SAFE-FBTIME / SAFE-EXAM	Flood after first blank
Tutor HITL	Prompt; name mode in brief	"Let them suffer" cosplay
Parent view	"Attempt before coach" honesty	"Never stuck" or "max struggle minutes"
Marketing	Recoverable difficulty + mode-smart help	Struggle theater / answer-dump hero
Analytics	`stall_mode`, `cue_shown`, `attempt_before_help`, `retry_120s`	Struggle Score™ / dwell-as-virtue

Competitive foil: ChatGPT = zero attempt dump. Duo = streak through shallow items. "Grit" apps = blank-time theater. MindCraft = **mode-contingent retrieval support** under FEI.

XCIII.8 Doctrine — SAFE-RETRIEVE (provisional)

1. **Name the mode** — TOT/partial ≠ blank/freeze ≠ PF design ≠ avoidance.
2. **Attempt before heavy help** — unsuccessful retrieval + feedback can potentiate (Kornell/Bjork line).
3. **TOT → wait + micro-cue** — preserve search; do not replace with lecture.
4. **Freeze → bounded soft scaffold** — anxiety–WM honesty; no endless stew.
5. **Progress marks productive struggle** — not clock alone (Warshauer / Hiebert).
6. **No Struggle Score™** — ban blank-seconds / dwell-XP / "maximize struggle" NS.
7. **No never-stuck dump brand** — ban executive rescue as identity.
8. **Proof** — `retry_120s`, near/solo transfer, lower freeze-binge peeks — not pep NPS.

Confidence: High that TOT is a studied metacognitive state distinct from mere absence of knowledge (Brown & McNeill; Schwartz & Metcalfe, 2011). High that failed retrieval + feedback can enhance later learning (Kornell, Hays & Bjork, 2009; Richland et al., 2009). High that productive struggle is sense-making effort, not duration (Hiebert & Grouws; Warshauer, 2015). High that anxiety taxes WM (Ashcraft line). Medium on automatic stall-mode classifiers in-product — run RETRIEVE-* + qual. High that one-size "always wait" or "always hint" will fail FEI.

XCIII.9 Experiments

ID	Question	Design	Primary
RETRIEVE-1	TOT micro-cue vs instant soft hint vs wait-only on partial-work stalls	A/B/C	`retry_120s`; solve without hard peek; near transfer
RETRIEVE-2	Freeze: short-bound soft scaffold vs long forced wait vs instant hard peek	A/B/C	Solo transfer; anxiety self-report; peek binge
RETRIEVE-3	Require attempt before any help vs optional skip-to-coach	A/B	Delayed retention; `attempt_before_help` rate
RETRIEVE-4	Progress-aware dwell (partial marks extend wait) vs fixed dwell timer	A/B	Transfer; freeze false-positives
RETRIEVE-5	Parent CBC: "attempt then coach" vs "never stuck" vs "max struggle time"	CBC	WTP; trust (SAFE-WTP)

RETRIEVE-QUAL	10 Maya: phenomenology of almost-have-it vs blank panic vs fruitful stuck	Qual	Mode codebook

Falsifier: Instant hard peek wins delayed solo transfer *and* 26w identity equally on freeze *and* TOT stalls → still ban as brand vs ChatGPT; allow emergency reveal off default.

Falsifier: Micro-cues never beat wait-only for TOT-class → keep wait bands; demote cue chrome.

Falsifier: Parents reject attempt-first CBC entirely → rewrite ("short try, then a nudge") before surrendering to never-stuck.

Pre-register: RETRIEVE- *before any "maximize productive struggle minutes / Struggle Score™ / never interrupt / never stuck" campaign (SAFE-LABMETA). Do not steal Kapur PF sequencing arms — those belong to id 94; this family owns* in-session stall response*.

XCIII.10 So what for MindCraft commercially

- **Copy:** "When you're stuck, we read the stuck — almost-there gets a nudge; freeze gets a next step." Never "unlimited struggle" or "never stuck."
- **Product:** Stall-mode proxies; TOT cues; freeze-bound soft hints; attempt-before-help gate; no blank-time XP.
- **Positioning:** Against answer-dump tutors *and* grit-theater apps; for recoverable, mode-smart difficulty.
- **Metric:** `stall_mode`, `cue_shown`, `attempt_before_help`, `retry_120s` — demote Struggle Score™ / dwell-as-virtue.
- **Kill list:** All-struggle≡productive; blank-time NS; always-wait; instant dump; never-stuck hero; errors-always-scar fear.
- **Growth:** Parent decks sell attempt→coach honesty; tutors brief on mode naming (SAFE-HITL / SAFE-TALK).
- **Vision:** Maya learns to stay with *almost* — and to accept a soft scaffold when the mind blanks — without becoming either helpless or performatively stuck.

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Part XCIV — Productive Failure vs Guided Success Sequencing

Chapter status: Living evidence + sequencing brief — Researcher tick 2026-08-07

Primary question: When should MindCraft put **generation / struggle-first** before instruction (productive failure / PS-I), and when should it put **guided success / instruction-first** (I-PS, fade, worked examples) — without mislabeling every miss as “PF” or selling fail-first as brand?

Owners: Product (Learn / Practice sequence flags) · Engine (mission templates) · HITL tutors · Brand · Red Team

Commercial job: Ship a **SAFE-PF** doctrine: PF is a *designed two-phase sequence* (generate → consolidate on student RSMs), not any failed attempt; fidelity and priors gate when struggle-first beats teach-first; prove with conceptual + transfer — never PF Score™, fail-first-always, or discovery-without-consolidation.

Builds on: Parts XXVI (CLT / fading), XLI (DD × anxiety), XLIX (productive misconceptions), LXXXVIII–XCIII (fade / help / hint / explain / retrieve). Sibling: XCIII owns *in-session stall response*; this chapter owns *mission-level generation→instruction order*.

XCIV.1 Why this chapter exists

Brand temptation: “We let students fail productively.” Competitor temptation: “We never let them struggle — answers now.” Both are costumes.

Kapur’s productive failure (PF) is a **learning design**: students generate representations and solution methods (RSMs) on a novel complex problem *before* teacher-led consolidation that compares student RSMs to canonical ones. That is not (a) blank-screen theater, (b) ChatGPT dump after three seconds, or (c) relabeling every Practice miss as science.

MindCraft already has guided rails (SAFE-FADE, SAFE-HINT, SAFE-EXPLAIN). The open commercial question is **when Learn-next / new-concept missions should invert the order** — generate first — and when that would waste WM, spike anxiety, or become discovery cosplay.

FOUNDER BELIEF under audit: A high-fidelity PF arm on *selected* novel concepts will beat tell-and-practice on conceptual understanding and transfer without becoming the default for every Practice item or exam drill.

Claims we refuse as doctrine:

1. Every unsuccessful attempt ≡ productive failure.
2. Fail-first / discovery-always as brand or North Star.
3. Generation without Phase-2 consolidation that builds on student work.
4. PF Score™ / generation-minutes / “maximize failure” KPIs.
5. Unguided discovery as identity (Kirschner/Sweller/Clark critique ignored).
6. Struggle-first for high-anxiety freeze states without SAFE-RETRIEVE exit.
7. PF as guaranteed ACT points or “2-sigma from failing first.”
8. Relabeling soft-wrong Practice as Kapur PF.

XCIV.2 Constructs (keep sequences distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
I-PS / DI	Instruction (or strong guidance) then problem solving	Coach/fade → practice; worked-example first	“Never let them try”
PS-I	Problem solving then instruction (umbrella)	Generate → teach	Any miss
Productive failure (PF)	High-fidelity PS-I: generation+exploration then consolidation assembling student RSMs → canonical	Opt-in Learn mission: invent methods → Map/coach C&C	Blank stall (XCIII)
Invention / PFL	Invent methods to prepare for future learning from resources (Schwartz & Martin)	Invent-a-measure / invent-a-rule cards before lecture	Unscaffolded forever
Guided success	Strong guidance so first successes are correct under support	SAFE-FADE completion → solo	Mastery fireworks
SAFE-PF	When to run PF vs guided success; fidelity gates	This chapter	Struggle Score™

Operational definition (HYPOTHESIS): A mission is *SAFE-PF compliant* when (a) the target concept is novel enough that students lack the canonical method, (b) Phase 1 requires multiple RSM attempts with prior-knowledge hooks (not empty invent-from-nothing), (c) Phase 2 **must** compare/contrast student-generated work with canonical forms (tutor or coach C&C — not a generic lecture that ignores what they produced), (d) anxiety/freeze exits to SAFE-RETRIEVE soft scaffold rather than endless generation, (e) outcomes judged on conceptual + transfer items (and later solo transfer), not on Phase-1 correctness or generation time, and (f) default Practice on already-taught skills stays guided/fade — not fail-first.

XCIV.3 PF is a design — not romanticized failure

FACT: Kapur & Bielaczyc (2012, *Journal of the Learning Sciences*, 21(1), 45–83, doi:10.1080/10508406.2011.591717) — design principles for PF across Singapore secondary math classrooms: collaborative complex problem solving *without* instructional scaffolds until teacher-led consolidation; PF students often "failed" to produce canonical solutions in Phase 1 yet outperformed direct-instruction peers on posttests, including representational flexibility.

FACT: Kapur (2014, *Cognitive Science*, 38(5), 1008–1022, doi:10.1111/cogs.12107) — controlled contrast of PF (problem solving then instruction) vs DI (instruction then problem solving) for learning standard deviation; PF outperformed DI; number of student-generated solutions predicted learning. PF is explicitly defined as reversing DI's teach-then-practice order while still providing instruction afterward (not pure discovery).

FACT (mechanisms summary): Kapur's program states four interdependent mechanisms — activation and differentiation of prior knowledge, attention to critical features, explanation/elaboration, and organization/assembly into the target concept — requiring both generation *and* consolidation that uses student RSMs (see Kapur design summaries; Kapur & Bielaczyc, 2012).

Kill: "Fail first" marketing with no consolidation that cites student work.

Survive: Two-phase design; generation diversity as process signal, not as KPI vanity.

XCIV.4 Meta-analytic grain — when PS-I wins, and when it does not

FACT: Sinha & Kapur (2021, *Review of Educational Research*, 91(5), 761–798, doi:10.3102/00346543211019105) — meta-analysis of 53 studies / 166 comparisons of PS-I vs I-PS: overall moderate advantage for PS-I (Hedge's $g \approx 0.36$, 95% CI ~0.20–0.51). Effects stronger when implementations match PF fidelity principles (g in roughly 0.37–0.58 range in high-fidelity subgroups). Contrasting trends: younger learners (roughly grades 2–5) and domain-general skill targets more often favored I-PS. Publication-bias–adjusted estimates in the paper are larger still — treat magnitude as uncertain, direction as the commercial takeaway.

FACT: Loibl, Roll & Rummel (2017, *Educational Psychology Review*, 29(4), 693–715) — theory of *when and how* problem solving followed by instruction supports learning: benefits hinge on preparation (activation, awareness of knowledge gaps) and on instruction that makes critical features discernable — often by building on learner solutions / contrasting cases.

FACT: Loibl & Rummel (2014, *Learning and Instruction*, 34, 74–85) — "knowing what you don't know" can make failure productive: gap awareness during/after generation is part of the mechanism, not mere time-on-wrong-answers.

Applied (HYPOTHESIS): MindCraft should **not** ship universal fail-first. Prefer PF arms when: (1) concept is new, (2) student has enough prior resources to generate *suboptimal* RSMs (SAFE-COLD / Map priors), (3) consolidation UI/tutor can show *their* attempts vs canonical, (4) age/domain fit secondary math-style conceptual targets — not every elementary procedure drill, (5) FEI co-primaries include conceptual/transfer, not Phase-1 accuracy.

Kill: One sequence for all missions; "science says always invent first."

Survive: Fidelity-gated PS-I; I-PS default when priors thin, anxiety high, or skill is procedural fluency under exam chrome (SAFE-EXAM).

XCIV.5 Invention prepares for future learning — assessments that include learning

FACT: Schwartz & Martin (2004, *Cognition and Instruction*, 22(2), 129–184, doi:10.1207/s1532690xci2202_1) — ninth-grade statistics invention activities looked inefficient on standard assessments (students often failed to invent canonical methods) but, when coupled with subsequent learning resources, yielded stronger preparation-for-future-learning outcomes than tell-and-practice — especially on transfer tests that included a learning resource.

Applied (FOUNDER BELIEF → testable): Parent and tutor copy must allow **ugly early work**. Measuring Learn missions only by Phase-1 correctness will kill PF before Phase 2. Product analytics: separate `pf_phase` = generate | consolidate; never score generation accuracy as mastery fireworks (SAFE-COLD / SAFE-INSTRUMENT).

Kill: Early-wrong shame on invent missions; gradebook cosplay of generation.

Survive: "Invent then compare" honesty; MoC after consolidation + transfer.

XCIV.6 The guided-instruction critique is a boundary, not a ban on PF

FACT: Kirschner, Sweller & Clark (2006, *Educational Psychologist*, 41(2), 75–86) — argue minimally guided instruction is generally less effective/efficient than strong guidance for novices, citing cognitive architecture and WM limits; guidance advantage recedes as prior knowledge rises. This is the load-bearing critique against pure discovery.

How PF answers without hand-waving (HYPOTHESIS + Kapur framing): Classic PF is *not* "minimal guidance forever." It is delayed strong guidance: Phase 1 generation uses prior knowledge on complex-but-accessible problems; Phase 2 is teacher-led assembly. Pure discovery without consolidation fails the Kirschner test *and* fails Kapur's own design. MindCraft must not ship Phase-1-only "explore the world" lore walls (also SAFE-STORYLOAD).

Bridge to XLI / XCIII: Under high math anxiety, forced generation without an exit can tax WM (Ashcraft line) and become freeze — use SAFE-RETRIEVE soft bound, then resume consolidation; do not romanticize panic as PF fidelity.

Kill: Discovery-without-teach brand; "minimal guidance = identity."

Survive: Hybrid PS-I with mandatory consolidation; guided success when priors or affect demand it.

XCIV.7 Product surface — SAFE-PF claim contract

Surface	Required behavior	Banned substitute
Learn-next (novel concept)	Optional PF mission flag: generate RSMs → C&C consolidate	Always invent-first for every topic

Practice (taught skill)	Guided/fade/retrieve rails (SAFE-FADE / HINT / RETRIEVE)	Relabel misses as PF
Coach / Solver	Phase 2 builds on *student* attempts (show their RSMs)	Generic lecture ignoring generation
Tutor HITL	Brief: elicit 2+ methods, then compare; no dump in Phase 1	“Let them suffer” or explain-first always
Map / ontology	Target novel join; priors must support generation	Cold invent with zero hooks
Exam rail	Prefer guided success / retrieval under time	Fail-first timed flood
Parent view	“Ugly invent → clear compare” timeline	Generation % / PF Score™
Marketing	Designed struggle-then-teach on new ideas	“We make you fail” / guaranteed points
Analytics	`pf_phase`, RSM count, consolidate_on_student_work, transfer	Generation-minutes NS

Competitive foil: ChatGPT = instruction/dump without generation. Pure discovery apps = generation without assembly. Duo = shallow success loops. MindCraft = **sequence honesty** — invent when priors allow; guide when they don’t; always consolidate before claiming learning.

XCIV.8 Doctrine — SAFE-PF (provisional)

1. **PF ≠ every miss** — reserve the label for designed generation→consolidation sequences (Kapur).
2. **Two phases or it is not PF** — Phase 2 must build on student RSMs (C&C), not ignore them.
3. **Fidelity over slogans** — generation diversity + gap awareness + feature discernment (Sinha & Kapur; Loibl et al.).
4. **Gate the arm** — novel concept + adequate priors + secondary-math-style conceptual targets; default I-PS/guided when thin priors, young procedural drills, or exam chrome.
5. **Anxiety exit** — freeze → SAFE-RETRIEVE soft scaffold; do not endless-stew.
6. **No PF Score™** — ban generation-minutes / fail-count / “maximize productive failure” NS.
7. **Proof** — conceptual understanding + near/far transfer + later `solo_transfer_pass` — not Phase-1 correctness.
8. **Positioning** — against answer-dump *and* discovery-cosplay; for inspectable sequence choice under FEI.

Confidence: High that Kapur PF is a documented two-phase design with classroom and lab evidence (Kapur & Bielaczyc, 2012; Kapur, 2014). High that meta-analytic PS-I advantage exists with fidelity and population moderators (Sinha & Kapur, 2021). High that invention can prepare future learning despite ugly early assessments (Schwartz & Martin, 2004). High that pure minimal guidance is a bad brand for novices (Kirschner et al., 2006). Medium on MindCraft’s ability to automate high-fidelity C&C in AI coach without tutor HITL — run PF-* with human-led arms first. Medium on anxiety × PF interaction in-product — co-design with SAFE-RETRIEVE / SAFE-EXPOSE.

XCIV.9 Experiments

ID	Question	Design	Primary
PF-1	High-fidelity PF (generate→C&C on student RSMs) vs DI (teach→practice) on one novel ACT-adjacent concept	A/B	Conceptual post; near transfer; delayed retention
PF-2	Consolidation that cites student RSMs vs generic lecture after same generation	A/B	Conceptual gain; `consolidate_on_student_work`
PF-3	PF vs guided-success when Map priors low (thin prior knowledge)	A/B	Transfer; freeze/peek binge; dropout
PF-4	PF with SAFE-RETRIEVE freeze exit vs PF with forced long generation	A/B	Solo transfer; anxiety self-report
PF-5	Parent CBC: "invent then compare" vs "never fail" vs "fail-first grit"	CBC	WTP; trust (SAFE-WTP)
PF-QUAL	10 Maya + 5 tutors: phenomenology of invent missions vs explain-first	Qual	Sequence codebook; shame markers

Falsifier: DI beats high-fidelity PF on conceptual *and* transfer for our secondary cohort on pre-registered concepts → keep PF as lab-only; do not market struggle-first.

Falsifier: Generic post-generation lecture equals student-RSM C&C → demote fidelity chrome; still ban Phase-1-only.

Falsifier: Parents reject invent-then-compare CBC → rewrite ("short try at inventing, then we show how experts organize it") before surrendering to never-fail brand.

Pre-register: PF- *before any "fail productively / discovery identity / PF Score™ / always invent first" campaign (SAFE-LABMETA). Do not steal RETRIEVE-* stall arms — those own mid-item stuckness, not mission order.

XCIV.10 So what for MindCraft commercially

- **Copy:** "On new ideas, try inventing a method — then we compare yours to the standard one." Never "we make you fail" or "never struggle."

• **Product:** `mission_sequence` = `pf` | `guided`; Phase flags; student-RSM capture for C&C; freeze exit into SAFE-RETRIEVE.

• **Positioning:** Against ChatGPT dumps *and* unguided discovery cosplay; for sequenced construction with inspectable consolidation.

• **Metric:** conceptual/transfer lifts under PF-1/2; `rsm_count`; `consolidate_on_student_work` — demote generation-minutes / PF Score™.

• **Kill list:** Every-miss≡PF; fail-first-always; generation-without-consolidation; PF Score™; discovery-as-identity; anxiety-forced stew; ACT-points-from-PF ads.

• **Growth:** Tutor playbooks for elicit→compare (feeds id 95 cognitive apprenticeship); parent decks sell invent-then-compare honesty.

• **Vision:** Maya learns that early wrong methods can be *raw material for understanding* — not proof she is not a math person — when the product finishes the job with consolidation.

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Part XCV — Cognitive Apprenticeship in Tutor Playbooks

Chapter status: Living evidence + HITL ops brief — Researcher tick 2026-08-07

Primary question: How should MindCraft encode **cognitive apprenticeship** (modeling → coaching → scaffolding → fading + articulation/reflection/exploration) into tutor playbooks and coach UX — without guild cosplay, demo-only "teaching," or never-fade permanent coaching?

Owners: HITL ops · Tutor training · Coach / Solver product · Brand · Red Team

Commercial job: Ship a **SAFE-APPRENTICE** doctrine: make expert *thinking* visible, then transfer responsibility on a fade schedule; prove with solo transfer and articulation quality — never Master Score™, modeling-as-lecture, or “we’re an apprenticeship” costume ads.

Builds on: Parts XXVI (CLT / tutoring calibration), LI (deliberate practice spine), LII (CoP careful transfer), LIV (AAR), LXVIII (HITL), LXXI (tutor grain), LXXXIV (talk-ratio), LXXXVIII–XCI (fade / help / hint), XCII–XCIV (explain / retrieve / PF). Sibling: SAFE-FADE owns *Solver example stage*; this chapter owns the **full method stack + sociology of the session** for human tutors and the AI coach that must not impersonate a guild.

XCV.1 Why this chapter exists

Brand temptation: “Learn like an apprentice under a master.” Competitor temptation: “Watch the AI solve it, then mimic.” Both fail the Collins test.

Cognitive apprenticeship was proposed because school often *hides* thinking: students see answers, not the control decisions, heuristics, and belief checks experts use. Collins, Brown, and Newman’s fix is not medieval cosplay — it is a design for making cognition visible, supporting performance, then withdrawing support so the learner owns the craft.

MindCraft already fades worked examples (SAFE-FADE), prices peeks (SAFE-HINT), and measures construction (SAFE-TALK). The open commercial question is whether tutor **playbooks** and coach **phase labels** implement the *whole* method stack — or whether we slap “apprenticeship” on explain-first sessions and call it science.

FOUNDER BELIEF under audit: Sessions briefed as model→coach→fade with forced student articulation will beat explain-pour + “got it?” on `solo_transfer_pass` and transfer without raising talk-% vanity.

Claims we refuse as doctrine:

1. Guild / master / “journeyman math” costume as pedagogy proof.
2. Modeling ≡ lecture or AI monologue without student attempt.
3. Scaffolding without fading (permanent coach dependency).
4. Talk-ratio alone ≡ cognitive apprenticeship (SAFE-TALK already killed airtime dogma).
5. Apprenticeship Score™ / “hours under master” North Stars.
6. Discord / community ≡ apprenticeship (SAFE-CoP boundary).
7. Reciprocal-teaching costume without turn-taking dialogue fidelity.
8. “Cognitive apprenticeship” marketing without inspectable method phases in the session log.

XCV.2 Constructs (keep methods distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Modeling	Expert externalizes thinking while performing	Think-aloud on one item; coach “watch my control moves” card	Dumping a full solution without process talk
Coaching	Observe learner; hints, feedback, reminders, new tasks in situ	Tutor prompts mid-attempt; soft→hard peeks	Pre-scripted lecture

Scaffolding	Temporary support controlling hard elements so learner can succeed on rest	Completion problems; Map-briefed hints; Wood functions	Permanent answer key
Fading	Gradual removal of support toward independence	SAFE-FADE E0→E3; less tutor airtime	Abrupt "you're on your own" shame
Articulation	Learner states knowledge/reasoning	SE prompts; postmortem why (SAFE-ANNOT cousin)	"Got it?" yes/no
Reflection	Compare own process to expert/peer	AAR light; film-clip of *their* attempt	Generic tip flood
Exploration	Learner sets goals / invents problems	Challenge accept; optional PF generate phase	Unguided forever
SAFE-APPRENTICE	When/how to run the stack in HITL + coach	This chapter	Guild branding

Operational definition (HYPOTHESIS): A session is *SAFE-APPRENTICE compliant* when (a) at least one short modeling segment makes *control/heuristic* moves audible (not only algebra steps), (b) the majority of problem time is student performance under coaching, (c) scaffolds are logged and *fade within the session or across the mission*, (d) the student articulates a principle or misconception before bottom-out reveal, (e) reflection compares *their* attempt to a canonical path (not a tip dump), (f) outcomes judged on solo transfer + articulation quality, not modeling minutes or master praise, and (g) AI coach phases use the same labels — monologue without attempt fails the gate.

XCV.3 The framework — methods that make thinking visible

FACT: Collins, Brown, & Newman (1989, in L. B. Resnick, Ed., *Knowing, learning, and instruction: Essays in honor of Robert Glaser*, pp. 453–494, Erlbaum) — propose *cognitive apprenticeship* to teach the crafts of reading, writing, and mathematics by adapting traditional apprenticeship (observation, coaching, practice / modeling, coaching, fading) so that otherwise invisible cognitive and metacognitive processes become observable. Core methods: modeling, coaching, scaffolding (with fading), articulation, reflection, exploration. Content types include domain knowledge, heuristic strategies, control strategies, and learning strategies. Sequencing principles include global before local skills, increasing complexity and diversity.

FACT: Collins, Brown, & Holum (1991, *American Educator*, Winter) — practitioner restatement: traditional apprenticeship's four aspects are modeling, scaffolding, fading, and coaching; cognitive apprenticeship extends these to school subjects and adds articulation, reflection, and exploration so students generalize and take autonomy. Emphasizes that in schooling, thinking is often invisible to both student and teacher — the design problem is visibility + transfer of responsibility.

FACT: Brown, Collins, & Duguid (1989, *Educational Researcher*, *18*(1), 32–42, doi:10.3102/0013189X018001032) — argue knowledge is situated in activity, context, and culture; propose cognitive apprenticeship as an alternative to didactic methods that treat concepts as context-free. Cite Schoenfeld's math problem-solving teaching and other classroom designs as exemplars of entering a *culture of practice* (tool use under authentic activity) rather than memorizing inert procedures.

Applied (HYPOTHESIS): MindCraft tutor briefs must name **phase** (`model` | `coach` | `scaffold` | `fade` | `articulate` | `reflect` | `explore`) in the session checklist — not "be Socratic" as a vibe. Product coach UI should show the same phase chip so AI and human share a fidelity language (SAFE-HITL / SAFE-TALK).

Kill: "Apprenticeship" as brand adjective with no phase log.

Survive: Inspectable method stack; thinking made audible; responsibility transferred.

XCV.4 Math exemplar — Schoenfeld's problem-solving teaching

FACT: Schoenfeld (1985, *Mathematical Problem Solving*, Academic Press) — framework for expert/novice math problem solving across resources, heuristics, control, and belief systems; documents teaching methods that make control and heuristic selection explicit rather than treating "problem solving" as extra practice items.

FACT (as cited in the CA literature): Collins et al. (1989/1991) and Brown et al. (1989) treat Schoenfeld's college problem-solving teaching as a cognitive-apprenticeship exemplar: modeling heuristic selection and control on live problems, coaching students as they attempt, scaffolding/fading support, and pushing reflection on process — aimed at entering mathematical *practice*, not only accumulating procedures.

Applied (FOUNDER BELIEF → testable): Tutor playbooks for ACT-adjacent sessions should script **one** short model of *control talk* ("I'm stuck — I'll try a simpler case / check units / draw the join") before coaching the student's attempt. Modeling only algebra steps without control talk fails Schoenfeld's point and becomes SAFE-EXPLAIN monologue risk.

Kill: Heuristic tip card flood without student control practice.

Survive: Audible control moves + student ownership of the next attempt.

XCV.5 Scaffolding is a function set — not a permanent crutch

FACT: Wood, Bruner, & Ross (1976, *Journal of Child Psychology and Psychiatry*, *17*(2), 89–100, doi:10.1111/j.1469-7610.1976.tb00381.x) — characterize tutoring as a *scaffolding* process: the tutor controls elements beyond the learner's capacity so the learner can complete what is within reach. Functions include recruitment, reduction of degrees of freedom, direction maintenance, marking critical features, frustration control, and demonstration.

FACT: Palincsar & Brown (1984, *Cognition and Instruction*, *1*(2), 117–175, doi:10.1207/s1532690xci0102_1) — reciprocal teaching: tutor and students take turns leading dialogue (summarize, question, clarify, predict); adult models then fades leadership; improves comprehension and transfer relative to typical classroom practice in reported studies. Collins et al. treat this as a canonical CA method stack for reading — turn-taking + fade of expert lead.

FACT (review grain): Dennen (2004, "Cognitive apprenticeship in educational practice," in *Handbook of Research on Educational Communications and Technology*) — synthesizes research on scaffolding, modeling, mentoring, and coaching as CA strategies; notes modeling/coaching/fading as predominant methods, with scaffolding as part of coaching; fading is gradual (hints become less frequent/detailed), not abrupt abandonment. Warns that implementations vary — label ≠ fidelity.

Bridge to SAFE-FADE / SAFE-HINT / SAFE-HELP: Worked-example fading and contingent peeks are *product instantiations* of scaffolding+fading. Cognitive apprenticeship adds the **sociology**: who leads the dialogue, when the expert models, when the learner must articulate. Reciprocal-style turn-taking on a hard item (student leads a "what do we know / what's the join?" beat) is a HITL technique, not a Discord forum.

Kill: Permanent scaffold as kindness brand; reciprocal-teaching costume without turn-taking.

Survive: Wood-function scaffolds with scheduled fade; student-led beats.

XCV.6 Boundaries — situated culture ≠ lore franchise; CA ≠ discovery

FACT / boundary (Brown et al., 1989): Situated cognition argues for authentic activity and culture of practice — not that any immersive story world equals apprenticeship. SAFE-STORYLOAD already budgets narrative load; CA does not license lore walls as "culture."

Boundary with SAFE-PF: Exploration in Collins's sense can include learner-set goals; Kapur PF is a *specific* generate→consolidate design. Do not rename every explore beat as PF, and do not run invent-first when the playbook called for modeling first on thin priors.

Boundary with Kirschner/Sweller/Clark (2006) critique (already in XCIV): Cognitive apprenticeship is *not* minimal guidance forever. Modeling and scaffolding are strong guidance; fading withdraws them as competence grows. Branding "apprenticeship" while shipping unguided stew fails both CA and the guidance literature.

Kill: Lore≡culture-of-practice; CA as discovery cosplay; never-guide brand.

Survive: Guided visibility → fade; authenticity = real problem work + audible thinking, not franchise trivia.

XCV.7 Product surface — SAFE-APPRENTICE claim contract

Surface	Required behavior	Banned substitute
Tutor playbook	Phase checklist: model (short) → coach → fade; articulation required	"Be warm / Socratic" vibes only
Live session QA	Log `ca_phase`; sample for modeling-without-attempt / never-fade	Talk-% only QA
Coach / Solver	Phase chip aligned with fade×hint; SE before wrap	AI monologue labeled "modeling"
Map brief	Heuristic/control focus for the target join	Tip flood / lore dump

AAR / Notes	Reflection on *student* process vs canonical	Tip-of-the-day dump
Parent copy	“We show how experts think, then you take the wheel”	Master/guild / hours-under-tutor
Marketing	Method stack + solo proof	“Cognitive apprenticeship™” without phases
Analytics	Phase dwell, fade events, articulation rate, solo transfer	Apprenticeship Score™ / modeling-minutes NS

Competitive foil: ChatGPT = fluent modeling without coaching/fade/articulation. Khan = content without audible control craft. Duo = practice without expert thinking made visible. MindCraft = **visible craft + transferred responsibility** under FEI.

XCV.8 Doctrine — SAFE-APPRENTICE (provisional)

1. **Visibility first** — model *control/heuristic* moves, not only steps (Collins; Schoenfeld).
2. **Performance under coaching** — most minutes = student attempt with in-situ support (coaching ≠ lecture).
3. **Scaffold then fade** — Wood functions + SAFE-FADE/HINT; no permanent crutch.
4. **Articulation required** — student-generated why before bottom-out (SAFE-EXPLAIN / SE).
5. **Reflection on their work** — compare learner process to canonical (AAR/ANNOT light).
6. **Exploration gated** — challenge/PF only when priors/affect allow (SAFE-PF / RETRIEVE).
7. **No costume metrics** — ban Apprenticeship Score™, guild cosplay, modeling-minutes NS.
8. **Shared language** — human tutor and AI coach use the same phase labels; fidelity > brand word.

Confidence: High that Collins et al. (1989/1991) define a method stack (model/coach/scaffold/fade + articulate/reflect/explore) aimed at making cognition visible. High that Brown et al. (1989) situate CA as an alternative to inert didactic knowledge. High that Schoenfeld (1985) supplies a math-relevant control/heuristic/belief frame used as a CA exemplar in that literature. High that Wood et al. (1976) and Palincsar & Brown (1984) specify scaffolding and reciprocal fade-of-leadership mechanisms. Medium that MindCraft HITL can hit phase fidelity at scale without over-scripting warmth away — run APPRENTICE-* with QA sampling. Medium that AI coach “modeling” can externalize control without becoming monologue — co-test with SAFE-EXPLAIN length caps.

XCV.9 Experiments

ID	Question	Design	Primary
APPRENTICE-1	Playbook with explicit model→coach→fade+ articulate vs explain-pour control	A/B sessions	`solo_transfer_pass`; articulation rubric

APPRENTICE-2	Short control-talk model vs steps-only model before same coaching	A/B	Near transfer; student control moves coded
APPRENTICE-3	Within-session fade schedule vs constant high scaffold	A/B	Solo transfer; peek binge; dependency
APPRENTICE-4	Reciprocal student-led beat (1 item) vs tutor-led whole session	A/B	Construction share; transfer
APPRENTICE-5	Parent CBC: “show thinking then you take over” vs “master tutor explains” vs “AI always models”	CBC	WTP; trust (SAFE-WTP)
APPRENTICE-QUAL	10 tutors + 10 Maya: phenomenology of phase checklist vs vibe coaching	Qual	Fidelity codebook; costume markers

Falsifier: Explain-pour equals or beats model→coach→fade on solo transfer *and* articulation for our secondary cohort → keep CA language internal; do not market apprenticeship.

Falsifier: Steps-only model equals control-talk model → demote control-talk chrome; still ban monologue-as-modeling.

Falsifier: Parents prefer master-explains CBC and reject take-the-wheel → rewrite copy before surrendering to explain-first brand.

Pre-register: APPRENTICE- *before any “cognitive apprenticeship / guild / master tutor / Apprenticeship Score™” campaign (SAFE-LABMETA). Do not steal FADE-*/HINT-*/TALK-*/PF-* arms — those own example stage, peek pricing, construction telemetry, and invent-first sequencing; this family owns **playbook phase fidelity** across the stack.

XCV.10 So what for MindCraft commercially

- **Copy:** “We make the thinking visible — then you take the wheel.” Never “study under a master” guild cosplay or “watch the AI solve everything.”
- **Product:** `ca_phase` on session/coach events; playbook checklist; articulation gate before hard reveal; fade schedule linked to SAFE-FADE/HINT.
- **Positioning:** Against ChatGPT monologue-as-tutor *and* content libraries without craft visibility; for inspectable apprenticeship *methods*, not costumes.
- **Metric:** solo transfer + articulation quality + fade events — demote modeling-minutes / Apprenticeship Score™ / talk-% alone.
- **Kill list:** Guild cosplay; modeling≡lecture; never-fade; talk-%≡CA; Apprenticeship Score™; Discord≡apprenticeship; reciprocal costume without turns; CA ads without phase logs.

• **Growth:** Tutor ops hire/train on phase fidelity (feeds SAFE-HITL / WORKFORCE); parent decks sell take-the-wheel honesty.

• **Vision:** Maya hears how a competent solver *manages* stuckness — then does it herself — so identity shifts from "I need someone to show me" to "I can run the craft."

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Part XCVI — Attention Residue & Device Distraction in Practice

Chapter status: Living evidence + practice UX brief — Researcher tick 2026-08-07

Primary question: How should MindCraft treat **attention residue** and **device/media distraction** during practice and tutoring so FEI attempts stay cognitively available — without Focus Score™ theater, phone-shame branding, or pretending "mobile-first" erases switching costs?

Owners: Practice UX · Coach / session design · Parent honesty · Brand · Red Team

Commercial job: Ship a **SAFE-ATTN** doctrine: protect working memory for the attempt; make task-switch tax inspectable; offer opt-in focus modes and tech breaks — never monetize multitasking heroics or sell distraction-free as guaranteed ACT points.

Builds on: Parts XXVI (CLT), XLI (DD × anxiety), XLVII (sleep/stress), LXIII (exam pressure), LXXXV (story-load), XCIII (retrieval failure modes), XCV (apprenticeship phases need available attention). Sibling surfaces: habit cues (XLIII) without streak-as-focus; privacy (LXIX) if we ever log "distraction" signals.

XCVI.1 Why this chapter exists

Brand temptation: “Our app is so engaging students never look away.” Competitor temptation: “Practice anywhere — phone in hand, tabs open, notifications on.” Both ignore the load-bearing constraint: mathematical thinking is a limited-capacity resource, and unfinished or competing cognitions steal it.

Sophie Leroy’s *attention residue* names the leftover cognitions about Task A that persist after switching to Task B. Smartphones add a second tax: even *unused* phones can drain available capacity (Ward et al.). Media multitasking while studying is common and reliably associated with worse academic outcomes in reviews and meta-analyses — not because teens are morally weak, but because switching and divided attention interfere with encoding and control.

MindCraft already fights fake struggle theater (SAFE-RETRIEVE), lore load (SAFE-STORYLOAD), and monologue token tax (SAFE-EXPLAIN). The open question: do we **design for residue and device presence**, or ship notification “engagement” and call interrupted attempts grit?

FOUNDER BELIEF under audit: Fewer mid-attempt switches and lower phone salience will raise `solo_transfer_pass` and cut freeze/binge-peek vs notification-on defaults — without a Focus Score™.

Claims we refuse as doctrine:

1. Focus Score™ / Deep Work Score™ as North Star.
2. Phone-shame or “kids these days” moralizing as pedagogy.
3. “Mobile-first multitasking is fine for math” without residue accounting.
4. Always-on push notifications as engagement science.
5. Streak / DAU protection via interrupt-driven re-engagement during an attempt.
6. Presence of phone ≡ proof of modern learning.
7. Focus-mode ads that promise guaranteed ACT points or “cured distraction.”
8. Covert keystroke/gaze distraction surveillance without SAFE-PRIVACY gates.

XCVI.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Attention residue	Cognitions about prior Task A persist into Task B	Mid-mission chat ping; unfinished homework tab; tutor Slack	Mere time-on-app
Task-switch cost	Performance drop when changing task sets	Alt-tab mid-item; notification → return	Deliberate interleaving of *problem types* (Part XXXIX)
Media multitasking	Concurrent media streams / off-task tech while studying	Phone + practice; social tab + Solver	Dual coding *within* one math item (Part XCVII queue)
Phone presence / brain drain	Own phone nearby reduces available WMC/Gf even unused	Phone on desk during timed set	Using phone *as* the practice device intentionally

SAFE-ATTN	When/how to protect attempt windows	This chapter	Productivity-guru cosplay

Operational definition (HYPOTHESIS): A mission is *SAFE-ATTN compliant* when (a) the product does not fire engagement pushes during an active attempt, (b) optional focus mode suppresses non-essential chrome and suggests phone-away for desk sessions, (c) tech-break / resume affordances exist so unfinished social cognitions can close cleanly (Leroy completion logic), (d) analytics separate `attempt_interrupted` from productive struggle modes (SAFE-RETRIEVE), (e) outcomes judged on transfer and construction — not Focus Score™ or minutes-without-switch vanity, and (f) any distraction telemetry is opt-in, non-biometric, and privacy-bound.

XCVI.3 Attention residue — unfinished cognitions steal the next task

FACT: Leroy (2009, *Organizational Behavior and Human Decision Processes*, *109*(2), 168–181, doi:10.1016/j.obhdp.2009.04.002) — defines *attention residue* as the persistence of cognitive activity about Task A after one has stopped working on Task A and is performing Task B. Two experiments show people struggle to fully transition attention away from unfinished prior work; subsequent Task B performance suffers. Finishing Task A before switching helps, but is not always enough: time pressure while finishing the prior task aids disengagement and contributes to higher subsequent performance. The mechanism is not “lazy switching” folklore — it is residual cognitions occupying capacity needed for the next task.

Applied (HYPOTHESIS): Mid-attempt notifications, coach chat bubbles, and “come back!” streaks create *unfinished social/app tasks* that leave residue on the math item. Product should prefer **close-the-loop** patterns: finish or park the interrupt (snooze with explicit resume) before demanding germane load on the problem.

Kill: Interrupt-driven engagement during active attempts as “habit science.”

Survive: Protect attempt windows; allow clean completion/park of interrupts.

XCVI.4 Device distraction & media multitasking — field pattern, not vibes

FACT: Rosen, Carrier, & Cheever (2013, *Computers in Human Behavior*, *29*(3), 948–958, doi:10.1016/j.chb.2012.12.001) — observed 263 middle-school, high-school, and university students studying 15 minutes at home. Participants averaged **less than six minutes** on task before switching, most often to technological distractions (social media, texting) or task-switching preference. Students who accessed Facebook had lower GPAs than those who avoided it; higher use of study strategies predicted more on-task behavior. Authors discuss short “technology breaks” and metacognitive strategies about when interruptions hurt — not total abstinence theater.

FACT: May & Elder (2018, *International Journal of Educational Technology in Higher Education*, *15*, 13, doi:10.1186/s41239-018-0096-z) — literature review: media multitasking interferes with attention and working memory and is associated with worse GPA, test performance, recall, reading comprehension, note-taking, self-regulation, and efficiency, in class and while studying. Students often **misjudge** how much multitasking will cost them. Self-regulation support is a promising intervention direction.

FACT: Kates, Wu, & Coryn (2018, *Computers & Education*, *127*, 107–112, doi:10.1016/j.compedu.2018.08.012) — meta-analysis of mobile phone use (non-educational-improvement use) and academic outcomes over 2008–2017:

overall average effect $r = -0.162$ (95% CI –0.196 to –0.128) — a **small negative** association, not a cartoon catastrophe and not zero. Effect sizes vary by design and construct; do not market "phones destroy learning" as a 2-sigma claim.

Applied (HYPOTHESIS): MindCraft should treat off-task media as a **moderated tax** on FEI attempts: design for fewer switches and better metacognition, instrument interruption, and avoid both denial ("multitask freely") and moral panic ("ban phones or fail").

Kill: Phone-apocalypse marketing; "multitasking digital natives" as free pass.

Survive: Small-to-medium evidenced tax; self-regulation affordances; tech breaks.

XCVI.5 Mere presence — the phone can tax you without a tap

FACT: Ward, Duke, Gneezy, & Bos (2017, *Journal of the Association for Consumer Research*, *2*(2), 140–154, doi:10.1086/691462) — "brain drain" hypothesis: mere presence of one's own smartphone can occupy limited-capacity cognitive resources. Across experiments, available working memory capacity and fluid intelligence measures were worse when the phone was more salient (e.g., desk vs other room), even when participants were not using the phone and did not report thinking about it. Costs were highest for those highest in smartphone dependence.

Applied (FOUNDER BELIEF → testable): For desk/laptop practice and tutor sessions, default copy can invite "phone in another room / face-down out of reach" as a *capacity* tip — framed as working-memory hygiene, not character judgment. On phone-as-device sessions, the tradeoff flips: the practice surface *is* the phone — then minimize *competing* apps/notifications rather than pretending presence-drain vanishes.

Boundary: Ward et al. are lab cognitive-capacity tasks, not ACT RCTs. Do not claim "phone in bag = +N ACT points."

Kill: Guaranteed-score focus kits; shaming dependent users in parent dashboards.

Survive: Optional salience reduction; honesty that unused phones can still cost capacity.

XCVI.6 Cognitive control — heavy multitaskers are not automatically better switchers

FACT: Ophir, Nass, & Wagner (2009, *Proceedings of the National Academy of Sciences, 106*(37), 15583–15587, doi:10.1073/pnas.0903620106) — heavy media multitaskers (vs light) showed greater susceptibility to interference from irrelevant environmental stimuli and memory representations, and *worse* performance on a task-switching test — consistent with a breadth-biased control style that filters poorly. Association, not proven causation from multitasking volume alone; still kills the folk claim "kids who multitask all day are better at switching for homework."

Bridge to SAFE-RETRIEVE / SAFE-EXPLAIN: Blanking and monologue wraps already burn WM. Residue + media interference compound those failure modes. Do not label notification-induced stalls as productive struggle.

Kill: Multitasking-as-talent brand.

Survive: Filter-friendly session design; one primary cognitive thread during attempts.

XCVI.7 Product surface — SAFE-ATTN claim contract

Surface	Required behavior	Banned substitute
Practice attempt	No marketing/streak pushes mid-item; visible “attempt in progress” lock	DAU pings during problem
Focus mode (opt-in)	Suppress non-essential chrome; suggest phone-away for desk; timed tech break	Focus Score™; forced lockout shame
Interrupt handling	Snooze/park with resume; log `attempt_interrupted`	Silent tab-switch ignored in analytics
Tutor sessions	Brief “devices down for attempt windows”; tech break between items	Phone-confiscation cosplay; emotion-AI attention cops
Parent copy	“Deep attempts need quiet WM — here’s an optional focus tip”	Shame ranks for “phone addiction”; Anxiety Score™
Marketing	Capacity honesty; self-regulation aids	“Never distracted again” / guaranteed points
Analytics	Interrupt rate, resume latency, solo transfer	Focus Score™ / minutes-on-task NS alone
Privacy	No covert gaze/mic distraction scoring	Empathy-camera attention (SAFE-PRIVACY ban)

Competitive foil: Duo = streak/notification engagement that may interrupt encoding. ChatGPT = infinite tab switching without attempt protection. Khan = video + phone parallel common. MindCraft = **attempt windows protected for construction**, with optional focus hygiene — still FEI, not productivity-guru SKU.

XCVI.8 Doctrine — SAFE-ATTN (provisional)

1. **Residue is real** — unfinished cognitions tax Task B (Leroy); close or park interrupts before demanding the join.
2. **Media multitasking costs learning on average** — reviews/meta show negative academic associations (May & Elder; Kates et al.); size is often small-to-moderate, not apocalyptic.
3. **Presence can drain** — unused phones may reduce available capacity (Ward et al.); offer desk hygiene tips without shame.
4. **Heavy multitasking ≠ better switching** — Ophir et al. association cuts against “digital native switcher” ads.
5. **Protect FEI attempt windows** — no engagement pushes mid-item; separate interrupt telemetry from productive struggle.
6. **Tech breaks > total war** — Rosen et al. style short breaks + metacognition beat abstinence theater.
7. **No Focus Score™ NS** — prove with solo transfer / construction, not vanity focus minutes.
8. **Privacy bound** — no biometric attention policing; opt-in only for any distraction signals.

Confidence: High that Leroy (2009) establishes attention residue as a measured Task A→B performance mechanism in experiments. High that Ward et al. (2017) show mere phone presence can reduce available WMC/Gf in lab tasks. High that May & Elder (2018) and Kates et al. (2018) document negative media-multitasking / phone-use associations with academic outcomes at review/meta grain. High that Rosen et al. (2013) observe frequent short on-task bouts and tech-driven switches in naturalistic studying. Medium that Ophir et al. (2009) generalize to MindCraft HS cohorts — association study, replication debates exist in the broader MMI literature; use as anti-folk-claim, not as trait diagnosis. Medium that product focus-mode + push suppression will move FEI metrics — run ATTN-* before Focus brand campaigns.

XCVI.9 Experiments

ID	Question	Design	Primary
ATTN-1	Mid-attempt push suppression vs default engagement pushes	A/B	`solo_transfer_pass`; interrupt rate; peek binge
ATTN-2	Opt-in focus mode (chrome quiet + phone-away tip) vs control	A/B	Transfer; resume latency; self-report residue
ATTN-3	Scheduled tech-break between items vs no break / vs mid-item free switch	A/B	On-task bout length; transfer
ATTN-4	Desk session: phone other-room tip vs phone-on-desk (observational + randomized tip)	A/B / hybrid	Capacity proxies; FEI events
ATTN-5	Parent CBC: “protect attempt windows” vs “streaks keep them coming back” vs “Focus Score™”	CBC	WTP; trust (SAFE-WTP)
ATTN-QUAL	10 Maya: phenomenology of residue after chat/social interrupt during a hard item	Qual	Interrupt codebook; shame markers

Falsifier: Push-on during attempts equals or beats push-off on solo transfer *and* retention for secondary cohort → keep ATTN internal; do not market focus windows.

Falsifier: Focus mode harms anxious students (avoidance/shame) more than it helps transfer → demote forced focus; keep optional + anxiety-gated (XLI).

Falsifier: Parents prefer streak-interrupt CBC and reject attempt-protection copy → rewrite GTM before surrendering to notification-growth brand.

Pre-register: ATTN- *before any "Focus Score™ / deep work / phone-free guaranteed gains / distraction-cured" campaign (SAFE-LABMETA). Do not steal RETRIEVE-*/HABIT-*/PRIV-* arms — those own struggle-mode taxonomy, cue design, and affect telemetry ethics; this family owns **attempt-window protection and device/residue hygiene**.

XCVI.10 So what for MindCraft commercially

- **Copy:** "Hard math needs a clear head — we protect your attempt window." Never "multitask like a pro" or "we'll shame the phone out of you."
- **Product:** Mid-attempt push lock; opt-in focus mode; tech-break + resume; `attempt_interrupted` event distinct from freeze/TOT modes.
- **Positioning:** Against notification-growth edtech *and* productivity-guru Focus Score™ SKUs; for capacity-honest practice that still ships FEI proof.
- **Metric:** solo transfer + interrupt/resume quality — demote Focus Score™ / raw minutes-on-task / streak pings during items.
- **Kill list:** Focus Score™ NS; phone-shame; multitasking-as-talent; mid-attempt DAU pushes; presence≡modern; guaranteed ACT from focus kits; covert attention surveillance.
- **Growth:** Parent decks sell attempt hygiene as care, not control; tutor ops brief device norms between items.
- **Vision:** Maya can finish a join without half her working memory still arguing with a group chat — so evidence of competence can land as identity, not as another interrupted almost.

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Part XCVII — Dual Coding & Diagram FormatId Load

Chapter status: Living evidence + FormatId / figure UX brief — Researcher tick 2026-08-07

Primary question: When do words+pictures (dual coding / multimedia principles) help MindCraft FormatId surfaces (`diagram`, `coordinate_graph`, `number_line`, …), and when do figures add **extraneous load**, split attention, or learning-styles costume — so we ship conversion competence without “visual learning” theater?

Owners: Practice / Solver figure UX · Bank & generation (GENQ figures) · Map format CTAs · Brand · Red Team

Commercial job: Ship a **SAFE-DUAL** doctrine: structural dual-channel design under CLT budgets; diagrams earn their keep via integration and conversion evidence; never sell Dual Coding Score™, always-more-visuals, animation-as-default, or visual-learner meshing as science.

Builds on: Parts XXVI (CLT), LXXXII (SAFE-FORMAT), LXXXV (SAFE-STORYLOAD), XCII (SAFE-EXPLAIN), LXXII (SAFE-GENQ). Sibling: XCVI (device residue ≠ pictorial channel). Product seams: `FormatId`, format-tagged bank, figure/render specs, coach wrap next to stems.

XCVII.1 Why this chapter exists

Brand temptation: “We use dual coding — words and pictures — so students learn twice as deep.” Competitor temptation: decorate every stem with stock art, autoplay animations, or a “visual learner” quiz. Both confuse **Paivio/Mayer dual-channel design** with **pretty multimedia** and **learning-styles meshing** (already banned under SAFE-FORMAT).

MindCraft’s format axis already treats vessels as diagnosis objects. Open load question: a `diagram` FormatId item can be the *target competence* (read/produce the vessel) or a *coach scaffold* (words+figure explaining a symbolic item). Those jobs have different design rules. Stacking narration + on-screen text + busy figure can *hurt*. Separating legend far from the figure forces split attention. Decorative story chrome next to a graph re-imports SAFE-STORYLOAD failures.

FOUNDER BELIEF under audit: Contiguous, coherent, structurally labeled words+diagrams (plus conversion practice across FormatIds) will raise `format_hop_solo_transfer_pass` vs decoration-heavy or styles-matched “visual track” defaults — without a Dual Coding Score™.

Claims we refuse as doctrine:

1. Dual Coding Score™ / Visual Engagement Score™ as North Star.
2. Always-add-a-picture ≡ dual coding science.
3. Decorative art / franchise chrome ≡ FormatId coverage or learning.
4. Learning-styles meshing (“visual learners get diagrams”) as product law.
5. Animation/video always better than static annotated diagrams.
6. Stacked animation + narration + on-screen text as “thorough dual coding.”
7. Split text beside distant figure as “more information.”
8. Unverified generated figures as diagnostic ground truth (SAFE-GENQ).
9. Guaranteed ACT points from “dual coding method” ads.

XCVII.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Dual coding (Paivio)	Separate verbal + nonverbal systems; referential links strengthen memory	Stem + structural figure; coach labels tied to parts	Learning-styles preference
CTML dual-channel	Limited pictorial + verbal channels; select–organize–integrate	Contiguous labels; modality-aware coach	Unlimited media stack
Multimedia principle	Words+pictures can beat words alone *when designed*	Dual-channel coach cards	Always-more-visuals
Split attention	Mentally integrating separated sources burns WM	Legend far from graph; footnote under fold	Deliberate conversion practice
Redundancy / coherence	Duplicate channels or seductive extras can hurt	Narration + full on-screen dump + figure	Thoroughness theater
FormatId vessel	Input representation as diagnosis object	`diagram` / `coordinate_graph` / …	Decoration
Conversion	Translate across registers (Duval; SAFE-FORMAT)	Diagram↔equation hops	Treatment within one vessel
SAFE-DUAL	When/how figures load under dual-channel + CLT	This chapter	Dual-coding costume ads

Operational definition (HYPOTHESIS): A figure treatment is *SAFE-DUAL compliant* when (a) the pictorial element is **structurally necessary** for the learning object (vessel reading, conversion, or constrained MER function — Ainsworth), not decorative, (b) corresponding words and picture parts are **spatially/temporally contiguous**, (c) extraneous words/sounds/chrome are cut (coherence), (d) redundant on-screen text is not stacked on narration+animation without cause, (e) outcomes are judged on retention/transfer and `format_hop_solo_transfer_pass` — not Dual Coding Score™ or "visual minutes," and (f) any generated figure used diagnostically is verify-gated (SAFE-GENQ).

XCVII.3 Dual coding is a memory architecture — not a marketing sticker

FACT: Paivio (1986, *Mental Representations: A Dual Coding Approach*, Oxford University Press) — performance on memory and related tasks is mediated by interconnected **verbal** and **nonverbal (imagery)** systems; concrete, imageable materials and referential links between systems support richer encoding than verbal-only routes alone. Dual coding is a theory of representation, not a license to wallpaper every worksheet.

FACT: Clark & Paivio (1991, *Educational Psychology Review*, *3*(3), 149–210, doi:10.1007/BF01320076) — review dual coding implications for education: concreteness, imagery, and dual verbal–nonverbal routes matter for learning design; still not equivalent to diagnosing "visual learners" and matching preference.

Applied (HYPOTHESIS): MindCraft should use dual coding as a **design constraint** (referential links between words and structural pictures) when the math object is spatial/graphic — and refuse dual-coding *brand* when the "picture" is unrelated atmosphere.

Kill: "Science-backed dual coding" ads that mean stock illustration.

Survive: Referential word↔structure links on FormatId surfaces that require them.

XCVII.4 Multimedia learning: limited channels, active integration

FACT: Mayer & Moreno (2003, *Educational Psychologist*, *38*(1), 43–52, doi:10.1207/S15326985EP3801_6) — cognitive theory of multimedia learning: dual-channel, limited-capacity, and active-processing assumptions; overload scenarios and theory-based load-reduction methods. Cognitive load is central — more media ≠ more learning.

FACT: Moreno & Mayer (1999, *Journal of Educational Psychology*, *91*(2), 358–368, doi:10.1037/0022-0663.91.2.358) — modality and contiguity: better learning when corresponding words and pictures are coordinated. Do not treat "add text everywhere" as dual coding.

FACT: Mayer, Heiser, & Lonn (2001, *Journal of Educational Psychology*, *93*(1), 187–198, doi:10.1037/0022-0663.93.1.187) — extraneous/seductive material can impair learning (**coherence**). Seductive details are not enrichment.

Bridge to SAFE-STORYLOAD / SAFE-EXPLAIN: Lore beside a graph and long monologue wraps compete for the same limited channels. Coach text next to a figure must be short, principle-linked, and contiguous.

Kill: Seductive detail as engagement; monologue+busy figure as thoroughness.

Survive: Weeded, contiguous multimedia under capacity limits.

XCVII.5 Split attention — separated sources tax integration

FACT: Chandler & Sweller (1991, *Cognition and Instruction*, *8*(4), 293–332, doi:10.1207/s1532690xci0804_2) — when learners must mentally integrate mutually referring sources (diagram + separate text), **split-source** formats can impose heavy extraneous load; **physically integrating** related information often improves learning when integration is necessary. Nonessential add-ons can hurt even when "integrated."

Applied (HYPOTHESIS): Keep axis labels, key lengths, and callouts **on or adjacent to** the figure; avoid legend-in-sidebar / explanation-three-scrolls-away for novices. Coach cards that reference "the red segment" must point or highlight.

Kill: Split-pane "more information density" as pedagogy.

Survive: Integrated labeling; necessity test before adding commentary layers.

XCVII.6 Diagrams are active constructions — not free downloads

FACT: Hegarty (1992, *Journal of Experimental Psychology: Learning, Memory, and Cognition, 18*(5), 1084–1102, doi:10.1037/0278-7393.18.5.1084) — *mental animation*: inferring kinematics from static mechanical diagrams is an active, capacity-limited process (piecemeal causal sequence); harder against causal order. Diagrams do not pour understanding in; learners build models.

FACT: Hegarty, Kriz, & Cate (2003, *Cognition and Instruction*, *21*(4), 209–249, doi:10.1207/s1532690xci2104_1) — mental vs external animation in understanding mechanical systems; spatial ability and knowledge shape what is learned from displays; comprehension is jointly determined by display and learner. Animations are not automatically superior to static displays that support mental animation.

FACT: Mayer, Hegarty, Mayer, & Campbell (2005, *Journal of Experimental Psychology: Applied, 11*(4), 256–265, doi:10.1037/1076-898X.11.4.256) — across four topics, paper-based **static diagrams + text** often matched or beat computer **narrated animations** on retention/transfer comparisons — supporting a *static media* hypothesis: static annotated illustrations can reduce extraneous processing and foster germane processing relative to narrated animation under those materials.

Applied (FOUNDER BELIEF → testable): Default FormatId scaffolds for geometry/graph items should prefer **static, annotated, contiguous** figures; treat animation as an opt-in for specific dynamic systems, not brand "modern = moving." Coordinate-graph FormatId competence is reading/producing the static vessel ACT uses — not watching a fly-through.

Kill: Animation-first Solver / "immersive diagram video" as default dual coding.

Survive: Static annotated figures; animation only when motion is the learning object and load is managed.

XCVII.7 Learning styles meshing is not dual coding

FACT: Pashler, McDaniel, Rohrer, & Bjork (2008, *Psychological Science in the Public Interest*, *9*(3), 105–119, doi:10.1111/j.1539-6053.2009.01038.x) — review finds inadequate evidence for the **meshing hypothesis** (match instruction to preferred style, e.g., visual vs verbal learners). Preferring pictures is not the same as needing pictures for a content-appropriate dual-channel design.

Bridge to SAFE-FORMAT: Format Personality™ / visual-learner tracks remain killed. Dual coding says: for *this math object*, use words and structural pictures under capacity rules — not "Maya self-reports visual, so only diagrams forever."

Kill: Style quiz → vessel lock.

Survive: Task-demand FormatId diagnosis + conversion practice.

XCVII.8 Product surface — SAFE-DUAL claim contract

Surface	Required behavior	Banned substitute
Diagram / graph stems	Structural figure; labels contiguous; no decorative chrome in critical area	Stock art “for dual coding”
Coach wrap beside figure	Short principle/misconception; points to parts; fades with expertise (SAFE-EXPLAIN)	Monologue + figure dump
Conversion practice	Explicit diagram↔symbolic hops with supports (SAFE-FORMAT)	Re-teach topic in strong vessel only
Animation	Opt-in; motion must be the object; segment + weed	Autoplay narrated video default
Story wrap	Structure-encoding only; lore off the critical path (SAFE-STORYLOAD)	Franchise sticker on axes
Generation	Verify figure+key before diagnostic use (SAFE-GENQ)	Unchecked LLM diagrams as ground truth
Marketing	“We design words+figures so the structure is readable”	Dual Coding Score™ / visual-learner SKU / guaranteed points
Analytics	`format_hop_solo_transfer_pass`; figure-load A/Bs	Visual Engagement Score™ / minutes-with-picture NS

Competitive foil: Brilliant = delight visuals that may under-specify conversion. Khan = video+worksheet split attention common. ChatGPT = text-only fluency without vessel diagnosis. Duo = decoration without dual-channel discipline. MindCraft = **FormatId honesty + contiguous structural dual-channel design** — still FEI, not multimedia theater.

XCVII.9 Doctrine — SAFE-DUAL (provisional)

1. **Dual coding ≠ decoration** — Paivio/Clark referential links require structural nonverbal content, not wallpaper.
2. **Channels are limited** — Mayer/Moreno overload rules; weed seductive detail (coherence).
3. **Integrate or pay the tax** — Chandler/Sweller split attention; keep words and picture parts contiguous when mutual reference is required.
4. **Diagrams demand construction** — Hegarty mental animation; do not treat viewing as mastery.
5. **Static annotated often wins** — Mayer et al. (2005) static media results; animation is conditional, not default modernity.
6. **No styles meshing** — Pashler et al.; task demand + FormatId gaps, not preference quizzes.
7. **Conversion remains the object** — SAFE-FORMAT: dual-channel scaffolds serve hops, not single-vessel comfort forever.
8. **No Dual Coding Score™** — prove with transfer and format hops; GENQ figures verify-gated.

Confidence: High that Paivio (1986) and Clark & Paivio (1991) establish dual verbal–nonverbal systems as a serious framework. High that Mayer & Moreno (2003) and Moreno & Mayer (1999) ground limited-capacity dual-channel principles. High that Mayer, Heiser, & Lonn (2001) show extraneous material can hurt. High that Chandler & Sweller (1991) demonstrate split-attention effects when sources must combine. High that Hegarty (1992) and Hegarty et al. (2003) show diagram comprehension is active and capacity-/knowledge-sensitive. High that Mayer et al. (2005) support static annotated media vs narrated animation on several science topics — **medium** generalization to ACT math FormatIds (run DUAL-*). High that Pashler et al. (2008) undermine learning-styles meshing as product law. Medium that contiguous-figure defaults will move `format_hop_solo_transfer_pass` — instrument before dual-coding brand campaigns.

XCVII.10 Experiments

ID	Question	Design	Primary
DUAL-1	Contiguous on-figure labels vs split legend/sidebar	A/B	Transfer; time-to-first-correct; peek binge
DUAL-2	Structural figure + short coach vs decorative figure + same coach vs text-only	A/B	Retention; `format_hop_solo_transfer_pass`
DUAL-3	Static annotated diagram vs narrated animation for same geometry/graph lesson	A/B	Transfer; extraneous-load self-report; interrupt rate
DUAL-4	Coherence weed (no seductive story chrome on figure) vs chrome-on	A/B	Transfer; story-trivia lure errors (SAFE-STORYLOAD)
DUAL-5	Parent CBC: "contiguous dual-channel figures" vs "visual learner dual coding" vs Dual Coding Score™	CBC	WTP; trust (SAFE-WTP)
DUAL-QUAL	10 Maya: where figures help vs overwhelm on ACT-like diagram items	Qual	Load codebook; conversion language

Falsifier: Decorative ≈ structural on format-hop transfer → demote figure strictness; keep FORMAT diagnosis.

Falsifier: Narrated animation reliably beats static annotated on ACT vessels without peek/avoidance → allow animation for those families only.

Falsifier: Parents prefer visual-learner CBC over contiguity copy → rewrite GTM; do not ship styles meshing.

Pre-register: DUAL- *before dual-coding / visual-learner / Dual Coding Score™ / animation-first / multimedia ACT-guarantee campaigns (SAFE-LABMETA). Do not steal FORMAT-*/STORYLOAD-/*EXPLAIN-*/GENQ-/*ATTN-* arms.

XCVII.11 So what for MindCraft commercially

- **Copy:** "Figures that carry structure — labels where you look." Never "dual coding for visual learners" or Dual Coding Score™.
- **Product:** Contiguous structural diagrams; weeded chrome; static-first; conversion hops; verify-gated GENQ figures.
- **Positioning:** Against decoration-edtech and styles quizzes; for FormatId-honest dual-channel design under CLT.
- **Metric:** `format_hop_solo_transfer_pass` + figure A/Bs — demote visual-minutes / Dual Coding Score™.
- **Kill list:** Dual Coding Score™; always-add-picture; decoration≡learning; styles meshing; animation-default; redundancy stack; split legends; unverified figure diagnostics; ACT guarantees from multimedia ads.
- **Growth:** Parent decks show contiguous vs split figure hygiene, not "we're a visual platform."
- **Vision:** Maya can read the graph vessel and convert it to symbols without WM spent on chrome, distant legends, or a false "visual learner" identity that never leaves her comfort vessel.

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Part XCVIII — Metacognitive Monitoring in Gap-Scan

Chapter status: Living evidence + Diagnostic / gap-scan UX brief — Researcher tick 2026-08-07 (UTC hour 12 ≡ Red Team slot, but ch98 never written → prefer Researcher per rotation; researcher count since synthesizer v1.12 = 6 → Researcher)

Primary question: How should MindCraft treat *metacognitive monitoring* inside gap-scan and related diagnostics — as a Monitoring Score™ to gamify, a confidence vibe to inflate, or a **control signal** that classifies knowledge vs confident misconception vs uninformed miss and routes FEI without corrupting the graph?

Owners: Product (Diagnostic C4 / gap-scan / soft-wrong) · Engine (`evidence_update_policy` / confidence tiers) · Brand · Red Team

Commercial job: Ship a **SAFE-MONITOR** doctrine that densifies SAFE-CALIB (Part L) and SAFE-COLD (Part LXXXI): elicit the right judgment type, separate absolute vs relative accuracy, use certainty to sort misconception from lack-of-knowledge, refuse Monitoring Score™ / "raise metacognition" theater, and prove value with `confidence_miss_tier` + weakness stability — not Belief Score™.

Builds on: Parts L (SAFE-CALIB), XLIX (SAFE-MISCON), LXXXI (SAFE-COLD), XXXIII (AI overtrust as *system* miscalibration), XC (FEI instrumentation). Product seams: gap-scan confidence ratings, hide-correctness probes, `/seed-assessment`, `/record-outcomes`, Layer-4 event schemas.

XCVIII.1 Why this chapter exists

Part L already bans "raise confidence" as North Star and wires item-level confidence to feedback tiers. Gap-scan still tempts a second costume: **metacognition theater** — a Monitoring Score™, a "know thyself" splash, or treating a single easy/kinda/hard slider as proof of self-regulated learning.

MindCraft's commercial wedge is not that students *feel* more metacognitive. It is that the product **uses monitoring judgments as diagnostic fuel**: distinguishing confident wrong (misconception / hypercorrection path) from low-confidence wrong (uninformed / scaffold path), and refusing to mint mastery fireworks from either. Competitors can ship a confidence emoji; few ship monitoring → classification → graph policy → FEI.

FOUNDER BELIEF under audit: Honest gap-scan requires *elicited* monitoring under destaked conditions (C4), not louder certainty — and monitoring without control (no change to routing/updates) is empty UX.

Claims we refuse as doctrine:

1. Monitoring Score™ / Metacognition % as North Star or parent KPI.
2. "We teach metacognition" ads without judgment→control wiring.
3. Global easy/kinda/hard alone ≡ calibrated monitoring science.
4. Immediate post-item confidence ≡ delayed judgment-of-learning accuracy.
5. Confidence slider without miss-class routing (Foster CA-alone null — Part L).

6. Raise-metacognition / raise-confidence as identity transformation.
7. Dunning–Kruger shaming as brand voice (Part L wound).
8. Guaranteed ACT points from “metacognitive gap scan” packaging.

XCVIII.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Monitoring	Judging own knowledge/performance (Nelson & Narens)	Item confidence; concept ease ratings	Identity survey
Control	Acting on those judgments (study, skip, restudy)	Soft-wrong route; challenge unlock; graph weight	Decorative slider
EOL / JOL / FOK / RCJ	Ease-of-learning; judgment of learning; feeling-of-knowing; retrospective confidence	Gap-scan ease ≈ EOL-ish; post-answer surety ≈ RCJ; restudy prompts ≈ JOL	One fused “metacog” number
Absolute accuracy	Match of mean confidence to mean accuracy	Bias / overconfidence index	Discrimination
Relative accuracy	Rank-order: higher conf → more likely correct	Resolution / Gamma-style discrimination	Calibration % vanity
CRI / three-tier logic	Certainty separates misconception vs don’t-know	High-conf wrong vs low-conf wrong classes	Always-reveal keys
SAFE-MONITOR	Gap-scan monitoring doctrine under SAFE-CALIB	This chapter	Metacognition costume ads

Operational definition (HYPOTHESIS): Gap-scan monitoring is *SAFE-MONITOR compliant* when (a) judgments are **item- or concept-grain**, not only global vibe, (b) judgment **type** is labeled (prospective ease vs retrospective confidence), (c) outcomes + confidence jointly classify miss types for routing and graph updates, (d) C4 hide-correctness prevents answer-hunting that collapses monitoring into key-chasing, (e) marketing never sells Monitoring Score™, and (f) success is measured by weakness stability, miss-class hit rate, and later FEI — not by rising self-rated “metacognition.”

XCVIII.3 Monitoring and control are a system — not a sticker

FACT: Nelson & Narens (1990, in Bower, *The Psychology of Learning and Motivation*, Vol. 26, Academic Press) — metamemory framework separates **monitoring** (EOL, JOL, FOK, retrospective confidence) from **control** (allocation of study, termination, strategy). Judgment types are not highly interchangeable; they can index different aspects of memory.

Product translation: Gap-scan's concept-level easy/kinda/hard is closer to a coarse **EOL / expected competence** signal for seeding (`/seed-assessment`). Practice soft-wrong "how sure?" is closer to **retrospective confidence**. Do not market them as the same "metacognition feature," and do not assume one slider fixes both.

Kill: One Metacognition Score™ merging EOL + RCJ + identity Likert.

Survive: Named judgment types → named control actions.

XCVIII.4 Delayed judgments beat fluency illusions (when the object is future recall)

FACT: Nelson & Dunlosky (1991, *Psychological Science*, *2*(4), 267–270, doi:10.1111/j.1467-9280.1991.tb00147.x) — judgments of learning made after a short delay (cue-only) predict later recall far better than immediate JOLs (**delayed-JOL effect**). Immediate JOLs often track short-term accessibility, not durable learning.

FACT: Rhodes & Tauber (2011, *Psychological Bulletin*, *137*(5), 791–813, doi:10.1037/a0021705) — meta-analysis: delaying JOLs yields large gains in **relative** accuracy (gamma; $g \approx 0.93$) and a small memorial benefit ($g \approx 0.08$). Delay is not magic for every product metric; it is a robust monitoring-accuracy lever for recall prediction.

Applied (HYPOTHESIS): MindCraft should not treat *immediate* post-answer confidence as a durable-mastery oracle. For restudy / spacing decisions (SAFE-SCHED / SAFE-FORGET), prefer delayed or cue-only "will you still get this?" prompts over fluency-right-after-feedback. For **miss classification** in gap-scan, retrospective confidence at answer time remains useful (CRI logic) even when it is a weak JOL for long-term retention.

Kill: Instant "I'm sure" after green check ≡ durable learning.

Survive: Separate diagnostic RCJ (classify miss) from delayed JOL (schedule return).

XCVIII.5 Math students mis-monitor — often toward overconfidence

FACT: Erickson & Heit (2015, *Frontiers in Psychology*, *6*, 742, doi:10.3389/fpsyg.2015.00742) — high-school and undergrad samples showed **overconfidence** predicting math performance, with greater overconfidence in math than in comparison subjects in their designs; anxiety and overconfidence can co-exist and both impair adaptive monitoring/control.

FACT: Lingel, Lenhart & Schneider (2019, *ZDM*, ERIC EJ1222655) — seventh-graders showed **pervasive overconfidence**; monitoring precision differed by scale format and prospective vs retrospective timing; absolute, relative, and diagnostic accuracy indices are not interchangeable.

FACT: Foster (2021/2022, *IJSME*, doi:10.1007/s10763-021-10207-9) — confidence assessment alone is **non-inferior** on attainment and **not** a proven attainment booster (Part L load-bearing null).

Commercial implication: Gap-scan that only asks "how confident are you in algebra?" without item probes will inherit overconfidence bias and mint false greens (anti-SAFE-COLD). Item/concept probes + hide-correctness + confidence×outcome classification are the wedge; "metacognitive onboarding" copy without that spine is costume.

Kill: Confidence-only placement ≡ diagnosis.

Survive: Monitoring judgments as *inputs* to classification under C4.

XCVIII.6 Certainty of response — misconception vs don't-know

FACT: Hasan, Bagayoko & Kelley (1999, *Physics Education, 34*(5), 294–299, doi:10.1088/0031-9120/34/5/304) — **Certainty of Response Index (CRI)** with MC items distinguishes misconceptions (wrong + high certainty) from lack of knowledge (wrong + low certainty) and from knowledgeable correct responses.

HYPOTHESIS for MindCraft: Gap-scan and FEI probes should treat high-confidence wrongs as **SAFE-MISCON / hypercorrection** candidates (Butterfield & Metcalfe lineage — Part L), and low-confidence wrongs as **uninformed / scaffold** candidates — never identical mastery deltas (Layer-4 `evidence_update_policy`).

Wound: CRI was developed in physics MC contexts; HS ACT items and Likert easy/kinda/hard are coarser. Survive the *logic*, not worksheet cosplay.

Kill: Equal graph update for every wrong.

Survive: Confidence×correctness classes as first-class events.

XCVIII.7 Practice-test feedback does not automatically fix miscalibration

FACT: Persistent miscalibration literature (e.g. introductory biology practice-test feedback studies — Niedjelski & colleagues line summarized in open reports such as PMC8442020 / *CBE—Life Sciences Education* calibration work) shows practice testing can improve average calibration for many students while **low performers often remain overconfident** and high performers can tip underconfident — feedback alone is incomplete.

Product translation: Do not promise "gap-scan calibrates you." Promise: gap-scan **surfaces** mismatch under destaked conditions, then practice springboards + FEI do the hard work. Parent copy: "we separate sure-and-wrong from unsure," not "we fix metacognition in one scan."

Kill: One diagnostic ⇒ calibrated learner ads.

Survive: Scan as evidence intake; calibration as longitudinal process (SAFE-CALIB + SAFE-LONGID).

XCVIII.8 Gap-scan product spine (SAFE-MONITOR)

1. **Label the judgment** — Concept ease (seed) ≠ item retrospective confidence (probe) ≠ delayed JOL (return schedule).
2. **Destake** — C4 hide-correctness during scan; no public shame ranks (SAFE-COMPARE).
3. **Classify** — correct/incorrect × low/med/high → guess / slip / confident misconception / underconfident correct.
4. **Control** — route: hypercorrection / contradicting-reason for high-conf miss; hug-then-dose for low-conf miss; challenge unlock for underconf correct (SAFE-CALIB stack).

5. **Weight evidence** — high-conf wrong ≠ low-conf wrong in mastery updates (engine).
6. **Refuse fireworks** — no day-one greens from ease ratings alone (SAFE-COLD).
7. **Measure** — `confidence_miss_tier`, bias Δ, `worstWeakness` stability, later `solo_transfer_pass` — never Monitoring Score™.

Marketing language that survives: "Know what you know," "sure-and-stuck vs guessing," "honest scan," "confidence that matches skill."

Marketing language that dies: "Metacognition Score™," "we raise metacognition," "calibrated in one quiz," "guaranteed ACT from monitoring method."

XCVIII.9 Competitive wedge (brief)

Khan (explain-first, weak miss classification), Duo (binary correctness + streaks), Brilliant (scaffold dopamine), ChatGPT tutors (fluent **system** certainty → student overtrust — Part XXXIII) all under-instrument *student* monitoring→control. MindCraft's L1-safe line: gap-scan is not a vibe check — it is **monitoring that changes what happens next**.

XCVIII.10 Claim ladder

Claim	Max ladder without new data
Monitoring ≠ control; judgment types dissociate (Nelson & Narens)	L1
Delayed JOLs improve relative accuracy vs immediate (Nelson & Dunlosky; Rhodes & Tauber)	L1
Math learners often overconfident; indices are plural (Erickson & Heit; Lingel et al.)	L1
CRI/confidence×accuracy separates misconception vs don't-know (Hasan et al.)	L1
CA alone does not raise attainment (Foster 2021) — monitoring must wire to control	L1
C4 + confidence×outcome improves weakness ranking vs confidence-only or reveal-as-you-go	L1 product bet (MONITOR-* / CAL-4)
Monitoring Score™ / one-scan calibration transforms identity	**Banned**

XCVIII.11 Experiments

ID	Question	Design	Primary

MONITOR-1	Item RCJ + outcome classification vs ease-only seed	2-arm	`worstWeakness` stability; next-session accuracy
MONITOR-2	High-conf-miss hypercorrection path vs uniform feedback	2-arm	delayed isomorphic correction; `retry_120s`
MONITOR-3	Delayed cue-only JOL for return scheduling vs immediate post-feedback “sure?”	2-arm	spacing adherence; delayed probe accuracy (SAFE-SCHED)
MONITOR-4	Hide-correctness + confidence vs reveal-as-you-go (shared CAL-4 / MIS-4)	2-arm	completion; affect; weakness stability
MONITOR-5	Parent CBC: “sure-and-wrong diagnosis” vs “Metacognition Score™” vs “raises confidence”	CBC	WTP; trust (SAFE-WTP)
MONITOR-QUAL	10 Maya: meaning of easy/kinda/hard vs item “how sure?”; shame after high-conf miss	Qual	codebook: threat vs curiosity

Falsifier: Ease-only seed matches item-classified routing on weakness stability → demote probe cost.

Falsifier: Uniform feedback ≥ tiered after high-conf miss → demote MONITOR/CAL tiering.

Falsifier: Parents prefer Metacognition Score™ CBC → rewrite GTM; do not ship score NS.

Pre-register: MONITOR- *before metacognition / Monitoring Score™ / one-scan-calibration / guaranteed-ACT-from-monitoring campaigns (SAFE-LABMETA). Do not steal CAL-*/COLD-/*MIS-*/SCHED-* arms — densify them.

XCVIII.12 So what for MindCraft commercially

- **Copy:** “Honest scan — sure-and-stuck vs guessing.” Never Monitoring Score™ or “we raise metacognition.”
- **Product:** Named judgments; C4; confidence×outcome classes; tiered control; no day-one fireworks from ease alone.
- **Positioning:** Against confidence-theater edtech and fluent AI certainty; for monitoring that changes routing and graph weight.
- **Metric:** `confidence_miss_tier` + weakness stability + FEI — demote Metacognition %.
- **Kill list:** Monitoring Score™; metacognition ads without control wiring; ease-only ≡ calibration science; immediate surety ≡ delayed JOL; equal wrong updates; one-scan calibration; ACT guarantees from monitoring

packaging.

• **Growth:** Parent decks show a 2×2 (correct/wrong × sure/unsure) with routing consequences, not a metacog leaderboard.

• **Vision:** Maya leaves gap-scan knowing the product heard *how sure she was*, not that she became a “metacognitive person” by finishing a quiz.

XCVIII.13 Doctrine — SAFE-MONITOR (provisional)

1. **Monitoring without control is costume** — Nelson & Narens: judgments must change study/routing/updates.
2. **Name the judgment** — EOL-ish ease ≠ RCJ ≠ delayed JOL.
3. **Math overconfidence is expected** — design for bias, do not shame (Erickson & Heit; Lingel).
4. **Certainty sorts miss types** — Hasan CRI logic → SAFE-MISCON vs uninformed paths.
5. **Delay when predicting future recall** — Nelson & Dunlosky; Rhodes & Tauber — not for every UI click.
6. **CA/slider alone ≠ attainment** — Foster null; wire to SAFE-CALIB.
7. **No Monitoring Score™** — prove with classification quality + FEI.
8. **Densifies SAFE-COLD** — humble seed + probes; no fireworks from self-ratings alone.

Confidence: High that Nelson & Narens (1990) establish monitoring/control and dissociable judgment types. High that Nelson & Dunlosky (1991) and Rhodes & Tauber (2011) establish delayed-JOL relative-accuracy benefits. High that Erickson & Heit (2015) and Lingel et al. (2019) document math monitoring/overconfidence patterns and measure plurality. High that Hasan et al. (1999) justify certainty×accuracy misconception sorting. High that Foster (2021) blocks CA-alone attainment claims (Part L). Medium that MindCraft’s specific gap-scan UX will move weakness stability — instrument MONITOR-* before metacognition brand campaigns.

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Part XCIX — Opportunity Cost of Tutor Minutes vs AI Scaffold

Chapter status: Living evidence + HITL economics / session-design brief — Researcher tick 2026-08-07 (UTC hour 18 ≡ Red Team slot, but ch99 never written → prefer Researcher per rotation; researcher count since synthesizer v1.12 = 7 → Researcher)

Primary question: When should MindCraft spend scarce *human tutor minutes* versus AI coach / Solver / Map scaffolds — and how do we sell that allocation honestly without "AI replaces tutors," "unlimited human," or hours-booked vanity?

Owners: Tutor ops · Product (session brief / coach routing) · Pricing / GTM · Brand · Red Team

Commercial job: Ship a **SAFE-ROI** doctrine that densifies SAFE-HITL / SAFE-TUTORGRAIN / SAFE-WORKFORCE / SAFE-HELP / SAFE-REPAIR: treat tutor time as a high-dosage *scarce input* with measurable opportunity cost; default AI for rehearsal and contingent soft scaffolds; reserve humans for diagnosis handoff, affective destake, bridge articulation, and repair escalation; prove value with FEI per tutor-minute — never Session ROI Score™ theater or "AI = free tutoring."

Builds on: Parts LXVIII (SAFE-HITL), LXXI (SAFE-TUTORGRAIN), LXXV (SAFE-WORKFORCE), XXXIII (AI trust / Bastani), LX (SAFE-REPAIR), LXXXIX (SAFE-HELP), XCI (SAFE-HINT), XC (SAFE-INSTRUMENT), LXVII (SAFE-ONTOLOGY). Product seams: Map brief before pour, coach cards, Solver fade, human escalate packets, session length / booking UX.

XCIX.1 Why this chapter exists

SAFE-HITL already says humans are operators on the Map, not warmth stickers. SAFE-TUTORGRAIN already says near-peer fidelity beats Ivy pour. Neither answers the commercial knife-fight every parent and district will force:

> "If the AI is so good, why am I paying for tutors?"

> "If tutors matter so much, why not unlimited human chat?"

Wrong answers kill the company twice: (a) **AI-replaces-tutors** cosplay that Bastani-style crutches and hallucination repair already falsify as learning strategy; (b) **human-minutes-as-unlimited entitlement** that bankrupts unit economics and drives pour-relapse under parent pressure (SAFE-WORKFORCE).

MindCraft's wedge is not cheaper chat or more booked hours. It is **honest allocation**: AI for high-frequency, low-stakes, Map-aware practice loops; humans for scarce moments where contingent prompting + diagnosis handoff + accountability change FEI — then fade back to solo.

FOUNDER BELIEF under audit: Tutor minutes are the dearest inventory in the product. Spending them on re-explaining what the Map already named, or on answer-dumping that Solver could gate, is the real product failure — not "not enough AI."

Claims we refuse as doctrine:

1. AI replaces tutors / "tutoring is free now."
2. Unlimited human minutes as brand promise or default SKU.

3. Hours booked / talk time / session count as North Star.
4. Session ROI Score™ as gamified ops KPI.
5. VanLehn ITS≈human as license for unguarded ChatGPT "tutors."
6. Bastani GPT Tutor safeguards ≡ FEI without solo transfer.
7. High-dosage school RCT ES as guaranteed ACT points from thin marketplace sessions.
8. Warmth, Ivy résumé, or "always-on AI" as substitutes for allocation honesty.

XCIX.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
High-dosage tutoring	Frequent, structured 1:1/small-group instruction	Near-peer sessions + Map brief	Occasional homework chat
Opportunity cost	Next-best use of scarce tutor minutes	FEI events per tutor-minute; deferred students	Raw cost/hour only
AI scaffold	Step/hint/coach without human	Solver fade, soft hints, Map cards	Unguarded answer dump
Guardrailed generative tutor	Pedagogy wrap that blocks crutches	SAFE-HINT / SAFE-FADE / SAFE-HELP stack	GPT Base / ChatGPT paste
Complementarity	Human + system beat either alone when roles differ	HITL on Map; AI for rehearsal	Human reading AI script
SAFE-ROI	Allocation doctrine for minutes vs AI	This chapter	Efficiency theater score

Operational definition (HYPOTHESIS): A MindCraft session stack is *ROI-honest* when (a) pre-session Map brief names the 1–3 human-worthy moves (bridge gap, soft-wrong prompt, exposure destake, repair escalate), (b) AI owns rehearsal / soft peeks / fade steps outside that list, (c) tutor talk favors construction prompts over re-lecture (SAFE-TALK), (d) marketing never claims AI≡tutor or unlimited humans, and (e) success is `solo_transfer_pass` and related FEI **per paid tutor-minute**, not hours booked or tokens generated.

XCIX.3 Tutoring works — and dosage / labor model set the cost curve

FACT: Nickow, Oreopoulos & Quan (2020, NBER WP 27476) — experimental PreK–12 tutoring meta-analysis: pooled ~**0.37 SD**; teacher/paraprofessional programs tend to beat nonprofessional/parent models; in-school tends to beat after-school.

FACT: Guryan et al. (2023, *AER*, *113*(3), 738–765, doi:10.1257/aer.20210434) — two Chicago RCTs of Saga-style **high-dosage** secondary math tutoring with paraprofessionals: about **0.18–0.40 SD** math gains, course-grade benefits, later-year persistence; cost held via paras (~$3.2–4.8k/student-year in related Saga/NBER WP 28531 writeups), not daily certified teachers.

FACT: Kraft & Falken (2021, *AERA Open*, doi:10.1177/23328584211042858) — individualized tutoring is powerful but **supply- and budget-constrained**; targeted national K–8 Title I scale scenarios are **billions**/year — software does not make human minutes free.

Wound: School-embedded high-frequency programs ≠ remote ACT marketplace 1–2×/week. Borrow dosage + structure + labor-arbitrage logic — not pooled ES as score insurance.

Kill: "Our tutors are 0.37σ." Kill "AI makes high-dosage free."

Survive: Humans work when dosage/structure are real — so do not waste the minutes you can afford.

XCIX.4 Cheap / thin human time is not automatic FEI

FACT: Kraft, List, Livingston & Sadoff (2022, *AEA Papers & Proceedings*, *112*, 614–618, doi:10.1257/pandp.20221038) — randomized pilot of **online college-volunteer** tutoring for middle schoolers: ITT estimates about **+0.07 SD math** and **+0.04 SD reading**, positive but **not statistically significant**; authors emphasize much **lower cost** than high-dosage in-person models and a **dosage** interpretation (more hours → larger effects) under COVID constraints.

Product translation: Near-peer online supply can be cost-efficient and still under-deliver if dosage, structure, and Map fidelity are thin. SAFE-TUTORGRAIN + SAFE-HITL are not optional garnish on "we have college tutors." Opportunity cost cuts both ways: expensive pour sessions *and* cheap unstructured chats can both waste the learning budget.

Kill: Marketplace volume / volunteer availability ≡ high-impact tutoring.

Survive: Fidelity × dosage × Map brief as the human product.

XCIX.5 ITS parity is not ChatGPT license

FACT: VanLehn (2011, *Educational Psychologist*, *46*(4), 197–221, doi:10.1080/00461520.2011.611369) — review finds adult human tutoring effect vs no tutoring around **d ≈ 0.79** and step-based intelligent tutoring systems around **d ≈ 0.76** — roughly **comparable** in that literature, undermining Bloom-style 2σ folklore and crude "humans always crush software" claims. Interaction appears to plateau around step-based granularity.

Wound: VanLehn's ITS are **constrained pedagogical systems**, not unconstrained LLM chat. Treating ChatGPT paste as "ITS-proven" is cite-wash.

HYPOTHESIS: MindCraft AI value is closest to VanLehn when it is **step/hint/Map-constrained** (SAFE-HINT / SAFE-FADE / SAFE-ONTOLOGY). Open generative dump is a different object — often a crutch.

Kill: "Science says AI = human tutor" as ChatGPT marketing.

Survive: Constrained scaffolds can carry large shares of practice load; humans still own accountability, diagnosis negotiation, and repair.

XCIX.6 Unguarded generative AI can buy practice scores and sell learning debt

FACT: Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122; earlier SSRN "Generative AI Can Harm Learning") — ~1,000 HS math students; GPT Base and GPT Tutor both **raise practice performance** sharply; when AI is removed, **GPT Base students do ~17% worse** than never-AI controls; GPT Tutor safeguards largely **mitigate the harm** but do not automatically produce large positive exam gains vs control. Students often use Base as a **solution crutch**; many do not perceive the learning loss.

Commercial implication: Selling "AI tutor that gets homework done" can raise short-run satisfaction while destroying solo transfer — the opposite of FEI / SAFE-PROOF. Human minutes spent *checking copied AI work* are negative ROI. Human minutes spent *enforcing attempt→soft scaffold→solo* are high ROI.

Kill: Practice-with-AI accuracy ≡ learning. Kill unlimited-AI-homework hero.

Survive: Guardrails + solo gates; humans as anti-crutch operators when AI fails or when dependency forms (SAFE-HELP / SAFE-REPAIR).

XCIX.7 When human minutes beat AI (allocation spine)

HYPOTHESIS — spend human minutes when ≥1 holds: Map diagnosis handoff (bridge/format/high-conf miss); affective destake / graded exposure (not therapy cosplay); construction prompts on soft-wrong (SAFE-TALK); repair escalate after AI fail (SAFE-REPAIR); witness solo transfer (SAFE-PROOF); fade / "your turn" gate (SAFE-FADE).

HYPOTHESIS — default AI / async Practice when: Spaced rehearsal on named weaknesses; soft contingent peeks after attempt; faded completion examples; FormatId conversion drills; parent MoC without live stalk.

Kill: Universal "human for every emotion" or "AI for every stuck." **Survive:** Pre-session **minute budget** tied to Map tasks.

XCIX.8 Unit economics without cruelty

FOUNDER BELIEF: Parents buy a bundle — trusted diagnosis + practice that doesn't fake mastery + humans for hard joins. Pure-AI underprices scarce input; pure-human overprices pour time.

SPECULATION: SKU ladder = Map/Practice/guardrailed coach → scheduled near-peer FEI sessions → expert escalate packets — never unlimited everything. SAFE-WTP CBC must test "tutors for joins; AI for reps" vs unlimited-AI vs unlimited-human.

Kill: Hours-booked discounting as the only lever. **Survive:** Outcome-linked session design + honest scarcity.

XCIX.9 SAFE-ROI product spine

1. **Minute budget on the brief** — Top 1–3 human moves named before start.
2. **AI owns the rest** — rehearsal / soft peeks / fade outside the budget.
3. **Anti-crutch gate** — no human or AI bottom-out before attempt (SAFE-HELP × Bastani).
4. **Repair path** — escalate packet, not infinite free human (SAFE-REPAIR).
5. **Instrument** — FEI per tutor-minute; pour-relapse; escalate precision; solo after AI.
6. **Refuse vanity** — no hours NS; no Session ROI Score™ that punishes destake time.
7. **Copy** — “Humans for the hard joins. AI for the reps.” Never tutoring-is-free / AI-replaces-tutors.

Survives: “Tutor time on what the Map says matters,” “reps without crutches,” “solo after the scaffold.”

Dies: “Unlimited AI tutor,” “unlimited human help,” “AI = private tutor,” “0.37σ guaranteed,” “hours = progress.”

XCIX.10 Competitive wedge (brief)

Khan/Brilliant: content + solo scaffolds (weak human dosage). Duo: streaks (weak FEI). ChatGPT tutors: fluent availability (Bastani crutch; SAFE-REPAIR debt). Marketplace experts: résumé + booked hours. MindCraft L1 line: **allocate scarce humans with Map honesty; constrained AI for volume without learning debt.**

XCIX.11 Claim ladder

Claim	Max ladder without new data
Experimental tutoring often large (~0.37 SD pooled; Nickow et al.)	L1
High-dosage para tutor models can move secondary math (Guryan et al. 2023)	L1
Scale is budget/supply constrained (Kraft & Falken 2021)	L1
Thin online volunteer dosage can be underpowered (Kraft et al. 2022)	L1
Step-based ITS ≈ human in VanLehn (2011) review — not LLM chat	L1
Unguarded GPT can harm post-AI exam performance (Bastani et al. 2025)	L1
Map-briefed minute budgets raise FEI per tutor-minute vs unstructured pour	L1 product bet (ROI-*)
Session ROI Score™ / AI-replaces-tutors transforms learning	**Banned**

XCIX.12 Experiments

ID	Question	Design	Primary
ROI-1	Map-briefed minute budget vs unstructured “help with homework” session	2-arm tutors	FEI per tutor-minute; pour-relapse
ROI-2	AI-first rehearsal + human join vs human-first explain-all vs AI-only	3-arm	`solo_transfer_pass`; help-abuse rate
ROI-3	Bastani-style hard-dump blocked vs allowed in Solver during practice week	2-arm	delayed no-AI probe (shared HELP/HINT)
ROI-4	Escalate-to-human packet precision vs open “book more hours” after AI fail	2-arm	repair success; cost per resolved gap
ROI-5	Parent CBC: “humans for joins / AI for reps” vs “unlimited AI tutor” vs “unlimited human chat”	CBC	WTP; trust (SAFE-WTP)
ROI-QUAL	10 Maya + 10 parents: when they want a human vs AI; shame if human “only” 20 focused minutes	Qual	codebook: entitlement vs relief

Falsifier: Unstructured human sessions ≥ Map-budgeted on FEI/minute → demote brief ritual cost.

Falsifier: AI-only ≥ AI+human-join on solo transfer for diagnosed bridge gaps → narrow human SKU (do not overclaim).

Falsifier: Parents prefer unlimited-AI CBC and show equal retention → rewrite GTM; still do not ship crutch UX.

Pre-register: ROI-* before “AI replaces tutors,” “unlimited human,” hours-as-NS, or Session ROI Score™ campaigns (SAFE-LABMETA). Do not steal HITL/HELP/HINT/REPAIR/TALK arms — densify them.

XCIX.13 So what for MindCraft commercially

- **Copy:** “Humans for the hard joins. AI for the reps.” Never tutoring-is-free or AI≡private-tutor.
- **Product:** Map minute budget; AI for rehearsal; human for diagnosis/destake/construction/repair/witness.
- **Positioning:** Against ChatGPT crutches and hours vanity; for scarce-human + constrained-AI complementarity.
- **Metric:** FEI per tutor-minute; escalate precision; solo-after-AI — demote hours / Session ROI Score™.

• **Kill list:** AI replaces tutors; unlimited human; hours NS; VanLehn-as-ChatGPT; Bastani practice≡learning; 0.37σ ACT guarantee.

• **Growth:** Parent decks show 3 human moves + AI reps plan — not a timesheet.

• **Vision:** Maya experiences aimed human minutes, then owns AI reps without crutches.

XCIX.14 Doctrine — SAFE-ROI (provisional)

1. **Tutor minutes are scarce inventory** — Nickow/Guryan/Kraft: tutoring works and costs real labor.
2. **Thin human time ≠ automatic impact** — Kraft et al. 2022 dosage wound.
3. **Constrained AI ≠ ChatGPT dump** — VanLehn ITS parity does not license crutches.
4. **Unguarded generative help can create learning debt** — Bastani et al. 2025.
5. **Allocate by Map task** — humans for joins/destake/repair/witness; AI for reps/fade/soft peeks.
6. **Measure FEI per tutor-minute** — not hours booked.
7. **No Session ROI Score™ / AI-replaces-tutors / unlimited-human brand**.
8. **Densifies SAFE-HITL × SAFE-HELP × SAFE-REPAIR** — complementarity with honesty.

Confidence: High that Nickow et al. (2020) and Guryan et al. (2023) establish large experimental tutoring impacts under structured dosage. High that Kraft & Falken (2021) establish scale/cost constraints. High that Kraft et al. (2022) show thin online volunteer models can be underpowered. High that VanLehn (2011) places step-based ITS near human averages without rescuing LLM dumps. High that Bastani et al. (2025) show unguarded GPT can harm post-removal performance. Medium that MindCraft's specific minute-budget UX raises FEI/minute — instrument ROI-* before AI-replaces-tutors or unlimited-human campaigns.

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Part C — Worked-Example vs Problem-Solving Timing Across Expertise

Chapter status: Living evidence + Practice/Solver schedule brief — Researcher tick 2026-08-08 (UTC hour 0 ≡ Red Team slot, but ch100 never written → prefer Researcher per rotation; researcher count since synthesizer v1.13 = 0 → Researcher)

Primary question: When should MindCraft schedule full worked examples vs completion vs conventional problem solving — especially when a fixed E0→E3 fade ladder collides with *prior knowledge that is already partial, format-specific, or misdiagnosed*?

Owners: Product (Practice / Solver fade machine) · Engine (Map grain / level / format) · HITL tutors · Brand · Red Team

Commercial job: Ship **SAFE-EXPTIME** densifying SAFE-FADE / SAFE-COLD / SAFE-FORMAT / SAFE-PF / SAFE-HINT: timing is diagnosed, not calendar cosplay; fixed ladders yield to rapid expertise checks; adaptive > fixed when grain is real — never Timing Score™, universal E0→E3 destiny, or "struggle-first always builds grit."

Builds on: Parts XXVI (CLT), LXXXVIII (SAFE-FADE), LXXXI (SAFE-COLD), LXXXII (SAFE-FORMAT), XCIV (SAFE-PF), XCI (SAFE-HINT), XCV (SAFE-APPRENTICE), XCIII (SAFE-RETRIEVE). Product seams: `fade_stage` E0–E3, gap-scan / Map eventCount, FormatId routing, completion blanks, solo transfer proof.

C.1 Why this chapter exists

SAFE-FADE already killed always-full-worked / never-fade / view≡mastery. What remains is the *schedule collision*: gap-scan "hard" → E0 full worked while Maya already half-knows the procedure but blanks on diagram FormatId → expertise-reversal tax + peek binge; or Map "kinda" → early E3 → freeze / help-abuse.

Competitors sell always-every-step (ChatGPT) or always-struggle-first grit theater. MindCraft's wedge is **when** — examples when schemas are thin; solving when they exist; completion as bridge; re-time when concept ≠ format ≠ bridge expertise.

FOUNDER BELIEF under audit: A fixed E0→E3 ladder without prior-knowledge checks is expertise-blind pedagogy with a curriculum skin. Honesty is "help matches *this* grain," not "we have a science fade sequence."

Claims we refuse as doctrine:

1. One universal E0→E3 schedule for all concepts/formats/students.
2. Always-example-first or always-problem-first as brand method.
3. Fixed fade ≡ adaptive fade (calendar costume without checks).
4. Gap-scan confidence alone ≡ fade stage (confidence≠schema; SAFE-CALIB / SAFE-MONITOR).
5. Example Timing Score™ / %worked as North Star.
6. Expertise reversal as license for zero guidance day one.
7. "We adapt perfectly" black-box AI timing hero.
8. Problem-solving first ≡ PF without consolidation gates (SAFE-PF).

C.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Worked-example effect	Studying full solutions beats unguided search for novices	E0 full coach / Solver study	Permanent dump
Expertise reversal	Novice-helpful guidance hurts / wastes experts	Same E0 after demonstrated grain	“They got bored” only
Completion / fading	Successive blanks until full solve	E1–E2 missing steps	Fake blanks + instant peek
Fixed fade	Same omit schedule for all	Default E0→E3 by level	Diagnosed timing
Adaptive fade	Fade point tracks measured understanding	Map×FormatId×recent solo	Black-box “AI knows”
SAFE-EXPTIME	Diagnose prior grain → choose example/completion/solve; recheck	This chapter	Timing Score™

Operational definition (HYPOTHESIS): A path is *EXPTIME-honest* when it (a) sets initial fade from **grain-local evidence** (concept×format×bridge), (b) **rechecks** via completion/SE/solo to move E up/down, (c) prefers **adaptive over fixed** when instrumented, (d) never equates gap-scan Likert with schema readiness, and (e) proves with `solo_transfer_pass` / delayed mixed accuracy — not %worked or explanation NPS.

C.3 Novices: worked examples beat unguided problem solving

FACT: Sweller & Cooper (1985, *Cognition and Instruction*, *2*(1), 59–89, doi:10.1207/s1532690xci0201_3) — algebra novices who studied worked examples as a substitute for conventional problem solving showed reduced time and errors on subsequent similar problems; conventional search imposed heavy load that retarded schema acquisition.

FACT: Sweller (1988, *Cognitive Science*, *12*(2), 257–285, doi:10.1207/s15516709cog1202_4) — problem-solving search can consume working memory needed for schema construction.

Applied (HYPOTHESIS): Day-one E3-only (“just try”) for true novices on a new grain violates CLT. SAFE-EXPTIME inherits SAFE-FADE’s stage-not-struggle-theater rule: start complete enough to build schema.

Kill: Problem-solving-first as universal brand for acquisition.

Survive: Full worked study as *early-stage* tool on thin grain.

C.4 Experts: problem solving can beat more examples

FACT: Kalyuga, Chandler, Tuovinen & Sweller (2001, *Journal of Educational Psychology*, *93*(3), 579–588, doi:10.1037/0022-0663.93.3.579) — as learners gain domain expertise, problem solving can become superior to studying worked examples (classic reverse pattern under CLT).

FACT: Kalyuga, Ayres, Chandler & Sweller (2003, *Educational Psychologist*, *38*(1), 23–31, doi:10.1207/s15326985ep3801_4) — methods highly effective for inexperienced learners can lose effectiveness or become detrimental as expertise grows because guidance becomes redundant processing.

FACT (review): Kalyuga (2007, *Educational Psychology Review*, *19*(4), 509–539, doi:10.1007/s10648-007-9054-3) — expertise reversal is a robust ATI-style pattern within CLT; learner-tailored systems must match support to knowledge state, not average syllabus week.

Commercial implication: Marketing "we walk every step with you" past partial mastery is ChatGPT costume with a curriculum skin. Map must *lower* default guidance after demonstrated competence on that grain.

Kill: Expertise-blind always-example brand.

Survive: Problem solving / sparse completion when grain evidence shows schemas exist.

C.5 Transition tool: fading / completion, not cliff jumps

FACT: Renkl & Atkinson (2003, *Educational Psychologist*, *38*(1), 15–22, doi:10.1207/S15326985EP3801_3) — theoretical CLT account: intrinsic load falls across skill acquisition so problem-solving elements can be *gradually* integrated into example study; propose fading from complete examples → incomplete → independent solving.

FACT: Renkl, Atkinson, Maier & Staley (2002, *Journal of Experimental Education*, *70*(4), 293–315, doi:10.1080/00220970209599510) — field + lab: successive fading of worked steps outperformed traditional example–problem pairs on near transfer; mediation via fewer errors during learning under fading.

FACT: Renkl, Atkinson & Große (2004, *Instructional Science*, *32*(1–2), 59–82, doi:10.1023/B:TRUC.0000021815.74806.f6) — fading associated with fewer *unproductive* learning events; position of faded steps mattered less than that fading triggered principle-focused processing on faded content.

Applied (HYPOTHESIS): MindCraft E1–E2 are not "half a ChatGPT dump." They are **completion problems** with named missing grain (procedure step, bridge join, format conversion). Cliff jumps E0→E3 without completion are the UX that creates freeze or peek binge.

Kill: Binary help-on / help-off without completion bridge.

Survive: Backward-leaning completion ladder as default bridge (SAFE-FADE), re-timed per C.6–C.7.

C.6 Fixed fade is not enough — adaptive beats fixed in tutored settings

FACT: Salden, Aleven, Schwonke & Renkl (2010, *Instructional Science*, *38*(3), 289–307, doi:10.1007/s11251-009-9107-8) — lab + classroom Geometry Cognitive Tutor studies: example-enhanced tutors with **adaptive** fading of worked steps outperformed **fixed** fading, which outperformed standard tutored problem solving alone on learning/retention outcomes discussed in the paper.

Wound: Cognitive Tutor grain ≠ MindCraft Map grain — transfer the *principle* (fade to understanding), not CMU costume ads. Shipping a fixed E0→E3 animation labeled "adaptive" is cite-wash.

Kill: Fixed ladder ≡ adaptive science.

Survive: Default ladder + instruments that move stage from evidence.

C.7 Rapid expertise checks — how to time without a Timing Score™

FACT: Kalyuga & Sweller (2004, *Journal of Educational Psychology*, *96*(3), 558–568, doi:10.1037/0022-0663.96.3.558) — schema-based rapid knowledge measures (algebra/geometry) correlate highly with traditional tests at much shorter time; used to select instructional design matched to expertise (expertise-reversal mitigation).

FACT: Kalyuga & Sweller (2005, *Educational Technology Research and Development*, *53*(3), 83–93, doi:10.1007/BF02504800) — learner-adapted instruction using rapid tests + cognitive-load indices yielded higher knowledge and cognitive-efficiency gains than a yoked non-adapted control in an elementary algebra tutor.

Applied (HYPOTHESIS): MindCraft already has cheap proxies — completion success, soft-wrong class, FormatId miss pattern, delayed solo, SE before wrap — that can play the *role* of rapid assessment without inventing Example Timing Score™. Gap-scan confidence is a *seed prior*, not a fade oracle (SAFE-COLD / SAFE-MONITOR).

Kill: One Likert → permanent E stage. Kill black-box "perfect interval of fading."

Survive: Inspectable rules: e.g., two consecutive completion successes on grain → fade up; freeze/TOT with no progress → soft bound then temporary fade down (SAFE-RETRIEVE), not instant hard dump.

C.8 Collision cases MindCraft will actually hit

HYPOTHESIS — prior knowledge ≠ zero on concept, zero on format: Student solves symbolic `linear_equations` but fails `coordinate_graph` isomorphic items. Correct timing: E0–E1 on *format conversion* grain, not re-teaching full algebra example dump (SAFE-FORMAT × SAFE-DUAL).

HYPOTHESIS — endpoints OK, bridge weak: Concept nodes look "easy"; join fails. Timing: co-present supports + completion on the *relation*, not more isolated worked examples (SAFE-BRIDGE).

HYPOTHESIS — overconfident cold start: High confidence, low probe accuracy. Timing: hide-correctness densifies evidence; do not gift E3 because they said "easy" (SAFE-CALIB / SAFE-MONITOR / SAFE-COLD).

HYPOTHESIS — exam dual rail: Late ACT prove week may need more problem solving / mixed retrieval than worked study — without relabeling cram as mastery (SAFE-EXAM / SAFE-DURABLE / SAFE-SCHED).

C.9 Product rules (MindCraft)

1. **Initial stage from grain-local evidence** — concept×format×bridge, not global belt.
2. **Default ladder exists** — E0 study → E1–E2 completion → E3 solo — but is a *prior*, not destiny.
3. **Adaptive overrides fixed** — success fades up; diagnosed freeze/miss-class can fade down one step, not flood.
4. **Completion is the commercial middle** — blank named join beats monologue.
5. **Attempt grain before reveal** — SE / soft-wrong before unlocking fuller worked (SAFE-FADE × SAFE-HINT).
6. **Proof = solo transfer / delayed mixed** — never %time-on-examples or explanation NPS.
7. **Tutor ops:** HITL brief names current fade stage + why; humans do not re-pour E0 after Map says E2 (SAFE-APPRENTICE / SAFE-ROI).
8. **Copy:** “Help that steps down as you step up” — not “always every step” or “always struggle first.”

C.10 Confidence table

Claim	Label	Confidence
Worked examples beat unguided PS for novices (Sweller & Cooper 1985; Sweller 1988)	FACT	High
Problem solving can beat examples as expertise rises (Kalyuga et al. 2001; 2003)	FACT	High
Fading/completion bridges example→solve better than cliff or static pairs (Renkl & Atkinson line)	FACT	High
Adaptive fade > fixed fade > PS-only in Cognitive Tutor studies (Salden et al. 2010)	FACT	High (wound: tutor context)
Rapid dynamic assessment can improve efficiency of adapted instruction (Kalyuga & Sweller 2004/2005)	FACT	High
MindCraft Map×FormatId proxies can substitute for lab rapid tests	HYPOTHESIS	Medium — EXPTIME-*
Fixed E0→E3 without recheck harms partial-knowledge Maya	HYPOTHESIS	Medium–High product bet
Example Timing Score™ / always-struggle-first improves FEI	**Banned**	—

C.11 Experiments

ID	Question	Design	Primary
EXPTIME-1	Fixed E0→E3 vs adaptive fade (completion success + miss-class rules) on same concept	2-arm	`solo_transfer_pass`; time-to-solo; peek binge
EXPTIME-2	Concept-matched vs concept×format grain for initial stage	2-arm	format conversion accuracy; redundant-example rate
EXPTIME-3	Gap-scan confidence→E stage vs probe-evidence→E stage	2-arm	calibration of stage; early quit; FEI
EXPTIME-4	Adaptive fade vs always-E0 worked vs always-E3 solo on mixed prior-knowledge cohort	3-arm	delayed mixed accuracy; mental effort proxy
EXPTIME-5	Parent CBC: "steps down as you step up" vs "always every step" vs "struggle-first grit"	CBC	WTP; trust (SAFE-WTP)
EXPTIME-QUAL	10 Maya: when full examples felt helpful vs insulting; when solo felt fair vs flooding	Qual	codebook: reversal pain / freeze

Falsifier: Fixed ladder ≥ adaptive on solo transfer across prior-knowledge strata → keep fixed as ops default; demote adaptive cost.

Falsifier: Always-E0 ≥ adaptive for partial-knowledge students on delayed tests → wound expertise-reversal product claim for our bank.

Falsifier: Parents prefer always-every-step CBC with equal retention → rewrite copy; still do not ship permanent dump.

Pre-register: EXPTIME-* before "perfect adaptive fade AI," struggle-first, or Timing Score™ campaigns. Densify FADE/HINT/COLD/FORMAT — do not fork a second fade machine.

C.12 So what for MindCraft commercially

- **Copy:** "Help that steps down as you step up — on *this* skill, not a fake level." Never always-every-step or grit-first acquisition.
- **Product:** Grain-local initial E stage; completion bridge; adaptive up/down from evidence; inspectable why.
- **Positioning:** Against ChatGPT permanent worked solutions *and* against struggle-theater apps that skip schema building.

- **Metric:** Stage appropriateness (miss-class × stage), time-to-honest-solo, `solo_transfer_pass` — demote %worked and Timing Score™.
- **Kill list:** Universal E0→E3 destiny; fixed≡adaptive; confidence≡stage; always-example / always-PS brand; black-box perfect fade AI.
- **Growth / vision:** Parent demo shows the same problem at E0 vs E2 for different Map states — Maya feels guided when thin and trusted when ready; identity is the handoff, not dump or flood.

C.13 Doctrine — SAFE-EXPTIME (provisional)

1. **Timing tracks expertise grain** — concept×format×bridge, not global belt.
2. **Examples early, solving late — on that grain** — Sweller/Kalyuga line.
3. **Completion fading is the bridge** — Renkl/Atkinson; not cliff jumps.
4. **Adaptive > fixed when evidence exists** — Salden et al. 2010 principle.
5. **Rapid rechecks beat Likert destiny** — Kalyuga & Sweller rapid assessment role.
6. **Densifies SAFE-FADE** — schedule collision chapter, not a second chrome.
7. **Proof = solo / delayed mixed** — not explanation time share.
8. **No Example Timing Score™ / always-struggle-first / black-box perfect-fade hero**.

Confidence: High on worked-example effect, expertise reversal, fading, and adaptive>fixed in published tutor studies. Medium that Map×FormatId rules will match Salden-class gains — instrument EXPTIME-* before adaptive-fade marketing.

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Part CI — Self-Regulated Learning Cycles (Zimmerman) in Practice UX

Chapter status: Living evidence + Practice session UX brief — Researcher tick 2026-08-08 (UTC hour 3; hour%6≠0; researcher count since synthesizer v1.13 = 1 → Researcher)

Primary question: How should MindCraft embed Zimmerman's cyclical SRL phases — forethought, performance, self-reflection — into Practice without minting an SRL Score™, reflection-journal theater, or streak-as-self-regulation costume?

Owners: Product (Practice / Notes / Map handoffs) · HITL tutors · Brand · Red Team

Commercial job: Ship **SAFE-SRL** densifying SAFE-MONITOR / SAFE-CALIB / SAFE-ATTN / SAFE-HELP / SAFE-APPRENTICE / SAFE-AAR: phase-timed micro-prompts that change control (goal grain, strategy choice, attribution→next plan), prove value with FEI + strategy-adaptive retries — never "we teach metacognition" splash without wiring.

Builds on: Parts XXV (self-efficacy), XXX (attribution), L (SAFE-CALIB), LIV (SAFE-AAR), XCV (SAFE-APPRENTICE), XCVI (SAFE-ATTN), XCVIII (SAFE-MONITOR), XC (FEI instrumentation). Product seams: mission launch, attempt windows, soft-wrong, AAR/annotation, Map return CTAs.

CI.1 Why this chapter exists

Competitors sell either autopilot (streak → next item) or post-hoc journaling ("reflect on your learning"). Both skip the load-bearing claim in social-cognitive SRL: regulation is a **cycle** that must fire *around* a real task — goals and plans *before* effort, monitoring and control *during*, judgment and adaptive inference *after* — and the after must rewrite the before.

MindCraft already has pieces (gap-scan ease, soft-wrong surety, film/AAR cards, Map). What is missing is doctrine that forbids fusing them into one vanity SRL meter while requiring **phase→control** wiring so Maya's next attempt actually changes.

FOUNDER BELIEF under audit: Identity transformation needs students who can run a thin cycle without a tutor present — but the product earns that claim by scaffolding *events*, not by advertising Self-Regulation Scores.

Claims we refuse as doctrine:

1. SRL Score™ / Metacognition % / Reflection streak as North Star.

2. “We teach self-regulated learning” ads without phase-timed control actions.
3. Global SRL surveys ≡ event regulation on *this* problem.
4. Forethought theater (goal picker) that never constrains strategy or difficulty.
5. Endless end-of-session journals as proof of reflection.
6. Streak / DAU ≡ self-regulation.
7. Ability attributions (“you’re not a math person”) as brand voice; defensive withdrawal as grit.
8. Guaranteed ACT points from “SRL-powered Practice.”

CI.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Forethought	Task analysis, goals, strategic planning, motivational beliefs before effort	Mission brief: grain goal + plan chip + efficacy check	Vague “get better at math”
Performance	Self-control + self-observation during the task	Attempt window; soft hints; monitoring prompts mid-stall	Mid-attempt DAU push
Self-reflection	Self-judgment (eval, attribution) + self-reaction (adaptive vs defensive)	Soft-wrong class + one adaptive inference → next plan	Shame reel / Calm Score™
Cyclical link	Reflection alters subsequent forethought	Miss → strategy attribution → new plan chip on retry	XP toast that resets nothing
Microanalysis	Brief, task-timed questions at phase boundaries	1–2 prompts pre/during/post attempt	20-item SRL inventory
SAFE-SRL	Practice doctrine for phase→control cycles under FEI	This chapter	SRL costume ads

Operational definition (HYPOTHESIS): A Practice path is *SAFE-SRL compliant* when (a) each FEI-relevant attempt has at least thin forethought (specific goal + strategy intent), (b) performance-phase support protects construction (SAFE-ATTN / SAFE-HINT) rather than interrupting for surveys, (c) reflection elicits **controllable** attributions and one adaptive next step, (d) that next step is *written into* the following forethought (visible plan change), and (e) success is measured by strategy-adaptive `retry_120s`, `challenge_accept` motives, and `solo_transfer_pass` — not by reflection completion rates.

CI.3 The cyclical model is company vocabulary — not a sticker

FACT: Zimmerman (2000, in Boekaerts, Pintrich & Zeidner, *Handbook of Self-Regulation*, Academic Press, pp. 13–39, doi:10.1016/B978-012109890-2/50031-7) — social-cognitive account of attaining self-regulation; self-regulatory processes organized as interdependent **forethought** → **performance** → **self-reflection** cycles, with motivational beliefs (e.g., self-efficacy) energizing the loop.

FACT: Zimmerman (2002, *Theory Into Practice*, *41*(2), 64–70, doi:10.1207/s15430421tip4102_2) — practitioner overview of the same three-phase structure: processes *before*, *during*, and *after* learning efforts; reflections from prior efforts feed subsequent forethought (e.g., dissatisfaction → lowered efficacy → diminished later effort unless attributions and plans are repaired).

FACT: Zimmerman & Moylan (2009, in Hacker, Dunlosky & Graesser, *Handbook of Metacognition in Education*, Routledge, pp. 299–315) — refined cyclical model emphasizing the intersection of metacognition and motivation; performance-phase volitional/self-control processes sit alongside monitoring.

FACT (review): Panadero (2017, *Frontiers in Psychology*, *8*, 422, doi:10.3389/fpsyg.2017.00422) — places Zimmerman's cyclical phases among major SRL models; phases are descriptive process architecture, not a license for a single composite "SRL ability" score.

Kill: One SRL Score™ collapsing phases.

Survive: Named phase moments with named controls.

CI.4 Forethought: specific goals and strategies beat vibe goals

FACT: Cleary & Zimmerman (2001, *Journal of Applied Sport Psychology*, *13*(2), 185–206, doi:10.1080/104132001753149883) — adolescent basketball free-throw microanalysis: experts set more **specific** goals, selected more **technique-oriented** strategies, and reported higher self-efficacy than non-experts and novices; forethought processes intercorrelated; self-reflection attributions predicted later strategy selection (cyclical evidence).

Applied (HYPOTHESIS): Mission launch should ask for a *grain-local* goal ("finish this FormatId conversion without hard peek") and a strategy chip (diagram first / equation first / bridge checklist) — not "crush algebra." Efficacy prompts stay task-specific (SAFE-MONITOR / Bandura grain), never identity Likert.

Commercial implication: Copy "plan the attempt, not the streak." Competitors can autoplay the next item; MindCraft sells a 10-second forethought that changes what counts as success on *this* card.

Kill: Goal theater with no strategy constraint.

Survive: Specific process goals + strategy intent as first-class events.

CI.5 Performance: protect the attempt; monitor without survey spam

FACT: Zimmerman's performance phase includes self-control (attention, strategies) and self-observation (Zimmerman, 2002; Zimmerman & Moylan, 2009). Heavy instrumentation mid-task can become the task.

HYPOTHESIS (product): Mid-attempt UX follows SAFE-ATTN + SAFE-RETRIEVE + SAFE-HINT: protect the window; mode-contingent stalls; soft before hard. Metacognitive monitoring prompts (if any) are single, stall-triggered, and routed to control — not a running Metacognition Score.

Wound: Cleary/Zimmerman athletic microanalysis pauses are interviewer-administered; app prompts must be thinner. Prefer silent telemetry + rare prompts over constant "how are you regulating?" popups.

Kill: Mid-attempt SRL questionnaire / Focus Score™.

Survive: Construction-first performance phase with contingent help.

CI.6 Self-reflection: controllable attributions → adaptive inferences

FACT: Zimmerman (2002) — attributing a poor math score to controllable process causes (wrong strategy) sustains motivation because a different strategy may work; ability attributions and defensive reactions (withdraw, skip tests) undermine further learning cycles.

FACT: Cleary, Zimmerman & Keating (2006, *Research Quarterly for Exercise and Sport*, *77*(2), 251–262, doi:10.1080/02701367.2006.10599358) — additive training across phases: groups trained in more phases showed better free-throw accuracy and more frequent self-correction after misses; three-phase trainees showed the most adaptive profile (strategic attributions, adaptive inferences, process criteria in self-evaluation).

Applied (HYPOTHESIS): Soft-wrong / post-mission reflection should offer **process/strategy** attribution chips ("picked wrong representation," "skipped bridge check") and one adaptive inference that becomes the next forethought plan — densifying SAFE-AAR / SAFE-ANNOT without military or grandmaster cosplay.

Kill: Ability shame; reflection without next-plan writeback; defensive skip framed as grit.

Survive: Strategy attribution → visible plan change on retry.

CI.7 Microanalysis > inventories for product truth

FACT: Cleary, Callan & Zimmerman (2012, *Education Research International, 2012*, Article 428639, doi:10.1155/2012/428639) — SRL **microanalysis**: brief, task-specific questions administered at temporal phase boundaries (before/during/after a defined task); targets subprocesses (goal-setting, strategic planning, monitoring, self-evaluation, attributions); designed as contextualized event assessment rather than decontextualized trait surveys.

Commercial implication: Instrument Practice with microanalytic *events* (`forethought_goal_type`, `strategy_intent`, `attribution_class`, `adaptive_inference_applied`) under SAFE-INSTRUMENT. Do not ship a 40-item "Are you a self-regulated learner?" onboarding that parents confuse with proof.

Kill: Survey SRL ≡ FEI readiness.

Survive: Phase-boundary micro-prompts as telemetry + light UX.

CI.8 Interventions work when strategy + motivation + reflection combine — with caveats

FACT: Dignath & Büttner (2008, *Metacognition and Learning, 3*(3), 231–264, doi:10.1007/s11409-008-9029-x) — meta-analysis of SRL interventions at primary and secondary levels: training can improve academic performance, strategy use, and motivation; effects vary by components (metacognitive reflection often important) and context (subject and grade moderators).

FACT: Dignath, Büttner & Langfeldt (2008, *Educational Research Review, 3*(2), 101–129, doi:10.1016/j.edurev.2008.02.003) — primary-school focused meta-analysis: SRL programmes can be effective even early; combinations of metacognitive and motivational elements matter; mathematics effects notable in primary samples in their synthesis — not a blank check for HS ACT apps.

Wound (HYPOTHESIS): Lab and school interventions ≠ 12-minute app sessions. MindCraft should claim **cycle scaffolding**, not "SRL training effect sizes transfer to our DAU." Secondary-school moderators in Dignath & Büttner warn against copying primary-math packages wholesale into ACT prep.

Kill: "Research shows SRL raises grades — therefore our reflection journal works."

Survive: Multi-phase micro-scaffolds + FEI outcomes as the honesty bar.

CI.9 Where competitors fail the cycle

Competitor pattern	Broken phase	MindCraft counter
Duo-style streak autopilot	Weak forethought; reflection = XP	Goal+strategy chips; adaptive inference writeback
ChatGPT dump	No student performance phase	Attempt-first; SAFE-FADE / SAFE-HINT
Brilliant delight streak	Reflection as dopamine	Process attribution; solo proof
School portal journals	Reflection detached from next task	One inference → next mission plan
"Metacognition module" quiz	Survey ≠ event	Microanalysis at phase boundaries

FOUNDER BELIEF: The wedge is not more content about SRL. It is a Practice loop that *is* an SRL cycle thin enough for Maya and rich enough for tutors to coach (SAFE-APPRENTICE articulate step).

CI.10 Decision table (ship / wound / kill)

Claim	Verdict	Notes
Three-phase cycle as UX architecture	**Ship (provisional)**	Zimmerman line; wire controls
Microanalytic events > trait SRL surveys	**Ship**	Cleary et al. 2012

Specific process goals + strategy intent	**Ship**	Cleary & Zimmerman 2001 analogue
Controllable attributions → plan writeback	**Ship**	Zimmerman 2002; Cleary et al. 2006
Multi-phase training additive	**Wound→test**	Motor-skill RCT; math transfer open
Meta-analytic SRL → ACT points	**Kill as ad**	Context/moderator wound
SRL Score™ / reflection streak NS	**Kill**	Standing ban family
Streak ≡ self-regulation	**Kill**	HID / SAFE-INSTRUMENT

CI.11 Experiments

ID	Question	Design	Primary
SRL-1	Forethought chips (goal+strategy) vs autopilot next-item on same bank	2-arm	`solo_transfer_pass`; peek binge; time-to-honest-attempt
SRL-2	Post-miss strategy attribution + plan writeback vs XP-only reflection	2-arm	adaptive `retry_120s`; strategy-change rate; early quit
SRL-3	Microanalytic 3-prompt cycle vs 8-item end journal	2-arm	FEI; completion without fatigue; tutor QA fidelity
SRL-4	Ability-attribution UI vs process-attribution UI after soft-wrong	2-arm	challenge_accept motives; defensive skip
SRL-5	Parent CBC: "plans the attempt" vs "builds metacognition habits" vs streak hero	CBC	WTP; trust (SAFE-WTP)
SRL-QUAL	10 Maya: when planning felt useful vs naggy; when reflection changed next try	Qual	codebook: theater vs control

Falsifier: Autopilot ≥ chip forethought on solo transfer and peek binge → demote forethought cost; keep optional.

Falsifier: Journal arm ≥ microanalytic cycle on FEI with less friction → wound microanalysis product claim; keep event telemetry only.

Falsifier: Parents prefer metacognition-habit CBC with equal retention → rewrite copy; still do not ship SRL Score™.

Pre-register: SRL-* before "AI metacognition coach," reflection streaks, or SRL Score™ campaigns. Densify MONITOR/AAR/HELP/ATTN — do not fork a second dashboard religion.

CI.12 So what for MindCraft commercially

- **Copy:** "Plan the attempt. Adjust the strategy. Prove it alone." Never "unlock your inner self-regulated learner" or reflection-streak hero.
- **Product:** Phase-boundary micro-prompts with writeback; strategy attribution chips; protect mid-attempt.
- **Positioning:** Against streak autopilot and against journal theater — MindCraft is the FEI gym with a visible cycle.
- **Metric:** Strategy-adaptive retries, motive-coded challenge accepts, solo transfer — demote reflection completion % and SRL Score™.
- **Kill list:** SRL Score™; survey≡event; streak≡SRL; forethought without control; ability-shame reflection; ACT points from metacognition modules.
- **Growth / vision:** Parent/tutor demo shows Maya miss → process attribution → new strategy chip → cleaner retry — identity is owning the cycle, not collecting journal badges.

CI.13 Doctrine — SAFE-SRL (provisional)

1. **SRL is a cycle around a real attempt** — forethought → performance → reflection → rewritten forethought.
2. **Phase→control or it is costume** — every prompt must change goal, strategy, help, or next plan.
3. **Specific process goals beat vibe goals** — grain-local, technique-oriented.
4. **Performance protects construction** — no mid-attempt survey religion (SAFE-ATTN).
5. **Controllable attributions → adaptive inferences** — ability shame and defensive skip are fails.
6. **Microanalysis > inventories** for product truth (Cleary lineage).
7. **Proof = FEI + strategy change** — not reflection NPS or SRL Score™.
8. **No SRL Score™ / streak-as-SRL / metacognition-module ACT guarantees**.

Confidence: High on cyclical architecture, microanalysis vs surveys, and attribution/adaptive-inference logic. Medium that thin app chips will match multi-phase training gains — instrument SRL-* before metacognition marketing.

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Part CII — Generative Learning Activities (Fiorella/Mayer) in Practice

Chapter status: Living evidence + Practice UX brief — Researcher tick 2026-08-08 (UTC hour 6 ≡ Red Team slot, but ch102 never written → prefer Researcher per rotation; researcher count since synthesizer v1.13 = 2 → Researcher)

Primary question: How should MindCraft ship Fiorella/Mayer generative learning activities — summarize, map, draw, imagine, self-test, self-explain, teach, enact — inside Practice without minting a Generative Learning Score™, journal theater, or “learning-by-doing” costume that never forces select–organize–integrate?

Owners: Product (Practice / Notes / Map / Solver handoffs) · Brand · HITL tutors · Red Team

Commercial job: Ship **SAFE-GENERATE** densifying SAFE-EXPLAIN / SAFE-DUAL / SAFE-SRL / SAFE-SE / SAFE-RETRIEVE / SAFE-ATTN: thin, timed generative asks that change what the student *constructs*, prove value with solo/delayed transfer — never activity-checklist ads or AI monologue sold as “generation.”

Builds on: Parts XXVI (CLT), XL (self-explanation), XCII (SAFE-EXPLAIN), XCVII (SAFE-DUAL), XCVIII (SAFE-MONITOR), CI (SAFE-SRL), XCIII (SAFE-RETRIEVE). Product seams: soft-wrong, coach wrap, FormatId conversion, teach-back after grain, Map return.

CII.1 Why this chapter exists

Competitors sell either (a) passive consumption (watch solution → green check) or (b) activity theater (summarize this lesson / draw anything / teach a friend for XP). Generative learning research says the load-bearing move is

sense-making: the learner selects relevant structure, organizes it, and integrates it with prior knowledge so transfer becomes possible. MindCraft's brand risk is claiming "active learning" while shipping either AI essays the student reads or checkbox activities that never reorganize math structure.

FOUNDER BELIEF under audit: Identity as a mathematical thinker requires *generating* explanations, sketches, and teach-backs — not collecting them from the model. The product earns that claim by wiring generative asks to FEI attempts, not by advertising eight strategies as a curriculum module.

Claims we refuse as doctrine:

1. Generative Learning Score™ / Activity Completion % as North Star.
2. "We use generative learning science" ads without SOI-forcing prompts.
3. AI wrap / monologue ≡ student generation (already SAFE-EXPLAIN / kill #8).
4. Verbatim copy-summary / highlight theater as summarizing.
5. Unguided free-draw as dual coding / format mastery.
6. Teach-back as social XP or influencer cosplay without content quality.
7. Eight-strategy checklist before every item (germane-load bomb).
8. Guaranteed ACT points from "generative Practice."

CII.2 Constructs (keep mechanisms distinct)

Construct	Research meaning	MindCraft analogue	Not the same as
Generative learning	Reorganize + integrate with prior knowledge for transfer (Wittrock; Fiorella & Mayer)	Student-produced structure after/during attempt	Passive solution view
SOI	Select → Organize → Integrate (Mayer)	Prompt that forces pick/structure/link	"Reflect in 3 sentences" vibe
GLA / eight strategies	Summarize, map, draw, imagine, self-test, self-explain, teach, enact	Sparse toolkit chips, grain-timed	Full eight before every card
Constructive (ICAP)	Generate ideas beyond given material	Soft-wrong SE; teach-back; structural draw	Clicking / underlining / rewatching
SAFE-GENERATE	Practice doctrine for SOI-forcing micro-GLAs under FEI	This chapter	Activity costume ads

Operational definition (HYPOTHESIS): A Practice path is *SAFE-GENERATE compliant* when (a) generative asks require student production that cannot be satisfied by pasting the coach text, (b) timing protects the attempt window (SAFE-ATTN) — generation is pre-brief, stall-contingent, or post-attempt, never mid-solve spam, (c) strategy choice is grain-local (e.g., draw for diagram FormatId; teach-back after bridge grain), (d) quality gates exist (structure check, key idea coverage) without a vanity Generative Score, and (e) success is measured by `solo_transfer_pass` / delayed mixed / format-hop — not GLA completion rates.

CII.3 Generative learning is company vocabulary — not a sticker pack

FACT: Wittrock (1974, *Educational Psychologist, 11*(2), 87–95, doi:10.1080/00461527409529129) — generative model: learners actively construct meanings consistent with prior knowledge; learning with understanding is generation and transfer of meaning from background and experience, not passive reception.

FACT: Wittrock (1989, *Educational Psychologist, 24*(4), 345–376, doi:10.1207/s15326985ep2404_2) — generative processes of comprehension: generation, motivation, attention, memory; mind as active constructor, not passive consumer.

FACT: Mayer (1996, *Educational Psychology Review, 8*(4), 357–371, doi:10.1007/BF01463939) — SOI model: select relevant information, organize into coherent structure, integrate with prior knowledge for sense-making from expository text.

FACT: Fiorella & Mayer (2016, *Educational Psychology Review, 28*(4), 717–741, doi:10.1007/s10648-015-9348-9; online 2015) — eight generative strategies with median effect sizes ≥ ~0.40 across reviewed literatures (summarizing *d*≈0.50; mapping ~0.62–1.07 by organizer type; drawing ~0.40; imagining ~0.65; self-testing ~0.57; self-explaining ~0.61; teaching ~0.77; enacting ~0.51); grounded in Wittrock + Mayer SOI; boundary conditions emphasized per strategy.

FACT (book): Fiorella & Mayer (2015, *Learning as a Generative Activity*, Cambridge University Press, doi:10.1017/CBO9781107707085) — same eight strategies with practical recommendations and boundary conditions for implementation.

FACT: Chi & Wylie (2014, *Educational Psychologist, 49*(4), 219–243, doi:10.1080/00461520.2014.965823) — ICAP: Passive < Active < Constructive < Interactive for learning outcomes; generative processing aligns with constructive/interactive modes, not passive receipt or mere manipulative “active” behaviors (underlining, verbatim notes).

Kill: Sticker pack of eight activities without SOI.

Survive: Named generative moves that force construction.

CII.4 Self-explain and self-test already live — do not rebrand them as a new religion

FACT: Self-explaining and self-testing are two of the eight GLAs with strong support (Fiorella & Mayer, 2016). Self-explanation science is already Part XL / SAFE-EXPLAIN densifier; retrieval/self-testing is SAFE-RETRIEVE / SAFE-SCHED.

HYPOTHESIS (product): SAFE-GENERATE does **not** fork a parallel SE or retrieval stack. It adds the *missing* GLAs (summarize-in-own-words after grain, structural map/draw, teach-back) and forbids calling coach monologue “self-explanation.”

Commercial implication: Marketing may say “students generate, not consume” — never “powered by eight generative strategies™.” Competitors can list Dunlosky techniques; MindCraft sells *when* generation happens relative to FEI attempts.

Kill: Eight-box curriculum module.

Survive: Sparse GLA chips densifying existing SE/retrieval doctrine.

CII.5 Summarize and teach: production quality beats presence

FACT: Effective summarizing requires restating main ideas in one’s own words — not verbatim copying (Fiorella & Mayer, 2016). Verbatim notes/highlighting map to ICAP *active*, not *constructive* (Chi & Wylie, 2014).

FACT: Fiorella & Mayer (2013, *Contemporary Educational Psychology*, *38*(4), 281–288, doi:10.1016/j.cedpsych.2013.06.001) — teaching expectancy improved immediate comprehension vs study-for-test ($d \approx 0.59$ in reported contrast); preparing-to-teach alone did not reliably sustain 1-week delay benefits; students who both prepared and actually taught showed delayed benefits — explanation act matters for durability.

FACT: Fiorella & Mayer (2014, *Contemporary Educational Psychology*, *39*(2), 75–85, doi:10.1016/j.cedpsych.2014.01.001) — expectations and explanations in learning-by-teaching: long-term benefits tied to generating explanations, not expectancy alone; quality of explanation (knowledge-building vs knowledge-telling) is critical.

Applied (HYPOTHESIS): After a bridge/grain win (or soft-wrong repair), offer a **teach-fictive-peer** card: 2–4 sentences or voice note explaining the *structure* (what to select, how pieces connect) without the solution stem visible. Reject paste-from-coach; optional rubric: names the bridge / FormatId conversion / key constraint. Summarize prompts are grain-local (“one sentence: what changed when you moved formats?”), not end-of-session essay.

Commercial implication: Parent/tutor demo shows Maya teaching the bridge back — competence proof (SAFE-PROOF) without influencer “teach TikTok” cosplay.

Kill: Teach-for-XP / expectancy-only theater.

Survive: Actual explanation production with delayed-transfer proof.

CII.6 Draw / map / imagine: guidance and FormatId, or cognitive tax

FACT: Fiorella & Zhang (2018, *Educational Psychology Review*, *30*(3), 1115–1137, doi:10.1007/s10648-018-9444-8) — learner-generated drawing helps vs reading-only or text-focused strategies; effects vs instructor-provided illustrations are mixed and often require high guidance (partially provided figures, compare-to-model); drawing is cognitively demanding; accuracy and monitoring mediate benefits.

FACT (age/support boundary): Brod (2021, *Educational Psychology Review*, *33*, 1295–1318, doi:10.1007/s10648-020-09571-9) — reviews which generative strategies have evidence at which school ages; support and prerequisites matter; drawing benefits vs provided illustrations often need substantial instructional support.

HYPOTHESIS (product): Drawing/mapping is **FormatId-contingent** and scaffolded: blank canvas only when grain is spatial/diagrammatic and prior model exists; prefer partially provided skeletons (axes, triangle template, number-line ticks) + "complete the structure" — aligns SAFE-DUAL (structural figures) and SAFE-EXPTIME (expertise-sensitive generation). Imagining (mental sketch) is a low-friction alternative when motor/UI cost is high — still require a check (label three parts aloud).

Kill: Always-draw / decoration doodle ≡ dual coding.

Survive: Guided structural generation on diagram/graph grains.

CII.7 Boundary conditions: germane load, anxiety, and timing

FACT: Generative strategies can fail under high element interactivity, poor training, or when compared to strong provided models (Fiorella & Mayer, 2016; Fiorella & Zhang, 2018). CLT / expertise reversal (Parts XXVI, C / SAFE-EXPTIME) imply novices may need worked structure before free generation.

HYPOTHESIS: SAFE-GENERATE inherits SAFE-DD × anxiety and SAFE-EXPOSE: destake generation (no public shame reel); dose GLAs (one micro-ask, not eight); never flood anxious Maya with teach-back on day-one ACT items. Mid-attempt generation spam is banned (SAFE-ATTN). Prefer post-attempt SOI prompts or stall-contingent SE already in SAFE-EXPLAIN.

Contradiction to watch: Activity completion can *feel* productive while transferring poorly if generation is shallow (knowledge-telling). Red Team must kill any GLA completion NS.

Kill: Generation-everywhere / struggle-as-generation.

Survive: Timed, scaffolded, grain-local GLAs.

CII.8 Competitive contrast (mechanism lens)

Competitor pattern	Mechanism	MindCraft counter
Solution video → green	Passive / active ICAP	Attempt-first + post generative ask
Unlimited AI essay	Coach generates; student consumes	SE/teach-back *by student* before wrap (SAFE-EXPLAIN)
"Active learning" checklist	Activity theater	SOI-forcing chips + FEI proof
Free-draw sticker	Unguided drawing tax	Guided FormatId skeletons (SAFE-DUAL)
Teach-for-streak	Social XP	Teach-fictive-peer + delayed transfer

Positioning line (FOUNDER BELIEF → copy test): "You generate the structure. The coach doesn't do it for you."

CII.9 Confidence matrix

Claim	Label	Confidence	Notes
Generative ≠ passive consumption; SOI is the spine	FACT (theory) + HYPOTHESIS (product map)	High	Wittrock; Mayer SOI; Fiorella & Mayer
Eight strategies median $d \geq \sim 0.4$ in reviewed literatures	FACT (review summary)	High on direction; medium on portability to ACT Practice UI	Do not quote as guaranteed product lift
Teach act > expectancy alone for delay	FACT	High	Fiorella & Mayer 2013/2014
Unguided draw often loses to provided illustrations	FACT (boundary)	High	Fiorella & Zhang 2018
Thin app GLAs → solo transfer lift	HYPOTHESIS	Medium	Needs GENACT-*
Generative Learning Score™ helps learning	SPECULATION (reject)	—	Kill

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CII.10 Experiments

ID	Question	Design	Primary
GENACT-1	Post-grain teach-fictive-peer vs solution-view-only	2-arm	delayed `solo_transfer_pass`; explanation quality rubric
GENACT-2	Guided structural draw (skeleton) vs free-draw vs provided figure only on diagram FormatId	3-arm	format-hop accuracy; time-on-tax; draw accuracy
GENACT-3	One-sentence SOI summarize after soft-wrong vs XP-only continue	2-arm	misconception recurrence; adaptive `retry_120s`
GENACT-4	Eight-strategy menu every item vs sparse grain-triggered GLA	2-arm	FEI; quit; peek binge (load)
GENACT-5	Parent CBC: “generates the explanation” vs “active learning strategies” vs streak hero	CBC	WTP; trust (SAFE-WTP)

GENACT-QUAL	10 Maya: which generative ask felt useful vs busywork; teach-back shame	Qual	codebook: SOI vs costume

Falsifier: Solution-view ≥ teach-back on delayed solo transfer → demote teach-back cost; keep optional Notes export only.

Falsifier: Free-draw ≥ guided skeleton on format-hop with less time → wound guidance claim; still ban decoration doodle ads.

Falsifier: Eight-menu ≥ sparse triggers on FEI with equal quit → still kill menu-as-brand; instrument load.

Pre-register: GENACT-* before “generative learning OS,” activity Score™, or eight-strategy splash campaigns. Densify EXPLAIN/DUAL/SRL/SE — do not fork a second pedagogy religion.

CII.11 So what for MindCraft commercially

- **Copy:** “Generate the structure. Then prove it alone.” Never “complete all eight generative strategies” or Generative Learning Score™.
- **Product:** Grain-triggered teach-back, one-sentence SOI summary, guided FormatId skeletons; protect attempt windows.
- **Positioning:** Against passive AI tutors and against busywork activity apps — MindCraft is the FEI gym where *Maya* constructs meaning.
- **Metric:** Solo/delayed transfer, format-hop, explanation quality samples — demote GLA completion %.
- **Kill list:** Generative Learning Score™; AI monologue≡generation; verbatim summary theater; unguided doodle≡dual coding; teach-for-XP; eight-menu load bomb; ACT guarantees from GLA branding.
- **Growth / vision:** Parent proof artifact = student’s teach-back or structural sketch on a bridge grain — identity evidence without star-walls.

CII.12 Doctrine — SAFE-GENERATE (provisional)

1. **Generation = SOI** — select, organize, integrate; else costume.
2. **Student produces; coach does not substitute** — densifies SAFE-EXPLAIN.
3. **Sparse, grain-triggered GLAs** — not eight-before-every-item.
4. **Teach-back requires the explanation act** — expectancy/XP alone insufficient.
5. **Draw/map need guidance on FormatId grains** — densifies SAFE-DUAL; no decoration.
6. **Protect performance windows** — no mid-solve generative spam (SAFE-ATTN).
7. **Proof = transfer / format-hop / quality sample** — not activity completion.
8. **No Generative Learning Score™ / AI≡generation / eight-strategy ACT ads**.

Confidence: High on SOI/ICAP constructive requirement, teach-back durability evidence, and drawing boundary conditions. Medium that thin mobile GLAs match lab effect sizes — instrument GENACT-* before generative-learning marketing.

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